airservices australia

AERONAUTICAL INFORMATION PUBLICATION AUSTRALIA

CONSULT NOTAM AND AIP SUPPLEMENT FOR LATEST INFORMATION

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AIS:	AUSTRALIA	AIP BOOK
AFTN:YSHOYOYX **DISTRIBUTION:** Tel:1300 306 630 (Local call Aust. wide except from mobile)	AERONAUTICAL INFORMATION SERVICE AIRSERVICES AUSTRALIA GPO BOX 367 CANBERRA ACT 2601	**Amendment No 129** **26 NOV 2026** **202611251600 UTC** **AIRAC** **(Next 18 MAR 27)**

AMENDMENT INSTRUCTIONS

1. Insert the attached replacement pages for AIP Book (129) **effective 26 NOV 2026** in accordance with the CHECKLIST OF AIP PAGES (GEN 0.4).
2. Record the amendment GEN 0.2.

Note1: Recipients are encouraged to carry out a full page check after amending the Book to check the integrity of the document.

Note2: A complete or amendment text can be ordered through the CanPrint Communications online AIP Shop:

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Personal purchase: the locations of reseller outlets in each state and territory can be obtained from the website identified above.

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Note: This summary does not include all details of the amendment and should not be substituted for reading the amended text.

Some Noteworthy Amendments in A/L 129:

Where possible, references to noteworthy amendments have been included (references may not indicate all amendments relating to that topic). Minor editorial and format changes are not shown in this summary.

Legislation: Updates to legislative instrument references, including year and section citations, *GEN 1.2 para 1.2.6* and *GEN 1.6 para 1.1.1.*

New Designated International Airport: Karratha (YPKA) and Sydney/Nancy-Bird Walton (YSWS) added as a designated international airport, *GEN 1.2 para 2.1, 2.2* and *2.3.4* and *GEN 1.3 para 6.2.4 .*

Communication and Navigation: Updates to phraseology associated with addition of follow readback requirements, restricted areas, CPDLC and alignment to ICAO-standard terms and CASR Part 91 MOS, *GEN 3.4 para 5.4, 6.3, 6.4, 6.9, 6.14, 6.15, 6.16, and 7.5*.

Terminology Amendment: replaced “active runway” with “runway”, *ENR 1.1 para 6.1.10, 9.1.14*

Flying Considerately: New section added. *ENR 1.5 para 9.3*.

Ground Delay Program (GDP): Revision of run time information, *ENR 1.9 para 3.1* and *3.4.*

AAR Tracks: Updates to W946 AAR. Amended to terminate at GUKVI, *ENR 5.2 para 4.*

PART 1 – GENERAL (GEN)

GEN 0.

GEN 0.1 PREFACE

1. NAME OF PUBLISHING AUTHORITY

Airservices Australia is authorised under regulation 175.055 of Civil Aviation Safety Regulations 1998 (CASR) Part 175 as the provider of the Aeronautical Information Service (AIS) in Australia.

2. APPLICABLE ICAO DOCUMENTS

The AIP is prepared in accordance with the Standards and Recommended Practices (SARP) of Annex 15 to the Convention on International Civil Aviation, the ICAO Aeronautical Information Services Manual *(Doc 8126)* and Procedures for Air Navigation Services - Aeronautical Information Management *(Doc 10066)*. Charts contained in the AIP are produced in accordance with Annex 4 to the Convention on International Civil Aviation and the ICAO Aeronautical Chart Manual *(Doc 8697)*. Differences from ICAO SARP are detailed in subsection *GEN 1.7*.

3. PUBLICATION MEDIA

The AIP is made available on Airservices AIS website:
www.airservicesaustralia.com/aip/aip.asp

The AIP is available in printed paper format from the AIP Shop
www.aipshop.canprint.com.au

4. THE AIP STRUCTURE AND ESTABLISHED REGULAR AMENDMENT INTERVAL

4.1 AIP Structure

The AIP forms part of the Integrated Aeronautical Information Package (IAIP), details of which are given in subsection *GEN 3.1*. The principal AIP structure is shown in graphic form at the conclusion of this section.

The AIP consists of three parts, General (GEN), En Route (ENR) and Aerodromes (AD), each divided into sections and subsections as applicable.

4.1.1 **Part 1 - General (GEN)**

Part 1 consists of five sections containing information as briefly described hereafter.

GEN 0. Preface; Record of AIP Amendments; Record of AIP Supplements; Checklist of AIP Pages; List of Hand Amendments to the AIP; and the Table of Contents to Part 1.

GEN 1. National Regulations and Requirements - Designated Authorities; Entry, Transit and Departure of Aircraft; Entry, Transit and Departure of Passengers and Crew; Entry, Transit and Departure of Cargo; Aircraft Instruments, Equipment and Flight Documents; Summary of National Regulations and International Agreements/Conventions; and Differences from ICAO Standards, Recommended Practices and Procedures.

GEN 2. Tables and Codes - Measuring System, Aircraft Markings, Holidays; Definitions and Abbreviations used in AIS Publications; Chart Symbols; Location Indicators; List of Radio Navigation Aids; Conversion Tables and First and Last Light Computations.

GEN 3. Services - Aeronautical Information Services; Aeronautical Charts; Air Traffic Services; Communication Services; Meteorological Services and Search and Rescue.

GEN 4. Charges for Aerodromes/Heliports and Air Navigation Services - Aerodrome/Heliport Charges; and Air Navigation Services Charges.

4.1.2 **Part 2 - En Route (ENR)**

Part 2 consists of seven sections containing information as briefly described hereafter.

ENR 0. Table of Contents to Part 2.

ENR 1. General Rules and Procedures - General Rules; Visual Flight Rules; Instrument Flight Rules; ATS Airspace Classification; Holding, Approach and Departure Procedures; Surveillance Services and Procedures; Altimeter Setting Procedures; Regional Supplementary Procedures; Air Traffic Flow Management; Flight Planning; Addressing of Flight Plan Messages; Interception of Civil Aircraft; Unlawful Interference; and Air Traffic Incidents.

ENR 2. Air Traffic Services Airspace - Detailed description of Flight Information Regions (FIR); Upper Flight Information Regions (UIR); Terminal Control Areas (TMA); and Other Regulated Airspace.

ENR 3. ATS Routes - Detailed description of Lower ATS Routes; Upper ATS Routes; Area Navigation Routes; Helicopter Routes; Other Routes; and En Route Holding.

Note: Other types of routes which are specified in connection with procedures for traffic to and from aerodromes/heliports are described in the relevant sections and subsections of Part 3 - Aerodromes.

ENR 4. Radio Navigation Aids/Systems - Radio Navigation Aids; En Route; Special Navigation Systems; Name - Code Designators for Significant Points; and Aeronautical Ground Lights En Route.

ENR 5. Navigation Warnings - Special Use Airspace - Prohibited, Restricted, Danger and Military Operating Areas; Military Exercise and Training Areas; Other Activities of a Dangerous Nature and Other Potential Hazards; Air Navigation Obstacles - En Route; Aerial Sporting and Recreational Activities; and Bird Migration and Areas with Sensitive Fauna.

ENR 6. En Route Charts - En Route Charts - ICAO.

4.1.3 **Part 3 - Aerodromes (AD)**

Part 3 consists of four sections containing information as briefly described in the following paragraphs.

AD 0. Table of Contents to Part 3.

AD 1. Aerodromes/Heliports - Introduction - Aerodromes/ Heliports Availability and Conditions of Use; Rescue and Fire-fighting Services and Snow Plan; Index to Aerodromes and Heliports; and Grouping of Aerodromes/Heliports; and the Status of the Certification of Aerodromes.

AD 2. Aerodromes - Detailed information about aerodromes.

AD 3. Heliports - Detailed information about heliports.

4.2 **Regular Amendment Interval**

Regular amendments to the IAIP are issued approximately every three months, excluding DAH and charts, which are updated every six months.

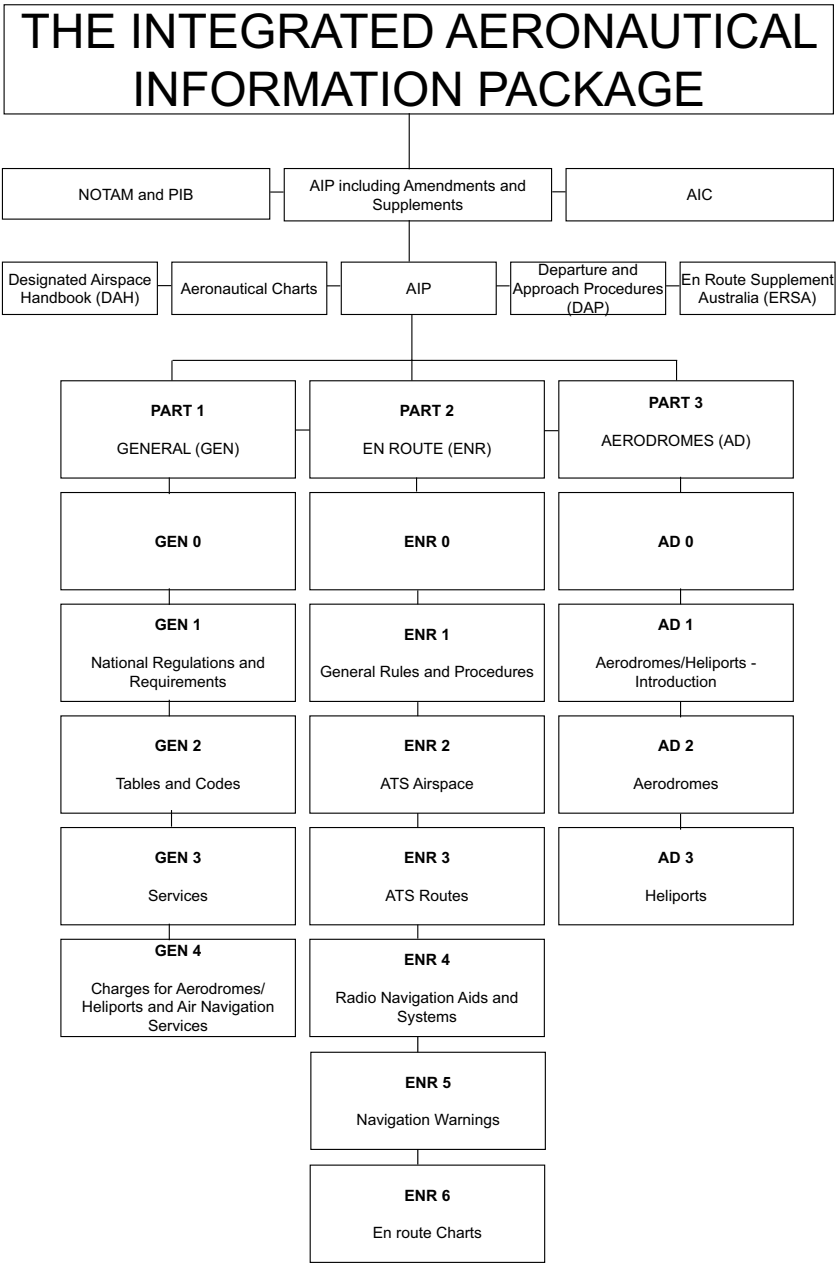
The IAIP is amended in accordance with the AIRAC schedule prescribed by ICAO. Information containing all publication 'cut off' timetables is available on the Airservices website.

5. COPYRIGHT

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6. SERVICE TO CONTACT IN CASE OF DETECTED AIP ERRORS OR OMISSIONS

In the compilation of the IAIP, care has been taken to ensure that the information contained therein is accurate and complete. Any errors and omissions which may nevertheless be detected, as well as any correspondence concerning the IAIP, should be emailed to: docs.amend@airservicesaustralia.com.


THE INTEGRATED AERONAUTICAL INFORMATION PACKAGE
NOTAM and PIB
AIP including Amendments and Supplements
AIC
Designated Airspace Handbook (DAH)
Aeronautical Charts
AIP
Departure and Approach Procedures (DAP)
En Route Supplement Australia (ERSA)
PART 1
GENERAL (GEN)
PART 2
EN ROUTE (ENR)
PART 3
AERODROMES (AD)
GEN 0
ENR 0
AD 0
GEN 1
National Regulations and Requirements
ENR 1
General Rules and Procedures
AD 1
Aerodromes/Heliports - Introduction
GEN 2
Tables and Codes
ENR 2
ATS Airspace
AD 2
Aerodromes
GEN 3
Services
ENR 3
ATS Routes
AD 3
Heliports
GEN 4
Charges for Aerodromes/ Heliports and Air Navigation Services
ENR 4
Radio Navigation Aids and Systems
ENR 5
Navigation Warnings
ENR 6
En route Charts

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GEN 0.2 RECORD OF AMENDMENTS

No	Date of AMDT	Date Inserted	Inserted by (Init.)	No	Date of AMDT	Date Inserted	Inserted by (Init.)
89				112			
90				113			
91				114			
92				115			
93				116			
94				117			
95				118			
96				119			
97				120			
98				121			
99				122			
100				123			
101				124			
102				125			
103				126			
104				127			
105				128			
106				129			
107				130			
108				131			
109				132			
110				133			
111				134			

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GEN 0.3 RECORD OF AIP SUPPLEMENTS

1. Australian AIP Supplements are promulgated under an Airservices Head Office (H) identifier, and given a number which is sequential for the year of distribution. For example, the fifth AIP Supplement (SUP) issued in the year 2024 is identified as H05/24. SUP which are required to be issued with the 28 days Aeronautical Information Regulation and Control (AIRAC) notice are identified further by “AIRAC” being printed to the left of the identifying number.
2. A Record of Supplements is published as a “Summary” of all current SUP/AIC for each monthly AIRAC date and published on the Airservices website.

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GEN 0.4 CHECKLIST OF AIP PAGES

AMENDMENT LIST 129 - EFFECTIVE 26 NOV 2026

Pages annotated with the change bars are new pages for this edition.

Sect - Page	Date
COVER PAGE	
Page 1 & 2	26 NOV 2026
GEN	
0.1 - 1 & 2	13 JUN 2024
0.1 - 3 & 4	13 JUN 2024
0.1 - 5 & 6	24 MAR 2022
0.2 - 1 & 2	10 NOV 2016
0.3 - 1 & 2	28 NOV 2024
0.4 - 1 & 2	26 NOV 2026
0.4 - 3 & 4	26 NOV 2026
0.4 - 5 & 6	26 NOV 2026
0.5 - 1 & 2	10 NOV 2016
0.6 - 1 & 2	26 NOV 2026
0.6 - 3 & 4	26 NOV 2026
0.6 - 5 & 6	26 NOV 2026
0.6 - 7 & 8	26 NOV 2026
0.6 - 9 & 10	26 NOV 2026
0.6 - 11 & 12	26 NOV 2026
0.6 - 13 & 14	26 NOV 2026
1.1 - 1 & 2	03 SEP 2026
1.1 - 3 & 4	09 JUL 2026
1.2 - 1 & 2	26 NOV 2026
1.2 - 3 & 4	03 SEP 2026
1.2 - 5 & 6	03 SEP 2026
1.2 - 7 & 8	03 SEP 2026
1.2 - 9 & 10	05 SEP 2024
1.2 - 11 & 12	19 MAR 2026
1.2 - 13 & 14	03 SEP 2026
1.2 - 15 & 16	03 SEP 2026
1.2 - 17 & 18	03 SEP 2026
1.2 - 19 & 20	05 SEP 2024
1.2 - 21 & 22	26 NOV 2026
1.2 - 23 & 24	26 NOV 2026
1.3 - 1 & 2	27 NOV 2025
1.3 - 3 & 4	30 NOV 2023
1.3 - 5 & 6	30 NOV 2023
1.3 - 7 & 8	03 SEP 2026
1.3 - 9 & 10	03 SEP 2026
1.3 - 11 & 12	03 SEP 2026
1.3 - 13 & 14	03 SEP 2026
1.3 - 15 & 16	05 SEP 2024
1.3 - 17 & 18	03 SEP 2026
1.3 - 19 & 20	09 JUL 2026
1.3 - 21 & 22	03 SEP 2026
1.3 - 23 & 24	30 NOV 2023
1.3 - 25 & 26	26 NOV 2026
1.3 - 27 & 28	03 SEP 2026
1.3 - 29 & 30	09 JUL 2026
1.3 - 31 & 32	09 JUL 2026
1.3 - 33 & 34	04 SEP 2025
1.4 - 1 & 2	03 SEP 2026
1.4 - 3 & 4	03 SEP 2026
1.4 - 5 & 6	03 SEP 2026
1.5 - 1 & 2	09 JUL 2026
1.5 - 3 & 4	09 JUL 2026
1.5 - 5 & 6	03 SEP 2026
1.5 - 7 & 8	09 JUL 2026
1.5 - 9 & 10	09 JUL 2026
1.6 - 1 & 2	26 NOV 2026
1.6 - 3 & 4	19 MAR 2026
1.6 - 5 & 6	09 JUL 2026
1.6 - 7 & 8	03 SEP 2026
1.6 - 9 & 10	03 SEP 2026
1.7 - 1 & 2	05 NOV 2020
2.1 - 1 & 2	20 MAR 2025
2.1 - 3 & 4	30 NOV 2023
2.2 - 1 & 2	27 NOV 2025
2.2 - 3 & 4	27 NOV 2025
2.2 - 5 & 6	26 NOV 2026
2.2 - 7 & 8	27 NOV 2025
2.2 - 9 & 10	27 NOV 2025
2.2 - 11 & 12	27 NOV 2025
2.2 - 13 & 14	27 NOV 2025
2.2 - 15 & 16	27 NOV 2025
2.2 - 17 & 18	27 NOV 2025

Sect - Page	Date
GEN	
2.2 - 19 & 20	27 NOV 2025
2.2 - 21 & 22	26 NOV 2026
2.2 - 23 & 24	26 NOV 2026
2.2 - 25 & 26	26 NOV 2026
2.2 - 27 & 28	26 NOV 2026
2.2 - 29 & 30	26 NOV 2026
2.2 - 31 & 32	26 NOV 2026
2.2 - 33 & 34	26 NOV 2026
2.2 - 35 & 36	26 NOV 2026
2.2 - 37 & 38	26 NOV 2026
2.2 - 39 & 40	26 NOV 2026
2.2 - 41 & 42	26 NOV 2026
2.2 - 43 & 44	26 NOV 2026
2.2 - 45 & 46	26 NOV 2026
2.2 - 47 & 48	26 NOV 2026
2.2 - 49 & 50	26 NOV 2026
2.2 - 51 & 52	26 NOV 2026
2.2 - 53 & 54	26 NOV 2026
2.2 - 55 & 56	26 NOV 2026
2.3 - 1 & 2	10 NOV 2016
2.4 - 1 & 2	05 NOV 2020
2.5 - 1 & 2	10 NOV 2016
2.6 - 1 & 2	15 JUN 2023
2.6 - 3 & 4	09 SEP 2021
2.6 - 5 & 6	21 MAR 2024
2.7 - 1 & 2	03 SEP 2026
2.7 - 3 & 4	08 SEP 2022
2.7 - 5 & 6	08 SEP 2022
2.7 - 7 & 8	10 NOV 2016
3.1 - 1 & 2	03 SEP 2026
3.1 - 3 & 4	05 SEP 2024
3.1 - 5 & 6	03 SEP 2026
3.1 - 7 & 8	03 SEP 2026
3.2 - 1 & 2	23 MAR 2023
3.2 - 3 & 4	23 MAR 2023
3.3 - 1 & 2	27 NOV 2025
3.3 - 3 & 4	27 NOV 2025
3.3 - 5 & 6	19 MAR 2026
3.3 - 7 & 8	21 MAR 2024
3.3 - 9 & 10	19 MAR 2026
3.4 - 1 & 2	21 MAR 2024
3.4 - 3 & 4	21 MAR 2024
3.4 - 5 & 6	27 NOV 2025
3.4 - 7 & 8	27 NOV 2025
3.4 - 9 & 10	21 MAR 2024
3.4 - 11 & 12	26 NOV 2026
3.4 - 13 & 14	26 NOV 2026
3.4 - 15 & 16	26 NOV 2026
3.4 - 17 & 18	26 NOV 2026
3.4 - 19 & 20	26 NOV 2026
3.4 - 21 & 22	26 NOV 2026
3.4 - 23 & 24	26 NOV 2026
3.4 - 25 & 26	27 NOV 2025
3.4 - 27 & 28	13 JUN 2024
3.4 - 29 & 30	13 JUN 2024
3.4 - 31 & 32	26 NOV 2026
3.4 - 33 & 34	27 NOV 2025
3.4 - 35 & 36	13 JUN 2024
3.4 - 37 & 38	13 JUN 2024
3.4 - 39 & 40	27 NOV 2025
3.4 - 41 & 42	27 NOV 2025
3.4 - 43 & 44	26 NOV 2026
3.4 - 45 & 46	13 JUN 2024
3.4 - 47 & 48	26 NOV 2026
3.4 - 49 & 50	26 NOV 2026
3.4 - 51 & 52	26 NOV 2026
3.4 - 53 & 54	26 NOV 2026
3.4 - 55 & 56	26 NOV 2026
3.4 - 57 & 58	26 NOV 2026
3.4 - 59 & 60	26 NOV 2026
3.4 - 61 & 62	26 NOV 2026
3.4 - 63 & 64	26 NOV 2026
3.4 - 65 & 66	26 NOV 2026
3.4 - 67 & 68	26 NOV 2026
3.4 - 69 & 70	26 NOV 2026
3.4 - 71 & 72	26 NOV 2026
3.4 - 73 & 74	26 NOV 2026
3.4 - 75 & 76	26 NOV 2026
3.4 - 77 & 78	26 NOV 2026
3.4 - 79 & 80	26 NOV 2026
3.4 - 81 & 82	26 NOV 2026
3.4 - 83 & 84	26 NOV 2026
3.4 - 85 & 86	26 NOV 2026
3.4 - 87 & 88	26 NOV 2026

Sect - Page	Date
GEN	
3.4 - 89 & 90	26 NOV 2026
3.4 - 91 & 92	26 NOV 2026
3.4 - 93 & 94	26 NOV 2026
3.4 - 95 & 96	26 NOV 2026
3.4 - 97 & 98	26 NOV 2026
3.4 - 99 & 100	26 NOV 2026
3.4 - 101 & 102	26 NOV 2026
3.4 - 103 & 104	26 NOV 2026
3.4 - 105 & 106	26 NOV 2026
3.4 - 107 & 108	26 NOV 2026
3.5 - 1 & 2	03 SEP 2026
3.5 - 3 & 4	03 SEP 2026
3.5 - 5 & 6	03 SEP 2026
3.5 - 7 & 8	03 SEP 2026
3.5 - 9 & 10	03 SEP 2026
3.5 - 11 & 12	05 SEP 2024
3.5 - 13 & 14	27 NOV 2025
3.5 - 15 & 16	27 NOV 2025
3.5 - 17 & 18	27 NOV 2025
3.5 - 19 & 20	27 NOV 2025
3.5 - 21 & 22	21 MAR 2024
3.6 - 1 & 2	24 MAR 2022
3.6 - 3 & 4	24 MAR 2022
3.6 - 5 & 6	02 DEC 2021
3.6 - 7 & 8	02 DEC 2021
4.1 - 1 & 2	03 SEP 2026
4.2 - 1 & 2	04 SEP 2025
ENR	
0.1 - 1 & 2	26 NOV 2026
0.1 - 3 & 4	26 NOV 2026
0.1 - 5 & 6	26 NOV 2026
0.1 - 7 & 8	26 NOV 2026
0.1 - 9 & 10	26 NOV 2026
0.1 - 11 & 12	26 NOV 2026
1.1 - 1 & 2	13 JUN 2024
1.1 - 3 & 4	27 NOV 2025
1.1 - 5 & 6	26 NOV 2026
1.1 - 7 & 8	19 MAR 2026
1.1 - 9 & 10	19 MAR 2026
1.1 - 11 & 12	19 MAR 2026
1.1 - 13 & 14	19 MAR 2026
1.1 - 15 & 16	19 MAR 2026

Sect - Page	Date
ENR	
1.1 - 17 & 18	19 MAR 2026
1.1 - 19 & 20	01 DEC 2022
1.1 - 21 & 22	23 MAR 2023
1.1 - 23 & 24	27 NOV 2025
1.1 - 25 & 26	24 MAR 2022
1.1 - 27 & 28	24 MAR 2022
1.1 - 29 & 30	23 MAR 2023
1.1 - 31 & 32	27 NOV 2025
1.1 - 33 & 34	15 JUN 2023
1.1 - 35 & 36	27 NOV 2025
1.1 - 37 & 38	27 NOV 2025
1.1 - 39 & 40	26 NOV 2026
1.1 - 41 & 42	21 MAR 2024
1.1 - 43 & 44	27 NOV 2025
1.1 - 45 & 46	27 NOV 2025
1.1 - 47 & 48	27 NOV 2025
1.1 - 49 & 50	21 MAR 2024
1.1 - 51 & 52	21 MAR 2024
1.1 - 53 & 54	21 MAR 2024
1.1 - 55 & 56	27 NOV 2025
1.1 - 57 & 58	26 NOV 2026
1.1 - 59 & 60	21 MAR 2024
1.1 - 61 & 62	27 NOV 2025
1.1 - 63 & 64	21 MAR 2024
1.1 - 65 & 66	21 MAR 2024
1.1 - 67 & 68	21 MAR 2024
1.1 - 69 & 70	21 MAR 2024
1.1 - 71 & 72	21 MAR 2024
1.1 - 73 & 74	27 NOV 2025
1.1 - 75 & 76	27 NOV 2025
1.1 - 77 & 78	21 MAR 2024
1.1 - 79 & 80	21 MAR 2024
1.1 - 81 & 82	21 MAR 2024
1.1 - 83 & 84	03 SEP 2026
1.1 - 85 & 86	21 MAR 2024
1.1 - 87 & 88	27 NOV 2025
1.1 - 89 & 90	27 NOV 2025
1.1 - 91 & 92	21 MAR 2024
1.1 - 93 & 94	27 NOV 2025
1.1 - 95 & 96	27 NOV 2025
1.1 - 97 & 98	27 NOV 2025
1.1 - 99 & 100	27 NOV 2025

Sect - Page	Date
ENR	
1.1 - 101 & 102	27 NOV 2025
1.1 - 103 & 104	21 MAR 2024
1.1 - 105 & 106	21 MAR 2024
1.2 - 1 & 2	02 DEC 2021
1.3 - 1 & 2	02 DEC 2021
1.4 - 1 & 2	13 JUN 2024
1.4 - 3 & 4	30 NOV 2023
1.4 - 5 & 6	30 NOV 2023
1.4 - 7 & 8	13 JUN 2024
1.4 - 9 & 10	27 NOV 2025
1.4 - 11 & 12	27 NOV 2025
1.4 - 13 & 14	27 NOV 2025
1.4 - 15 & 16	27 NOV 2025
1.4 - 17 & 18	27 NOV 2025
1.5 - 1 & 2	20 MAR 2025
1.5 - 3 & 4	20 MAR 2025
1.5 - 5 & 6	04 SEP 2025
1.5 - 7 & 8	20 MAR 2025
1.5 - 9 & 10	20 MAR 2025
1.5 - 11 & 12	20 MAR 2025
1.5 - 13 & 14	20 MAR 2025
1.5 - 15 & 16	20 MAR 2025
1.5 - 17 & 18	20 MAR 2025
1.5 - 19 & 20	20 MAR 2025
1.5 - 21 & 22	20 MAR 2025
1.5 - 23 & 24	20 MAR 2025
1.5 - 25 & 26	26 NOV 2026
1.5 - 27 & 28	20 MAR 2025
1.5 - 29 & 30	20 MAR 2025
1.5 - 31 & 32	20 MAR 2025
1.5 - 33 & 34	20 MAR 2025
1.5 - 35 & 36	19 MAR 2026
1.5 - 37 & 38	20 MAR 2025
1.5 - 39 & 40	20 MAR 2025
1.5 - 41 & 42	20 MAR 2025
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1.5 - 47 & 48	26 NOV 2026
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1.5 - 51 & 52	20 MAR 2025
1.6 - 1 & 2	27 NOV 2025
1.6 - 3 & 4	26 NOV 2026
1.6 - 5 & 6	27 NOV 2025
1.6 - 7 & 8	27 NOV 2025
1.6 - 9 & 10	26 NOV 2026
1.7 - 1 & 2	27 NOV 2025
1.7 - 3 & 4	30 NOV 2023
1.7 - 5 & 6	01 DEC 2022
1.7 - 7 & 8	01 DEC 2022
1.7 - 9 & 10	01 DEC 2022
1.8 - 1 & 2	10 NOV 2016
1.9 - 1 & 2	27 NOV 2025
1.9 - 3 & 4	26 NOV 2026
1.9 - 5 & 6	26 NOV 2026
1.9 - 7 & 8	26 NOV 2026
1.10 - 1 & 2	23 MAR 2023
1.10 - 3 & 4	23 MAR 2023
1.10 - 5 & 6	27 NOV 2025
1.10 - 7 & 8	05 SEP 2024
1.10 - 9 & 10	23 MAR 2023
1.10 - 11 & 12	02 DEC 2021
1.10 - 13 & 14	27 NOV 2025
1.10 - 15 & 16	27 NOV 2025
1.10 - 17 & 18	05 SEP 2024
1.10 - 19 & 20	30 NOV 2023
1.10 - 21 & 22	26 NOV 2026
1.10 - 23 & 24	20 MAR 2025
1.10 - 25 & 26	30 NOV 2023
1.10 - 27 & 28	28 NOV 2024
1.10 - 29 & 30	28 NOV 2024
1.10 - 31 & 32	09 JUL 2026
1.10 - 33 & 34	02 DEC 2021
1.10 - 35 & 36	27 NOV 2025
1.11 - 1 & 2	26 NOV 2026
1.12 - 1 & 2	27 NOV 2025
1.12 - 3 & 4	27 NOV 2025
1.12 - 5 & 6	13 JUN 2024
1.13 - 1 & 2	10 NOV 2016
1.14 - 1 & 2	23 MAR 2023
1.14 - 3 & 4	23 MAR 2023
1.14 - 5 & 6	23 MAR 2023
1.14 - 7 & 8	23 MAR 2023
1.14 - 9 & 10	20 MAR 2025
1.14 - 11 & 12	27 NOV 2025

Sect - Page	Date
ENR	
1.14 - 13 & 14	20 MAR 2025
1.14 - 15 & 16	20 MAR 2025
1.14 - 17 & 18	20 MAR 2025
1.14 - 19 & 20	20 MAR 2025
1.14 - 21 & 22	20 MAR 2025
1.14 - 23 & 24	20 MAR 2025
1.14 - 25 & 26	20 MAR 2025
1.14 - 27 & 28	20 MAR 2025
1.14 - 29 & 30	20 MAR 2025
1.14 - 31 & 32	20 MAR 2025
1.14 - 33 & 34	20 MAR 2025
1.14 - 35 & 36	20 MAR 2025
1.14 - 37 & 38	20 MAR 2025
1.14 - 39 & 40	20 MAR 2025
1.14 - 41 & 42	27 NOV 2025
1.14 - 43 & 44	20 MAR 2025
1.14 - 45 & 46	20 MAR 2025
1.14 - 47 & 48	20 MAR 2025
2.1 - 1 & 2	10 NOV 2016
2.2 - 1 & 2	12 JUN 2025
2.2 - 3 & 4	12 JUN 2025
2.2 - 5 & 6	12 JUN 2025
2.2 - 7 & 8	05 NOV 2020
2.2 - 9 & 10	05 NOV 2020
3.1 - 1 & 2	21 MAY 2020
3.2 - 1 & 2	10 NOV 2016
3.3 - 1 & 2	02 DEC 2021
3.4 - 1 & 2	10 NOV 2016
3.5 - 1 & 2	20 MAR 2025
3.6 - 1 & 2	25 MAR 2021
4.1 - 1 & 2	10 NOV 2016
4.2 - 1 & 2	10 NOV 2016
4.3 - 1 & 2	01 DEC 2022
4.4 - 1 & 2	05 NOV 2020
4.5 - 1 & 2	05 NOV 2020
5.1 - 1 & 2	23 MAR 2023
5.1 - 3 & 4	20 MAR 2025
5.2 - 1 & 2	13 JUN 2024
5.2 - 3 & 4	21 MAR 2024
5.2 - 5 & 6	21 MAR 2024
5.2 - 7 & 8	20 MAR 2025
5.2 - 9 & 10	21 MAR 2024
ENR	
5.2 - 11 & 12	04 SEP 2025
5.2 - 13 & 14	04 SEP 2025
5.2 - 15 & 16	04 SEP 2025
5.2 - 17 & 18	04 SEP 2025
5.2 - 19 & 20	20 MAR 2025
5.2 - 21 & 22	09 JUL 2026
5.2 - 23 & 24	04 SEP 2025
5.2 - 25 & 26	27 NOV 2025
5.2 - 27 & 28	04 SEP 2025
5.2 - 29 & 30	04 SEP 2025
5.2 - 31 & 32	04 SEP 2025
5.2 - 33 & 34	04 SEP 2025
5.2 - 35 & 36	09 JUL 2026
5.2 - 37 & 38	26 NOV 2026
5.2 - 39 & 40	21 MAR 2024
5.2 - 41 & 42	04 SEP 2025
5.2 - 43 & 44	04 SEP 2025
5.2 - 45 & 46	04 SEP 2025
5.2 - 47 & 48	04 SEP 2025
5.2 - 49 & 50	27 NOV 2025
5.3 - 1 & 2	28 NOV 2024
5.4 - 1 & 2	02 DEC 2021
5.5 - 1 & 2	02 DEC 2021
5.5 - 3 & 4	28 NOV 2024
5.6 - 1 & 2	10 NOV 2016
6.1 - 1 & 2	10 NOV 2016
AD	
0.1 - 1 & 2	26 NOV 2026
1.1 - 1 & 2	03 SEP 2026
1.1 - 3 & 4	26 NOV 2026
1.1 - 5 & 6	09 JUL 2026
1.1 - 7 & 8	19 MAR 2026
1.1 - 9 & 10	19 MAR 2026
1.1 - 11 & 12	19 MAR 2026
1.2 - 1 & 2	03 SEP 2026
1.2 - 3 & 4	03 SEP 2026
1.2 - 5 & 6	13 JUN 2024
1.2 - 7 & 8	13 JUN 2024
1.3 - 1 & 2	05 NOV 2020
1.4 - 1 & 2	10 NOV 2016
1.5 - 1 & 2	30 NOV 2023
2 - 1 & 2	05 NOV 2020

Sect - Page	Date
AD	
3 - 1 & 2	05 NOV 2020

GEN 0.5 LIST OF HAND AMENDMENTS

1. As a matter of principle, this document does not normally require handwritten amendment. Nevertheless, to alleviate printing costs, minor editorial changes are notified on this page, but will not be actioned as formal amendments until they can be accommodated with a significant change to the relevant section(s). **Inclusion of manuscript amendments is, therefore, at the AIP holder's discretion.**

2. Current minor amendments are: NIL

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GEN 0.6 TABLE OF CONTENTS

1. NAME OF PUBLISHING AUTHORITY ..... GEN 0.1 - 1
2. APPLICABLE ICAO DOCUMENTS ..... GEN 0.1 - 1
3. PUBLICATION MEDIA ..... GEN 0.1 - 1
4. THE AIP STRUCTURE AND ESTABLISHED REGULAR AMENDMENT INTERVAL ..... GEN 0.1 - 1
4.1 AIP Structure ..... GEN 0.1 - 1
4.2 Regular Amendment Interval ..... GEN 0.1 - 4
5. COPYRIGHT ..... GEN 0.1 - 4
6. SERVICE TO CONTACT IN CASE OF DETECTED AIP ERRORS OR OMISSIONS ..... GEN 0.1 - 4
GEN 0.2 RECORD OF AMENDMENTS ..... GEN 0.2 - 1
GEN 0.3 RECORD OF AIP SUPPLEMENTS ..... GEN 0.3 - 1
GEN 0.4 CHECKLIST OF AIP PAGES ..... GEN 0.4 - 1
GEN 0.5 LIST OF HAND AMENDMENTS ..... GEN 0.5 - 1
GEN 1. NATIONAL REGULATIONS AND REQUIREMENTS ..... GEN 1.1 - 1
GEN 1.1 DESIGNATED AUTHORITIES ..... GEN 1.1 - 1
1. DESIGNATED AUTHORITIES ..... GEN 1.1 - 1
1.1 Department of Infrastructure, Transport, Regional Development, Communications and the Arts (Department of Infrastructure) ..... GEN 1.1 - 1
1.2 Department of Home Affairs ..... GEN 1.1 - 1
1.3 Australian Border Force ..... GEN 1.1 - 2
1.4 Civil Aviation Safety Authority ..... GEN 1.1 - 2
1.5 Air Traffic Services ..... GEN 1.1 - 3
1.6 Department of Agriculture, Fisheries and Forestry (DAFF) ..... GEN 1.1 - 3
1.7 Department of Health and Aged Care – Human Biosecurity ..... GEN 1.1 - 3
1.8 Aircraft Accident Investigation ..... GEN 1.1 - 3
1.9 Meteorology ..... GEN 1.1 - 4
1.10 Search and Rescue ..... GEN 1.1 - 4
1.11 Department of Defence ..... GEN 1.1 - 4
GEN 1.2 ENTRY, TRANSIT AND DEPARTURE OF AIRCRAFT ..... GEN 1.2 - 1
1. ENTRY, TRANSIT AND DEPARTURE OF INTERNATIONAL FLIGHTS ..... GEN 1.2 - 1
1.1 Preamble ..... GEN 1.2 - 1

1.2 Scheduled International Commercial Services by Foreign Aircraft of Contracting States to the Chicago Convention..... GEN 1.2 - 2
1.3 Summary of Documents to be presented by Pilot in Command or Authorised Agent..... GEN 1.2 - 3
1.4 Non-Scheduled International Commercial Services by Foreign Aircraft of Contracting States to the Chicago Convention ..... GEN 1.2 - 6
1.5 Non-Scheduled International Commercial Services by Australian Aircraft..... GEN 1.2 - 7
1.6 Non-Scheduled International Commercial Services by Australian Aircraft and Foreign Aircraft of Non-Contracting States and Contracting States, which do not Require Prior Approval from the Department of Infrastructure..... GEN 1.2 - 8
1.7 International Flights by Foreign Aircraft not Possessing Nationality of Contracting State to the Chicago Convention..... GEN 1.2 - 10
1.8 Flights by Foreign State Aircraft..... GEN 1.2 - 10
1.9 International Private Flights ..... GEN 1.2 - 12
1.10 Aviation Security ..... GEN 1.2 - 14
1.11 Aviation Security: Transport Security Program requirements..... GEN 1.2 - 15
1.12 Carriage of Munitions of War and Implements of War..... GEN 1.2 - 17
1.13 Foreign Clearances - Australian Aircraft ..... GEN 1.2 - 18
1.14 Aircraft on International Flights to Comply with Laws ..... GEN 1.2 - 18
1.15 Section 22 of the Civil Aviation Act 1988..... GEN 1.2 - 18
1.16 Aircraft Noise Operating Restrictions..... GEN 1.2 - 19
1.17 Air Carriers' Liability and Insurance ..... GEN 1.2 - 20
1.18 Australian Operational Documents Available to Pilots Licensed by another State Proposing to Visit Australia ..... GEN 1.2 - 20
2. DESIGNATED INTERNATIONAL AIRPORTS - AUSTRALIA..... GEN 1.2 - 20

2.1 Major International Airports ..... GEN 1.2 - 21
2.2 Restricted Use International Airports ..... GEN 1.2 - 21
2.3 Alternate Airports to International Airports ..... GEN 1.2 - 22
2.4 International Non-Scheduled Flight Airports ..... GEN 1.2 - 24
2.5 External Territory International Airport ..... GEN 1.2 - 24
GEN 1.3 ENTRY, TRANSIT AND DEPARTURE OF PASSENGERS AND CREW ..... GEN 1.3 - 1
1. INTRODUCTION ..... GEN 1.3 - 1
2. BIOSECURITY REQUIREMENTS ..... GEN 1.3 - 1
2.1 General ..... GEN 1.3 - 1
2.2 Disinsection ..... GEN 1.3 - 1
2.3 Pre-arrival Reporting (PAR) Requirements ..... GEN 1.3 - 3
2.4 Ill Passengers or Death On Board ..... GEN 1.3 - 4
2.5 Pratique ..... GEN 1.3 - 5
2.6 Biosecurity Waste ..... GEN 1.3 - 6
2.7 Biosecurity In–flight Announcement ..... GEN 1.3 - 7
2.8 Required Vaccination and Health Documents ..... GEN 1.3 - 7
2.9 Biosecurity screening of baggage ..... GEN 1.3 - 8
2.10 Conditionally Non-prohibited Goods ..... GEN 1.3 - 8
3. IMMIGRATION/EMIGRATION REQUIREMENTS ..... GEN 1.3 - 9
3.1 General ..... GEN 1.3 - 9
3.2 Advance Passenger Processing and Reporting of Passengers and Crew ..... GEN 1.3 - 10
3.3 Inwards Clearance – Passports or Other Travel Document ..... GEN 1.3 - 10
3.4 Visa Requirements ..... GEN 1.3 - 12
3.5 Returning Non-citizen Permanent Residents of Australia ..... GEN 1.3 - 16
3.6 Incoming Passenger Cards ..... GEN 1.3 - 17
3.7 Inward Clearance - Documentation ..... GEN 1.3 - 17
3.8 Outwards Clearance – Documentation ..... GEN 1.3 - 18
4. CUSTOMS REQUIREMENTS ..... GEN 1.3 - 19
4.1 Inward Clearance – Documentation ..... GEN 1.3 - 19
4.2 Examination of Crew and Passengers – Arrivals ..... GEN 1.3 - 19
4.3 Outward Clearance – Documentation ..... GEN 1.3 - 20

4.4 Examination of Crew and Passengers – Departures ..... GEN 1.3 - 20
5. TRANSIT PASSENGERS – CLEARANCE REQUIREMENTS AND PROCEDURES WITHIN AUSTRALIA ..... GEN 1.3 - 21
5.1 Immigration Requirements ..... GEN 1.3 - 21
5.2 Customs and Biosecurity Requirements ..... GEN 1.3 - 22
5.3 Requirements for the airline representative ..... GEN 1.3 - 22
6. LANDINGS AT DESIGNATED ALTERNATE AIRPORTS OR ELSEWHERE THAN AT DESIGNATED INTERNATIONAL AIRPORTS WITHIN AUSTRALIA ..... GEN 1.3 - 23
6.1 General ..... GEN 1.3 - 23
6.2 Designated Alternate Airports to International Airports ..... GEN 1.3 - 23
6.3 Landing made other than at a Designated Alternate Airport ..... GEN 1.3 - 26
7. CHARTER FLIGHTS – GUIDELINES FOR CLEARANCE ..... GEN 1.3 - 27
7.1 General ..... GEN 1.3 - 27
7.2 Arrival/Departure at International Airports ..... GEN 1.3 - 27
7.3 Arrival/Departure at Non–International Airports, and Restricted Use International Airports ..... GEN 1.3 - 27
7.4 Processing of the Application ..... GEN 1.3 - 28
8. DESIGNATED INTERNATIONAL AIRPORTS – AUSTRALIAN EXTERNAL TERRITORIES – ENTRY AND DEPARTURE REQUIREMENTS AND PROCEDURES ..... GEN 1.3 - 29
8.1 Biosecurity Requirements for External Territories ..... GEN 1.3 - 29
8.2 External Territory International Airports ..... GEN 1.3 - 29
8.3 Territory of Christmas Island ..... GEN 1.3 - 30
8.4 Territory of Cocos (Keeling) Islands ..... GEN 1.3 - 31
8.5 Territory of Norfolk Island ..... GEN 1.3 - 32
GEN 1.4 ENTRY, TRANSIT AND DEPARTURE OF CARGO ..... GEN 1.4 - 1
1. CUSTOMS REQUIREMENTS CONCERNING CARGO AND OTHER ARTICLES ..... GEN 1.4 - 1
1.1 Cargo Report ..... GEN 1.4 - 1

1.2 List of Stores and Prohibited Goods (e.g. narcotic drugs, beer, wine, spirits and tobacco products) ..... GEN 1.4 - 2
1.3 Import Declarations and Carnets ..... GEN 1.4 - 2
1.4 Treatment of Diplomatic and Consular Goods ..... GEN 1.4 - 2
1.5 Departure Report ..... GEN 1.4 - 3
1.6 Outward Manifest ..... GEN 1.4 - 3
1.7 Export Reporting ..... GEN 1.4 - 3
1.8 Certificate of Clearance ..... GEN 1.4 - 4
2. BIOSECURITY REQUIREMENTS ..... GEN 1.4 - 4
3. AIR CARGO SECURITY REQUIREMENTS ..... GEN 1.4 - 5
GEN 1.5 AIRCRAFT INSTRUMENTS, EQUIPMENT AND FLIGHT DOCUMENTS ..... GEN 1.5 - 1
1. RADIO COMMUNICATIONS SYSTEM REQUIREMENTS ..... GEN 1.5 - 1
2. RADIO NAVIGATION SYSTEM REQUIREMENTS ..... GEN 1.5 - 1
2.3 Rated Coverages ..... GEN 1.5 - 2
3. EMERGENCY LOCATOR TRANSMITTER (ELT) ..... GEN 1.5 - 3
4. AIRBORNE WEATHER RADAR ..... GEN 1.5 - 3
5. GROUND PROXIMITY WARNING SYSTEM (GPWS) ..... GEN 1.5 - 4
6. RADIO ALTIMETERS ..... GEN 1.5 - 4
7. SURVEILLANCE EQUIPMENT ..... GEN 1.5 - 5
7.1 General requirements and capabilities ..... GEN 1.5 - 5
7.2 Flight without serviceable required surveillance equipment ..... GEN 1.5 - 5
7.3 ADS-B equipment in State Aircraft ..... GEN 1.5 - 6
8. AIRBORNE COLLISION AVOIDANCE SYSTEM (ACAS) / TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS) ..... GEN 1.5 - 6
8.1 Overview ..... GEN 1.5 - 6
8.2 Use of ACAS/TCAS Indicators ..... GEN 1.5 - 6
8.3 Responsibility for Separation ..... GEN 1.5 - 7
8.7 High Vertical Rate (HVR) Encounters ..... GEN 1.5 - 8
9. AREA NAVIGATION SYSTEMS APPROVAL AND OPERATIONS ..... GEN 1.5 - 8

9.3 Notification of failure or operations outside of tolerance ..... GEN 1.5 - 9
10. RVSM APPROVAL AND OPERATIONS ..... GEN 1.5 - 9
11. AOC TO BE CARRIED ON BOARD ..... GEN 1.5 - 9
GEN 1.6 SUMMARY OF NATIONAL REGULATIONS AND INTERNATIONAL AGREEMENTS/CONVENTIONS ..... GEN 1.6 - 1
1. KEY CIVIL AVIATION LEGISLATION AND AIR NAVIGATION REGULATIONS IN FORCE IN AUSTRALIA AND ITS TERRITORIES ..... GEN 1.6 - 1
1.1 Legislation by the Parliament of the Commonwealth of Australia ..... GEN 1.6 - 1
1.2 Other Relevant Legislation ..... GEN 1.6 - 5
2. SECURITY OF GENERAL AVIATION OPERATIONS ..... GEN 1.6 - 5
2.1 Introduction ..... GEN 1.6 - 5
2.2 Activities Which May Arouse Suspicion ..... GEN 1.6 - 5
2.3 Actions and Contacts ..... GEN 1.6 - 6
3. AUSTRALIAN SANCTIONS ..... GEN 1.6 - 7
3.1 Introduction ..... GEN 1.6 - 7
3.2 Sanctions Regimes under Australian sanction laws ..... GEN 1.6 - 7
3.3 Further information ..... GEN 1.6 - 8
4. AUSTRALIAN AIR CARGO PROHIBITIONS ..... GEN 1.6 - 8
4.1 Introduction ..... GEN 1.6 - 8
4.2 Air Cargo Prohibitions under Australian Law ..... GEN 1.6 - 8
4.3 Further Information ..... GEN 1.6 - 9
GEN 1.7 DIFFERENCES FROM ICAO STANDARDS, RECOMMENDED PRACTICES AND PROCEDURES ..... GEN 1.7 - 1
GEN 2. TABLES AND CODES ..... GEN 2.1 - 1
GEN 2.1 MEASURING SYSTEM, AIRCRAFT MARKINGS, HOLIDAYS ..... GEN 2.1 - 1
1. UNITS OF MEASUREMENT ..... GEN 2.1 - 1
2. TIME SYSTEM ..... GEN 2.1 - 1
2.1 Coordinated Universal Time (UTC) ..... GEN 2.1 - 1
2.2 Date and time ..... GEN 2.1 - 2
3. GEODETIC REFERENCE DATUM ..... GEN 2.1 - 2
4. PUBLIC HOLIDAYS ..... GEN 2.1 - 2
4.1 National ..... GEN 2.1 - 2

GEN 2.2 DEFINITIONS AND ABBREVIATIONS ..... GEN 2.2 - 1
1. DEFINITIONS ..... GEN 2.2 - 1
2. GENERAL AND METEOROLOGICAL ABBREVIATIONS ..... GEN 2.2 - 34
GEN 2.3 CHART SYMBOLS ..... GEN 2.3 - 1
GEN 2.4 LOCATION INDICATORS ..... GEN 2.4 - 1
1. CODE ALLOCATION ..... GEN 2.4 - 1
2. LIST OF CODES ..... GEN 2.4 - 1
GEN 2.5 RADIO NAVIGATION AIDS ..... GEN 2.5 - 1
GEN 2.6 CONVERSION OF UNITS OF MEASUREMENT ..... GEN 2.6 - 1
1. FUEL WEIGHT TABLES ..... GEN 2.6 - 1
2. GENERAL CONVERSIONS ..... GEN 2.6 - 2
3. DECIMAL MINUTES OF AN ARC TO SECONDS OF AN ARC ..... GEN 2.6 - 3
4. SECONDS OF AN ARC TO DECIMAL MINUTES OF AN ARC ..... GEN 2.6 - 4
5. WIND COMPONENT ..... GEN 2.6 - 5
GEN 2.7 FIRST LIGHT AND LAST LIGHT COMPUTATIONS ..... GEN 2.7 - 1
5. LOCAL TIME ..... GEN 2.7 - 2
GEN 3. SERVICES ..... GEN 3.1 - 1
GEN 3.1 AERONAUTICAL INFORMATION SERVICES ..... GEN 3.1 - 1
1. RESPONSIBLE SERVICE ..... GEN 3.1 - 1
1.2 AIS Headquarters ..... GEN 3.1 - 1
1.3 International NOTAM Office ..... GEN 3.1 - 1
1.4 Hours of Operations ..... GEN 3.1 - 1
2. AREA OF RESPONSIBILITY ..... GEN 3.1 - 2
3. AERONAUTICAL PUBLICATIONS ..... GEN 3.1 - 2
3.2 Aeronautical Information Publication (AIP) ..... GEN 3.1 - 2
3.3 Amendment Service to the AIP (AIP AMDT) ..... GEN 3.1 - 2
3.4 Supplement to the AIP (AIP SUP) ..... GEN 3.1 - 3
3.5 NOTAM ..... GEN 3.1 - 3
3.6 Aeronautical Information Circular (AIC) ..... GEN 3.1 - 4
3.7 Aeronautical charts ..... GEN 3.1 - 4
3.8 Sale of Publications ..... GEN 3.1 - 5
4. AIRAC SYSTEM ..... GEN 3.1 - 5
5. PREFLIGHT INFORMATION SERVICE ..... GEN 3.1 - 6
6. DIGITAL DATA SETS ..... GEN 3.1 - 7

GEN 3.2 AERONAUTICAL CHARTS ..... GEN 3.2 - 1
1. RESPONSIBLE SERVICES ..... GEN 3.2 - 1
2. MAINTENANCE OF CHARTS ..... GEN 3.2 - 1
3. PURCHASE ARRANGEMENTS ..... GEN 3.2 - 1
4. AERONAUTICAL CHART SERIES AVAILABLE ..... GEN 3.2 - 1
4.2 General description of each series ..... GEN 3.2 - 2
5. LIST OF AERONAUTICAL CHARTS AVAILABLE ..... GEN 3.2 - 3
6. INDEX TO THE WORLD AERONAUTICAL CHART (WAC) - 1:1 000 000 ..... GEN 3.2 - 3
7. TOPOGRAPHIC CHARTS ..... GEN 3.2 - 3
8. CORRECTIONS TO CHARTS NOT CONTAINED IN THE AIP ..... GEN 3.2 - 3
GEN 3.3 AIR TRAFFIC SERVICES ..... GEN 3.3 - 1
1. RESPONSIBLE SERVICE ..... GEN 3.3 - 1
2. AREA OF RESPONSIBILITY ..... GEN 3.3 - 2
3. TYPES OF SERVICE ..... GEN 3.3 - 2
3.1 General ..... GEN 3.3 - 2
3.2 Air Traffic Control Service ..... GEN 3.3 - 2
3.3 Flight Information Service (FIS) ..... GEN 3.3 - 3
3.4 Alerting Service ..... GEN 3.3 - 6
3.5 Safety Alerts and Avoiding Action ..... GEN 3.3 - 6
4. CALCULATION OF LOWEST SAFE ALTITUDE ..... GEN 3.3 - 6
4.6 For Routes Defined by Radio Navigation Aids or to be Navigated by DR ..... GEN 3.3 - 7
4.7 For Routes Operated Under the RNP 2 Navigation Specification ..... GEN 3.3 - 8
4.8 For Other Area Navigation Operations ..... GEN 3.3 - 8
4.9 VFR Flights at Night ..... GEN 3.3 - 8
5. CONTINGENCY PROCEDURES - AIR TRAFFIC SERVICES TEMPORARILY NOT AVAILABLE ..... GEN 3.3 - 9
5.1 Introduction ..... GEN 3.3 - 9
6. CONTROL OF VEHICULAR AND PEDESTRIAN MOVEMENT ON AERODROMES ..... GEN 3.3 - 9
GEN 3.4 COMMUNICATION AND NAVIGATION SERVICES ..... GEN 3.4 - 1
1. RESPONSIBLE SERVICE ..... GEN 3.4 - 1
1.3 Applicable ICAO Documents ..... GEN 3.4 - 1
2. AREA OF RESPONSIBILITY ..... GEN 3.4 - 1
2.1 Continental and Oceanic Areas ..... GEN 3.4 - 1

2.2 External Territories ..... GEN 3.4 - 2
2.3 Extra Territorial Services ..... GEN 3.4 - 2
3. RADIO NAVIGATION SERVICE ..... GEN 3.4 - 2
3.1 General ..... GEN 3.4 - 2
3.2 Identification ..... GEN 3.4 - 2
3.3 Non-Directional Beacons ..... GEN 3.4 - 3
3.4 VHF Omni-Directional Radio Range (VOR) ..... GEN 3.4 - 3
3.5 Distance Measuring Equipment (DME) ..... GEN 3.4 - 3
3.6 Tactical Air Navigation (TACAN) ..... GEN 3.4 - 3
3.7 Public Broadcasting Stations ..... GEN 3.4 - 4
3.8 Abnormal Operation of Radio Navigation Aids ..... GEN 3.4 - 4
3.9 Remote facilities ..... GEN 3.4 - 4
3.10 Monitoring of Identifier ..... GEN 3.4 - 5
3.11 Global Navigation Satellite System (GNSS) ..... GEN 3.4 - 5
3.12 Global Positioning System (GPS) ..... GEN 3.4 - 5
4. COMMUNICATIONS SERVICES ..... GEN 3.4 - 5
4.1 Radio Frequencies ..... GEN 3.4 - 5
4.2 Certified Air/Ground Radio Service (CA/GRS) ..... GEN 3.4 - 6
4.3 UNICOM ..... GEN 3.4 - 8
4.4 Aerodrome Frequency Response Unit ..... GEN 3.4 - 9
4.5 Aeronautical Fixed Telecommunication Network (AFTN) ..... GEN 3.4 - 9
4.6 Special Broadcast Services ..... GEN 3.4 - 10
4.7 Aeronautical Codes ..... GEN 3.4 - 10
5. RADIOTELEPHONY PROCEDURES ..... GEN 3.4 - 10
5.1 Introduction ..... GEN 3.4 - 10
5.2 General ..... GEN 3.4 - 11
5.3 Transmission Format ..... GEN 3.4 - 11
5.4 Readback Requirements ..... GEN 3.4 - 12
5.5 Conditional Clearances ..... GEN 3.4 - 13
5.6 Route Terminology ..... GEN 3.4 - 13
5.7 Amended Route or Level ..... GEN 3.4 - 13
5.8 Language ..... GEN 3.4 - 14
5.9 Phonetic Alphabet ..... GEN 3.4 - 14
5.10 Pronunciation of numbers ..... GEN 3.4 - 15

5.11 Transmission of numbers ..... GEN 3.4 - 15
5.12 Aeronautical Station call signs ..... GEN 3.4 - 18
5.13 Aircraft call signs ..... GEN 3.4 - 19
5.14 Flight Number call signs - Using Group Form ..... GEN 3.4 - 20
5.15 Selection of Aircraft Identification Numbers and Suffixes ..... GEN 3.4 - 20
5.16 Ground Vehicles ..... GEN 3.4 - 21
5.17 Interchange and Leased Aircraft ..... GEN 3.4 - 21
5.18 Unmanned Aerial Vehicles ..... GEN 3.4 - 21
5.19 Call signs - Full and Abbreviated Formats ..... GEN 3.4 - 21
5.20 Registration of Radiotelephony Designators ..... GEN 3.4 - 23
5.21 Call signs - Special Task Operations ..... GEN 3.4 - 23
6. PHRASEOLOGIES ..... GEN 3.4 - 25
6.1 Standard Words and Phrases ..... GEN 3.4 - 25
6.2 Emergency ..... GEN 3.4 - 28
6.3 Traffic Alert and Collision Avoidance System (TCAS), Safety Alerts and Avoiding Action and Wind Shear Escape ..... GEN 3.4 - 30
6.4 Status of Restricted Areas and Military Operating Areas ..... GEN 3.4 - 33
6.5 SARWATCH ..... GEN 3.4 - 34
6.6 General Phrases ..... GEN 3.4 - 35
6.7 Frequency Management ..... GEN 3.4 - 39
6.8 Traffic Information ..... GEN 3.4 - 42
6.9 Meteorological Information ..... GEN 3.4 - 43
6.10 Reports and Information ..... GEN 3.4 - 45
6.11 Clearances ..... GEN 3.4 - 48
6.12 SID ..... GEN 3.4 - 52
6.13 STAR ..... GEN 3.4 - 56
6.14 Approach and Area Control Services ..... GEN 3.4 - 58
6.15 Phraseologies to be used related to CPDLC ..... GEN 3.4 - 62
6.16 Vicinity of the Aerodrome ..... GEN 3.4 - 63
6.17 ATS Surveillance Service Phraseologies ..... GEN 3.4 - 83

7. ATS DATA LINK SERVICES ..... GEN 3.4 - 95
7.1 General ..... GEN 3.4 - 95
7.2 Pilot Responsibilities ..... GEN 3.4 - 95
7.3 Controller Pilot Data Link Communication (CPDLC) ..... GEN 3.4 - 97
7.4 Logon Procedures ..... GEN 3.4 - 99
7.5 Position Reporting Requirements ..... GEN 3.4 - 101
7.6 CPDLC Level Reporting Following Change of Level ..... GEN 3.4 - 101
7.7 Reporting Back on Route by CPDLC ..... GEN 3.4 - 101
7.8 Block Level Clearances ..... GEN 3.4 - 102
7.9 Revision of Estimates Using CPDLC ..... GEN 3.4 - 102
7.10 CPDLC Message Restrictions ..... GEN 3.4 - 102
7.11 CPDLC CONTACT and MONITOR messages ..... GEN 3.4 - 102
7.12 CPDLC route clearances ..... GEN 3.4 - 103
7.13 CPDLC Failure ..... GEN 3.4 - 103
7.14 Notification Of Emergency ..... GEN 3.4 - 103
7.15 CPDLC Message Set ..... GEN 3.4 - 104
GEN 3.5 METEOROLOGICAL SERVICES ..... GEN 3.5 - 1
1. RESPONSIBLE SERVICE ..... GEN 3.5 - 1
1.1 Contact details: ..... GEN 3.5 - 1
1.2 Applicable ICAO Documents ..... GEN 3.5 - 1
1.3 Resources ..... GEN 3.5 - 2
2. AREA OF RESPONSIBILITY ..... GEN 3.5 - 2
3. METEOROLOGICAL OBSERVATIONS AND REPORTS ..... GEN 3.5 - 2
3.1 METAR/SPECI ..... GEN 3.5 - 2
3.2 Other Meteorological reports ..... GEN 3.5 - 2
4. TYPES OF SERVICES ..... GEN 3.5 - 3
4.1 ATS meteorological information service ..... GEN 3.5 - 4
4.2 Interpretation and use of forecasts ..... GEN 3.5 - 4
4.3 Forecasts for operations above 10,000FT ..... GEN 3.5 - 5
4.4 Significant Weather (SIGWX) charts ..... GEN 3.5 - 5
4.5 Grid Point Wind and Temperature (GPWT) forecasts ..... GEN 3.5 - 5
4.6 Graphical Area Forecasts (GAF) for operations surface to 10,000FT ..... GEN 3.5 - 6

4.7 Aerodrome Forecasts (TAF) ..... GEN 3.5 - 7
4.8 Airport Weather Briefings (AWB) ..... GEN 3.5 - 8
4.9 Wind shear warnings ..... GEN 3.5 - 8
4.10 Microburst and wind shear alerts ..... GEN 3.5 - 9
4.11 Volcanic Ash Advisories (VAA) ..... GEN 3.5 - 9
4.12 Space Weather (SWX) ..... GEN 3.5 - 10
4.13 Tropical Cyclones (TC) ..... GEN 3.5 - 10
5. NOTIFICATION REQUIRED FROM OPERATORS ..... GEN 3.5 - 11
5.1 For international operations ..... GEN 3.5 - 11
5.2 For domestic operations ..... GEN 3.5 - 11
5.3 Forecasts for flights - valid GAF not available ..... GEN 3.5 - 11
6. AIRCRAFT REPORTS ..... GEN 3.5 - 11
6.1 Routine aircraft observations ..... GEN 3.5 - 11
6.2 AIREP Special ..... GEN 3.5 - 12
6.3 Wind shear - reporting ..... GEN 3.5 - 12
6.4 Volcanic activity reports ..... GEN 3.5 - 13
7. VOLMET SERVICE ..... GEN 3.5 - 13
8. SIGMET AND AIRMET SERVICE ..... GEN 3.5 - 20
8.1 SIGMET ..... GEN 3.5 - 20
8.2 AIRMET ..... GEN 3.5 - 21
9. OTHER AUTOMATED METEOROLOGICAL SERVICES ..... GEN 3.5 - 21
9.1 Aerodrome Weather Information Service (AWIS) and Weather and Terminal Information Reciter (WATIR) ..... GEN 3.5 - 21
GEN 3.6 SEARCH AND RESCUE ..... GEN 3.6 - 1
1. PURPOSE ..... GEN 3.6 - 1
2. RESPONSIBLE AUTHORITY ..... GEN 3.6 - 1
3. ORGANISATION ..... GEN 3.6 - 1
4. RESCUE COORDINATION CENTRES ..... GEN 3.6 - 1
5. PROCEDURES ..... GEN 3.6 - 2
5.1 Pilot Notification ..... GEN 3.6 - 2
5.2 Advice to Aircraft Operators ..... GEN 3.6 - 2
5.3 In-Flight Emergency Response Action ..... GEN 3.6 - 2
5.4 Medical Emergency ..... GEN 3.6 - 3
5.5 Participation In Searches ..... GEN 3.6 - 4

6. EMERGENCY LOCATOR TRANSMITTER (ELT) ..... GEN 3.6 - 4
6.2 Monitoring 121.5MHz ..... GEN 3.6 - 4
6.3 Inadvertent Activations ..... GEN 3.6 - 4
6.4 Testing ELT ..... GEN 3.6 - 4
6.5 ELT Registration ..... GEN 3.6 - 5
7. COSPAS-SARSAT ..... GEN 3.6 - 5
8. SEARCH AND RESCUE TRANSPONDER (SART) ..... GEN 3.6 - 5
8.5 Example of SART - 40° left at 19NM ..... GEN 3.6 - 6
9. SAR ACTION ..... GEN 3.6 - 6
10. SRR BOUNDARY ..... GEN 3.6 - 7
GEN 4. CHARGES FOR AERODROMES/HELIPORTS AND AIR NAVIGATION SERVICES ..... GEN 4.1 - 1
GEN 4.1 AERODROME/HELIPORT CHARGES ..... GEN 4.1 - 1
1. LANDING CHARGES ..... GEN 4.1 - 1
2. PASSENGER MOVEMENT CHARGE ..... GEN 4.1 - 1
GEN 4.2 AIR NAVIGATION SERVICES CHARGES ..... GEN 4.2 - 1
1. AIRCRAFT CHARGES ..... GEN 4.2 - 1

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GEN 1. NATIONAL REGULATIONS AND REQUIREMENTS

GEN 1.1 DESIGNATED AUTHORITIES

1. DESIGNATED AUTHORITIES

1.1 **Department of Infrastructure, Transport, Regional Development, Communications and the Arts (Department of Infrastructure)**

a. Facilitation and aviation policy:

Assistant Secretary
COVID Aviation Reforms Branch
Aviation and Airports Group
Department of Infrastructure, Transport, Regional Development, Communications and the Arts
GPO Box 594
CANBERRA ACT 2601 AUSTRALIA
PH: 61 2 6274 7760
Email: internationalaviation@infrastructure.gov.au

b. Aircraft noise operating restrictions:

First Assistant Secretary - COVID Aviation Response Programs Division
Aviation and Airports Group
Department of Infrastructure, Transport, Regional Development, Communications and the Arts
GPO Box 594
CANBERRA ACT 2601 AUSTRALIA
PH: 61 2 6274 7892
Email: aircraftpermits@infrastructure.gov.au

1.2 **Department of Home Affairs**

a. **Aviation Security**
First Assistant Secretary
Critical Infrastructure Risk and Regulation Division
Department of Home Affairs
GPO Box 25
BELCONNEN ACT 2616 AUSTRALIA
PH: 61 2 6195 9584
Email: national.coordinator@homeaffairs.gov.au

b. Critical Centre - Critical Infrastructure Risk and Regulation Division Transport Security Incident Reporting
PH: 1300 791 581 (Option 0)
61 2 5127 8995 (outside Australia)
Email: transport.security@homeaffairs.gov.au

c. Critical Centre - Critical Infrastructure Risk and Regulation Division Transport Security Applications (e.g. Transport Security Programs) - Regulatory Assessment Operations
PO Box 25
BELCONNEN ACT 2616 AUSTRALIA
Email: national.coordinator@homeaffairs.gov.au

1.3 **Australian Border Force**

Director - Traveller Policy Section
Australian Border Force
PO Box 25
MAJURA PARK ACT 2609 AUSTRALIA
PH: 61 2 6264 1385
Email: travellerpolicy@abf.gov.au

1.4 **Civil Aviation Safety Authority**

a. Head Office:
Civil Aviation Safety Authority
GPO Box 2005
CANBERRA ACT 2601 AUSTRALIA
PH: 131 757 (within Australia)
61 2 6217 1449 (from overseas)
Web: www.casa.gov.au

b. International Operations:
International Operations
GPO Box 2005
CANBERRA ACT 2601 AUSTRALIA
PH: 61 7 3144 7400
Fax: 61 7 3144 7555
Email: international_ops@casa.gov.au
Web: https://www.casa.gov.au/licences-and-certificates/air-operators/foreign-operators

1.5 **Air Traffic Services**

Chief Executive Officer - Airservices Australia
Alan Woods Building
25 Constitution Ave
CANBERRA CITY ACT 2601, or
PO Box 367
CANBERRA ACT 2601
PH: 61 2 6268 4111
Fax: 61 2 6268 5693

1.6 **Department of Agriculture, Fisheries and Forestry (DAFF)**

Director – Conveyance, Passenger and Mail (CPM) Systems
Digital Systems, Solutions and Sustainment
PH: 61 2 6272 3933
Email: cpmsadmin@aff.gov.au

1.7 **Department of Health and Aged Care – Human Biosecurity**

Director - Border Health Section
Emergency Management Division
Chief Medical Officer Group Department of Health and Aged Care
GPO Box 9848
CANBERRA ACT 2601
PH: 1800 020 103
(after hours emergency number 61 2 6289 3030)
Email: humanbiosecurity@health.gov.au
Web: www.health.gov.au/humanbiosecurity

1.8 **Aircraft Accident Investigation**

Australian Transport Safety Bureau (ATSB)
PO Box 321
CANBERRA ACT 2601
PH: 1800 011 034
61 2 6122 1602
Email: atsbinfo@atsb.gov.au
Web: www.atsb.gov.au

1.9 **Meteorology**

Group Executive Enterprise Services and Chief Operating Officer
Australian Bureau of Meteorology
GPO Box 1289
MELBOURNE VIC 3001
PH: 61 3 9669 4000
Email: metauthority@bom.gov.au
Web: www.bom.gov.au/met-authority/

1.10 **Search and Rescue**

a. Australian Maritime Safety Authority (AMSA)
GPO Box 2181
CANBERRA ACT 2601
Email enquiries: www.amsa.gov.au/about/contact-us
Web: www.amsa.gov.au

b. For Search and Rescue General Enquiries:
PH: 1800 627 484
61 2 6279 5000

c. For the Australian Joint Rescue Coordination Centre (JRCC):
PH: 1800 815 257
61 2 6230 6899

1.11 **Department of Defence**

a. Chief of Air Force
Department of Defence
Building R1, Russell Offices
RUSSELL ACT 2600

b. Chief of Joint Operations
HQ Joint Operations Command
Department of Defence
PO Box 7928
CANBERRA BC ACT 2610

c. Diplomatic Clearance Authorisation Cell
PH: 61 2 6128 4819
Email enquiries: dipa.hqac@defence.gov.au

GEN 1.2 ENTRY, TRANSIT AND DEPARTURE OF AIRCRAFT

1. ENTRY, TRANSIT AND DEPARTURE OF INTERNATIONAL FLIGHTS

1.1 Preamble

1.1.1 All flights into, from, or over Australian territory, and landings in such territory must be carried out in accordance with the legislation of Australia regarding civil aviation. This includes obtaining separate approvals pertaining to air safety, aviation security, aircraft licencing and landing permits, border and biosecurity controls from a number of independent Australian Government Departments detailed in section *GEN 1.1,* including:

a. The Department of Home Affairs.

b. The Civil Aviation Safety Authority (CASA).

c. The Department of Infrastructure.

d. The Department of Agriculture, Fisheries and Forestry.

1.1.2 In accordance with *Section 10* of the *Air Navigation Act 1920*, aircraft arriving in or departing from any part of Australian territory must land at and depart from airports designated for that purpose.

1.1.3 The *Customs Act 1901 (s58(1))* and the *Migration Act 1958 (s247)* also require that aircraft arriving into Australian territory from overseas must arrive at an airport appointed under the *Customs Act*. Information on airports for entry and departure are shown in *GEN 1.2 para 2.* and *GEN 1.3 para 8.*

1.1.4 Aircraft which are completely cleared by the Australian Border Force (ABF) at a designated international airport are permitted to land at other airports within Australian territory. Nevertheless, when such aircraft depart from Australian territory, they can only do so from a designated international airport.

1.1.5 Aircraft that are not fully cleared by Department of Agriculture, Fisheries and Forestry (DAFF) - Biosecurity and formally released from biosecurity control, at their first airport of call, continue to be subject to biosecurity control and restrictions for their movements to other locations, until released.

1.2 Scheduled International Commercial Services by Foreign Aircraft of Contracting States to the Chicago Convention

1.2.1 **International Airline Licence:** An international airline of a country other than Australia must not operate a scheduled international air service over or into Australian territory except in accordance with an International Airline Licence.

1.2.2 The requirement to hold an International Airline Licence does not apply to the following categories of scheduled international air services:

a. scheduled international air services that fly over Australian territory but do not land in Australian territory;

b. scheduled international air services that land in Australian territory but do not set down or take on passengers or cargo for reward or hire;

c. scheduled international air services operated in accordance with an agreement, between an international airline and the holder of an International Airline Licence, which the Secretary of the Department of Infrastructure has approved in writing.

1.2.3 An International Airline Licence shall not be granted to an international airline of a country other than Australia unless that country and Australia are parties to an agreement or arrangement under which the scheduled international air services of that other country may be operated over or into Australian territory.

1.2.4 Applicants for an International Airline Licence may obtain a copy of Guidance Notes for Applicants at: https://www.infrastructure.gov.au/infrastructure-transport-vehicles/aviation/international-aviation/international-airline-licences/guide

1.2.5 All holders of International Airline Licences are required to notify the Secretary of the Department of Infrastructure in an approved form of any new or changed international (scheduled) air service.

1.2.6 **Lodging of timetables:** The requirements for lodging of timetables by international airlines are set out in *Air Navigation Regulations 2026, Section 25*. This is available on the Federal Register of Legislation at: www.legislation.gov.au

1.2.7 An application for approval of a timetable must be lodged with the Secretary not less than 35 days before the date from which the airline proposes to operate the service in accordance with the timetable, or within such shorter period as the Secretary allows.

1.2.8 Should an airline wish to lodge an application in less than the 35 day period, it should provide its reasons for wanting to do so. For practical purposes, lodgement "with the Secretary" means lodgement with the Assistant Secretary, International Aviation, Department of Infrastructure, at the address shown in *GEN 1.1.*

1.2.9 Further information on timetable applications is available on the Department of Infrastructure's website: https://www.infrastructure.gov.au/infrastructure-transport-vehicles/aviation/international-aviation/international-airline-licences/application#anc_7

1.2.10 **Australian Foreign Air Transport Air Operator's Certificate (FATAOC):** In addition to an Australian International Airline Licence, an operator of a proposed scheduled service to Australia is required to hold an FATAOC issued by CASA.

1.2.11 Application forms for FATAOC are available from CASA International Operations (see contact details in *GEN 1.1*)

1.3 **Summary of Documents to be presented by Pilot in Command or Authorised Agent**

Note: All required documents must be furnished in English, and originals and all copies must be completely legible. Names should be shown in block letters, and with regard to names of passengers, initials at least are to be inserted. Documents must be fully and accurately completed.

1.3.1 **At First Airport of Call in Australia**

a. **Impending Arrival Report**

The aircraft operator must report the impending arrival of the aircraft to the Australian Border Force, regardless of whether or not the aircraft is carrying cargo. If the aircraft is carrying cargo, the Impending Arrival Report must be lodged electronically in the Integrated Cargo System (ICS):

(i) not more than 10 days before the estimated time of arrival of the aircraft, and:

(1) not later than three hours before the estimated time of arrival of the aircraft if the flight from the airport is likely to take not less than three hours; or

(2) one hour if the flight from the airport is likely to take less than three hours.

If the aircraft is not carrying cargo, the Impending Arrival Report may be lodged either electronically in the ICS, or by Form B364, available online: www.abf.gov.au/form-listing/forms/b364.pdf, within the above time frames.

b. **Actual Arrival Report**

The aircraft operator must report the particulars of the arrival of the aircraft and the time of arrival to the Australian Border Force. Where an aircraft is carrying cargo, the Actual Arrival Report must be lodged electronically in the ICS within three hours of the arrival of the aircraft or before the certificate of clearance is issued, whichever occurs first.

If the aircraft is not carrying cargo, the Actual Arrival Report may be lodged electronically in the ICS, or by Form B358, available online: www.abf.gov.au/form-listing/forms/b358.pdf, providing the reporter satisfies an Evidence of Identity and the form is signed in the manner specified on the form.

The Actual Arrival Report in ICS is made in place of a General Declaration. However, some countries still require General Declarations for arriving aircraft. ABF Officers will stamp the General Declaration on departure of the aircraft in these circumstances.

Note: ABF officers may still require a General Declaration be produced as part of general mandatory disclosures.

c. **List of Stores and Prohibited Goods (e.g. narcotic drugs, beer, wine, spirits and tobacco products)**

The aircraft operator must report the particulars of the aircraft stores and of any prohibited goods contained in those stores within the same timeframe as the Actual Arrival Report to Australian Border Force. A report may be made using form B367, available online:
www.abf.gov.au/form-listing/forms/b367.pdf.
See *GEN 1.4 para 1.2* for more information.

d. **Cargo Report**

The cargo report is used to report the particulars of all cargo (including mail, in-transit and transhipment cargo) on board an aircraft. The carrier is required to report the full detail of cargo, including any cargo carried on behalf of another cargo reporter to the Australian Border Force and the details of the depot operator who will first receive the cargo after it has been unloaded from the aircraft at a place in Australia. Private charter flights which are carrying cargo (air freight) are not exempt from these requirements. See *GEN 1.4 para 1.1* for more information.

e. **Crew Declaration – One copy per crew member**

Each individual crew member must complete a copy of Crew Declaration form B465, available online:
https://www.abf.gov.au/form-listing/forms/b465.pdf.

1.3.2 **At Airports other than First Airport of Call in Australia**

The requirements for the following reports are the same as those outlined in *para 1.3.1*.

a. Impending Arrival Report;

b. Actual Arrival Report;

c. List of Stores and Prohibited Goods (e.g. narcotic drugs, beer, wine, spirits and tobacco products).

Further, at stops other than the first airport of call, the pilot in command or authorised agent will present the copy of the Certificate of Clearance signed and stamped by the ABF Officer prior to departure from the previous stop.

If the aircraft is due to arrive at its first airport of call since its last departure airport outside Australia, the pilot or authorised agent must report to the Department of Home Affairs, in accordance with this section, particulars of all cargo:

a. the pilot in command or the authorised agent has arranged to be carried on the aircraft on the flight; and

b. that is intended to be unloaded from the aircraft at an airport in Australia (whether the first airport or any subsequent airport on the same flight).

1.3.3 **At First Airport of Departure from Australia**

a. **Export Permits (where required)**

Note: One copy of each manifest must be initialled by the pilot in command or authorised agent.

b. **List of Stores that are prohibited narcotic drugs** – See *GEN 1.4 para 1.2*.

c. **Departure Report –** The departure report is a prerequisite that must be satisfied before a certificate of clearance can be granted by the Australian Border Force (ABF). See *GEN 1.4 para 1.5*.

d. **Outwards (Main) Manifest (electronic, lodged in the ICS) –** The outwards manifest is used to notify the ABF of all goods that were loaded on board the aircraft. See *GEN 1.4 para 1.5*.

If a departing aircraft is not carrying any export cargo, a manifest must still be lodged. This manifest states that no cargo was loaded and is called a 'Nil Manifest'.

e. **Certificate of Clearance –** The pilot of an aircraft must not depart from any airport without receiving a signed and stamped Certificate of Clearance in respect to the aircraft from an ABF Officer.

1.4 Non-Scheduled International Commercial Services by Foreign Aircraft of Contracting States to the Chicago Convention

1.4.1 **Definition of Non-scheduled Services.** Non-scheduled service, in relation to an aircraft that possesses the nationality of a Contracting State, means a flight by that aircraft over or into Australian territory other than under the authority of an International Airline Licence.

1.4.2 Where an aircraft makes a commercial non-scheduled flight into or out of Australian territory, it must have prior permission from the Department of Infrastructure and CASA.

Some categories of non-scheduled flights have standing Department of Infrastructure approvals. See *para 1.6*.

1.4.3 Department of Infrastructure international freight and charter policy guidelines, and permission for flights, may be obtained from the address shown in *GEN 1.1* or at: https://www.infrastructure.gov.au/infrastructure-transport-vehicles/aviation/international-aviation. When applying for permission for flights, applicants must advise the Department of Infrastructure of permission for slot allocation from Airport Coordination Australia (see *ENR 1.9 para 2*.)

An application form (Form 094) for CASA permission for non-scheduled flights can be obtained from the address in *GEN 1.1.*

1.4.4 At the same time that permission is sought from Department of Infrastructure and CASA, an aircraft operator, through the designated representative, must obtain and agree to the conditions of use (including aircraft charges) or similar policies as applied by each airport owner or operator. Airport Owners and Operators are listed in *ERSA FAC*.

1.4.5 Applications for permission should be accompanied by the prescribed fees. Details of application fees under Part 5, Division 3 of the *Air Services Act 1995* may be obtained from Airservices at the addresses shown at *GEN 1.1*.

1.5 Non-Scheduled International Commercial Services by Australian Aircraft

1.5.1 Department of Infrastructure permission must be sought as per the requirements of *para 1.4* or via standing approvals outlined in *para 1.6*.

1.5.2 CASA permission is required under *Section 26* of the *Civil Aviation Act 1988*. Operators should apply to CASA.

Note: Separate CASA permission is not required if the operator's Air Operator Certificate is specifically endorsed to authorise such international flights.

1.5.3 At the same time that permission is sought from Department of Infrastructure and CASA, an aircraft operator, through the designated representative, must obtain and agree to the conditions of use (including aircraft charges) or similar policies as applied by each aerodrome owner or operator. Aerodrome owners and operators are listed in *ERSA FAC*.

1.6 Non-Scheduled International Commercial Services by Australian Aircraft and Foreign Aircraft of Non-Contracting States and Contracting States, which do not Require Prior Approval from the Department of Infrastructure

1.6.1 A delegate of the Secretary of the Department of Infrastructure pursuant to subsection 15A(3) of the *Air Navigation Act 1920* has determined that permission is not required in respect of the following categories charter flights:

Note: "charter flight" means a non-scheduled flight by an aircraft which takes on or discharges passengers, cargo or mail for carriage for reward in Australian territory.

a. A single charter flight which does not form part of a program, subject to the following condition;

   (i) while the Biosecurity (Human Biosecurity Emergency) (Human Coronavirus with Pandemic Potential) Declaration 2020 remains in force, this exemption does not apply to a charter flight conducted by an aircraft that is carrying one or more passengers and has a seating capacity exceeding 80 persons

b. programs of passenger charter flights by aircraft with a maximum seating capacity of 10 or fewer seats;

c. programs of five or less passenger charter flights with aircraft having a maximum seating capacity of 40;

d. programs of two or less passenger charter flights with aircraft having a maximum seating capacity of 80;

e. own-use charter flights, whether or not forming part of a program, where there is a single charterer;

f. charter flights for carriage of homogenous cargo - a charter flight, whether or not forming part of a program, where there is a single charterer, and the dominant purpose of the flight is the transport of cargo. The cargo must not consist of or include a consignment consolidated by a freight forwarder. The cargo is homogenous. Examples of homogenous cargo include cargo consisting of electronic equipment; cargo consisting of meat of any number of kinds; cargo consisting of animals (including livestock) of any number of species.

1.6.2 Operators of any flight included in one of the categories above are required under subsection 15A(7) of the *Air Navigation Act 1920*, within 14 days after the end of the flight, to give a written notice setting out details in relation to the flight and the passengers, cargo and mail. This notice is to be provided to the Assistant Secretary of International Aviation, Department of Infrastructure, Transport, Regional Development, Communications and the Arts at the address in *GEN 1.1*. The required details are:

a. the name and address of the charterer;

b. the name and address of the charter operator;

c. the type and capacity of the aircraft;

d. whether the flight was a single charter flight or part of a program of charter flights;

e. if the aircraft carried cargo, the type of cargo;

f. the following particulars of the flight:

(i) the place where the flight began;

(ii) the place where the flight ended;

(iii) any intermediate stopping places, specifying at which places passengers, cargo or mail were taken on or discharged;

(iv) the dates of departure from, and arrival at, the places mentioned in the preceding subparagraphs;

g. whether the flight was a “charter flight for carriage of homogenous cargo” and whether the flight was an “own-use charter flight”.

1.6.3 Operators may provide details of more than one flight in a single notification provided that the Department of Infrastructure receives notification of all flights within 14 days after the end of each individual flight. If Australian authorities find that charter flights coming within the above categories are being repeated in close proximity, operators may be required to submit applications for assessment of any further flights.

1.6.4 All operators should note the separate requirements for transport security program approval and aircraft noise certification.

1.6.5 At the same time that permission is sought from Department of Infrastructure and CASA, an aircraft operator, through the designated representative, must obtain and agree to the conditions of use (including aircraft charges) or similar policies as applied by each aerodrome owner or operator. Aerodrome owners and operators are listed in *ERSA FAC*.

1.7 International Flights by Foreign Aircraft not Possessing Nationality of Contracting State to the Chicago Convention

1.7.1 For international flights over or into Australian territory where the carrier is registered in a State which is not a party to the Chicago Convention, the operator must obtain prior permission from the Department of Infrastructure, Transport, Regional Development, Communications and the Arts, and CASA through diplomatic channels. All such requests must be in writing.

1.8 Flights by Foreign State Aircraft

1.8.1 Aircraft Diplomatic Clearance Application

Diplomatic clearance is required for foreign state aircraft to enter and operate in Australian territory, including the airspace above Australian offshore islands. Applications for foreign state aircraft to conduct flying operations within Australian territory must be submitted at least five working days before the proposed date of entry into Australian airspace.

At least two weeks' notice is required if military services are requested (such as parking at a Royal Australian Air Force base). A separate application must be submitted for each aircraft, or formation of aircraft. The Department of Foreign Affairs web pages provide aircraft diplomatic clearance application forms and instructions, at: www.dfat.gov.au/about-us/foreign-embassies/protocol/Pages/diplomatic-clearances-aircraft-and-ships

1.8.2 Applications are submitted by completing the application form and emailing it to: dipa.hqac@defence.gov.au. Diplomatic clearance applicants will be required to provide the following information:

a. Requesting Country;

b. Point of Contact details;

c. Purpose of the flight;

d. Aircraft Operator (if civil registered, the address and nationality);

e. Aircraft Type;

f. Aircraft Registration Mark;

g. Aircraft call sign;

h. Itinerary (including previous and next destinations);

i. Flight Routes;

j. Aircraft Captain Details;

k. Crew and Passenger Numbers;

l. VIP Details (if applicable);

m. Weapons Details;

n. Dangerous Cargo Details;

o. Ground Handling Details (for aircraft landing at Defence Establishments); and

p. Ground Handling Agent (for aircraft landing at Civilian Airfields).

1.8.3 In the event of any changes to the flight details, the request must be updated and be resent as soon as possible to: dipa.hqac@defence.gov.au.

1.8.4 Any questions relating to diplomatic clearances should be addressed to the Diplomatic Clearance Authorisation Cell on +61 2 6128 4819 or dipa.hqac@defence.gov.au.

1.8.5 **Compliance**

Foreign State aircraft operating under diplomatic clearance are required to comply with applicable aerodrome procedures and air traffic control directions.

1.8.5.1 Diplomatic clearance does not exempt the requesting government's responsibility to meet other Australian Government requirements such as customs, biosecurity and immigration or the requirement for permission to carry or import munitions or implements of war. It is the responsibility of the foreign government to ensure all necessary paperwork and/or clearances from Australian Government agencies are arranged prior to arrival.

1.8.5.2 A *CASR 91.975* approval from the Australian Civil Aviation Safety Authority (CASA) is also required. In certain circumstances, diplomatic clearance may only be issued subject to the aircraft, cargo and passengers undertaking additional checks and searches. Failure to comply with any conditions on a diplomatic clearance, or with other government agencies' procedures, could result in penalties and affect issuance of future diplomatic clearances.

1.8.6 **Foreign Military Aircraft Participating in Exercises within Australia**

Foreign State aircraft visiting Australia for a combined exercise are required to seek diplomatic clearance and CASA permission for their transit to and from their operating location. All exercise flights are covered under the exercise arrangements. However, any transits from their deployed location to another location outside the exercise schedule will require approval. Certain military aircraft will require permission to import from the Australian Border Force. Export permission from the Department of Defence may also be required.

1.8.7 **Alternate Routes and In-flight Emergencies**

Approval will not be issued for alternate or diversion airfield requests inside Australia. Aircraft diverting in response to an in-flight emergency or poor weather do not require diplomatic approval and should select the nearest appropriate airfield. Notification of any emergency diversions should be sent to the Diplomatic Clearance Cell at the earliest opportunity thereafter landing (dipa.hqac@defence.gov.au).

1.9 **International Private Flights**

1.9.1 A private flight means a flight by an aircraft carrying passengers or cargo whose costs are met by the owner and/or operator of the aircraft.

1.9.2 *Section 14* of the *Air Navigation Act 1920* allows an aircraft that possesses the nationality of a Contracting State undertaking a private flight to enter or leave Australia or fly in transit across Australia without the requirement of obtaining prior permission.

1.9.3 No specific operational assessment is required for such flights, but pilots are advised of the following:

a. in addition to the requirements of *Division 91.C.3* of *CASR*, a journey log must be carried which details particulars of the aircraft, its crew and of each journey.

b. the rules and regulations pertaining to the flight and manoeuvring of aircraft must comply with the following:

(i) Australian rules, where applicable;

(ii) ICAO rules, when operating over the high seas; and

(iii) foreign State rules, where applicable, while flying over a foreign State's territory.

c. Also note the provisions of *para 1.16* in relation to aircraft noise certification.

1.9.4 The pilot in command must assume and exercise responsibility for the operation and safety of the aircraft from the moment that the doors of the aircraft are closed until the moment when it comes to rest at the end of the flight and the engines used as primary propulsion units are shut down.

1.9.5 If an emergency endangers the safety of the aircraft or persons on board and requires action in violation of the law of a foreign State, the pilot in command must:

a. as soon as practicable, tell the foreign State authority responsible for that law; and

b. if required by the authority, give a written report of the violation to the authority; and

c. send a copy of the report to CASA.

1.9.6 The pilot in command must assume and exercise responsibility to tell the nearest appropriate State authority, by the quickest means possible, of any accident involving the aircraft that has resulted in:

a. death or serious injury to any person; or

b. substantial damage to the aircraft or any property.

1.9.7 The pilot in command may not operate at an aerodrome using lower operating minima than those established for the aerodrome by the responsible authority, without approval from the responsible authority. The pilot in command may not utilise operating minima lower than 200FT above ground level without approval from CASA.

1.9.8 Before a flight, the pilot in command must ensure that the aircraft is carrying the following:

a. an accessible first aid kit;

b. current and suitable charts for the route of the proposed flight and for all routes along that route to which it is reasonable to expect the flight may be diverted;

c. procedures for pilots in command of intercepted aircraft, as described in *Annex 2* to the *Chicago Convention*;

1.9.9 Before departure from Australian territory, the pilot in command must ensure that:

a. the certificate of airworthiness for the aircraft will remain valid while the aircraft is outside Australian territory; and

b. either:

(i) the maintenance release for the aircraft will not expire while the aircraft is outside Australian territory; or

(ii) before the maintenance release expires, an authorised person will issue a maintenance release; and

c. if the pilot in command is not the registered operator of the aircraft - arrangements are in place with the registered operator for the pilot in command to be notified about any urgent maintenance or operational requirements while the aircraft is outside Australian territory.

Note: Urgent maintenance or operational requirements include, for example, Airworthiness Directives.

1.10 **Aviation Security**

1.10.1 All aviation security incidents must be reported to the Department of Home Affairs, Critical Infrastructure Risk and Regulation Division (see *GEN 1.1*). Refer to: https://www.homeaffairs.gov.au/help-and-support/departmental-forms/online-forms/aviation-security-incident-report

1.10.2 Under the *Aviation Transport Security Act 2004*, aviation security incidents are defined as:

a. a threat of unlawful or operational interference with aviation; or

b. an unlawful or operational interference with aviation.

1.11 Aviation Security: Transport Security Program requirements

1.11.1 The *Aviation Transport Security Act 2004* (Division 2 of Part 2) requires that all operators of a security controlled airport or a prescribed air service operating within, to, or from Australia have a Transport Security Program approved by the Department of Home Affairs.

Note: A prescribed air service is defined in regulation 1.06 of the Aviation Transport Security Regulations 2005 and includes:

a. a regular public transport operation; or

b. an air service in which a jet is used; or

c. an air service in which an aircraft with a certificated maximum take-off weight greater than 5,700 kilograms is used.

1.11.2 A Transport Security Program is required to articulate the security measures and procedures that will be implemented by the aircraft operator of a prescribed air service to prevent unlawful interference with aviation when operating from, to or within Australia and its territorial limits. The Transport Security Program must demonstrate that the operator understands its role, its local risk context, obligations and responsibilities in managing aviation security.

1.11.3 Transport Security Program assessment is subject to a 60 day consideration period. Operators of a prescribed air service are strongly encouraged to submit a Transport Security Program for consideration at least 60 days prior to the intended commencement of air services into or within Australia to ensure that planned aviation services are not impacted. The submitted Transport Security Program must meet all requirements set out in *sections 16* and *17* of the *Aviation Transport Security Act 2004* and *Part 2 Division 2.3* of the *Aviation Transport Security Regulations 2005*.

1.11.4 The Australian Government does not recognise aviation security programs approved by other nations. It is an offence, under the *Aviation Transport Security Act 2004*, to operate a prescribed air service into or within Australia without an approved Transport Security Program in force. Exemptions to these requirements are not generally available.

1.11.5 Guidance for the production and submission of a Transport Security Program is available on the Department of Home Affairs website at: https://www.cisc.gov.au/information-for-your-industry/transport/aviation/aircraft-operators-security-requirements

1.11.6 Transport Security Program submissions are to be lodged by operators of prescribed air services by email to: national.coordinator@homeaffairs.gov.au

1.11.7 Operators of prescribed air services that are registered in a nation/country other than Australia are required to provide proof of legal entity name from their country of business registration as part of their submission.

1.11.8 The Department of Home Affairs prefers to communicate directly with the prescribed aircraft operator making application by an authorised direct employee representative of the aircraft operator/company at time of application.

1.11.9 Enquiries around regulatory requirements or submissions under development should be directed to national.coordinator@homeaffairs.gov.au within the Critical Infrastructure Risk and Regulation Division, Department of Home Affairs (see GEN 1.1). Where an operator of a prescribed air service proposes to operate directly into Australia from a last port of call that is not already recognised and approved, a comprehensive aviation security risk assessment of that location and appropriate risk controls is required to accompany the Transport Security Program.

1.11.10 Enquiries whether a Transport Security Program is required are to include:

a. whether an application has been made or approved for “Flights by Foreign State Aircraft” – see *para 1.8*.

b. if an application has not been made or approved then the following information is required:

(i) the legal entity name of the aircraft operator;

(ii) the name and address of the aircraft operator;

(iii) the name and address of any engaged or intended ground handling agent;

(iv) the type and capacity of the aircraft;

(v) whether the flight is a single charter or part of scheduled or unscheduled program of flights and the period in which it is intended to operate;

(vi) the nature of the aircraft operations (purpose of flight) including if the aircraft is carrying cargo, the type of cargo;

(vii) the place/places where the flight/flights will originate; (both external to Australia and within Australia);

(viii) any intermediate stopping places specifying at which places passengers or cargo (including mail) were taken on board or discharged;

(ix) the intended date of commencement of the service/ flight; and

(x) details of any wet or dry lease arrangements.

1.12 **Carriage of Munitions of War and Implements of War**

1.12.1 In accordance with *Section 19(2)* of the *Air Navigation Act 1920*, munitions of war or implements of war must not be carried by an aircraft in or over Australian territory, or by an Australian aircraft outside Australian territory, except with the permission in writing of the delegate of the Minister for Infrastructure, Transport and Regional Development.

1.12.2 Applications for transportation of munitions of war by State aircraft of a country other than Australia in Australian territory, or by an Australian aircraft outside Australian territory should be lodged with the Chief of Air Force or Chief of Joint Operations, Department of Defence (see *GEN 1.1*).

1.12.3 Applications for carriage of munitions of war or implements of war in all other circumstances should be lodged with the Secretary of the Department of Infrastructure, through the Assistant Secretary Aviation Industry Policy, Aviation and Airports Division (see *GEN 1.1*).

1.12.4 Certain firearms and weapons will require permission to import from the Australian Border Force, the Attorney-General's Department, or state/territory police - depending on the firearm or weapon type and intended use in Australia. Export permission from the Department of Defence may also be required.

1.13 **Foreign Clearances - Australian Aircraft**

1.13.1 Australian aircraft operators are responsible for obtaining foreign clearances when necessary for overflights of, or landings in, the territory of another State.

1.13.2 For guidance on this process, contact the Diplomatic Clearance Officer, Department of Defence via email at: dipa.hqac@defence.gov.au

1.13.3 Pilots are advised that their flight plans will not be considered by some countries unless documentation of onward foreign clearance is produced.

1.14 **Aircraft on International Flights to Comply with Laws**

1.14.1 The owner, operator, hirer, pilot in command and any other pilot of any aircraft granted a licence, permission or approval that enters or departs Australian territory must comply with the provisions of all applicable laws of the Commonwealth or of a State or Territory. This includes laws relating to entry and departure or clearance of passengers, crew and/or cargo, immigration, passports, customs and biosecurity. See *Section 16* of the *Air Navigation Act 1920* for more information.

1.15 **Section 22 of the Civil Aviation Act 1988**

1.15.1 *Section 22* of the *Civil Aviation Act 1988* gives effect in Australian law to the provisions of *Article 3* of the *Chicago Convention*. This prohibits the use of force against civil aircraft and provides for the regulation of civil aircraft flying over the territory of foreign countries without authorisation or for any purpose that is inconsistent with the aims of the *Chicago Convention*.

1.15.2 Aircraft under Australian jurisdiction shall not be flown over the territory of a foreign country without authorisation or for a purpose that is prejudicial to the security, public order or safety of air navigation in that country. If an aircraft is being flown in these circumstances, the pilot in command must comply with an order to land or any other instruction that is given by an authorised official of the foreign country.

1.15.3 The pilot in command who is found guilty of an offence under *Section 22* shall be subject to severe penalties such as imprisonment. Ancillary offenders (for example, the operator) may be prosecuted under the *Crimes Act 1914*.

1.15.4 Nothing in this legislation relieves obligations which any other law, including the law of a foreign country, might impose. An Australian aircraft which is flying with proper authorisation over the territory of a foreign country is required to obey a direction legally given by the aviation authorities of the country concerned, just as a foreign aircraft flying over Australian territory is required to obey a lawful direction which may be given by the relevant Australian authorities.

1.16 **Aircraft Noise Operating Restrictions**

1.16.1 Under the *Air Navigation (Aircraft Noise) Regulations 2018*, international and domestic aircraft operating in Australia are required to be certified as compliant with the relevant ICAO *Annex 16* noise standards. Subsonic jets must be certified as Chapter 3, Chapter 4 or Chapter 14. Aircraft with Chapter 2 noise certification are not permitted to operate.

1.16.2 Large, Marginally noise Compliant Chapter 3 (MCC3) jet aircraft are prohibited from operating at: Sydney, Melbourne, Brisbane, Perth, Adelaide, Hobart, Canberra, Darwin, Cairns, Gold Coast, Williamtown, Essendon Fields and Avalon airports. For a full list of potentially affected aircraft types contact the Assistant Secretary, Airports Branch. The department email address is aircraft.permits@infrastructure.gov.au (see *GEN 1.1*).

1.16.3 Operators of MCC3 aircraft who want to operate at these airports must apply to the Department of Infrastructure for a (noise) exemption to operate. The department email address is aircraft.permits@infrastructure.gov.au.

1.16.4 Aircraft owners and pilots requiring information about aircraft noise operating restrictions or the *Air Navigation (Aircraft Noise) Regulations 2018* should contact the Assistant Secretary, Airports Branch.

IMPORTANT - International operators must also contact CASA to apply for a safety permission to operate in Australian Territory. Contact details available online:
www.casa.gov.au/aircraft/foreign-operators

International operators must also contact the Department of Home Affairs to apply for aviation security approval to operate in Australian Territory. Contact details available online: www.homeaffairs.gov.au/about-us/our-portfolios/transport-security/air-cargo-and-aviation/aviation/aircraft-operators.

1.17 **Air Carriers' Liability and Insurance**

1.17.1 Australia's carriers' liability and insurance arrangements are outlined in the *Civil Aviation (Carriers' Liability) Act 1959* (the CACL Act). The Act gives the force of law to a number of passenger liability frameworks, including those arising under the 'Warsaw System', the Montreal Convention, and a separate system of liability for domestic travel.

1.17.2 Commercial passenger transport operations are required to maintain appropriate insurance arrangements in accordance with the requirements of the Act and the operator's licence(s).

1.18 **Australian Operational Documents Available to Pilots Licensed by another State Proposing to Visit Australia**

1.18.1 For pilots licensed by another State proposing to come to Australia, Airservices Australia has available the following documents which relate to the proposed flight. These are available on a payment basis:

a. **IFR Flight Documents** Australian AIP Book; En Route, Planning and Terminal Charts; Departure and Approach Procedures and ERSA.

b. **VFR Flight Documents** Australian AIP Book; En Route, Planning and Visual Terminal Charts, and ERSA.

1.18.2 A login using a valid Aviation Reference Number is required to order AIP products. The documents are ordered through www.aipshop.canprint.com.au

2. DESIGNATED INTERNATIONAL AIRPORTS - AUSTRALIA

Note: Operations by aircraft at all of the airports listed in the following section are limited to the pavement strength shown against the aerodrome in AIP ERSA. Prior application must be made to the airport operator for a pavement concession where this is necessary.

2.1 **Major International Airports**

2.1.1 "Major International Airport" means an airport of entry and departure for international air traffic where there is an ongoing border agency presence to conduct all formalities incident to Customs, Immigration and Biosecurity clearance.

Airport	Clearances Available
Adelaide	Customs, immigration and biosecurity.
Brisbane	Customs, immigration and biosecurity.
Cairns	Customs, immigration and biosecurity.
Darwin	Customs, immigration and biosecurity.
Melbourne	Customs, immigration and biosecurity.
Perth	Customs, immigration and biosecurity.
Sydney/Kingsford Smith	Customs, immigration and biosecurity.
Sydney/Nancy-Bird Walton	Customs, immigration and biosecurity.

Note: Reasonable notification required for non-scheduled traffic. Mandatory biosecurity reporting requirements apply to all non-scheduled flights landing in Australian Territory.

2.2 **Restricted Use International Airports**

2.2.1 "Restricted Use International Airport" means an airport of entry and departure at which the formalities incident to Customs, immigration, and biosecurity and similar procedures are made available on a restricted basis, to coincide with flights with prior approval only.

2.2.2 Restricted Use International Airports are:

a. Brisbane West Wellcamp

*Note: A First Port of Entry for biosecurity for overseas freighter aircraft **only** and goods carried on a freighter aircraft, except live horses. All other aircraft require prior DAFF - Biosecurity approval to land.*

b. Broome

c. Canberra

d. Coffs Harbour

Note: Coffs Harbour is not determined as a first point of entry for overseas aircraft - requires prior DAFF - Biosecurity approval to land.

e. Gold Coast

Note: Although not categorised as a major international airport, Gold Coast Airport has an ongoing border agency presence to conduct customs, immigration and biosecurity clearances.

f. Hobart

g. Karratha

Note: Karratha is not determined as a biosecurity first point of entry for overseas aircraft - requires prior DAFF - Biosecurity approval to land.

h. Learmonth

Note: Learmonth is a determined first point of entry for biosecurity for overseas aircraft, but goods are not permitted to be unloaded - requires prior DAFF - Biosecurity approval.

i. Lord Howe Island

Note: Lord Howe Island is not determined as a biosecurity first point of entry for overseas aircraft - requires prior DAFF - Biosecurity approval to land.

j. Melbourne/Avalon

k. Port Hedland

l. Sunshine Coast

m. Townsville

n. Williamtown/Newcastle

2.2.3 The airline or its agent/representatives may be responsible for covering additional expenses relating to the positioning of resources from another border agency base to a Restricted Use International Airport.

2.3 **Alternate Airports to International Airports**

2.3.1 “Alternate Airport” means an airport specified in the flight plan to which a flight may proceed when it becomes inadvisable to land at the airport of intended landing (see also *GEN 1.3 para 6.*).

2.3.2 The Airline or its agent/representatives may be responsible for covering additional expenses relating to the positioning of resources from another border agency base to an Alternate Airport.

2.3.3 The following designated airports have customs, immigration and biosecurity clearances available if reasonable notification of diversion is given:

a. Canberra

Note: May be nominated as an International alternate subject to first obtaining Canberra Airport consent. Canberra Airport may require a separate agreement as capacity is limited.

b. Gold Coast

c. Port Hedland

d. Townsville - may be nominated as an international alternate for wide bodied aircraft subject to the following conditions:

– Use of the military apron will be subject to the requirements of the RAAF.

– Taxiway “K” may be used when the military apron is not available.

2.3.4 In the event of emergency/distress, an aircraft engaged in an international flight may land at the following alternate airports:

a. Alice Springs

b. Busselton Margaret River

c. Coffs Harbour

d. Geraldton

e. Kalgoorlie

f. Karratha

g. Launceston

h. Learmonth

i. Melbourne/Avalon

j. Rockhampton

k. Tindal

When safe to do so, the aircraft is then required to proceed directly on to a designated international airport where customs, immigration and biosecurity clearances can be completed. Further information is at *GEN 1.3 Section 6.*

2.4 **International Non-Scheduled Flight Airports**

2.4.1 An "International Non-scheduled Flight Airport" is an airport where approval may be granted, for international non-scheduled flights only, if the prescribed notice is provided in advance. No other international operations are permitted.

Horn Island	Customs, immigration and biosecurity clearances are available if reasonable prior notice is given.

2.5 **External Territory International Airport**

2.5.1 "External Territory International Airport" means an airport of entry and departure for international air traffic located upon an Australian External Territory, where all formalities incident to Immigration, Biosecurity and Territory Customs, and similar procedures are available.

2.5.2 Australian external territory international airports are as follows:

a. Christmas Island

b. Cocos (Keeling) Islands

c. Norfolk Island

2.5.3 At these airports, customs, immigration and biosecurity clearance services are provided to coincide with approved flights only.

GEN 1.3 ENTRY, TRANSIT AND DEPARTURE OF PASSENGERS AND CREW

1. INTRODUCTION

1.1 Travellers and aircrew need to ensure they understand and are fully compliant with all relevant requirements for travel to Australia.

The two stages of examination of passengers:

a. biosecurity processing, and

b. customs and immigration processing.

1.2 Standard procedures have been designed to facilitate the clearance of passengers through the two stages of examination.

2. BIOSECURITY REQUIREMENTS

2.1 General

2.1.1 Australia is free from many diseases, pests and weeds which cause serious damage in other parts of the world. Air crew and passengers are required to comply with legislative requirements under the *Biosecurity Act 2015*.

2.1.2 Detailed information about the biosecurity requirements for overseas aircraft can be found on the Department of Agriculture, Fisheries and Forestry website at: www.agriculture.gov.au/biosecurity/avm/aircraft/

2.2 Disinsection

2.2.1 All aircraft are required to meet Australia's disinsection requirements. The operator of an overseas aircraft will make arrangements for the disinsection of the aircraft in a manner, and within a time, approved by the Director of Human Biosecurity.

2.2.2 The following disinsection options are available to airline operators to meet Australia's cabin and hold disinsection requirements:

Method	Cabin Chemicals	Hold Chemicals
Residual	2% permethrin	2% permethrin
Pre-embarkation	Pre-flight - 2% permethrin	1 shot - 2% permethrin and 2% d-phenothrin
*Pre-departure method	Pre-departure - 2% d-phenothrin or 1R-trans-phenothrin 2%	1 shot - containing d-phenothrin 2% or 1R-trans-phenothrin 2% and permethrin 2%
On arrival	2% d-phenothrin	1 shot - 2% permethrin and 2% d-phenothrin

**The World Health Organization (WHO) no longer recommends the 'pre-flight and top-of-descent' aircraft disinsection method. This method is no longer an approved method for aircraft arrivals into Australia and may be replaced by the 'pre-departure' method or one of the other approved methods listed above. Aircraft operators intending to arrive in Australia should check the Department of Agriculture, Fisheries and Forestry (DAFF) website to confirm the availability of disinsection options, www.agriculture.gov.au/biosecurity/avm/aircraft/disinsection, or contact biosecurity officials at the intended arrival airport.*

2.2.3 The Residual and Pre-embarkation methods can only be used when an Approved Arrangement with the DAFF or a Compliance Agreement with the Ministry for Primary Industries, New Zealand (MPI) has been administered.

2.2.4 The *'Schedule of aircraft disinsection procedures for flights into Australia and New Zealand'* has been prepared in cooperation between the DAFF and the MPI. Further information is available at: www.agriculture.gov.au/biosecurity/avm/aircraft/disinsection

2.2.5 If an aircraft has not completed the required disinsection measures prior to arrival, a pre-arrival report must be given stating that fact - see sections *2.3* and *2.5*.

2.3 Pre-arrival Reporting (PAR) Requirements

2.3.1 Prior to arrival at a first point of entry in Australia, the operator of an international aircraft is required to advise the DAFF - Biosecurity of the following:

a. Details of any person on board the aircraft who has, or had signs or symptoms of a listed human disease during the flight (see *section 2.4*);

b. Details of any person on board the aircraft who died during the flight;

c. If there are animals or plants (or both) in the cabin of the aircraft;

d. If any animal in the cabin of the aircraft died during the flight - that fact; and

e. If the aircraft is an incoming aircraft and the prescribed disinsection measures for the aircraft have not been taken, or will not have been taken, before the aircraft arrives at its first landing place in Australian territory.

Note: The report may be delivered through an airline authorised representative at the arrival airport. However, it remains the liability of the aircraft operator to ensure any such reports are delivered to a biosecurity officer.

2.3.2 The operator, or commander of the aircraft on a non-scheduled flight must also report the following additional information without exception:

a. Information identifying the aircraft;

b. The intended first landing place of the aircraft in Australian territory;

c. The estimated day and time of arrival of the aircraft;

d. The name and contact details of:

(i) the operator of the aircraft; and

(ii) if the operator is not the owner of the aircraft - the owner of the aircraft

e. Details about any animals or plants in the cabin of the aircraft.

Note: The report may be delivered through an airline authorised representative at the arrival airport. However, it remains the liability of the aircraft operator to ensure any such reports are delivered to a biosecurity officer.

2.3.3 The pre-arrival report must be given:

a. at the earlier of:

(i) as close to the top of descent as is operationally practicable before the aircraft is estimated to arrive at its first landing place in Australian territory; and

(ii) 30 minutes before the aircraft is estimated to come to a standstill after arriving at its first landing place in Australian territory; or

b. at the time specified by a biosecurity official.

Note: The commander of an aircraft on a non-scheduled flight may provide the additional information to the DAFF - Biosecurity prior to the departure of the aircraft from the last port before entering Australian territory.

2.3.4 The pre-arrival report must be made to biosecurity officers located at the intended first landing place (or at the department office responsible for biosecurity clearances at the intended first landing place) either orally or in writing (including electronically).

2.3.5 After this information is reported, if the Commander becomes aware the information is incomplete or incorrect they will provide additional information or correct the information as soon as practicable.

2.4 **Ill Passengers or Death On Board**

2.4.1 Any traveller showing signs of a listed Human Disease or a serious illness and needing medical assistance must be reported to a biosecurity officer in a pre-arrival report (see *section 2.3*). Airlines/Commanders are required to report any changes to this status that occur after submitting the pre-arrival report.

2.4.2 The operator of the aircraft is responsible for requesting medical or ambulance services.

2.4.3 Any of the following are considered possible signs or symptoms of a Listed Human Disease (LHD) or reportable illness:

a. fever or suspected fever;

b. jaundice;

c. a new rash;

d. unusual bleeding;

e. a new coughing illness; and

f. any illness that required prompt medical assistance.

2.4.4 If there is any doubt whether an ill traveller needs to be reported, the commander should contact a biosecurity officer. Commanders do not need to report travellers whose illness is a result of:

a. drug or alcohol use;

b. an injury or a pre-existing physical condition; or

c. motion sickness.

d. Commanders are not required to report a traveler who has a pre-arranged medical transfer prior to boarding the flight.

2.4.5 The operator of an aircraft is legally responsible for ensuring the DAFF - Biosecurity is notified. Failure to report ill travellers, as per *para 2.4.3* or death on board an aircraft is an offence under the *Biosecurity Act 2015* and can lead to the application of additional reporting requirements, fines or civil penalties.

2.4.6 Any contaminants on the aircraft from dead or sick people must be cleaned or disinfected in accordance with post-event disinfection procedures for aircraft, as outlined in the World Health Organization 2009, Guide to Hygiene and Sanitation in Aviation, third edition.

2.4.7 For ill passengers that are in transit through Australia the requirements are the same as for those entering Australia. Passengers in transit are not permitted to leave the transit area other than for actual boarding of their outward flight. If the time between arrival and departure allows, and if it is determined to be advisable, the person may either be isolated in a biosecurity facility (i.e. an airport health room) or be allowed restricted access to airport facilities and wait areas as advised by biosecurity officers.

2.5 **Pratique**

2.5.1 Pratique is the granting of permission to disembark and unload baggage and cargo based on the absence of disease in the passengers and crew. Aircraft entering Australia operate under a system of (automatic) positive pratique. Under this system permission to disembark and unload cargo and baggage is automatically granted, unless any of the following applies:

a. the prescribed disinsection measures for the aircraft have not been undertaken;

b. the aircraft has reported an individual as having, or having had, signs or symptoms of a listed human disease, or an individual has died during the flight;

c. a human biosecurity official or a biosecurity official believes an individual on the flight is displaying signs or symptoms of a listed human disease, has been exposed to a listed human disease; or has died during the flight; or

d. a pre-arrival report consistent with *para 2.3* was not provided.

2.5.2 Any aircraft not entering under (automatic) positive pratique, will be met on arrival by a biosecurity officer. All passengers and crew must remain on board until pratique is granted by this officer. When the biosecurity officer is satisfied that there are no further biosecurity issues, the officer will verbally grant pratique and advise that disembarkation and the unloading of baggage and other goods may commence.

2.6 **Biosecurity Waste**

2.6.1 All cabin, galley and hold biosecurity waste on board the aircraft must only be collected, transported, stored and/or treated by either a service provider that has entered into an approved arrangement with the DAFF - Biosecurity or under the supervision of the DAFF - Biosecurity on a fee for service basis.

2.6.2 Biosecurity waste may include refuse and sweepings from areas of the aircraft, any unconsumed and partly consumed foods, any non-washable items, other waste or materials that may have come in contact with biosecurity waste, animal or plant waste or materials used to pack or stabilise imported goods.

2.6.3 Unmanaged waste, non-compliant handling of waste or inadvertent contamination detected on an aircraft may result in action being required to be taken by the aircraft operator. This can result in delays to the servicing and turn-around of the aircraft.

2.6.4 The operator/commander of an aircraft arriving in Australian territory must ensure that waste has been removed from the aircraft before the aircraft is moved further within Australian territory, unless prior approval has been given by the DAFF - Biosecurity.

2.7 **Biosecurity In–flight Announcement**

2.7.1 Prior to arrival in Australia (at top of descent), operator/ commander of all international aircraft must provide, to all travelling passengers and crew, the approved in-flight announcement in audio or video format which outlines Australia's biosecurity requirements. If the audio message cannot be played, commanders must ensure that their crew make a verbal in-flight announcement prior to arrival in Australia.

2.7.2 The delivery of the announcement is a legal requirement under Australian law. The audio or video announcement is available in a number of formats and languages on the DAFF website at: https://www.agriculture.gov.au/biosecurity-trade/aircraft-vessels-military/aircraft/traveller-announcement-resources.
The approved announcement must not be edited.

2.8 **Required Vaccination and Health Documents**

2.8.1 All persons (passengers and crew) arriving in Australia must have the following documents ready for examination by a biosecurity officer and an ABF Officer when requested:

a. An Incoming Passenger Card or crew declaration (as applicable).

b. A valid International Certificate of Vaccination or Prophylaxis against yellow fever (if in the last six days a person has visited a yellow fever declared country for overnight or longer). A valid certificate is consistent with the requirements in Annex 6 of the *International Health Regulations (2005).* A list of yellow fever declared countries can be found in the Australian *Biosecurity (Entry Requirements) Determination 2016*.

Note: If the person does not have a valid certificate, entry into Australia will be permitted after assessment by a biosecurity officer.

2.8.2 Passengers and crew will pass from the aircraft to the ABF Entry Control Point where the vaccination certificates will be assessed.

2.8.3 Australia has no vaccination requirements for departure. However, travellers will have to satisfy the requirements of countries to, or through which, they travel.

2.9 **Biosecurity screening of baggage**

2.9.1 Biosecurity screening of the baggage of passengers and crew, where required, normally occurs at the airport at which passengers and crew disembark the aircraft.

2.9.2 The baggage of passengers and crew disembarking at a particular airport will be unloaded from the aircraft and collected by passengers and crew after they have passed the Entry Control Point.

2.9.3 The baggage of passengers and crew requiring biosecurity screening will be screened by x-ray, detector dog or manual inspection. Screening of baggage by x-ray and/or detector dog may result in subsequent manual inspection.

2.9.4 After any required biosecurity screening of passenger and crew baggage has been completed and after any biosecurity risk goods and/or non-compliance have been addressed, the passengers and crew will remove their baggage from the secondary examination area.

2.10 **Conditionally Non-prohibited Goods**

2.10.1 The following goods must not be imported into Australia unless the relevant import conditions have been met as outlined in the DAFF - Biosecurity Import Conditions Database (BICON), available online https://bicon.agriculture.gov.au/BiconWeb4.0.

a. all animals (including, but not limited to, mammals, birds, reptiles, amphibians and insects) and animal products;

b. cultures of micro–organisms capable of causing human disease and goods of biological origin and other infectious agents;

c. foodstuffs of animal origin, including meat, poultry, sausages, eggs, cheese and milk;

d. plants and plant products (e.g. wooden articles, flowers, seeds, fruit and vegetables);

e. fungi;

f. human remains, fluids and tissues;

g. bioremedial agents and fertilizers.

Note: Any goods brought or imported into Australia not meeting import conditions may be treated, exported or forfeited to the Commonwealth for disposal.

2.10.2 *Biosecurity (First Point of Entry) Determinations* detail the first points of entry into Australia through which aircraft and goods may enter Australia. The determination may designate locations within the first point of entry, called biosecurity entry points, where specific classes of goods must be unloaded.

2.10.3 A list of first points of entry is available on the DAFF - Biosecurity website at: www.agriculture.gov.au/import/before/sending/airports.

3. IMMIGRATION/EMIGRATION REQUIREMENTS

3.1 General

3.1.1 Information in this section is based on the *Migration Act 1958,* the *Migration Regulations 1994* and the *Customs Act 1901*.

3.1.2 Airline operators should ensure that their staff are fully aware of Australia's immigration and customs requirements. The Travel Information Manual (TIM) and the online Timaticweb (www.iata.org/timatic-web) provides a regularly updated, ready–reference for information on Australia's requirements. The master, owner, agent, charterer and operator of a vessel on which a non–citizen is brought into Australia are guilty of an offence against *Section 229(1)* of the *Migration Act 1958* unless the non–citizen when entering Australia:

a. is in possession of evidence of a visa that is in effect and that permits them to travel to Australia;

b. is deemed to be a person having a prescribed status and holds a special purpose visa;

c. is a transit passenger who meets certain criteria;

d. is eligible for a special category visa; or

e. is entering by sea and has a maritime crew visa.

3.1.3 A person who is guilty of an offence against *Section 229(1)* of the *Migration Act* is liable, upon conviction, to a fine not exceeding AUD10,000 for each non-citizen who is brought to Australia. In lieu of prosecution, airlines may pay a penalty of AUD8,250.

3.2 **Advance Passenger Processing and Reporting of Passengers and Crew**

3.2.1 Airlines flying into Australia who provide a 'regular international passenger air service' are required to report all inbound passengers and crew, including all transit passengers, to Home Affairs though the electronic Advance Passenger Processing (APP) system. The information is collected at check–in through the APP system and transmitted to Australia for use by border agencies prior to the arrival of the aircraft. Airlines seeking detailed information concerning the legislative and system requirements can contact Australian Border Force (ABF) at: appwebsite@abf.gov.au or travellerpolicy@abf.gov.au.

3.2.2 Airlines flying into Australia who do not provide a 'regular international passenger air service', as defined under the *Migration Act 1958* are required to report all inbound passengers and crew, including all transit passengers, to the ABF through form 2A and 2B for passengers and form 3 and 3B for crew.

3.3 **Inwards Clearance – Passports or Other Travel Document**

3.3.1 All persons seeking to enter Australia, whether for a visit or temporary or permanent residence, must identify themselves by presenting a valid evidence of identity document such as a valid passport. In the case of non–citizens, they must hold a valid visa in effect or be eligible for a grant of visa at the border. Although, passports are the most common and preferred type of travel document for identification purposes, other forms of identity documents specified below in section 3.3.3 are acceptable travel documents. A valid passport is required for travel to Australia from all overseas destinations.

3.3.2 Some countries still issue family group passports which may cover, for example, a husband and/or wife and children. For entry control purposes, Australia accepts dependents that are endorsed on such passports only when accompanied by the principal holder.

3.3.3 Passports are not required for holders of the following:

a. Certificates of Identity, Documents of Identity or Australian Migration Status ImmiCards issued by Australian authorities.

b. Documents of Identity, issued by a country other than Australia. (Documents must have a photograph of the bearer and re–entry authority to the country of issue).

c. Military identity documents and movement orders issued to members of:

(i) armed forces that have a Status of Forces Agreement with Australia (France, Japan, Papua New Guinea, Republic of the Philippines, Singapore, Turkey, USA, Malaysia and New Zealand)

(ii) Asia–Pacific armed forces (Brunei, Fiji, Malaysia, Thailand and Tonga); or

(iii) Commonwealth forces of Antigua, Bahamas, Barbados, Belize, Canada, Grenada, Jamaica, Mauritius, New Zealand, Papua New Guinea, Saint Lucia, Saint Vincent and the Grenadines, Solomon Islands, St Christopher and Nevis, Tuvalu, and the United Kingdom of Great Britain and Northern Ireland.

(iv) More information is available at: www.abf.gov.au/entering-and-leaving-australia/crossing-the-border/military-personnel

3.3.4 Certificates for air crew members, travelling on duty as operational or positioning crew:

a. operational crew must carry a current identity document issued by the airline by which the crew member is employed (“Airline ID card”) and a valid passport; and

b. positioning crew not listed as crew members must carry a valid passport and a letter from their employer certifying air crew status and setting out the purpose of the persons’ travel to Australia and the arrangements for them to leave Australia.

3.3.5 Documents issued to stateless persons as follows:

a. Certificate of Identity, provided it holds proof to re–enter the country of residence; or

b. Titre de Voyage issued to persons recognised as refugees under the 1951 Convention Relating to the Status of Refugees.

3.3.6 Domestic passengers who are only travelling on domestic sectors of international flights within Australia must carry acceptable forms of photo-identification.

3.4 **Visa Requirements**

3.4.1 With the exception of New Zealand passport holders and prescribed classes of persons under the Special Purpose visa framework - see *para 3.4.12* and *3.4.13*, all non–citizens are required to have a visa for travel to Australia.

3.4.2 Carriers are required to ensure non–citizen passengers travelling to Australia hold, or are eligible to hold, a valid visa. Persons in possession of expired visas must not be carried to Australia. Visas for travel to Australia may be for either single or multiple journeys within the validity of the visa. Visas facilitate travel, but do not guarantee entry. Visitors who fail to satisfy border checks can be refused entry to Australia.

3.4.3 The majority of visas granted will not be evidenced by a stick-in visa label in a passport as visa labels ceased to be issued from 01/09/2015. Existence of a visa should be verified by airlines at check-in via the Advanced Passenger Processing (APP) system.

3.4.4 There are two types of visas: temporary and permanent.

3.4.5 **Electronic Travel Authorities (ETA)** are temporary visas for short term stays for tourism or business visitor activities.

3.4.6 Individuals, travellers, airline representatives or registered travel agents may apply for an ETA, through the ETA APP, but the applicant or traveller must be physically present in order to meet the 'LIVE' image requirement for the ETA APP.

3.4.7 Bearers of the following passports are eligible to apply for an ETA issued by the following countries:

Andorra	Greece	Malta	Sweden
Austria	Hong Kong	Monaco	Switzerland
Belgium	Iceland	Netherlands	Taiwan (excluding official diplomatic passports)
Brunei	Ireland	Norway	United Kingdom (British Citizens and British Nationals Overseas)
Canada	Italy	Portugal	
Denmark	Japan	San Marino	
Finland	Liechtenstein	Singapore	
France	Luxembourg	South Korea	USA
Germany	Malaysia	Spain	Vatican City

3.4.8 Further information regarding these visas and the visa application process can be found at https://immi.homeaffairs.gov.au/visas/getting-a-visa/visa-listing/electronic-travel-authority-601#Overview

3.4.9 For industry users (travel agents and airline representatives), additional guidance is available through the Australian ETA APP Guidance notes (pdf).

3.4.10 The Australian ETA APP can be downloaded from the Apple Store or Google Play at Australian ETA APP.

3.4.11 The industry channel for ETA lodgment is no longer available to airlines to apply for ETA on behalf of passengers.

3.4.12 **Special Purpose Visas (SPV)** are a class of temporary visa taken to have been granted by operation of law to certain persons or classes of persons.

3.4.13 SPV holders are not required to complete an application form provided they belong to any of the following classes of persons:

a. Members of the armed forces of France, Japan, Papua New Guinea, Republic of the Philippines, Singapore, Turkey, USA, Malaysia and New Zealand under the Status of Forces Agreement (SOFA) travelling on duty (movement orders issued from an official source of the relevant country) and holding military identity documents;

b. Members of Asia Pacific armed forces (Brunei, Fiji, Malaysia, Thailand or Tonga) travelling on duty and holding military identity documents and movement orders;

c. Members of the following Commonwealth armed forces travelling on duty (movement orders issued from an official source of the relevant country) and holding military identity documents (Antigua, Bahamas, Barbados, Belize, Canada, Grenada, Jamaica, Mauritius, New Zealand, Papua New Guinea, Saint Lucia, Saint Vincent and the Grenadines, Solomon Islands, St Christopher and Nevis, Tuvalu, and the United Kingdom of Great Britain and Northern Ireland);

d. Members of the civilian component of SOFA provided they hold passports and certificates stating that the person is a member of the civilian component of the armed forces of the relevant country;

e. Dependents of SOFA, (excluding dependents accompanying a Japanese SOFA member) Asia Pacific and Commonwealth armed forces listed above provided they hold a passport and either movement orders or certificates stating they are a spouse or a dependent of a member of the armed forces, or the civilian component of the armed forces of the relevant country, and are accompanying or joining that member;

f. Airline crew members travelling as passengers in the course of employment, who will be departing Australia as crew of an aircraft ("positioning crew"), provided they are in possession of a letter from the employer regarding aircrew status, purpose of travel and arrangements for departure from Australia within five days ("Certificate of Status"). This SPV arrangement is not for air crew seeking to work specifically on domestic sectors or to perform other duties in Australia. An appropriate visa should be applied for in these cases; and

g. Transit passengers (not applicable to stateless persons and refugees) who are direct transit passengers arriving and departing by aircraft are taken to hold a SPV provided they:

   (i) will be continuing their journey to a third country by the same or a connecting aircraft within eight hours of arrival in Australia;

   (ii) do not leave the airport transit lounge except to continue their journey;

   Note: If a person in this class seeks to leave the transit lounge, i.e. seek immigration clearance, the SPV will cease.

   (iii) are in possession of confirmed onward reservations and hold correct documentation for entry to their destination; and

   (iv) be a citizen of the following countries:

Andorra	Greece	Netherlands	Slovakia
Argentina	Hungary	New Zealand	Slovenia
Austria	Iceland	Norway	Solomon Islands
Belgium	Indonesia	Oman	South Korea
Brunei	Ireland	Palau	Spain
Canada	Italy	Papua New Guinea	Sweden
Chile	Japan	Philippines	Switzerland
Croatia	Kiribati	Poland	Thailand
Cyprus	Latvia	Portugal	Tonga
Czech Republic	Liechtenstein	Qatar	Tuvalu
Denmark	Lithuania	Republic of Bulgaria	United Arab Emirates
Estonia	Luxembourg	Republic of South Africa	United Kingdom (including its colonies)
Federated States of Micronesia	Malaysia	Republic of Marshall Islands	United States of America
Fiji	Malta	Romania	Uruguay
Finland	Mexico	Samoa	Vanuatu
France	Monaco	San Marino	Vatican
Germany	Nauru	Singapore	

(v) are a resident of Hong Kong holding Hong Kong Special Administrative Region (HKSAR) passports or British National Overseas (BNO) passports;

(vi) are a resident of Taiwan holding a passport issued by the Authorities of Taiwan (other than passports purported to be official or diplomatic passports);

(vii) are official passport holders from India;

(viii) are diplomatic passport holders, excluding holders of:

– Arab Non-National Passports; and

– diplomatic passports from the following countries:

Afghanistan	Iran	Madagascar	Sierra Leone
Algeria	Iraq	Morocco	Somalia
Angola	Jordan	Pakistan	Sudan
Bahrain	Kuwait	Republic of Yemen	Syria
Comoros	Lebanon	Russian Federation	Tunisia
Democratic People's Republic of Korea	Libya	Saudi Arabia	Zimbabwe
Egypt	Mauritania		

h. are members of the Royal family or a Royal Party;

i. are official guests of the Australian Government and accompanying members of their immediate family.

3.4.14 **Special Category Visa (SCV).** A New Zealand citizen, who holds and produces a valid New Zealand Passport to an officer or authorised system and answers the health and character questions either on the Incoming Passenger Card (IPC) or via the SmartGate, may be eligible to be granted the SCV at the border.

3.4.15 **Merchant Seaman.** The visa regulations for merchant seamen if they arrive in Australia by air are the same as for holders of normal passports. The Australian Maritime Crew Visa (MCV) does not permit travel or entry to Australia by air. Merchant Seamen holding an MCV seeking to fly to Australia must hold a second visa issued by the Australian Government.

3.5 **Returning Non-citizen Permanent Residents of Australia**

3.5.1 Non-citizens who are or were permanent residents of Australia wishing to travel overseas after their travel facility period has expired must hold a permanent visa in effect if they wish to return to Australia as permanent residents. This may include holding:

a. an old style authority to return or Return Endorsement issued between 1976 and 1986;

b. a Resident Return Visa (RRV); or

c. another type of permanent visa in effect that allows travel to Australia.

For more information, see: Overseas travel as a permanent resident at: https://immi.homeaffairs.gov.au/visas/permanent-resident/overseas-travel

3.5.2 Permanent residents who hold an "Authority to Return" or "Return Endorsement" which are in the form of a wet stamp in their passport may not be recorded electronically in Home Affairs Departmental systems. Airlines will need to confirm with the Department of Home Affairs whether holders of these wet stamps are returning to Australia within three years of their most recent departure. These old style wet stamps issued between 1976 and 1986 are also acceptable in expired or cancelled passports or other travel documents provided the holder also has a valid foreign travel document.

3.6 **Incoming Passenger Cards**

3.6.1 Incoming Passenger cards are required to be completed by all passengers except for:

a. airline crew members who are on duty; and

b. direct transit passengers described in *sub-para 3.4.13g*.

3.6.2 Supplies of Incoming Passenger Cards should be maintained on aircraft and issued to passengers in ample time for completion before their arrival for immigration clearance at an airport in Australia. If passengers cannot complete their own cards because of age or physical infirmity, the cards must be completed by the accompanying parent, guardian or attendant.

3.6.3 Incoming Passenger Cards are available in English and a number of foreign languages. All written fields on incoming passenger cards must be completed in the English language.

3.7 **Inward Clearance - Documentation**

3.7.1 Immigration examination of passengers is generally done at the point of final disembarkation in Australia, except when special arrangements to the contrary have been made.

3.7.2 For the purposes of examination, the following documents must be ready for presentation to the ABF Officer:

a. **Aircraft Crew**

(i) For operational flight crew, a valid passport and a certificate of status from their employer in the form of an airline ID card.

(ii) Positioning crew not listed as crew members must carry a valid passport and a letter from their employer certifying air crew status and setting out the purpose of the persons' travel to Australia and the date for them to leave Australia.

(iii) A completed Form B465 Crew Declaration, which can be accessed via: www.abf.gov.au/help-and-support/forms

b. **Passengers**

(i) In the case of Australian citizens, valid Australian passports, or other valid documents of identity having the characteristics of passports, and Incoming Passenger Cards.

(ii) In the case of New Zealand citizens, valid New Zealand passports and Incoming Passenger Cards.

(iii) In the case of all other persons, valid national passports, or other acceptable documents listed in *para 3.4.2*, with visa as required for entry into Australia, and Incoming Passenger Cards.

3.8 Outwards Clearance – Documentation

3.8.1 **APP reporting of passengers and crew.** Airlines flying from Australia who provide a 'regular international passenger air service' are required to report all departing passengers and crew, including all transit passengers, to the Department of Home Affairs through the electronic APP system. The information is collected at check-in through the APP system and transmitted to Australia for use by border agencies prior to the departure of the aircraft. Airlines seeking detailed information concerning the legislative and system requirements can contact ABF at: travellerpolicy@abf.gov.au, or appwebsite@abf.gov.au.

4. CUSTOMS REQUIREMENTS

4.1 Inward Clearance – Documentation

The pilot in command (or authorised agent) of an aircraft landing at a designated international airport which is the first airport of call in Australia shall furnish the following documents to the Department of Home Affairs (see *GEN 1.2 para 1.3*):

a. Impending Arrival Report;

b. Actual Arrival Report;

c. Cargo Report;

d. Crew Declaration;

e. List of Stores and Prohibited Goods.

4.2 Examination of Crew and Passengers – Arrivals

4.2.1 Complete ABF examination of the baggage of passengers and crew members is normally made at the airport where the passengers and crew members finally disembark from the aircraft.

4.2.2 Baggage of all persons (including crew) destined for that airport will be unloaded and brought into the baggage examination area of the terminal building for examination. Crew baggage will be separated from passenger baggage. Cargo will be unloaded for immediate delivery to a depot licensed by the Australian Border Force.

4.2.3 ABF Officers may maintain surveillance over the unloading of all baggage and cargo and ensure that it is taken directly to the baggage examination area and depot respectively. An officer may also check goods owned by, or in possession of, the crew against the List of Stores and Crew Declaration. See *GEN 1.2 para 1.3*.

4.2.4 Passengers and crew, after disembarking, must proceed to the Entry Control Point for completion of Customs, Immigration and Biosecurity requirements.

4.2.5 All persons entering Australia who are in possession of Australian currency, foreign currency, cheques, travellers cheques, money orders or bearer negotiable instruments where the value of all combined is AUD10,000 or more, must complete a Cross-Border Movement-Monetary Instrument reporting form. Reporting forms for this purpose are available from air and sea ports or from the Australian Transaction Reports and Analysis Centre (AUSTRAC). There is no limit to the amount of currency that can be brought into Australia, but failure to declare the currency may result in seizure and prosecution.

4.2.6 After the examination of crew and passenger baggage has been completed and customs duty and tax (if any) paid, the persons concerned will be authorised to remove their baggage from the secondary examination area.

4.3 **Outward Clearance – Documentation**

4.3.1 The pilot in command (or authorised agent) of an aircraft departing from a designated international airport, which is the first airport of departure from Australia, shall furnish the following documents to the ABF Officer (see *GEN 1.2 para 1.3*):

a. Export Permits (when required);

b. Departure Report;

c. Outward Manifest.

4.3.2 When aircraft landing in Australia are in transit, the pilot in command or authorised agent will present documents for inward clearance. The ABF Officer will, where requested, sign and stamp the General Declaration presented on arrival in Australia and return it to the pilot in command. A certificate of clearance will be provided to the pilot upon completion of all reporting formalities and permits the aircraft to depart the airport.

4.4 **Examination of Crew and Passengers – Departures**

4.4.1 All persons leaving Australia who are in possession of Australian currency, foreign currency, cheques, travellers cheques, money orders or bearer negotiable instruments where the value of all combined is AUD10,000 or more, must complete a Cross-Border Movement-Monetary Instrument reporting form. Reporting forms for this purpose are available from air and sea ports or from AUSTRAC. There is no limit to the amount of currency that can be taken out of Australia, but failure to declare the currency may result in seizure and prosecution.

4.4.2 The baggage of outward passengers may be subject to Customs examination.

4.4.3 Passengers departing Australia are required to, subject to exemptions, pay the Passenger Movement Charge whether ticketed or not. See *GEN 4.1*

4.4.4 Information for visitors can be found at: https://www.abf.gov.au/entering-and-leaving-australia/crossing-the-border/passenger-movement/passenger-movement-charge-(pmc)

5. TRANSIT PASSENGERS – CLEARANCE REQUIREMENTS AND PROCEDURES WITHIN AUSTRALIA

5.1 Immigration Requirements

5.1.1 Passengers who are in direct transit on through–flights will not be required to complete Incoming Passenger Cards. This concession applies irrespective of whether the passengers are transiting on the same or different aircraft.

5.1.2 Direct transit passengers will need to hold an appropriate visa unless they meet requirements in *para 3.4.13*, in which case they will be taken to hold a special purpose visa.

5.1.3 Passengers disembarking in Australia (i.e. leaving the transit area) from such a through–flight must present passports/travel documents, visas, Incoming Passenger Cards and airline tickets evidencing confirmed onward booking to a third country.

5.1.4 Where international passengers leave the transit area and transfer to another international flight at the same airport, Incoming Passenger Cards, passport/travel documents and visas (unless within the exempt categories) are required to be presented.

5.1.5 A “through-flight” in this context is as defined in *Chapter 1 - Definitions and Applicability*, of *ICAO Annex 9 (Facilitation)* to the Convention on International Civil Aviation:

“Through-Flight”. A particular operation of aircraft, identified by the operator by the use throughout of the same symbol from point of origin via any intermediate points to point of destination”.

5.1.6 The “through–flight” definition implies a single operator and does not prevent the use of more than one aircraft for a through–flight. The recording of dual flight numbers, when applicable, should satisfy, for local purposes, the requirements of the definition in so far as retention of “the same symbol” from point of origin to point of destination is concerned.

5.2 **Customs and Biosecurity Requirements**

5.2.1 Transit passengers (either proceeding on the same aircraft or another aircraft) are not required to make a customs or biosecurity declaration provided they do not pass through a customs control point.

5.2.2 Personal hand baggage covers only normal personal requisites needed by the passenger for the period of the stopover. Such baggage, however, is liable to inspection.

5.2.3 Passengers having other articles may be required, at the discretion of an ABF or biosecurity officer, to make a customs declaration.

5.3 **Requirements for the airline representative**

5.3.1 **Passenger and Crew Manifests.** The airline representative should hand over the following manifests to an ABF officer:

a. a manifest of travellers and crew in transit who are joining the flight, having departed from other ports in Australia;

b. a manifest of travellers in transit who have arrived on an international flight and are connecting with another international flight;

c. a flight interruption manifest or manifest of travellers/crew who have been offloaded onto alternative international flights rather than not departing. This will assist in reconciling traveller movements post processing.

5.3.2 **Coordinating Traveller/Crew Processing.** The ABF officer and airline representative should coordinate passenger/crew processing by separating the passenger and crew groups of:

a. international traveller/crew transiting Australia;

b. international traveller/crew originating from port of departure;

c. international traveller/crew originating from another port in Australia;

d. domestic travellers;

e. unlawful non-citizens, including removees.

5.3.3 **Cancelled or Aborted Flights Departing Australia.** Following the decision to cancel or abort a flight, an airline representative must advise an ABF officer.

6. LANDINGS AT DESIGNATED ALTERNATE AIRPORTS OR ELSEWHERE THAN AT DESIGNATED INTERNATIONAL AIRPORTS WITHIN AUSTRALIA

6.1 General

6.1.1 Landings elsewhere than at major international, restricted use international, and international non–scheduled flight airports may be divided into two categories:

a. landings at designated alternate airports to international airports; and

b. landings elsewhere than at a designated alternate airport which are made as a result of an emergency.

6.1.2 Under the requirements of the *Customs Act 1901* and the *Migration Act 1958*, an aircraft engaged on an international flight which has landed elsewhere than at a designated international airport is required to proceed directly to a designated international airport where Customs and Immigration clearances can be completed. Biosecurity clearance is normally undertaken at the airport of entry.

6.1.3 Where safe to do so, it is preferable that crew and passengers remain on board the aircraft so they do not have to undergo security clearance when the flight resumes to the original airport of destination. The pilot may disembark to perform the necessary safety inspections. An airline representative may board with the relevant documents.

6.1.4 For charter aircraft and other flights arriving at alternate airports or non–international airports see *para 7.3*.

6.2 Designated Alternate Airports to International Airports

6.2.1 A list of the designated Australian alternate airports (see also *GEN 1.2 para 2.3*).

6.2.2 General procedures effective for when a landing is to be made by an international aircraft at a designated alternate airport are as follows:

a. **Biosecurity.** Biosecurity officers of the DAFF are not stationed permanently at the airport, and the airport is serviced for international arrivals under a request for service arrangement.

b. **Customs and immigration.** ABF Officers of the Department of Home Affairs are not stationed permanently at the airport. The relevant District Office or a Local Area Command in the ABF will be established and be responsible for coordinating the border agency response. It will establish ongoing communication with the airport, airline, biosecurity officers and police if required

6.2.3 The Airline or its agent/representatives may be responsible for covering additional expenses relating to the positioning of resources from another border agency base to an Alternate Airport.

6.2.4 Additional or varied procedures are in place at the alternate airports outlined in the following table:

Airport	Additional or Varied Biosecurity, Customs and Immigration Procedures
Alice Springs	An officer of the Northern Territory police (ABF representative) will exercise surveillance over the aircraft while it is at the airport.
Gold Coast	Biosecurity and ABF officers are stationed at Gold Coast Airport and will be available to grant overnight or full clearances, as required, provided reasonable advance notice of the diversion is given by Airservices Australia.
Kalgoorlie	An officer of the Western Australian police (ABF representative) will have the responsibility to exercise surveillance over the aircraft while it is at the airport.
Karratha and Learmonth	Goods must not be landed at this airport unless an application has been made under *Section 146* of the *Biosecurity Act 2015* and approval granted. An officer of the Western Australian police (ABF representative) will have the responsibility to exercise surveillance over the aircraft while it is at the airport.
Port Hedland	Biosecurity and ABF officers are stationed at Port Hedland but not at the Airport.
Tindal	Goods must not be landed at this airport unless an application has been made under *Section 146* of the *Biosecurity Act 2015* and approval granted. Aircraft other than jet fighter aircraft must not be landed at this airport unless an application has been made under *Section 239* of the *Biosecurity Act 2015* and approval granted. An airline representative/RAAF personnel may board with the relevant documents.
Townsville	Biosecurity and ABF officers are stationed at Townsville but not at the Airport.

6.3 **Landing made other than at a Designated Alternate Airport**

If a landing is made other than at a designated international or international alternate airport, the pilot in command or the next senior crew member available shall report the landing as soon as practicable to the National Operations Management Centre (NOMC) in Airservices Australia. This notification may be made through aeronautical channels (or by other means if this method of communication is not available).

6.3.1 Procedures for such landings are outlined below.

6.3.2 The relevant District Office or a Local Area Command in the ABF will be established and be responsible for coordinating the border agency response. It will establish ongoing communication with the airport, airline, biosecurity officers and police if required.

6.3.3 The pilot in command is responsible for ensuring the following:

a. If pratique has not been granted to the aircraft at any previous landing in Australian territory, contact between passengers, crew and other persons is avoided.

b. Cargo, stores, baggage and mail, if required to be removed from the aircraft for safety reasons, must be deposited in a nearby area and remain there pending completion of the necessary formalities. Mail must be disposed of as is required pursuant to para 7.4.4 of ICAO Annex 9.

c. Any foodstuffs of overseas origin, or any plant material, are not removed from the aircraft except where local food is unobtainable. All food refuse (including peelings, cores, and stones of fruit) must be collected by the aircraft operator and returned to the galley refuse container. The contents of the galley refuse container should not be removed from the aircraft except for hygiene reasons, in which case they must be disposed of, as directed by a biosecurity official.

6.3.4 Notwithstanding the provisions set out above, the pilot in command, while awaiting the instructions of the District Office or Local Area Command, or if unable to get in touch with such authorities, may take such emergency measures as deemed necessary for the health and safety of the passengers and crew. This may include the securing of suitable accommodation and the avoiding or minimising of loss or destruction to the aircraft itself and its load.

7. CHARTER FLIGHTS – GUIDELINES FOR CLEARANCE

7.1 General

7.1.1 The Department of Home Affairs coordinates the activities of Government border agencies involved in passenger processing for charter flights through the National Passenger Processing Committee (NPPC).

Note: Separate application to the Department of Home Affairs may be required to satisfy the requirements of the Aviation Transport Security Act 2004 are shown in GEN 1.2 para 1.10 onwards and application at the address shown in GEN 1.1.

7.2 Arrival/Departure at International Airports

7.2.1 Charter flights will be treated as normal commercial flights when they arrive or depart from designated major international airports (see *GEN 1.2 para 2.1)* and the Gold Coast International Airport.

7.3 Arrival/Departure at Non–International Airports, and Restricted Use International Airports

7.3.1 As there is no full–time border agency presence at non–international airports or at Restricted Use International Airports (other than Gold Coast), aircraft operators are required to obtain prior approval from the NPPC for flights into and out of these airports. Aircraft operators should comply with the following procedures to enable timely consideration by border agencies for approval and clearances:

a. **Submit Application.** The Application process can be found at: www.abf.gov.au/entering-and-leaving-australia/aircraft-requirements
Give at least 10 business day's notification to the following address prior to the arrival of the charter:

The Chairman
National Passenger Processing Committee
Department of Home Affairs
PH: 61 2 6246 1210
Email: nppc@abf.gov.au

The application should include itinerary, aircraft type, and estimated number of passengers/crew.

b. **Provide an Advance Passenger Information.** If the charter operator does not hold an International Air Licence, air operators should deliver, email or fax a listing of passengers and crew to the ABF office at the first port of call at least four working days prior to the arrival of the charter. This listing should include family and given names, date of birth, gender, nationality and passport numbers. If the charter operator holds an International Air License the operator should provide passenger and crew data through the APP system at check-in. For more information, contact Home Affairs at: appwebsite@abf.gov.au. For departures, deliver, email or fax similar Advance Passenger Information data to the ABF office at the last port of call at least one working day prior to the departure of the charter.

c. **Remit Passenger Movement Charge (PMC).** Collect AUD70 PMC from all liable passengers and remit to the Department of Home Affairs in accordance with the conditions contained within the PMC Arrangement. See *GEN 4.1 para 2.*

d. **Pay Other Costs.** Pay any border agency clearance costs which may be applicable and which fall outside those covered by PMC. This will vary from charter to charter and will be dependent on airport location and arrival/departure time, e.g. officer travel, accommodation, overtime and potential airport infrastructure changes which may be necessary to provide a secure processing environment. Separate accounts will be submitted by the Department of Home Affairs after completion of the charter operation.

7.4 **Processing of the Application**

7.4.1 As soon as practicable, but normally within five working days after receipt of the application, the Department of Home Affairs will formally notify the air operator of the NPPC decision and any conditions, as well as an estimation of the border agency costs.

7.4.2 The appropriate regional ABF office will then contact the charter operator to discuss processing arrangements.

8. DESIGNATED INTERNATIONAL AIRPORTS – AUSTRALIAN EXTERNAL TERRITORIES – ENTRY AND DEPARTURE REQUIREMENTS AND PROCEDURES

8.1 Biosecurity Requirements for External Territories

8.1.1 The *Biosecurity Act 2015* extends to the external Territories of Christmas Island, Cocos (Keeling) Islands and Norfolk Island.

8.1.2 All aircraft (including aircraft from Australia) arriving at Christmas Island, Cocos (Keeling) Islands and Norfolk Island are required to meet the first point of entry, disinfection, pre-arrival reporting, mandatory passenger announcement and practical requirements outlined in the *Biosecurity Act 2015* and subordinate legislation, including the *Biosecurity Regulations 2016* and the *Biosecurity (Human Health) Regulation 2016*. See *para 2.1.2* for more information.

8.1.3 Christmas Island, Cocos (Keeling) Islands and Norfolk Island each have their own goods determination which outlines the import conditions for goods that are to be brought or imported into these external Territories, including:

a. Biosecurity (Prohibited and Conditionally Non-prohibited Goods – Norfolk Island) Determination 2016

b. Biosecurity (Conditionally Non-prohibited Goods – Christmas Island) Determination 2016

c. Biosecurity (Conditionally Non-prohibited Goods – Cocos (Keeling) Islands) Determination 2016.

These determinations can be found on the Federal Register of Legislative Instruments website: www.legislation.gov.au

8.2 External Territory International Airports

8.2.1 The following is a list of the designated Australian External Territory International Airports (see also *GEN 1.2):*

a. Christmas Island

b. Cocos (Keeling) Islands

c. Norfolk Island.

Note: Operations by aircraft at the above airports are limited to the pavement strengths shown against these airports in AIP ERSA. Prior application must be made for a pavement concession when this is necessary.

8.2.2 Aircraft operators should use the contact details listed in ERSA FAC to obtain specific information and/or obtain copies of appropriate documents for these airports.

8.3 Territory of Christmas Island

8.3.1 Summary of Documents to be Presented by Pilot or Authorised Agent

a. **On Arrival:**

General Declaration (showing names of crew)	2 copies
Passenger Manifest	2 copies
Cargo Manifest	2 copies
Customs Clearance (from last airport)	2 copies

b. **On Departure:**

General Declaration (showing names of crew)	1 copy
Cargo Manifest	1 copy
Customs Clearance	1 copy

8.3.2 Immigration Requirements – Christmas Island

a. Normal Australian immigration procedures apply when entry is made from outside Australia. All non-citizens must hold visas prior to arrival at Christmas Island when entering from outside Australia.

b. No passports or visas are required when arriving on Christmas Island from the Australian mainland or Tasmania; however, some form of government–issued identification must be produced for clearance through Customs/ Immigration; e.g. Medicare Card or Driver Licence.

8.3.3 Customs Requirements – Christmas Island

a. Inwards. Passengers must complete an Incoming Passenger Card for both Customs and Immigration purposes. The card includes a Customs declaration, which includes a requirement to declare all prohibited imports.

b. **Outwards.** Each passenger must declare all prohibited exports.

8.3.4 Passenger Movement Charge – Christmas Island

a. **Inwards**. Passengers travelling to Christmas Island (from Australia) and departing Christmas Island for another country are required to pay the Passenger Movement Charge whether ticketed or not subject to valid exemptions.

b. **Outwards.** Passengers departing from an Indian Ocean Territory other than mainland Australia are required to pay the Passenger Movement Charge whether ticketed or not subject to valid exemptions.

8.4 Territory of Cocos (Keeling) Islands

8.4.1 Immigration Requirements – Cocos (Keeling) Islands

a. Normal Australian immigration procedures apply when entry is made from outside Australia. All non-citizens must hold visas prior to arrival at Cocos (Keeling) Islands when entering from outside Australia.

b. No passports or visas are required when arriving on Cocos (Keeling) Islands from the Australian mainland or Tasmania; however, some form of identification must be produced for clearance through Customs/Immigration in Perth (e.g. Medicare Card) unless intending to depart Cocos (Keeling) Islands for a foreign country.

c. There are no statutory restrictions on visits to Cocos (Keeling) Islands. It is preferred that accommodation is confirmed prior to departure.

8.4.2 Customs Requirements – Cocos (Keeling) Islands

a. **Inwards.** Passengers must complete an Incoming Passenger Card for both Customs and Immigration purposes. The card includes a Customs declaration, which includes a requirement to declare all prohibited imports.

b. **Outwards.** Each passenger must declare all prohibited exports.

8.4.3 Passenger Movement Charge – Cocos (Keeling) Islands

a. **Inwards.** Passengers travelling to Cocos (Keeling) Islands (from Australia) and who intend to depart from there for a country other than Australia are required to, subject to valid exemptions, pay the Passenger Movement Charge whether ticketed or not.

b. **Outwards.** Passengers departing for other than mainland Australia are required to pay the Passenger Movement Charge whether ticketed or not, subject to exemptions.

8.5 Territory of Norfolk Island

8.5.1 Summary of Documents to be Presented by Pilot or Authorised Agent

a. **On Arrival**

General Declaration (showing names of crew)	2 copies
Passenger Manifest	2 copies
Cargo Manifest	2 copies
Customs Clearance (from last airport)	2 copies

b. **On Departure**

General Declaration (showing names of crew)	1 copy
Cargo Manifest	1 copy
Customs Clearance	1 copy

8.5.2 Immigration Requirements - Norfolk Island

a. Normal Australian immigration procedures apply when entry is made from outside Australia. All non-citizens must hold visas prior to arrival at Norfolk Island when entering from outside Australia.

b. No passports or visas are required when arriving on Norfolk Island from the Australian mainland or Tasmania; however, some form of government issued identification must be produced for clearance through Customs/Immigration, e.g. Medicare Card or Driver Licence.

c. Flights from an overseas location to Norfolk Island via mainland Australia (including Tasmania) must complete immigration clearance when the aircraft arrives in mainland Australia.

d. Airlines flying from an overseas location should not provide boarding passes to a passenger all the way through to Norfolk Island if the flight first arrives in mainland Australia. Passengers should collect bags and check-in at the airport to travel to Norfolk Island.

8.5.3 Customs Requirements - Norfolk Island

a. **Inwards.** Passengers must complete an Incoming Passenger Card for both Customs and Immigration purposes. The card includes a Customs declaration, which includes a requirement to declare all prohibited imports.

b. **Outwards.** Each passenger must declare all prohibited exports.

8.5.4 **Passenger Movement Charge - Norfolk Island**

a. **Inwards.** Passengers travelling to Norfolk Island (from Australia) and departing Norfolk Island for another country are required to pay the Passenger Movement Charge whether ticketed or not, subject to valid exemptions.

b. **Outwards.** Passengers departing from a Pacific Ocean Territory other than mainland Australia are required to pay the Passenger Movement Charge whether ticketed or not subject to valid exemptions.

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GEN 1.4 ENTRY, TRANSIT AND DEPARTURE OF CARGO

1. CUSTOMS REQUIREMENTS CONCERNING CARGO AND OTHER ARTICLES

1.1 Cargo Report

1.1.1 The cargo report is used to report the particulars of all goods (including mail, in-transit and transhipment cargo) that a cargo reporter has arranged to be carried on board regardless of whether the goods will be offloaded in Australia. Private charter flights which are carrying cargo (air freight) are not exempt from these requirements.

1.1.2 The carrier (airline) is required, as the first cargo reporter, to report the full detail of cargo for which they are directly responsible to the Australian Border Force. They are also required to notify the Australian Border Force of any cargo carried on behalf of another cargo reporter and the details of the depot operator who will first receive the cargo after it has been unloaded from the aircraft at a place in Australia. This includes airline mail items (sometimes referred to as ‘on company service’).

1.1.3 The cargo report must be lodged electronically in the ICS at least two hours prior to the estimated time of arrival of the aircraft (as indicated on the Impending Arrival Report). If the charter business is not a regular cargo carrier, and does not have the facilities to make a cargo report using the ICS, they are recommended to engage a Freight Forwarder or a bureau to submit the required electronic reports to the ABF.

1.1.4 When a cargo report is submitted showing the cargo has a discharge port as an Australian port, but the destination port is not an Australian port, the ICS recognises that ultimately the cargo is destined for a place outside Australia. The ICS assigns the cargo report transhipment status and will automatically generate a Transhipment Number. A Transhipment Number is a valid Customs Authority Number for the purposes of export and must be quoted in the outwards manifest.

1.2 **List of Stores and Prohibited Goods (e.g. narcotic drugs, beer, wine, spirits and tobacco products)**

1.2.1 When an aircraft has arrived at an airport in Australia, the operator must report the particulars of the aircraft's stores and of any prohibited goods contained in those stores at the time of arrival to the Australian Border Force via Form B367, Stores and Prohibited Goods Report. See:
www.abf.gov.au/form-listing/forms/b367.pdf

1.2.2 The report of aircraft stores and prohibited goods must be made within three hours of the arrival of the aircraft or before the certificate of clearance is issued, whichever happens first.

1.3 **Import Declarations and Carnets**

1.3.1 Import declarations are used to clear goods with a value exceeding AUD1,000 from ABF control. Import declarations are communicated to the Australian Border Force electronically via the ICS or by lodgement of a completed import declaration form (B650) - www.abf.gov.au/form-listing/forms/b650.pdf, at an ABF counter.

1.3.2 A self-assessed clearance declaration must be made for imported goods arriving by air cargo valued not exceeding AUD1,000. Form B650:
www.abf.gov.au/form-listing/forms/b650.pdf, can also be used to make this declaration.

1.3.3 Temporarily imported goods subject to a carnet may be cleared without an import declaration. Temporarily imported goods must be re-exported from Australia within the time period specified by the ABF.

1.4 **Treatment of Diplomatic and Consular Goods**

1.4.1 All air cargo consignments of diplomatic and consular goods require a cargo report. However, an import declaration is not required to be submitted to the Australian Border Force (for diplomatic/consular/mail/pouch/bags).

1.4.2 Most diplomatic and consular goods are exempt from duty and taxes provided the goods meet all the Australian Border Force and the DAFF - biosecurity legislative requirements including the *Customs (Prohibited Imports) Regulations 1956* and the *Biosecurity Regulations 2016*. Where prohibited goods are to be imported, permission must be obtained prior to the importations of the goods. Further details are available online:
www.abf.gov.au/importing-exporting-and-manufacturing/prohibited-goods.

1.4.3 The B615 form and other reporting requirements must be provided to the Australian Border Force for the release of privileged imports from customs control.

1.5 **Departure Report**

1.5.1 The departure report is a prerequisite that must be satisfied before a Certificate of Clearance (CoC) can be granted by the Australian Border Force.

1.5.2 A departure report is a statement made by the pilot or owner of the aircraft, or an agent, to the Australian Border Force providing information concerning the proposed date and time of departure of the aircraft.

1.5.3 The departure report must be lodged electronically in the ICS.

1.5.4 For airlines that co-load/codeshare, a Departure Report is required for each flight number even though it is the same aircraft departing.

1.6 **Outward Manifest**

1.6.1 Air cargo being exported from Australia must be reported in the form of an outward manifest. The outwards manifest must specify all goods that were loaded on board the aircraft.

1.6.2 The pilot or owner of the aircraft must provide the outward manifest electronically in the ICS, no later than three days after the day of departure of the aircraft.

1.6.3 If a departing aircraft is not carrying any export cargo, a manifest must still be lodged. This manifest states that no cargo was loaded and is called a 'Nil Manifest'.

1.7 **Export Reporting**

1.7.1 The exporter must lodge an export declaration and obtain a “clear” Export Declaration Number before the cargo may be loaded for export. An export declaration can be communicated by document or electronically through the ICS. An Export Declaration Number is required for the following:

a. goods requiring a permit (regardless of value);

b. goods on which a drawback is to be claimed;

c. customable and excisable goods on which duty/excise has not been paid; and

d. goods with a value greater than AUD2,000, except exempt goods.

1.7.2 Goods that are exempt from requiring an Export Declaration include:

a. personal effects;

b. pets;

c. goods with a value of AUD2,000 or less;

d. some goods temporarily imported under section *162A* of the *Customs Act 1901*;

e. Australia Post or diplomatic bags of mail;

f. Australian aircraft's stores;

g. military goods of any value that are the property of Australian Government, for use overseas by Australian Defence Forces;

h. Australian domestic cargo;

i. containers for the international carriage of cargo.

1.7.3 These goods are reported to ABF by citing an exemption code on the outwards manifest.

1.7.4 A Main Manifest Number is supplied by ABF.

1.8 **Certificate of Clearance**

The pilot of an aircraft must not depart from any airport without receiving an aircraft Certificate of Clearance from an ABF Officer.

2. BIOSECURITY REQUIREMENTS

2.1 Australia is free from many diseases, pests and weeds which cause serious damage in other parts of the world. Air crew and passengers are required to comply with legislative requirements under the *Biosecurity Act 2015* to help preserve this.

2.2 Imported air cargo of biosecurity interest (such as fruit and vegetables, live plants, seeds, animal, avian and aquatic species or commodities derived from these products) must be reported via the ABF ICS System.

2.3 The relevant type of import declaration is required to be lodged for each imported consignment of biosecurity interest prior to release of cargo.

2.4 The import conditions for cargo of biosecurity interest are outlined in the DAFF - Biosecurity Import Conditions Database (BICON): https://bicon.agriculture.gov.au/BiconWeb4.0.

2.5 Transport and packaging requirements for live animal, avian and aquatic species are specified in the International Air Transport Association – Live Animal Regulations, available at: www.iata.org/whatwedo/cargo/live-animals/Pages/index.aspx

2.6 Importers should note that all biosecurity inspections, permits and entries carry a fee for service charge. More information is available at: www.agriculture.gov.au/fees

3. AIR CARGO SECURITY REQUIREMENTS

3.1 The *Aviation Transport Security Act 2004* (ATSA) and the *Aviation Transport Security Regulations 2005* (the Regulations) establish a regulatory framework to safeguard against unlawful interference with aviation and maintain and improve aviation security.

3.2 Supply chain security for air cargo is regulated under the ATSA and the Regulations, which require certain air cargo supply chain industry participants to hold and maintain approved security programs. Operating under the Regulated Air Cargo Agent scheme, the Accredited Air Cargo Agent scheme and the Known Consignor scheme, security programs set out the measures and procedures industry participants need to implement to meet their obligations under the ATSA and Regulations.

3.3 All outbound international air cargo is required to be examined at piece-level by a Regulated Air Cargo Agent or originate from a Known Consignor. Piece-level examination means that each individual box, carton or other item in a shipment must be examined at a deconsolidated level. To originate from a Known Consignor means goods must be produced, packaged, stored, controlled, transported and handled in a manner that ensures their integrity and protects them from unlawful interference from their point of origin through to loading onto an aircraft.

3.4 In order for an item of cargo to be loaded onto an international aircraft, that cargo must be issued with a security declaration from a Known Consignor or Regulated Air Cargo Agent.

3.5 Domestic air cargo must be examined at the piece level by a Regulated Air Cargo Agent, or originate from a Known Consignor and have a security declaration if it is loaded onto aircraft departing any designated or tier 1 security controlled airport. This applies only to aircraft with a seating capacity of 40 persons or more and/or a maximum take-off weight of 20,000KG or more. Closed charter flights are excluded from the domestic air cargo examination requirements. Refer to *Regulation 1.03* of the *Regulations* for the definition of 'closed charter operation'.

3.6 Enquiries around regulatory requirements should be directed to the Transport Security Guidance Centre within the Cyber and Infrastructure Security Centre, Department of Home Affairs (see *GEN 1.1*).

3.7 Refer to *GEN 1.6 para 4.*, for information on the current air cargo prohibitions that apply to cargo origination from certain countries.

GEN 1.5 AIRCRAFT INSTRUMENTS, EQUIPMENT AND FLIGHT DOCUMENTS

1. RADIO COMMUNICATIONS SYSTEM REQUIREMENTS

1.1 The underpinning rule for all radio communication equipment requirements is that an aircraft, in controlled or uncontrolled airspace, must have radio communication systems capable of:

a. collectively communicating on all frequencies necessary to meet the reporting, broadcast and listening watch requirements under the civil aviation legislation, from any point on the route of the flight, including in the event of any diversions; and

b. 2-way voice communications; and

c. communicating on the aeronautical emergency frequency 121.5 MHz.

1.2 Specific radio equipment carriage requirements (other rules discuss radio usage) are contained in the following rules:

a. for *Part 103* operations – *section 11.09* of the *Part 103 MOS*.

b. for *Part 121, 133* or *135* operations – *section 11.08* of the respective *CASR Part MOS.*

c. for *Part 131* operations – *sections 26.07* and *26.08* of the *Part 131 MOS*.

d. for all other operations – *sections 26.18* and *26.19* of the *Part 91 MOS*.

1.3 VHF voice frequency allocations are generally based on 50 kHz channel spacing. However, there are also VHF frequency allocations based on 25 kHz spacing – often for Aerodrome Weather Information Service (AWIS) installations, but also for certain Air Traffic Service (ATS) units. Australia has no intention at this time to implement 8.33 kHz spacing.

2. RADIO NAVIGATION SYSTEM REQUIREMENTS

2.1 The navigation equipment requirements for aeroplane VFR flights by night and IFR flights are contained in the following rules:

a. *sections 26.07* and *26.08* of the *Part 91 MOS*; or

b. for *Part 121* or *135* operations – *section 11.09* of the respective *CASR Part MOS*.

2.2 The navigation equipment requirements for rotorcraft VFR flights by night and IFR flights are contained in the following rules:

a. s*ection 26.11* and *26.12* of the *Part 91 MOS*; or

b. for *Part 133* operations – *section 11.09* of the *Part 133 MOS*.

2.3 **Rated Coverages**

The following ranges are quoted for planning purposes. Actual ranges obtained may sometimes be less than these due to facility and site variations (see *ERSA*). The localiser ranges are for those installations that have been nominated for position fixing at ranges beyond 25NM:

a. NDB (published in *ERSA*);

b. VOR and DME

Aircraft Altitude (FT)	**Range (NM)**
Below 5,000	60
5,000 to below 10,000	90
10,000 to below 15,000	120
15,000 to below 20,000	150
20,000 and above	180

c. localiser

Aircraft Altitude (FT)	**Range (NM)**
At 2,000 AGL within $\pm 10^{\circ}$ of course line	25
Below 5,000	30
5,000 and above	50

d. **GBAS course deviation limitation**

For GLS course deviation information see *ERSA Facilities (FAC)* section for each location under the heading RADIO NAVIGATION AND LANDING AIDS.

3. EMERGENCY LOCATOR TRANSMITTER (ELT)

3.1 International flights must be equipped with ELT in accordance with the relevant ICAO Standard and Recommended Practices (SARPS) which are:

(1) for International Commercial Air Transport, *para 6.17 of ICAO Annex 6, Part I, International Commercial Air Transport – Aeroplanes*;

(2) for International General Aviation, *para 2.4.12 of ICAO Annex 6, Part II, International General Aviation – Aeroplanes*; and

(3) for Helicopters, *para 4.7 of ICAO Annex 6, Part III International Operations – Helicopters*.

For these purposes, the whole of Australia is a designated area.

3.2 Certain Australian aircraft are required to be fitted with, or carry an ELT. The rules for these requirements are contained in:

a. for *Part 103* operations – *division 6 of chapter 6* of the *Part 103 MOS*;

b. for *Part 121, 133 or 135* operations – *division 10* of *chapter 11* of the respective *CASR Part MOS*;

c. for other operations (except *Part 131* which has no mandatory requirements) – *division 26.12* of the *Part 91 MOS*.

3.3 It is requested that pilots monitor 121.5MHz as often as possible during flight. Reception of an ELT transmission is requested to be reported to ATS or the RCC as soon as possible.

4. AIRBORNE WEATHER RADAR

4.1 Certain aircraft carrying out Australian air transport operations are required to be fitted with airborne weather radar equipment. The rules for this equipment are contained in:

a. for *Part 121* operations – *sections 11.26* and *11.27* of the *Part 121 MOS*;

b. for *Part 133* operations – *sections 11.20* and *11.21* of the *Part 133 MOS*;

c. for *Part 135* operations – *sections 11.27* and *11.28* of the *Part 135 MOS*.

5. GROUND PROXIMITY WARNING SYSTEM (GPWS)

5.1 Certain aircraft carrying out Australian air transport operations are required to be fitted with GPWS or TAWS. The rules for this equipment are contained in:

a. for *Part 121* operations – *sections 11.24* and *11.25* of the *Part 121 MOS*;

b. for *Part 133* operations – *sections 11.18* and *11.19* of the *Part 133 MOS*;

c. for *Part 135* operations – *sections 11.25* and *11.26* of the *Part 135 MOS*.

6. RADIO ALTIMETERS

6.1 Radio altimeters fitted to aircraft used for the conduct of low-visibility approaches need to comply with relevant FAA airworthiness directives unless an exemption applies. This ensures the radio altimeters are better able to withstand wireless broadband transmissions in nearby frequency bands during a critical phase of flight.

6.2 These requirements apply to both Australian and foreign-registered aicraft when operating within Australian territory.

6.3 A low-visibility approach is an instrument approach procedure with a minima below category I precision approach minima.

6.4 The requirements are specified in the CASR Manual of Standards as follows:

a. Part 91 MOS - Section 26.26A

b. Part 121 MOS - Section 11.20A

c. Part 133 MOS - Section 11.17A

d. Part 135 MOS - Section 11.21A

6.5 A Direction under Part 11 of CASR is in force for foreign air transport Air Operator's Certificate (AOC) holders and New Zealand AOC holders with Australia New Zealand Aviation (ANZA) privileges.

6.6 Temporary interim relief from these requirements may apply at certain specific aerodromes. Further details may be found on the CASA website: *www.casa.gov.au/operations-safety-and-travel/safety-advice/5g-and-aviation-safety*

7. SURVEILLANCE EQUIPMENT

7.1 General requirements and capabilities

7.1.1 The requirements for the carriage and use of surveillance equipment – including transponders and ADS-B transmitting equipment – are detailed in:

a. for *Part 121, 133* or *135* operations – *division 13 of chapter 11* of the respective *CASR Part MOS*;

b. for operations under *Part 103 – division 8 of chapter 6* of the *Part 103 MOS*;

c. for operations under *Part 131 – division 26.10* of the *Part 131 MOS*;

d. for other operations – *division 26.16* of the *Part 91 MOS*.

7.1.2 Further advice, including advice about installing equipment, can be found in *AC 91-23* and on the CASA website.

7.2 Flight without serviceable required surveillance equipment

7.2.1 Where an aircraft's required surveillance equipment (transponder or ADS-B OUT) is unserviceable, the civil aviation legislation requires that a flight may only take place if:

a. the flight begins from an aerodrome at which there is no facility for the surveillance equipment to be repaired or replaced;

b. the flight ends not more than 72 hours after the time the surveillance equipment was found to be inoperative; and

c. before the flight commences, the pilot in command informs ATS about the unserviceability.

7.2.2 Flights of an aircraft without the required surveillance equipment being serviceable or fitted to the aircraft may experience delays or restrictions to air traffic control clearances. Pilots in command should contact ATC for planning advice before the flight commences.

Note: Airservices Australia contact number: 03 9235 7550 (recorded line).

7.2.3 If surveillance equipment becomes unserviceable in flight, aircraft may experience delays or ATC assignment of non-optimal cruising levels or routing.

7.3 **ADS-B equipment in State Aircraft**

7.3.1 State aircraft fitted with non-compliant ADS-B transmitting equipment should set the equipment to output a value of zero for the NUCp, NACp, NIC or SIL. If unable to apply these settings, the ADS-B equipment non-compliance must be detailed in the flight notification. Non-ADS-B equipped state aircraft have equal priority with ADS-B-equipped aircraft.

Note 1: The operation of non-compliant ADS-B equipment can significantly affect and degrade ATS surveillance capability.

Note 2: State aircraft in flight that have not flight planned as above may request operation under the provisions of ENR 1.7 paragraph 3. Cruising Levels - "DUE OPERATIONAL REQUIREMENT".

8. AIRBORNE COLLISION AVOIDANCE SYSTEM (ACAS) / TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)

8.1 **Overview**

8.1.1 Aircraft fitted with a serviceable ACAS/TCAS, and with a crew trained in its use, are permitted to operate that system while in Australian airspace.

8.1.2 Pilots of transponder-equipped aircraft should ensure their transponder is switched to ON/ALT (Mode C) at all times.

Note: TCAS will neither track or display:

a. *non-transponder-equipped aircraft;*

b. *aircraft with an inoperable transponder; or*

c. *aircraft operating a Mode A transponder.*

8.2 **Use of ACAS/TCAS Indicators**

8.2.1 **Traffic Advisory (TA).** In the event of a TA, the pilot should use all available information to prepare for appropriate action if an RA occurs including:

a. attempt to establish visual contact; and

b. change the flight path only if a collision risk is established visually.

Note: RA collision avoidance manoeuvres will not be provided to an aircraft with TA-only mode selected, e.g. during engine failure or operating in known close proximity to other traffic such as approaches to closely spaced parallel runways.

8.2.2 **Resolution Advisory (RA).** In the event of an RA, pilots must:

a. immediately conform to the RA indication, even if this conflicts with an ATC instruction, unless doing so would jeopardise the safety of the aircraft;

b. limit the alterations of the flight path to the minimum extent necessary to comply with the RA; and

c. notify ATC, as soon as permitted by workload, of an RA which requires a deviation from the current ATC instruction or clearance.

8.3 **Responsibility for Separation**

8.3.1 Once an aircraft manoeuvres in response to an RA, ATS is not responsible for providing separation between that aircraft and any other aircraft, airspace, terrain or obstruction.

8.3.2 When the conflict is resolved, pilots must:

a. promptly return to the terms of the latest ATC instruction or clearance and notify ATC of the manoeuvre; or

b. comply with an amended ATC clearance or instruction issued.

8.3.3 ATC responsibility for separation resumes when separation is re-established after:

a. the responding aircraft has returned to its assigned level;

b. the pilot advises ATC that the TCAS manoeuvre is complete; or

c. the responding aircraft has executed an alternate clearance.

8.4 The Australian Transport Safety Bureau (ATSB) requires that all TCAS Resolution Advisories are treated as Routine Reportable Matters.

8.5 The ATSB reporting requirements apply to all TCAS-equipped aircraft operating in Australian-administered airspace. The requirements also apply to Australian registered aircraft operating outside Australian-administered airspace.

8.6 The above information is required for proactive systems analysis in relation to accident prevention.

8.7 **High Vertical Rate (HVR) Encounters**

8.7.1 A TCAS Resolution Advisory (RA) may result from having a high vertical rate when approaching an assigned altitude or flight level when another aircraft is maintaining, or approaching, an adjacent altitude or flight level.

To avoid RAs in these circumstances, the pilot of the climbing or descending aircraft should, where practicable, reduce the vertical rate to less than 1,500FPM when within the last 1,000FT of the assigned altitude or flight level, unless otherwise directed by ATC.

Note 1: Pilots are not required to modify vertical speed for every level-off. This is not necessary and would introduce a significant increase in pilot workload.

Note 2: Pilots may become aware of the presence of an adjacent aircraft by several means, including:

a. visual acquisition;

b. information provided by ATC; or

c. TCAS Traffic Advisory (TA).

9. AREA NAVIGATION SYSTEMS APPROVAL AND OPERATIONS

9.1 Requirements, advice and information on area navigation, PBN and GNSS aspects can also be found on the CASA website at: www.casa.gov.au/airspace/standard-page/cns-atm-navigation.

9.1.1 Area navigation systems rely on data in the navigation database to be current, valid and correct for the navigation system to provide the correct information. Navigation databases should be regularly checked for integrity and if any discrepancy in the data is discovered, report the discrepancy as soon as practicable to the database provider.

Note: Errors may occur in navigation databases from time to time. Pilots should subscribe to any alerting services offered by the navigation database provider to receive notifications of any navigation database errors.

9.2 The pilot in command or operator of an aircraft must hold an approval to conduct RNP-AR procedures in Australia (see *CASR 91.660*).

9.3 **Notification of failure or operations outside of tolerance**

9.3.1 Pilots using area navigation systems for navigation must notify ATC

a. about navigation equipment failure; or

b. of operations of the equipment outside the approved tolerances; or

c. for inertial systems, if the times between updates, or from departure, exceeding three (3) hours for single units or five (5) hours for multiple units for flights in controlled airspace other than OCA, and five (5) hours for a single unit or 12 hours for multiple units for flights in OCA.

Note: ATC may discontinue applying certain area navigation standards after receipt of the advice.

10. RVSM APPROVAL AND OPERATIONS

10.1 RVSM is an ICAO standard, which allows the use of 1,000FT separation between RVSM-approved aircraft operating from FL290 to FL410 inclusive. In Australia, RVSM is applied in accordance with the ICAO standard. Operators and aircraft must be approved by the NAA of the State of registry or the State of the operator (see *CASR 91.655*).

11. AOC TO BE CARRIED ON BOARD

11.1 In accordance with *ICAO Annex 6 Parts I*, an aeroplane conducting an international commercial air transport operation shall carry a certified true copy of the AOC and a copy of the associated Operations Specifications relevant to the aeroplane type, issued in conjunction with the Certificate.

11.2 This provision for aeroplanes is outlined in *Annex 6, Part I, Chapter 6, para 6.1.2.*

11.3 The provision for helicopters is outlined in *Annex 6 Part III, Section II, Chapter 4, para 4.1.2*.

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GEN 1.6 SUMMARY OF NATIONAL REGULATIONS AND INTERNATIONAL AGREEMENTS/CONVENTIONS

1. KEY CIVIL AVIATION LEGISLATION AND AIR NAVIGATION REGULATIONS IN FORCE IN AUSTRALIA AND ITS TERRITORIES

1.1 Legislation by the Parliament of the Commonwealth of Australia

1.1.1 ***Air Navigation Act 1920*** **and** ***Air Navigation Regulations 2026.*** Ratification of Chicago Convention and Protocols amending Articles 45, 48, 49 and 61 of Chicago Convention and Air Transit Agreement:

a. Control of Foreign Aircraft on Flights in Australia;

b. International Airline Licences;

c. Carriage of Munitions in Aircraft.

1.1.2 ***Air Services Act 1995*** **and** ***Air Services Regulations 2019.*** Legislation establishing Airservices Australia.

1.1.3 ***Aviation Transport Security Act 2004*** **and** ***Aviation Transport Security Regulations 2005.*** Establishes a regulatory framework to safeguard against unlawful interference with aviation.

1.1.4 ***Biosecurity Act 2015***. Explains how biosecurity threats to plant, animal and human heath in Australia and its external territories are managed.

1.1.5 ***Civil Aviation Act 1988.*** An Act establishing the Civil Aviation Safety Authority with regulatory functions relating to the safety of civil aviation and related purposes, including:

a. Air Operators Certificates;

b. Use of Defence Aerodromes;

c. Interference with Navigation Aids.

1.1.6 ***Civil Aviation Regulations.*** Australia's aviation regulations are contained in two instruments: the *Civil Aviation Regulations 1988 (CAR 1988)* and the *Civil Aviation Safety Regulations 1998 (CASR 1998)*. In due course, the regulations contained in *CAR 1988* will be replaced by *CASR 1998*. In the interim, both *CAR 1988* and *CASR 1998* apply.

CAR 1988

Part 1	Preliminary
Part 2	Administration and Organisation
Part 4	Airworthiness Requirements
Part 4A	Maintenance
Part 4B	Defect Reporting
Part 4C	Flight Manuals
Part 4D	Removal of Data Plates and Aircraft Registration Identification Plates
Part 5	Balloon Flight Crew Licencing
Part 9	Aerodromes
Part 10	Air Traffic Services and Other Services
Part 14	Air Service Operations
Part 16	Refusal to Grant, and Suspension and Cancellation of, Licences, Certificates and Authorities
Part 17	Penal Provisions and Prosecutions
Part 18	Evidence
Part 19	Miscellaneous
Part 20	Transitional

CASR 1998

Part 1	Preliminary
Part 11	Regulatory Administrative Procedures
Part 13	Enforcement Procedures
Part 21	Certification and Airworthiness Requirements for Aircraft and Parts
Part 22	Airworthiness Standards for Sailplanes and Powered Sailplanes
Part 23	Airworthiness Standards for Aeroplanes in the Normal, Utility, Acrobatic or Commuter Category
Part 25	Airworthiness Standards for Aeroplanes in the Transport Category
Part 26	Airworthiness Standards for Aircraft in the Primary Category or Intermediate Category
Part 27	Airworthiness Standards for Rotorcraft in the Normal Category

Part 29	Airworthiness Standards for Rotorcraft in the Transport Category
Part 31	Airworthiness Standards for Manned Free Balloons
Part 32	Airworthiness Standards for Engines for Very Light Aeroplanes
Part 33	Airworthiness Standards for Aircraft Engines
Part 35	Airworthiness Standards for Aircraft Propellers
Part 39	Airworthiness Directives
Part 42	Continuing Airworthiness Requirement for Aircraft and Aeronautical Products
Part 45	Display of nationality marks, registration marks and aircraft registration identification plates
Part 47	Registration of Aircraft and related matters
Part 60	Synthetic Training Devices
Part 61	Flight Crew Licensing
Part 64	Authorisations for Non-Licensed Personnel
Part 65	Air Traffic Services Licensing
Part 66	Continuing airworthiness - aircraft engineer licences and ratings
Part 67	Medical
Part 90	Additional Airworthiness Requirements
Part 91	General Operating and Flight Rules
Part 92	Consignment and carriage of dangerous goods by air
Part 99	Drug and Alcohol Management Plans and Testing
Part 101	Unmanned Aircraft and Rockets
Part 103	Sport and Recreation Aircraft
Part 105	Parachuting from aircraft
Part 117	Representations and Surveys
Part 119	Australian air transport operations – certification and management
Part 121	Australian air transport operations – larger aeroplanes
Part 129	Foreign Air Transport Operators - certification and operating requirements
Part 131	Balloons and hot air airships
Part 132	Limited category aircraft

Part 133	Australian air transport operations – rotorcraft
Part 135	Australian air transport operations – smaller aeroplanes
Part 137	Aerial application operations - other than rotorcraft
Part 138	Aerial Work Operations
Part 139	Aerodromes
Part 141	Recreational, Private and Commercial Pilot Flight Training, other than Certain Integrated Training Courses
Part 142	Integrated and multi-crew pilot flight training, contracted training and contracted checking
Part 143	Air Traffic Services Training Providers
Part 145	Continuing Airworthiness - Part 145 Approved Maintenance Organisations
Part 147	Continuing airworthiness - maintenance training organisations
Part 149	Approved self-administrating aviation organisations
Part 171	Aeronautical Telecommunication Service and Radio Navigation Service Providers
Part 172	Air Traffic Service Providers
Part 173	Instrument Flight Procedure Design
Part 175	Aeronautical Information Management
Part 200	Modified application of Regulations to certain aircraft
Part 201	Miscellaneous
Part 202	Transitional

1.1.7 ***Damage by Aircraft Act 1999.*** Facilitates the recovery of damages for certain injury, loss, damage or destruction caused by aircraft, or by people, animals or things that are dropped, or that fall, from aircraft that are in flight.

1.1.8 ***Civil Aviation (Carriers' Liability) Act 1959.*** Establishes carriers' liability arrangements as they apply to the carriage of passengers, baggage and cargo. This includes giving effect to the provisions of the 1999 Montreal Convention and the Warsaw Convention as amended by relevant instruments. This Act also establishes carrier's liability for domestic carriage by air.

1.1.9 ***Transport Safety Investigation Act 2003 (TSI Act).*** Provides guidance for the investigation of transport accidents and other matters affecting transport safety in aviation, marine and rail modes of transport.

1.2 **Other Relevant Legislation**

Airspace Act 2007

Australian Maritime Safety Authority Act 1990 Section 6

Customs Act 1901

Migration Act 1958

Customs Regulation 2015

Transport Safety Investigation Regulations 2003 (TSI Regulations) Parts 1, 2 and 5

Convention on International Civil Aviation (Chicago Convention)

International Convention for the Safety of Life at Sea (SOLAS)

International Convention on Maritime Search and Rescue (SAR)

2. SECURITY OF GENERAL AVIATION OPERATIONS

2.1 **Introduction**

2.1.1 Since late 2001, the Australian Government has circulated warnings concerning the possibility of terrorist attacks against Australian civil aviation interests, including general aviation.

2.1.2 The general security environment in relation to general aviation has not changed in recent years. Nevertheless, general aviation industry personnel are urged to be particularly vigilant and alert to any activities in the industry which may arouse suspicions in regard to possible terrorist actions.

2.2 **Activities Which May Arouse Suspicion**

2.2.1 General aviation operators should be alert to the importance of reporting and/ or resolving any event or activity occurring in their operating environment that appears to be unusual or suspicious.

2.2.2 Examples of suspicious behaviour can include:

a. unusual enquiries regarding flight training (including the use of, or training in, crop dusters or helicopters);

b. enquiries concerning aircraft configurations and capabilities;

c. loading and unloading of unusual or unauthorised cargo;

d. unusual activity related to the use or acquisition of dangerous chemicals;

e. watching, observing, photographing, sketching, measuring and note taking;
f. examining or enquiring about security systems and guarding;
g. visiting airports and not conducting 'normal' airport business;
h. avoiding notice around critical airport infrastructure or assets;
i. being airside (or in other non public areas) without a verifiable excuse [be aware of seemingly convincing cover stories];
j. enquiring about airport associated work that does not exist;
k. entering or leaving the airport precinct in an unusual manner;
l. creating distractions at times critical to aircraft operations;
m. insisting on urgency in apparent benign conditions;
n. attempting to circumvent security measures or procedures;
o. attempting to conceal baggage or avoiding scrutiny of it;
p. using or carrying innocent items that could mask more sinister items; and
q. presenting suspect ID (it might be worn or photocopied/ photographed).

2.3 **Actions and Contacts**

2.3.1 All members of the aviation community are reminded to continue to maintain and, if necessary, enhance their vigilance and security arrangements.

2.3.2 Any unusual behaviour that cannot be satisfactorily explained should be reported to:

a. the National Security Hotline by phone on 1800 123 400;
b. the local police;
c. the Department of Home Affairs, Critical Infrastructure Risk and Regulation Division by phone on 1300 791 581; or
d. company security officers.

2.3.3 Particular attention should be paid to details such as names and descriptions of suspicious persons, and vessel/vehicle identification markers.

3. AUSTRALIAN SANCTIONS

3.1 Introduction

3.1.1 Australian sanction laws implement United Nations Security Council sanctions regimes and Australian autonomous sanctions regimes.

3.1.2 Contravening an Australian sanction law is a serious criminal offence. Penalties for sanctions offences include up to 10 years imprisonment and substantial fines.

3.1.3 All operators should check the sanctions measures which apply, including before bringing goods into or out of Australia or dealing with persons who are subject to targeted financial sanctions or whose entry into or transit through Australia is prohibited.

3.2 Sanctions Regimes under Australian sanction laws

3.2.1 The sanctions regimes currently implemented under Australian sanction laws are:

Central African Republic	Guinea-Bissau	Mali	Syria
Counter-terrorism	Iran	Myanmar	The Taliban
Crimea and Sevastopol	Iraq	Russia	Ukraine
Democratic People's Republic of Korea (North Korea)	ISIL (Da'esh) and Al-Qaida	Somalia	Yemen
Democratic Republic of the Congo	Lebanon	South Sudan	Zimbabwe
Former Federal Republic of Yugoslavia	Libya	Sudan	

3.2.2 For an updated list of individuals and entities that may be subject to targeted financial sanctions and travel bans, which might include nationals and entities from countries other than the ones listed above, please consult the Consolidated List available at: www.dfat.gov.au/international-relations/security/sanctions/Pages/consolidated-list

3.3 **Further information**

3.3.1 The Department of Foreign Affairs and Trade website provides detailed information on the sanctions regimes and the specific restrictions applicable under each regime: www.dfat.gov.au/international-relations/security/sanctions/Pages/sanctions-regimes

3.3.2 The Minster for Foreign Affairs or their delegate may be able to grant a permit authorising an activity that would otherwise contravene an Australian sanction law. The Minister for Home Affairs may be able to grant a visa authorising travel that would otherwise contravene a travel ban. For further information see the website listed above.

4. AUSTRALIAN AIR CARGO PROHIBITIONS

4.1 **Introduction**

4.1.1 Air cargo prohibitions are a preventive security measure, based on the Government's assessment of the threat and risk environment in certain countries. There is no information to suggest that there is any specific threat for flights to or from Australia.

4.1.2 The prohibitions apply equally to air cargo carried on passenger and freighter aircraft.

4.2 **Air Cargo Prohibitions under Australian Law**

4.2.1 Australia has in place air cargo prohibitions from identified high risk countries – Afghanistan, Bangladesh, Egypt, Somalia, Syria, Turkey and Yemen.

4.2.2 Airlines are prevented from carrying air cargo that has originated from, or transited through Somalia, Syria or Yemen, except for items that are currently exempt from screening under Australian regulations such as diplomatic bags and smaller items of international mail.

4.2.3 Air cargo that has originated from, or transited through, Afghanistan or Egypt is permitted but only if the cargo has been examined at a specified last port of call prior to arrival in Australia.

4.2.4 Air cargo that has originated from, or transited through, Bangladesh is prohibited, unless it has undergone security examination at a European Union approved air cargo examination facility (RA3) or examined at an approved last port of call before travelling to Australia or is otherwise exempt from examination under Australian regulations.

4.2.5 Air cargo that has originated from, or transited through, Turkey is prohibited, but only if it contains an electromechanical device that weighs over 1KG.

4.3 **Further Information**

4.3.1 Updates and further information is available at: https://www.cisc.gov.au/information-for-your-industry/transport/air-cargo-security/prohibitions

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GEN 1.7 DIFFERENCES FROM ICAO STANDARDS, RECOMMENDED PRACTICES AND PROCEDURES

1. The differences that exist between Australian national aviation regulations and those specified by ICAO as Standards and Recommended Practices (SARPS) are identified on the Airservices Australia website at:

www.airservicesaustralia.com/aip/aip.asp

Read and agree to the Copyright Notice, then click on the current “”Differences from ICAO Standards. Recommended Practices and Procedures” link.

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GEN 2. TABLES AND CODES

GEN 2.1 MEASURING SYSTEM, AIRCRAFT MARKINGS, HOLIDAYS

1. UNITS OF MEASUREMENT

1.1 Units of measurement to be used in airways operations and air-ground communications are as follows:

Measurement	Units
Distances used in navigation (generally in excess of 2NM*)	nautical miles and tenths*
Short distances	metres
Altitudes, elevations and heights	feet
Horizontal speed, including wind speed	knots
Vertical speed	feet per minute
Wind direction for runway operations	degrees magnetic
Wind direction except for runway operations	degrees true
Visibility, including runway visual range	kilometres or metres
Altimeter setting	hectopascals
Temperature	degrees celsius
Weight (Mass) Metric	tonnes or kilograms
Time	hours and minutes
**Miles must be read as meaning nautical miles unless otherwise stated. The word "nautical" may be omitted from air-ground communications.*	

1.1.1 An aircraft which is temporarily unable to use these units must so advise and request the ground station to transmit in units useable by the aircraft.

2. TIME SYSTEM

2.1 **Coordinated Universal Time (UTC)**

Is used for civil aviation.

2.2 **Date and time**

Is indicated in a combination of the date and time in a single six figure group. However, a 10 figure group comprising the year, month, date, hours and minutes is used for NOTAM and SUP. This is reduced to an eight figure group (nil year) for SPFIB.

Note: When the days of the week are referenced in a NOTAM, the day refers to the local day with timings in UTC i.e. SUN 2300-0300 refers to SAT 2300 UTC - SUN 0300 UTC.

3. GEODETIC REFERENCE DATUM

3.1 All published geographical coordinates are expressed in term of the World Geodetic System – 1984 (WGS-84). Most geographical coordinates have been surveyed; however, those geographical coordinates that have been mathematically derived are indicated by an asterisk.

3.2 Geographical coordinates published in AIP documents/charts and NOTAM are expressed as degrees, minutes, seconds (if required), and if more precision is required, tenths/hundredths of a second with the cardinal point last; e.g. 3635S 14626E or 050721.2S 0652522.6E.

4. PUBLIC HOLIDAYS

4.1 **National**

New Year's Day	1 January
Australia Day	26 January
Good Friday	Friday before Easter
Easter Monday	Monday after Easter Sunday
Anzac Day	25 April
King's Birthday	According to the published date for the relevant State/Territory
Christmas Day	25 December
Boxing Day	26 December

Note: When New Year's Day, Australia Day, Christmas Day and Boxing Day falls on a Saturday or Sunday, the next following working day is declared the public holiday. In this case, both the actual day and the following declared public holiday are considered to be public holidays.

4.2 Some services may be affected on public holidays. Operators should check NOTAM and/or contact the relevant aerodrome owner/operator.

4.2.1 Airspace specified as active or not active on public holidays refers only to National holidays as listed in *para 4.1*. Any other holidays affecting activation will be specified by NOTAM.

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GEN 2.2 DEFINITIONS AND ABBREVIATIONS

1. DEFINITIONS

Active LAHSO Runway: The runway used during LAHSO for arriving aircraft issued with a hold short instruction.

Administering Authority: With respect to a Danger Area or Military Operating Area this is the agency nominated to exercise the conditions specified for the area and, where authorised, the Air Traffic Service provider for that area. See also Controlling Authority.

ADS-C Agreement: A reporting plan which establishes the conditions of ADS-C data reporting (i.e. data required by the air traffic services unit and frequency of ADS-C reports which have to be agreed to prior to the provision of air traffic services).

Aerodrome: An area of land or water (including any buildings, installations and equipment), the use of which as an aerodrome is authorised under the regulations, being such an area intended for use wholly or partly for the arrival, departure and movement of aircraft. *Civil Aviation Act 1988 (Cth).*

Aerodrome Beacon: An aeronautical beacon, used to indicate the location of an aerodrome from the air.

Aerodrome Control Service: ATC service for aerodrome traffic.

Aerodrome Control Tower: A unit established to provide ATC service to aerodrome traffic.

Aerodrome Elevation: The elevation of the highest point of the landing area.

Aerodrome Meteorological Minima (Ceiling and Visibility Minima): The minimum heights of cloud base (ceiling) and minimum values of visibility which are prescribed in pursuance of *CASR Part 91* for the purpose of determining the useability of an aerodrome either for take-off or landing.

Aerodrome Meteorological Office: An office designated to provide meteorological service for aerodromes serving international air navigation.

Aerodrome Proprietor: Any Owner, Licensee, Authority, Corporation, or any other body which has a legal responsibility for a particular aerodrome.

Aerodrome Reference Point (ARP): The designated geographical location of an aerodrome.

Aerodrome Traffic: All traffic on the manoeuvring area of an aerodrome and all aircraft flying in, entering, or leaving the traffic circuit.

Aerodrome Traffic Circuit: The specified path to be flown by aircraft flying in, entering, or leaving the traffic circuit.

Note: At a controlled aerodrome, an aircraft is in the traffic circuit when it is within the CTR and established on a leg of the circuit.

Aeronautical Beacon: An aeronautical ground light visible at all azimuths, either continuously or intermittently, to designate a particular point on the surface of the earth.

Aeronautical Information Circular (AIC): A notice containing information that does not qualify for the origination of a NOTAM, or for inclusion in the AIP, but which relates to flight safety, air navigation, technical, administrative or legislative matters.

Aeronautical Information Publication (AIP): A publication issued by or with the authority of a State and containing aeronautical information of a lasting character essential to air navigation.

AIP Supplement (SUP): Temporary changes to the information contained in the AIP which are published by means of special pages.

Airborne Collision Avoidance System (ACAS): An aircraft system based on secondary surveillance radar (SSR) transponder signals which operates independently of ground-based equipment to provide advice to the pilot on potential conflicting aircraft that are equipped with SSR transponders.

Aircraft Address: A unique combination of 24 bits available for assignment to an aircraft for the purpose of air-ground communications, navigation and surveillance. Expressed as a six character hexadecimal code.

Aircraft Classification Number (ACN): A number expressing the relative effect of an aircraft on a pavement for a specific standard sub-grade category.

Aircraft Identification: An identification of up to seven (7) alpha-numeric characters used to identify the aircraft in flight notifications and in Mode S transponders/ADS-B transmitters.

Note: The Aircraft Identification entered into the Mode S Transponder, or ADS-B Transmitter, must match the Aircraft Identification entered into Item 7 of the Flight Notification or, when no flight notification has been filed, the aircraft registration. Hyphens or symbols may not be used within the identification.

Aircraft Parking Position Taxilane: A portion of an apron designated as a taxiway and intended to provide access to aircraft parking positions only.

Air-Ground Communications: Two-way communications between aircraft and stations on the surface of the earth.

AIRMET Information: Information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en route weather phenomena which may affect the safety of low-level aircraft operations and which was not already included in the forecast issued for low-level flights in the flight information region concerned or sub-area thereof.

Airport Collaborative Decision Making (A-CDM): A set of processes which allows aerodromes, aircraft operators, air traffic controllers, ground handling agents, pilots and air traffic flow managers to exchange operational information and work together to efficiently manage operations at aerodromes.

Air-Report (AIREP): A report from an aircraft in flight prepared in conformity with requirements for position and operational and/or meteorological reporting.

Airspace Release: A defined volume of airspace normally under the jurisdiction of one controlling or administering authority that is temporarily released, by common agreement, for exclusive use of another.

Airspace Speed Limitation: A speed limit specified for a particular class of airspace.

Air Taxiing: Movement of a helicopter/VTOL above the surface of an aerodrome, normally in ground effect and at a speed normally less than 20KT.

Air Traffic Control Clearance: Authorisation for aircraft to proceed under conditions specified by an ATC unit.
Note: For convenience, the term "Air Traffic Control Clearance" is normally abbreviated to "Clearance" when used in appropriate context.

Air Traffic Control Instructions: Directives issued by ATC for the purpose of requiring a pilot to take a specific action.

Air Traffic Control Service: A service provided for the purpose of:

a. preventing collisions:
   (1) between aircraft; and
   (2) on the manoeuvring area between aircraft and obstructions; and
b. expediting and maintaining an orderly flow of air traffic.

Air Traffic Control Speed Restriction: An ATC traffic management speed or an ATC-issued speed control instruction.

Air Traffic Service (ATS): A generic term meaning variously, flight information service, alerting service, air traffic advisory service, ATC service (area control service, approach control service, or aerodrome control service).

Air Transit: The airborne movement of a helicopter that is:

a. for the expeditious transit from one place within an aerodrome to another place within the aerodrome;

b. at or below 100FT above the surface; and

c. at speeds greater than those used in air taxiing.

Airways Clearance: A clearance, issued by ATC, to operate in controlled airspace along a designated track or route at a specified level to a specified point or flight planned destination.

Alerted See-and-Avoid: A procedure where flight crew, having been alerted to the existence and approximate location of other traffic in their immediate vicinity, seek to sight and avoid colliding with those known aircraft.

Alerting Post: An agency designated to serve as an intermediary between a person reporting an aircraft in distress and a rescue coordination centre.

Alerting Service: A service provided to notify appropriate organisations regarding aircraft in need of search and rescue aid, and to assist such organisations as required.

Alternate Aerodrome: An aerodrome to which an aircraft may proceed when it becomes either impossible or inadvisable to proceed to or to land at the aerodrome of intended landing.

Altimeter Setting: A pressure datum which when set on the sub-scale of a sensitive altimeter causes the altimeter to indicate vertical displacement from that datum. A pressure-type altimeter calibrated in accordance with Standard Atmosphere may be used to indicate altitude, height or flight levels, as follows:

a. when set to **QNH** or **Area QNH** it will indicate **altitude;**

b. when set to **Standard Pressure** (1013.2 HPA) it may be used to indicate **flight levels.**

Altitude: The vertical distance of a level, a point or an object, considered as a point, measured from mean sea level.

Approach Control Service: ATC service for arriving or departing flights.

Approach Sequence: The order in which two or more aircraft are cleared to approach to land at the aerodrome.

Apron: A defined area on a land aerodrome, intended to accommodate aircraft for purposes of loading or unloading passengers, mail, cargo, fuelling, parking or maintenance.

Apron Service: A traffic regulatory and information service provided to aircraft using the apron area of an aerodrome.

Apron Taxiway: A portion of a taxiway system located on an apron and intended to provide a through taxi route across the apron.

Area Control Service: ATC service for controlled flights in control areas.

Area Navigation: A method of navigation which permits aircraft operation on any desired flight path within the coverage of ground or space-based navigation aids, or within the limits of the capability of self-contained aids, or a combination of these.

Area Navigation Route: An ATS route established for the use of aircraft capable of employing area navigation.

Area Navigation Systems: Navigation systems supporting area navigation.

Area QNH: A forecast altimeter setting which is representative of the QNH of any location within a particular area.

Area VHF: The appropriate FIA VHF channel for a location.

ATS Route: A specified route designed for channelling the flow of traffic as necessary for the provision of air traffic services.

ATS Surveillance Service: Term used to indicate an air traffic service provided directly by means of an ATS surveillance system.

ATS Surveillance System: A generic term meaning variously, ADS-B, PSR, SSR or any comparable ground-based system that enables the identification of aircraft.

Note: A comparable ground-based system is one that has been demonstrated, by comparative assessment or other methodology, to have a level of safety and performance equal to, or better than, monopulse SSR.

Automatic Dependent Surveillance Broadcast (ADS-B): A means by which aircraft, aerodrome vehicles and other objects can automatically transmit or receive data such as identification, position and additional data, as appropriate, in a broadcast mode via a data link.

Automatic Dependent Surveillance - Contract (ADS-C): A means by which the terms of an ADS-C agreement will be exchanged between the ground system and the aircraft, via a data link, specifying under what conditions ADS-C reports would be initiated, and what data would be contained in the reports.

Note: The abbreviated term "ADS contract" is commonly used to refer to ADS event contract, ADS demand contract, ADS periodic contract or an emergency mode.

Automatic Terminal Information Service (ATIS): The provision of current, routine information to arriving and departing aircraft by means of continuous and repetitive broadcasts during the hours when the unit responsible for the service is in operation.

Aviation Reference Number (ARN): A unique identifier assigned to a person by CASA for the purposes of CASA records.

Base Turn (Instrument Approach): A turn executed by the aircraft during the initial approach between the end of the outbound track and the beginning of the intermediate or final approach track. The tracks are not reciprocal.

Note: Base turns may be designated as being made either in level flight or while descending, according to the circumstances of each individual procedure.

Blanket Clearance: A pre-arranged clearance originated for specific activities or events and specified in a letter of agreement.

Blind Transmission: A transmission from one station to another station in circumstances where two-way communication cannot be established, but where it is believed that the called station is able to receive the transmission.

Block Level: A section of airspace with specified upper and lower limits on a specific track, in which cleared aircraft are permitted to manoeuvre.

Break-out Procedure(s): Immediate evasive manoeuvres, which are performed on instruction by air traffic control.

Note: In the context of simultaneous parallel operations, break-out procedures are used to direct a threatened aircraft and a deviating aircraft away from each other.

Briefing: The act of giving in advance, specific pre-flight instructions or information to aircrew.

Broadcast: A transmission of information relating to air navigation for which an acknowledgement is not expected.

Calculated Off-Block Time (COBT): A time calculated and issued by an ATFM unit, as a result of pre-tactical slot allocation, at which a flight is expected to pushback/vacate its parking position so as to meet a CTOT, taking into account start-up and taxi time.

Calculated Take-Off Time (CTOT): A time calculated and issued by an ATFM unit, as a result of pre-tactical slot allocation, at which a flight is expected to become airborne.

Ceiling: The height above the ground or water of the base of the lowest layer of cloud below 20,000FT covering more than one-half of the sky.

CENSAR: An automated centralised SARTIME database software package used by ATS to manage SARTIMEs.

Centre: A generic call sign which can include ATC, Advisory, Flight Information and Alerting services, depending on the classification of airspace in which the service is provided.

Certified Aerodrome: A place that is certified as an aerodrome under the Civil Aviation Safety Regulations.

Circling Approach: An extension of an instrument approach procedure which provides for visual circling of the aerodrome prior to landing.

Clearance Limit: The point to which an aircraft is granted an ATC clearance.

Clearance Expiry Time: A time specified by an ATC unit at which a clearance ceases to be valid.

Clearway: A defined rectangular area on the ground or water under the control of the appropriate authority, selected or prepared as a suitable area over which an aeroplane may make a portion of its initial climb to a specified height.

Closely spaced runways: Runways that are parallel or near-parallel and spaced less than 1,525M but not less than 1,035M apart.

Collocated (Navigation) Aids: En route waypoints or navigation aids that are within 600M of each other.

Common Traffic Advisory Frequency (CTAF): A designated frequency on which pilots make positional broadcasts when operating in the vicinity of a non-controlled aerodrome or within a Broadcast Area.

Communicable Diseases: Communicable diseases include cholera, typhus (epidemic), smallpox, yellow fever, plague, and such other diseases as the contracting States shall, from time to time, decide to designate.

Community Service Flight (CSF): A flight that

a. involves:

(i) the transport of one or more individuals (a patient) to a destination for the purpose of each such individual receiving non-emergency medical treatment or services at the destination; or

(ii) the transport of a patient from a destination mentioned in *para (i).* (the treatment destination) to another treatment destination; or

(iii) the transport of a patient from a treatment destination:

(1) back to a place from which the patient departed for a treatment destination; or

(2) to a destination at which the patient resides; and

b. is provided to a patient, and any person who accompanies the patient to provide support and assistance, without a charge being made to any of those persons for their carriage; and

c. medical treatment is not provided on board the aircraft for the flight, other than the administering of medication or in response to an unexpected medical emergency; and

d. is coordinated, arranged or facilitated by an entity for a charitable purpose or community service purpose.

Company Operations Representative: The representative of an operating agency who is authorised to act in the capacity of liaison officer between ATC and the operating agency in respect of the control of an aircraft of that agency.

Conditional Route (CDR): An ATS route or portion thereof which can be planned and used under specified conditions.

Contaminated Runway: A runway that has more than 25% of the runway surface area within the required length and width being used covered by:

a. water, or slush, more than 3mm deep; or

b. loose snow more than 20mm deep; or

c. compacted snow or ice, including wet ice.

Note: A contaminated runway report should identify the type of contaminant using the relevant runway surface condition descriptor (most commonly 'standing water' in Australia), and if possible and relevant – the extent of contamination. 'CONTAMINATED RUNWAY' is not normally used in radiotelephony.

Continuous Descent Final Approach (CDFA): A technique, consistent with stabilised approach procedures, for flying the final approach segment of a non-precision instrument approach procedure as a continuous descent, without level-off, from an altitude/height at or above the final approach fix altitude/height to a point approximately 50FT above the landing runway threshold or the point where the flare manoeuvre should begin for the type of aircraft flown.

Control Area (CTA): A controlled airspace extending upwards from a specified limit above the earth.

Controlled Aerodrome: An aerodrome at which ATC service is provided to aerodrome traffic.

Controlled Airspace: Airspace of defined dimensions within which ATC service is provided in accordance with the airspace classification.

Controller: An air traffic controller, operating within an organisation approved under *CASR Part 172* and qualified in accordance with *CASR Part 65*.

Controller Pilot Data Link Communications (CPDLC): A means of communication between controller and pilot using data link for ATC communications.

Controlling Authority:

a. With respect to airspace classifications, this is the Air Traffic Service provider for that area.

b. With respect to Restricted Areas, this is the agency nominated to exercise the conditions of entry and, where authorised, the Air Traffic Service provider for that area. *See also Administering Authority.*

Control Zone (CTR): A controlled airspace extending upwards from the surface of the earth to a specified upper limit.

CPDLC message: Information exchanged between an airborne system and its ground counterpart. A CPDLC message consists of a single message element or a combination of message elements conveyed in a single transmission by the initiator.

CPDLC Message Set: A list of standard message elements and free text message elements.

Cruise Climb: An aeroplane cruising technique resulting in a nett increase in altitude as the aeroplane weight decreases.

Cruising Level: A level maintained during a significant portion of a flight.

Current flight plan: The flight plan that reflects changes to the filed flight plan, if any, by subsequent ATC clearances.

Danger Area: An airspace of defined dimensions within which activities dangerous to the flight of aircraft may exist at specified times.

Day: The period between the beginning of morning civil twilight (first light) and the end of evening civil twilight (last light).

Dead Reckoning (DR) Navigation: The estimating or determining of position by advancing an earlier known position by the application of direction, time and speed data.

Decision Altitude/Height (DA/H): A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Note 1: DA is referenced to mean sea level and DH is referenced to the threshold elevation.

Note 2: The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path. In Category III operations with a decision height the required visual reference is that specified for the particular procedure and operation.

Defined Point After Take-off (DPATO): The point within the take-off and initial climb phase before which the helicopter's ability to continue the flight safely, with one engine inoperative, is not assured and a forced landing may be required.

Density Height: An atmospheric density expressed in terms of height which corresponds to that density in the Standard Atmosphere.

Dependent Parallel Approaches: Simultaneous instrument approaches to parallel or near-parallel instrument runways where ATS surveillance system separation minima between aircraft on adjacent extended runway centrelines are prescribed.

Distance Measuring Equipment (DME): Equipment which measures in nautical miles, the slant range of an aircraft from the selected DME ground station.

DME Distance: The slant range from the source of a DME signal to the receiving antenna.

Domestic Flight: A flight between two points within the Australian FIR.

Dry Runway: A runway that is free of visible moisture and not contaminated within the area intended to be used.

East Coast SSR Coverage: The area of Australia east-coast secondary surveillance radar (SSR) coverage within approximately 200NM of a line Cairns - Brisbane - Sydney - Melbourne - Adelaide.

Elevation: The vertical distance of a point or a level, on or affixed to the surface of the earth, measured from mean sea level.

Emergency Fuel: The term used to describe a situation when the calculated usable fuel predicted to be available upon landing at the nearest aerodrome where a safe landing can be made is less than the final reserve fuel for the flight.

Note: The emergency fuel declaration is a distress message.

Emergency Phases:

a. Uncertainty Phase: A situation wherein uncertainty exists as to the safety of an aircraft and its occupants.

b. Alert Phase: A situation wherein apprehension exists as to the safety of an aircraft and its occupants.

c. Distress Phase: A situation wherein there is reasonable certainty that an aircraft and its occupants are threatened by grave and imminent danger or require immediate assistance.

Equivalent Single Isolated Wheel Load: The equivalent load that would be imposed on a pavement by a single wheel if any wheel group on an aircraft were replaced by a single wheel using the same tyre pressure.

Essential Radio Navigation Service. A radio navigation service whose disruption has a significant impact on operations in the affected airspace or aerodrome.

Estimate: The time at which it is estimated that an aircraft will be over a position reporting point or over the destination.

Estimated Elapsed Time (EET): The estimated time required to proceed from one significant point to another.

Estimated Off Block Time: The estimated time at which the aircraft will commence movement associated with departure.

Estimated Time of Arrival (ETA): For IFR flights, the time at which it is estimated that the aircraft will arrive over that designated point, defined by reference to navigation aids, from which it is intended that an instrument approach procedure will be commenced, or, if no navigation aid is associated with the aerodrome, the time at which the aircraft will arrive over the aerodrome. For VFR flights, the time at which it is estimated that the aircraft will arrive over the aerodrome.

Expected Approach Time (EAT): The time at which ATC expects that an arriving aircraft, following a delay, will leave the holding fix to complete its approach for a landing.

Note: The holding fix referred to in the EAT is that shown on the instrument approach chart from which the instrument approach is prescribed to commence.

Filed Flight Plan (FPL): The latest flight plan as submitted by the pilot, an operator or a designated representative for use by ATS units.

Final Approach: That part of an instrument approach procedure which commences at the specified final approach fix or point, or where such a fix or point is not specified:

a. at the end of the last procedure turn, base turn or inbound turn of a racetrack procedure, if specified; or
b. at the point of interception of the last track specified in the approach procedure; and
c. ends at a point in the vicinity of an aerodrome from which a landing can be made, or a missed approach is initiated.

Final Approach Altitude: The specified altitude at which final approach is commenced.

Final Approach Course: Where the aircraft is aligned with the runway centreline and established on the same lateral navigational guidance as the GLS approach procedure.

Final Approach Fix (FAF): A specified point on a non-precision instrument approach which identifies the commencement of the final segment.

Final Approach Point (FAP): A specified point on the glide path of a precision instrument approach which identifies the commencement of the final segment.

Note: The FAP is co-incident with the FAF of a localiser based non-precision approach.

Final Approach Segment: That segment of an instrument approach procedure in which alignment and descent for landing are accomplished.

Final Approach and Take-off Area (FATO): A defined area over which the final phase of the approach manoeuvre to hover or landing is completed and from which the take-off manoeuvre is commenced. Where the FATO is to be used by performance Class 1 helicopters, the defined area includes the rejected take-off area available.

Final Leg: The path of an aircraft in a straight line immediately preceding the landing (alighting) of the aircraft.

Fix: A geographical position of an aircraft at a specific time determined by visual reference to the surface, or by navigational aids.

Flight File: A file stored on the NAIPS system which contains stored briefings, or a stored flight notification. Flight files are owned by pilots and/or operators, and updated at their request.

Flight Following: The provision of an ongoing Surveillance Information Service (SIS).

Flight Information: Information useful for the safe and efficient conduct of flight, including information on air traffic, meteorological conditions, aerodrome conditions and airways facilities.

Flight Information Area (FIA): An airspace of defined dimensions, excluding controlled airspace, within which flight information and SAR alerting services are provided by an ATS unit.

Note: FIAs may be sub-divided to permit the specified ATS unit to provide its services on a discrete frequency or family of frequencies within particular areas.

Flight Information Region (FIR): An airspace of defined dimensions within which flight information service and SAR alerting service are provided.

Flight Information Service (FIS): A service provided for the purpose of giving advice and information useful for the safe and efficient conduct of flights.

Flight Level (FL): A surface of constant atmospheric pressure which is related to a specific pressure datum, 1013.2HPA, and is separated from other such surfaces by specific pressure intervals.

Flight plan: Specified information relative to an intended flight or portion of a flight of an aircraft.
Note 1: The term flight plan may be prefixed by the words "preliminary", "filed", "current" or "operational" to indicate the context and different stages of a flight.
Note 2: When the word "message" is used as a suffix to this term, it denotes the content and format of the flight plan data as transmitted.

Flight Procedure Authorisation (FPA): Authorisations which allow a pilot holding a Private IFR rating to use additional types of navigation aids as well as night flying, instrument approaches and instrument departures.

Flight Note: Details of the route and timing of a proposed flight provided by the pilot in command of an aircraft, which is other than notification submitted to Airservices Australia, and which is required to be left with a person who could be expected to notify appropriate authorities in the event that the flight becomes overdue.

Flight Notification (within Australian FIR): Specified information provided to ATS units, relative to the intended flight or portion of flight of an aircraft.

Flight Path Monitoring: The use of ATS surveillance systems for the purpose of providing aircraft with information and advice relative to significant deviations from nominal flight path including deviations from the terms of their ATC clearances.

Note: Some applications may require a specific technology e.g. radar, to support the function of flight path monitoring.

Flight Visibility: The visibility forward from the cockpit of an aircraft in flight.

Forecast: A statement of expected meteorological conditions for a specified period, and for a specified area or portion of airspace.

Formation: Two or more aircraft flown in close proximity to each other and operating as a single aircraft with regard to navigation, position reporting and control.

Note: A CASR Part 61 flight activity endorsement is required when aeroplane or helicopter pilots are flying in formation (flying in formation is defined slightly differently to the definition of formation in the AIP).
CASR 91.205 details conditions on formation flights.

Free text message element: Part of a message that does not conform to any standard message element in the PANS-ATM (DOC 4444).

Glide Path (GP): A descent profile determined for vertical guidance during a final approach.

Global Navigation Satellite System (GNSS): A satellite-based radio navigation system that uses signals from orbiting satellites to determine precise position and time.
Note: While the term "GNSS" covers a variety of systems such as GPS, GLONASS, Galileo etc, Australia requires the use of GPS for aviation purposes.

Global Positioning System (GPS): A GNSS constellation operated by the United States Government.

Gross Weight: The weight of the aircraft together with the weight of all persons and goods (including fuel) on board the aircraft at that time.

Ground Based Augmentation System (GBAS): An augmentation system in which the user receives augmentation information directly from a ground-based transmitter.

Ground Based Augmentation System (GBAS) Landing System (GLS): A system for approach and landing operations using a GBAS, as the primary navigational reference.

Ground Based Navigation Aid: Means NDB, VOR, DME.

Ground Taxiing: The movement of a helicopter under its own power and on its undercarriage wheels.

Ground Visibility: The visibility at an aerodrome, as reported by an accredited observer.

Hazardous Conditions: Meteorological conditions which may endanger aircraft or adversely affect their safe operation, particularly those phenomena associated with volcanic ash cloud and thunderstorms - icing, hail and turbulence.

Head of State: Heads of State or of Government, or other selected dignitaries on official visits to Australia (as provided by Department of Prime Minister and Cabinet Protocol and International Visits Branch), or the personal transport of the Governor-General or the Prime Minister.

Heading (HDG): The direction in which the longitudinal axis of an aircraft is pointed, usually expressed in degrees from North (true, magnetic, compass or grid).

Height: The vertical distance of a level, a point or an object considered as a point measured from a specified datum.

Helicopter Access Corridor: A corridor wholly within controlled airspace designed for the exclusive use of helicopters in VMC. The extent and alignment of the corridor is related to and delineated by prominent geographical/topographical features.

Helicopter Landing Site (HLS): A place that is used as an aerodrome for the purposes of the landing and taking-off of helicopters.

Helicopter Lane: A lane, outside controlled airspace, designed for use by helicopters to facilitate traffic flow.

Helicopter Movement Area: The movement area for helicopters is that part of an aerodrome that can safely be used for the hovering, taxiing, take-off and landing of helicopters and consists of the manoeuvring area and aprons, but excluding those areas reserved for unrestricted use by the general public.

Helicopter Reference Point (HRP): The designated location of a heliport or a landing location.

Helideck: An area intended for use wholly or partly for the arrival or departure of rotorcraft on a ship, or a floating or fixed structure on water.

High Capacity Aircraft: An aircraft that is certified as having a maximum seating capacity exceeding 38 seats or a maximum payload exceeding 4,200KG.

Hold Short Line/Lights: A line marked across a runway, with associated lights, in accordance with the requirements of *AIP AD 1.1*, at which landing aircraft must stop when required during Land and Hold Short Operations (LAHSO).

Holding Bay: A defined area where aircraft can be held, or bypassed, to facilitate efficient surface movement of aircraft.

Holding Fix: A specified location identified by visual or other means in the vicinity of which the position of an aircraft in flight is maintained in accordance with ATC Instructions.

Holding Procedure: A predetermined manoeuvre which keeps an aircraft within a specified airspace whilst awaiting further clearance.

Hospital Aircraft: (see Medical Flight).

Hot Spot: A location on an aerodrome movement area with a history or potential risk of collision or runway incursion, and where heightened attention by pilots/drivers is necessary.

Identification: The situation which exists when the position indication of a particular aircraft is seen on a situation display and positively identified by ATC.

IFR Pick-up: A pilot procedure whereby a flight operating to the IFR in Class G airspace changes to VFR upon entering Class E airspace whilst awaiting an airways clearance.

Independent Parallel Approaches: Simultaneous instrument approaches to parallel or near-parallel instrument runways where ATS surveillance system separation minima between aircraft on adjacent extended runway centrelines are not prescribed.

Independent Parallel Departures: Simultaneous departures in the same direction from parallel or near-parallel instrument runways.

Independent Visual Approach: Visual approach operations to parallel or near-parallel instrument runways where the distance between runway centrelines and use of particular procedures allows a visual approach to one runway independently of approaches occurring on an adjacent parallel or near-parallel runway.

Inertial Navigation/Reference System (INS/IRS): A self-contained navigation system that continually measures the accelerations acting upon the vehicle of which it is part. Suitably integrated, these forces provide velocity and thence position information.

Initial Approach Fix (IAF): The fix at the commencement of an instrument approach.

Initial Approach Segment: That segment of an instrument approach procedure between the initial approach fix and the intermediate approach fix or, where applicable, the final approach fix or point.

Initial Departure Fix (IDF): The terminal fix for the visual segment and the fix where the instrument phase of the PinS departure begins.

Instrument Approach Operations: An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

a. a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and
b. a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance.

Note 1: Lateral and vertical navigation guidance refers to the guidance provided either by:

a. ground-based radio navigation aids; or

b. computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these.

Note 2: The classification of instrument approach operations is outlined in AIC H26/14.

Instrument Approach Procedure (IAP): A series of predetermined manoeuvres by reference to flight instruments with specified protection from obstacles from the initial approach fix, or where applicable, from the beginning of a defined arrival route to a point from which a landing can be completed and thereafter, if a landing is not completed, to a position at which holding or en route obstacle clearance criteria apply. Instrument approach procedures are classified as follows:

a. Non-precision approach (NPA) procedure. An instrument approach procedure designed for 2D instrument approach operations Type A.

Note: Non-precision approach procedures may be flown using a Continuous Descent Final Approach technique (CDFA). CDFA with advisory. VNAV guidance calculated by on-board equipment are considered 3D instrument approach operations. CDFA with manual calculation of the required rate of descent are considered 2D instrument approach operations.

b. Approach Procedure with Vertical guidance (APV). A Performance Based Navigation (PBN) instrument approach procedure designed for 3D instrument approach operations Type A.

c. Precision Approach (PA) procedure. An instrument approach procedure based on navigation systems (ILS, MLS, GLS and SBAS CAT I) designed for 3D instrument approach operations Type A or B.

Note: Refer to AIC H26/14 for instrument approach operation types.

Instrument Landing System (ILS): A precision instrument approach system which normally consists of the following electronic components: VHF Localiser, UHF Glideslope, VHF Marker Beacons.

Instrument Runway: One of the following types of runways intended for the operation of aircraft using instrument approach procedures:

a. Non-precision approach runway. An instrument runway served by visual aids and a non-visual aid providing at least directional guidance adequate for a straight-in approach.

b. Precision approach runway, CAT I. An instrument runway served by a precision approach procedure and visual aids intended for operations with a decision height not lower than 60M (200FT) and either a visibility not less than 800M, or a RVR not less than 550M.

c. Precision approach runway, CAT II. An instrument runway served by ILS and visual aids intended for operations with a decision height lower than 60M (200FT), but not lower than 100FT, and a RVR not less than 300M.

d. Precision approach runway, CAT III. An instrument runway served by ILS to and along the surface of the runway and:

   (i) for CAT IIIA – intended for operations with a decision height lower than 30M (100FT), or no decision height, and a RVR not less than 175M;

   (ii) for CAT IIIB – intended for operations with a decision height lower than 15M (50FT), or no decision height, and a RVR less than 175M, but not less than 50M;

   (iii) for CAT IIIC – intended for operations with no decision height and no RVR limitations.

Integrated Aeronautical Information Package: A package which consists of the following elements: AIP, including amendment service; supplements to the AIP; NOTAM and Preflight Information Bulletins (PIBs); AIC; and checklists and summaries.

Integrity: That quality which relates to the trust which can be placed in the correctness of information supplied by a system. It includes the ability of a system to provide timely warnings to users when the system should not be used for navigation.

Intermediate Approach Segment: That segment of an instrument approach procedure between either the intermediate approach fix and the final approach fix or point, or between the end of the reversal, race track or dead reckoning track procedure and the final approach fix or point, as appropriate.

Intermediate Fix (IF): A fix that marks the end of an initial segment and the beginning of the intermediate segment.

In the Vicinity: An aircraft is in the vicinity of a non-controlled aerodrome if it is within a horizontal distance of 10 miles; and within a height above the aerodrome reference point that could result in conflict with operations at the aerodrome.

Land And Hold Short Operations (LAHSO): A procedure involving dependent operations conducted on two intersecting runways whereby aircraft land and depart on one runway while aircraft landing on the other runway hold short of the intersection.

Landing Area: That part of the movement area intended for the landing or take-off of aircraft.

Land Rescue Unit: A land party equipped to undertake a search for an aircraft within the region of its responsibility.

Level: A generic term relating to the vertical position of an aircraft in flight and meaning variously, height, altitude or flight level.

Localiser (LOC): The component of an ILS which provides azimuth guidance to a runway. It may be used as part of an ILS or independently.

Logon address: A specified code used for data link logon to an ATS unit.

Low Jet Route (LJR): A route, or part of a route, at or below 5,000FT AGL used by MLJ aircraft for low level, high speed operations.

Low Visibility Operation: An operation involving:

a. an approach with minima less than precision approach category I; or

b. a take-off with visibility below 550M.

Low Visibility Procedures: Procedures applied at an aerodrome for protecting aircraft operations during conditions of reduced visibility or low cloud.

Lowest Safe Altitude (LSALT): The lowest altitude which will provide safe terrain clearance at a given place.

Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

Marker: An object displayed above ground level in order to indicate an obstacle or delineate a boundary.

Marker Beacon: A type of radio beacon, the emissions of which radiate in a vertical pattern.

Markings: A symbol or group of symbols displayed on the surface of the movement area in order to convey aeronautical information.

Maximum Take-off Weight (MTOW): The maximum take-off weight of an aircraft as specified in its Certificate of Airworthiness.

Medical Flight: A flight providing transport of medical patients, personnel, and/or equipment, prioritised as follows:

MEDEVAC: a life critical medical emergency evacuation e.g. An aircraft proceeding to pick up, or carrying, a severely ill patient, or one for whom life support measures are being provided.

HOSP: a medical flight declared by medical authorities e.g. An aircraft transporting or proceeding to pick up medical personnel and/or equipment urgently required for the treatment of a severely ill patient, or returning urgently required medical personnel and/or equipment at the termination of a MEDEVAC flight.

METBRIEF (Automated Meteorological Telephone Briefing): Self help system which delivers meteorological information on the telephone using a computer generated voice, in response to a tone generated telephone request.

Meteorological Information: Meteorological report, analysis, forecast, and any other statement relating to existing or expected meteorological conditions.

Meteorological Office (MO): An office designated to provide meteorological service for air navigation.

Meteorological Warning: A statement or meteorological report of the occurrence or expectation of a deterioration or improvement in meteorological conditions or of any meteorological phenomenon which may seriously affect the safe operation of aircraft.

Military Airspace: A collective term encompassing military controlled airspace/control zones, military Restricted Areas and Military Operating Areas (MOA).

Military Low Jet (MLJ): Military aircraft operating on LJR

Military Operating Area (MOA): An airspace of defined dimensions, with specified conditions, established for hazardous military activities.

Minimum Crossing Altitude (MCA): The minimum IFR altitude that aircraft may cross the IDF.

Minimum Descent Altitude/Height (MDA/H): A specified altitude or height in a 2D instrument approach operation or circling approach operation below which descent must not be made without the required visual reference.

Note 1: MDA is referenced to Mean Sea Level (MSL) and MDH is referenced to the aerodrome elevation or to the threshold elevation if that is more than 7FT below the aerodrome elevation. A minimum descent height for a circling approach is referenced to the aerodrome elevation.

Note 2: The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path. In the case of a circling approach the required visual reference is the runway environment.

Minimum Fuel: The term used to describe a situation when an aircraft's fuel supply has reached a state where having committed to land at a specific aerodrome, the pilot calculates that any change to the existing clearance to that aerodrome may result in landing with less than final reserve fuel for the flight.

Note: The minimum fuel state is not an emergency situation but an indication that an emergency situation is possible should any additional delay occur.

Minimum Sector Altitude (MSA): The lowest altitude which may be used which will provide a minimum clearance of 1,000FT above all objects located in an area contained within a circle or a sector of a circle of 25NM or 10NM radius centred on a significant point, the ARP, or the HRP.

Minimum Vector Altitude: The lowest altitude which a controller may assign to a pilot in accordance with the *Radar Terrain Clearance* chart.

Missed Approach Holding Fix (MAHF): A fix used in area navigation applications that marks the end of the missed approach segment and the centre point for the missed approach holding.

Missed Approach Point (MAPt): That point in an instrument approach procedure at or before which the prescribed missed approach procedure must be initiated in order to ensure that the minimum obstacle clearance is not infringed.

Missed Approach Procedure: The procedure to be followed if the approach cannot be continued.

Missed Approach Turning Fix (MATF): A fix different from MAPt that marks a turn in the missed approach segment.

Movement Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).

NAIPS: The National Aeronautical Information Processing System, which provides briefings and flight notification functions.

Navigation Specification: A set of aircraft and flight crew requirements needed to support performance based navigation operations within a defined airspace. There are two kinds of navigation specifications:

RNP Specification: A navigation specification based on area navigation that includes the requirement for on board performance monitoring and alerting, designated by the prefix RNP, e.g. RNP4, RNP APCH.

RNAV Specification: A navigation specification based on area navigation that does not include the requirement for on board performance monitoring and alerting, designated by the prefix RNAV, e.g. RNAV 5, RNAV 1.

Note: The Performance-based Navigation Manual (Doc 9613), Volume II, contains detailed guidance on navigation specifications.

Night: The period between the end of evening civil twilight (last light) and the beginning of the following morning civil twilight (first light).

Night Vision Goggles (NVG): A self-contained binocular night vision enhancement device, usually helmet mounted or otherwise worn by a person, that can detect and amplify light in both the visual and near infra-red bands of the electromagnetic spectrum.

Night Vision Imaging System (NVIS): A self-contained binocular night vision enhancement device, usually including goggles, that:

a. is helmet mounted or otherwise worn by a person; and

b. can detect and amplify light in both the visual and near infra-red bands of the electromagnetic spectrum.

Non-Controlled Aerodrome: An aerodrome at which ATC is not operating.

Non-Directional Beacon (NDB): A special radio station, the emissions of which are intended to enable a mobile station to determine its radio bearing or direction with reference to that special radio station.

Normal Operating Zone (NOZ): Airspace of defined dimensions extending to either side of a published instrument approach procedure final approach course or track. Only that half of the normal operating zone adjacent to a No-Transgression Zone (NTZ) is taken into account in independent parallel approaches.

NOTAM: A notice distributed by means of telecommunication containing information concerning the establishment, condition or change in any aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential to personnel concerned with flight operations.

No-Transgression Zone (NTZ): In the context of independent parallel approaches, a corridor of airspace of defined dimensions located centrally between the two extended runway centrelines, where a penetration by an aircraft requires a controller intervention to manoeuvre any threatened aircraft on the adjacent approach.

One Way Route: A route with limitations for use in one direction, depicted on *ERC-H*, *ERC-L* and/or *TAC* charts by an arrow in the direction that can be used without limitation (see *ERSA GEN-FPR* for additional details).

Operator: A person, organisation or enterprise engaged in or offering to engage in aircraft operation.

Overshoot Shear: A wind shear occurrence which produces an INITIAL effect of overshooting the desired approach path and/or increasing airspeed.

Parking Area: A specially prepared or selected part of an aerodrome within which aircraft may be parked.

Passive LAHSO Runway: The runway used during LAHSO for arriving and departing aircraft that have the full length available.

Pavement Classification Number (PCN): A number expressing the bearing strength of a pavement for unrestricted operations.

Performance-Based Navigation (PBN). Area navigation based on performance requirements for aircraft operating along an ATS route, on an instrument approach procedure or in a designated airspace.

Note: Performance requirements are expressed in navigation specifications (RNAV specification, RNP specification) in terms of accuracy, integrity, continuity and functionality needed for the proposed operation in the context of a particular airspace concept. Availability of a global navigation satellite system signal in space and/or some other navigation aid infrastructure is considered within the airspace concept in order to enable the navigation application.

Performance Class 1 (PC1): PC1 is the class of helicopter performance such that in the event of failure of the critical power-unit the helicopter is able either to land within the rejected take-off distance available, or to safely continue the flight to an appropriate landing area, depending on when the failure occurs.

Performance Class 2 (PC2): PC2 is the class of helicopter performance such that in the event of critical power-unit failure performance is available to enable the helicopter to safely continue the flight except when the failure occurs early during the take-off manoeuvre or late in the landing manoeuvre, in which cases a forced landing may be required.

Permissible All-Up-Weight: The weight to which an aircraft is limited by virtue of the physical characteristics of an aerodrome.

Pilot in Command: The pilot designated by the operator, or in the case of general aviation, the owner, as being in command and charged with the safe conduct of a flight.

Point-in-Space Reference Point: Reference point for the point-in-space approach as identified by the latitude and longitude of the MAPt.

Precision Approach Procedure: An instrument approach procedure utilising lateral and vertical guidance provided by an ILS or GLS.

Precision Runway Monitor (PRM): An ATS surveillance system and associated procedures used for independent parallel approaches to closely spaced runways.

Pre-Departure Clearance (PDC): A means of delivering an unsolicited, text-based airways clearance to eligible aircraft via an ATC data link.

Preferred Runway: A runway nominated by ATC or listed in the AIP as the most suitable for the prevailing wind, surface conditions or noise sensitive areas in the proximity of the aerodrome.

Primary Means Navigation System: A navigation system that, for a given operation or phase of flight, must meet accuracy and integrity requirements, but need not meet full availability and continuity of service requirements. Safety is achieved by either limiting flights to specific time periods, or through appropriate procedural restrictions and operational requirements.

Private IFR: The Private IFR Rating (PIFR) authorises the holder to act as pilot in command of flights under the IFR by day in single pilot aircraft having a MTOW not greater than 5,700KG.

Procedural Service: Term used to indicate that information derived from an ATS surveillance system is not required for the provision of ATS.

Procedure Altitude/Height: A specified altitude/height flown at or above the minimum altitude/height, and established to accommodate a stabilised descent at a prescribed descent gradient/angle in the intermediate/final approach segment.

Prohibited Area: An airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is prohibited. Designation is appropriate only for reasons of military necessity.

Published Speed: A speed restriction shown on a Standard Instrument Departure (SID), Standard Instrument Arrival (STAR), or other instrument flight procedure.

QNH Altimeter Setting: That pressure setting which, when placed on the pressure setting sub-scale of a sensitive altimeter of an aircraft located at the reference point of an aerodrome, will cause the altimeter to indicate the vertical displacement of the reference point above mean sea level.

Radio Altimeter (RA) Height. An indication of vertical distance between a point on the normal glidepath at DA and the terrain directly beneath this point.

Radio Navigation Service. A service providing guidance information or position data for the efficient and safe operation of aircraft supported by one or more radio navigation aids.

Rapid-Exit Taxiway: A taxiway connected to a runway at an acute angle and designed to allow landing aeroplanes to turn off at high relative speeds.

Receiver Autonomous Integrity Monitoring (RAIM): A system whereby an airborne GNSS receiver/processor autonomously monitors the integrity of the navigation signals from GNSS satellites.

Reduced Vertical Separation Minimum (RVSM): The vertical separation minimum of 1,000FT between FL 290 and FL 410 inclusive.

Reference Datum Height (RDH): The height of the measured ILS glide path at the threshold. It will provide a similar value to Threshold Crossing Height.

Reporting Point: A specified geographical location in relation to which the position of an aircraft can be reported.

Required Navigation Performance (RNP): A statement of the navigation performance necessary for operation within a defined airspace.

RNP Type: A containment value expressed as a distance in nautical miles from the intended position within which flights would be for at least 95 per cent of the total flying time.

Rescue Coordination Centre (RCC): A unit established for promoting efficient organisation of search and rescue service and for coordinating the conduct of search and rescue operations within a search and rescue region.

Resolution Advisory (RA): An indication given to the flight crew recommending a manoeuvre or a manoeuvre restriction to avoid collision.

Restricted Area: An airspace of defined dimensions above the land areas or territorial waters of a State, within which the flight of aircraft is restricted in accordance with certain specified conditions.

Note: This designation is used when necessary in the interests of public safety or the protection of the environment.

Route: A way to be taken in flying from a departure to a destination aerodrome, specified in terms of track and distance for each route segment.

Runway (RWY): A defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft.

Runway-Holding Position: A designated position intended to protect a runway, an obstacle limitation surface, or an ILS critical/sensitive area at which taxiing aircraft and vehicles must stop and hold, unless otherwise authorised by the aerodrome control tower.

Note: In radiotelephony phraseologies, the expression "holding point" is used to designate the runway-holding position.

Runway Number: The runway identification associated with the runway direction end.

Runway Strip: The defined area, including the runway (and stopway if provided), intended both to reduce the risk of damage to aircraft inadvertently running off the runway and to protect aircraft flying over it during take-off, landing or missed approach.

Runway Surface Condition(s): A description of the condition(s) of the runway surface, from the following:

a. Dry Runway – See 'Dry Runway' definition.

b. Wet Runway – See 'Wet Runway' definition.

c. Slippery wet runway – A wet runway where the surface friction characteristics of a significant portion of the runway have been determined to be degraded.

Note: Slippery wet runway surface condition is reported as a SLIPPERY WET runway with RWYCC 3.

d. Contaminated Runway – See the definitions for 'Contaminated Runway' and 'Runway Surface Condition Descriptors'.

Runway Surface Condition Descriptors: One of the following elements on the surface of the runway:

a. **Compacted Snow:** Snow that has been compacted into a solid mass such that aeroplane tires, at operating pressures and loadings, will run on the surface without significant further compaction or rutting of the surface.

b. **Dry Snow:** Snow from which a snowball cannot readily be made.

c. **Frost:** Frost consists of ice crystals formed from airborne moisture on a surface whose temperature is below freezing. Frost differs from ice in that the frost crystals grow independently and therefore have a more granular texture.

Note: Under certain conditions frost can cause the surface to become very slippery and it is then reported appropriately as reduced braking action.

d. **Ice:** Water that has frozen or compacted snow that has transitioned into ice, in cold and dry conditions.

e. **Slush:** Snow that is so water-saturated that water will drain from it when a handful is picked up or will splatter if stepped on forcefully.

f. **Standing Water:** Water of depth greater than 3mm.

Note: This is the most likely runway contaminant to be experienced in Australia. Running water of depth greater than 3mm is reported as standing water.

g. **Wet Ice:** Ice with water on top of it or ice that is melting.

h. **Wet Snow:** Snow that contains enough water content to be able to make a well-compacted, solid snowball, but water will not squeeze out.

Runway Visibility (RV): The distance along a runway over which a person can see and recognise a visibility marker or runway lights.

Note: The term RUNWAY VISIBILITY is used by ATC or ground personnel to report visibility along a runway as determined by a ground observer.

Runway Visual Range (RVR): The range over which the pilot of an aircraft on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identifying its centre line. (*ICAO*)

Note: Within Australia, the term "RUNWAY VISUAL RANGE" or "RVR" is used exclusively in relation to RVR measured by an instrumented system.

SARTIME: The time nominated by a pilot for the initiation of SAR action if a report has not been received by the nominated unit.

SARWATCH: A generic term covering SAR alerting based either on full position reporting procedures, scheduled reporting times (SKEDS), or SARTIME.

Search and Rescue (SAR): The act of finding and returning to safety, aircraft and persons involved in an emergency phase.

Search and Rescue Region (SRR): The specified area within which search and rescue is coordinated by a particular Rescue Coordination Centre.

Segment Minimum Safe Altitude: The lowest altitude at which the minimum obstacle clearance is provided.

Segregated Parallel Operations: Simultaneous operations on parallel or near-parallel instrument runways in which one runway is used exclusively for approaches and the other runway is used exclusively for departures.

Self Contained Navigation Systems: Area navigation systems based on INS, IRS or GNSS.

Significant Point: A specified geographical location used in defining an ATS route or the flight path of an aircraft and for other navigation and ATS purposes.

Note: There are three categories of significant points: ground-based navigation aid, intersection and waypoint. In the context of this definition, intersection is a significant point expressed as radials, bearings and/or distances from ground-based navigation aids.

Significant Weather: Any weather phenomenon which might affect flight visibility or present a hazard to an aircraft.

Simultaneous Opposite Direction Parallel Runway Operations (SODPROPS): A condition whereby arriving aircraft will approach and land on one runway, concurrent with aircraft departures from the parallel runway using the opposite direction to that being used for approach and landing.

Situation Display: An electronic display depicting the position and movement of aircraft and other information as required.

Sole Means Navigation System: A navigation system that, for a given phase of flight, must allow the aircraft to meet all four navigation system performance requirements - accuracy, integrity, availability and continuity of service.

Space Weather Centre (SWXC): A centre designated to monitor and provide advisory information on space weather phenomena expected to affect high-frequency radio communications, communications via satellite, GNSS-based navigation and surveillance systems and/or pose a radiation risk to aircraft occupants.

Note: A space weather centre is designated as global and/or regional.

Special Air-Report (AIREP Special): An AIREP containing the report of special meteorological conditions, i.e. SIGMET phenomenon, or any other MET phenomenon which is likely to affect the safety or efficiency of other aircraft.

Special Authorisation Category I (SA CAT I) Operation: A precision approach CAT I operation with a DH lower than 200FT, but not lower than 150FT; and an RVR not less than 450M.

Special Authorisation Category II (SA CAT II) Operation: A precision approach operation to a runway where some or all of the elements of the precision approach CAT II lighting system are not available, with:

a. a DH lower than 200FT but not lower than 100FT; and

b. RVR of not less than 350M.

Special Use Airspace (SUA): A generic term used for airspace volumes designated for specific operations that may impose limitations on airspace access or use for non-participating aircraft. SUA includes Prohibited, Restricted, Danger and Military Operating Areas, and airspace reservations.

SSR Code: The number assigned to a particular multiple-pulse reply signal transmitted by a transponder in Mode A or Mode C.

Standard Instrument Arrival (STAR): A designated IFR arrival route linking a significant point, normally on an ATS route, with a point from which a published instrument approach procedure can be commenced.

Standard Instrument Departure (SID): A designated IFR departure route linking the aerodrome or a specified runway of the aerodrome with a specified significant point, normally on a designated ATS route, at which the en route phase of a flight commences.

Standard Message Element: Part of a message defined in the PANS-ATM (DOC 4444) in terms of display format, intended use and attributes.

Standard Pressure: The pressure of 1013.2 Hectopascals which, if set upon the pressure sub-scale of a sensitive altimeter, will cause the latter to read zero when at mean sea level in a standard atmosphere.

State Aircraft: An aircraft of any part of the Defence Force (including any aircraft that is commanded by a member of that force in the course of their duties as such a member), and aircraft used in the military, customs, or police services of a foreign country.

Stop-and-Go Landing: A procedure whereby an aircraft lands, comes to a complete stop on the runway and then commences take-off from that point.

Stopway: A defined rectangular area on the ground at the end of the take-off run available prepared as a suitable area in which an aircraft can be stopped in the case of an abandoned take-off.

Supplemental Means Navigation System: A navigation system that must be used in conjunction with a sole means navigation system.

Surveillance Information Service (SIS): An on-request service provided to assist pilots of VFR flights, within ATS surveillance system coverage in Class E and Class G airspace, to avoid other aircraft or to assist in navigation.

Tactical Air Navigation (TACAN): An ultra-high frequency navigation aid which provides a continuous indication of bearing and slant range, in nautical miles, to the selected ground station.

TAF3: An aerodrome forecast (TAF) routinely issued every 3 hours.

Target Off-Block Time (TOBT): The time that the aircraft operator and/or its ground handling agent estimate that an aircraft will be ready for departure - aircraft doors are closed, boarding bridge or stairs disconnected, pushback equipment in place and ready for start-up or pushback upon receiving an ATC clearance.

Target Start-up Approval Time (TSAT): The time at which the flight crew can expect start-up or pushback approval from ATC.

Taxiway (TWY): A defined path on a land aerodrome established for the taxiing of aircraft and intended to provide a link between one part of the aerodrome and another.

Terrain Clearance: The vertical displacement of an aircraft's flightpath from the terrain.

Threshold: The beginning of that portion of the runway usable for landing.

Threshold Crossing Height (TCH): The calculated height of the procedure nominal approach path at the threshold. For ILS or GLS, the TCH will be similar to the Reference Datum Height.

Total Estimated Elapsed Time: For IFR flights, the estimated time required from take-off to arrive over that designated point, defined by reference to navigation aids, from which it is intended that an instrument approach procedure will be commenced, or if no navigation aid is associated with the destination aerodrome, to arrive over the destination aerodrome. For VFR flights the estimated time required from take-off to arrive over the destination aerodrome.

Touch-and-Go Landing: A procedure whereby an aircraft lands and takes off without coming to a stop.

Track: The projection on the earth's surface of the path of an aircraft, the direction of which path at any point is usually expressed in degrees from North (true, magnetic or grid).

Traffic Advisory (TA): An indication given to the flight crew that a certain intruder is a potential threat.

Transition Altitude: The altitude at or below which the vertical position of an aircraft is controlled by reference to altitudes.

Transition Layer: The airspace between the transition altitude and the transition level.

Transition Level: The lowest flight level available for use above the transition altitude.

Transitional Surface: An inclined plane associated with the runway strip and the approach surfaces.

Transponder: A receiver/transmitter which will generate a reply signal upon proper interrogation; the interrogation and reply being on different frequencies.

Unalerted See-and-Avoid: A procedure where flight crew, who have no specific knowledge of other aircraft in their vicinity, rely solely on their ability to physically sight and avoid colliding with aircraft that may be in their vicinity.

Undershoot Shear: A wind shear occurrence which produces an INITIAL effect of undershooting the desired approach path and/or decreasing air speed.

UNICOM (Universal Communications): UNICOM is a non-ATS communications service provided to enhance the value of information normally available about a non-controlled aerodrome.

Unmanned Free Balloon: A non-power-driven, unmanned, lighter-than-air aircraft in free flight.

Note: Unmanned Free Balloons are classified as small, light, medium and heavy. For further details regarding these classifications and for approvals to operate Unmanned Free Balloons, refer to CASR Subpart 101.E

Unserviceable Area: A portion of the movement area not available for use by aircraft because of the physical condition of the surface, or because of any obstruction on the area.

Vectoring: Provision of navigational guidance to aircraft in the form of specific headings, based on the use of an ATS surveillance system.

VFR Climb and Descent: ATC authorisation for an IFR flight in VMC, in classes D and E airspace, to conduct a visual climb or descent.

VFR-on-Top: ATC authorisation for an IFR flight to operate in VMC, in Class E airspace at any appropriate VFR altitude or flight level.

VHF Omni-directional Radio Range (VOR): A VHF radio navigational aid which provides a continuous indication of bearing from the selected VOR ground station.

Visibility: Visibility for aeronautical purposes is the greater of:

a. the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognised when observed against a bright background; or

b. the greatest distance at which lights in the vicinity of 1,000 candelas can be seen and identified against an unlit background.

Visibility Marker: A dark object of suitable dimensions for use as a reference in evaluating runway visibility.

Visual (ATC usage): Used by ATC to instruct a pilot to see and avoid obstacles while conducting flight below the MVA or MSA/LSALT.

Visual (Pilot usage): Used by a pilot to indicate acceptance of responsibility to see and avoid obstacles while operating below the MVA or MSA/LSALT.

Visual Approach Slope Indicator System (VASIS): A system of lights so arranged as to provide visual information to pilots on approach of their position in relation to the optimum approach slope for a particular runway.

VOLMET: Meteorological information for aircraft in flight.

Data link-VOLMET (D-VOLMET): Provision of current aerodrome routine meteorological reports (METAR) and aerodrome special meteorological reports (SPECI), aerodrome forecasts (TAF), SIGMET, special air-reports not covered by a SIGMET and, where available, AIRMET via data link.

VOLMET broadcast: Provision, as appropriate, of current METAR, SPECI, TAF and SIGMET by means of continuous and repetitive voice broadcasts.

V_{s1g} means the one-g stall speed at which the aeroplane can develop a lift force (normal to the flight path) equal to its weight.

Waypoint: A specified geographical location used to define an area navigation route or the flight path of an aircraft employing area navigation. Waypoints are identified as either:

a. **Fly-by Waypoint:** A waypoint which requires turn anticipation to allow tangential interception of the next segment of a route or procedure, or

b. **Flyover Waypoint:** A waypoint at which a turn is initiated in order to join the next segment of a route or procedure.

Wet Runway: A runway that:

a. is covered by surface water not more than 3mm deep; or

b. is covered by slush or loose snow equivalent to surface water not more than 3mm deep; or

c. has sufficient moisture on the surface to cause it to appear reflective, but without significant areas of standing water.

Note: Wet Runway now covers the previous Australian description – 'Damp'.

World Area Forecast Centre (WAFC): A meteorological centre designated to prepare and issue significant weather forecasts and upper-air forecasts in digital form on a global basis direct to States using the aeronautical fixed service internet-based services.

World Area Forecast System (WAFS): A worldwide system by which world area forecast centres provide aeronautical meteorological en route forecasts in uniform standardised formats.

2. GENERAL AND METEOROLOGICAL ABBREVIATIONS

This list covers abbreviations which may be found throughout the AIP and on associated charts, or which are used in NOTAM, AIP Supplements (SUP) and in meteorological messages and documentation.

Abbreviations marked "+" may be used as spoken words in radio telephony. Abbreviations marked "#" may be spoken using the individual letters rather than the phonetic alphabet.

Abbreviations marked "•" are not included in *ICAO Doc 8400* and must not be used in international NOTAM.

#	2D	Two-dimensional
#	3D	Three-dimensional
	A/A	Air to Air
•	AA	Audio Acknowledgment
•	AAIS	Automatic Aerodrome Information Service
	AAL	Above Aerodrome Level
	AAR	Air to Air Refuelling
	ABM	Abeam
	ABN	Aerodrome Beacon
	ABV	Above...
	AC	Altocumulus
•	AC	Advisory Circular
+	ACARS	Aircraft Communication Addressing and Reporting System (pronounced "AY-CARS")
+	ACAS	Airborne Collision Avoidance System
#	ACC	Area Control Centre
•	ACD	Airways Clearance Delivery
#	A-CDM	Airport Collaborative Decision Making
	ACFT	Aircraft
	ACK	Acknowledge
	ACN	Aircraft Classification Number
	ACPT	Accept, Accepted
	ACT	Active, Activated, Activity
	AD	Aerodrome
	ADDN	Addition, Additional
#	ADF	Automatic Direction Finding Equipment
+	ADIZ	Air Defence Identification Zone
	ADJ	Adjacent
#	ADS-B	Automatic Dependent Surveillance-Broadcast
#	ADS-C	Automatic Dependent Surveillance-Contract
	ADZ	Advise
•	AEP	Aerodrome Emergency Plan

	AFIL	Flight notification: - filed in the air, or - indicating the position at which ATS services will first be required
	AFM	Yes, Affirm, Affirmative, That is correct
•+	AFRU	Aerodrome Frequency Response Unit
	AFT	After....
•	AFZ	Australian Fishing Zone(s)
	A/G	Air-to-Ground
	AGA	Aerodromes, Air Routes and Ground Aids
#	AGL	Above Ground Level
	AGN	Again
•	AH	After Hours
#	AIC	Aeronautical Information Circular
#	AIP	Aeronautical Information Publication
+	AIRAC	Aeronautical Information Regulation and Control
+	AIREP	Air-Report
+	AIRMET	Information concerning weather significant to aircraft operations at or below 10,000FT not contained in a valid GAF
#	AIS	Aeronautical Information Service
•#	ALA	Aircraft Landing Area
+	ALERFA	Alert Phase
	ALS	Approach Lighting System
	ALT	Altitude
	ALTN	Alternate, Alternating (light alternates in colour)
	ALTN	Alternate (aerodrome)
•	ALTRV	Altitude Reservation
	AMD	Amend, Amended
•	AMDAR	Aircraft Meteorological Data Relay
	AMDT	Amendment (AIP Amendment)
•	AMSA	Australian Maritime Safety Authority
#	AMSL	Above Mean Sea Level
•	AOC	Air Operator's Certificate
	AP	Airport
+	APAPI	Abbreviated Precision Approach Path Indicator (pronounced "AY-PAPI")
	APCH	Approach
	APN	Apron
	APP	Approach Control, Approach Control Office, Approach Control Service
	APR	April
	APRX	Approximate, Approximately

#	APU	Auxiliary Power Unit
	APV	Approach Procedure with Vertical guidance
•	AQZ	Area QNH Zone
•	ARN	Aviation Reference Number
	ARNG	Arrange
	ARP	Aerodrome Reference Point
	ARR	Arrive, Arrival
	AS	Altostratus
#	ASAP	As Soon as Possible
	ASDA	Accelerate-Stop Distance Available
	ASE	Altimetry System Error
•	A-SMGCS	Advanced Surface Movement Guidance and Control System
	ASPH	Asphalt
#	ATA	Actual Time of Arrival
#	ATC	Air Traffic Control (in general)
#	ATD	Actual Time of Departure
	ATFM	Air Traffic Flow Management
•	ATFMX	Exemption from ATFM measures by ATC
+	ATIS	Automatic Terminal Information Service
	ATM	Air Traffic Management
	ATP	At... (time or place)
#	ATS	Air Traffic Services
•	ATSAS	Automatic Thunderstorm Alert Service
	ATTN	Attention
+	AT-VASIS	Abbreviated “T” Visual Approach Slope Indicator System (pronounced “AY-TEE-VASIS”)
	AUG	August
	AUTH	Authorised, Authorisation
	AUTO	Automatic
	AUW	All Up Weight
	AUX	Auxiliary
	AVBL	Available
+•	AVFAX	Meteorological and NOTAM Facsimile Service
	AVG	Average
+	AVGAS	Aviation Gasoline
•	AVM	Abrupt Vertical Manoeuvres (by the MIL)
•+	AWIS	Aerodrome Weather Information Service
•	AWR	Aerodrome Weather Report
•	AWS	Automatic Weather Station
	AWY	Airway
	AZM	Azimuth
	B	Blue
+	BARO-VNAV	(to be pronounced “BAA-RO-VEENAV”) Barometric Vertical Navigation

D

+	BASE	Cloud Base
	BCFG	Fog Patches
	BCN	Beacon (aeronautical ground light)
	BCST	Broadcast
•	BCTA	Base of CTA (used only on charts)
	BDRY	Boundary
	BECMG	Becoming
	BFR	Before
	BKN	Broken (cloud descriptor)
	BL...	Blowing (followed by DU=dust, SA=sand or SN=snow)
	BLDG	Building
	BLW	Below
•	BOF	Briefing Office
	BOMB	Bombing
	BR	Mist
	BRG	Bearing
	BRKG	Braking
	BS	Broadcasting Station (Commercial)
	BTN	Between
	C	Degrees Celsius (Centigrade)
	C	Centre (Runway)
•	CA/GRS	Certified Air/Ground Radio Service
•	CAO	Civil Aviation Order
•	CAR	Civil Aviation Regulation
•	CASA	Civil Aviation Safety Authority
+	CAT	Category
	CAT	Clear Air Turbulence
+	CAVOK	Visibility, cloud and present weather better than prescribed values or conditions (pronounced "KAV-OH-KAY")
#	CB	Cumulonimbus
	CC	Cirrocumulus
•	CCTS	Circuits
•	CDFA	Continuous Descent Final Approach technique
	CDR	Conditional Route
•	CEN	En Route and Area ATC Unit
	CFM	Confirm, I confirm
	CH	Channel
	CHEM	Chemical
	CI	Cirrus
	CIV	Civil
	CK	Check
	CL	Centre Line
	CLA	Clear type of ice formation
	CLBR	Calibration
	CLD	Cloud
	CLG	Calling
•	CLIAS	Climbing Indicated Airspeed
	CLR	Clear, Cleared to..., Clearance
	CLSD	Closed, Close, Closing
	CM	Centimetre
	CMB	Climb to or Climbing to

	CMPL	Completion, Completed, or Complete
•	CMSD	Commissioned
	CNL	Cancel, Cancelled
	CNS	Communications, Navigation and Surveillance
	COBT	Calculated Off Blocks Time
	COM	Communications
	CONC	Concrete
	COND	Condition
	CONS	Continuous
	CONST	Construction, Constructed
	CONT	Continue(s), Continued
	COOR	Coordinate, Coordinated
	COORD	Coordinates
	COR	Correct, Corrected, Correction
•	COS	Conical Surface
	COT	At the Coast, Coastal
	COV	Cover, Covered, Covering
#	CPDLC	Controller-Pilot Data Link Communication
	CRU	Control and Reporting Unit (RAAF)
	CRZ	Cruise
	CS	Cirrostratus
	CS	Call sign
#	CTA	Control Area
•+	CTAF	Common Traffic Advisory Frequency
	CTC	Contact
	CTL	Control
	CTN	Caution
	CTOT	Calculated Take-off Time
	CTR	Control Zone
	CU	Cumulus
	CUF	Cumuliform
	CUST	Customs
	CWY	Clearway
#	**D...**	Danger Area (followed by identification)
	D	Downward (tendency in RVR during previous 10 minutes)
	DA	Decision Altitude
•	DAH	Designated Airspace Handbook
•	DAP	Departure and Approach Procedures
	DCKG	Docking
•	DCMSD	Decommissioned
#	DCPC	Direct Controller-Pilot Communications
	DCT	Direct (in relation to flight plan clearances and type of approach)
	DEC	December
	DEG	Degrees
	DEP	Depart, Departure, Departed, Departing, Departure Message
	DER	Departure End of Runway
	DEST	Destination

+	DETRESFA	Distress Phase
	DEV	Deviation, Deviating
#	DF	Direction Finder/ Finding
	DFDR	Digital Flight Data Recorder
	DH	Decision Height
	DIF	Diffuse
•	DISP	Displaced
	DIST	Distance
	DIV	Diversion, Divert, Diverting
	DL	Distant lightning
•	DLE	Delay En route
	DLY	Daily
#	DME	Distance Measuring Equipment
	DNG	Danger, Dangerous
•	DOC	Documents
+	DOF	Date of Flight
	DOM	Domestic
	DP	Dew Point Temperature
	DPT	Depth
#	DR	Dead Reckoning
	DR...	Low drifting (followed by DU=dust,SA=sand or SN=snow)
	DRG	During
	DS	Duststorm
	DTG	Date-Time Group
	DTHR	Displaced Runway Threshold
	DTRT	Deteriorate, Deteriorating
	DU	Dust
	DUC	Dense Upper Cloud
	DUR	Duration
	DVOR	Doppler VOR
	DZ	Drizzle
	E	East, East Longitude
	EAT	Expected Approach Time
	EB	Eastbound
#	EET	Estimated Elapsed Time
	ELEV	Elevation
#	ELT	Emergency Locator Transmitter
	EM	Emission
	EMBD	Embedded in a Layer (to indicate cumulonimbus embedded in layers of other clouds)
	EMERG	Emergency
•	ENDCE	Endurance
	ENE	East North-East
	ENG	Engine
	ENR	En Route
	ENRC	En Route Chart *(followed by name/ title)*
	EOBT	Estimated off Block Time
•+	EPIRB	Electronic Position Indicating Radio Beacon (marine term)
	EQPT	Equipment
•#	ERC	En Route Chart
•+ #	ERSA	En Route Supplement Australia
	ESE	East South-East

	EST	Estimate *or* estimated *or* estimate (message type designator)
#	ETA	Estimated Time of Arrival, Estimating Arrival
#	ETD	Estimated Time of Departure or Estimating Departure
	ETO	Estimated Time Over significant point
	EV	Every
	EVS	Enhanced Vision System
	EXC	Except
	EXER	Exercises, Exercising, to exercise
	EXP	Expect, Expected, Expecting
	EXTD	Extend, Extending, Extended
	F	Fixed (chart symbol)
	FAC	Facility, Facilities
	FAF	Final Approach Fix
•	FANS 1/A	The term used to describe the initial future air navigations system
	FAP	Final Approach Point
	FAS	Final Approach Segment
	FATO	Final Approach and Take-off Area
+	FAX	Facsimile Transmission
	FBL	Light (used to indicate the intensity of WX phenomena, interference or static reports, e.g. FBL RA = light rain)
	FC	Funnel Cloud (tornado or water spout)
	FCST	Forecast
•	FDE	Fault Detection and Exclusion
	FDPS	Flight Data Processing System
	FEB	February
	FEW	Few (cloud descriptor)
•	FFR	Flood or Fire Relief; Fire Fighting
	FG	Fog
•#	FIA	Flight Information Area
#	FIR	Flight Information Region
#	FIS	Flight Information Service
	FL	Flight Level
	FLD	Field
	FLG	Flashing
	FLR	Flares
	FLT	Flight
	FLTCK	Flight Check For Calibration of Navaids
	FLUC	Fluctuating, Fluctuation, Fluctuated
	FLW	Follow(s), Following
	FLY	Fly, Flying

	FM	From
	FM...	From (followed by time weather change is forecast to begin)
•	FMC WPR	The term used to describe flight management computer waypoint position reporting
#	FMS	Flight Management System
	FMU	Flow Management Unit
•	FN	Fly Neighbourly Area
	FNA	Final Approach
•	FPA	Flight Procedure Authorisation
	FPL	Filed Flight Plan
	FPM	Feet per Minute
	FR	Fuel Remaining
	FREQ	Frequency
	FRI	Friday
	FRNG	Firing
	FRQ	Frequent
•#	FS	Flight Service (in general)
•	FSP	Fish Spotting
	FST	First
	FT	Feet
	FU	Smoke
•	FXD	Fixed
	FZ	Freezing
	FZDZ	Freezing Drizzle
	FZFG	Freezing Fog
•	FZLVL	Freezing Level (in AIRMET products)
	FZRA	Freezing Rain
	G	Green
	G	Variation from mean wind speed (gusts) (MET - used in METAR/SPECI and TAF code forms)
#	GA	General Aviation
+	GAF	Graphical Area Forecast
#	GBAS	Ground Based Augmentation System
#	GDP	Ground Delay Program
	GEN	General
	GEO	Geographic, true
	GES	Ground Earth Station
•	GFY	Glider Flying
	GLD	Glider
+	GLONASS	Global Orbiting Navigation Satellite System (pronounced “GLO-NAS”)
#	GLS	GBAS Landing System
	GND	Ground
	GNDCK	Ground Check
#	GNSS	Global Navigation Satellite System
	GP	Glide Path
#	GPS	Global Positioning System
#	GPU	Ground Power Unit
#	GPWS	Ground Proximity Warning System
•	GPWT	Grid Point Wind and Temperature
	GR	Hail

•	GRAD	Minimum Required Climb Gradient
+	GRASS	Grass Landing Area
•	GRF	Global Reporting Format
+	GRIB	Processed meteorological data in the form of grid point values expressed in binary form (meteorological code)
	GRVL	Gravel
	GS	Groundspeed
	GS	Small Hail and/or Snow Pellets
	H	High pressure area or the centre of high pressure (MET)
#	H24	Continuous day and night service
•	HAZMAT	Hazardous Material
	HBN	Hazard Beacon
	HDG	Heading
•	HDS	Hours of Daylight Saving
•	HEAD	Head of State
	HEL	Helicopter
#	HF	High Frequency (3,000 to 30,000 kHz)
	HGT	Height, Height Above
•	HH	Time of commencement of a meteorological report validity period
•+	HIAL	High Intensity Approach Lighting
•	HIOL	High Intensity Obstacle Lights
•	HIRL	High Intensity Runway Lighting
#	HJ	Sunrise to Sunset
	HLDG	Holding
	HLP	Heliport
	HLS	Helicopter Landing Site
#	HN	Sunset to Sunrise
	HO	Service available to meet operational requirements
	HOSP	Hospital Aircraft
	HPA	Hectopascal
	HR	Hours
	HRP	Heliport Reference Point
•	HS	Homestead
	HS	Service available during hours of scheduled operations
	HSL	Hold Short Lights
	HUD	Head-up display
	HUM	Humanitarian
	HVY	Heavy
	HVY	Heavy (used to indicate the intensity of WX phenomena, e.g. HVY RA = heavy rain)
	HX	No specific working hours
	HZ	Haze
	HZ	Hertz (cycle per second)
•	HZS	Horizontal Surface

	IAC	Instrument Approach Chart (*followed by name/title*)
	IAF	Initial Approach Fix
•#	IAL	Instrument Approach and Landing
	IAP	Instrument Approach Procedure
#	IAS	Indicated Airspeed
+	ICAO	International Civil Aviation Organization
	ICE	Icing
	ID	Identifier, identify
+	IDENT	Identification
•+	IDEP	Instrument Departure (FPA)
	IF	Intermediate Fix or Intermediate Approach Fix
#	IFR	Instrument Flight Rules
#	ILS	Instrument Landing System
	IM	Inner Marker
#	IMC	Instrument Meteorological Conditions
	IMG	Immigration
	IMPR	Improve, Improving, Improvement
	INBD	Inbound
+	INCERFA	Uncertainty Phase
+	INFO	Information
+	INOP	Inoperative
#	INS	Inertial Navigation System
	INSTL	Install, Installed, Installation
	INSTR	Instrument
	INT	Intersection
•+	INTER	Intermittent (i.e. lasting less than 30 minutes) Fluctuations from forecast prevailing conditions
	INTL	International
	INTRP	Interrupt, Interruption, Interrupted
	INTSF	Intensify, Intensifying
	INTST	Intensity
+	ISA	International Standard Atmosphere
	ISOL	Isolated
•	IVA	Independent Visual Approach
•	IWI	Illuminated Wind Indicator
	JAN	January
•	JF	Saturday, Sunday and PH
•	JO	Monday to Friday except PH
	JTST	Jet Stream
	JUL	July
	JUN	June
	KG	Kilograms
	kHz	Kilohertz
	KIAS	Knots Indicated Airspeed
	KM	Kilometres
	KMH	Kilometres per Hour
	kPa	Kilopascals
	KT	Knots

	KW	Kilowatts
	L	Left (runway identification)
	L	Litre
	L	Low pressure area or the centre of low pressure (MET)
•	LAHSO	Land and Hold Short Operations
+	LAT	Latitude
	LCA	Locally, Location, Located, Local
	LDA	Landing Distance Available
	LDG	Landing
	LEN	Length
	LGT	Light, Lighting
	LGTD	Lighted
	LIH	Light Intensity High
	LIL	Light Intensity Low
	LIM	Light Intensity Medium
•	LIOL	Low Intensity Obstacle Lights
•	LIRL	Low Intensity Runway Lights
•	LJR	Low Jet Route
•	LL	Lower Limits
•	LLN	Low Level Navigation (by the MIL)
•	LLO	Low Level Operations (by the MIL)
	LMT	Local mean time
+	LNAV	Lateral Navigation
	LOC	Localiser
•	LOE	Lane of Entry
+	LONG	Longitude
•	LP	Localiser Performance
	LPV	Localiser Performance with Vertical Guidance
•	LSALT	Lowest Safe Altitude
	LTD	Limited
•	LUL	Lowest Usable Level
	LV	Light and Variable (relating to wind)
	LVL	Level
•	LVO	Low Visibility Operation(s)
#	LVP	Low Visibility Procedure(s)
	LYR	Layer, Layered
	M	Metres (preceded by figures)
	M	Mach number (followed by figures)
	M...	Military Operating Area (followed by identification)
	MAG	Magnetic
	MAINT	Maintenance
•	MAN	Manual
	MAP	Aeronautical Maps and Charts
	MAPT	Missed Approach Point
	MAR	At Sea
	MAR	March
•+	MARSA	Military Assumes Responsibility for Separation of Military Aircraft
+	MAX	Maximum

•	MBA	Mandatory Broadcast Area
	MBST	Microburst
	MDA	Minimum Descent Altitude
	MDH	Minimum Descent Height
#	MEA	Minimum En Route Altitude
+	MEDEVAC	Medical Evacuation Flight
+	MET	Meteorological, Meteorology
+	METAR	Aviation routine weather report (in aeronautical meteorological code)
	MET REPORT	Aviation routine weather report
	MF	Medium Frequency (300 to 3,000 kHz)
	MHz	Megahertz
•	MI	Shallow (MET)
	MIFG	Shallow Fog
	MIL	Military
	MIN	Minutes
•	MIOL	Medium Intensity Obstacle Lights
•	MIRL	Medium Intensity Runway Lights
•	MISC	Miscellaneous
•	MLJ	Military Low Jet
#	MLS	Microwave Landing System
	MM	Middle Marker
	MNM	Minimum
	MNT	Monitor, Monitoring, Monitored
	MNTN	Maintain, Maintained, Maintaining
•	MO	Meteorological Office
	MOA	Military Operating Area
	MOD	Moderate (used to indicate the intensity of WX phenomena, interference or static reports, e.g. MOD RA = moderate rain).
	MON	Monday
+	MOPS	Minimum Operational Performance Standards
	MOV	Move, Moved, Moving, Movement
•	MOWP	Method of Working Plan
	MS	Minus
#	MSA	Minimum Sector Altitude
	MSG	Message
	MSL	Mean Sea Level
	MSSR	Monopulse Secondary Surveillance Radar
	MT	Mountain
•	MTOW	Maximum Take-off Weight
•	MTP	Maximum Tyre Pressure
	MTW	Mountain waves
•	MVA	Minimum Vector Altitude

	MWO	Meteorological Watch Office
	MX	Mixed type of ice formation (white and clear)
	N	No distant tendency (in RVR during previous 10 minutes)
	N	North, North Latitude
•	NAA	National Aviation Authority
•	NAIPS	National Aeronautical Information Processing System
•	NAP	Noise Abatement Procedures
•	NAT	NAVAID Training
	NAV	Navigation
	NAVAID	Navigation Aid
	NB	Northbound
	NC	No Change
	NCD	No Cloud Detected (by ceilometer) [*used in automated* METAR/SPECI]
#	NDB	Non-Directional Radio Beacon
	NE	North-East
	NEG	Negative, No, permission not granted, or, that is not correct
	NGT	Night
+	NIL	None
	NM	Nautical Miles
	NML	Normal
	NN	No name, unnamed
	NNE	North North-East
	NNW	North North-West
	NOF	International NOTAM Office
•	NOMC	National Operations Management Centre
	NONSTD	Non-Standard
+	NOSIG	No Significant Change
+	NOTAM	Notice to Airmen (A notice containing information concerning the establishment, condition or change in facility, service, procedure or hazard which is essential to personnel concerned with flight operations)
	NOTAMC	Cancelling NOTAM
	NOTAMN	New NOTAM
	NOTAMR	Replacing NOTAM
	NOV	November
#	NOZ	Normal Operating Zone
	NPA	Non-Precision Approach
	NR	Number
	NS	Nimbostratus
	NSC	Nil Significant Cloud
	NSW	Nil Significant Weather
•	NTA	No TAF Amendment
	NTL	National
#	NTZ	No Transgression Zone
•#	NVG	Night Vision Goggles

•+	NVIS	Night Vision Imaging System (pronounced "EN-VIZ")
	NW	North-West
	NXT	Next
	OBS	Observe, Observed, Observation
	OBSC	Obscure, Obscured, Obscuring
	OBST	Obstacle
•	OBSTR	Obstruction
#	OCA	Oceanic Control Area
	OCA	Obstacle Clearance Altitude
	OCC	Occulting (light)
	OCH	Obstacle Clearance Height
	OCNL	Occasional, Occasionally
	OCT	October
•#	OCTA	Outside Control Area
•#	OCTR	Outside Control Zone
	OFZ	Obstacle Free Zone
	OHD	Overhead
	OK	We agreed, *or*, It is correct
•	OLS	Obstacle Limitation Surface
	OM	Outer Marker
	OPA	Opaque. White type of ice formation
+	OPMET	Operational Meteorological (information)
	OPN	Open, Opening, Opened
	OPR	Operator, Operate, Operative, Operating, Operational
	OPS	Operations
	O/R	On Request
•	OT	Other Times
	OVC	Overcast
•	OW	Over Water
#	**P....**	Prohibited Area (followed by identification)
	PA	Precision Approach
•+	PAL	Pilot Activated Lighting
	PANS	Procedures for Air Navigation Services
+	PAPI	Precision Approach Path Indicator
	PARL	Parallel
	PAX	Passengers
•	PBG	Pilot Boarding Ground
	PBN	Performance-based navigation
	PCD	Proceed, Proceeding
	PCL	Pilot Controlled Lighting
	PCN	Pavement Classification Number
#	PDC	Pre-Departure Clearance
	PER	Performance
	PERM	Permanent
•	PFR	Preferred Route
•	PH	Public Holiday
	PIB	Pre-flight Information Bulletin

•	PIFR	Private IFR (rating)
•	PILS	Practice ILS
•	PinS	Point-in-Space
	PJE	Parachute Jumping Exercise
	PL	Ice Pellets
•	PLN	Flight Plan
	PN	Prior Notice Required
#	PNR	Point of No Return
	PO	Dust Devils
#	POB	Number of Persons on Board
	POSS	Possible
	PPR	Prior Permission Required
	PPSN	Present Position
•	PRD	Prohibited, Restricted and Danger Areas *See also SUA*
	PRFG	Aerodrome Partially Covered by Fog (MET code)
	PRI	Primary
	PRKG	Parking
•	PRM	Precision Runway Monitoring
+	PROB	Probability
	PROC	Procedure
	PROV	Provisional
	PRP	Point-in-space Reference Point
	PS	Plus
	PSG	Passing
	PSN	Position
	PSP	Pierced Steel Plank
#	PSR	Primary Surveillance Radar
•	PTBL	Portable
	PTN	Procedure Turn
•	PTT	Press to Talk
•	PVT	Private
	PWR	Power
#	**QNH**	Altimeter subscale setting to obtain elevation or altitude
	QUAD	Quadrant
	R	Red
	R	Right (runway identification)
	R	Runway (followed by figures in METAR/ SPECI)
	R...	Radial from VOR (followed by three figures)
	R....	Restricted Area (followed by number)
•	RA	Radio Altimeter
	RA	Rain
	RA	Resolution Advisory
•	RA	Restricted Area
•	RAD	Radius
+	RAIM	Receiver Autonomous Integrity Monitoring
•+	RAPIC	Radar Picture (MET)
+	RASC	Regional AIS System Centre
•	RCAM	Runway Condition Assessment Matrix
#	RCC	Rescue Coordination Centre
•	RCGL	Runway Circling Guidance Lights

	RCH	Reach, Reaching
	RCL	Runway Centre Line
	RCLL	Runway Centre Line Lights
•	RCR	Runway Condition Report
	RDH	Reference Datum Height
	RDL	Radial
	RDO	Radio
	RE...	Recent (used to qualify weather phenomena, e.g. RERA = recent rain)
	REC	Receive, Receiver, Received
	REDL	Runway Edge Lights
	REF	Reference to..., Refer to...
	REG	Registration
	RENL	Runway End Lights
	REP	Report, Reported, Reporting, Reporting Point
	REQ	Request, Requested
	RERTE	Re Route
•	RES	Reserve Fuel
	RESA	Runway End Safety Area
•	RESTR	Restrictions
•	REV	Review
#	RF	Constant Radius Arc to Fix
•	RFC	Regional Forecasting Centre (MET)
	RFFS	Rescue and Fire Fighting Services
	RHC	Right Hand Circuit
	RIF	Reclearance in Flight
	RL	Report Leaving
	RLA	Relay to
	RLLS	Runway Lead-in Lighting System
	RMK	Remark
+	RNAV	Navigation specification designator
#	RNP	Required Navigation Performance (Navigation Specification prefix)
+	ROBEX	Regional OPMET Bulletin Exchanges
	ROC	Rate of Climb
	ROD	Rate of Descent
#	RP	Remote Pilot
#	RPA	Remotely Piloted Aircraft
•+	RPAS	Remotely Piloted Aircraft System (pronounced “AR-PAZ”)
	RPT	Repeat, I Repeat
	RQ	Require(d)
	RQMNTS	Requirements
	RSCD	Runway Surface Condition
	RSP	Responder Beacon
	RTE	Route
	RTF	Radio Telephone
	RTHL	Runway Threshold Light(s)
•	RTIL	Runway Threshold Identification Lights

	RTN	Return, Returned, Returning
	RTS	Return to Service
	RTZL	Runway Touchdown Zone Light(s)
#	RVR	Runway Visual Range
	RVSM	Reduced Vertical Separation Minimum
•	RWS	Runway Strip
	RWY	Runway
•	RWYCC	Runway Condition Code
	S	South, South Latitude
	SA	Sand
#•	SA	Special Authorisation
	SALS	Simple Approach Lighting System
+	SAR	Search and Rescue
	SARPS	Standards and Recommended Practices (ICAO)
•+	SARTIME	Time search action required
	SAT	Saturday
+	SATCOM	Satellite Communication (used only when referring generally to both voice and data satellite communication or only data satellite communication)
+	SATVOICE	Satellite Voice Communication
	SB	Southbound
+	SBAS	Satellite-Based Augmentation System
	SC	Stratocumulus
	SCT	Scattered
	SDBY	Standby
	SE	South East
	SEA	Sea (used in connection with sea-surface temperature and state of the sea)
	SEC	Seconds
	SECT	Sector
+	SELCAL	Selective Calling System
	SEP	September
	SER	Service, Serviced, Servicing
	SEV	Severe (used e.g. to qualify icing and turbulence report)
	SFC	Surface
•	SFL	Sequenced Flashing Lights
	SG	Snow Grains
	SH...	Showers (followed by RA=rain, SN=snow, PL=ice pellets, GR=hail, GS=small hail and or snow pellets or combinations thereof, e.g. SHRASN = showers of rain and snow)
+	SID	Standard Instrument Departure

+	SIGMET	Information concerning en route weather and other phenomena in the atmosphere that may affect the safety of aircraft operations
	SIGWX	Significant Weather
	SIMUL	Simultaneous, or Simultaneously
•+	SIS	Surveillance Information Service
+	SKED	Schedule, Scheduled
	SLP	Speed Limiting Point
	SLW	Slow
#	SMC	Surface Movement Control
•#	SMCV	Surface Movement Control Vehicles
	SMR	Surface Movement Radar
	SN	Snow
+	SNOWTAM R	A special series NOTAM given in a standard format providing a surface condition report notifying the presence or cessation of hazardous conditions due to snow, ice, slush, frost, standing water or water associated with snow, slush, ice or frost on the movement area
•	SOT	Start of TORA (take-off)
•	SP	Single Pilot
•	SPA	Sport Aviation
+	SPECI	Aviation Special Weather (in aeronautical meteorological code)
•	SPFIB	Specific Preflight Information Bulletin
+	SPOT	Spotwind
	SQ	Squall
	SQL	Squall Line
	SR	Sunrise
#	SRR	Search and Rescue Region
	SRY	Secondary
	SS	Sandstorm
	SS	Sunset
	SSB	Single Sideband
	SSE	South South-East
	SSR	Secondary Surveillance Radar
	SST	Supersonic Transport
	SSW	South South-West
	ST	Stratus
+	STAR	Standard Instrument Arrival
	STD	Standard
	STF	Stratiform
	STN	Station
	STNR	Stationary
•	STODA	Supplementary Take-off Distance
	STOL	Short Take-off and Landing
	STS	Status
	STWL	Stopway Light(s)
•	SUA	Special Use Airspace *See also PRD*

	SUBJ	Subject to
	SUN	Sunday
	SUP	Supplement (AIP Supplement)
	SUPPS	Regional Supplementary Procedures
	SVCBL	Serviceable
•	SVY	Survey Operations
	SW	South-West
•	SWS	Soft Wet Surface
	SWY	Stopway
	T	Bearing (true)
	T	Temperature
	TA	Traffic Advisory
	TA	Transition Altitude
•+	TAC	Terminal Area Chart
+	TACAN	UHF Tactical Air Navigation Aid
+	TAF	Aerodrome Forecast
#	TAS	True Airspeed
	TAX	Taxiing, Taxi
•	TBA	To be advised
	TC	Tropical Cyclone
+	TCAS	(tee-kas) Traffic Alert and Collision Avoidance System
	TCH	Threshold Crossing Height
•	TCTA	Trans-Continental Control Area
•	TCU	Terminal Control Unit
	TCU	Towering Cumulus
•	TDA	Temporary Danger Area
	TDO	Tornado
	TDZ	Touchdown Zone
	TECR	Technical Reason
	TEL	Telephone
+	TEMPO	Temporary, Temporarily
	TFC	Traffic
•	TFR	Terrain Following Radar (by the MIL)
	TGL	Touch & Go Landing
	THR	Threshold
	THRU	Through
	THU	Thursday
	TIBA	Traffic Information Broadcasts by Aircraft
+	TIL	Until
	TKOF	Take-off
	TL...	Till (followed by time by which weather change is forecast to end)
•	TLW	Time Limited WIP (work in progress)
#	TMA	Terminal Control Area
•	TM...	Temporary Military Operating Area (followed by identification)
	TN	Indicator for Minimum Temperature (MET - used in TAF code form)
•	TNS	Transitional Surface
#	TOBT	Target Off-Block Time
	TODA	Take-off Distance Available
	TOP	Cloud Top
+	TORA	Take-off Run Available

	TOX	Toxic
	TP	Turning Point
	TR	Track
•#	TRA	Temporary Restricted Area
	TRANS	Transmits, Transmitter
	TROP	Tropopause
	TS...	Thunderstorm (followed by RA=rain, SN=snow, PE=ice pellets, GR=hail, GS=small hail and/or snow pellets or combinations thereof, e.g. TSRASN = thunderstorm with rain and snow)
+	TSAT	Target Start-up Approval Time
•	TSO	Technical Standard Order
+	TSUNAMI	Tsunami (used in aerodrome warnings)
	TUE	Tuesday
	TURB	Turbulence
+	T-VASIS	"T" Visual Approach Slope Indicator System (pronounced "TEE-VASIS")
•	TW	Tailwind
	TWR	Aerodrome Control Tower *or* Aerodrome Control
	TWY	Taxiway
	TX	Indicator for Maximum Temperature (MET - used in TAF code form)
	TXL	Taxilane
	TYP	Type of Aircraft
	TYPH	Typhoon
	U	Upward (tendency in RVR during previous 10 minutes)
	UA	Unmanned aircraft
	UAS	Unmanned aircraft system
	UFN	Until Further Notice
#	UHF	Ultra High Frequency (300 to 3,000 MHz)
•	UL	Upper Limits
	UNA	Unable
	UNL	Unlimited
	UNREL	Unreliable
	UP	Unknown Precipitation
	U/S	Unserviceable
#	UTC	Coordinated Universal Time
	V	Variation from mean wind speed (MET - used in METAR/ SPECI code forms)
	VA	Volcanic Ash
	VAAC	Volcanic Ash Advisory Centre
	VAR	Magnetic Variation
+	VASIS	Visual Approach Slope Indicator System

	VC	Vicinity of the aerodrome (followed by FG=fog, TS=thunderstorm, FC=funnel cloud, PO=dust/sand whirls, BLDU=blowing dust, BLSA=blowing sand or BLSN=blowing snow; e.g. VCFG=vicinity fog
	VCY	Vicinity
#	VFR	Visual Flight Rules
#	VHF	Very High Frequency (30 to 300 MHz)
•	VIA	By way of...
#	VIP	Very Important Person
	VIS	Visibility
#	VMC	Visual Meteorological Conditions
#	VNAV	Vertical Navigation
•	VNC	Visual Navigation Chart
+	VOLMET	Meteorological Information for Aircraft in Flight
#	VOR	VHF Omnidirectional Radio Range
	VRB	Variable
	VSA	by Visual reference to the ground
•	VTC	Visual Terminal Chart
	VTOL	Vertical Take-off and Landing
	W	West,West Longitude
	W	White
	WAC	World Aeronautical Chart - ICAO 1:1,000,000 (*followed by name/title*)
•	WAE	Workers and Equipment
	WAFC	World Area Forecast Centre
•	WAFS	World Area Forecast System
•	WATIR	Weather and Terminal Information Reciter
	WB	Westbound
	WDI	Wind Direction Indicator
	WDSPR	Widespread
	WED	Wednesday
	WEF	With Effect From, Effective From
	WGS-84	World Geodetic System - 1984
	WI	Within
	WID	Width
	WIE	With Immediate Effect, Effective Immediately
+	WILCO	Will Comply
	WIP	Work in Progress
	WKN	Weaken, Weakening
	WNW	West North-West
	WO	Without
	WPT	Waypoint
	WRNG	Warning
	WS	Wind Shear
	WSW	West South-West
	WT	Weight

	WTSPT	Water Spout
	WWW	World Wide Web
	WX	Weather
	WXR	Weather Radar
	X	Cross
•	XW	Crosswind
	Y	Yellow
	YCZ	Yellow Caution Zone
	YR	Your/s
	Z	Coordinated Universal Time (in meteorological messages)

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GEN 2.3 CHART SYMBOLS

1. Symbols used for Australian aeronautical charts are identified on the legend of individual charts.

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GEN 2.4 LOCATION INDICATORS

1. CODE ALLOCATION

1.1 Australia follows international conventions in the allocation of codes. For locations where a landing area exists (fixed-wing or helicopter), a four-letter location indicator is assigned with the first letter being a “Y”. This is referred to as the “Y” code. Locations other than those given a “Y” code are identified by two, three, four or five letter codes, but to avoid confusion with location indicators, visual waypoint codes do not begin with the letter “Y”. The following table summarises code allocation:

Type	Code	Example
Certified Aerodrome Aircraft Landing Area Helicopter Landing Site	4 letters (first letter “Y”)	Renmark (YREN)
Navigation Aid	2 or 3 letters (2 letter codes no longer issued)	Caiguna VOR (CAG)
Visual Waypoint (as depicted on chart)	4 letters (first letter other than “Y”) (3 letter codes no longer issued)	Cranbourne (CRAN)
IFR Waypoint	5 letters (3 letter codes no longer issued)	DADOP

2. LIST OF CODES

2.1 Codes are listed in *ERSA GEN.*

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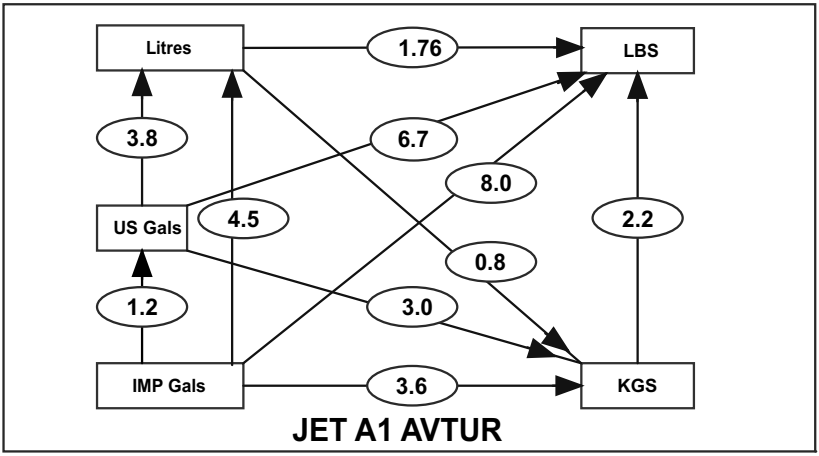
GEN 2.5 RADIO NAVIGATION AIDS

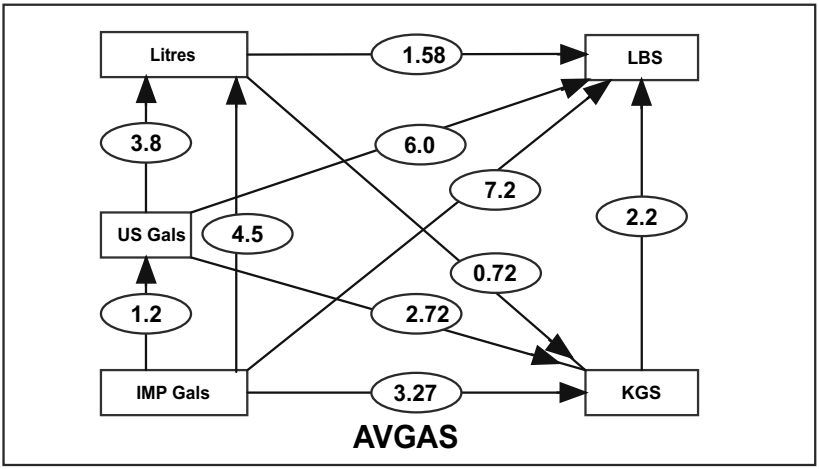
1. Australian radio navigation aids are identified in ERSA Facilities (FAC) section for each location under the heading NAVIGATION AIDS. Listings conform to the following sequence:
Aid Ident Frequency Lat/Long Remarks

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GEN 2.6 CONVERSION OF UNITS OF MEASUREMENT

1. FUEL WEIGHT TABLES





1. To convert:
   multiply by the factor in the "balloon" when moving in the direction of the arrow, or
   divide by that factor if converting in the opposite direction.

2. Fuel SG (0.8 AVTUR and 0.72 AVGAS) is based on ISA temperature at MSL. Therefore, fuel weights will be approximate for other than 15DEG Celsius.

2. GENERAL CONVERSIONS

TO CONVERT	INTO	MULTIPLY BY
Celsius	Fahrenheit	1.8 and add 32
Centimetres	Inches	0.394
Feet	Metres	0.3048
Fahrenheit	Celsius	Subtract 32 and multiply by 0.555
IMP Gallons	US Gallons	1.200
IMP Gallons	Litres	4.546
Inches	Centimetres	2.540
Kilograms	Pounds	2.2046
Kilometres	Nautical Miles	0.540
Kilometres	Statute Miles	0.621
Kilopascals	Pound/Square Inch	0.145
Litres	IMP Gallons	0.220
Litres	US Gallons	0.264
Metres	Feet	3.281
Metres	Yards	1.094
Pounds	Kilograms	0.4536
Pounds/Square Inch	Kilopascals	6.895
Nautical Miles	Kilometres	1.852
Nautical Miles	Metres	1852
Nautical Miles	Statute Miles	1.151
Statute Miles	Kilometres	1.609
Statute Miles	Nautical Miles	0.868
US Gallons	IMP Gallons	0.833
US Gallons	Litres	3.79
Yards	Metres	0.914

3. DECIMAL MINUTES OF AN ARC TO SECONDS OF AN ARC

MIN	SEC	MIN	SEC	MIN	SEC	MIN	SEC
0.01	0.6	0.26	15.6	0.51	30.6	0.76	45.6
0.02	1.2	0.27	16.2	0.52	31.2	0.77	46.2
0.03	1.8	0.28	16.8	0.53	31.8	0.78	46.8
0.04	2.4	0.29	17.4	0.54	32.4	0.79	47.4
0.05	3.0	0.30	18.0	0.55	33.0	0.80	48.0
0.06	3.6	0.31	18.6	0.56	33.6	0.81	48.6
0.07	4.2	0.32	19.2	0.57	34.2	0.82	49.2
0.08	4.8	0.33	19.8	0.58	34.8	0.83	49.8
0.09	5.4	0.34	20.4	0.59	35.4	0.84	50.4
0.10	6.0	0.35	21.0	0.60	36.0	0.85	51.0
0.11	6.6	0.36	21.6	0.61	36.6	0.86	51.6
0.12	7.2	0.37	22.2	0.62	37.2	0.87	52.2
0.13	7.8	0.38	22.8	0.63	37.8	0.88	52.8
0.14	8.4	0.39	23.4	0.64	38.4	0.89	53.4
0.15	9.0	0.40	24.0	0.65	39.0	0.90	54.0
0.16	9.6	0.41	24.6	0.66	39.6	0.91	54.6
0.17	10.2	0.42	25.2	0.67	40.2	0.92	55.2
0.18	10.8	0.43	25.8	0.68	40.8	0.93	55.8
0.19	11.4	0.44	26.4	0.69	41.4	0.94	56.4
0.20	12.0	0.45	27.0	0.70	42.0	0.95	57.0
0.21	12.6	0.46	27.6	0.71	42.6	0.96	57.6
0.22	13.2	0.47	28.2	0.72	43.2	0.97	58.2
0.23	13.8	0.48	28.8	0.73	43.8	0.98	58.8
0.24	14.4	0.49	29.4	0.74	44.4	0.99	59.4
0.25	15.00	0.50	30.0	0.75	45.0		

4. SECONDS OF AN ARC TO DECIMAL MINUTES OF AN ARC

SEC	MIN	SEC	MIN	SEC	MIN	SEC	MIN
1	0.02	16	0.27	31	0.52	46	0.77
2	0.03	17	0.28	32	0.53	47	0.78
3	0.05	18	0.30	33	0.55	48	0.80
4	0.07	19	0.32	34	0.57	49	0.82
5	0.08	20	0.33	35	0.58	50	0.83
6	0.10	21	0.35	36	0.60	51	0.85
7	0.12	22	0.37	37	0.62	52	0.87
8	0.13	23	0.38	38	0.63	53	0.88
9	0.15	24	0.40	39	0.65	54	0.90
10	0.17	25	0.42	40	0.67	55	0.92
11	0.18	26	0.43	41	0.68	56	0.93
12	0.20	27	0.45	42	0.70	57	0.95
13	0.22	28	0.47	43	0.72	58	0.97
14	0.23	29	0.48	44	0.73	59	0.98
15	0.25	30	0.50	45	0.75		

Note: Climb/Descent and Altitude Correction versus Temperature tables are published in AIP DAP.

5. WIND COMPONENT

		For crosswind component Angle Between Wind Direction and Runway Heading								
		10	20	30	40	50	60	70	80	90
Wind speed (KT)	5	1	2	2	3	4	4	4	5	5
	10	2	3	5	6	7	8	9	9	10
	15	3	5	7	9	11	13	14	14	15
	20	3	7	10	13	15	17	18	19	20
	25	4	8	12	16	19	22	23	24	25
	30	5	10	15	19	23	26	28	29	30
	35	6	12	17	22	26	30	32	34	35
	40	7	14	20	25	30	35	37	39	40
	45	8	15	22	29	34	39	42	44	45
	50	9	17	25	32	38	43	47	49	50
	55	10	19	27	35	42	48	52	54	55
	60	10	20	30	38	46	52	56	59	60
	65	11	22	32	42	50	56	61	64	65
	70	12	24	35	45	54	60	66	69	70
	75	13	26	37	48	57	64	70	73	75
	80	14	27	40	51	60	69	75	78	80
		80	70	60	50	40	30	20	10	0
		For Headwind Component Angle Between Wind Direction and Runway Heading								

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GEN 2.7 FIRST LIGHT AND LAST LIGHT COMPUTATIONS

1. For all intents and purposes, 'first light' should be construed as the beginning of morning civil twilight, and 'last light' as the end of evening civil twilight.

2. To compute first light and last light using the graphs contained in this section:

   a. enter the top or bottom of the scale at the appropriate date;

   b. move vertically up or down to the curve for the latitude of the place concerned (interpolating for intermediate latitudes if necessary);

   c. move horizontally to the left or right and read local mean time on the vertical scale at the side;

   d. to convert to UTC, subtract (in E longitudes) from the LMT obtained, the time increment corresponding to the longitude of the place concerned in the "Conversion of Arc to Time" table.

   e. to convert to EST, add 10 hours to UTC;

   f. to convert to CST, add $9^1/_2$ hours to UTC;

   g. to convert to WST, add 8 hours to UTC.

   Example: To determine last light at Echuca
   (360900S 1444600E) on 20th November.
   Using the graph, enter at 20th November at the top of the page and follow downwards to latitude 36° (by interpolation), then horizontally to the left and read off LMT = 1919. To convert to UTC, enter the "Conversion of Arc to Time" table, at longitude 144° (9 hours 36 minutes). Add the increment corresponding to 46' in the right hand column
   = 3' 04" + 0936 = 0939.
   Subtract this from the LMT found: 1919 – 0939 = 0940 UTC.
   To find EST add 10 hours to UTC = 1940 EST.

   Users of these graphs should note that the parameters used in compiling the Time of First Light/Last Light Graphs do not include the nature of the terrain surrounding a location, or the presence of other than a cloudless sky and unlimited visibility at that location.

Consequently, the presence of cloud cover, poor visibility or high terrain to the west of an aerodrome will result in last light occurring at a time earlier than that extracted from the appropriate graph. Allowance should be made for these factors when planning a flight having an ETA near the time of last light.

3. NAIPS automatically computes first light and last light. This information can be provided through pilot access, as part of a telephone briefing, or from FLIGHTWATCH.

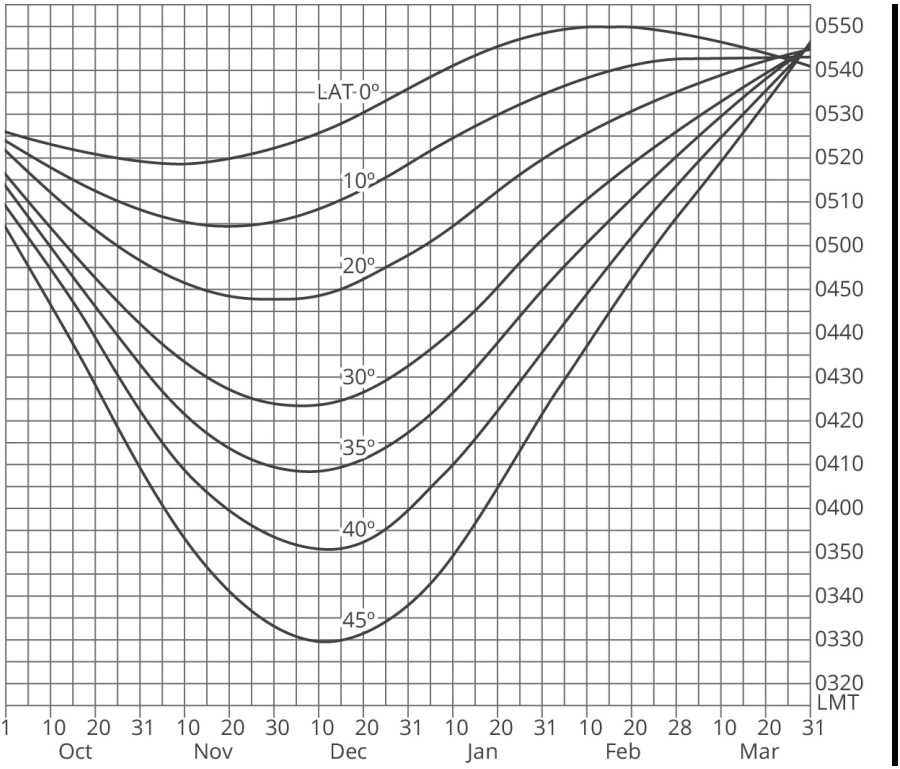
4. Sunrise, sunset and civil twilight times (first and last light) can also be obtained from www.ga.gov.au/geodesy/astro/sunrise.jsp

5. **LOCAL TIME**

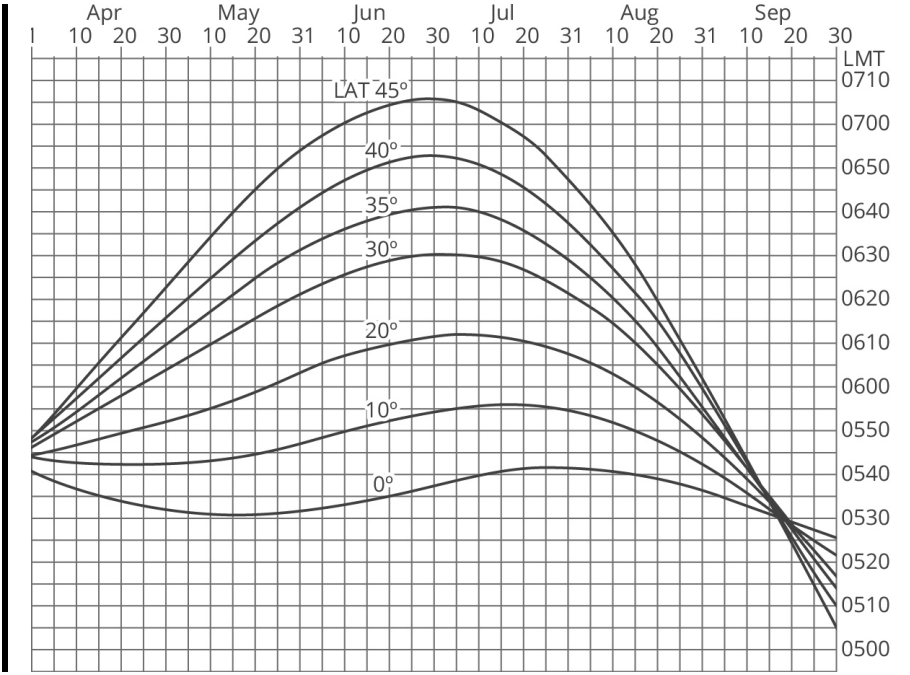
Local time in Australia falls into three separate zones:

Time Zone	Local Time	States
Eastern Standard Time (EST)	UTC + 10 hours	New South Wales* (except the Broken Hill area), Queensland, Victoria*, Tasmania* and the Australian Capital Territory*
Central Standard Time (CST)	UTC + 9 ½ hours	South Australia*, the Northern Territory and the Broken Hill area*
Western Standard Time (WST)	UTC + 8 hours	Western Australia
Note:* *Daylight Saving Time applies between October of one year and April of the following year, which adds an additional hour to the local time of that State or area. NOTAM or AIP Supplements will be issued detailing revised hours of operation during Daylight Saving Time where such hours are not already promulgated in AIP.*		

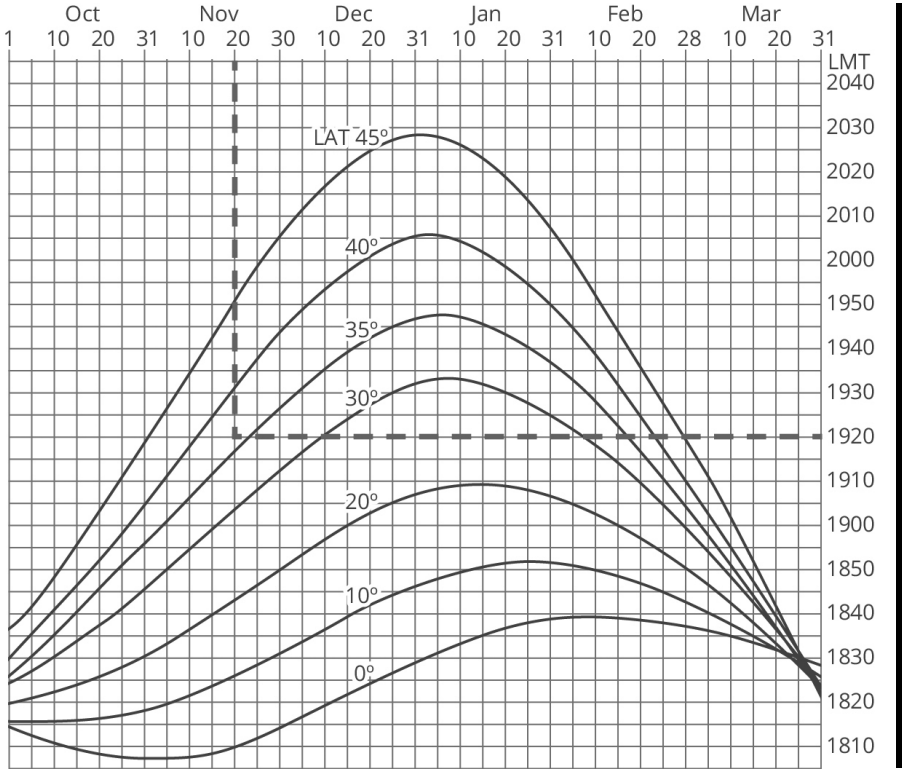
TIME OF FIRST LIGHT



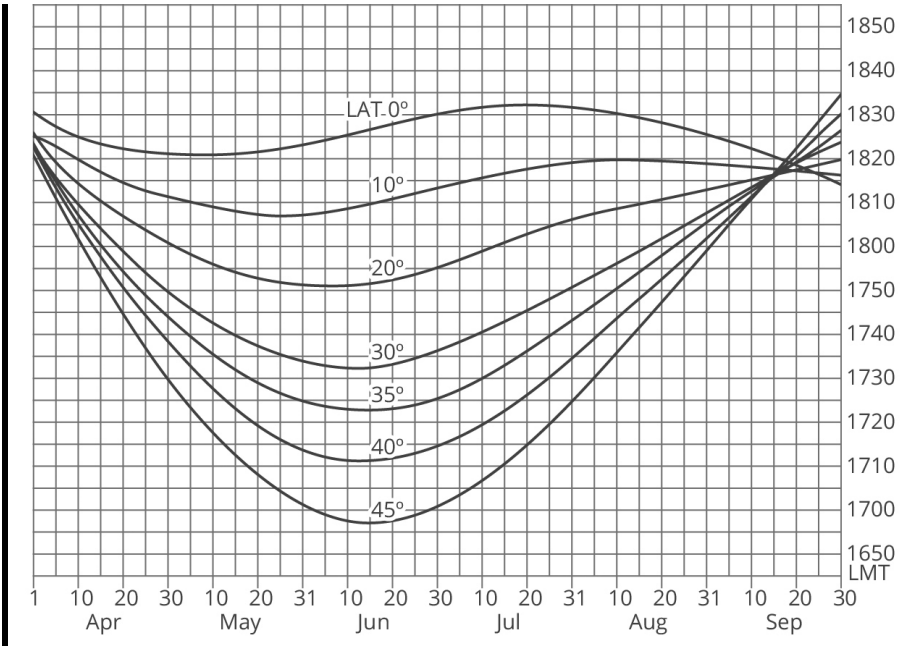
TIME OF FIRST LIGHT



TIME OF LAST LIGHT



TIME OF LAST LIGHT



CONVERSION OF ARC TO TIME

DEGREES				MINUTES				
Long Deg	Time Hours Min	Long Deg	Time Hours Min	Long Min	Time Min Sec	Long Min	Time Min Sec	
110	7 20	140	9 20	0	0 00	30	2 00	
111	7 24	141	9 24	1	0 04	31	2 04	
112	7 28	142	9 28	2	0 08	32	2 08	
113	7 32	143	9 32	3	0 12	33	2 12	
114	7 36	144	9 36	4	0 16	34	2 16	
115	7 40	145	9 40	5	0 20	35	2 20	
116	7 44	146	9 44	6	0 24	36	2 24	
117	7 48	147	9 48	7	0 28	37	2 28	
118	7 52	148	9 52	8	0 32	38	2 32	
119	7 56	149	9 56	9	0 36	39	2 36	
120	8 00	150	10 00	10	0 40	40	2 40	
121	8 04	151	10 04	11	0 44	41	2 44	
122	8 08	152	10 08	12	0 48	42	2 48	
123	8 12	153	10 12	13	0 52	43	2 52	
124	8 16	154	10 16	14	0 56	44	2 56	
125	8 20	155	10 20	15	1 00	45	3 00	
126	8 24	156	10 24	16	1 04	46	3 04	
127	8 28	157	10 28	17	1 08	47	3 08	
128	8 32	158	10 32	18	1 12	48	3 12	
129	8 36	159	10 36	19	1 16	49	3 16	
130	8 40			20	1 20	50	3 20	
131	8 44			21	1 24	51	3 24	
132	8 48			22	1 28	52	3 28	
133	8 52			23	1 32	53	3 32	
134	8 56			24	1 36	54	3 36	
135	9 00			25	1 40	55	3 40	
136	9 04			26	1 44	56	3 44	
137	9 08			27	1 48	57	3 48	
138	9 12			28	1 52	58	3 52	
139	9 16			29	1 56	59	3 56	

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GEN 3. SERVICES

GEN 3.1 AERONAUTICAL INFORMATION SERVICES

1. RESPONSIBLE SERVICE

1.1 The Aeronautical Information Service provided by Airservices Australia ensures the flow of information necessary for the safety, regularity and efficiency of air navigation within the area of its responsibility as indicated in *para 2.* It consists of AIS Headquarters and an International NOTAM Office (NOF).

1.2 **AIS Headquarters**

Postal Address:	Aeronautical Information Service GPO Box 367 CANBERRA ACT 2601 AUSTRALIA
Aeronautical/ Telegraphic Address:	YSHOYOYX
Email:	docs.amend@airservicesaustralia.com
Website:	www.airservicesaustralia.com/industry-info/aeronautical-information-management/

1.3 **International NOTAM Office**

Postal Address:	Airservices NOTAM Office GPO Box 367 CANBERRA ACT 2601 AUSTRALIA
Aeronautical/ Telegraphic Address:	YBBBYNYX
Email:	nof@airservicesaustralia.com
Website:	www.airservicesaustralia.com/flight-briefing

1.4 **Hours of Operations**

AIS:	Business hours - Canberra time
NOTAM Office:	H24

2. AREA OF RESPONSIBILITY

The AIS is responsible for the collection and dissemination of aeronautical information for the entire territory of Australia and its associated airspace, and for the airspace over the high seas encompassed by the Brisbane and Melbourne FIR and the Australian external territories of Norfolk Island and Christmas Island.

3. AERONAUTICAL PUBLICATIONS

3.1 Aeronautical Information is provided in the form of the Integrated Aeronautical Information Package (IAIP) consisting of the following:

- The Aeronautical Information Publication (AIP);
- AIP Amendments (AIP AMDT);
- AIP Supplements (AIP SUP);
- NOTAM and pre-flight information bulletins;
- Aeronautical Information Circulars (AIC)

3.2 Aeronautical Information Publication (AIP)

The AIP is the basic aviation document intended primarily to satisfy international requirements for the exchange of permanent aeronautical information and long duration temporary changes essential for air navigation.

AIP Australia is published in:

- Aeronautical Information Publication (AIP Book);
- Departure and Approach Procedures (DAP);
- En Route Supplement Australia (ERSA);
- Designated Airspace Handbook (DAH) and
- Maps and Charts.

The AIP is published in English only for use in international and domestic operations whether commercial or private.

3.3 Amendment Service to the AIP (AIP AMDT)

Amendments to:

- AIP Book;
- DAP and
- ERSA

are normally issued quarterly in accordance with AIRAC and include check lists of all current effective pages of the document (excluding ERSA).

Amendments to:

– DAH and

– Maps and Charts (other than WAC)

are normally issued biannually around May and November.

New information and changes are identified by a vertical black line (revision bar). Deletions within the AIP Book have a "**D**" added to the vertical line. Amendments to the Table of Contents are not identified by a revision bar. New or revised information published in DAP charts will be advised in the chart margin.

Amendments to DAH and ERSA are issued as a separate complete product.

3.4 Supplement to the AIP (AIP SUP)

Temporary changes of long duration (three months and longer) and information of short duration which consists of extensive text and/or graphics, supplementing the permanent information contained in the AIP, are published as AIP SUP. Operationally significant temporary changes to the AIP are published in accordance with the AIRAC system and its established effective dates and are identified clearly by the acronym AIRAC AIP SUP. AIP SUP are distributed via the Airservices website with an associated Trigger NOTAM valid for 14 days.

A check list (summary) of current AIP SUP is issued monthly.

3.5 NOTAM

NOTAM supplements AIP, providing rapid distribution of information of direct operational significance.

Note: For information about date and time used in NOTAM, see GEN 2.1 para 2.2.

In Australian Domestic distribution only NOTAM series and their respective categories are:

'B' series for Domestic NOTAM - ATS category only

'C' series for Domestic NOTAM – All other categories.

3.5.1 In the international environment, Australia issues NOTAM against a series of registers. These registers are:

- individual FIR
- multiple FIR or
- Australia General.

The individual FIR and multiple FIR registers are further subdivided by NOTAM category. The series are as follows:

Brisbane FIR - SUA NOTAM category	D
Brisbane FIR - AD NOTAM category	J
Brisbane FIR - ATS NOTAM category	K
Brisbane FIR - all other NOTAM category	N
Melbourne FIR - SUA NOTAM category	E
Melbourne FIR - AD NOTAM category	H
Melbourne FIR - ATS NOTAM category	L
Melbourne FIR - all other NOTAM category	F
Australia General FIR	G
Jakarta FIR - all NOTAM categories for Christmas Island	I
Auckland Oceanic FIR - all NOTAM categories for Norfolk Island	M

3.5.2 **NOTAM and valid AIP element checklist**

A checklist of valid NOTAM is issued monthly (on the 1st of the month) via NAIPS. This NOTAM also identifies the elements of AIP effective on that date, including a list of AIP SUP and AIC.

3.6 **Aeronautical Information Circular (AIC)**

AIC are used to disseminate aeronautical information of an administrative nature which does not qualify for promulgation in AIP or by NOTAM.

A checklist (summary) of current AIC is issued monthly.

3.7 **Aeronautical charts**

Aeronautical charts are a visual representation of a portion of the earth specifically designated to meet the needs of air navigation. Detailed information regarding aeronautical charts for Australia is available in *GEN 3.2.*

3.8 **Sale of Publications**

Details on how to purchase AIP Products is available from CanPrint Communications, online:

www.aipshop.canprint.com.au

4. AIRAC SYSTEM

4.1 In order to control and regulate the operationally significant changes requiring amendments to aeronautical charts, route listings, etc, such changes will generally be issued on predetermined dates according to the Aeronautical Information Regulation and Control (AIRAC) system. This type of information will normally be published as a planned AIP amendment or an AIRAC SUP. In circumstances in which AIRAC notification should be given, but a normal AIP amendment or an SUP cannot be produced due to a lack of time, a non-AIRAC SUP or NOTAM will be issued.

4.2 The following table indicates AIRAC effective dates for the coming years. AIRAC information will be issued so that the information will be received by the user not later than 28 days before the effective date. In some circumstances of major change, AIRAC notice of 56 days may be given.

4.3 Documents and charts issued in accordance with the AIRAC cycle become effective at 1600 hours UTC on the day prior to the nominated date unless otherwise notified.

4.4 At each AIRAC effective date, a Head Office Trigger NOTAM will be issued which identifies the elements of the AIP effective on that date and their respective 'with effect' dates.

AIRAC Effective Dates			
2026	**2027**	**2028**	**2029**
22 JAN	21 JAN	20 JAN	18 JAN
19 FEB	18 FEB	17 FEB	15 FEB
19 MAR	18 MAR	16 MAR	15 MAR
16 APR	15 APR	13 APR	12 APR
14 MAY	13 MAY	11 MAY	10 MAY
11 JUN	10 JUN	08 JUN	07 JUN
09 JUL	08 JUL	06 JUL	05 JUL
06 AUG	05 AUG	03 AUG	02 AUG
03 SEP	02 SEP	31 AUG	30 AUG
01 OCT	30 SEP	28 SEP	27 SEP
29 OCT	28 OCT	26 OCT	25 OCT
26 NOV	25 NOV	23 NOV	22 NOV
24 DEC	23 DEC	21 DEC	20 DEC

5. PREFLIGHT INFORMATION SERVICE

A preflight information service is provided from the National Operations Management Centre (NOMC) Pilot Briefing Office. Detailed information is published in the Preflight Information and Flight Planning Manual, available online: www.airservicesaustralia.com/aip/aip.asp and www.airservicesaustralia.com/industry-info/flight-briefing/,

Postal Address: Airservices Pilot Briefing Office
GPO Box 367
CANBERRA ACT 2601
AUSTRALIA

Phone: +61 3 9235 7518

Fax: +61 2 6268 5033

Website: www.airservicesaustralia.com/naips/

6. DIGITAL DATA SETS

6.1 Airservices provides Vertical Obstacle Data (VOD) for:

- Area 1 – Australian data collected and maintained as defined in *ICAO Annex 15*, and
- Area 2 – All designated international aerodromes (excluding Horn Island), as listed in *GEN 1.2. Section 2.*

Information on how to order this data is available from the Airservices website:
www.airservicesaustralia.com/industry-info/aeronautical-information-management/electronic-data/

Terrain data is available from Geoscience Australia via their online store: https://elevation.fsdf.org.au/

Obstacle data in the vicinity of aerodromes is collated and provided by the airport operator and is published as ICAO Type A/B charts. Terrain data for precision approach runways is also collected by the airport operator and published as Precision Approach Terrain Charts. The aerodromes where this information is available, and the airport contact details are available on the Airservices website:
www.airservicesaustralia.com/aip/aip.asp

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GEN 3.2 AERONAUTICAL CHARTS

1. RESPONSIBLE SERVICES

Aeronautical charts for the Australian FIR are provided by the Aeronautical Information Services unit within Airservices Australia, see *GEN 3.1 Para 1.* Charts are produced in accordance with the provisions contained in Annex 4 - Aeronautical Charts. Differences to these provisions are available in *GEN 1.7*.

2. MAINTENANCE OF CHARTS

The aeronautical charts are kept up-to-date by amendments to the AIP, publication dates for products is available online: www.airservicesaustralia.com/industry-info/aeronautical-information-management/document-amendment-calendar/

3. PURCHASE ARRANGEMENTS

Charts are available for purchase through CanPrint Communication, online: www.aipshop.canprint.com.au.

4. AERONAUTICAL CHART SERIES AVAILABLE

4.1 The following series aeronautical charts are produced:

- Planning Chart Australia (PCA);
- World Aeronautical Chart (WAC);
- Visual Terminal Chart (VTC);
- Visual Navigation Chart (VNC);
- En Route Chart - Low (ERC Low);
- En Route Chart - High (ERC High);
- Terminal Area Chart (TAC);
- Aerodrome (AD) chart;
- Apron chart;
- Noise Abatement Procedures;
- Standard Instrument Departure (SID) chart;
- Standard Instrument Arrival (STAR) chart;
- DME and GNSS Arrival chart;
- Instrument Approach and Landing (IAL) chart; and
- Obstruction chart (Type A) (available from aerodrome operator).

4.2 **General description of each series**

a. PCA

- Briefing/Area QNH boundaries;
- GAF boundaries;
- WAC coverage and chart titles;
- Location names and abbreviations;
- Estimated FIS VHF coverage at 5,000FT and 10,000FT; and
- HF network boundaries.

b. **WAC (scale 1:1 000 000).** Designed for preflight planning and pilotage. They are constructed on Lambert Conformal Conic projection. Australian coverage is shown on the front of each chart.

c. **VNC (scale 1:500 000).** Designed for operations under the VFR. They contain an aeronautical overlay of controlled airspace over a topographical base, and contain some radio communication and other navigational data appropriate for visual navigation. Map coverage is shown on the front of each chart.

d. **VTC (scale 1:250 000)**. Designed for visual operations near terminal areas. They contain some topographical detail and appropriate airspace, radio communication and navigation aid information. These charts are intended for use up to and including FL180.

Note: When planning visual navigation outside the coverage of VTC, pilots will need to refer to the appropriate VNC (if available) or IFR chart ERC Low for depiction of controlled airspace and Special Use Airspace.

e. **ERC Low.** Published in various scales to accommodate significant air traffic route areas and shows controlled airspace, Special Use Airspace, air routes, ATS and radio navigation services. Aeronautical information within terminal areas may not be complete and pilots should use a TAC or VTC.

f. **ERC High.** Designed for use by aircraft operating on transcontinental and inter-capital routes above FL200.

g. **TAC.** Designed for use in terminal areas, these charts provide controlled airspace, Special Use Airspace, air routes, navigation aids and radio frequencies. They are designed to display aeronautical information at a larger scale for easier use in congested areas. Scale varies for each chart.

h. Aerodrome charts, Apron charts, Noise Abatement Procedures, SID charts, STAR charts, DME and GNSS Arrival charts, and IAL charts are published in DAP East and DAP West.

i. Obstruction charts (Type A), when required to be produced, are available from respective aerodrome owners/operators.

5. LIST OF AERONAUTICAL CHARTS AVAILABLE

A list of currently available charts and a user guide to Airservices aeronautical charts are available online: www.airservicesaustralia.com/aip/aip.asp

6. INDEX TO THE WORLD AERONAUTICAL CHART (WAC) - 1:1 000 000

A list of current WAC including Chart and Edition numbers and diagram depicting WAC coverage areas is available online: www.airservicesaustralia.com/aip/aip.asp

7. TOPOGRAPHIC CHARTS

The topographic data and information used in the production of Airservices aeronautical charts is sourced from Geoscience Australia. Information about this data is available online: www.ga.gov.au.

8. CORRECTIONS TO CHARTS NOT CONTAINED IN THE AIP

If required any corrections to charts will be notified through NOTAM or AIP SUP.

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GEN 3.3 AIR TRAFFIC SERVICES

1. RESPONSIBLE SERVICE

1.1 The responsible authorities for the provision and administration of Air Traffic Services in Australia are:

a. Airservices Australia

GPO BOX 367
CANBERRA ACT, 2601
Australia
Email: satss@airservicesaustralia.com
Website: www.airservicesaustralia.com/about-us/our-services/air-traffic-management/

b. Department of Defence - Royal Australian Air Force (RAAF)

No. 44 Wing
RAAF Base Williamtown
WILLIAMTOWN NSW, 2314
Australia
Email: hqsrga7.anspstand@defence.gov.au

1.2 The services are provided in accordance with the following ICAO documents:

a. Annex 2

b. Annex 11

c. Annex 10; Vol II

d. DOC 4444 (PANS-ATM)

e. DOC 7030 (Regional Supplementary Procedures)

f. DOC 8168 (PANS-OPS)

g. DOC 9426 (ATS Planning Manual

1.3 Differences from ICAO Standards and Recommended Practices are detailed in *GEN 1.7*.

2. AREA OF RESPONSIBILITY

2.1 Air traffic services are provided as follows:

Authority	Area of Responsibility
Airservices Australia	The territory of Australia, including territorial waters and the airspace over the high seas within the Brisbane FIR and Melbourne FIR, except: Norfolk Island and Christmas Island and associated territorial waters; and Where the Department of Defence provide services as below.
Department of Defence (RAAF)	Aerodrome services at Edinburgh and Richmond. Aerodrome and Approach services at Amberley, Darwin, East Sale, Nowra, Oakey, Pearce, Tindal, Townsville and Williamtown.

3. TYPES OF SERVICE

3.1 General

3.1.1 The following types of services are provided to ensure the safe, orderly and expeditious flow of air traffic:

a. Air Traffic Control Service:

(i) En Route Control (Surveillance and Procedural);

(ii) Approach Control (Surveillance and Procedural);

(iii) Aerodrome Control (Tower);

b. Flight Information Service; and

c. Alerting Service.

3.1.2 Hours of operation, services provided, and identification of ATS units are published in *ERSA*.

3.2 Air Traffic Control Service

3.2.1 An Air Traffic Control service is provided to:

a. all IFR flights in classes A, B, C, D and E airspace;

b. all VFR flights in classes B, C and D airspace;

c. all Special VFR flights in classes B, C and D airspace; and

d. all aerodrome traffic at controlled aerodromes.

3.3 Flight Information Service (FIS)

3.3.1 Pilot Responsibility

3.3.1.1 Pilots are responsible for obtaining information necessary to make operational decisions.To ensure that accurate information is obtained in adequate time, pilots must take into consideration that ATC initiated FIS is limited to aircraft within one hours' flight time of the condition or destination at time of receipt of the information by ATC. The only exception to this is SIGMET information, which shall cover a portion of the route up to two hours' flight time ahead of the aircraft.

3.3.2 Preflight Information

3.3.2.1 The Preflight Briefing Service is available online via the NAIPS Internet Service or via telephone to the Airservices Pilot Briefing Office on 1800 805 150. Detailed information is published in the Preflight Information and Flight Planning Manual, available online: www.airservicesaustralia.com/aip/aip.asp and www.airservicesaustralia.com/industry-info/flight-briefing/

3.3.2.2 Pilots must obtain an appropriate preflight briefing before departure from places where the above facilities are accessible. Where those facilities are not accessible, a briefing may be obtained from FLIGHTWATCH as soon as practicable after the flight commences. The information requested should be confined to data considered essential for the safe conduct of the flight to the first point of intended landing where additional information can be obtained.

3.3.2.3 Preflight briefing will not normally be provided on ATC communications channels.

3.3.3 In-flight Information

3.3.3.1 The in-flight information service consists of three elements:

a. ATC Initiated FIS;

b. Automatic Broadcast Services; and

c. On-Request Service.

3.3.4 ATC Initiated FIS

3.3.4.1 ATC provides pilots with pertinent information that will affect flight within one hour's flight time (two hours for SIGMET). At the time the information is identified, information will be directed to pilots maintaining continuous communications and broadcast on appropriate ATS frequencies.

3.3.5 Automatic Terminal Information Service (ATIS)

3.3.5.1 Operational information required by aircraft for take-off or landing is broadcast on a dedicated frequency and/or on the voice channel of radio navigation aids.

3.3.5.2 Outside the hours of tower activation, operational information of an unchanging nature may be broadcast over ATIS.

3.3.6 Aerodrome Flight Information Service (AFIS)

3.3.6.1 An AFIS provides pilots with an alerting service, local traffic and operational information on the CTAF assigned to the particular aerodrome.

3.3.6.2 Essential aerodrome information is provided by an Automatic Aerodrome Information Service (AAIS) broadcast on a dedicated frequency (similar to ATIS) during AFIS HRS.

3.3.7 Surveillance Flight Information Service (SFIS)

3.3.7.1 An SFIS provides a pilot with traffic information within a mandatory broadcast area using the aerodrome's Common Traffic Advisory Frequency (CTAF). The SFIS is not a separation or sequencing service.

3.3.7.2 SFIS is dependent on detection by the ATS surveillance system. This means the service cannot be provided for or about aircraft below ATS surveillance system coverage or not operating transponder or ADS-B OUT equipment.

3.3.7.3 Traffic information may be summarised or limited to generic statements e.g. number of known aircraft in the circuit. The SFIS may not be aware of the position of aircraft within the circuit.

3.3.7.4 SFIS does not relieve pilots of their see-and-avoid, SARWATCH and SARTIME responsibilities.

3.3.8 Traffic Information

3.3.8.1 A traffic information service is provided, depending on higher priority duties of the controller or other limitations; e.g. surveillance limitations, volume of traffic and/or frequency congestion. Additionally, controllers may not be able to provide traffic information concerning all traffic in the aircraft's proximity. Traffic information does not relieve pilots of their responsibility to see and avoid other aircraft.

3.3.8.2 In Class G airspace, a traffic information service is provided to IFR flights about other conflicting IFR and observed VFR flights except:

a. An IFR flight reporting taxiing or airborne at a non-controlled aerodrome will be advised of conflicting IFR traffic that is not on the CTAF; and

b. An IFR flight inbound to a non-controlled aerodrome will be advised of conflicting IFR traffic until the pilot reports changing to the CTAF.

3.3.8.3 A Surveillance Information Service (SIS) is available on request, to VFR flights in classes E and G airspace within ATS surveillance system coverage, subject to ATC workload. Pilots receiving a SIS are provided with traffic information, an alerting service and on request position or navigation information.

Note: All information is advisory in nature. The pilot remains responsible for terrain clearance, aircraft-to-aircraft separation and obtaining clearances into controlled airspace.

3.3.8.4 Pilots wishing to receive a SIS must be in direct VHF communications with ATC and equipped with a serviceable SSR transponder or ADS-B transmitter. The pilot must maintain a continuous listening watch with ATC, advise ATC prior to any changes to track or level and advise prior to leaving the frequency.

3.3.8.5 SIS may be terminated at any time by the controller, or by pilot advice.

3.3.9 **On-Request Service - ATC and FLIGHTWATCH**

3.3.9.1 An On-Request FIS is available to aircraft in all classes of airspace on ATC VHF or HF frequencies.

3.3.9.2 Information derived from BoM weather radar sites is available to pilots on request, subject to ATS workload.

3.4 **Alerting Service**

3.4.1 An Alerting Service will be provided:

a. for all aircraft provided with ATC service;

b. in so far as practicable, to all other aircraft having filed a flight plan or otherwise known to the air traffic services; and

c. to any aircraft known or believed to be the subject of unlawful interference.

3.5 **Safety Alerts and Avoiding Action**

3.5.1 ATC will issue a Safety Alert to aircraft, in all classes of airspace, when they become aware that an aircraft is in a situation that is considered to place it in unsafe proximity to:

a. terrain;

b. obstruction;

c. active restricted, prohibited or military operating areas; or

d. other aircraft.

3.5.2 When providing an ATS surveillance service, ATC will issue avoiding action advice as a priority, when they become aware that an aircraft is at risk of collision with another aircraft.

4. CALCULATION OF LOWEST SAFE ALTITUDE

4.1 A pilot using Grid LSALT for obstacle clearance is responsible for determining the allowance for navigation error that should be applied, considering the limitations of the navigation aids or method of navigation being used for position fixing. This navigation error allowance must be applied to the proposed track. The highest Grid LSALT falling within the area covered by the determined navigation error must be used.

4.2 For routes and route segments not shown on AIP aeronautical charts, the lowest safe altitude must not be less than that calculated in accordance with *para 4.3* within an area defined in the following *paras 4.6, 4.7, 4.8 and 4.9.*

4.3 Unreported obstacles up to 360FT may exist in navigation tolerance areas. The LSALT must be calculated using the following method:

a. where the highest obstacle is more than 360FT above the height determined for terrain, the LSALT must be 1,000FT above the highest obstacle; or

b. where the highest obstacle is less than 360FT above the terrain, or there is no charted obstacle, the LSALT must be 1,360FT above the elevation determined for terrain; except that

c. where the elevation of the highest terrain or obstacle in the tolerance area is not above 500FT, the LSALT must not be less than 1,500FT.

4.4 An aircraft must not be flown under the IFR, lower than the published lowest safe altitude or the lowest safe altitude calculated in accordance with this section, unless permitted by *CASR 91.265, 91.267* and *91.305* or another civil aviation legislation provision.

4.5 If the navigation of the aircraft is inaccurate, or the aircraft is deliberately flown off-track, or where there is a failure of any radio navigation aid normally available, the area to be considered is a circle centred on the DR position, with a radius of 5NM plus 20% of the air distance flown from the last positive fix.

4.6 **For Routes Defined by Radio Navigation Aids or to be Navigated by DR**

The area to be considered must be within an area of 5NM surrounding and including an area defined by lines drawn from the departure point or en route radio aid, 10.3° each side of the nominal track (where track guidance is provided by a radio navigation aid), or 15° each side of the nominal track (where no track guidance is provided) to a limit of 50NM each side of track, thence paralleling track to abeam the destination and then converging by a semicircle of 50NM radius centred on the destination.

On shorter routes, where these lines are displaced by less than 50NM abeam the destination, they shall converge by a radius based on that lesser distance. Where the lines thus drawn come at any time within the coverage of an en route or destination radio aid the aircraft is equipped to use, they will converge by straight lines to that aid. The minimum angle of convergence which must be used in this case is 10.3° each side of track.

4.7 **For Routes Operated Under the RNP 2 Navigation Specification**

The area to be considered must be within an area of 5NM surrounding and including the departure point, the destination and each side of the nominal track.

4.8 **For Other Area Navigation Operations**

The area to be considered must be within an area of 5NM surrounding and including an area defined by lines drawn from the departure point not less than 15° each side of the nominal track to a maximum of:

a. 8NM for a flight under the RNP 4 navigation specification;

b. 7NM for flight under an RNAV navigation specification having a GNSS input; or

c. 30NM for flight under a non GNSS area navigation specification.

Thence paralleling track to abeam the destination and converging by a semicircle of the same radius centered on the destination.

4.9 **VFR Flights at Night**

The area to be considered must be:

a. the area specified in *para 4.6, 4.7* or *4.8* for aircraft navigated by means of a radio navigation system; or

b. within a radius of 10NM from any point along the aircraft's nominal track.

However, the pilot of an aircraft who has positively determined by visual fix that a critical obstruction has been passed may nevertheless descend immediately to a lower altitude, provided that the required obstacle clearance above significant obstructions ahead of the aircraft is maintained.

4.10 An aircraft must not be flown under the VFR at night, lower than the published lowest safe altitude or the lowest safe altitude calculated in accordance with this section unless permitted by CASR *91.265, 91.267* and *91.277* or another civil aviation legislation provision.

5. CONTINGENCY PROCEDURES - AIR TRAFFIC SERVICES TEMPORARILY NOT AVAILABLE

5.1 Introduction

5.1.1 When Air Traffic Services are temporarily not available traffic management, access control and/or communication procedures enhancing situational awareness and protecting against traffic saturation and frequency congestion within the affected airspace will be promulgated by NOTAM.

Note: Refer to ENR 1.1 for Procedures which may be implemented when ATS is temporarily not available.

6. CONTROL OF VEHICULAR AND PEDESTRIAN MOVEMENT ON AERODROMES

6.1 Drivers of vehicles which need to operate on the manoeuvring area must not seek permission to do so from ATC, unless the driver and vehicle holds the appropriate Driver Authority for Use Airside as issued by the airport operator and/or owner.

6.2 ATC has the authority to issue or to withhold for reasons of traffic, permission for vehicular or pedestrian movement on the manoeuvring area. When it is permitted, such movement will be rigidly controlled.

6.3 All persons, including drivers of vehicles, will be instructed to stop and hold position for radio clearances, or light signal clearances, from the tower before crossing any runway or taxiway, unless they are on a portion of the manoeuvring area marked off by lights, flags or other conventional warning signs. In radio advices to aircraft, ATC will identify as distinctly as possible, persons or vehicles on the manoeuvring area.

6.4 All persons, including drivers of vehicles, on the manoeuvering area must stop and hold at all illuminated stop bars and may only proceed further when a verbal ATC clearance to enter or cross the runway has been received and the stop bar lights have been switched off. (See also *ENR 1.1 Section 2.3.6*)

6.5 All persons, including drivers of vehicles, required to hold short of a runway, must hold at the appropriate runway-holding position for that runway, or the runway strip edge.

6.6 All persons, including drivers of vehicles, in receipt of a clearance from ATC to enter a runway, must hold short of an intersecting runway except when an instruction “CROSS RUNWAY (number)” has been issued and the stop bar lights, where fitted, have been switched off. (See also *ENR 1.1 Section 2.3.6*.)

GEN 3.4 COMMUNICATION AND NAVIGATION SERVICES

1. RESPONSIBLE SERVICE

1.1 Airservices Australia (Airservices) is responsible for aeronautical communication and navigation services in Australia.

Postal Address:	Airservices Australia 25 Constitution Avenue CANBERRA ACT 2600 AUSTRALIA
Phone:	+61 2 6268 4111
Fax:	+61 2 6268 5693
Website:	www.airservicesaustralia.com

1.2 A small number of assets within the aeronautical communications and navigation systems are owned by entities other than Airservices.

1.3 **Applicable ICAO Documents**

ICAO Standards, Recommended Practices and Procedures described in the following are applied to communication and navigation services provided within Australia, subject to the differences identified at *GEN 1.7*.

- Annex 10 - Aeronautical Telecommunications
- Annex 11 - Air Traffic Services
- Doc 4444 - Procedures for Air Navigation Services - Air Traffic Management (PANS-ATM)
- Doc 7030 - Regional Supplementary Procedures
- Doc 7910 - Location Indicators
- Doc 8400 - ICAO Abbreviations and Codes
- Doc 8585 - Designators for Aircraft Operating Agencies, Aeronautical Authorities and Services
- Doc 9869 - Performance Based Communications and Surveillance (PBCS) Manual
- Doc 10037 - Global Operational Data Link (GOLD) Manual

2. AREA OF RESPONSIBILITY

2.1 **Continental and Oceanic Areas**

Airservices provides communication and navigation services throughout the Brisbane (YBBB) and Melbourne (YMMM) FIR.

2.2 **External Territories**

Airservices provides communication and navigation services beyond the YBBB and YMMM FIR to the Australian external territories of Norfolk Island and Christmas Island.

2.3 **Extra Territorial Services**

Airservices provides communication services in Class A airspace within the Honiara (AGGG) and Nauru (ANAU) FIR.

3. RADIO NAVIGATION SERVICE

3.1 **General**

The following types of radio aids to navigation operate within Australian FIR:

a. MF Non-Directional Beacon (NDB);
b. VHF Omni-Directional Radio Range (VOR);
c. Distance Measuring Equipment (DME);
d. Instrument Landing System (ILS);
e. Primary Surveillance Radar (PSR);
f. Secondary Surveillance Radar (SSR);
g. Global Navigation Satellite System (GNSS) including GPS, GLONASS, Galileo etc;
h. GBAS Landing System (GLS) - (Polarisation GBAS/H);
i. Tactical Air Navigation System (TACAN) (military locations);
j. Automatic Dependent Surveillance – Contract (ADS-C) (FANS-1/A); and
k. Automatic Dependent Surveillance – Broadcast (ADS-B) (1090MHz Extended Squitter).

3.2 **Identification**

3.2.1 Radio navigation aids serving the same location normally radiate the same identification codes. Further, at ILS/localiser installations, the normal identification is preceded by the letter “I”; and at private non-accredited NDBs, a four letter identifier is radiated, the first letter being “X”.

3.2.2 A GBAS station is identified by the relevant Aerodrome’s 4 letter ICAO code where the GBAS is installed (e.g. YSSY). Approaches provided from the GBAS station are identified by the Reference Path ID field in the Final Approach Segment (FAS) data blocks. The Reference Path ID is unique to the approach to be used and published on the Approach Plate for the runway. Each Reference Path ID commences with a letter “G”.

3.3 **Non-Directional Beacons**

3.3.1 Operate in the frequency band 190Khz - 1750Khz. Due to the frequency range available within Australia, a number of beacons share the same frequency. Where this occurs, the beacons are widely spaced geographically.

3.3.2 Some NDB have been sited in mountainous country and reflections of the signal can cause bearing fluctuations which may occasionally exceed 10°. Since these fluctuations may tend to obscure the “over the top” indications, other aids should be used as a check. The rated coverage of these NDBs is generally reduced to no more than 30NM.

3.4 **VHF Omni-Directional Radio Range (VOR)**

3.4.1 These aids operate in the frequency band 112 – 118MHz. The track radials are designated by their magnetic bearing away from the station.

3.5 **Distance Measuring Equipment (DME)**

3.5.1 Australia uses the international 1000MHz system. The system uses the channels designated in *ICAO Annex 10* for operation with the VOR frequency selected for the same site. This “pairing” facilitates compatible airborne equipment to display both the DME and VOR information by the selection of only the VOR frequency.

3.5.2 When specific ICAO requirements are met, collocated DME and VOR are said to be “associated” and are shown in AIP documents as VOR/DME with the VOR frequency. In other cases, a bracket will be used to indicate collocation of navigation aids.

3.5.3 When a DME is not “associated” with the VOR at the same site, it is identified in NOTAM and AIP documents by its channel number and suffix with the VOR frequency in parenthesis - e.g. DME 111X (116.4).

3.6 **Tactical Air Navigation (TACAN)**

TACAN systems are installed at a number of military/joint-user aerodromes. The DME element of the TACAN can be obtained by using its “paired” VOR frequency.

Note: TACAN and VOR installations at joint-user airfields are never “associated”.

TACAN is shown on AIP aeronautical charts by the channel number with the “paired” VOR frequency - e.g. TACAN88 (114.1).

3.7 **Public Broadcasting Stations**

Some broadcast stations are shown on visual navigation charts when they may be of value as supplementary aids to navigation.

3.8 **Abnormal Operation of Radio Navigation Aids**

3.8.1 Users must notify ATS of any abnormal condition in the operation of any radio navigation facility.

3.8.2 **Aids not Available for Navigation**. Sometimes a facility that is not suitable for navigation has to be operated for test purposes. To provide a warning to pilots in such cases, in addition to NOTAM or verbal advice, the station identifier will either:

a. be suppressed; or

b. for ILS:

(1) if the localiser is out of service, the glide path will not be radiated and there will be no identifier; or

(2) if the glide path is out of service but radiating test signals, the localiser will not be radiated; or

(3) if the glide path is out of service and switched off, the localiser may be radiated together with the station identification; or

c. for GLS:

(1) when required for maintenance or on failure the system will be withdrawn from service; or

(2) if the audible Reference Path ID is corrupt or not received, the system will be withdrawn from service.

d. for newly installed NDBs or experimental facilities, the identifier XP will be used.

Note: When a disabled GLS approach is selected, the Reference Path ID may still be displayed on the Primary Flight Display and the audible IDENT heard. No deviation indications or steering commands will be displayed.

3.9 **Remote facilities**

Sometimes circumstances necessitate the introduction of a station identification before the aid is notified as being available for operational use. Whilst use of the aid in these circumstances does not present an operational hazard, navigation by use of the aid must not be planned until its availability is notified by NOTAM.

3.10 **Monitoring of Identifier**

If a station identification is not received or is corrupt, the aid should not be used as the primary means of navigation except when its serviceability in this condition is confirmed by NOTAM.

3.11 **Global Navigation Satellite System (GNSS)**

3.11.1 GNSS is a worldwide satellite navigation service comprising of one or more satellite constellations, including GPS, aircraft receivers and integrity monitoring, augmented as necessary to achieve specific navigational performance.

3.12 **Global Positioning System (GPS)**

3.12.1 GPS is a satellite-based radio navigation system, based on the World Geodetic System - 1984 (WGS-84) datum, which provides highly accurate position and velocity information.

3.12.2 Use of GPS for IFR operations is dependent on the US Department of Defence GPS Standard Positioning Service operating to its defined full operating capability. Use of this service to meet the requirements for a sole means navigation system must be in accordance with *GEN 1.5 Section 2.*

4. COMMUNICATIONS SERVICES

4.1 **Radio Frequencies**

4.1.1 **General.** Air ground communications in Australian FIRs are conducted by radiotelephony in the VHF, UHF and HF bands. Air-to-air communications are normally conducted in the VHF band. The requirements for carriage of radio communications systems are contained in *GEN 1.5 Section 1.* Frequencies are published in *ERSA*. VHF is the primary frequency band, with HF only being used when outside VHF coverage.

Selected VHF frequencies are published in *ERSA* and are depicted on AIP Aeronautical Navigation Charts. The estimated coverage is shown on *Planning Chart Australia (PCA).*

4.1.2 **UHF** is intended primarily for use with military aircraft. At some locations, re-transmit facilities, which permit UHF and VHF aircraft to copy VHF/UHF transmissions conducted with a common ATS unit, are installed.

4.1.3 **HF.** The appropriate HF frequency for an area of operations is shown on Planning Chart Australia (PCA). In TCTA and OCA, ATS may nominate the appropriate frequency to be used by aircraft. For those operations outside VHF coverage, where the maintenance of a continuous two-way communications watch with an airways operations unit is mandatory, the carriage of suitable HF frequencies is required.

Note 1: ATC will provide the relevant frequencies to IFR flights when transferring from VHF to HF communication, and to other flights on request.

Note 2: SELCAL is available to appropriately equipped aircraft. Frequencies for this service are listed in ERSA.

4.1.4 **Interpilot Air-to-Air Communication.** In accordance with regional agreements, 123.45MHz is designated as the air-to-air VHF communications channel. Use of this channel will enable aircraft engaged in flights over remote and oceanic areas out of range of VHF ground stations and not in the vicinity of a non-controlled aerodrome depicted on an aeronautical chart to exchange necessary operational information and to facilitate the resolution of operational problems.

4.1.5 **SELCAL** is available to appropriately equipped aircraft. Frequencies for this service are listed in *ERSA*.

4.2 **Certified Air/Ground Radio Service (CA/GRS)**

4.2.1 A Certified Air/Ground Radio Service is an aerodrome-based radio information service, which may operate at non-controlled aerodromes. The service is a safety enhancement facility which provides pilots with operational information relevant to the particular aerodrome. The service is operated by or for the aerodrome operator to published hours, on the CTAF assigned to the particular aerodrome. It is not an Airservices or RAAF-provided air traffic service.

4.2.2 The CA/GRS does not provide any separation service.

4.2.3 The call sign of the service is the aerodrome location followed by “Radio”; e.g. “Ayers Rock Radio”. The radio operators of the service have been certified to meet a CASA standard of communication technique and aviation knowledge appropriate to the service being provided.

4.2.4 The CA/GRS is provided to all aircraft operating within the designated broadcast area for the specific location. Refer to *ERSA* for the location specific designated broadcast areas.

4.2.5 When a CA/GRS is operating, pilot procedures are unchanged from the standard non-controlled aerodrome operating and communication procedures. *ERSA* includes location specific information relating to procedures.

4.2.6 The operational information provided by a CA/GRS assists pilots in making informed operational decisions. Pilots retain authority and responsibility for the acceptance and use of the information provided.

4.2.7 Aircraft making the normal inbound or taxiing broadcast receive a responding broadcast from the CA/GRS operator, conveying the following information:

a. Confirmation of the correct CTAF.

b. Current, known, relevant traffic in the vicinity of the aerodrome and on the manoeuvring area of the aerodrome. Traffic information may include some or all of the following:

(1) the aircraft type, call sign, position and intention; or

(2) where circuit flying is in operation, general advice on the number of aircraft in the circuit, and position in the circuit if relevant.

Note: this information is provided as an advisory to pilots in VMC and to assist pilots in arranging self-separation in IMC.

c. Weather conditions and operational information for the aerodrome. The information which may be advised includes:

(1) runway favoured by wind or noise abatement,

(2) runway surface conditions,

(3) wind direction and speed,

(4) visibility and present weather,

(5) estimated cloud base,

(6) aerodrome surface temperature, and

(7) aerodrome QNH.

This information will be provided by means of an Automatic Aerodrome Information Service (AAIS) broadcast on a discrete frequency (similar to ATIS) during OPR HR of CA/GRS or on request to the CA/GRS operator. Pilots should monitor the published AAIS frequency before making the taxiing or inbound broadcast, and indicate that the AAIS information has been received when making the inbound or taxiing broadcast.

d. Other operational information of a local nature, relevant to the safety of operations at the aerodrome.

4.2.8 The CA/GRS will provide emergency services call-out if requested by the pilot in an emergency or, if in the opinion of the operator, a call-out is warranted.

4.2.9 The weather information provided by the service is derived from approved measuring equipment, which meets BoM aeronautical precision standards. QNH provided by a CA/GRS or AAIS may be used to reduce landing, circling and alternate minima in accordance with *ENR 1.5 para 5.3* (QNH Sources).

4.2.10 The CA/GRS operator may act as a representative of an air operator (where formal agreement with the operator has been established) for the purposes of holding SARWATCH.

4.3 **UNICOM**

4.3.1 UNICOM (Universal Communications) is a non-ATS communications service provided to enhance the value of information normally available about a non-controlled aerodrome.

4.3.2 The primary function of the frequency used for UNICOM services where the UNICOM is the CTAF is to provide pilots with the means to make standard positional broadcasts when operating in the vicinity of an aerodrome. Participation in UNICOM services must not inhibit the transmission of standard positional broadcasts.

4.3.3 Participation in UNICOM services relates to the exchange of information concerning:

a. fuel requirements;
b. estimated times of arrival and departure;
c. aerodrome information;
d. maintenance and servicing of aircraft including the ordering of parts and materials urgently required;
e. passenger requirements;
f. unscheduled landings to be made by aircraft;
g. general weather reports; and
h. basic information on traffic.

4.3.4 This information is available to all aircraft during the times that the UNICOM is operating.

4.3.5 Weather reports, other than simple factual statements about the weather, may not be provided by UNICOM operators unless they hold an aviation weather observer qualification issued by the Bureau of Meteorology.

4.3.6 The UNICOM operator is solely responsible for the accuracy of any information passed to an aircraft, while the use of information obtained from a UNICOM is at the discretion of the pilot in command.

4.3.7 Stations providing a UNICOM service are required to be licensed by the Australian Communications and Media Authority (ACMA). Detailed information regarding the licensing and use of equipment may be obtained by contacting the ACMA in the appropriate State or Territory capital city.

4.4 **Aerodrome Frequency Response Unit**

4.4.1 To assist pilots' awareness of inadvertent selection of an incorrect VHF frequency when operating into non-controlled aerodromes, a device known as an Aerodrome Frequency Response Unit (AFRU) may be installed. An AFRU will provide an automatic response when pilots transmit on the CTAF for the aerodrome at which it is installed.

4.4.2 The features of the AFRU are as follows:

a. When the aerodrome traffic frequency has not been used for the past five (5) minutes, the next transmission over two (2) seconds long will cause a voice identification to be transmitted in response, e.g. "GOULBURN CTAF".

b. When the aerodrome traffic frequency has been used within the previous five (5) minutes, a 300 millisecond tone will be generated after each transmission over two (2) seconds long.

4.4.3 A series of three (3) microphone clicks within a period of five (5) seconds will also cause the AFRU to transmit a voice identification for the particular aerodrome.

4.4.4 In the event that the transmitter in the AFRU becomes jammed for a period of greater than one minute, the unit will automatically shut down.

4.4.5 The operation of the AFRU provides additional safety enhancements by confirming the operation of the aircraft's transmitter and receiver, the volume setting, and that the pilot has selected the correct frequency for use at that aerodrome.

4.5 **Aeronautical Fixed Telecommunication Network (AFTN)**

4.5.1 The AFTN is established primarily for ATS unit intercommunication. However, subject to certain provisos, the AFTN may be used to transmit messages concerning flight safety, flight regularity, reservation and general operating agency aspects. Details are available from ATS units.

4.5.2 A matrix of the Australian AFTN circuitry is contained at *APPENDIX 1*.

4.6 **Special Broadcast Services**

4.6.1 **ATIS** is a continuous transmission, on a discrete frequency or on the IDENT channel of a VOR or NDB, of the operational information. Details are in *GEN 3.3 Section 3.3* and frequencies in *ERSA.*

4.6.2 **VOLMET** broadcasts contain selected meteorological information on discrete frequencies. Details are in *GEN 3.5 Section 7.* and frequencies in *ERSA*.

4.6.3 **AWIS** broadcasts actual weather conditions on navigation aids from AWS sites which use Bureau of Meteorology approved AWS equipment. Details are in *GEN 3.5 Section 9.* and frequencies in *ERSA*.

4.7 **Aeronautical Codes**

4.7.1 Aeronautical codes, including location indicators for Australian aerodromes, are published in *ERSA* and on aeronautical charts.

5. RADIOTELEPHONY PROCEDURES

5.1 **Introduction**

5.1.1 The communication procedures, phraseologies and requirements contained in this section have been selected to harmonise with ICAO, and international practices where applicable. Additional phrases to supplement where ICAO is silent have been included.

5.1.2 Primary reference documents on radiotelephony are *ICAO Doc 4444, Doc 9432, and Annex 10*. ATS and pilots should refer to these documents to obtain additional information as necessary. Only procedures appropriate to Australia and commonly used phrases are contained in this section.

5.1.3 Use of standard phrases for radio telephony communication between aircraft and ground stations is essential to avoid misunderstanding the intent of messages and to reduce the time required for communication.

5.1.4 Where circumstances warrant, and no phraseology is available, clear and concise plain language should be used to indicate intentions.

5.1.5 Phraseologies contained in this section are generic and, although primarily reflecting a controlled airspace environment, pilots operating in Class G airspace should use these generic phrases unless specific Class G phrases are shown.

5.2 **General**

5.2.1 **Symbols and Parentheses**

Words in parentheses "()" indicate that specific information, such as a level, a place, or a time, etc, must be inserted to complete the phrase, or alternatively, that optional phrases may be used. Words in square parentheses "[]" indicate optional additional words or information that may be necessary in specific instances.

5.2.2 The following symbols indicate phraseologies which may differ from those used in an international aviation environment, but are necessitated by Australian requirements.

- UNIQUE TO AUSTRALIA (ICAO SILENT)

▲ MILITARY SPECIFIC PHRASEOLOGIES

5.2.3 Phraseologies show the text of message components without call signs. They are not intended to be exhaustive, and when circumstances differ, pilots, ATS and Air Defence personnel, and other ground personnel will be expected to use appropriate subsidiary phraseologies which should be clear, concise, and designed to avoid any possible confusion.

5.2.4 For convenience, the phraseologies are grouped according to types of air traffic service. However, users should be familiar with and use, as necessary, phraseologies from groups other than those referring specifically to the type of air traffic service being provided. All phraseologies must be used in conjunction with call signs (aircraft, ground vehicle, ATC or other) as appropriate.

5.2.5 Phraseologies for the movement of vehicles are not listed separately as the phraseology associated with the movement of aircraft is applicable. The exception is for taxi instructions, in which case the word "PROCEED" will be substituted for the word "TAXI" when ATC communicates with vehicles.

5.3 **Transmission Format**

5.3.1 When initiating a transmission to ATS, pilots will commence the transmission with the call sign of the unit being addressed followed by the aircraft call sign.

5.3.2 For aircraft in the Super or Heavy wake turbulence categories, the word 'SUPER' or 'HEAVY' respectively must be included immediately after the aircraft call sign in the initial radiotelephony contact between such aircraft and each approach, departures, director, ground and tower controller.

5.3.3 The ATS unit will respond using the station's call sign followed by their call sign. In the absence of an instruction to "STAND BY", this response by the ATS unit is an invitation for the aircraft calling to pass their message.

Note: The use of the words "GO AHEAD" is no longer considered appropriate due to the possibility of misconstruing "GO AHEAD" as an authorisation for an aircraft to proceed.

5.3.4 A readback of an ATS message will be terminated with the aircraft's call sign.

5.4 **Readback Requirements**

5.4.1 As required by Section 11.12 of the Part 91 MOS, the pilot in command must ensure a readback of the safety-related parts of any ATC clearance or instruction received by voice. Without limiting this requirement, the following parts of an ATC clearance or instruction must always be read back:

a. ATC route clearances, including any amendments;

b. en route holding instructions;

c. route and runway-holding positions specified in a taxi clearance;

d. clearances, conditional clearances and instructions to taxi on, enter, line up on, wait on, land on, take off from, hold short of, cross, or backtrack on, any runway;

e. the assigned runway or HLS, altimeter settings, Mode A transponder codes, data link logon addresses, altitude instructions, heading and speed instructions; and

f. radio frequency instructions.

Notes:

1. *ATC will expect a readback of:*

   a.''VIA SID/STAR TO' instructions;

   b.instructions, or acknowledgement of instructions, to maintain terrain clearance visually i.e. 'VISUAL';

   c.traffic sequencing instructions, including 'FOLLOW'; and

   d.clearances, conditional clearances and instructions relating to an HLS.

2. *ATC route clearances include departure, en route, arrival and approach clearances.*
3. *Other clearances, instructions, or information should be read back in sufficient detail to clearly indicate that they have been understood and will be complied with.*
4. *For subparagraph 5.4.1 (e), 'altitude instructions' includes flight level instructions.*
5. *"Rest of clearance unchanged" or general broadcasts do not need to be read back.*
6. *'Expectation' about a runway, a STAR etc is not a clearance and should not be read back.*
7. *Readback requirements for direct communications also apply when an ATC clearance, information or instruction is relayed from ATC to a flight via another flight.*

5.4.2 The controller will listen to the readback to ascertain that the clearance or instruction has been correctly acknowledged and will take immediate action to correct any discrepancies revealed by the readback.

5.4.3 Reported level figures of an aircraft must be preceded by the words "FLIGHT LEVEL" when related to standard pressure and may be followed by the word "FEET" when related to QNH.

5.5 **Conditional Clearances**

5.5.1 In all cases, a conditional clearance will be given in the following order and consist of:

a. identification (call sign);
b. the condition (including position of the subject of the condition);
c. the clearance; and
d. brief reiteration of the condition.

5.6 **Route Terminology**

5.6.1 The phrase "FLIGHT PLANNED ROUTE" may be used to describe any route or portion thereof that is identical to that filed in the flight notification and sufficient routing details are given to definitely establish the aircraft on its route.

5.7 **Amended Route or Level**

5.7.1 When ATS provide an initial airways clearance that is not in accordance with the details currently held by ATC system, ATS will prefix the route and/or level details with the word "AMENDED".

5.7.2 When an issued airways clearance needs to be changed ATS will prefix the new route and/or level details with the word "RECLEARED". The level will be stated in all clearance changes regardless of whether a change to the cleared level is made or not.

5.7.3 The prefixes "AMENDED" and "RECLEARED" will not be used:

a. for SID or STAR clearances; or

b. during normal progressive climb/descent instructions

5.8 **Language**

5.8.1 English language must be used for all air-ground RTF communications within Australian FIRs unless use of an alternative language has been arranged with ATS prior to any specific flight.

5.9 **Phonetic Alphabet**

5.9.1 When proper names, service abbreviations and words of which the spelling is doubtful are spelled out in radiotelephony, the alphabet shall be used as follows:

Letter/Word	Pronunciation	Letter/Word	Pronunciation
A ALFA	AL fah	**B** BRAVO	BRAH voh
C CHARLIE	CHAR lee or SHAR lee	**D** DELTA	DELL tah
E ECHO	ECK ho	**F** FOXTROT	FOKS trot
G GOLF	GOLF	**H** HOTEL	ho TELL
I INDIA	IN dee A	**J** JULIETT	JEW lee ETT
K KILO	KEY loh	**L** LIMA	LEE mah
M MIKE	MIKE	**N** NOVEMBER	no VEM ber
O OSCAR	OSS cah	**P** PAPA	pah PAH
Q QUEBEC	keh BECK	**R** ROMEO	ROW me oh
S SIERRA	see AIR rah	**T** TANGO	TANG go
U UNIFORM	YOU nee form or OO nee form	**V** VICTOR	VIK tah
W WHISKEY	WISS key	**X** X-RAY	ECKS ray
Y YANKEE	YANG key	**Z** ZULU	ZOO loo

Note: For pronunciation, syllables to be emphasised are in all capitals.

5.10 **Pronunciation of numbers**

5.10.1 Radiotelephony pronunciation of numbers shall be as follows:

0	ZE-RO	**5**	FIFE	**Decimal**	DAY-SEE-MAL
1	WUN	**6**	SIX	**Hundred**	HUN-dred
2	TOO	**7**	SEV-en	**Thousand**	TOU-SAND
3	TREE	**8**	AIT		
4	FOW-er	**9**	NIN-er		

Note: The syllables printed in capital letters in the above list are to be stressed.

5.11 **Transmission of numbers**

Note: See section 5.10 for pronunciation.

5.11.1 All numbers, other than those described in 5.11.2 - 5.11.8, should be transmitted by pronouncing each digit separately.

Element	**Transmission format**
Heading	
100 degrees	HEADING **ONE ZERO ZERO**
080 degrees	HEADING **ZERO EIGHT ZERO**
Wind direction and speed	
200 degrees 70 knots	WIND **TWO ZERO ZERO** DEGREES **SEVEN ZERO** KNOTS
160 degrees 18 knots gusting 30 knots	WIND **ONE SIX ZERO** DEGREES **ONE EIGHT** KNOTS GUSTING **THREE ZERO** KNOTS
Runway	
27	RUNWAY **TWO SEVEN**
30	RUNWAY **THREE ZERO**

5.11.2 Flight levels should be transmitted by pronouncing each digit separately except for the case of flight levels in whole hundreds, which should be transmitted by pronouncing the digit of the whole hundred followed by the word HUNDRED.

Flight levels	**Transmission format**
FL 180	FLIGHT LEVEL **ONE EIGHT ZERO**
FL 200	FLIGHT LEVEL **TWO HUNDRED**

5.11.3 The altimeter setting should be transmitted by pronouncing each digit separately except for the case of a setting of 1,000HPA which must be transmitted as ONE THOUSAND.

Altimeter setting	Transmission format
1,010	QNH **ONE ZERO ONE ZERO**
1,000	QNH **ONE THOUSAND**
993	QNH **NINE NINE THREE**

5.11.4 All numbers used in the transmission of transponder codes should be transmitted by pronouncing each digit separately except that transponder code containing whole thousands should be transmitted by pronouncing the digit in the number of thousands followed by the word THOUSAND.

Transponder codes	Transmission format
2,400	SQUAWK **TWO FOUR ZERO ZERO**
1,000	SQUAWK **ONE THOUSAND**
2,000	SQUAWK **TWO THOUSAND**

5.11.5 All numbers used in the transmission of altitude, cloud height, visibility and RVR, which contain whole hundreds and whole thousands, should be transmitted by pronouncing each digit in the number of hundreds or thousands followed by the word HUNDRED or THOUSAND as appropriate. Combinations of thousands and whole hundreds should be transmitted by pronouncing each digit in the number of thousands followed by the word THOUSAND followed by the number of hundreds followed by the word HUNDRED.

Element	Transmission format
Altitude	
800	**EIGHT HUNDRED**
3,400	**THREE THOUSAND FOUR HUNDRED**
12,000	**ONE TWO THOUSAND**
Cloud height	
2,200	**TWO THOUSAND TWO HUNDRED**
4,300	**FOUR THOUSAND THREE HUNDRED**

Element	Transmission format
Visibility	
1,000	VISIBILITY **ONE THOUSAND**
700	VISIBILITY **SEVEN HUNDRED**
RVR	
600	RVR **SIX HUNDRED**
1,700	RVR **ONE THOUSAND SEVEN HUNDRED**

5.11.6 When providing information regarding relative bearing in terms of the 12-hour clock, the information should be transmitted by pronouncing the double digits as TEN, ELEVEN, or TWELVE [O'CLOCK].

5.11.7 Numbers containing a decimal point should be transmitted with the decimal point in appropriate sequence being indicated by the word DECIMAL. For VHF radio frequencies, only the first five digits of the frequency should be transmitted or acknowledged. However, if both the fifth and - if used - sixth digits are zero, only the first four digits should be transmitted or acknowledged.

Number	Transmitted as
100.3	ONE ZERO ZERO **DECIMAL** THREE
134.325	ONE THREE FOUR **DECIMAL** THREE TWO
38,143.9	THREE EIGHT ONE FOUR THREE **DECIMAL** NINE
29.93 *(inches QNH)*	QNH TWO NINE **DECIMAL** NINE THREE
M0.84 (Mach)	MACH **DECIMAL** EIGHT FOUR

5.11.8 When transmitting time, only the minutes of the hour should normally be required. Each digit should be pronounced separately. However, the hour should be included when any possibility of confusion is likely to result.

Element	Transmitted as
Time	
0920	TIME **TWO ZERO** or **ZERO NINE TWO ZERO**
1643	TIME **FOUR THREE** or **ONE SIX FOUR THREE**
Accurate time check (nearest half minute)	
0945 and 25 seconds	TIME **FOUR FIVE AND A HALF** or **ZERO NINE FOUR FIVE AND A HALF**

5.12 **Aeronautical Station call signs**

5.12.1 **ATS call signs**

ATS units are identified by the name of the location followed by the service available as follows:

En route area control, SIS and FIS	CENTRE
Approach control where provided as a separate function	APPROACH
Departure control where provided as a separate function	DEPARTURES
Surveillance control providing vectors onto final approach	FINAL/DIRECTOR
Aerodrome control or aerodrome and approach control where these services are provided from an aerodrome control tower, e.g. Coffs Harbour	TOWER
Surface movement control	GROUND
Clearance delivery to departing aircraft	DELIVERY
Flight Information Service	FLIGHTWATCH
Aerodrome or Surveillance Flight Information Service	INFORMATION
HF	RADIO

5.12.2 **Other Station call signs**

Other non-ATS stations are identified by the name of the location followed by the service available as follows:

Apron management service	APRON
Certified Air/Ground Radio Service (CA/GRS)	RADIO
Universal communications service (UNICOM)	UNICOM

5.12.3 The name of the location or the service may be omitted provided that satisfactory communication has been established and no risk of misidentification exists.

5.13 **Aircraft call signs**

5.13.1 Improper use of call signs can result in pilots executing a clearance intended for another aircraft. call signs should never be abbreviated on an initial contact or at any time when other aircraft call signs have similar numbers/sounds or identical letters/numbers,
e.g. CHARLIE WHISKEY ZULU - WHISKEY CHARLIE ZULU.

5.13.2 Pilots must be certain that aircraft identification is complete and clearly identified before taking action on an ATC clearance. ATS will use full or abbreviated call signs in accordance with section *5.19*. The pilot may only use an abbreviated call sign when initiated by ATS. When aware of similar/identical call signs, ATS will take action to minimise errors by:

a. emphasising certain numbers/letters,

b. repeating the entire call sign, e.g. QANTAS451 QANTAS451, or

c. repeating the prefix, e.g. QANTAS451 QANTAS, or

d. asking pilots to use a different call sign, either temporarily or for the duration of the flight.

Pilots should use the phrase “VERIFY CLEARANCE FOR (complete call sign)” if doubt exists concerning proper identity.

5.13.3 Civil aircraft pilots may state the aircraft type, model or manufacturer’s name, followed by the alpha numeric characters of the registration number, e.g.
Bonanza CHARLIE ALPHA FOUR
Cherokee ALPHA BRAVO CHARLIE.

5.14 **Flight Number call signs - Using Group Form**

5.14.1 Within Australian airspace, “group form” is the preferred means of transmitting call sign/flight number. Group form should also be used with military and other aircraft using a rootword call sign with numeric suffix.

5.14.2 Group form is the grouping of numbers into pairs, or where a number ending in “00” is spoken in hundreds *(refer para 5.15)*. For three digit numbers, the second and third numbers are grouped. Examples are as follows:

QLINK 122	QLINK ONE TWENTY TWO
QANTAS 1220	QANTAS TWELVE TWENTY
CAR 21	CAR TWENTY ONE
CLASSIC 12	CLASSIC TWELVE
VIRGIN 702	VIRGIN SEVEN ZERO TWO
BIRDDOG 021	BIRDDOG ZERO TWENTY ONE

5.14.3 Pilots and ATS should be aware that the preference to use “group form” does not invalidate any transmissions made in conventional formats. However, to retain the integrity in the communication between ATS and operators, the identification format used should be consistent.

5.14.4 A pilot not using “group form” in establishing communication, but subsequently addressed by ATS in this format, should adopt the use of “group form” for the remainder of the flight in Australian airspace.

5.14.5 There is no additional abbreviated form when using flight number call signs. The airline designator and all digits of the call sign, including leading zeros, must be pronounced.

5.15 **Selection of Aircraft Identification Numbers and Suffixes**

5.15.1 When selecting an aircraft identification number or call sign suffix, operators should avoid using numbers that correlate with:

a. ending in “zero” or “five”, to avoid confusion with headings;

b. potential level utilisation (e.g. 3000, 500, 350 etc);

c. emergency codes (e.g. 7600, 7700 etc); and

d. numerical aircraft types (e.g. 767, 330 etc).

5.15.2 Flight numbers and call sign suffix numbers should be limited to 2 or 3 characters and take into account flight numbers already in use by the operator and other agencies in the intended control environment, operational area or nearby.

5.15.3 Operators with approval to use a two-letter designator (e.g. AN) must use a numerical suffix of at least 2 numbers.

5.16 **Ground Vehicles**

5.16.1 Ground vehicles shall be identified by the type of vehicle; e.g. car, truck, tractor, tug, etc, or an ATS approved format, followed by the assigned vehicle number spoken in group form. e.g-

TRUCK 12	“TRUCK TWELVE”
CAR 23	“CAR TWENTY THREE”.

5.17 **Interchange and Leased Aircraft**

5.17.1 Controllers issue traffic information based on familiarity with airline equipment and colour/markings. When an airline dispatches a flight using another company’s aircraft and the pilot does not advise the terminal ATC facility, the possible confusion in aircraft identification can compromise safety.

5.17.2 Pilots flying an “interchange” or “leased” aircraft, not bearing the normal colours/markings of the company operating the aircraft, should inform the terminal ATC facility (on first contact) of the name of the operating company and aircraft call sign, followed by the company name as displayed on the aircraft, and aircraft type, e.g. VELOCITY THREE ELEVEN, AIR NEW ZEALAND INTERCHANGE (or LEASE), BOEING SEVEN FOUR SEVEN.

5.18 **Unmanned Aerial Vehicles**

5.18.1 Unless assigned a flight number call sign or another approved call sign, operators of Unmanned Aerial Vehicles (UAV) should select an identification based on the aircraft’s manufacturer or model name, using no more than three syllables. Numbers may be added. UAV flight plan identification is detailed in *ENR 1.10 APPENDIX 2*

5.18.2 Communications on any frequency must use the prefix “UNMANNED” before the call sign. When the UAV operation is conducted in controlled airspace ATC may vary this requirement after initial contact.

5.19 **Call signs - Full and Abbreviated Formats**

5.19.1 When establishing two-way communications and for subsequent communications on any frequency, Australian registered aircraft on domestic flights (i.e. within Australian FIR) must use one of the following call signs:

a. for VH-registered aircraft, the last 3 characters of the registration marking (e.g. VH-TQK “TANGO QUEBEC KILO”); or

b. the approved telephony designator of the aircraft operating agency, followed by the last 3 characters of a VH registration marking (e.g. “QLINK TANGO QUEBEC KILO”); or

c. the approved telephony designator of the aircraft operating agency, followed by the flight identification (e.g. “VELOCITY EIGHT FIFTY SIX DELTA”); or

d. for recreation-category aircraft, the aircraft type followed by the last 4 characters of the aircraft’s registration number (e.g. “JABIRU THIRTEEN FORTY SIX”); or

e. for certain special task operations - see *section 5.21*.

5.19.2 When establishing two-way communications on any frequency, Australian registered aircraft on international flights and foreign registered aircraft must use one of the following call signs:

a. the characters corresponding to the registration marking of the aircraft (e.g. N35826 “NOVEMBER THREE FIFTY EIGHT TWENTY SIX”, or VH-TQK “VICTOR HOTEL TANGO QUEBEC KILO”); or

b. the approved telephony designator of the aircraft operating agency, followed by the last 4 characters of the registration marking of the aircraft (e.g. “UNITED FIFTY EIGHT TWENTY SIX”); or

c. the approved telephony designator of the aircraft operating agency, followed by the flight identification (e.g. “SPEEDBIRD FIFTY FIVE”).

Note: The name of the aircraft manufacturer or aircraft model may be used as a radiotelephony prefix to the call sign type mentioned in sub-para 5.19.1a. and 5.19.2a.

5.19.3 After establishing two-way communications, ATS may initiate abbreviated call signs for the type stated in *5.19.2a*. and *5.19.2b*. These call signs may be abbreviated to:

a. the first character of the registration and at least the last 2 characters of the registration marking (e.g. N35826 “NOVEMBER EIGHT TWENTY SIX”, or VH-TQK “VICTOR QUEBEC KILO”);

b. the telephony designator of the aircraft operating agency, followed by at least the last 2 characters of the registration marking (e.g. “UNITED TWENTY SIX”).

Note: For flight planning, all call signs are limited to 7 characters.

5.19.4 In addition to the requirements of *5.19.1* to *5.19.3*, the prefix “HELICOPTER” must be used by rotary wing aircraft when first establishing two-way communications on any frequency (e.g. VH-WSO “HELICOPTER WHISKEY SIERRA OSCAR”).

5.19.5 Civil formation flights of Australian registered aircraft may use the suffix “FORMATION” after one of the following call signs:

a. the registration of the formation leader, e.g. “ALPHA BRAVO CHARLIE FORMATION”, or

b. the approved telephony designator of the formation leader, with or without an alpha or numeric, e.g. “JETSPEED FORMATION” or “JETSPEED ONE FORMATION” or “JETSPEED BRAVO FORMATION”.

5.20 **Registration of Radiotelephony Designators**

5.20.1 Operators wishing to use flight number call signs must obtain approval from Airservices Australia. Application information and materials are available from Airservices Australia website: www.airservicesaustralia.com/about-us/our-services/flight-number-call-signs.

5.21 **Call signs - Special Task Operations**

5.21.1 Aircraft engaged in special task operations may use a call sign indicative of the nature of the task with a numerical suffix (if applicable) (see *5.15* and *5.21.2*). The following table specifies the available special task designators and call signs. To obtain approval to use a special task call sign and to register a numerical suffix, contact Airservices Australia by email: satss@airservicesaustralia.com.

Type of Operation	Radio Telephony Designator (call sign)	Flight Plan Designator
Ambulance	AMBULANCE	AM
Coordination of Firebombing Aircraft	BIRDDOG	BDOG
Fire Bombing	BOMBER	BMBR

Type of Operation	Radio Telephony Designator (call sign)	Flight Plan Designator
Powerline and pipeline survey and construction	ENERGY	ENRG
Federal Police	FEDPOL	FPL
Federal Police (Priority)	FEDPOL RED	FPLR
Night-time NVIS firefighting operations	FIREAIR	FYRA
General Fire Support Tasks (light rotary)	FIREBIRD	FBIR
Remote Sensing Fire Operations	FIRESCAN	FSCN
Fire Intelligence Gathering	FIRESPOTTER	SPTR
General Fire Support Tasks (medium rotary)	HELITAK	HLTK
Gliding operations	GLIDER	GLDR
Lifesaver Operations	LIFESAVER	LIFE
Media Operations	MEDIA	MDIA
Validation of instrument procedures	NAVCHECK	NVCK
Parks and Wildlife Service	PARKAIR	PKAR
Police	POLAIR	POL
Police (Priority)	POLAIR RED	POLR
Rescue Mission	RESCUE	RSCU
Aerial Survey (mapping, geo-graphical, etc.)	SURVEY	SVY

5.21.2 To ensure aircraft transiting state borders utilising the same call sign prefix do not duplicate an existing call sign suffix number or flight plan, call sign suffix numbers are allocated as follows:

NSW/ACT – commencing with 2 (e.g. 201, 214, 223)

VIC – commencing with 3 **QLD** – commencing with 4

SA – commencing with 5 **WA** – commencing with 6

TAS – commencing with 7 **NT** – commencing with 8

Defence – commencing with 9

Gliding operations - alphanumeric characters

6. PHRASEOLOGIES

6.1 Standard Words and Phrases

6.1.1 The following words and phrases are to be used in radiotelephony communications, as appropriate, and have the meaning given:

Word/Phrase	Meaning
ACKNOWLEDGE	Let me know that you have received and understood this message.
AFFIRM	Yes.
APPROVED	Permission for proposed action granted.
BREAK	I hereby indicate the separation between portions of the message (to be used where there is no clear distinction between the text and other portions of the message).
BREAK BREAK	I hereby indicate separation between messages transmitted to different aircraft in a very busy environment.
CANCEL	Annul the previously transmitted clearance.
CHECK	Examine a system or procedure (no answer is normally expected).
CLEARED	Authorised to proceed under the conditions specified.

Word/Phrase	Meaning
CONFIRM	I request verification of: (clearance, instruction, action, information).
CONTACT	Establish communication with...
CORRECT	True *or* Accurate.
CORRECTION	An error has been made in this transmission (or message indicated) the correct version is...
DISREGARD	Ignore.
HOW DO YOU READ	What is the readability of my transmission? The readability scale is: 1. Unreadable 2. Readable now and then 3. Readable but with difficulty 4. Readable 5. Perfectly readable.
I SAY AGAIN	I repeat for clarity or emphasis.
MAYDAY	My aircraft and its occupants are threatened by grave and imminent danger and/or I require immediate assistance.
MAINTAIN	Continue in accordance with the condition(s) specified or in its literal sense, e.g. “Maintain VFR”.
MONITOR	Listen out on (frequency).
NEGATIVE	No *or* Permission is not granted *or* That is not correct *or* Not capable.
OUT	This exchange of transmissions is ended and I expect no response from you (*not normally used in VHF or satellite communication*).
OVER	My transmission is ended and I expect a response from you (*not normally used in VHF or satellite communication*).

Word/Phrase	Meaning
PAN PAN	I have an urgent message to transmit concerning the safety of my aircraft or other vehicle or of some person on board or within sight but I do not require immediate assistance.
READBACK	Repeat all, or the specified part, of this message back to me exactly as received.
RECLEARED	A change has been made to your last clearance and this new clearance supersedes your previous clearance or part thereof.
REPORT	Pass me the following information.
REQUEST	I should like to know or I wish to obtain.
ROGER	I have received all of your last transmission (*under NO circumstances to be used in reply to a question requiring READBACK or a direct answer in the affirmative or negative*).
SAY AGAIN	Repeat all or the following part of your last transmission
SPEAK SLOWER	Reduce your rate of speech.
STANDBY	Wait and I will call you.
UNABLE	I cannot comply with your request, instruction or clearance *(normally followed by a reason).*
VERIFY	Check and confirm with originator.
WILCO	I understand your message and will comply with it.

Word/Phrase	Meaning
WORDS TWICE	a. as a request: Communication is difficult. Please send every word or group of words twice. b. as information: Since communication is difficult every word or group of words in this message will be sent twice.

6.2 **Emergency**

Circumstances	***Phraseologies*** ** Denotes pilot transmission*
1. Distress message	a.* MAYDAY [MAYDAY, MAYDAY] *followed as necessary by:* (i) *(station addressed)* (ii) *(aircraft identification)* (iii) *(nature of distress condition e.g. FUEL or EMERGENCY DESCENT)* (iv) *(intentions)* (v) *(position, level and heading)* (vi) *(any other useful information).*
2. Acknowledgement of distress message	
ATC acknowledgement of MAYDAY call	a. ROGER MAYDAY
ATC acknowledgement of MAYDAY on frequency transfer	b. MAYDAY [*(type of emergency)*] ACKNOWLEDGED
Imposition of radio silence	c. STOP TRANSMITTING. MAYDAY

Circumstances		*Phraseologies* ** Denotes pilot transmission*	
	ATC broadcast for emergency descent traffic	d.	EMERGENCY DESCENT AT *(significant point or location)* ALL AIRCRAFT BELOW *(level)* WITHIN *(distance)* OF *(significant point or navigation aid)* [LEAVE IMMEDIATELY] [*(specific instructions as to direction, heading or track, etc)*]
	Cancellation of distress condition	e.*	CANCEL DISTRESS *(information)*
	Termination of distress and radio silence	f.	DISTRESS TRAFFIC ENDED
3.	Urgency message	a*.	PAN PAN [PAN PAN, PAN PAN] *followed as necessary by:* (i) *(station addressed)* (ii) *(aircraft identification)* (iii) *(nature of urgency condition e.g. MEDICAL PRIORITY REQUIRED or WEATHER DEVIATION REQUIRED)* (iv) *(intentions)* (v) *(position, level and heading)* (vi) *(any other useful information).*
	ATC acknowledgement of PAN call	b.	ROGER PAN
	ATC acknowledgement of PAN on frequency transfer	c.	PAN [*(type of emergency)*] ACKNOWLEDGED

6.3 Traffic Alert and Collision Avoidance System (TCAS), Safety Alerts and Avoiding Action and Wind Shear Escape

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Level Changes, Reports/ Rates	
a flight crew starts to deviate from any ATC clearance or instruction to comply with an ACAS resolution advisory (RA) (pilot and controller interchange)	a.* TCAS RA b. ROGER
after the response to an ACAS RA is completed and a return to the ATC clearance or instruction is Initiated (pilot and controller interchange)	c.* CLEAR OF CONFLICT RETURNING TO (*assigned clearance*) d. ROGER *(or alternative instructions*)
after the response to an ACAS RA is completed and the assigned ATC clearance or instruction has been resumed (Pilot and Controller interchange)	e.* CLEAR OF CONFLICT (*assigned clearance*) RESUMED f. ROGER (*or alternative instructions*)
after an ATC clearance or instruction contradictory to the ACAS RA is received, the flight crew will follow the RA and inform ATC directly (Pilot and Controller interchange)	g.* UNABLE TO COMPLY, TCAS RA h. ROGER *Note: Pilots are required to comply with any TCAS RA manoeuvre irrespective of ATC traffic advisories or instructions*

Circumstances	*Phraseologies* * Denotes pilot transmission
2. Safety Alert and Avoiding Action	a. SAFETY ALERT, *followed as necessary by*:
low altitude warning	(i) LOW ALTITUDE WARNING, CHECK YOUR ALTITUDE IMMEDIATELY, QNH IS (*number*) [(*units*)]. [THE MINIMUM SAFE ALTITUDE IS (*altitude*)].
terrain alert	(ii) TERRAIN, CHECK YOUR ALTITUDE IMMEDIATELY (*suggested pilot action if possible*)
traffic alert	(iii) TRAFFIC (*number*) MILES OPPOSITE DIRECTION/ CROSSING LEFT TO RIGHT/ RIGHT TO LEFT (*level information*).
unauthorised entry into active Restricted Area or Military Operating Area has occurred or is imminent	(iv) RESTRICTED AREA (*or* MILITARY OPERATING AREA) [(*number*)] ACTIVE. [SUGGEST] TURN LEFT/ RIGHT IMMEDIATELY HEADING *(three digits)*
	(v) RESTRICTED AREA (*or* MILITARY OPERATING AREA) [(*number*)] ACTIVE. [SUGGEST] CLIMB/DESCEND IMMEDIATELY TO *(level)*
when unauthorised deviation into Restricted Area or Military Operating Area is unavoidable	(vi) UNABLE TO ISSUE CLEARANCE. PROCEED AT YOUR OWN RISK, SQUAWK 7700.

Circumstances	*Phraseologies* * Denotes pilot transmission
avoiding action	b. AVOIDING ACTION, *followed as necessary by:* (i) [SUGGEST] TURN LEFT/ RIGHT IMMEDIATELY HEADING (*three digits*) TRAFFIC ([LEFT/RIGHT] *number*) O'CLOCK (*distance*) MILES OPPOSITE DIRECTION/ CROSSING LEFT TO RIGHT/ RIGHT TO LEFT (*level information*). (ii) [SUGGEST] CLIMB/ DESCEND IMMEDIATELY TO (*level*) TRAFFIC [LEFT/ RIGHT] (*number*) O'CLOCK (*distance*) MILES OPPOSITE DIRECTION/ CROSSING LEFT TO RIGHT/ RIGHT TO LEFT (*level information*). *Note 1: Where clock codes are used to provide the relative bearing, the prefix left/right is optional.* *Note 2: In high density traffic scenarios it may be impractical for ATC to use the full phraseologies for safety alerts and avoiding action. ATC will provide information that conveys the immediacy of the situation and relevant instructions to allow pilots the best opportunity to avoid a collision.* *Note 3: Pilots are required to comply with any TCAS RA manoeuvre irrespective of ATC traffic advisories or instructions.*

Circumstances	*Phraseologies* * Denotes pilot transmission
3. Wind Shear Escape Manoeuvre	•a.* WIND SHEAR ESCAPE
ATC acknowledgement	b. ROGER [SAFETY ALERT] [TRAFFIC (*distance*) MILES (*relevant information*)]
Mutual traffic information	c. [SAFETY ALERT] TRAFFIC (*distance*) MILES (*relevant information*) EXPERIENCING WIND SHEAR
Wind Shear Escape Manoeuvre complete	d. * CLEAR OF WIND SHEAR RETURNING TO (*assigned clearance, instruction and/or procedure etc*)
ATC acknowledgement	e. ROGER [*alternative instructions*]
Wind shear prevents compliance with an ATC clearance or instruction	f. * UNABLE, WIND SHEAR ESCAPE
4. Wind Shear or Microburst Alert	a. RUNWAY (*number*) ARRIVAL (*or* DEPARTURE) WIND SHEAR (*or* MICROBURST) ALERT (*windspeed*) KNOT GAIN (*or* LOSS) (*location*)

6.4 Status of Restricted Areas and Military Operating Areas

Circumstances	*Phraseologies* * Denotes pilot transmission
When active:	a. RESTRICTED AREA (*or* MILITARY OPERATING AREA) (*number*) ACTIVE, *followed as necessary by*: (i) CLEARANCE REQUIRED (ii) AVAILABLE FOR TRANSIT (iii) AVAILABLE UNTIL TIME *(time)* (iv) CLEARANCE NOT AVAILABLE (v) *...(other qualification as appropriate)*

Circumstances	*Phraseologies* * Denotes pilot transmission
Released to civil ATC	b. RESTRICTED AREA (*or* MILITARY OPERATING AREA) (*number*) RELEASED TO (*civil* ATS *unit*), *followed as necessary by*: i) CLEARANCE NOT AVAILABLE ii) (*clearance*) iii) APPROVED TO OPERATE IN RESTRICTED AREA (*or* MILITARY OPERATING AREA) (*number*) [CLASS (*airspace category*) PROCEDURES APPLY]

6.5 **SARWATCH**

6.5.1 **SARTIME**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. SARTIME nomination	a.* SARTIME *details* b. STANDBY *or* (*call sign*) c.* SARTIME FOR DEPARTURE (*or* ARRIVAL) [*location*] (*time*)
2. SARTIME cancellation	a.* SARTIME *details* b. STANDBY *or* (*call sign*) c.* (*position*/*location*) CANCEL SARTIME
3. SARTIME amendment	a.* SARTIME *details* b. STANDBY *or* (*call sign*) c. *As required, including specific phrases nominated above if applicable*.

6.5.2 SARWATCH Other Than SARTIME

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Departure Reports to initiate a SARWATCH when communication on the ground is not available.	•a.* AIRBORNE (*location*)
2. Flight & Arrival Reports	•a.* (*position*) CANCEL SARWATCH [ADVISE (*unit*) *if appropriate*]
	•b. SARWATCH CANCELLED [WILCO (*unit*)]
form of acknowledgement to CANCEL SARWATCH	•c. [*location*] SARWATCH TERMINATED
when the ATS unit accepting the arrival report is other than the unit addressed	•d. ROGER (*identity of unit acknowledging*)

6.6 General Phrases

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Description of Levels (subsequently referred to as "(*level*)"	a. FLIGHT LEVEL (*number*) *or* b. (*number*) [FEET]
2. Level Instructions	a. CLIMB (*or* DESCEND) *followed as necessary by:*
when there is an expectation that the aircraft will maintain the level or to eliminate confusion, the instruction "AND MAINTAIN" shall be included	(i) TO (*level*) (ii) TO AND MAINTAIN (*level*) (iii) TO REACH (*level*) AT (*or* BY) (*time or significant point*) (iv) TO (*level*) REPORT LEAVING (*or* REACHING *or* PASSING *or* APPROACHING) (*level*) (v) AT (*number*) FEET PER MINUTE [MINIMUM (*or* MAXIMUM)]

Circumstances	*Phraseologies* * Denotes pilot transmission
when rate is required to be in accordance with "STANDARD RATE" specifications	(vi) AT STANDARD RATE
when advising expectation of a level restriction	b. EXPECT A RESTRICTION TO REACH (*level*) BY *(time or position) followed as necessary by (a)*
	c. STEP CLIMB (*or* DESCENT) (*aircraft identification*) ABOVE (*or* BENEATH) YOU
	d. REQUEST LEVEL CHANGE FROM (*name of unit*) AT (t*ime or significant point*)
	e. STOP CLIMB (*or* DESCENT) AT (*level*)
	f. CONTINUE CLIMB (*or* DESCENT) TO [AND MAINTAIN] (*level*)
	g. EXPEDITE CLIMB (*or* DESCENT) [UNTIL PASSING (*leve*l)]
	h. EXPECT CLIMB (*or* DESCENT) AT (*time or location*)
pilot requesting a change of level	i.* REQUEST CLIMB (*or* DESCENT) [AT (*time or location*)] [TO (*level*)]
to require action at a specific time or place	j. IMMEDIATELY
	k. AFTER PASSING (*significant point*)
	l. AT (*time or significant point*)
to require action when convenient	m. WHEN READY (*instruction*)
when a pilot is unable to comply with a clearance or instruction	n.* UNABLE TO COMPLY
when a descent clearance is issued in relation to the DME (or GNSS) steps	•o. DESCEND TO (*level*) NOT BELOW DME (*or* GNSS) STEPS

Circumstances	*Phraseologies* * Denotes pilot transmission
when a pilot is assigned and required to maintain separation with a sighted aircraft	•p. FOLLOW (*or* MAINTAIN OWN SEPARATION WITH [AND PASS BEHIND]) (*aircraft type or identification*) [*instructions or restriction*]
ATC requesting confirmation of equipment, capability or approval e.g. RVSM, ADS-B, PRM.	•q.* CONFIRM (*equipment, capability or approval*) APPROVED (*or* EQUIPPED)
Pilot report of equipment, capability or approval status	•r.* AFFIRM (*or* NEGATIVE) (*equipment, capability or approval*) *(reason if applicable)*
Pilot of IFR flight requests to climb/descend VFR	s.* REQUEST VFR CLIMB (*or* DESCENT) [TO (*level*)]
	t.* CLIMB (*or* DESCEND) VFR *to* (*level*)
IFR separation is available for part of the climb/descent	u. CLIMB (*or* DESCEND) [TO (*assigned level*)] *followed as necessary by*: (i) CLIMB (*or* DESCEND) VFR BETWEEN (*level*) AND (*level*)] (ii) CLIMB (*or* DESCEND) VFR BELOW (*or* ABOVE) (*level*)
3. NVIS Operations	
Pilots requesting operations not above a published or pilot calculated LSALT using NVIS.	a.* REQUEST NOT ABOVE (*altitude*) (PILOT CALCULATED [or PUBLISHED] LOWEST SAFE) NVIS b. CLIMB (*or* DESCEND) TO (or OPERATE NOT ABOVE) (*altitude*) [PILOT CALCULATED LOWEST SAFE] NVIS
Pilots requesting operations at (or not above) a level which is below the published or pilot calculated LSALT using NVIS.	c.* REQUEST [NOT ABOVE] (*altitude*) NVIS

Circumstances	***Phraseologies*** * Denotes pilot transmission
	d. CLIMB (*or* DESCEND) TO (*or* OPERATE NOT ABOVE) (*altitude*) NVIS
When climbing to regain LSALT/MSA other than in accordance with assigned airways clearance	e.* CLIMBING TO (*level*), (*reason e.g.* NVIS *failure or inadvertent* IMC)
4. Pilots without NVIS requesting operations below the published LSALT at night or in IMC.	a.* REQUEST (DESCENT *or* CLIMB *or* OPERATIONS) [AT (NOT ABOVE)] *(altitude)* OWN TERRAIN CLEARANCE
	b. CLIMB TO (*or* DESCEND TO *or* OPERATE) [AT (NOT ABOVE)] *(altitude)* MAINTAIN OWN TERRAIN CLEARANCE
5. Maintenance of Specified Levels *Note: The term "MAINTAIN" must not to be used in lieu of "DESCEND" or "CLIMB" when instructing an aircraft to change level*	a. MAINTAIN (*level*) [TO (*significant point*)] [*condition*]
6. Use of Block Levels	•a.* REQUEST BLOCK LEVEL (*level*) TO (*level*)
	•b.* CLIMB (*or* DESCEND) TO AND MAINTAIN BLOCK (*level*) TO (*level*)
established in the level range	•c. MAINTAIN BLOCK (*level*) TO (*level*)
cancelling block level clearance	•d. CANCEL BLOCK CLEARANCE. CLIMB (*or* DESCEND) TO AND MAINTAIN (*level*)

Circumstances	*Phraseologies* * Denotes pilot transmission
7. Specification of Cruising Levels	a. CROSS (*significant point*) AT (*or* ABOVE, *or* BELOW) (*level*) b. CROSS (*significant point*) AT (*time*) OR LATER (*or* BEFORE) AT (*level*)
reply to cruise climb request	c. CRUISE CLIMB NOT AVAILABLE [*reason*]
8. Where an aircraft operation requires random climb and descent at and below (or at and above) a specified level.	•a. OPERATE NOT ABOVE (*or* BELOW) (*level*)
9. Termination of Identification and Control Services or Control Services only	a. [IDENTIFICATION AND] CONTROL SERVICE TERMINATED *followed as necessary by:* i) [DUE (*reason*)] ii) (*instructions)* iii) FREQUENCY CHANGE APPROVED
10. When instructing an aircraft to turn 180° or more when tracking instructions follow	a. TURN LEFT (*or* RIGHT) - I SAY AGAIN - LEFT (*or* RIGHT) *[tracking instructions]*

6.7 Frequency Management

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Transfer of Control and/or Frequency Change *Note: An aircraft may be requested to "STAND BY" on a frequency when the intention is that the ATS unit will initiate communications, and to "MONITOR" a frequency when information is being broadcast thereon.*	a. CONTACT (*unit call sign*) (*frequency*) b.* (*frequency*) c. AT (*or* OVER) (*time or place*) CONTACT (*unit call sign*) (*frequency*) d. IF NO CONTACT (*instructions*) e.* REQUEST CHANGE TO (*frequency*) (*service*) f. FREQUENCY CHANGE APPROVED

Circumstances	*Phraseologies* * Denotes pilot transmission	
pilot requesting to maintain radio silence for a specific time or event (e.g. fuel dump)	g.*	REQUEST TO MAINTAIN RADIO SILENCE DUE *(reason)* [UNTIL *(time)*]
	h.	MONITOR (*unit call sign*) (*frequency*)
	i.*	MONITORING (*frequency*)
	j.	REMAIN THIS FREQUENCY
nominating scheduled reporting times	k.	REPORT *followed as necessary by:* (i) *(situation)* (ii) AT (iii) BY (iv) TIME *(time)*
	l.	STAND BY FOR (*unit call sign*) (*frequency*)
an IFR pilot changing to the CTAF	•m.*	CHANGING TO (*location*) CTAF (*frequency*)
a pilot contacting next frequency when on a heading	n.*	HEADING (*as previously assigned*)
when a pilot/ATC broadcasts general information	o.*	ALL STATIONS (*appropriate information*)
when a pilot broadcasts location specific general information	•p.*	(*location*) TRAFFIC (*appropriate information*) (*location*)
notifying wake turbulence category to approach, departures, director, ground and tower	q.*	SUPER (*or* HEAVY)
ATC acknowledgment	r.	SUPER (*or* HEAVY)

Circumstances	***Phraseologies*** * Denotes pilot transmission
2. Flights Contacting Approach Control not identified or procedural tower	•a.(i)* (*distance*) MILES (GNSS *or* DME) [FROM] (*aerodrome*) (ii) (*GNSS track*) TRACK ((*or* (VOR *radial*) RADIAL) *or* (STAR designator) *or* (*compass quadrant from aerodrome)*) (iii) MAINTAINING (*or* DESCENDING TO) (*level*) (iv) VISUAL *if visual approach can be made* (v) INFORMATION (ATIS *identification*)
3. Change of call sign to instruct an aircraft to change call sign to advise an aircraft to revert to the call sign indicated in the flight notification to ATS	a. CHANGE YOUR CALL SIGN TO (*new call sign*) [UNTIL FURTHER ADVISED] b. REVERT TO FLIGHT PLAN CALL SIGN (*call sign*) (AT (*significant point*))
4. After landing	a. CONTACT GROUND [*frequency*] b. WHEN VACATED CONTACT GROUND [*frequency*]
5. To request a station relay a clearance or information to a third party	FOR [RELAY TO] *(third party call sign) (clearance or information)*

6.8 Traffic Information

Circumstances		*Phraseologies* * Denotes pilot transmission
1.	Traffic Information	
	pilot request for traffic information	a.* REQUEST TRAFFIC
	to pass traffic information	•b. NO REPORTED (IFR) TRAFFIC
		c. [IFR] TRAFFIC (*relevant information*) [REPORT SIGHTING]
		d. [ADDITIONAL] [IFR] TRAFFIC (*direction*) BOUND (t*ype of aircraft*) (*level*) ESTIMATED (*or* OVER) (*significant point*) AT (*time*)
	to acknowledge traffic information	e.* LOOKING
		f.* TRAFFIC IN SIGHT
		g.* NEGATIVE CONTACT (*reasons*)
	interception of relevant traffic information transmitted by other aircraft or ATS facility	•h.* COPIED (*call sign of traffic intercepted*)
2.	Advice of Military Aircraft Conducting Abrupt Vertical Manoeuvres	•a. ABRUPT VERTICAL MANOEUVRES AT (*position*) UP TO (*level*)
3.	Advice of Military Low Jet Operations Known to be Taking Place	•a. MILITARY LOW JET OPERATIONS (*relevant information*)

6.9 Meteorological Information

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Meteorological Conditions *Note: Wind is always expressed by giving the mean direction and speed and any significant variations.*	a. [THRESHOLD] WIND (*number*) DEGREES (*number*) KNOTS b. WIND AT (*height/altitude/flight level*) (*number*) DEGREES (*number*) KNOTS c. WIND AT UP WIND END (*number*) DEGREES (*number*) KNOTS d. VISIBILITY (*distance*) (*direction*) e. RUNWAY VISUAL RANGE (RVR) *or* RUNWAY VISIBILITY (RV) [RUNWAY (*number*)] (*distance*) (*for* RV *assessments* – ASSESSED AT (*time*))
During RVR/RV operations where an assessment is not available or not reported.	f. RUNWAY VISUAL RANGE (RVR) *or* RUNWAY VISIBILITY (RV) [RUNWAY (*number*)] NOT AVAILABLE (*or* NOT REPORTED).
Where multiple RVR/RV observations are made. *Note 1. Multiple RVR/RV observations are always representative of the touchdown zone, midpoint zone and the roll-out/stop end zone, respectively.* *Note 2. Where reports for three locations are given, the indication of these locations may be omitted, provided that the reports are passed in the order of touchdown zone, followed by the midpoint zone and ending with the roll-out/ stop end zone report.*	g. RUNWAY VISUAL RANGE (RVR) *or* RUNWAY VISIBILITY (RV) RUNWAY (*number*) (*first position*) (*distance*) (*units*), (*second position*) (*distance*) (*units*), (*third position*) (*distance*) (*units*) (*for* RV *assessments* – ASSESSED AT (*time*))

Circumstances	***Phraseologies*** * Denotes pilot transmission
When RVR/RV information on any one position is not available this information will be included in the appropriate sequence.	h. RUNWAY VISUAL RANGE (RVR) *or* RUNWAY VISIBILITY (RV) RUNWAY (*number*) (*first position*) (*distance*) (*units*), (*second position*) NOT AVAILABLE, (*third position*) (*distance*) (*units*) (*for* RV *assessments* – ASSESSED AT (*time*))
	i. PRESENT WEATHER (*details*)
	j. CLOUD (*amount,* [*type*] *and height of base*) (*or* SKY CLEAR)
	k. CAVOK
	l. TEMPERATURE [MINUS] (*number*) (*and/or* DEWPOINT [MINUS] (*number*))
	m. QNH (*number*) (*units*)
	n. MODERATE (*or* SEVERE) ICING (*or* TURBULENCE) [IN CLOUD] (*area*)
	o. REPORT FLIGHT CONDITIONS
unless responding to a request for turbulence or icing information	•p.* IMC (*or* VMC)

6.10 **Reports and Information**

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Position Reporting *Note: Phrases for use in en route position and MET reports are listed in ENR 1.1 APPENDIX 1.*	a. NEXT REPORT AT (*significant point*)
2. Additional Reports	a. REPORT PASSING (*significant point*)
to request a report at a specified place or distance	b. REPORT (*distance*) MILES [GNSS (*or* DME)] FROM (*significant point*)
	c. REPORT PASSING (*three digits*) RADIAL (*name of VOR*) VOR
to request a report of present position	d. REPORT [GNSS (*or* DME)] DISTANCE FROM (*significant point*)
when descending a non-DME equipped aircraft to LSALT above CTA steps	•e. REPORT PASSING CONTROL AREA STEPS FOR FURTHER DESCENT
the pilot will give this only when satisfied that the CTA step has been passed, allowing for navigational tolerances.	•f.* INSIDE (*distance of a CTA step*) MILES
3. GNSS tracking	a. CONFIRM (*or* REPORT) ESTABLISHED ON THE [(*three digits*)] GNSS TRACK [BETWEEN (*significant point*) AND (*significant point*)]
	b. MAINTAIN TRACK BETWEEN (*significant point*) AND (*significant point*). REPORT ESTABLISHED ON THE TRACK
	c. * ESTABLISHED ON THE [(*three digits*)] TRACK [BETWEEN (*significant point*) AND (*significant point*)]

Circumstances	*Phraseologies* * Denotes pilot transmission
4. GNSS navigation	a. CONFIRM GNSS NAVIGATION
	b. * AFFIRM GNSS NAVIGATION
GNSS unavailable	c. * GNSS UNAVAILABLE [DUE TO (*reason e.g.* LOSS OF RAIM *or* RAIM ALERT)]
Resuming GNSS operation	d. * GNSS AVAILABLE [DUE TO (*reason*)]
5. Aerodrome Information *Note 1: See AD 1.2 Section 3 for information about runway condition reports.*	a. RUNWAY (*number*) SURFACE CONDITION [CODE (three-digit number)] followed as necessary by:
	(i) ISSUED AT (date and time UTC)
Note: Additional surface descriptors may apply in countries with polar weather conditions.	(ii) WET [or DRY, or SLIPPERY WET, or STANDING WATER, or FROST, or DRY SNOW, or WET SNOW or SLUSH]
Only for contaminants	(iii) DEPTH ((*depth of deposit*) MILLIMETERS or NOT REPORTED)
	(iv) COVERAGE ((number) PER CENT or NOT REPORTED)
Taxiway conditions only reported if operationally significant.	(v) TAXIWAY (identification of taxiway) POOR
Apron conditions only reported if operationally significant.	(vi) APRON (identification of apron) POOR
Where appropriate, any additional significant operational information about the surface conditions	(vii) *Plain language remarks*
D	b. [(location)] RUNWAY SURFACE CONDITION RUNWAY (number) NOT CURRENT
	c. LANDING SURFACE (*condition*)
	d. CAUTION (WORK IN PROGRESS) (OBSTRUCTION) (*position and any necessary advice*)

Circumstances	*Phraseologies* * Denotes pilot transmission
	e. BRAKING ACTION REPORTED BY (*aircraft type*) AT (*time*) GOOD (or GOOD TO MEDIUM, or MEDIUM, or MEDIUM TO POOR or POOR, or LESS THAN POOR) f. TAXIWAY WET [or STANDING WATER, FROST, or DRY SNOW, or WET SNOW or SLUSH] g. TOWER OBSERVES (weather information) h. PILOT REPORTS (weather information)
6. Information to Aircraft wake turbulence jet blast on apron or taxiway propeller-driven aircraft slipstream helicopter downwash	a. CAUTION (i) WAKE TURBULENCE (ii) JET BLAST (iii) SLIPSTREAM (iv) DOWNWASH
7. Pilot Initiated Waiver of Wake Turbulence Separation Standards	•a.* ACCEPT WAIVER
8. ATS relay of information from aircraft operator to flight crew	•a. YOUR COMPANY ADVISES (*information*)

6.11 Clearances

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Clearances	
	a.* REQUEST CLEARANCE
If the route and/or level issued in the initial airways clearance is not in accordance with the flight plan	b. CLEARED TO c. CLEARED TO (*destination*) [AMENDED ROUTE] (*route clearance details*) [AMENDED LEVEL] *(level)*
If an airways clearance is amended en route	d. RECLEARED (*amended clearance details*) [REST OF CLEARANCE UNCHANGED] [(*level*)] e. RECLEARED (*amended route portion*) TO (*significant point of original route*) [REST OF CLEARANCE UNCHANGED] (level)
when the clearance is relayed by a third party; e.g. pilot/ FLIGHT WATCH (ATC excepted)	f. (*name of unit*) CLEARS (*aircraft identification*)
when clearance will be issued subject to a delay	g. REMAIN OUTSIDE CLASS (*airspace class*/es) AIRSPACE (*or* RESTRICTED AREA *or* MILITARY OPERATING AREA [(*number*)]) AND STANDBY
when clearance will be issued at a specified time or place	h. REMAIN OUTSIDE CLASS (*airspace class/es*) AIRSPACE (*or* RESTRICTED AREA *or MILITARY OPERATING AREA* [(*number*)]) EXPECT CLEARANCE AT (*time*/*place*)

Circumstances	*Phraseologies* * Denotes pilot transmission
when a clearance will not be available	i. CLEARANCE NOT AVAILABLE, REMAIN OUTSIDE CLASS (*airspace class/es*) AIRSPACE (*or* RESTRICTED AREA *or MILITARY OPERATING AREA* [(*number*)])
when requesting a deviation from cleared route	j.* REQUEST TO DEVIATE UP TO (*distance*) MILES LEFT (*or* RIGHT) OF ROUTE DUE (*reason*)
when requesting a deviation from cleared track	k.* REQUEST TO DEVIATE UP TO (*distance*) MILES LEFT (*or* RIGHT) OF TRACK DUE (*reason*)
when a deviation from cleared route or track is requested	•l. DEVIATE UP TO (*distance*) MILES LEFT (*or* RIGHT) OF ROUTE (*or* TRACK)
when clearance for deviation cannot be issued	m. UNABLE, TRAFFIC (*direction*) BOUND (*type of aircraft*) (*level*) ESTIMATED (*or* OVER) (*significant point*) AT (*time*) CALL SIGN (*call sign*) ADVISE INTENTIONS
when a weather deviation has been completed and onwards clearance is requested	n. *CLEAR OF WEATHER [REQUEST (*route clearance*)]
when a weather deviation has been completed and the aircraft has returned to its cleared route	o. * BACK ON ROUTE (*or* TRACK)
when subsequent restrictions/ requirements are imposed in addition to previous restrictions to be complied with	•p. FURTHER RESTRICTION
	q. [RE] ENTER CONTROLLED AIRSPACE (*or* CONTROL ZONE) [VIA (*significant point or route*)] AT (*level*) [AT (*time*)]

Circumstances	*Phraseologies* * Denotes pilot transmission
	r. LEAVE CONTROLLED AIRSPACE (*or* CONTROL ZONE) [VIA (*significant point or route*)] AT (*level*) (*or* CLIMBING*, or* DESCENDING) s. LEAVE AND RE-ENTER CONTROLLED AIRSPACE AT (*level*) (*or* CLIMBING TO (*level*), *or* DESCENDING TO (*level*) *or* ON (*type of approach*)) t. JOIN (*specify*) AT (*significant point*) AT (*level*) [AT (*time*)]
2. Indication of Route and Clearance Limit	a. FROM (*location*) TO (*location*) b. TO (*location*) *followed as necessary by*: (i) DIRECT (ii) VIA (*route and/or significant points*) (iii) FLIGHT PLANNED ROUTE (iv) VIA (*distance*) DME ARC (*direction*) OF (*name of DME station*) c. (*level or route*) NOT AVAILABLE DUE (*reason*) ALTERNATIVE[S] IS/ARE (*levels or routes*) ADVISE
issuing a specific clearance limit	•d. CLEARANCE LIMIT (*place/aid*)
when a pilot requests, or ATC issues a visual departure in lieu of a SID	•e. [*clearance details*] VISUAL DEPARTURE
3. When a Clearance has been Cancelled	•a. CANCEL CLEARANCE •b.* CANCEL CLEARANCE
4. Change of Flight Rules	
canceling IFR (to be initiated only by the pilot)	•a.* CANCEL IFR, [REQUEST] (*intention*)

Circumstances	*Phraseologies* * Denotes pilot transmission
changing from VFR to IFR	•b. IFR CANCELLED, OPERATE VFR (*instruction or clearance*) •c.* CHANGE OF FLIGHT RULES, REQUEST IFR [CLEARANCE] [AT (*time or place*)] (IFR *level*) •d. [AT (*time or place*)] OPERATE IFR, [CLEARED] (*clearance or instructions*)
5. Requesting Clearance	
when notification of flight details has not been submitted to ATS	•a.* FLIGHT DETAILS [INBOUND *or* FOR (DEPARTURE *or* TRANSIT)]
flight details to be passed after ATS response	•b.* (*Aircraft type*) (*position*) (*route in controlled airspace and next estimate*) (*preferred level*)
if clearance cannot be issued immediately upon request)	•c. EXPECT CLEARANCE AT (*time or place*)
if giving warning of clearance requirement	•d.* EXPECT CLEARANCE REQUEST (*aircraft type*) VFR (*if appropriate*) FOR (*destination*) VIA (*point outside controlled airspace at which clearance will be requested*) ESTIMATE (*estimate at destination*) AT (*altitude proposed for entry to controlled airspace*)
when requesting IFR Pick-up	•e.* REQUEST IFR PICK-UP
6. Pilot of IFR flight requests to climb to VFR-on-top	a.* REQUEST VFR-ON-TOP b. CLIMB TO [(*level*)] AND REPORT REACHING VFR-ON-TOP, TOPS REPORTED (*level*), *or* NO TOPS REPORTS
Pilot of an IFR flight is established VFR-on-top	c.* VFR-ON-TOP d. MAINTAIN VFR ON TOP
Where vertical restrictions apply	e. MAINTAIN VFR-ON-TOP AT OR BELOW/ABOVE/BETWEEN (*level*(*s*))

Circumstances	*Phraseologies* * Denotes pilot transmission
Pilot request to cancel VFR-on-top	f.* REQUEST (IFR *level*)
Note: Full IFR separation is applied when ATC re-clears the aircraft to maintain an IFR level	g. MAINTAIN (IFR *level*)
7. VFR Departure:	
Pilot of IFR flight requests VFR departure	•a. *REQUEST VFR DEPARTURE •b. VFR DEPARTURE APPROVED
Pilot of IFR flight approved to depart VFR wishing to revert to IFR	•c. *REQUEST IFR CLEARANCE [AT (*time or place*)] (IFR *level*)
Note: The pilot is responsible for separation until IFR separation can be applied by ATC	
Pilot of IFR flight having departed VFR, on first contact with ATC entering Class G airspace	•d. *RESUMING IFR
Note: Pilots wishing to continue VFR should CANCEL IFR. See sub-para 4. above.	
8. Parachute Operations: Clearance for parachutists to exit the aircraft and transit Restricted Area(s), Military Operating Area(s) or Classes A, C or D airspace	a. CLEAR TO DROP

6.12 **SID**

Circumstances	*Phraseologies* *Denotes pilot transmission
1. Issuing a SID	a. CLEARED (*SID designator)* DEPARTURE (*level instruction)*

Circumstances	***Phraseologies*** *Denotes pilot transmission
2. Clearance to climb on a SID a. comply with published level restrictions b. follow the lateral profile of the SID c. comply with published speed restrictions and ATC-issued speed control instructions	a. CLIMB VIA SID TO (*level)*
3. During a SID climb: a. published level restrictions are cancelled b. follow the lateral profile of the SID c. comply with published speed restrictions and ATC-issued speed control instructions	a. [CLIMB VIA SID TO *(level)*], CANCEL LEVEL RESTRICTION(S)
4. During a SID climb: a. published level restrictions at the specified point(s) are cancelled b. follow the lateral profile of the SID c. comply with published speed restrictions and ATC-issued speed control instructions	a. [CLIMB VIA SID TO *(level)*], CANCEL LEVEL RESTRICTION(S) AT *(point(s))*
5. During a SID climb: a. comply with published level restrictions b. follow the lateral profile of the SID c. published speed restrictions and ATC-issued speed control instructions are cancelled	a. [CLIMB VIA SID TO *(level)*], CANCEL SPEED RESTRICTION(S)

Circumstances	***Phraseologies*** *Denotes pilot transmission
6. During a SID climb: a. comply with published level restrictions b. follow the lateral profile of the SID c. published speed restriction are cancelled at the specific point(s) d. comply with ATC-issued speed control instructions	a. [CLIMB VIA SID TO *(level)*], CANCEL SPEED RESTRICTION(S) AT *(point(s))*
7. During a SID climb: a. published level restrictions are cancelled b. follow the lateral profile of the SID c. published speed restrictions and ATC-issued speed control instructions are cancelled *Note: the phrase 'CLIMB UNRESTRICTED TO...' is not used*	a. [CLIMB TO *(level)*], CANCEL LEVEL AND SPEED RESTRICTIONS
8. Clearance to proceed direct during a SID: a. track direct to the specified waypoint and then follow the lateral profile of the SID b. published level and speed restrictions for bypassed waypoints are cancelled c. comply with published speed and level restrictions at and after the specified waypoint *Note: direct tracking on a SID does not require a rejoin instruction*	a. CLEARED DIRECT *(waypoint)*

Circumstances	***Phraseologies*** *Denotes pilot transmission
9. Initiation of vectoring during SID	a. TURN LEFT (*or* RIGHT) HEADING (*three digits)* [(*reason)*], *(level instruction)*, [EXPECT TO REJOIN SID] [AT (*waypoint)*]
Clearance to proceed direct to a waypoint that is not on a SID	b. CLEARED DIRECT *(waypoint) (level instruction)*
10. Rejoining a SID	a. REJOIN SID *(or (SID designator))* [AT *(waypoint)*] *[(transition restrictions)]*
11. When a SID has been cancelled	a. CANCEL SID *(instructions)*

6.13 **STAR**

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Notification of STAR clearance availability (on first contact) when associated with a frequency transfer	a. EXPECT STAR CLEARANCE
Notification of STAR clearance availability when NOT associated with a frequency transfer	b. STAR CLEARANCE AVAILABLE
2. Issuing a STAR clearance	a. CLEARED (STAR *designator*) ARRIVAL [(*name*) TRANSITION] [RUNWAY(*number*)] (*level instruction*)
3. Descend to the cleared level: a. comply with published level restrictions b. follow the lateral profile of the STAR c. comply with published speed restrictions and ATC-issued speed control instructions	a. DESCEND VIA STAR TO *(level)*
4. During a STAR descent: a. published level restrictions are cancelled b. follow the lateral profile of the STAR c. comply with published speed restrictions and ATC-issued speed control instructions	a. [DESCEND VIA STAR TO *(level)*], CANCEL LEVEL RESTRICTION(S)

Circumstances	***Phraseologies*** * Denotes pilot transmission
5. During a STAR descent: a. published level restrictions at the specified point(s) are cancelled b. follow the lateral profile of the STAR c. comply with published speed restrictions and ATC-issued speed control instructions	a. [DESCEND VIA STAR TO *(level)*], CANCEL LEVEL RESTRICTION(S) AT *(point(s))*
6. During a STAR descent: a. comply with published level restrictions b. follow the lateral profile of the STAR c. published speed restrictions and ATC-issued speed control instructions are cancelled	a. [DESCEND VIA STAR TO *(level)*], CANCEL SPEED RESTRICTION(S)
7. During a STAR descent: a. comply with published level restrictions b. follow the lateral profile of the STAR c. published speed restrictions are cancelled at the specific point(s) d. comply with ATC-issued speed control instructions	a. [DESCEND VIA STAR TO *(level)*], CANCEL SPEED RESTRICTION(S) AT *(point(s))*
8. During a STAR descent: a. published level restrictions are cancelled b. follow the lateral profile of the STAR c. published speed restrictions and ATC-issued speed control instructions are cancelled *Note: the phrase 'DESCEND UNRESTRICTED TO...' is not used*	a. DESCEND TO *(level),* CANCEL LEVEL AND SPEED RESTRICTIONS

Circumstances	***Phraseologies*** * Denotes pilot transmission
9. Clearance to proceed direct during a STAR: a. track direct to the specified waypoint and then follow the lateral profile of the STAR b. published level and speed restrictions for bypassed waypoints are cancelled c. comply with published speed and level restrictions at and after the specified waypoint *Note: direct tracking on a STAR does not require a rejoin instruction*	a. CLEARED DIRECT *(waypoint)*
10. Initiation of vectoring after STAR has been issued	a. TURN LEFT (*or* RIGHT) HEADING *(three digits)* [(*reason*)], *(level instruction)*, [EXPECT TO REJOIN STAR] [AT*(waypoint)*]
Clearance to proceed direct to a waypoint that is not on a STAR	b. CLEARED DIRECT *(waypoint) (level instruction)*
11. Rejoining a STAR	a. REJOIN STAR *(or (STAR designator))* [AT *(waypoint)*] [*(transition restrictions)*]
12. When a STAR clearance is cancelled	a. CANCEL STAR *(instructions)*

6.14 Approach and Area Control Services

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Departures Instructions	a. TRACK (*three digits*) DEGREES [MAGNETIC] TO (*or* FROM) (*significant point*) [UNTIL (*time*) (*or* REACHING) (*fix or significant point or level*)]

Circumstances		***Phraseologies*** * Denotes pilot transmission
2. Approach Instructions *Note: See ENR 1.1 para 2.11.2.5 for further approach phraseology information.*	•a.	CLEARED DME (*or* GNSS) ARRIVAL [SECTOR (*identifying letter of the sector*)]
	b.	CLEARED *(type of approach)* APPROACH [RUNWAY *(number)*]
	c.*	REQUEST STRAIGHT-IN [(*type of approach*)] APPROACH [RUNWAY (*number*)]
	d.	CLEARED STRAIGHT-IN [(*type of approach*)] APPROACH [RUNWAY (*number*)]
	e.	CLEARED (*type of approach*) APPROACH RUNWAY (*number*) FOLLOWED BY CIRCLING TO RUNWAY (*number*)
RNP APCH via an IAF or IF	f.*	REQUEST (*type of approach*) APPROACH VIA (*last two letters of the IAF or IF designator*) [RUNWAY (*number*)]
	•g.	RECLEARED DIRECT *(last two letters of the IAF or IF designator)* CLEARED (*type of approach*) APPROACH [RUNWAY (*number*)]
	h.	COMMENCE APPROACH AT (*time*)

Circumstances	*Phraseologies* * Denotes pilot transmission
RNP APCH or RNP (AR) APCH where an aircraft has been subject to vectoring or random tracking and is subsequently re-cleared direct to the IAF.	i. RECLEARED DIRECT (*IAF/ Latest Intercept Point designator*) *followed as necessary by:* (i) TRACK VIA (*type of approach*) APPROACH [RUNWAY (*number*)] MAINTAIN (*or* DESCEND TO) (*level*) (ii) WHEN ESTABLISHED, CLEARED (*type of approach)* APPROACH [RUNWAY (*number*)]
3. Degradation of aircraft navigation performance	a. UNABLE RNP (*or* RNAV) (*specify type*) [DUE TO (*reason, e.g.* LOSS OF RAIM or RAIM ALERT)].
4. Where a temporary level restriction is to be imposed. (Applicable to civil aircraft during practice approaches in VMC; or MIL aircraft NPA, or precision if clearance will allow descent in accordance with procedure)	•a. TRACK VIA (*type of approach*) APPROACH [RUNWAY (*number*)] NOT BELOW (*level*)
pilot to advise when able to conduct a visual approach	b. REPORT VISUAL c. REPORT RUNWAY [LIGHTS] IN SIGHT d. REPORT (*significant point*) [OUTBOUND *or* INBOUND]
visual approach (by day or night)	e. CLEARED VISUAL APPROACH [RUNWAY (*number*)] [TRACKING VIA THE STAR]

Circumstances	*Phraseologies* * Denotes pilot transmission
visual approaches by night	•f. WHEN ESTABLISHED (*position*) CLEARED VISUAL APPROACH [RUNWAY (*number*)]
when including a VFR climb/ descent instruction:	
– when VFR descent clearance applies for the entire approach	g. DESCEND VFR, CLEARED (*type of approach*) APPROACH [RUNWAY (*number*)]
– when VFR descent clearance applies for a portion of the approach	h. CLEARED (*type of approach*) APPROACH [RUNWAY (*number*)] DESCEND VFR ABOVE (or BETWEEN) (*level(s)*)
5. Holding Instructions	
visual	a. HOLD VISUAL [OVER] (position)
published holding procedure over a waypoint, facility or fix	b. HOLD AT (*waypoint, facility or fix*) (*level*) EXPECT APPROACH (*or* FURTHER CLEARANCE) AT (*time*)
when pilot requires an oral description of holding procedure based on a facility	c.* REQUEST HOLDING INSTRUCTIONS
	d. HOLD AT (*waypoint, facility or fix*) (*call sign and frequency, if necessary*) (*level*) INBOUND TRACK (*three digits*) DEGREES RIGHT (*or* LEFT) HAND PATTERN, OUTBOUND TIME (*number*) MINUTES (*additional instructions, if necessary*)

Circumstances	*Phraseologies* * Denotes pilot transmission
	e. HOLD ON THE (*three digits*) RADIAL OF THE (*name*) VOR/ TACAN (*call sign and frequency, if necessary*) AT (*distance*) DME (*or* BETWEEN (*distance*) AND (*distance*) DME) (*level*) INBOUND TRACK (*three digits*) DEGREES RIGHT (*or* LEFT) HAND PATTERN (*additional instructions, if necessary*)
6. To advise ATC of Minimum Fuel status	a.* MINIMUM FUEL
ATC acknowledgment of Minimum Fuel status	b. MINIMUM FUEL ACKNOWLEDGED [NO DELAY EXPECTED *or* EXPECT (*delay information*)]
Note: Advice of fuel status must be made to each subsequent ATC sector on frequency transfer and ATC will acknowledge the status.	
7. Expected Approach Time	a. NO DELAY EXPECTED b. EXPECTED APPROACH TIME (*time*)

6.15 **Phraseologies to be used related to CPDLC**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Incorrect CPDLC message	
to disregard an incorrect CPDLC message	a. DISREGARD CPDLC (message type) MESSAGE
to instruct the flight crew to manually initiate a logon to the subsequent ATSU	b. DISCONNECT CPDLC THEN LOGON TO [facility designation]

Circumstances	*Phraseologies* * Denotes pilot transmission
to advise the flight crew prior to the commencement of CPDLC shutdown and instruct them to continue on voice	c. CPDLC WILL BE SHUT DOWN DISCONNECT CPDLC. CONTINUE ON VOICE

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6.16 **Vicinity of the Aerodrome**

6.16.1 **Visual Identification**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Identification of Aircraft	a. SHOW LANDING LIGHT
2. Acknowledgment by Visual Means	a. ACKNOWLEDGE BY MOVING AILERONS (*or* RUDDER) b. ACKNOWLEDGE BY ROCKING WINGS c. ACKNOWLEDGE BY FLASHING LANDING LIGHTS

6.16.2 **Starting and Initial Clearance Issue**

Circumstances		***Phraseologies*** * Denotes pilot transmission
1. Starting Procedures		
to request permission to start engines	a.*	*[aircraft location]* REQUEST START-UP
	b.*	[*aircraft location*] REQUEST START-UP INFORMATION (ATIS *identification*)
ATC response	c.	START-UP APPROVED
	d.	START-UP AT (*time*)
	e.	EXPECT START-UP AT (*time*)
	f.	EXPECT DEPARTURE (*time*) START-UP AT OWN DISCRETION
2. When clearance delivery is in operation.	•a.*	(*flight number, if any*) TO (*aerodrome of first intended landing*) REQUEST CLEARANCE
if runway other than runway nominated is required.	•b.*	REQUIRE RUNWAY (*number*)

6.16.3 **Pushbacks**

Circumstances		***Phraseologies*** * Denotes pilot transmission
1. Pushback Procedures for Aircraft	a.*	[*aircraft location*] REQUEST PUSHBACK
Note: Where a pushback does not enter the manoeuvring area, ERSA will specify the frequency on which apron service is provided.	b.	PUSHBACK APPROVED [TAIL (*direction e.g. North or Right*)]
	c.	PUSHBACK AT OWN DISCRETION [TAIL (*direction e.g. Left or West*)]
	d.	EXPECT (*number*) MINUTES DELAY DUE (*reason*)

Circumstances	*Phraseologies* * Denotes pilot transmission
2. Towing Procedures	a. # REQUEST TOW [*company name*] (*aircraft type*) FROM (*location*) TO (*location*)
ATC response	b. TOW APPROVED VIA (*specific routing to be followed*)
	# <u>*Denotes transmission from aircraft/ tow vehicle combination*</u>
3. To request time check and/or aerodrome data for departure when no ATIS broadcast is available	a.* REQUEST TIME CHECK
	b. TIME (*time*)
	c.* REQUEST DEPARTURE INFORMATION
Note: If multiple visibility, RVR or RV observations are available, those that represent the roll-out/ stop end zone should be used for take-off.	d. RUNWAY (*number*), WIND (*direction and speed*)(*units*) QNH (*number*)[(*units*)] TEMPERATURE [MINUS] (*number*), [VISIBILITY (*or* RUNWAY VISUAL RANGE *or* RVR *or* RUNWAY VISIBILITY) (*distance*)(*units*)], [TIME(*time*)]
4. Calculated Off Block Time (COBT) non-compliance - early request for clearance	a.* PUSHBACK (or TAXI) NOT AVAILABLE DUE FLOW MANAGEMENT. EXPECT PUSHBACK (or TAXI) AT (COBT - 5 minutes)
COBT non-compliance - late request for clearance	b. YOU ARE NON-COMPLIANT WITH FLOW MANAGEMENT. EXPECT AIRBORNE DELAY.
Target Off-Block Time (TOBT) and Target Start-up Approval Time (TSAT) compliant	c. PUSHBACK (or START-UP) APPROVED
	d. STAND BY FOR (or CONTACT) GROUND (frequency)
TOBT and/or TSAT non-compliant - early request for clearance	e. PUSHBACK (or START-UP) NOT AVAILABLE DUE FLOW MANAGEMENT, EXPECT TRANSFER (or START-UP or PUSHBACK) AT (time)

Circumstances	*Phraseologies* * Denotes pilot transmission
TOBT non-compliant – late request for clearance	f. YOU ARE NON-COMPLIANT WITH FLOW MANAGEMENT, CONTACT COMPANY FOR NEW TOBT

6.16.4 **Taxi Procedures**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Taxi Procedures	
for departure at a controlled aerodrome	a.* [*flight number*] [*aircraft type*] [*wake turbulence category if "Super or Heavy"*] [POB (*number*)] [DUAL (*or* SOLO)] INFORMATION (*ATIS identification*) [SQUAWK (*SSR code*)] [*aircraft location*] [*flight rules, if IFR*] [TO (*aerodrome of destination*)] REQUEST TAXI [*intentions*]
for departure at a non-controlled aerodrome	• b.* (*aircraft type*) [POB (*number*)] [IFR (*if operating IFR*)] TAXIING (*location*) FOR (*destination or intentions*) RUNWAY (*number*)
military pilots on local sorties when ready to taxi (*include details of flight if not already notified*)	▲c.* (*number of aircraft*) FOR (*area of operation*) POB (*number*) (DANGEROUS CARGO) INFORMATION (*ATIS identification*) REQUEST TAXI
	d. TAXI TO (HOLDING POINT [*identifier*] *or intermediate point*) [RUNWAY (*number*)] [TIME (*minutes*)]
	e.* ([HOLDING POINT] (*identifier*) *or intermediate point*), RUNWAY (*number*)
where detailed taxi instructions are required	f.* [*aircraft type*] REQUEST DETAILED TAXI INSTRUCTIONS

Circumstances	*Phraseologies* * Denotes pilot transmission
	g. TAXI VIA (*specific routing to be followed*) TO HOLDING POINT (*identifier*) [RUNWAY (*number*)] [TIME (*minutes*)]
	h.* [HOLDING POINT] (*identifier*), RUNWAY (*number*)
where aerodrome information is not available from an alternative source such as ATIS	i. TAXI TO HOLDING POINT (*identifier*) (*followed by aerodrome information as applicable*) [TIME (*minutes*)]
	j.* [HOLDING POINT] (*identifier*)
for arrival at a controlled aerodrome	k.* (*aircraft call sign*) [*parking area or bay number*]
	l. TAXI TO [TERMINAL *or other location; e.g.* GENERAL AVIATION AREA] [STAND (*number*)]
2. Intersection Departures when a pilot requests an intersection departure	• a.* REQUEST INTERSECTION DEPARTURE FROM (t*axiway identifier*)
	• b. TAXI TO HOLDING POINT (*taxiway identifier*) [RUNWAY (*number*)]
when a pilot is offered an intersection departure	• c. INTERSECTION DEPARTURE AVAILABLE FROM (*taxiway identifier*) (*distance*) REMAINING (*if this information is not readily available to the pilot*)
when a pilot accepts an intersection departure	• d. TAXI TO HOLDING POINT (*taxiway identifier*) [RUNWAY (*number*)]
3. Specific Routing	a. TAKE (*or* TURN) FIRST (*or* SECOND) LEFT (*or* RIGHT)
	b. TAXI VIA (*identification of taxiway*)
	c. TAXI VIA RUNWAY (*number*)

Circumstances	***Phraseologies*** * Denotes pilot transmission
4. Manoeuvring on Aerodrome	a.* REQUEST BACKTRACK
	b. BACKTRACK APPROVED
	c. BACKTRACK RUNWAY (*number*)
general	d.* (*aircraft location*) REQUEST TAXI TO (*destination on aerodrome*)
	e. TAXI STRAIGHT AHEAD
	f. TAXI WITH CAUTION (*reason*)
	g. GIVE WAY TO (*description and position of other aircraft*)
	h.* GIVING WAY TO (traffic)
	i. TAXI INTO HOLDING BAY
	j. FOLLOW (*description of other aircraft or vehicle*)
	k. VACATE RUNWAY
Note: The pilot must, when requested, report RUNWAY VACATED" when the aircraft is well clear of the runway.	l.* RUNWAY VACATED
	m. EXPEDITE TAXI [*reason*]
	n.* EXPEDITING
5. When aerodrome works or OLS obstructions create a temporary reduction in TORA from the departure holding point	• a. TAXI TO HOLDING POINT (taxiway identifier), REDUCED TORA RUNWAY (number)

6.16.5 **Aerodrome Movements**

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Holding *Note: The procedure words ROGER and WILCO are insufficient acknowledgment of the instructions HOLD, HOLD POSITION and HOLD SHORT OF (position). In each case, the acknowledgment must be by the phraseology HOLDING or HOLDING SHORT, as appropriate.*	a. HOLD (*direction*) OF (*position, runway number, etc*) b. HOLD POSITION c. HOLD SHORT OF (*position*) d.* HOLDING e.* HOLDING SHORT
2. To Cross a Runway *Note: If the control tower is unable to see the crossing aircraft (e.g. night, low visibility, etc), the instruction should always be accompanied by a request to report when the aircraft has vacated and is clear of the runway.*	a.* [AT (*or* ON) (*location*)] REQUEST CROSS RUNWAY (*number*) b. AT (*or* ON) (*location*) CROSS RUNWAY (*number*) [REPORT VACATED] c.* AT (*or* ON) (*location*) CROSSING RUNWAY (number) d. EXPEDITE CROSSING RUNWAY (*number*) TRAFFIC (*aircraft type*) (*distance*) MILES FINAL
3. To Enter a Runway (not used in conjunction with clearance to line-up or enter the Operational Readiness Platform). *Note: If the control tower is unable to see the relevant aircraft (e.g. night, low visibility, etc), the instructions should always be accompanied by a request to report when the aircraft has vacated and is clear of the runway.*	a.* [AT (*or* ON) (*location*)] REQUEST ENTER RUNWAY (*number)* b. AT (*or* ON) (*location*) ENTER RUNWAY (*number*) [REPORT VACATED] c.* AT (*or* ON) (*location*) ENTER RUNWAY (*number*)

6.16.6 **Runway Operations**

Note: During multiple runway operations where the possibility of confusion exists, the runway number will be stated. The runway number may be stated if the caller wishes to emphasise the runway to be used. For parallel runway operations on discrete frequencies, at Class D aerodromes, the runway number may be omitted.

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Preparation for Take-off	a. REPORT WHEN READY [FOR DEPARTURE]
at Class D aerodromes when reporting ready and parallel runway operations are in progress	b.* READY, RUNWAY *(runway)*
when reporting ready for operations wholly within Class D CTR or departure from Class D CTR not in receipt of airways clearance for operations	c.* READY [*intentions*] *[(circuit operations, tracking details, departure procedures, etc.)]*
outside Class D airspace.	d. ARE YOU READY FOR IMMEDIATE DEPARTURE?
	e.* READY
2. Clearance to Enter Runway and Await Take-off	
when the pilot desires to enter the runway and assume take-off position for checks before departure	•a.* REQUEST LINE-UP [REQUIRE (*required number of seconds delay in lined-up position before departure*) SECONDS ON RUNWAY]
	b. LINE UP [RUNWAY (*number*)] [AND WAIT] [BE READY FOR IMMEDIATE DEPARTURE]
conditional clearances	c. (*condition*) LINE UP [(RUNWAY (*number*)] (*brief reiteration of condition*)

Circumstances		*Phraseologies* * Denotes pilot transmission
acknowledgment of a conditional clearance	d.*	(*condition*) LINE UP [RUNWAY (*number*)] [AND WAIT]
when stop bar contingency procedures are in force	•e.	AT (*holding point*), CROSS THE ILLUMINATED STOP BAR, LINE UP (*or* CLEARED FOR TAKE-OFF *or* ENTER *or* CROSS) RUNWAY (number)
	f.*	AT (*holding point*), CROSS THE ILLUMINATED STOP BAR, LINE UP (*or* CLEARED FOR TAKE-OFF *or* ENTER *or* CROSS) RUNWAY (number)
3. Take-off Clearance	a.	CLEARED FOR TAKE-OFF [REPORT AIRBORNE]
multiple runway operations, other than Class D aerodromes where aircraft are operating on parallel runways using discrete frequencies	b.	RUNWAY (*number*) CLEARED FOR TAKE-OFF
when take-off clearance has not been complied with	c.	TAKE-OFF IMMEDIATELY OR VACATE RUNWAY
	d.	TAKE-OFF IMMEDIATELY OR HOLD SHORT OF THE RUNWAY
when LAHSO are in use	•e.	(*aircraft type*) LANDING ON CROSSING RUNWAY WILL HOLD SHORT - RUNWAY (*number*) CLEARED FOR TAKE-OFF
when a radar SID has been issued	•f.	ASSIGNED HEADING [LEFT (*or* RIGHT)] (*three digits*) [(*altitude restriction*)] [RUNWAY (*number*)] CLEARED FOR TAKE-OFF

Circumstances	*Phraseologies* * Denotes pilot transmission	
	g.*	HEADING (*or* LEFT *or* RIGHT) (*three digits)* [(*altitude restriction*)] [RUNWAY (*number*)] CLEARED FOR TAKE-OFF
	h.	TRACK EXTENDED CENTRE LINE (*three digits)* DEGREES [*(altitude restriction)*] [RUNWAY *(number)*] CLEARED FOR TAKE-OFF
when an IFR aircraft is cleared for a visual departure to a level at or above the MVA or MSA/ LSALT	•i.	(*instructions*) [RUNWAY (*number*)] CLEARED FOR TAKE-OFF, [MAKE LEFT (*or* RIGHT) TURN]
	•j.*	(*instructions*) [RUNWAY (*number*)] CLEARED FOR TAKE-OFF, [LEFT (*or* RIGHT) TURN]
when a VFR aircraft, or an IFR aircraft cleared for a visual departure is issued radar heading instructions	•k.	(*instructions*) MAINTAIN RUNWAY HEADING (*or* TURN LEFT (*or* RIGHT) HEADING (*three digits*)) VISUAL, [(*altitude restriction)*] [RUNWAY (*number*)] CLEARED FOR TAKE-OFF
	•l.*	(*instructions*) RUNWAY HEADING (*or* LEFT (*or* RIGHT) HEADING (*three digits*)) VISUAL, [(*altitude restriction)*] [RUNWAY (*number*)] CLEARED FOR TAKE-OFF
when an IFR aircraft cleared for a visual departure is assigned a level below the MVA or MSA/LSALT in the departure instructions.	•m.	(*instructions*) CLIMB TO (*level*) VISUAL, [RUNWAY (*number*)] CLEARED FOR TAKE-OFF

Circumstances	*Phraseologies* * Denotes pilot transmission
	•n.* (*instructions*) (*level*) VISUAL, [RUNWAY (*number*)] CLEARED FOR TAKE-OFF
when the airways clearance issued to an IFR aircraft includes a visual departure and a level below the MVA or MSA/ LSALT and no turn on departure required	•o. [RUNWAY (*number*)] CLEARED FOR TAKE-OFF, VISUAL
4. Take-off Clearance Cancellation	a. HOLD POSITION, CANCEL, I SAY AGAIN CANCEL TAKE-OFF (*reasons*)
	b.* HOLDING
to stop a take-off in emergency conditions *Note: Used only when an aircraft is in imminent danger.*	c. STOP IMMEDIATELY (*repeat aircraft call sign*) STOP IMMEDIATELY (*reason*)
	d.* STOPPING RUNWAY (*number*)
5. When aerodrome works or OLS obstructions create a temporarily displaced landing threshold	a. THRESHOLD [RUNWAY (*number*)] DISPLACED
When LDA is reduced at the upwind end of the runway	•b. REDUCED LANDING DISTANCE [RUNWAY (*number*)]

6.16.7 **Helicopter Operations**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Helicopter Operations air taxi or air transit for departure and arrival	a.* REQUEST AIR TAXI (*or* AIR TRANSIT *or* GROUND TAXI) FROM (*or* VIA) TO (*location or routing as appropriate*)

Circumstances	***Phraseologies*** * Denotes pilot transmission
	b. AIR TAXI (*or* AIR TRANSIT *or* GROUND TAXI) TO (*or* VIA) (*location, parking position, stand, or routing as appropriate*) [CAUTION (*dust, loose debris, taxiing light aircraft, personnel, wake turbulence, etc*)]
	c. AIR TAXI (*or* AIR TRANSIT *or* GROUND TAXI) VIA (*direct, as requested, or specified route*) TO (*location, heliport, parking position, stand, operating or movement area, or runway*) AVOID (*aircraft or vehicles or personnel*)
2. Departure from: a) a RWY, or b) HLS visible to the tower and located on a manoeuvring area subject to ATC.	a. (*instructions as appropriate, position or runway*) CLEARED FOR TAKE-OFF
Departure other than above	b. (*instructions as appropriate*) [DEPARTURE APPROVED] REPORT AIRBORNE
3. Arrival to: a) a RWY, or b) HLS visible to the tower and located within a manoeuvring area subject to ATC.	a. (*instructions as appropriate, position or runway*) CLEARED TO LAND
Arrival other than above	b. CLEARED VISUAL APPROACH (*instructions as appropriate*), REPORT ON THE GROUND

6.16.8 **After Take-off**

Note: ALL "level" reports within ATS surveillance system coverage must be to the nearest 100FT.

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Tracking After Take-off	a.* REQUEST RIGHT (*or* LEFT) TURN [WHEN AIRBORNE]
	b. LEFT (*or* RIGHT) TURN APPROVED
	c. AFTER PASSING (*level*) (*instructions*)
when instructing an aircraft to turn 180° or more after take-off	d. MAKE LEFT (*or* RIGHT) - I SAY AGAIN - LEFT (*or* RIGHT) TURN
heading to be followed	e. CONTINUE ON (*magnetic direction of runway*) (*instructions*)
when a specific track is to be followed	f. TRACK (*magnetic direction of runway*) (*instructions*)
	g. CLIMB STRAIGHT AHEAD (*instructions*)
2. Airborne Report - where an ATS surveillance service is provided unrestricted turn to track (including SID) heading specified by ATC	•a.* PASSING (*level*) CLIMBING TO (*level*)
heading specified by ATC	•b.* TURNING LEFT (*or* RIGHT) (*three digits*) PASSING (*level)* CLIMBING TO (*level*) *or*
	•c.* MAINTAINING RUNWAY HEADING PASSING (*level*) CLIMBING TO (*level*)

Circumstances	*Phraseologies* * Denotes pilot transmission
confirmation of an assigned Radar SID heading when establishing contact with ATC and unable to execute turn immediately due procedural requirements	•d.* ASSIGNED HEADING LEFT (*or* RIGHT) (*three digits*) PASSING (*level*) CLIMBING TO (*level*)
when assigned heading approximates runway bearing	e.* HEADING (*three digits*) PASSING (*level*) CLIMBING TO (*level*)
3. Departure Report - when notifying departure report to a Class D control tower	•a.* TRACKING (*track being flown*) (FROM (*reference aid used to establish track) or VIA SID identifier*) CLIMBING TO (*level*)
non-controlled aerodromes - non-surveillance	•b.* DEPARTED (*location*) (*time in minutes*) TRACKING [TO INTERCEPT] (*track*) CLIMBING TO (*intended level*) ESTIMATING (*first reporting point*) AT (*time*)
non-controlled aerodromes-surveillance when notifying departure and identification is expected with the departure report	•c.* (*location reference departure aerodrome*) PASSING (*current level*) CLIMBING TO (*intended level*) ESTIMATING (*first reporting point*) AT (*time*)

6.16.9 **Arrival at Aerodrome**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Entering an Aerodrome Traffic Circuit	a.* [*aircraft type*] (*position*) (*level*) (*intentions*)
when ATIS information is available	b.* [*aircraft type*] (*position*) (*level*) INFORMATION (ATIS i*dentification*) (*intentions*)
	c. JOIN (*instruction*) RUNWAY (*number*) [(level)] [QNH (*detail*)] [TRAFFIC (*detail*)] [TRACK (*requirements*)]

Circumstances	*Phraseologies* * Denotes pilot transmission
	d. OVERFLY [(circuit direction) RUNWAY (*number*) [(level)] [QNH (*detail*)] [TRAFFIC (*detail)*] [TRACK (*requirements*)]
2. In the Circuit	a.* (*position in circuit, e.g.* DOWNWIND/ FINAL)
when advising or requesting a non-standard circuit	b.* (*position in circuit, e.g.* DOWNWIND/FINAL) [GLIDE APPROACH, FLAPLESS APPROACH])
	c. [NUMBER (*sequence number*)] FOLLOW (*aircraft type and position*) [*additional instructions if required*]
	d.* [NUMBER (*sequence number*)] FOLLOW THE (*aircraft type and position*)
	e.* BASE (*or* CROSSWIND)
	f.* FINAL (*or* LONG FINAL)
nearing position at which approach must be aborted if not cleared to land	g.* SHORT FINAL
Abnormal Operations/Doubt Exists – (additional phrases)	
For a civil aircraft, when doubt exists as to whether the gear is fully extended, or when a general aviation aircraft with retractable undercarriage has experienced abnormal operations	h. CHECK GEAR DOWN [AND LOCKED]
	i.* GEAR DOWN [AND LOCKED]
Military Pilots – (additional phrases)	

Circumstances	*Phraseologies* * Denotes pilot transmission	
routine circuit reports must be made as and when arranged	▲ j.*	LEFT (*or* RIGHT) INITIAL
to sequence for downwind	▲ k.	PITCH LONG (*or* SHORT)
base call and wheel check (on reaching the base leg of a circuit, each aircraft, whether in stream landing or single, is to call tower and advise undercarriage down)	▲ l.*	BASE GEAR GREEN (*or* THREE GREENS *or* THREE WHEELS)
	▲ m.	(*instruction*) CHECK WHEELS
	▲ n.*	(*readback*) (*activate beeper*) *or*
	▲ o.*	(*readback*) GEAR GREEN (*or* THREE GREENS *or* THREE WHEELS)
ATC wheels check will include hook check for all hook cable operations	▲ p.	APPROACH/DEPARTURE END CABLE UP (*instruction*) CHECK WHEELS AND HOOK
	▲ q.*	(*readback*) HOOK DOWN (*activate beeper*)
3. Arriving at an Aerodrome – Military Formations Circuit Area	▲ a.*	(*formation call sign*), BASE THREE GREENS (*or* GEAR GREEN *or* THREE WHEELS)
	▲ b.	(*formation call sign*) (*instruction*) CHECK WHEELS
Lead aircraft undercarriage status report	▲ c.*	(*individual call sign*) (*activate beeper*) *or*
	▲ d.*	THREE GREENS (*or* GEAR GREEN *or* THREE WHEELS) (*individual call sign*)
Subsequent formation aircraft undercarriage status report	▲ e.*	(*individual call sign*), THREE GREENS (*or* GEAR GREEN *or* THREE WHEELS) (*activate beeper if fitted*)
	▲ f.	*(formation call sign)*

Circumstances	*Phraseologies* * Denotes pilot transmission	
Pairs Landing	▲ g.*	(*formation call sign*), OUTER MARKER (*or* FINAL APPROACH FIX) SIX GREENS (*or* GEAR GREEN *or* SIX WHEELS)
	▲ h.	(*formation call sign*) CLEARED TO LAND, CHECK WHEELS
	▲ i.*	LAND (*individual call sign*) (*activate beeper if fitted*)
	▲ j.*	(*individual call sign*) (*activate beeper if fitted*)
	▲ k.*	*(formation call sign)*
In-trail Landing	▲ l.*	(*formation call sign*) IN TRAIL, OUTER MARKER (*or* FINAL APPROACH FIX) THREE GREENS (*or* GEAR GREEN *or* THREE WHEELS)
	▲ m.*	(*formation call sign*) IN TRAIL CLEARED TO LAND, CHECK WHEELS
	▲ n.*	LAND (*call sign*) 1 (*activate beeper if fitted*)
	▲ o.*	(*call sign*) 2 OUTER MARKER (*or* FINAL APPROACH FIX) THREE GREENS (or GEAR GREEN *or* THREE WHEELS) (*activate beeper if fitted*)
	▲ p.*	(*call sign*) 3 OUTER MARKER (or FINAL APPROACH FIX) THREE GREENS (*or* GEAR GREEN *or* THREE WHEELS) (*activate beeper if fitted*)
	▲ q.	CALL SIGN
4. Speed Adjustments - <u>Military Aircraft</u> military ATC instruction	▲ a.	REDUCE TO CIRCUIT SPEED

Circumstances	*Phraseologies* * Denotes pilot transmission
	▲ b. REDUCE TO APPROACH SPEED ▲ c. REDUCE TO MINIMUM SAFE SPEED
5. Approach Instructions *Note: The report "LONG FINAL" is made when aircraft turn on to final approach at a distance greater than 4NM from touchdown or when an aircraft on a straight-in approach is 8NM from touchdown. In both cases, a report "FINAL" is required at 4NM from touchdown.*	a. MAKE SHORT APPROACH b. MAKE LONG APPROACH (*or* EXTEND DOWNWIND) c. REPORT BASE (*or* FINAL *or* LONG FINAL) d. CONTINUE APPROACH
6. Landing	a. CLEARED TO LAND (*or* TOUCH AND GO) (*or* STOP AND GO)
multiple runway operations, other than Class D aerodromes where aircraft are operating on parallel runways using discrete frequencies.	b. RUNWAY (*number*) CLEARED TO LAND (*or* TOUCH AND GO) (*or* STOP AND GO)
where the aircraft cannot be sighted by ATC	•c. [RUNWAY (*number*)] NOT IN SIGHT - CLEARED TO LAND
pilot requesting option for touch and go, full stop, stop and go, or go around	•d.* (*position in circuit*) REQUEST THE OPTION
advising the pilot the option to touch and go, full stop, stop and go, or go around	•e. [RUNWAY (*number*)] CLEARED FOR THE OPTION
where ATC require the aircraft to make a full stop landing during the conduct of circuit operations	f. MAKE FULL STOP (*reason*) CLEARED TO LAND

Circumstances		*Phraseologies* * Denotes pilot transmission
when runway is occupied and ATC assessment is that the runway will not become available.	▲ g.	AT THE MINIMA GO AROUND
7. When Landing Approved and LAHSO Are in Use	a.	(*aircraft type*) DEPARTING (*or* LANDING) ON CROSSING RUNWAY, HOLD SHORT RUNWAY (*number*) CLEARED TO LAND RUNWAY (*number*)
required readback	•b.*	HOLD SHORT RUNWAY (*number*) CLEARED TO LAND RUNWAY (*number*)
When the full length of the landing runway subsequently becomes available	c.	FULL RUNWAY LENGTH NOW AVAILABLE *Note: The HOLD SHORT lights will remain illuminated even though the full length of the RWY is available*
Where an aircraft operating on a flight number call sign cannot participate in LAHSO	d.	* NEGATIVE ACTIVE (*or* PASSIVE *or* ACTIVE AND PASSIVE) LAHSO
8. When a Pilot Advises That an “Autoland”, “Coupled” or Similar Approach is Being Made (note not applicable for GLS) and the ILS Critical Area is Not Protected.	•a.	ILS CRITICAL AREA NOT PROTECTED
9. Delaying Aircraft	a.	ORBIT RIGHT (*or* LEFT) [FROM PRESENT POSITION]

Circumstances	*Phraseologies* * Denotes pilot transmission
10. Pilot Request for Low Approach or Pass	
to make an approach along a runway descending to an agreed minimum level	a.* REQUEST LOW APPROACH (*reasons*)
	b. CLEARED LOW APPROACH [RUNWAY (*number*)] [(*altitude restriction*)] [(g*o around instructions*)]
to fly past the control tower or other observation point for the purpose of visual inspection by persons on the ground	•c.* REQUEST LOW PASS (r*easons*)
	•d. CLEARED LOW PASS [RUNWAY (*number*)] [(*altitude restriction*)] [(*go around instructions*)]
11. Missed Approach	
to discontinue an approach	a. GO AROUND [TRACK EXTENDED CENTRE LINE *(three digits)* DEGREES *(or instructions)*]
	b.* GOING AROUND
multiple runway operations	c.* GOING AROUND RUNWAY (*number*)

6.17 ATS Surveillance Service Phraseologies

6.17.1 General Phrases

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Identification of Aircraft	a. REPORT HEADING [AND FLIGHT LEVEL (*or* ALTITUDE)] b. FOR IDENTIFICATION TURN LEFT (*or* RIGHT) HEADING (*three digits*) c. IDENTIFIED [*position*] d. NOT IDENTIFIED [*reason*], [RESUME (*or* CONTINUE) OWN NAVIGATION]
2. Termination of ATS Surveillance Service	a. IDENTIFICATION TERMINATED [DUE (*reason*)] [(*instructions*)] [FREQUENCY CHANGE APPROVED] b. WILL SHORTLY LOSE IDENTIFICATION (*appropriate instructions or information*) c. IDENTIFICATION LOST [*reasons*] (*instructions*)
3. ATS Surveillance System To request traffic, position, and/or navigation information	•a.* REQUEST (i) ATS SURVEILLANCE ASSISTANCE (*reason*) (ii) POSITION [WITH REFERENCE TO (*aid or location*)] (iii) TRAFFIC (*or* POSITION *or* NAVIGATION) ADVISORY [BY SURVEILLANCE]

Circumstances	*Phraseologies* * Denotes pilot transmission
to provide position information	b. POSITION (*distance*) (*direction*) OF (*significant point*) (*or* OVER *or* ABEAM (*significant point*))
4. To request initiation or continuation of an ongoing SIS	a.* REQUEST [HAND-OFF FOR] FLIGHT FOLLOWING
to terminate an ongoing SIS	b.* CANCEL FLIGHT FOLLOWING
Where ongoing service is not available	c. SURVEILLANCE SERVICE NOT AVAILABLE
5. ATS surveillance system ground equipment unserviceability	a. SURVEILLANCE SYSTEM OUT OF SERVICE (*or* DEGRADED) (*appropriate information as necessary*)
6. To request the aircraft's SSR or ADS-B capability	a. ADVISE TRANSPONDER CAPABILITY
	b. ADVISE ADS-B CAPABILITY
7. To advise the aircraft's SSR or ADS-B capability	a.* TRANSPONDER (ALPHA, CHARLIE *or* SIERRA *as shown in the Flight Plan*)
	b.* ADS-B TRANSMITTER [TEN NINETY DATALINK]
	c.* ADS-B RECEIVER [TEN NINETY DATALINK]
	d.* NEGATIVE TRANSPONDER

6.17.2 **ATS Surveillance Service Communication and Navigation**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Communications	a. [IF] RADIO CONTACT LOST (*instructions*)
	b. IF NO TRANSMISSIONS RECEIVED FOR (*number*) MINUTES (*or* SECONDS) (*instructions*)

Circumstances	*Phraseologies* * Denotes pilot transmission
	c. REPLY NOT RECEIVED (*instructions*)
if loss of communication is suspected	d. IF YOU READ [*manoeuvre instructions or* SQUAWK (*code or* IDENT)]
	e. (*manoeuvre or* SQUAWK) OBSERVED, POSITION (*position of aircraft*), WILL CONTINUE TO PASS INSTRUCTIONS
2. Aircraft Directional Indicator Failure	
notify pilot of intention to use directional indicator failure procedures	•a. ATS SURVEILLANCE SERVICE WILL CONTINUE, MAKE ALL TURNS RATE ONE (*or* RATE HALF *or (number)* DEGREES PER SECOND), EXECUTE INSTRUCTIONS IMMEDIATELY UPON RECEIPT
when suspected by ATC	•b. CONFIRM HEADING
if heading response appears at variance with the track of the ATS Surveillance symbol	•c. SUSPECT YOUR DIRECTIONAL INDICATOR HAS FAILED
turn instructions	d. TURN LEFT (*or* RIGHT) NOW
	e. STOP TURN NOW

6.17.3 **ATS Surveillance System Manoeuvres**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. General Manoeuvres	a. LEAVE (*significant point*) HEADING (*three digits*) [INBOUND] [AT (*time*)]
	b. CONTINUE HEADING (*three digits*)

Circumstances	*Phraseologies* * Denotes pilot transmission
	c. CONTINUE PRESENT HEADING
	d. FLY HEADING (*three digits*)
	e. TURN LEFT (*or* RIGHT) (*number*) DEGREES (*or* HEADING (*three digits*)) [*reason*]
	f. ORBIT LEFT (*or* RIGHT) [*reason*]
when an ACFT is assigned a level below the MVA or MSA/LSALT	g. CLIMB (*or* DESCEND) TO (*level*) VISUAL
when an ACFT is issued a heading instruction below the MVA or MSA/LSALT *Note: Where both heading and altitude instructions are issued, VISUAL need only be appended to the second part of the instruction.*	h. TURN LEFT (*or* RIGHT) (*number*) DEGREES (*or* HEADING (*three digits*)) [CLIMB (*or* DESCEND) TO (*level*)] VISUAL
	i. STOP TURN HEADING (*three digits*)
when instructing an aircraft to turn 180° or more and in order to emphasize the direction of turn	j. TURN LEFT (*or* RIGHT) - I SAY AGAIN - LEFT (*or* RIGHT) HEADING (*three digits*) [*reason*]
when necessary to specify a reason for a manoeuvre, the following phraseologies should be used:	(i) DUE TRAFFIC (ii) FOR SPACING (iii) FOR DELAY (iv) FOR DOWNWIND (or BASE, *or* FINAL)
2. Aircraft Vectoring by ATS Surveillance Service	a.* REQUEST VECTORS [TO (*or* FROM) (*aid, location or reason*)]
	b. DO YOU WANT VECTORS?

Circumstances	***Phraseologies*** * Denotes pilot transmission
3. To transfer responsibility to the pilot for navigation on termination of vectoring	a. RESUME OWN NAVIGATION (*position of aircraft*) (*specific instructions*)

6.17.4 **Speed Control**

Circumstances	***Phraseologies*** * Denotes pilot transmission
1. Speed *Note: All speed communications shall relate to* ***INDICATED AIRSPEED*** *unless otherwise stipulated. Where applicable, Mach Number may be nominated as the basis of a speed statement.*	a.* SPEED (*number*) KNOTS (*or Mach Number*)
	b. REPORT SPEED *or* ([CLIMB *or* CRUISE] MACH NUMBER)
	c. MAINTAIN (*number*) KNOTS (*or* MACH (*number*)) [OR GREATER (*or* LESS)] [UNTIL (*significant point*)]
	d. MAINTAIN PRESENT SPEED
	e. INCREASE (*or* REDUCE) SPEED TO (*or* BY) (*number*) KNOTS [OR GREATER (*or* LESS)]
	f. REDUCE TO MINIMUM APPROACH SPEED
	g. CROSS *(significant point)* [AT *(time)*] [OR LATER (*or* OR BEFORE)] [AT (*number*) KNOTS]
when an aircraft is required to increase to, or maintain, the maximum speed at which it can safely operate given the phase of flight.	h. INCREASE TO (*or* MAINTAIN) MAXIMUM SPEED [UNTIL (*significant point)*]

Circumstances	*Phraseologies* * Denotes pilot transmission
when an aircraft is required to reduce speed to the minimum possible in a clean configuration	•i. REDUCE TO MINIMUM CLEAN SPEED
when an ATC-issued speed restriction no longer applies and the aircraft is on a published procedure for which a published speed currently applies	•j. RESUME PUBLISHED SPEED (*or* LEVEL RESTRICTIONS *or* SPEED AND LEVEL RESTRICTIONS)
when ATC-issued speed restriction no longer applies and no published speed currently applies. Comply with airspace speed limitations and all subsequent published speed restrictions. (*Note: Normal speed is the speed the aircraft would be maintaining had it not been issued an ATC speed restriction.)*	k. RESUME NORMAL SPEED
when aircraft speed is pilot's discretion. ATC speed restrictions are cancelled. Comply with airspace speed limitations. *(Note: Not used with SID or STAR instructions).*	l. NO ATC SPEED RESTRICTIONS
when aircraft speed is pilot's discretion. All airspace and ATC speed restrictions are cancelled. (*Note: Not used with SID or STAR instructions)*.	m. NO SPEED RESTRICTIONS
Note: Airspace speed limitations are at ENR 1.4 para 4.	

6.17.5 Traffic Information

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Traffic Information	a. TRAFFIC (*number*) O'CLOCK (*distance*) (*direction of flight*) [*any other pertinent information*]
	(i) UNKNOWN
	(ii) SLOW MOVING
	(iii) FAST MOVING
	(iv) CLOSING
	(v) OPPOSITE (*or* SAME) DIRECTION
	(vi) OVERTAKING
	(vii) CROSSING LEFT TO RIGHT (*or* RIGHT TO LEFT)
aircraft type to be passed if known	(viii) (*type*)
	(ix) (*level*)
	(x) CLIMBING (*or* DESCENDING)
	b. CLEAR OF TRAFFIC [*appropriate instructions*]

6.17.6 Secondary Surveillance Radar (SSR) and ADS-B

Circumstances	*Phraseologies* * Denotes pilot transmission
1. To Instruct Setting of Transponder	a. SQUAWK (*code*) [AND IDENT *if required*]
(The word "code" is not used in transmissions.)	b.* [SQUAWK] (*code*) [AND IDENT *if instructed by* ATS]
	c. SQUAWK NORMAL
<u>to request</u>:	
reselection of the assigned mode and code	d. RESET [(*mode*)] (*code*)
	e.* RESETTING [(*mode*)] (*code*)

Circumstances	*Phraseologies* * Denotes pilot transmission
reselection of aircraft identification	f. RE-ENTER MODE S (*or* ADS-B) AIRCRAFT IDENTIFICATION
confirmation of Mode A Code selection	g. CONFIRM SQUAWK (*code*)
	h.* SQUAWKING (*code*)
operation of the IDENT feature	i. SQUAWK IDENT
	j. TRANSMIT ADS-B IDENT
temporary suspension of transponder operation *Note: ADS-B and SSR are linked in many aircraft and terminating one will terminate the other.*	k. SQUAWK STANDBY [TRANSMIT ADS-B ONLY]
emergency code selection	l. SQUAWK MAYDAY
termination of SSR transponder or ADS-B transmitter operation *Note: ADS-B and SSR are linked in many aircraft and terminating one will terminate the other.*	m. STOP SQUAWK [TRANSMIT ADS-B ONLY] n. STOP ADS-B TRANSMISSION [SQUAWK (*code*) ONLY]
transmission of pressure altitude	o. SQUAWK CHARLIE
	p. TRANSMIT ADS-B ALTITUDE
pressure setting check and confirmation of level	q. CHECK ALTIMETER SETTING AND CONFIRM LEVEL
termination of pressure altitude transmission because of faulty operation *Note: ADS-B and SSR are linked in many aircraft and terminating one will terminate the other.*	r. STOP SQUAWK CHARLIE, WRONG INDICATION s. STOP ADS-B ALTITUDE TRANSMISSION [(WRONG INDICATION, *or reason*)]
altitude check	t. VERIFY LEVEL

Circumstances	*Phraseologies* * Denotes pilot transmission
confirmation of ADS-B operation	•u. ADS-B TRANSMISSIONS NOT RECEIVED, CONFIRM ADS-B OPERATIONAL
change to secondary transponder	•v. SELECT SECONDARY TRANSPONDER
2. Advice on Traffic Level Where the Pressure Altitude Derived Level Information Has Not Been Verified	•a. UNVERIFIED LEVEL (*level*)

6.17.7 **General ADS-C Phraseologies**

Circumstances	*Phraseologies* * Denotes pilot transmission
ADS-C DEGRADATION	ADS-C (*or* ADS-CONTRACT) OUT OF SERVICE (*appropriate information as necessary*).

6.17.8 **Approach Radar Services**

Circumstances	*Phraseologies* * Denotes pilot transmission
1. Vectoring for Approach *Note: See ENR 1.1 para 2.11.2.5 for further approach phraseology information.*	a. VECTORING FOR (*type of approach*) APPROACH RUNWAY (*number*) b. VECTORING FOR VISUAL APPROACH RUNWAY (*number*) REPORT FIELD (*or* RUNWAY) IN SIGHT c. VECTORING FOR (*positioning in the circuit*) d. (*type of approach*) APPROACH [RUNWAY (*number*)] NOT AVAILABLE DUE (*reason*) (*alternative instructions*)

Circumstances	*Phraseologies* * Denotes pilot transmission
2. Vectoring for ILS and other approach procedures	a. POSITION (*number*) MILES FROM (*fix*), TURN LEFT (*or* RIGHT) HEADING (*three digits*)
	b. YOU WILL INTERCEPT (FINAL APPROACH COURSE *or radio aid*) (*distance*) FROM (*significant point or* TOUCHDOWN)
	c.* REQUEST (*distance*) FINAL
instructions and information	d. CLEARED FOR (*type of approach*) APPROACH [RUNWAY (*number*)]
	e. REPORT ESTABLISHED ON LOCALISER (*or* ON [GLS/RNP] [FINAL] APPROACH [COURSE])
	f. CLOSING FROM LEFT (*or* RIGHT) [REPORT ESTABLISHED]
	g. TURN LEFT (*or* RIGHT) HEADING (*three digits*) [TO INTERCEPT] *or* [REPORT ESTABLISHED]
	h. EXPECT VECTOR ACROSS THE (LOCALISER *or* [GLS/RNP] FINAL APPROACH COURSE *or radio aid*) (*reason)*
	i. THIS TURN WILL TAKE YOU THROUGH THE (LOCALISER *or* [GLS/RNP] FINAL APPROACH COURSE *or radio aid*) [*(reason)*]
	j. TAKING YOU THROUGH THE (LOCALISER *or* [GLS/RNP] FINAL APPROACH COURSE *or radio aid*) [(*reason)*]

Circumstances	*Phraseologies* * Denotes pilot transmission
	k. MAINTAIN (*level*) UNTIL GLIDE PATH INTERCEPTION l. REPORT ESTABLISHED ON GLIDE PATH m. INTERCEPT (LOCALISER *or* [GLS/RNP] [FINAL] APPROACH [COURSE] *or radio aid*) [RUNWAY (*number*)] [REPORT ESTABLISHED]
3. Independent and Dependent Parallel Approaches	a. CLEARED FOR (*type of approach)* APPROACH RUNWAY (*number*) LEFT (*or* RIGHT) b. YOU HAVE CROSSED THE LOCALISER (*or* GLS/RNP FINAL APPROACH COURSE). TURN LEFT (*or* RIGHT) IMMEDIATELY AND RETURN TO THE LOCALISER (*or* GLS/RNP FINAL APPROACH COURSE) RUNWAY (*number*) c. TURN LEFT (*or* RIGHT) HEADING *(three digits)* JOIN FINAL RUNWAY *(number)* FROM THAT HEADING CLEARED INDEPENDENT VISUAL APPROACH
Independent parallel approaches to confirm correct runway selection	•d. [EXPECT] ILS (*or* GLS) [RUNWAY *(number)* LEFT (*or* RIGHT),] IDENT IS *(ILS or GLS ident)*
When aircraft will operate within 1NM of traffic on the adjacent final approach	e. TRAFFIC (*aircraft type)* [RUNWAY LEFT (*or* RIGHT)] BEHIND (*or* AHEAD *or* ADJACENT)

Circumstances	*Phraseologies* * Denotes pilot transmission
Independent Parallel Approaches when an aircraft is observed to be deviating towards the NTZ	f. YOU ARE DEVIATING FROM THE FINAL APPROACH COURSE. TURN LEFT (*or* RIGHT) IMMEDIATELY AND RETURN TO YOUR CLEARED APPROACH
Break-out instruction issued when an aircraft penetrates, or is likely to penetrate, the NTZ	g. BREAK-OUT ALERT, (*call sign*) TURN LEFT (*or* RIGHT) IMMEDIATELY HEADING (*three digits*) CLIMB (*or* DESCEND) TO (*level*)
4. Position	a. (*distance*) FROM TOUCHDOWN

6.17.9 **Pilot Actions for Speechless Radar Approach**

Circumstances	*Phraseologies* ** Denotes pilot transmission*
1. Pilot Request for Speechless Radar Approach When Microphone/s Unserviceable	▲a.* *Pilot transmits four* (4) *separate and distinct unmodulated transmissions of one second duration*
2. Pilot Response to Subsequent Control Questions:	
(i) affirmative or acknowledgment	▲a. *(i) one distinct transmission*
(ii) negative	▲b.* *(ii) two separate and distinct transmissions*
(iii) say again	▲c.* *(iii) three separate and distinct transmissions*
3. Pilot Indication of a Further and Pertinent Unserviceability or an Emergency	▲a.* *Five* (5) *separate continuous transmissions*
4. Pilot Indication of Abandoning the Aircraft	▲a.* *A single continuous transmission as long as practicable*

Circumstances	*Phraseologies* ** Denotes pilot transmission*
5. Controller Requires Pilot to Indicate When an Instruction Has Been Completed	▲a. WHEN (*condition or instruction is completed*) MAKE A TWO SECOND TRANSMISSION

7. ATS DATA LINK SERVICES

7.1 General

7.1.1 FANS 1/A data link services (CPDLC and ADS-C) are available within airspace controlled by Australian ATC within the Melbourne (YMMM), Brisbane (YBBB), Honiara (AGGG) and Nauru (ANAU) FIRs.

7.1.2 Pre-Departure Clearances (PDC) utilising FANS 1/A data link are not available within Australian administered airspace. (See *ENR 1.1 para 2.2.9*).

7.1.3 The data link procedures described herein are in accordance with international procedures. For more information concerning other data link procedures, refer to Chapter 5 of the *ICAO Global Operational Data Link (GOLD) Manual (Doc 10037).*

7.1.4 FMC Waypoint Position Reporting (FMC WPR) is available within Oceanic airspace east of Australia for operators whose AOC supports the conversion of ACARS position reports into a suitable format (ARP) and the forwarding of these messages to ATC.

Operators wishing to participate in FMC WPR should submit such a request to Airservices Australia.

7.2 Pilot Responsibilities

7.2.1 Participating flight crews must be trained in the use of data link equipment to a level approved by the State of Registry of the operator, and the aircraft must meet all State of Registry ATC data link requirements.

7.2.2 A logon request must be received from the aircraft before any data link connections can be initiated by the ground system. The initial logon request must be initiated by the flight crew in accordance with company and ATS procedures.

7.2.3 When using CPDLC for communications and/or ADS-C or FMC WPR for position reporting, flight crews must advise ATC when any fault occurs (e.g. loss of SATCOM) that may result in the degraded performance or non-availability of CPDLC, ADS-C and/or FMC WPR.

7.2.4 To ensure reliable Automatic Dependant Surveillance-Contract (ADS-C) is available, flight crews must ensure that the ADS-C application remains ARMED. Flight crews must also ensure that the ADS-C emergency mode has not been selected inadvertently.

7.2.5 CPDLC latency monitor functionality is used in Australian administered airspace.

7.2.5.1 If the aircraft is equipped with variable CPDLC latency monitor functionality, the pilot must, on receipt of the relevant message, enter the ATC specified latency time value into the aircraft avionics.

Note: Some avionics will automatically set the latency time value.

7.2.5.2 If the aircraft is not equipped with latency monitor functionality, or the functionality is not available, the pilot must still respond 'ROGER' and append the message 'TIMER NOT AVAILABLE'.

7.2.5.3 On receipt of a CPDLC indication that an uplink message has been delayed, the pilot must not act on the delayed uplink message and must seek clarification from ATC via voice.

Note: Some aircraft systems will discard the delayed message without indication to the flight crew and will automatically notify ATC of the network delay rejection.

7.2.5.4 When ATC receive an indication of a delayed uplink message, they will either:

a. resend the delayed CPDLC message;

b. contact the pilot via voice to clarify the situation; or

c. instruct the pilot to disregard the CPDLC message, disconnect CPDLC and continue communication by voice.

7.2.5.5 When exiting Australian administered airspace, pilots should reset the latency time value as appropriate to their company procedure, location and aircraft system.

7.3 **Controller Pilot Data Link Communication (CPDLC)**

7.3.1 In controlled airspace beyond the range of VHF voice, CPDLC is the primary means of communications between ATC and flight crews operating FANS 1/A equipped aircraft. HF voice will be used as the backup communications medium. In Class G airspace HF is the primary means of communication, although controllers may initiate the use of CPDLC. The controller communicating by CPDLC holds responsibility for SAR and communications alerting.

7.3.2 In Australian continental airspace, VHF voice is normally the primary medium for communication. Within East Coast SSR Coverage flight crews should not initiate CPDLC messaging except:

a. when transmitting position reports in accordance with *para 7.5.1* or

b. when authorised by the controller; or

c. in an emergency

7.3.3 Outside East Coast SSR coverage, ATC and pilots may use CPDLC to augment VHF communications.

7.3.4 Where CPDLC is to be used as the primary means of communications, flight crews will be instructed to transfer to CPDLC using the phraseology:
"TRANSFERTO(MELBOURNE/BRISBANE)CENTREONDATA LINK. MONITOR [frequency]."

7.3.5 CPDLC messages must be constructed using standard message elements, free text message elements or a combination of both. Standard message elements are contained in *PANS-ATM (DOC 4444), Appendix 5* and I*CAO Doc 10037 - Global Operational Data Link (GOLD) Manual*.

7.3.6 When CPDLC is being used and the intent of the message is included in the CPDLC message set contained in *PANS-ATM (Doc 4444), Appendix 5* or *ICAO Doc 10037 - Global Operational Data Link (GOLD) Manual*, the associated standard message elements must be used.

7.3.7 The use of free text message elements should be kept to a minimum. When the CPDLC message set contained in the *PANS-ATM (Doc 4444)* or *ICAO Doc 10037 - Global Operational Data Link (GOLD) Manual* does not provide for specific circumstances, the free text message elements may be used. These free text message elements should be sorted for selection within the aircraft system or ground system to facilitate their use.

7.3.8 To ensure the correct synchronisation of messages, dialogues initiated by voice must, wherever possible, be closed by voice. Dialogues initiated by CPDLC must be closed by CPDLC.

7.3.9 To avoid ambiguity, a CPDLC downlink message should not contain more than one clearance request.

7.3.10 If a CPDLC downlink message contains multiple clearance requests and not all of the requested clearances are available, the controller will respond with the uplink message UNABLE. The controller may subsequently uplink a separate CPDLC message containing any requested clearances that are available.

7.3.11 If any ambiguity exists as to the intent of a CPDLC message, clarification should be sought by voice.

7.3.12 A clearance issued by CPDLC requires only the appropriate CPDLC response, not a readback as would be required if the clearance had been delivered by voice.

7.3.13 The downlink response WILCO indicates that the flight crew has accepted the full terms of the entire uplink message including any clearance or instruction.

7.3.14 The downlink response UNABLE indicates that the flight crew is unable to accept the uplinked clearance or instruction. The flight crew must continue to operate in accordance with the most recently accepted clearance and/or instructions.

7.3.15 The use of the uplink STANDBY message element provides advice that the downlink request is being assessed and a short-term delay of less than 10 minutes can be expected until a response will be sent. The original message remains open.

7.3.16 The use of the uplink REQUEST DEFERRED message element provides advice that the downlink request is being assessed and a longer-term delay of greater than 10 minutes can be expected until a response will be sent. The original message remains open.

7.4 **Logon Procedures**

7.4.1 Before entering an airspace where data link applications are used by the ATS unit, data link communications shall be initiated between the aircraft and the ATS unit in order to register the aircraft and, when necessary, allow the start of a data link application. This shall be initiated by the aircraft, either automatically or by the pilot, or by the ATS unit on address forwarding.

Note: Provisions concerning the data link initiation capability (DLIC) are contained in Annex 10, Volume II, Chapter 8.

7.4.2 Brisbane ATC provides data link services in the Honiara and Nauru FIRs. The logon addresses for the appropriate FIRs are:
Brisbane YBBB
Honiara YBBB
Nauru YBBB
Melbourne YMMM.

7.4.3 To avoid automatic rejection of the logon request:

a. the aircraft identification and the aircraft registration contained in the logon request must be identical to the aircraft details filed in the flight plan, and

b. aircraft departing from airports either within Australia (YBBB or YMMM FIR) or the AGGG and ANAU FIRs should not logon earlier than 45 minutes prior to the flight planned EOBT, and

c. aircraft inbound to the YBBB, YMMM, AGGG or ANAU FIRs should not logon:

(i) prior to departure; or

(ii) earlier than 45 minutes prior to the FIR boundary estimate

7.4.4 Aircraft departing from airports either within Australia (YBBB or YMMM FIR) or the AGGG and ANAU FIRs, requesting data link services, should logon as described below:

a. On the ground using the logon address for the FIR that the departure airport is located within; or

b. Airborne at any time after passing 10,000FT, using the logon address for the FIR in which the aircraft is currently operating, with the following exceptions:

(i) Aircraft approaching an FIR boundary should logon to the next ATS unit, rather than the current ATS unit; and

(ii) Airborne aircraft departing YSSY, YSWS, YSBK or YSRI should logon in accordance with the following table:

Airborne aircraft departing Sydney/Bankstown/Richmond	
Departure track	**Logon address**
Airborne aircraft departing on tracks south of a line from YSSY-OPTIC through west to YSSY-YSWS-BENBU-SPLIZ (inclusive).	YMMM
Airborne aircraft departing on tracks north of a line from YSSY-YSWS-BENBU-SPLIZ through east to YSSY-OPTIC (inclusive).	YBBB

Note: The above table applies to logons from airborne aircraft only. Flight crews who logon whilst on the ground at YSSY, YSWS, YSBK or YSRI must logon to YMMM.

7.4.5 When instructed to logon by voice, the flight crew must readback the logon address.

7.4.6 Unless otherwise instructed, aircraft remaining wholly within East Coast SSR Coverage are not required to logon for data link services.

7.4.7 Flight crews of operators participating in PDC at Australian airports must not initiate a logon until after the PDC has been received.

7.4.8 Aircraft requesting data link services on entering the AGGG, ANAU, YBBB or YMMM FIRs from a non-data link FIR should logon to the ATS appropriate unit between 15 and 45 minutes prior to crossing the FIR boundary. CPDLC and ADS-C connections will be established automatically by the ATS unit concerned.

7.4.9 For aircraft entering the AGGG, ANAU, YBBB or YMMM FIR from a data link FIR, the CPDLC connection will be either transferred automatically by the Address Forwarding process, or the transferring ATS unit will instruct the flight crew to logon manually at an appropriate time/distance prior to the FIR boundary.

7.4.10 Aircraft transiting between the YBBB and YMMM FIR will be automatically Address Forwarded to the appropriate ATS unit. Aircraft departing Australian airspace (or the AGGG and ANAU FIRs) and directly entering an adjoining data link FIR can expect to be Address Forwarded to the appropriate ATS unit prior to the FIR boundary.

7.5 **Position Reporting Requirements**

7.5.1 For aircraft connected by data link, ADS-C reporting fulfils position reporting requirements. In the event of ADS-C failure, the pilot will be instructed to resume position reporting by either voice or CPDLC as appropriate.

7.6 **CPDLC Level Reporting Following Change of Level**

7.6.1 For non-identified aircraft, ADS-C reports fulfil level reporting requirements for a flight. Where ATC requires a CPDLC report, following or during a change of level, an appropriate instruction, will be uplinked; e.g. “CLIMB TO AND MAINTAIN 370. REPORT LEVEL 370” or “REPORT LEAVING 350”

7.6.2 Upon receipt of this CPDLC uplink, flight crews must ensure that the correct downlink report is sent.

7.6.3 If no instruction to make a CPDLC report is received, the flight crew has no requirement to report either maintaining the assigned level, or leaving the previously assigned level.

7.7 **Reporting Back on Route by CPDLC**

7.7.1 If an aircraft is subject to a weather deviation and has been instructed to report back on route, the CPDLC BACK ON ROUTE message must not be downlinked until either:

- The aircraft has rejoined its previously cleared route; or
- The aircraft has requested and has been cleared direct to a subsequent waypoint and is proceeding directly to that waypoint

7.7.2 If an aircraft is off track, but proceeding directly to a subsequent waypoint, the BACK ON ROUTE message must not be downlinked until after the aircraft has sequenced that waypoint.

Note: Downlinking BACK ON ROUTE before the aircraft is actually established on the cleared route may result in ATC applying incorrect separation tolerances to the aircraft.

7.8 **Block Level Clearances**

7.8.1 Subject to traffic, ATC may issue block level clearances to facilitate operations in adverse weather or to allow flight crews to optimise fuel burn for an aircraft.

7.8.2 A block level clearance is cancelled or amended by the issuing of a new vertical clearance.

7.8.3 To request a cancellation of a block clearance when it is no longer required, flight crews should downlink a CPDLC request for the preferred level in order to enable ATC to issue the new vertical clearance.

7.9 **Revision of Estimates Using CPDLC**

7.9.1 Under normal circumstances, an aircraft position reporting via ADS-C is not required to advise ATC of any revised waypoint estimates. Exceptions are:

a. revisions of greater than two (2) minutes to a previous flight crew advised estimate (i.e. by voice or CPDLC); or

b. following a flight crew-initiated action (e.g. speed change) resulting in an amended estimate of greater than two (2) minutes.

7.9.2 If required, flight crews shall advise a revised estimate by one of the following methods:

a. by voice report; or

b. a CPDLC position report containing the revised estimate; or

c. the CPDLC free text message element, “REVISED ETA [position][time]”

7.10 **CPDLC Message Restrictions**

7.10.1 Controllers will not issue uplink instructions relating to cruise climbs, or the message element “CRUISE (altitude)”. Additionally controllers will not issue uplink instructions containing the “AT PILOT DISCRETION” message element.

7.11 **CPDLC CONTACT and MONITOR messages**

7.11.1 The format of the [frequency] variable in the CPDLC “CONTACT” and “MONITOR” message elements permits the inclusion of only a single frequency. Due to this limitation, only the primary frequency will be notified to flight crews. The secondary frequency - uplinked as a free text - will be available on request.

7.11.2 The format of the [unit name] variable in the CPDLC "CONTACT" and "MONITOR" message elements does not support the ATS unit "RADIO". Due to this limitation, "CENTRE" will be used (i.e. "CONTACT BRISBANE CENTRE 13261", rather than "CONTACT BRISBANE RADIO 13261").

7.12 **CPDLC route clearances**

7.12.1 On occasions, controllers may issue amended route clearances by CPDLC in order to optimise an aircraft's route, or to separate with either traffic or restricted airspace. Operators must ensure that flight crews are familiar with the functionality associated with the display and loading of CPDLC route clearances.

7.13 **CPDLC Failure**

7.13.1 Flight crews becoming aware of a CPDLC connection failure when CPDLC is being used as the primary means of communication must immediately re-establish primary communications on the appropriate voice frequency.

7.13.2 In the event of either a programmed or unexpected CPDLC shutdown, ATC will advise all data link connected aircraft to re-establish primary communications by voice. The return of the system to an operational state may require a new logon from affected aircraft. Voice will continue to be used as the primary means of communication until CPDLC connections are re-established and ATC has authorised a return to data link communications.

7.14 **Notification Of Emergency**

7.14.1 Depending on the nature of the emergency condition experienced, flight crews should notify ATC of the circumstances by the most efficient means (voice, CPDLC or ADS-C).

7.14.2 If a CPDLC MAYDAY or PAN message is received by the ATS unit, ATC will respond with the free text uplink message ROGER MAYDAY (or ROGER PAN). ATC will not expect a ROGER response to the uplink until being notified that the emergency situation has been cancelled or stabilised to the extent that messages are able to continue being exchanged (if CPDLC is considered to be the best communications medium for the situation).

7.14.3 ATC may respond via CPDLC to a report indicating unlawful interference with the uplink message ROGER 7500.

7.14.4 If an ADS-C report containing emergency mode is received by the ground system without voice or CPDLC confirmation of an emergency situation, ATC will check for inadvertent activation of emergency mode by voice or CPDLC using the phraseology "CONFIRM ADS-C EMERGENCY".

a. If the emergency mode has been inadvertently activated, the flight crew must respond by voice or CPDLC using the phraseology "ADS-C RESET" and cancel the ADS-C emergency mode.

b. ATC will assume an emergency situation exists if there is no confirmation that the activation of emergency mode was inadvertent and ADS-C emergency reports are still being received.

7.14.5 When an emergency situation no longer exists, the flight crew should cancel the ADS-C emergency mode (if activated).

7.15 **CPDLC Message Set**

7.15.1 Standard message elements are listed in *ICAO Doc 10037 - Global Operational Data Link (GOLD) Manual.*

7.15.2 The following table contains CPDLC free text message elements used in Australia.

Uplinks - Free text message elements

MESSAGE ELEMENT	MESSAGE INTENT	RESPONSE
SET MAX UPLINK DELAY VALUE TO [sss] SEC	Instruction to set the CPDLC Latency Monitor time value to the value provided.	R
REPORT DISTANCE (to/from) (position)	Instruction to report the present distance to or from the specified position.	NE
CONFIRM POSITION	Instruction to report the present position.	NE
CONFIRM ALTITUDE	Instruction to report the present level.	NE
CONFIRM TIME OVER REPORTED WAYPOINT	Instruction to confirm the previously reported time over the last reported waypoint.	NE

MESSAGE ELEMENT	MESSAGE INTENT	RESPONSE
CONFIRM REPORTED WAYPOINT	Instruction to confirm the identity of the previously reported waypoint.	NE
CONFIRM NEXT WAYPOINT	Instruction to confirm the identity of the next waypoint.	NE
CONFIRM NEXT WAYPOINT ETA	Instruction to confirm the previously reported estimated time at the next waypoint.	NE
CONFIRM HEADING	Instruction to report the present heading.	NE
IDENTIFICATION TERMINATED	ATS advisory that the radar and/or ADS-B service is terminated.	R
CONFIRM ADS-C ARMED	Instruction for the flight crew to check that the ADS-C function is armed.	R
ADVISE PREFERRED LEVEL TO CROSS [position]	Instruction to advise the preferred flight level at the specified position for the flight.	R
REPORT ETA [position]	Instruction to report the estimated time of arrival at the specified position.	R
ADS-C INDICATES OFF ROUTE. ADVISE INTENTIONS	Instruction to notify of receipt of any ADS-C report indicating a deviation from cleared route and to request the flight crew to advise of intentions.	R
ADS-C INDICATES ROUTE NON-CONFORMANCE. ADVISE INTENTIONS	Instruction to notify of receipt of any ADS-C report indicating a future deviation from cleared route and to request the flight crew to advise of intentions.	R
ADS-C ESTIMATES APPEAR INACCURATE. CHECK FMS	Instruction to notify of receipt of any ADS-C report that appears to contain inaccurate time estimates and to request the flight crew to check FMS.	R

MESSAGE ELEMENT	MESSAGE INTENT	RESPONSE
CHECK AND RESPOND TO OPEN CPDLC MESSAGES	Instruction to check the status of CPDLC messages and to respond to unanswered uplink messages.	R
EXPECT SELCAL CHECK HF [frequency]	Notification that a SELCAL check on the specified HF frequency should be expected.	R
EXPECT CPDLC TRANSFER AT [time]	Notification that the CPDLC transfer process will not be completed at the FIR boundary and will be delayed until the specified time. If the CPDLC transfer is not completed by the specified time, the flight crew should manually disconnect and logon to the next centre.	R
EXPECT NEXT CENTRE [facility designation]. CONTACT WITH [facility designation] NOT REQUIRED	Notification that a CPDLC connection is not required by the next FIR (e.g. due to short transition time of the next FIR) and CPDLC connection will be transferred to the subsequent FIR.	R
REQUEST RECEIVED RESPONSE WILL BE VIA VOICE	Notification that the CPDLC downlink request was received by the controller and that any further communication about the request will be via voice.	R
ADS-C INDICATES LEVEL DEVIATION. ADVISE INTENTIONS	Instruction to notify of receipt of any ADS-C report indication a deviation from cleared level and to request the flight crew to advise of intentions.	R
REACH [level] BY [time]	Instruction that a change of level is to continue, but at a rate such that the specified level is reached at or before the specified time.	R

MESSAGE ELEMENT	MESSAGE INTENT	RESPONSE
REACH [level] BY [position]	Instruction that a change of level is to continue, but at a rate such that the specified level is reached at or before the specified position.	R
LEAVE CONTROL AREA DESCENDING	Instruction to leave controlled airspace on descent.	R
CRUISE CLIMB PROCEDURE NOT AVAILABLE IN AUSTRALIAN ADMINISTERED AIRSPACE	Response to a request for a cruise climb. This procedure in not available.	R
IDENTIFIED	ATS advisory that the aircraft has been identified on radar and/or ADS-B.	R
REACH [level] BY [distance] NM [before/after] [position]	Instruction that a change of level is to continue, but at a rate such that the specified level is reached at or before the specified distance before or after the specified position.	R
MELBOURNE [or BRISBANE] CENTRE AT REDUCED SYSTEM CAPACITY. DISCONNECT CPDLC	Notification that the specified ATS Unit is operating at reduced capacity.	R
RE-ENTER ADS-B IDENTIFICATION [flight identification]	Instruction to check and to re-enter the correct ADS-C flight identification for the flight.	R
ADS-C REPORTS NOT RECEIVED. REQUEST CPDLC POSITION REPORTS	Notification that ADS-C reports are not being received from a flight and that the flight crew is to report position using CPDLC position reports.	R
DO NOT DISCONNECT CPDLC. LOGON TO [unit name]	Instruction to logon to the specified ATS Unit without disconnecting CPDLC.	R
DISCONNECT CPDLC	Instruction to disconnect CPDLC.	R

APPENDIX 1

AUSTRALIAN AFTN CIRCUITRY: NATIONAL COMMUNICATIONS CENTRE MELBOURNE- *YMMLYFYX*

ATS UNITS IN AUSTRALIA

Adelaide	(YPAD)
Adelaide/Parafield	(YPPF)
Albury	(YMAY)
Alice Springs	(YBAS)
Ballina/Byron Gateway	(YBNA)
Brisbane FIR	(YBBB)
Brisbane	(YBBN)
Brisbane/Archerfield	(YBAF)
Broome/INTL	(YBRM)
Cairns/Cairns INTL	(YBCS)
Camden	(YSCN)
Canberra	(YSCB)
Christmas Island =	(YPXM)
Coffs Harbour	(YCFS)
Darwin *	(YPDN)
Gold Coast	(YBCG)
Hamilton Island	(YBHM)
Head Office	(YSHO)
Hobart	(YMHB)
Karratha	(YPKA)
Launceston	(YMLT)
Mackay	(YBMK)
Melbourne/Avalon	(YMAV)
Melbourne FIR	(YMMM)
Melbourne *COMC/BOF/NOF	(YMML)
Melbourne/Essendon	(YMEN)
Melbourne/Moorabbin	(YMMB)
Perth	(YPPH)
Perth/Jandakot	(YPJT)
Port Hedland	(YPPD)
Rockhampton	(YBRK)
Sunshine Coast	(YBSU)
Sydney/Bankstown	(YSBK)
Sydney/Kingsford Smith	(YSSY)
Sydney/Nancy-Bird Walton	(YSWS)
Tamworth	(YSTW)
Townsville/Townsville INTL *	(YBTL)
Woomera	(YPWR)

EXTERNAL ORGANISATIONS

Australian Maritime Safety Authority (AMSA)	(YSMO)
Joint Rescue Coordination Centre (JRCC)	(YSARYCYX)
Australian Transport Safety Bureau (ATSB)=	(see Note 2)
Civil Aviation Safety Authority (CASA)	(YSCA)
Customs	(YSCBCUST)
Darwin District Airport Inspector	(YPDNYDYX)
Bureau of Meteorology (MET)	(YMMC)
Airline Company Network (SITA)	(WSSSSITX)

LEGEND

* Military/civil use

= Fax connection

INTERNATIONAL CENTRES

Fiji - Nadi	(NFFF)
Indonesia - Jakarta	(WIII)
Johannesburg	(FAJS)
Nauru Is - Nauru	(ANAU)
New Guinea - Port Moresby	(AYPY)
New Zealand - Christchurch	(NZCH)
Singapore Is - Singapore	(WSSS)
Solomon Islands - Honiara	(AGGH)
USA - Salt Lake City	(KSLC)
Vanuatu - Port Vila	(NVVV)

MILITARY UNITS

Amberley	(YAMB)
East Sale	(YMES)
Edinburgh	(YPED)
Nowra	(YSNW)
Oakey	(YBOK)
Pearce	(YPEA)
RAAF AIS	(YMMX)
Richmond	(YSRI)
Tindal *	(YPTN)
Williamtown *	(YWLM)

AIRLINE COMPANIES

Japan Airlines	(YBCSJALX)
Qantas Airlines	(YSSYQFAO)
United Airlines	(YSSYUALO)
Virgin Australia	(YBBBVOZX)

Note 1: There are many supplementary inter-unit AFTN circuits available which are not indicated.

Note 2: Messages for the ATSB should be directed by Fax to 61-2-6274 6434.

GEN 3.5 METEOROLOGICAL SERVICES

1. RESPONSIBLE SERVICE

The Australian Bureau of Meteorology (BoM) is responsible for the provision of meteorological services within the Australian FIR and external Australian territories.

1.1 Contact details:

General Manager Aviation Land and Maritime Transport Program
Australian Bureau of Meteorology
GPO Box 1289
MELBOURNE VIC 3001
PH: 61 3 9669 4000
Email: webav@bom.gov.au
Website: www.bom.gov.au/aviation/

1.2 Applicable ICAO Documents

The meteorological service is provided in accordance with provisions in the following ICAO documents and registered ICAO differences:

- *Annex 3 – Meteorological Service for International Air Navigation*
- *Doc 10157 – PANS – Meteorology*
- *Doc 7030 – Regional Supplementary Procedures*
- *Doc 8400 – PANS – ICAO Abbreviations and Codes*
- *Doc 8896 – Manual of Aeronautical Meteorological Practice*
- *Doc 9673 – Air Navigation Plan Asia and Pacific Regions*
- *Doc 10100 – Manual on Space Weather Information in Support of International Air Navigation*
- *Annex 11: Air Traffic Services* (MET related issues only)
- *Annex 12: Search and Rescue* (MET related issues only)
- *Annex 15: Aeronautical Information Services* (MET related issues only)

Differences to *Annex 3, "Meteorological Service for International Air Navigation",* are listed in *GEN 1.7.*

1.3 Resources

BoM publishes online resources regarding aviation weather at www.bom.gov.au/aviation/.

2. AREA OF RESPONSIBILITY

The area of responsibility for the provision of aeronautical meteorological services set out in this publication covers the Australian FIR and Australian external territories in other FIR. An aeronautical meteorological watch service is provided to issue SIGMET for the Australian FIR.

Refer to the BoM Aviation Knowledge Centre for more information: www.bom.gov.au/aviation/knowledge-centre/.

3. METEOROLOGICAL OBSERVATIONS AND REPORTS

Details about services available at individual aerodromes are available in *ERSA FAC*. There is also some information provided by BoM online: www.bom.gov.au/aviation/location-info/.

3.1 METAR/SPECI

More information on METAR/SPECI can be found in the Product Information section of the BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

3.1.1 Routine Reports (METAR)

A METAR is a routine report of meteorological conditions at an aerodrome. METAR are normally issued on the hour and half hour.

3.1.2 Special Reports (SPECI)

SPECI are non-routine aerodrome reports issued whenever one or more observed meteorological elements meet specified criteria significant to aviation.

3.2 Other Meteorological reports

On request other Meteorological reports may be provided based on the whole horizon or only the area that will contain the probable flight path of an aircraft.

Other Meteorological reports are provided by:

a. Tower ATC at controlled aerodromes; or

b. A CA/GRS or UNICOM at certain non-controlled aerodromes.

4. TYPES OF SERVICES

BoM provides an English language H24 service from the Brisbane MET Office (MO) and Melbourne MO. The Melbourne MET Watch Office (MWO) is operated by the Melbourne MO and the BoM Hazardous Weather Unit (HWU). The Brisbane MWO is operated by the Brisbane MO and BoM HWU. Australia also operates a Volcanic Ash Advisory Centre and Tropical Cyclone Advisory Centre.

The following services are available from/for Australia:

a. Forecast upper-level wind and temperature charts;

b. Satellite imagery;

c. Significant weather charts;

d. Route forecasts (wind and temperature);

e. Grid point forecasts (wind and temperature);

f. OPMET (TAF, METAR/SPECI, Aerodrome Warning, Wind Shear Warning, Wind Shear Alert, Microburst Alert, SIGMET, AIRMET, GAF, VA Advisory, TC Advisory and SWX Advisory);

g. Aerodrome weather briefing; and

h. Area QNH.

For more information of these products, refer to the BoM aviation website: www.bom.gov.au/aviation/.

A consultative briefing service is available from both MO. Local numbers align with the 10 areas as detailed on the PCA and are given in the following table:

Brisbane MO		**Melbourne MO**	
QLD-North	07 3239 8721	NSW-West	02 9296 1527
QLD-South	07 3239 8737	NSW-East	02 9296 1526
NT	08 8920 3814	VIC	03 9669 4850
WA-North	08 9263 2259	TAS	03 6221 2026
		SA	08 8366 2773
		WA-South	08 9263 2255

Note: Conversations on these briefing services are recorded.

Meteorological information is available by phone and electronic briefing systems from the NOMC Pilot Briefing Office. Contact details are:

Airservices Pilot Briefing Office
Tower Road Melbourne Airport
MELBOURNE VIC 3045

PH: +61 3 9235 7518
Fax: +61 2 6268 5033

4.1 ATS meteorological information service

The ATS meteorological information service is contained within the Flight Information Service (FIS) described in *GEN 3.3 Section 3.3.*

4.2 Interpretation and use of forecasts

In TAF, cloud heights are given above the aerodrome elevations.

In other forecasts, heights are expressed:

a. as a flight level; or

b. with reference to mean sea level.

4.3 **Forecasts for operations above 10,000FT**

The meteorological information for operations above 10,000FT will consist of the following as appropriate:

a. Significant Weather Charts:

b. Wind and Temperature Charts:

c. Grid Point Forecasts (wind and temperature):

d. Route Sector Winds and Temperatures

This information is available from:

- ATS automates briefing systems: NAIPS-AVFAX or charts;
- ATS briefing offices by phone or fax; and
- BoM.

4.4 **Significant Weather (SIGWX) charts**

The significant weather expected in the airspace from FL250 to FL600 (SIGWX High) and FL100 to FL250 (SIGWX Medium) are depicted on the relevant chart using approved symbols and abbreviations. Significant features of the surface synoptic situation may also be shown.

More information on SIGWX Charts can be found in the Product Information section of the BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

Significant weather charts are available from NAIPS (under Charts) and from the BoM aviation web page.

4.5 **Grid Point Wind and Temperature (GPWT) forecasts** D

GPWT charts provide a display of wind and temperature data derived from weather model data

More information on GPWT forecasts can be found in the Product Information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

Low-level GPWT Reference Charts with PCA points can be found in the NAIPS Charts Directory.

4.6 Graphical Area Forecasts (GAF) for operations surface to 10,000FT

GAF are issued for aircraft operations at or below 10,000FT for the 10 areas as detailed on the PCA. They comprise of an image and supporting text detailing the meteorological conditions. GAF are prepared and issued at the following times (UTC):

a. 6 hour validity periods 2300 to 0500, 0500 to 1100, 1100 to 1700 and 1700 to 2300;

b. At each issue time two GAF will be issued covering a 12 hour period, for example at 2200, both 2300 to 0500 and 0500 to 1100 GAF will be issued;

c. GAF will be issued no later than 30MIN before the commencement of the validity period of the first GAF;

d. the issuance of a new GAF replaces the previously issued GAF for the same validity period.

Forecasts for critical locations are included in some GAF. These include:

- Bowral (BWL - NSW-E);
- Mt Victoria (MVI - NSW-E);
- Murrurundi (MUI - NSW-E); and
- Kilmore Gap (KMG - VIC).

Aerodrome Weather Reports (AWR), i.e. METAR/SPECI and AWIS, are only available for MUI and KMG. However, AWR from Moss Vale (MSV) and Mount Boyce (MTB) typically provide representative observations from BWL and MVI respectively.

GAF are available from the ATS automated briefing systems and briefing offices.

More information on GAF can be found in the Product Information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

A Flight Forecast (text based forecasts) may be issued for any part of a flight for which a routine GAF is not prepared - requests are to be made to the relevant MO.

4.7 **Aerodrome Forecasts (TAF)**

TAF are a statement of meteorological conditions expected for a specified period in the airspace within a radius of 8KM of the ARP.

TAF service is typically provided in accordance with the aerodrome's TAF category, as follows:

CAT	Aerodrome Type	Routine TAF Service
TAF3	Selected aerodromes specified in *ERSA*	Issued 3 hourly. Validity is either 18, 24 or 30 hours depending on which aerodrome type and category of TAF (A or B).
A	**International**	Issued 6 hourly, valid for 18, 24 or 30 hours. Commencement times 00, 06, 12 and 18 UTC.
B	**Large**	Issued 6 hourly, valid for 12 or 18 hours. Commencement times 00, 06, 12 and 18 UTC.
C	**Medium**	Issued 6 hourly, typically valid for 12 hours. Commencement times are 02, 08, 14 and/or 20 UTC, except in Western Australia where commencement times are 04, 10, 16 and/or 22 UTC.
D	**Small**	Issued 6 or 12 hourly, valid for up to 12 hours. Commencement times are typically 20 and/or 02 UTC, except in Western Australia where commencement times are typically 22 and/or 04 UTC.

Note 1: Commencement times for C and D TAF will be one hour earlier in states using daylight saving.

Note 2: TAF will be provided upon request for other locations in support of SAR and emergency flights.

Note 3: TAF3 aerodromes may not have H24 TAF3 service. MO hours for TAF3 watch and routine TAF3 validity periods are annotated in ERSA.

Note 4: TAF3 is identical in format to any other TAF and can be identified by the indicator "TAF3" in the RMK section of the TAF.

Amendments, labeled TAF AMD, are issued immediately once the need is recognised.

More information on TAF3 and TAF can be found in the product information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

Note: Flight planning requirements for TAF can be found at ENR 1.10 para 1.2.4.

4.8 Airport Weather Briefings (AWB)

AWB are provided for:

- Brisbane (YBBN);
- Melbourne (YMML);
- Adelaide (YPAD);
- Perth (YPPH);
- Sydney (YSSY); and
- Darwin (YPDN) - only available during the wet season

The AWB is provided to expand on the information provided in the TAF. Once issued, it is not amended or updated.

4.9 Wind shear warnings

Aircraft reports of wind shear encountered during climb and descent are the primary means of detecting wind shear. The MO provides advice, when possible, on the likely duration of the event and forecast low level winds.

Wind shear warnings for an event will specify a validity period, and sequence numbers will be assigned to each warning associated with an event. A wind shear warning will be cancelled when wind shear is no longer expected. Locations where this service is provided is published in *ERSA FAC*.

When wind shear is forecast or reported by pilots at an intensity greater than "light", this information, together with a forecast low level wind, will be included on the ATIS at any of the above aerodromes.

More information on wind shear can be found in the Hazardous Phenomena section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

4.10 Microburst and wind shear alerts

In addition to wind shear warnings, remote sensing technology has been deployed at some locations (see *ERSA FAC*) to provide alerts when low level wind shear and microburst activity has been detected.

Air traffic controllers will issue a microburst or wind shear alert to affected aircraft using standard phraseology when the following conditions are detected:

	Wind	Alert
Headwind gain	>15 KT	Wind shear
Headwind loss	>15 KT	Wind shear
	>30 KT	Microburst

If multiple instances of wind shear/microbursts are detected on the same runway corridor, an alert will only be issued for the instance with the greatest intensity.

If instances of both wind loss and wind gain are detected on the same runway corridor, an alert will only be issued for the wind loss instance.

4.11 Volcanic Ash Advisories (VAA)

Volcanoes to the north and east of Australia are monitored by the Darwin Volcanic Ash Advisory Centre (VAAC) operated by the BoM.

If required, the Darwin VAAC will issue a “Volcanic Ash Advisory” in both text and graphical form.

More information on VAA can be found in the Product Information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

The Melbourne MWO is responsible for issuing Volcanic Ash SIGMET in the Australian FIR based on information from VAA.

Refer to *para 8.1* for more information on SIGMET.

During times of prolonged volcanic activity, in conjunction with neighbouring States, temporary airspace and airways will be established to avoid hazardous areas and notified by NOTAM.

4.12 Space Weather (SWX)

The BoM is part of one of the ICAO designated aviation Space Weather Centres (SWXC) established to deliver space weather (SWX) services.

The SWXC monitor and provide current conditions, advisories and forecasts of space weather phenomena with the potential to impact:

a. high frequency radio communications (HF COM);

b. global navigation satellite systems (GNSS) based navigation and surveillance systems;

c. radiation effects on avionics and/or human health (RADIATION).

A space weather advisory is issued whenever space weather conditions exceed pre-defined ICAO thresholds as defined in *ICAO Doc 10100* for both moderate impacts (MOD) and severe impacts (SEV).

More information on Space Weather Advisories can be found in the Product Information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

Note: SIGMET are not issued for Space Weather.

4.13 Tropical Cyclones (TC)

The Darwin Tropical Cyclone Advisory Centre (TCAC) is operated by BoM and is responsible for issuing "Tropical Cyclone Advisories" in text and graphical formats for its area of responsibility.

A Tropical Cyclone Advisory (TCA) provides information concerning the position of the cyclone centre, its direction and speed of movement, central pressure and maximum surface wind near the centre.

The Melbourne MWO is responsible for issuing Tropical Cyclone SIGMET in the Australian FIR based on information from TCA.

Refer to *para 8.1* for more information on SIGMET.

5. NOTIFICATION REQUIRED FROM OPERATORS

5.1 For international operations

Forecast Required	Availability	Notice Required
a)Preliminary operational planning (to assist in the general planning of the following day's operations)	3 - 24 hours before EOBT	8 hours
b)Preflight	At least 3 hours before EOBT	8 hours
c)En route	As arranged	

5.2 For domestic operations

All meteorological information issued on a routine basis and held by the NOMC Pilot Briefing Office is available without prior notice. For non-routine forecasts the MO requires 8 hours notice.

5.3 Forecasts for flights - valid GAF not available

Flight Forecasts required for flights for which valid GAF are not available will be supplied subject to the request being received three days prior to departure and forecaster capacity to provide the service.

Note: Every effort will be made to expedite MET documentation for urgent flights.

Notification should include part or all of the following information as applicable:

a. departure aerodrome and EOBT;

b. destination and ETA;

c. route;

d. ETA and EOBT for intermediate stopping places;

e. heights for upper winds and temperatures;

f. time documentation required.

6. AIRCRAFT REPORTS

6.1 Routine aircraft observations

Routine Observations in Australian FIR are only required from aircraft equipped with Aircraft Meteorological Data Relay (AMDAR).

AMDAR equipped aircraft should transmit routine meteorological observations:

a. at waypoints indicated by the symbols  and  on en route charts (ERCL and ERCH).

b. at MET reporting points shown on charts which are used in lieu of Australian AIP aeronautical charts.

c. whenever conditions experienced are significantly above or below those forecast.

In areas where ground meteorological reports are infrequent or any hazardous weather is encountered, or observed either visually or by radar, pilots in command are encouraged to report observations of MET conditions which they consider will assist in the provision of meteorological services.

Routine weather observations should be reported in accordance with the AIREP Format shown in *ENR 1.1 APPENDIX 1*.

6.2 **AIREP Special**

A pilot in command should make an AIREP Special report when requested, or as soon as practicable after encountering or observing hazardous meteorological conditions which, in the opinion of the pilot are, or may become, severe enough to warrant a SIGMET, regardless of any reports from other aircraft and regardless of any SIGMET issued.

ENR 1.1 APPENDIX 1 (Position reports, AIREP Special and volcanic ash reports) details the conditions warranting an AIREP Special and the format of the report. The estimate of next position may be omitted from an AIREP Special report unless the report is made at a planned position reporting point.

6.3 **Wind shear - reporting**

Wind shear encountered by aircraft should be reported by pilots to ATS as aircraft following may not have the performance required to recover from the same wind shear encounter. The wind shear may also be increasing in intensity, making flight through the wind shear more dangerous for following traffic.

Due to flight deck workload, reports may be initially reported as Wind Shear Escape and a full report provided when workload allows.

The full report should include:

a. an assessment of the intensity as follows:

   (1) light - shear causing minor excursions from flight path and/or airspeed;

   (2) moderate - shear causing significant effect on control of the aircraft;

   (3) strong - shear causing difficulty in keeping the aircraft to desired flight path and/or airspeed; or

   (4) severe - shear causing hazardous effects to aircraft controllability; and

b. a factual plain language report regarding airspeed/ground speed changes (gain or loss) or undershoot/overshoot effects; and

c. the altitude or altitude band at which the adverse effect was experienced; and

d. where practicable, other relevant information such as significant changes in wind direction and/or speed may be included.

At non-controlled aerodromes, the report should also be broadcast to all aircraft on the CTAF and should include the name of the aerodrome.

6.4 Volcanic activity reports

Volcanic activity reports should be made by the pilot of an aircraft whenever observed, regardless of any previous reports from other aircraft. See *ENR 1.1 APPENDIX 1* for a sample form Volcanic Activity Reporting (Model VAR).

7. VOLMET SERVICE

Specific information regarding the VOLMET service is detailed in the following table:

Name of station	Call sign/ IDENT/ Abbreviation (EM)	Frequency	Broadcast period	Hours of Service	Aerodromes/ Heliports included	REP, SIGMET INFO, FCST and Remarks
1	2	3	4	5	6	7
Alice Springs	VKA-930 VKA-931 'Australian'	6676kHz 11387kHz	H24	H+00 to H+05 and H+30 to H+35	Adelaide Brisbane Cairns Darwin Melbourne Perth Sydney Townsville	– METAR/ SPECI – TAF where significant – Availability of SIGMET (see note)
Bellenden Kerr		119.75MHz	H24	Continuous	Amberley Brisbane Cairns Hamilton Island Mackay Rockhampton Townsville	– METAR/ SPECI – TAF where significant

Name of station	Call sign/ IDENT/ Abbreviation (EM)	Frequency	Broadcast period	Hours of Service	Aerodromes/ Heliports included	REP, SIGMET INFO, FCST and Remarks
1	2	3	4	5	6	7
Broken Hill		128.25MHz	H24	Continuous	Adelaide Alice Springs Amberley Brisbane Darwin Melbourne Sydney	– METAR/ SPECI – TAF where significant
Ceduna		128.45MHz	H24	Continuous	Adelaide Alice Springs Kalgoorlie Melbourne Perth Sydney	– METAR/ SPECI – TAF where significant
Derby		128.45MHz	H24	Continuous	Broome Darwin Kununurra Meekatharra Perth Port Hedland	– METAR/ SPECI – TAF where significant

Name of station	Call sign/ IDENT/ Abbreviation (EM)	Frequency	Broadcast period	Hours of Service	Aerodromes/ Heliports included	REP, SIGMET INFO, FCST and Remarks
1	2	3	4	5	6	7
Goochegoochera		128.45MHz	H24	Continuous	Alice Springs Cairns Darwin Tennant Creek Tindal Townsville	– METAR/ SPECI – TAF where significant
Kalgoorlie		128.25MHz	H24	Continuous	Adelaide Alice Springs Ceduna Kalgoorlie Laverton Perth	– METAR/ SPECI – TAF where significant
Meekatharra		128.45MHz	H24	Continuous	Broome Karratha Meekatharra Mount Magnet Paraburdoo Perth Port Hedland	– METAR/ SPECI – TAF where significant

Name of station	Call sign/ IDENT/ Abbreviation (EM)	Frequency	Broadcast period	Hours of Service	Aerodromes/ Heliports included	REP, SIGMET INFO, FCST and Remarks
1	2	3	4	5	6	7
Mt Blackwood		119.85MHz	H24	Continuous	Amberley Brisbane Cairns Hamilton Island Mackay Rockhampton Townsville	– METAR/ SPECI – TAF where significant
Mt Canobolas		127.7MHz	H24	Continuous	Adelaide Alice Springs Amberley Brisbane Melbourne Perth Sydney Williamtown	– METAR/ SPECI – TAF where significant
Mt Ginini		128.65MHz	H24	Continuous	Adelaide Canberra Hobart Melbourne Wagga Wagga	– METAR/ SPECI – TAF where significant

Name of station	Call sign/ IDENT/ Abbreviation (EM)	Frequency	Broadcast period	Hours of Service	Aerodromes/ Heliports included	REP, SIGMET INFO, FCST and Remarks
1	2	3	4	5	6	7
Mt Isa		120.35MHz	H24	Continuous	Alice Springs Amberley Brisbane Cairns Mt Isa Tindal Townsville	– METAR/ SPECI – TAF where significant
Mt Mowbullan		119.95MHz	H24	Continuous	Amberley Brisbane Gold Coast Mackay Rockhampton Sunshine Coast Sydney	– METAR/ SPECI – TAF where significant
Mt William		119.75MHz	H24	Continuous	Adelaide Hobart Launceston Melbourne Mildura Perth	– METAR/ SPECI – TAF where significant

Name of station	Call sign/ IDENT/ Abbreviation (EM)	Frequency	Broadcast period	Hours of Service	Aerodromes/ Heliports included	REP, SIGMET INFO, FCST and Remarks
1	2	3	4	5	6	7
Point Lookout		119.75MHz	H24	Continuous	Amberley Brisbane Canberra Gold Coast Melbourne Rockhampton Sydney Williamtown	– METAR/ SPECI – TAF where significant
Note: Due to broadcast time constraints TAF for a location will be included only if significant elements are forecast during the first three hours of validity. If no significant elements are forecast in this period the identifier NOSIG will be broadcast in place of the TAF details following the METAR/SPECI for that location. Cloud types, excepting cumulonimbus, will not be included in VOLMET broadcasts.						

8. SIGMET AND AIRMET SERVICE

8.1 SIGMET

SIGMET is a concise description of the occurrence or expected occurrence, in an area over which area meteorological watch is maintained, of specified phenomena which may affect the safety of aircraft operations.

SIGMET are issued by MWO and disseminated by ATS as an element of ATC initiated FIS to aircraft operating on routes or in areas likely to be affected. This information will normally relate the phenomenon reported to designated reporting points and where possible will indicate the area in which the phenomenon exists.

MWO	HR	FIR	Type and validity
Melbourne	H24	YBBB YMMM	SIGMET 4HR
Specific procedures: a. SIGMET for turbulence or icing above 10,000FT are issued north of 50°S. SIGMET for turbulence or icing above 10,000FT south of 50°S upon request b. SIGMET for thunderstorms are issued for north of 50°S. c. SIGMET for phenomena at and below 10,000FT (other than thunderstorms, tropical cyclones and volcanic ash) are issued for GAF areas, and outside GAF areas upon request.			
Melbourne	H24	YBBB YMMM	SIGMET VA/TC 6HR
Specific procedures: Nil			
Brisbane	H24	YBBB YMMM	SIGMET 4HR
Specific procedures: a. SIGMET for phenomena at and below 10,000FT (other than thunderstorms, tropical cyclones and volcanic ash) are issued for GAF areas, and outside GAF areas on request.			

SIGMET for volcanic ash cloud and tropical cyclones is issued for the whole of Melbourne and Brisbane FIR (YMMM and YBBB).

SIGMET are issued in both text and graphical format.

If a text SIGMET cannot be rendered graphically, it will be displayed in text format on the graphic.

More information on SIGMET can be found in the Product Information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

8.2 AIRMET

AIRMET information concerns the occurrence or expected occurrence, in an area over which meteorological watch is being maintained, of certain phenomena that have not been included in a current GAF.

AIRMET information, which concerns phenomena of a lesser degree of severity than SIGMET information, at or below 10,000FT.

AIRMET information is issued by MWO and disseminated by ATS as an element of ATC initiated FIS, to aircraft operating on routes or in areas likely to be affected. It will indicate the locality or area in which the phenomena exist or are expected to exist.

AIRMET are issued in both text and graphical format.

More information on AIRMET, including the phenomena that warrant issuance, can be found in the Product Information section of BoM Aviation Knowledge Centre: www.bom.gov.au/aviation/knowledge-centre/.

9. OTHER AUTOMATED METEOROLOGICAL SERVICES

9.1 Aerodrome Weather Information Service (AWIS) and Weather and Terminal Information Reciter (WATIR)

AWIS and WATIR transmit meteorological information from the Automatic Weather Station via a phone number, or at some aerodromes via a VHF radio broadcast. WATIR combines the meteorological information with additional terminal information from the airport operator. AWIS and WATIR information is classed as “real time” data.

Some, or all of the following information may be provided:

a. Message identifier e.g. “AWS AERODROME WEATHER” or “AUTOMATED WEATHER INFORMATION SERVICE”.
b. Station identifier as a plain language station name;
c. time (UTC);
d. wind direction in degrees magnetic and speed in knots;
e. visibility;
f. RVR (where available);
g. present weather;
h. cloud below 10,000FT, amount and height;
i. temperature in whole degrees Celsius;
j. dew point in whole degrees Celsius;
k. QNH in whole hectopascals;
l. rainfall (last 10 minutes).

When information is not available the relevant element of the broadcast will be identified as “[ELEMENT NAME] CURRENTLY NOT AVAILABLE”; e.g. “TEMPERATURE CURRENTLY NOT AVAILABLE”.

When the information from the AWIS is determined as being corrupt a NOTAM will be issued.

The QNH from a BoM managed or BoM approved AWS is an approved source of QNH and may be used in accordance with *ENR 1.5 para 5.3*.

When AWIS information is available after hours (AH), and the aerodrome is uncontrolled, reference will be made to its availability in ATIS UTC.

The availability of AWIS and WATIR is contained in *ERSA FAC.*

GEN 3.6 SEARCH AND RESCUE

1. PURPOSE

1.1 The purpose of the Search and Rescue (SAR) organisation is to provide assistance to aircraft in distress and to search for, provide aid to, and organise the rescue of survivors of aircraft accidents and forced landings.

2. RESPONSIBLE AUTHORITY

2.1 Airservices Australia is responsible for the provision of a SAR alerting and in-flight emergency response service. Joint Rescue Coordination Centre (JRCC) Australia is responsible for the conduct of SAR for missing aircraft, aircraft reported crashed and ELT searches within the Search and Rescue Region (SRR) under Australian jurisdiction.

3. ORGANISATION

3.1 The Australian Maritime Safety Authority (AMSA) have established a Joint Rescue Coordination Centre in Canberra (JRCC AUSTRALIA) which is responsible for the coordination of aviation and maritime SAR efforts within the Australian SRR. Refer to the chart at *para 10.*

4. RESCUE COORDINATION CENTRES

SAR Region	Location and Address
AUSTRALIA SRR	JRCC Australia 82 Northbourne Avenue BRADDON ACT 2601 Postal Address: GPO Box 2181 CANBERRA CITY ACT 2601 Ph: 1 800 815 257 Ph: +61 2 6230 6899 (overseas) Ph: +61 419 309 721 (text message) Fax: 1 800 622 153 Email: rccaus@amsa.gov.au Website: www.amsa.gov.au/safety-navigation/search-and-rescue AFTN: YSARYCYX

5. PROCEDURES

5.1 Pilot Notification

5.1.1 The efficacy of the SAR action by Airservices or JRCC Australia is directly related to the amount and accuracy of details notified in the flight notification or flight note, and to any position details reported in flight. When notifying of in-flight difficulties, early advice and the degree of apprehension felt by the pilot will enhance the assistance which can be provided by the ground organisation.

5.1.2 When a pilot lands at a place other than an aerodrome included in the flight notification, the pilot should report the fact to ATS, JRCC Australia or the police as soon as possible.

5.1.3 **Flight Notes**

Persons maintaining a SARWATCH by means of a Flight Note must contact JRCC Australia in the event that the flight becomes overdue.

5.2 Advice to Aircraft Operators

5.2.1 If an emergency phase is declared, where practicable, the aircraft operator will be advised as follows:

a. **Uncertainty or Alert phase** - the ATS unit will advise the operator of all relevant information passed to JRCC Australia; or

b. **Distress phase** – JRCC Australia will advise the operator of all SAR response actions.

5.3 In-Flight Emergency Response Action

5.3.1 **Radio Failure or Failure to Report.** On the basis that only the aircraft's transmitter may have failed and that ground transmissions can still be received, the following information may be broadcast from suitable stations, including some radio navigation aids:

a. lowest safe altitudes;

b. direction, bearings, DR position or headings to steer if practicable;

c. emergency aerodromes;

d. weather conditions at destination and alternates;

e. ditching weather report; and

f. separation action being taken by ATC.

5.3.2 **Navigational Assistance**. All available navigation aids, which may be of use to the aircraft, will be switched on and non-continuous aids may also be activated. Furthermore:

a. ATS surveillance system equipped units will keep a lookout for the aircraft;

b. aerodrome lighting may be activated; and

c. if the aircraft can advise its last known position, and headings, speed and times flown since that position, an air plot can be made, actual winds added, and a DR position and heading to steer passed to the aircraft on an advisory basis.

5.3.3 **Intercept and Escort.** Arrangements may be made to escort an aircraft experiencing abnormal operations.

5.3.4 **Ditching.** When a ditching is likely, the JRCC will obtain the positions of ships along the aircraft's route and advise the pilot, and:

a. attempts will be made to arrange a direct speech link or a rapid relay system between the aircraft and a selected ship;

b. weather reports, including sea conditions and a selected ditching heading will be passed to the aircraft; and

c. arrangements may be made for ships to provide navigational assistance, lighting, and reduce the effects of the sea swell or wind.

5.4 **Medical Emergency**

5.4.1 A pilot that is <u>not</u> engaged in the transport of patients under HOSP or MEDEVAC operations, should notify ATC of an on board medical emergency by declaring a PAN and appending the words 'MEDICAL PRIORITY REQUIRED'. The pilot should specify any service attendance requirements such as RFFS. ATC will provide the flight with MEDEVAC priority but will not arrange an ambulance or activate aerodrome emergency procedures unless requested. Declaring a medical emergency does not satisfy the biosecurity pre-reporting requirements specified in *GEN 1.3*.

5.4.2 The pilot of a medical flight should notify ATC of changes to priority requirements by changing the status of their flight e.g. by amending status from HOSP to MEDEVAC.

5.5 **Participation In Searches**

5.5.1 **General.** JRCC Australia may request aircraft and crew to assist with a SAR operation.

5.5.2 **Briefing.** Whenever possible the pilot of a search aircraft should contact the JRCC to confirm the briefing prior to the search commencing.

5.5.3 **Debriefing.** Pilots should present a post-flight report to the JRCC as soon as possible after completion of the flight.

5.5.4 **Safety.** Civil aircraft engaged in SAR operations are required to comply with the Civil Aviation Regulations and Civil Aviation Safety Regulations.

6. EMERGENCY LOCATOR TRANSMITTER (ELT)

6.1 Procedures detailing the use of ELT in an emergency are found in *ERSA EMERG*.

Note 1: While the installation and use of automatic ELT saves lives, improper use will lead to false alarms and a resultant strain on scarce SAR resources.

Note 2: An ELT with GPS is able to transmit an accurate GPS location during an emergency. Search and rescue authorities will be able to locate the aircraft much faster if the ELT has GPS.

6.2 **Monitoring 121.5MHz**

Distress beacons, including ELT, transmit on 121.5, 243 and 406MHz. Pilots are requested to monitor 121.5MHz before engine start and after shutdown. Reception of a distress beacon transmission at any time is requested to be reported to ATS or the JRCC immediately.

6.3 **Inadvertent Activations**

If an ELT has been inadvertently activated, this must also be reported to ATS or the JRCC immediately.

6.4 **Testing ELT**

Operational tests must be limited to 5SEC, and the preferred procedure is that they be conducted within the first 5MIN of the hour. JRCC Australia must be notified in advance of the test and, where the beacon is operated on 406MHz, its HexID must be provided. Detailed ELT testing procedures can be found on the AMSA website at www.amsa.gov.au/beacons.

6.5 **ELT Registration**

All ELT must be registered on the Australian Beacon Register. See www.amsa.gov.au/beacons for further details.

Registration allows the JRCC Australia to respond more quickly and effectively to real distress activations of an ELT, and also helps the JRCC Australia respond appropriately to inadvertent activations.

7. COSPAS-SARSAT

7.1 COSPAS (from the Russian acronym for "Space System for Search of Vessels in Distress") and SARSAT (from "Search and Rescue Satellite Aided Tracking") is an international distress beacon detection system. The system has the capability to detect and provide locations for Emergency Locator Transmitters (ELT), Personal Locator Beacons (PLBs) and Emergency Position Indicating Radio Beacons (EPIRBs).

The Australian Mission Control Centre (AUMCC) operated by the JRCC Australia in Canberra receives beacon data from satellite tracking stations in Australia and New Zealand, and from other Mission Control Centres (MCCs) around the world.

The COSPAS-SARSAT system detects and locates 406MHz ELT, PLB and EPIRB all over the world. The system uses three satellite constellations to provide continuous, global coverage.

COSPAS-SARSAT compatible beacons also transmit a local homing signal on 121.5MHz. Some defence force beacons transmit an additional homing signal on 243MHz.

8. SEARCH AND RESCUE TRANSPONDER (SART)

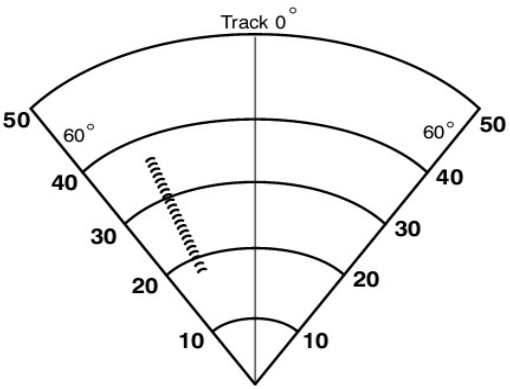
8.1 The International Maritime Organisation has introduced a requirement for all vessels over 300 tonnes gross weight, engaged in international trade, to be equipped with SARTs. These devices are intended to enhance the probability of location and rescue of survivors.

8.2 SARTs work by responding to radar transmissions in the 9 Gigahertz (GHZ) range, the frequencies used by most maritime and aviation navigation and weather radars (aircraft radars would need to be used in the mapping mode).

8.3 When a SART receives a radar pulse, it will transmit 10 - 20 pulses in reply. This will show on the radar screen as a series of 'returns' leading the way from the SARTs position. SART transmissions will not interfere with the primary role of aircraft radars, i.e: weather detection.

8.4 ICAO and Airservices do not intend to introduce SARTs to the aviation industry. However, pilots who detect SART transmissions should report them, with the position, to ATS, who will alert SAR authorities.

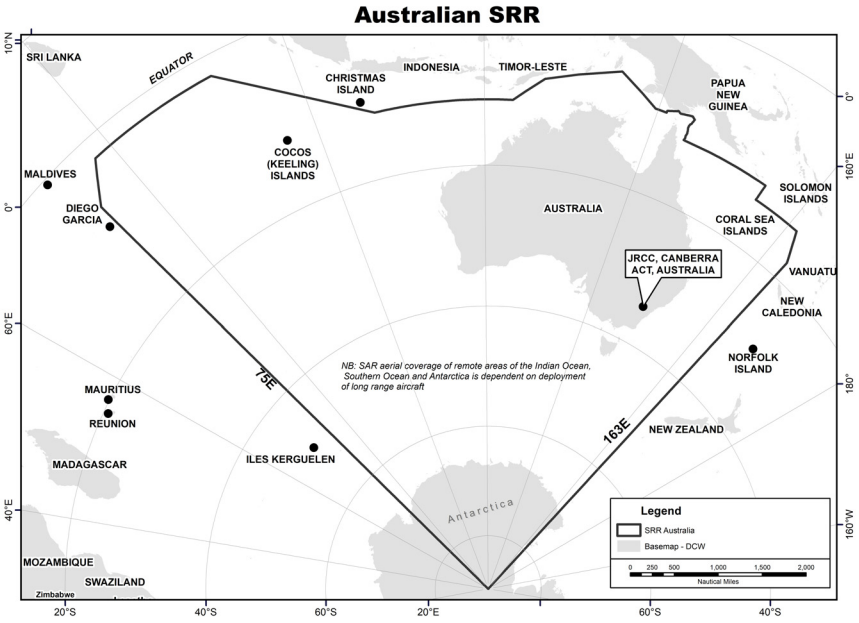
8.5 **Example of SART - 40° left at 19NM**



9. SAR ACTION

9.1 Events that will initiate SAR action include, but are not limited to: failure to cancel SARTIME, failure to cancel SARWATCH, aircraft declared overdue by Flight Note or concern from ATS, company, family or friends, an ELT activation, or other electronic distress notification device. As such, following safe arrival at destination, pilots are strongly encouraged to notify whomever is holding their SARWATCH as soon as practical after landing to avoid unnecessary SAR action.

10. SRR BOUNDARY



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GEN 4. CHARGES FOR AERODROMES/HELIPORTS AND AIR NAVIGATION SERVICES

GEN 4.1 AERODROME/HELIPORT CHARGES

1. LANDING CHARGES

Most aerodromes will have airport specific conditions of use (including access agreements) and landing charge regimes. Aircraft operators should use the contact details listed in *ERSA FAC* to obtain specific information and/or obtain copies of appropriate documents.

2. PASSENGER MOVEMENT CHARGE

The PMC is an AUD60 charge payable where a passenger departs Australia for another country, whether or not the person returns to Australia. The PMC is administered by the Department of Home Affairs and its operational arm, the Australian Border Force. The preferred method of collection is through voluntary Arrangements with carriers, where the passenger pays PMC as incorporated in the ticket price. Details relating to the charge including how PMC is collected, how to establish a PMC Arrangement and exemptions can be found online:
https://www.abf.gov.au/entering-and-leaving-australia/crossing-the-border/passenger-movement/passenger-movement-charge-(pmc)
Carriers operating out of Australia should contact the PMC Administration Office via email: pmcadministration@abf.gov.au for further information on PMC and to establish an Arrangement.

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GEN 4.2 AIR NAVIGATION SERVICES CHARGES

1. AIRCRAFT CHARGES

1.1 Airservices Australia is responsible for the provision of air navigation facilities and services in Australia. These services are provided on commercial terms and Airservices levies flight-based charges to recover the cost associated with these facilities and services, including ARFFS from aircraft operators. Detailed information on Airservices charges, including how they are calculated and methods of payment can be found online: www.airservicesaustralia.com/industry-info/aviation-charging/.

1.2 Airservices also recovers a number of charges on behalf of other entities, including:

a. The meteorological service charge on behalf of the Bureau of Meteorology, covering the provision of aviation meteorological services and;

b. En route charges for operations in the Honiara and Nauru Flight Information Regions.

1.3 For queries in relation to charges levied by Airservices, contact: billing@airservicesaustralia.com.

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PART 2 - EN ROUTE (ENR)
ENR 0.
ENR 0.1 TABLE OF CONTENTS

ENR 1. GENERAL RULES AND PROCEDURES ..... ENR 1.1 - 1
ENR 1.1 GENERAL RULES ..... ENR 1.1 - 1
1. INTRODUCTION ..... ENR 1.1 - 1
2. OPERATIONS IN CONTROLLED AIRSPACE ..... ENR 1.1 - 1
2.1 Separation ..... ENR 1.1 - 1
2.2 Air Traffic Control Clearances and Instructions ..... ENR 1.1 - 1
2.3 Ground Movement ..... ENR 1.1 - 6
2.4 Take-off ..... ENR 1.1 - 7
2.5 Visual Departure - IFR Flights ..... ENR 1.1 - 8
2.6 VFR Departure by an Aircraft Planned IFR ..... ENR 1.1 - 8
2.7 After Take-off ..... ENR 1.1 - 9
2.8 VFR Climb and Descent - IFR Flights ..... ENR 1.1 - 10
2.9 VFR-On-Top - IFR Flights ..... ENR 1.1 - 11
2.10 En Route ..... ENR 1.1 - 13
2.11 Descent and Approach ..... ENR 1.1 - 14
2.12 Landing ..... ENR 1.1 - 18
2.13 Circuit Operations ..... ENR 1.1 - 21
2.14 Go Around and Missed Approach Procedure in VMC ..... ENR 1.1 - 22
2.15 Taxiing After Landing ..... ENR 1.1 - 22
3. OPERATIONS IN CLASS E AIRSPACE ..... ENR 1.1 - 24
3.1 ATC Traffic Services ..... ENR 1.1 - 24
3.2 VFR Flights in Class E Airspace ..... ENR 1.1 - 24
4. NAVIGATION REQUIREMENTS ..... ENR 1.1 - 25
4.1 Flight under the IFR ..... ENR 1.1 - 25
4.2 Flight under the VFR ..... ENR 1.1 - 25
4.3 Time ..... ENR 1.1 - 26
4.4 Track Keeping ..... ENR 1.1 - 27
4.5 Position Fixing ..... ENR 1.1 - 27
4.6 Aircraft Deviations in Controlled Airspace - Advice to ATC ..... ENR 1.1 - 27
4.7 Long Over-Water Flights ..... ENR 1.1 - 28
4.8 GNSS - Operations Without RAIM ..... ENR 1.1 - 28
5. AIR ROUTE SPECIFICATIONS ..... ENR 1.1 - 29
6. RADIO COMMUNICATION AND NAVIGATION REQUIREMENTS ..... ENR 1.1 - 30
6.1 Summary of Report and Broadcast Requirements ..... ENR 1.1 - 30

6.2 Inoperative Radio and No Radio Procedures ..... ENR 1.1 - 40
6.3 GNSS Reporting Requirements and Procedures ..... ENR 1.1 - 42
7. LAND AND HOLD SHORT OPERATIONS (LAHSO) ..... ENR 1.1 - 43
7.1 Introduction ..... ENR 1.1 - 43
7.2 Locations Where LAHSO are Used ..... ENR 1.1 - 43
7.3 A Dependent Procedures ..... ENR 1.1 - 43
7.4 Participation ..... ENR 1.1 - 44
7.5 Conditions for LAHSO ..... ENR 1.1 - 45
7.6 Pilot Advise of LAHSO Approval ..... ENR 1.1 - 46
7.7 ATIS Broadcast ..... ENR 1.1 - 47
7.8 Directed Traffic Information ..... ENR 1.1 - 47
7.9 Readback Requirements ..... ENR 1.1 - 47
7.10 Landing Distance Assessments ..... ENR 1.1 - 48
7.11 Go Around During LAHSO ..... ENR 1.1 - 48
8. VERTICAL SEPARATION IN THE AUSTRALIAN FIR ..... ENR 1.1 - 48
8.1 Reduced Vertical Separation Minimum (RVSM) ..... ENR 1.1 - 48
8.2 RVSM Operations ..... ENR 1.1 - 49
8.3 Operational Procedures Before Entering the RVSM Flight Level Band - RVSM Approved Aircraft ..... ENR 1.1 - 50
8.4 Operational Procedures After Entering the RVSM Flight Level Band – RVSM Approved Aircraft ..... ENR 1.1 - 50
8.5 ATC Responsibilities ..... ENR 1.1 - 52
8.6 Weather and Wake Turbulence and System Alerts ..... ENR 1.1 - 52
8.7 Flight Level Deviation Reporting ..... ENR 1.1 - 53
9. OPERATIONS IN CLASS G AIRSPACE ..... ENR 1.1 - 54
9.1 Communications ..... ENR 1.1 - 54
9.2 Circuit Information ..... ENR 1.1 - 58
9.3 Separation Minima for Take-off ..... ENR 1.1 - 59
9.4 Circuit Height ..... ENR 1.1 - 60
9.5 Taxiing ..... ENR 1.1 - 60
9.6 Departure Information ..... ENR 1.1 - 61
9.7 Climb and Cruise Procedures ..... ENR 1.1 - 62
9.8 Position Reports ..... ENR 1.1 - 63
9.9 Descent from Controlled Airspace ..... ENR 1.1 - 63
9.10 Arrival Information ..... ENR 1.1 - 63
9.11 Landing Manoeuvres ..... ENR 1.1 - 64
9.12 Circuit Entry ..... ENR 1.1 - 65
9.13 Straight-in Approach ..... ENR 1.1 - 68
9.14 Joining on Base ..... ENR 1.1 - 69

9.15 Military Initial and Pitch Circuit Procedures ..... ENR 1.1 - 71
9.16 Separation Minima for Landing ..... ENR 1.1 - 72
9.17 The Traffic Mix and Other Hazards at Non-Controlled Aerodromes ..... ENR 1.1 - 73
10. OPERATIONAL REQUIREMENTS - GENERAL ..... ENR 1.1 - 73
10.1 Taxiing After Landing ..... ENR 1.1 - 73
10.2 SARWATCH ..... ENR 1.1 - 73
10.3 Radio Telephony Requirements Outside Controlled Airspace ..... ENR 1.1 - 75
10.4 Diversion to an Alternate Aerodrome ..... ENR 1.1 - 75
10.5 Operating Requirements for Transponders ..... ENR 1.1 - 75
10.6 Operating Requirements for ADS-B Transmitters ..... ENR 1.1 - 76
10.7 Alternate Aerodromes ..... ENR 1.1 - 76
10.8 Suitability of Aerodromes ..... ENR 1.1 - 81
10.9 Fuel Requirements ..... ENR 1.1 - 83
10.10 Information by Pilots ..... ENR 1.1 - 84
10.11 Flights Over Water ..... ENR 1.1 - 85
10.12 Procedures for Ground Operations of Turbo-Jet Aircraft ..... ENR 1.1 - 86
10.13 Clearances - Pilot Responsibility ..... ENR 1.1 - 86
10.14 Special Requirements ..... ENR 1.1 - 86
10.15 Helicopter Operations - At Aerodromes and Helicopter Access Corridors and Lanes ..... ENR 1.1 - 91
11. PROCEDURES WHEN ATS TEMPORARILY NOT AVAILABLE ..... ENR 1.1 - 95
11.1 Traffic Information Broadcast by Aircraft (TIBA) ..... ENR 1.1 - 95
11.2 Listening Watch ..... ENR 1.1 - 96
11.3 Time of Broadcasts ..... ENR 1.1 - 96
11.4 Acknowledgment of Broadcasts ..... ENR 1.1 - 96
11.5 Changes of Cruising Level ..... ENR 1.1 - 96
11.6 Collision Avoidance ..... ENR 1.1 - 97
11.7 Position Reporting ..... ENR 1.1 - 97
12. MANDATORY BROADCAST PROCEDURES (ATS TEMPORARILY NOT AVAILABLE) ..... ENR 1.1 - 97
ENR 1.2 VISUAL FLIGHT RULES ..... ENR 1.2 - 1
1. FLIGHT RULES ..... ENR 1.2 - 1
1.1 The Visual Flight Rules (VFR) ..... ENR 1.2 - 1
1.2 Special VFR ..... ENR 1.2 - 1
2. VISUAL METEOROLOGICAL CONDITIONS (VMC) - TAKE-OFF, EN ROUTE AND LANDING ..... ENR 1.2 - 2

ENR 1.3 INSTRUMENT FLIGHT RULES ..... ENR 1.3 - 1
1. THE INSTRUMENT FLIGHT RULES (IFR) ..... ENR 1.3 - 1
2. FLIGHT RULES NOMINATION ..... ENR 1.3 - 1
ENR 1.4 ATS AIRSPACE CLASSIFICATION..... ENR 1.4 - 1
1. CONTROLLED AIRSPACE ..... ENR 1.4 - 1
1.1 General.....ENR 1.4 - 1
1.2 Control Areas.....ENR 1.4 - 5
1.3 Control Zone.....ENR 1.4 - 6
1.4 Airspace Reservation .....ENR 1.4 - 6
1.5 Authorisation of Civil Flights in Military Airspace .....ENR 1.4 - 7
1.6 Release of Control Areas, Control Zones, Restricted or Military Operating Areas .....ENR 1.4 - 7
1.7 Deactivation of Control Zones or Special Use Airspace .....ENR 1.4 - 8
2. PROVISION OF SEPARATION IN CONTROLLED AIRSPACE..... ENR 1.4 - 8
2.1 General.....ENR 1.4 - 8
2.2 Special Provisions .....ENR 1.4 - 9
2.3 Performance Based Navigation (PBN) for separation ....ENR 1.4 - 10
2.4 Wake Turbulence Separation .....ENR 1.4 - 11
3. CLASS G AIRSPACE ..... ENR 1.4 - 12
3.1 Flight Information Areas .....ENR 1.4 - 12
3.2 Broadcast Areas .....ENR 1.4 - 12
4. CLASSES OF AIRSPACE-SERVICES AND REQUIREMENTS ..... ENR 1.4 - 13
5. LANES OF ENTRY (LOE)..... ENR 1.4 - 15
6. REGULATION OF FLIGHT - ASSESSMENT OF PRIORITIES..... ENR 1.4 - 15
ENR 1.5 HOLDING, APPROACH AND DEPARTURE PROCEDURES..... ENR 1.5 - 1
1. HOLDING AND APPROACH TO LAND PROCEDURES..... ENR 1.5 - 1
1.1 General.....ENR 1.5 - 1
1.2 Aircraft Performance Category .....ENR 1.5 - 1
1.3 Helicopters.....ENR 1.5 - 2
1.4 Minimum Route Altitudes.....ENR 1.5 - 2
1.5 Procedure Entry.....ENR 1.5 - 2
1.6 Circling Approaches and Visual Circling.....ENR 1.5 - 2

1.7 Visual Manoeuvring (non-Circling) Subsequent to Non-Precision Approaches (NPA) and Approaches with Vertical Guidance (APV) ..... ENR 1.5 - 8
1.8 Visual Approach Area - Helicopter ..... ENR 1.5 - 9
1.9 Missed Approach - Standard Procedures ..... ENR 1.5 - 10
1.10 Missed Approach Tracking ..... ENR 1.5 - 12
1.11 Missed Approach Requirements - GNSS ..... ENR 1.5 - 13
1.12 Missed Approach - Helicopter Procedures ..... ENR 1.5 - 13
1.13 Visual Segments ..... ENR 1.5 - 14
1.14 Visual approach requirements for all flights ..... ENR 1.5 - 14
1.15 Visual approach requirements for IFR flights ..... ENR 1.5 - 15
1.16 Visual approach requirements for VFR flights ..... ENR 1.5 - 17
1.17 Handling Speeds ..... ENR 1.5 - 18
1.18 Speed Restrictions ..... ENR 1.5 - 18
1.19 Obstacle Clearance Altitude (OCA) ..... ENR 1.5 - 19
1.20 Aerodrome Operating Minima (AOM) ..... ENR 1.5 - 19
1.21 Descent Gradients ..... ENR 1.5 - 20
1.22 Descent ..... ENR 1.5 - 20
1.23 Wind Effect ..... ENR 1.5 - 21
1.24 Bank Angle ..... ENR 1.5 - 21
2. APPROACH PROCEDURES ..... ENR 1.5 - 22
2.1 Operations On Parallel Runways At Class C Aerodromes ..... ENR 1.5 - 22
2.2 Use of Navigation Aids ..... ENR 1.5 - 22
2.3 Minimum Sector Altitude ..... ENR 1.5 - 22
2.4 Approach Design Concept ..... ENR 1.5 - 23
2.5 Procedure Entry ..... ENR 1.5 - 24
2.6 Approach Segments and Fixes ..... ENR 1.5 - 26
2.7 Missed Approach Procedure ..... ENR 1.5 - 27
2.8 Reversal Procedures ..... ENR 1.5 - 27
2.9 DME Arcs ..... ENR 1.5 - 30
3. HOLDING PROCEDURES ..... ENR 1.5 - 30
3.1 General ..... ENR 1.5 - 30
3.2 Holding in controlled airspace ..... ENR 1.5 - 31
3.3 Limitations ..... ENR 1.5 - 31
3.4 Entry Into the Holding Pattern ..... ENR 1.5 - 33
3.5 Standard Holding Pattern ..... ENR 1.5 - 34
3.6 DME limit ..... ENR 1.5 - 35
3.7 Shortening ..... ENR 1.5 - 35
3.8 Descent in Holding Pattern ..... ENR 1.5 - 35

4. AERODROME METEOROLOGICAL MINIMA ..... ENR 1.5 - 35
4.1 Ceiling and Visibility Minima ..... ENR 1.5 - 35
4.2 Runway Visual Range and Runway Visibility ..... ENR 1.5 - 35
4.3 IFR Take-off and Landing Minima ..... ENR 1.5 - 36
4.4 Correction of Instrument Procedure Minima for non-standard temperatures ..... ENR 1.5 - 37
5. APPLICATION OF AERODROME METEOROLOGICAL MINIMA ..... ENR 1.5 - 37
5.1 Pilot Responsibilities ..... ENR 1.5 - 37
5.2 ATC Assessment ..... ENR 1.5 - 37
5.3 QNH Sources ..... ENR 1.5 - 38
6. ALTERNATE WEATHER MINIMA ..... ENR 1.5 - 38
6.1 IFR Flights ..... ENR 1.5 - 38
6.2 Special Alternate Weather Minima ..... ENR 1.5 - 39
7. PRECISION APPROACH OPERATIONS ..... ENR 1.5 - 40
7.1 General ..... ENR 1.5 - 40
7.2 Failures ..... ENR 1.5 - 40
7.3 Altimeter Checks and Flight Tolerances ..... ENR 1.5 - 41
7.4 Protection of GLS Critical and Sensitive Areas ..... ENR 1.5 - 41
8. STANDARD INSTRUMENT DEPARTURES ..... ENR 1.5 - 41
8.1 General ..... ENR 1.5 - 41
8.2 SID Procedures ..... ENR 1.5 - 42
9. NOISE ABATEMENT PROCEDURES ..... ENR 1.5 - 44
9.1 Application ..... ENR 1.5 - 44
9.2 Curfews ..... ENR 1.5 - 46
9.3 Flying Considerately ..... ENR 1.5 - 46
10. STANDARD INSTRUMENT ARRIVALS (STAR) ..... ENR 1.5 - 47
10.1 General ..... ENR 1.5 - 47
10.2 STAR Procedures ..... ENR 1.5 - 47
11. DME OR GNSS ARRIVAL PROCEDURES ..... ENR 1.5 - 48
11.1 General ..... ENR 1.5 - 48
11.2 Operations in Controlled Airspace ..... ENR 1.5 - 49
12. SIGNALS FOR THE CONTROL OF AERODROME TRAFFIC ..... ENR 1.5 - 50
12.1 Light Signals to Aircraft ..... ENR 1.5 - 50
12.2 Light Signals to Vehicles and Pedestrians ..... ENR 1.5 - 51
12.3 Ground Signals to Aircraft ..... ENR 1.5 - 52
ENR 1.6 ATS SURVEILLANCE SERVICES AND PROCEDURES ..... ENR 1.6 - 1
1. RADIO COMMUNICATIONS PROCEDURES ..... ENR 1.6 - 1

2. IDENTIFICATION PROCEDURES ..... ENR 1.6 - 1
3. VECTORING PROCEDURES ..... ENR 1.6 - 1
4. ATC RESPONSIBILITIES IN RESPECT OF UNIDENTIFIED AIRCRAFT ..... ENR 1.6 - 4
5. SPEED CONTROL (ARRIVING AIRCRAFT) ..... ENR 1.6 - 5
6. EMERGENCY PROCEDURES ..... ENR 1.6 - 5
6.1 General ..... ENR 1.6 - 5
6.2 Radio Failure Procedure ..... ENR 1.6 - 6
6.3 ATS Surveillance System - Failure Procedure ..... ENR 1.6 - 6
6.4 SSR Emergency Codes ..... ENR 1.6 - 6
6.5 ADS-B Emergency Codes ..... ENR 1.6 - 7
7. AIRCRAFT TRANSPONDER ..... ENR 1.6 - 7
7.1 Operation of SSR Transponders ..... ENR 1.6 - 7
ENR 1.7 ALTIMETER SETTING PROCEDURES ..... ENR 1.7 - 1
1. PRE-FLIGHT ALTIMETER CHECK ..... ENR 1.7 - 1
1.1 General ..... ENR 1.7 - 1
1.2 IFR Altimeters ..... ENR 1.7 - 1
1.3 VFR Altimeters ..... ENR 1.7 - 2
1.4 Accurate QNH and Site Elevation ..... ENR 1.7 - 2
2. BASIC ALTIMETER SETTING PROCEDURES ..... ENR 1.7 - 2
2.1 Transition Altitude, Transition Layer and Transition Level ..... ENR 1.7 - 2
2.2 Area QNH ..... ENR 1.7 - 3
3. CRUISING LEVELS ..... ENR 1.7 - 4
3.1 Selection of Levels ..... ENR 1.7 - 4
3.3 Block Levels ..... ENR 1.7 - 5
4. CHANGE OF LEVELS ..... ENR 1.7 - 6
4.1 ATC Approval Required ..... ENR 1.7 - 6
4.2 ATC Approval Not Required ..... ENR 1.7 - 7
5. TABLES OF CRUISING LEVELS (NORTH OF 80° S) ..... ENR 1.7 - 8
6. TABLES OF CRUISING LEVELS (SOUTH OF 80° S) ..... ENR 1.7 - 9
ENR 1.8 REGIONAL SUPPLEMENTARY PROCEDURES ..... ENR 1.8 - 1
ENR 1.9 AIR TRAFFIC FLOW MANAGEMENT ..... ENR 1.9 - 1
1. FLOW MANAGEMENT STAGES ..... ENR 1.9 - 1
2. STRATEGIC - SCHEDULE COORDINATION ..... ENR 1.9 - 1
3. PRE-TACTICAL - GROUND DELAY PROGRAM ..... ENR 1.9 - 3
3.1 General ..... ENR 1.9 - 3
3.2 Calculated Off Blocks Time (COBT) ..... ENR 1.9 - 3

3.3 Non-scheduled flights ..... ENR 1.9 - 4
3.4 GDP run times ..... ENR 1.9 - 5
3.5 GDP revision ..... ENR 1.9 - 5
4. AIRPORT COLLABORATIVE DECISION MAKING (A-CDM) ..... ENR 1.9 - 6
4.1 General ..... ENR 1.9 - 6
4.2 TOBT and TSAT ..... ENR 1.9 - 6
4.3 GDP and A-CDM compliance ..... ENR 1.9 - 7
4.4 GDP revisions at A-CDM airports ..... ENR 1.9 - 7
5. TACTICAL ..... ENR 1.9 - 7
5.1 Aircraft sequencing near ATFM Airports ..... ENR 1.9 - 7
ENR 1.10 FLIGHT PLANNING ..... ENR 1.10 - 1
1. FLIGHT PLAN PREPARATION ..... ENR 1.10 - 1
1.2 Forecasts ..... ENR 1.10 - 1
1.3 GNSS Prediction Analysis - Flight in Oceanic and Remote Areas ..... ENR 1.10 - 3
2. FLIGHT NOTIFICATION ..... ENR 1.10 - 3
3. FLIGHT NOTIFICATION/NOTE CONTENTS ..... ENR 1.10 - 8
3.1 Forms ..... ENR 1.10 - 8
3.2 Flight Rules ..... ENR 1.10 - 8
3.3 PBN Notification ..... ENR 1.10 - 9
3.4 Military Flights ..... ENR 1.10 - 9
3.5 General ..... ENR 1.10 - 9
3.6 Location data ..... ENR 1.10 - 10
4. FLIGHT NOTIFICATION AMENDMENT ..... ENR 1.10 - 11
5. CARRIAGE OF FLIGHT DOCUMENTATION ..... ENR 1.10 - 12
ENR 1.11 ADDRESSING OF FLIGHT PLAN MESSAGES ..... ENR 1.11 - 1
1. GENERAL ..... ENR 1.11 - 1
ENR 1.12 INTERCEPTION OF CIVIL AIRCRAFT ..... ENR 1.12 - 1
1. INTERCEPTION PROCEDURES ..... ENR 1.12 - 1
1.1 Action by intercepted aircraft ..... ENR 1.12 - 1
1.2 Radio communications during interception ..... ENR 1.12 - 1
1.3 Signals initiated by intercepting aircraft and responses by intercepted aircraft ..... ENR 1.12 - 4
1.4 Signals initiated by intercepted aircraft and responses by intercepting aircraft ..... ENR 1.12 - 6
ENR 1.13 UNLAWFUL INTERFERENCE ..... ENR 1.13 - 1
1. PILOT ACTIONS ..... ENR 1.13 - 1

ENR 1.14 AIR TRAFFIC INCIDENTS ..... ENR 1.14 - 1
1. AUSTRALIAN TRANSPORT SAFETY BUREAU (ATSB) ..... ENR 1.14 - 1
1.1 About the ATSB ..... ENR 1.14 - 1
1.2 Contact information ..... ENR 1.14 - 1
2. REPORTING OF ACCIDENTS AND INCIDENTS ..... ENR 1.14 - 2
2.1 Introduction ..... ENR 1.14 - 2
2.2 Aircraft that must report reportable matters ..... ENR 1.14 - 2
2.3 Responsible persons for reporting safety matters to the ATSB ..... ENR 1.14 - 3
2.4 Reporting of accidents and incidents ..... ENR 1.14 - 4
3. REPORTABLE MATTERS ..... ENR 1.14 - 4
3.1 Introduction ..... ENR 1.14 - 4
3.2 Reportable matters for each type of operation category ..... ENR 1.14 - 5
3.3 Category A (passenger transport) aircraft operations ..... ENR 1.14 - 7
3.4 Category B (commercial non-passenger) aircraft operations ..... ENR 1.14 - 8
3.5 Category C (non-commercial) aircraft operations ..... ENR 1.14 - 8
3.6 Category D (Type 2 RPA and certain uncrewed balloons) aircraft operations ..... ENR 1.14 - 8
4. DEFINITIONS AND EXAMPLES OF TYPES OF REPORTABLE MATTERS ..... ENR 1.14 - 9
4.1 Introduction ..... ENR 1.14 - 9
4.2 Emergency declaration ..... ENR 1.14 - 9
4.3 Loss of separation standard between aircraft ..... ENR 1.14 - 9
4.4 Serious property damage incident (external) ..... ENR 1.14 - 10
4.5 Aircraft accident ..... ENR 1.14 - 10
4.6 Reportable serious aircraft incident ..... ENR 1.14 - 11
4.7 Aircraft incidents ..... ENR 1.14 - 15
4.8 Aircraft incident (external) ..... ENR 1.14 - 38
5. GUIDANCE FOR DETERMINING INJURY AND DAMAGE ..... ENR 1.14 - 41
5.1 Fatal aircraft-related injury ..... ENR 1.14 - 41
5.2 Serious aircraft-related injury ..... ENR 1.14 - 41
5.3 Guidance for the determination of aircraft damage ..... ENR 1.14 - 42
6. DETAILS TO BE REPORTED ..... ENR 1.14 - 44
6.1 Introduction ..... ENR 1.14 - 44

6.2 Telephone notification of Immediately reportable matters ..... ENR 1.14 - 44
6.3 Written reports (Immediately and Routine reportable matters) excluding bird/animal strikes ..... ENR 1.14 - 45
6.4 Written reports – aircraft incidents (external) bird/animal strikes ..... ENR 1.14 - 47
7. PROTECTING OF EVIDENCE FOR INVESTIGATIONS ..... ENR 1.14 - 47
ENR 2. AIR TRAFFIC SERVICES AIRSPACE ..... ENR 2.1 - 1
ENR 2.1 FIR, UIR, TMA ..... ENR 2.1 - 1
ENR 2.2 OTHER REGULATED AIRSPACE ..... ENR 2.2 - 1
1. OPERATIONS IN OCEANIC CONTROLLED AIRSPACE ..... ENR 2.2 - 1
1.1 Clearance Requirements ..... ENR 2.2 - 1
1.2 Mach Number Technique ..... ENR 2.2 - 1
2. SPECIAL PROCEDURES FOR IN-FLIGHT CONTINGENCIES ..... ENR 2.2 - 2
2.1 Introduction ..... ENR 2.2 - 2
2.2 General procedures ..... ENR 2.2 - 2
2.3 Actions to be taken once offset from track ..... ENR 2.2 - 4
2.4 Weather deviation procedures ..... ENR 2.2 - 5
3. STRATEGIC LATERAL OFFSET PROCEDURES (SLOP) IN OCA ..... ENR 2.2 - 7
4. USE OF GNSS IN OCEANIC AND REMOTE AREAS ..... ENR 2.2 - 8
5. OPERATIONS IN OCEANIC AIRSPACE REQUIRING PERFORMANCE-BASED COMMUNICATION AND SURVEILLANCE (PBCS) AUTHORISATION ..... ENR 2.2 - 9
ENR 3. ATS ROUTES ..... ENR 3.1 - 1
ENR 3.1 LOWER ATS ROUTES ..... ENR 3.1 - 1
ENR 3.2 UPPER ATS ROUTES ..... ENR 3.2 - 1
ENR 3.3 AREA NAVIGATION ROUTES ..... ENR 3.3 - 1
ENR 3.4 HELICOPTER ROUTES ..... ENR 3.4 - 1
ENR 3.5 OTHER ROUTES ..... ENR 3.5 - 1
2. CONDITIONAL ROUTES (CDR) ..... ENR 3.5 - 1
ENR 3.6 EN ROUTE HOLDING ..... ENR 3.6 - 1
ENR 4. RADIO NAVIGATION AIDS/SYSTEMS ..... ENR 4.1 - 1
ENR 4.1 RADIO NAVIGATION AIDS - EN ROUTE ..... ENR 4.1 - 1
ENR 4.2 SPECIAL NAVIGATION SYSTEMS ..... ENR 4.2 - 1
ENR 4.3 GLOBAL NAVIGATION SATELLITE SYSTEM (GNSS) ..... ENR 4.3 - 1

ENR 4.4 NAME-CODE DESIGNATORS FOR SIGNIFICANT POINTS....ENR 4.4 - 1
ENR 4.5 AERONAUTICAL GROUND LIGHTS – EN ROUTE....ENR 4.5 - 1
ENR 5. NAVIGATION WARNINGS....ENR 5.1 - 1
ENR 5.1 PROHIBITED, RESTRICTED, DANGER AND MILITARY OPERATING AREAS....ENR 5.1 - 1
1. GENERAL....ENR 5.1 - 1
2. FLIGHT WITHIN PRD AND MOA....ENR 5.1 - 2
ENR 5.2 MILITARY EXERCISE AND TRAINING AREAS AND AIR DEFENCE IDENTIFICATION ZONE (ADIZ)....ENR 5.2 - 1
1. MILITARY SENSITIVE AREAS....ENR 5.2 - 1
1.1 Brisbane FIR....ENR 5.2 - 1
1.2 Melbourne FIR....ENR 5.2 - 7
1.3 Permanent Drilling Rigs....ENR 5.2 - 8
2. MILITARY AAR AND AEW&C AIRSPACE....ENR 5.2 - 8
2.1 General....ENR 5.2 - 8
2.2 Explanation of terms....ENR 5.2 - 9
2.3 Amberley AAR and AEW&C airspace....ENR 5.2 - 11
2.4 Darwin AAR and AEW&C airspace....ENR 5.2 - 17
2.5 EAXA AAR and AEW&C airspace....ENR 5.2 - 19
2.6 Edinburgh AAR and AEW&C airspace....ENR 5.2 - 21
2.7 Tindal AAR and AEW&C airspace....ENR 5.2 - 22
2.8 Townsville AAR and AEW&C airspace....ENR 5.2 - 25
2.9 Williamtown AAR and AEW&C airspace....ENR 5.2 - 27
3. MILITARY AAR AND AEW&C AIRSPACE ANCHOR WAYPOINTS....ENR 5.2 - 34
3.1 Amberley....ENR 5.2 - 34
3.2 Darwin....ENR 5.2 - 35
3.3 EAXA....ENR 5.2 - 35
3.4 Edinburgh....ENR 5.2 - 35
3.5 Tindal....ENR 5.2 - 35
3.6 Townsville....ENR 5.2 - 36
3.7 Williamtown....ENR 5.2 - 36
4. MILITARY AAR TRACK WAYPOINTS....ENR 5.2 - 38
5. MILITARY AAR AND AEW&C PATTERNS....ENR 5.2 - 39
6. REFUELLING TRACKS....ENR 5.2 - 42
7. PROCEDURES FOR AIRCRAFT OPERATING IN AN AIR DEFENCE IDENTIFICATION ZONE....ENR 5.2 - 48
7.1 General....ENR 5.2 - 48
7.2 Special Requirements....ENR 5.2 - 49

7.3 Non-Compliance ..... ENR 5.2 - 49
7.4 Diversion of Aircraft for Defence Operations ..... ENR 5.2 - 50
ENR 5.3 OTHER ACTIVITIES OF A DANGEROUS NATURE AND OTHER POTENTIAL HAZARDS ..... ENR 5.3 - 1
ENR 5.4 AIR NAVIGATION OBSTACLES - EN ROUTE ..... ENR 5.4 - 1
1. MAN-MADE OBSTACLES GREATER THAN 360 FEET ..... ENR 5.4 - 1
2. MARKING OF POWER LINES AND OTHER OVERHEAD CABLES ..... ENR 5.4 - 1
3. NVIS AND OBSTACLE LIGHTING ..... ENR 5.4 - 1
ENR 5.5 AERIAL SPORTING AND RECREATIONAL ACTIVITIES ..... ENR 5.5 - 1
1. GLIDING OPERATIONS ..... ENR 5.5 - 1
1.1 General ..... ENR 5.5 - 1
1.2 Operations at Certified Aerodromes ..... ENR 5.5 - 1
2. PARACHUTING OPERATIONS ..... ENR 5.5 - 3
2.1 General ..... ENR 5.5 - 3
2.2 Conflicting Traffic ..... ENR 5.5 - 3
2.3 Additional Requirements in Controlled Airspace ..... ENR 5.5 - 3
2.4 Additional Requirements for Operations Above 10,000FT AMSL ..... ENR 5.5 - 3
3. BALLOON OPERATIONS ..... ENR 5.5 - 3
3.1 General ..... ENR 5.5 - 3
ENR 5.6 BIRD MIGRATION AND AREAS WITH SENSITIVE FAUNA ..... ENR 5.6 - 1
ENR 6. EN ROUTE CHARTS ..... ENR 6.1 - 1
ENR 6.1 AUSTRALIAN AIP AERONAUTICAL CHARTS ..... ENR 6.1 - 1

ENR 1. GENERAL RULES AND PROCEDURES

ENR 1.1 GENERAL RULES

1. INTRODUCTION

The air traffic rules and procedures applicable to air traffic in Australia and its Territories conform to *CASR Part 91* and the *Part 91 MOS*, and *CASR Part 172* and the *Part 172 MOS*. Differences to Annexes 2 and 11 to the Convention on International Civil Aviation are listed in *GEN 1.7.*

2. OPERATIONS IN CONTROLLED AIRSPACE

2.1 Separation

Note: Separation between aircraft is provided in accordance with ENR 1.4 Section 4.

2.2 Air Traffic Control Clearances and Instructions

2.2.1 A pilot in command of an aircraft must not enter a control zone or a control area that is Class A, B, C, D or E airspace without ATC clearance unless it is a VFR flight entering Class E airspace.

Note 1: An ATC clearance is valid:

a. for a flight wholly contained in controlled airspace – to the specified destination or clearance limit;

b. for a flight that is partly contained in controlled airspace – to the point where the aircraft first leaves controlled airspace; or

c. only if the flight enters controlled airspace in accordance with the clearance at or before the clearance expiry time, if issued.

Note 2: To request access to Restricted or Military Operating Areas contact the relevant controlling or administering authority - refer ENR 5.1.

2.2.2 A pilot must advise ATC immediately if issued a clearance with which the pilot cannot comply. If considered necessary, a pilot should request a different clearance from that issued.

2.2.3 When clearance has been issued to deviate from the cleared route the pilot must advise ATC when the deviation:

a. is no longer required; or

b. has been completed and the aircraft is established back on its cleared route.

2.2.4 If unable to obtain a clearance for a deviation and the pilot considers there is no safer alternative course of action:

1) select code 7700;

2) broadcast an urgency message specifying details of the deviation on the appropriate ATC and emergency frequencies; and

3) repeat the urgency message at regular intervals.

Note 1: Aircraft proceed at their own risk when entering an active Restricted or Military Operating Area without a clearance or approval.

Note 2: ATC will terminate control services and continue to provide flight information and alerting services.

2.2.4.1 Pilots must obtain a clearance to re-enter controlled airspace before exiting the Restricted or Military Operating Area.

2.2.5 Except for instructions related to SID and STAR operations when ATC issues a level clearance to an aircraft, any level restriction issued with an earlier clearance is automatically cancelled. ATC will issue (or re-issue) any required level restrictions with new level clearances by:

a. restating all restrictions; or

b. prefixing the subsequent instruction with “FURTHER RESTRICTION”.

2.2.6 **Entering Controlled Airspace**

2.2.6.1 When communication facilities permit, clearances will be passed direct to pilots by ATC. When direct communication on the published frequency is not possible the pilot should request a clearance through the ATS unit providing services in the preceding non-controlled airspace.

2.2.6.2 If proposing to fly into a control area from an aerodrome located so close to the entry point that making a full position report before entry is not practicable, the pilot should request a clearance:

a. prior to entering the runway, where direct communication is available;

b. after take-off, provided that the aircraft does not enter control area until cleared; or

c. prior to landing, when intending to depart for controlled airspace shortly after landing.

2.2.6.3 Where the en route LSALT is in controlled airspace, the pilot should request clearance prior to departure and include an estimated airborne time.

2.2.6.4 Excluding Class D airspace, when requesting clearance and flight details have not been provided, pilots must wait for ATC to respond with their call sign before providing the flight details.

2.2.6.5 When making first contact with Approach Control, the following apply:

a. Not Identified – report:

(i) DME or GNSS distance from aerodrome, if available;

(ii) VOR radial, GNSS track or compass quadrant from the aerodrome, or if issued a STAR clearance, the STAR designator;

(iii) assigned level;

(iv) flight conditions, if appropriate; and

(v) receipt of ATIS (code);

b. Identified – report:

(i) assigned level;

(ii) flight conditions, if appropriate; and

(iii) receipt of ATIS (code).

2.2.7 **Operations in Class D Airspace**

2.2.7.1 Within a Class D CTR, a clearance to take-off is a clearance to operate within or depart the CTR into Class G airspace in accordance with the ready report.

2.2.7.2 Two-way communications established between a pilot and ATC constitutes a clearance for the aircraft to enter Class D airspace.To establish two-way communications the pilot must:

a. initiate communications and advise current position, altitude, intention, ATIS received and any request(s); and

b. establish communications with ATC as follows in order to enter Class D airspace:

ATC Response	Communications Established	Pilot Action
(aircraft call sign)	Yes	– Fly the track, level and intentions stated when initiating two-way communications. – Comply with any subsequent ATC instructions. – When no level instruction is issued, descend as necessary to join the aerodrome traffic circuit.
(aircraft call sign) (instructions)	Yes	– Comply with ATC instructions. – When no level instruction is issued, descend as necessary to join the aerodrome traffic circuit.
Responds to the initial radio call without using the aircraft call sign, e.g. AIRCRAFT CALLING ARCHER TOWER, STANDBY, or AIRCRAFT CALLING ROCKY TOWER, SAY AGAIN	No	Remain outside Class D airspace

Note: See GEN 3.4 for generic phraseology used by ATC when a clearance is not immediately available.

2.2.8 **RNP AR Departures**

2.2.8.1 Pilots of aircraft that have included PBN/T1 in Field 18 of the flight notification form should request an RNP AR departure at clearance delivery unless there is a standing agreement between the Operator and the ATS provider to automatically assign RNP AR departures for eligible flights.

2.2.9 **Pre-departure Clearance (PDC)**

2.2.9.1 Use of PDC is limited to operations authorised by Airservices Australia. Operators wishing to participate in PDC should submit a request to Airservices Australia. Participating operators must not delete any component of the PDC message nor amend the order of the text.

2.2.9.2 When departing an airport participating in PDC, pilots must obtain the PDC, via ACARS or hard copy message, no later than 15 minutes prior to EOBT. If the PDC is not available by 15 minutes prior to EOBT, pilots must contact the ACD frequency for a verbal airways clearance.

Note: ATC will not send amended route clearances via PDC.

2.2.9.3 Pilots must readback the following items on the ACD frequency, or on the SMC frequency if ACD is not established, prior to a pushback or taxi request:

a. The SID, including runway and/or transition (if issued);

b. Transponder code;

c. Additional requirements specified in the PDC; and

d. Current parking position/bay.

2.2.10 **Flying Training Clearances**

2.2.10.1 Pilots of multi-engined aircraft must obtain ATC approval before conducting asymmetric training within 5NM of a controlled aerodrome.

2.2.11 **Clearances for operations at night or in IMC below published LSALT**

2.2.11.1 Some flights are specially authorised to operate below the published LSALT at night or in IMC (e.g. using NVIS). The pilot in command is solely responsible for avoiding terrain in such cases. Associated clearance requests must be expressly initiated by the pilot in command and should inform ATC about the nature of the operation (e.g. NVIS, OWN TERRAIN CLEARANCE, etc.).

2.3 Ground Movement

2.3.1 Pushback

2.3.1.1 The pilot in command must obtain an approval to pushback where this manoeuvre is necessary prior to taxiing. Information about other aircraft moving on the same apron will be provided by the apron service.

Note: See ENR 1.9 for start-up requirements at aerodromes that have implemented Airport Collaborative Decision Making.

2.3.2 Taxi Clearance

2.3.2.1 When operating from a controlled aerodrome where ATIS is in operation, a pilot in command must obtain the ATIS prior to taxi, and advise ATC of the ATIS code when requesting taxi clearance.

2.3.2.2 For IFR flights other than Australian air transport operations the pilot in command must provide ATC with the number of POB when requesting taxi clearance.

2.3.2.3 Pilots of civil VFR training flights should advise DUAL or SOLO, as appropriate, when requesting clearance.

2.3.2.4 The pilot in command must obtain a taxi clearance either prior to moving on the manoeuvring area, or in the case of *para 2.3.1*, at the completion of the pushback manoeuvre.

2.3.2.5 Avoidance of collision on apron areas is a joint responsibility of the pilot in command and any assisting company ground personnel.

2.3.2.6 VFR flights wishing to depart without submitting flight notification should provide the following information on first contact with ATC:

a. aircraft call sign and “FLIGHT DETAILS FOR DEPARTURE” (wait for a response from ATC);

b. destination and first tracking point;

c. preferred level; and

d. identification of ATIS code received.

2.3.3 Circuit Direction

A pilot in command must use the word “REQUIRE” to notify ATC if a particular turn or circuit is essential to the safe operation of the aircraft.

2.3.4 **Stop Bar Procedures**

2.3.4.1 At a controlled aerodrome, all runways are considered available for use unless aerodrome users are notified otherwise (*see para 2.3.5*). Where installed, stop bars are operated at all times to prevent entry onto a runway. Pilots must stop and hold at all illuminated stop bars and may only proceed when a verbal ATC clearance to enter or cross the runway has been received and the stop bar lights have been switched off.

2.3.5 **Closed Runways Local Procedures**

2.3.5.1 Closed runways are excluded from standardised stop bar procedures and are subject to local arrangements between the aerodrome operator, ATC and runway users (refer to *AIC/SUP, NOTAM* or the aerodrome operator).

2.3.6 **Stop Bar Contingency Procedures**

2.3.6.1 If stop bar lighting cannot be deselected, the activation of stop bar contingency procedures will be notified via voice or the ATIS. ATC may instruct pilots and drivers to cross an illuminated stop bar when stop bar contingency procedures are in force.

2.4 **Take-off**

2.4.1 **Tower Frequency and Information Transfer**

2.4.1.1 Domestic aircraft should change to tower frequency:

a. in the holding bay; or

b. close to, or at, the runway-holding position of the nominated runway, when ready for take-off.

2.4.1.2 At Class D aerodromes, pilots must include the following information when reporting ready:

a. The departure runway when parallel runway operations are in progress;

b. Their intentions when operating wholly within a Class D CTR; and

c. Their tracking details when departing the Class D CTR and not in receipt of an airways clearance.

2.4.2 **Runway Entry**

2.4.2.1 When a backtrack on the runway nominated for take-off is required, the pilot must obtain a clearance to backtrack prior to entering the runway.

2.4.2.2 Aircraft issued a conditional clearance to enter or cross a runway must identify the vehicle or aircraft causing the conditional clearance.

2.4.3 **Holding on Runway**

2.4.3.1 The pilot in command must obtain a clearance before holding on the runway in use.

D

2.5 **Visual Departure - IFR Flights**

2.5.1 By day in VMC, the pilot of an IFR flight may request a visual departure, or ATC may issue a visual departure.

2.5.2 **ATC Responsibilities**

2.5.2.1 ATC will only issue a visual departure to an IFR flight when the cloud base is such that the pilot can maintain flight in VMC below the MVA (ATS surveillance services) or the MSA/LSALT.

2.5.2.2 When an IFR aircraft is issued heading instructions and/or required to maintain a level below the MVA or MSA/LSALT during a visual departure, “VISUAL” will be appended to the departure instruction.

2.5.3 **Pilot Responsibilities**

2.5.3.1 The requirements of this section are the visual departure procedures applicable to IFR flights under *CASR 91.305(3)(b)(i).*

2.5.3.2 A pilot of an IFR flight may only request a visual departure when the cloud base will allow the aircraft to climb in VMC to the MSA/LSALT applicable to the departure. Additionally, if the intended cruising level is lower than route LSALT, the cloud base must permit flight in VMC at that level.

2.5.3.3 During the conduct of a visual departure, a pilot must:

a. maintain the track(s)/heading(s) authorised by ATC;

b. remain not less than 500FT above the lower limit of the CTA; and

c. visually maintain obstacle clearance.

2.6 **VFR Departure by an Aircraft Planned IFR**

2.6.1 The pilot of an IFR flight departing a Class D aerodrome may request a VFR departure with the expectation of obtaining an IFR clearance en route.

2.6.2 The pilot of an IFR flight conducting a VFR departure:

a. must comply with the VFR.

b. is responsible for separation with other aircraft within the Class D airspace.

c. must obtain ATC clearance prior to entering Class A or C airspace.

d. must obtain ATC clearance to resume IFR in Class A, C, D or E airspace.

e. must notify ATC when reverting to IFR once in Class G airspace.

2.6.3 When an IFR aircraft conducts a VFR departure, ATC will treat the aircraft as:

a. VFR for separation services in Classes C, D and E airspace until the pilot requests and is granted an IFR clearance.

b. VFR in Class C or D airspace and VFR in receipt of an SIS in Class E or G airspace for traffic information.

c. IFR for all other services, such as SAR, weather and NOTAM information, in all classes of airspace.

2.7 **After Take-off**

2.7.1 **Airborne Report in airspace with ATS Surveillance**

In Class C and Class D control zones where an ATS surveillance service is provided, on first contact with Centre, Approach or Departures, a pilot must report:

a. if assigned an initial heading - the direction of turn and assigned heading;

b. the altitude passing, to nearest 100FT; and

c. the last assigned level.

2.7.2 **Departure Report - certain Class D aerodromes**

2.7.2.1 At certain Class D aerodromes where the tower also provides a procedural approach control service (see *ERSA*), a pilot must report on the TWR frequency after take-off:

a. tracking information; and

b. the last assigned altitude

However, this report is not required:

a. for VFR aircraft departing the control zone directly into Class G airspace; or

b. for aircraft that have been instructed to contact Centre, Approach or Departures once airborne - in which case an airborne report will be made on the relevant frequency.

2.7.2.2 Tracking information must confirm the track established with reference to the appropriate navigation aid or, if tracking via a SID, confirm the SID designator.

2.7.3 **Establishment on Track**

Unless tracking via a SID or otherwise instructed by ATC, a pilot in command must remain within 5NM of the departure aerodrome to establish flight on the departure track as soon as practicable after take-off.

2.7.4 **Frequency Change**

2.7.4.1 When frequency change instructions are issued immediately preceding the take-off clearance, pilots must transfer automatically from Tower as soon as practicable after take-off, preferably within one mile of becoming airborne.

2.7.4.2 In all other situations, pilots of departing aircraft are required to remain on Tower frequency until specific frequency change instructions are issued. Pilots can generally expect an instruction to contact Departures Control prior to reaching 2,000FT and should, when advised, effect the change as soon as possible.

2.7.4.3 When contacting Area Control, pilots must advise the last assigned level and, if not maintaining the assigned level, the level maintaining or last vacated level.

Note: The "last vacated level" may be omitted by identified aircraft squawking pressure altitude derived level information.

2.8 **VFR Climb and Descent - IFR Flights**

2.8.1 **General**

2.8.1.1 A pilot of an IFR flight, operating in VMC, in classes D and E airspace, may request to climb/descend VFR.

2.8.1.2 When, in the controller's judgement, there is reason to believe that flight in VMC may become impracticable, the controller will issue an alternative clearance that ensures separation from all other aircraft for which they have separation responsibility.

2.8.1.3 The pilot of an IFR flight cleared to "Climb/Descend VFR" will receive a service in accordance with *para 2.8.3.2*.

2.8.1.4 An appropriate clearance must be obtained prior to entering a different class of controlled airspace.

2.8.2 **Pilot Procedures**

2.8.2.1 The pilot of an IFR flight requires a clearance to conduct a VFR climb/descent in VMC.

2.8.2.2 When operating in VMC with an ATC clearance to "Climb/Descend VFR", pilots of IFR flights must:

a. comply with the VFR visibility and distance from cloud criteria stipulated in *ENR 1.2 Section 2*.;

b. comply with instrument flight rules that are applicable to the flight; i.e. position reporting, radio communications, cleared route, adherence to ATC clearance, etc; and

c. visually maintain obstacle clearance.

2.8.2.3 The pilot of an IFR flight operating VFR climb/descent must maintain vigilance so as to see and avoid other aircraft. Additionally, the pilot accepts the responsibility for wake turbulence separation.

2.8.3 **ATC Procedures**

2.8.3.1 On receiving a request for VFR climb/descent, ATC may instruct the pilot to "Climb/Descend VFR" for a specified portion of the flight.

2.8.3.2 When the pilot is cleared to "Climb/Descend VFR", ATC will provide:

a. mutual traffic information service on IFR flights;

b. traffic information service on known VFR flights as far as practicable; and

c. a flight information service.

Note: IFR separation is not provided.

2.9 **VFR-On-Top - IFR Flights**

2.9.1 **General**

2.9.1.1 In Class E airspace, a pilot of an IFR flight may request VFR-on-top in lieu of an assigned altitude. This permits a pilot to select a VFR altitude or flight level of their choice subject to any ATC restrictions.

2.9.1.2 Pilots desiring to climb through cloud, haze, smoke, or other meteorological formation may request a climb to VFR-on-top.

2.9.1.3 ATC clearance to “Maintain VFR-on-top” is not intended to restrict pilots so that they must operate only above an obscuring meteorological formation (layer). Instead, the clearance permits operation above, below, between layers, or in areas where there is no meteorological obscuration.

2.9.1.4 When, in the controller’s judgement, there is reason to believe that flight in VMC may become impracticable, the controller must issue an alternative clearance that ensures separation from all other aircraft for which they have separation responsibility.

2.9.1.5 The pilot of an IFR flight cleared to “Maintain VFR-on-top” will receive a service in accordance with *para 2.9.3.2*.

2.9.1.6 An appropriate clearance must be obtained prior to entering a different class of controlled airspace.

2.9.1.7 ATC resumes separation responsibility when the aircraft is re-cleared to maintain an IFR level.

2.9.2 **Pilot Procedures**

2.9.2.1 The pilot of an IFR flight requires a clearance to operate VFR-on-top.

2.9.2.2 When operating with an ATC clearance to “Maintain VFR-on-top”, pilots on IFR flight plans must:

a. fly at the appropriate VFR levels as prescribed in *ENR 1.7 Section 5.*;

b. comply with the VFR visibility and distance from cloud criteria stipulated in *ENR 1.2 Section 2.*;

c. comply with instrument flight rules that are applicable to the flight; i.e. minimum IFR altitudes, position reporting, radio communications, cleared route, adherence to ATC clearance, etc; and

d. advise ATC prior to any altitude change to ensure the exchange of accurate traffic information.

2.9.2.3 The pilot of an aircraft operating VFR-on-top must maintain vigilance so as to see and avoid other aircraft. Additionally, the pilot accepts the responsibility for wake turbulence separation.

2.9.3 **ATC Procedures**

2.9.3.1 On receiving a request for VFR-on-top, ATC may instruct the pilot to climb to "VFR-on-top". This instruction will include:

a. if required, a clearance limit, routing, and an alternative clearance if VFR-on-top is not reached by a specified altitude;

b. the requirement to report reaching VFR-on-top; and

c. the reported height of the tops or that no tops reports are available.

2.9.3.2 When the pilot reports reaching VFR-on-top, ATC re-clears the aircraft to "Maintain VFR-on-top" and will provide:

a. mutual traffic information service on IFR flights,

b. traffic information service on known VFR flights as far as practicable, and

c. a flight information service.

Note: IFR separation is not provided.

2.9.3.3 ATC will not clear an aircraft to "Maintain VFR-on-top" at night to separate holding aircraft from each other or from en route aircraft unless restrictions are applied to ensure the appropriate IFR vertical separation.

2.10 **En Route**

2.10.1 All levels flown in classes A, C and D airspace, and IFR levels flown in Class E airspace, must be assigned by ATC. Levels flown by VFR aircraft or IFR flights maintaining VFR-on-top in Class E airspace must be in accordance with the VFR Table of Cruising Levels.

2.10.2 Except when identified, position reports are required for all aircraft in classes A, C and D airspace, and for IFR flights or flights using the IFR Pick-up procedure after initial contact with ATC in classes E and G airspace.

2.10.3 **Reports**

2.10.3.1 The position report format appears in *APPENDIX 1*. *Section 2* of the report should only be transmitted when required by the operator or when deemed necessary by the pilot. *Section 3* of the report is required for the situations described in *APPENDIX 1*.

2.10.3.2 Aircraft operating area-type flights and nominating scheduled reporting times may limit the report to “level” and the “present position” or the sector of the survey area in which the aircraft is currently operating.

2.10.3.3 Pilots must give ATS notice of an impending position report by use of the word “POSITION”; e.g. “MELBOURNE CENTRE (call sign) POSITION”. Pilots must wait for the ATS instruction before reporting position.

2.10.3.4 Pilots must report maintaining an assigned level, unless ATC has advised IDENTIFIED. An IFR flight operating VFR-on-top or requesting IFR Pick-up must advise level maintaining.

2.10.3.5 After any frequency change, pilots must advise the last assigned level and, if not maintaining the assigned level, the level maintaining or last vacated level.

Note: The “last vacated level” may be omitted by identified aircraft squawking pressure altitude derived level information.

2.10.3.6 Pilots should normally report ETA at the place of intended landing when at the last position report.

2.11 Descent and Approach

2.11.1 General

2.11.1.1 Most companies operating jet aircraft have agreed to a standard descent profile which may be specified in the operator’s operational document suite. Pilots must adhere to this profile unless operational reasons require, or ATC instructs or approves, otherwise. A sustained speed variation of more than ±10KT IAS or ±M0.02 must be advised to ATC.

2.11.1.2 Pilots are not required to nominate a descent point if identified.

2.11.1.3 Pilots of IFR flights leaving classes A, C, D or E airspace should, before entering Class G airspace, contact the ATS unit providing services in that airspace.

2.11.2 Instrument Approach

2.11.2.1 **ATC Authorisation.** Unless authorised to make a visual approach, an IFR flight must conform to the published instrument approach procedure nominated by ATC.

2.11.2.2 A pilot request to conduct a specific approach should be made prior to STAR clearance issue, or prior to top of descent for arriving aircraft not on a STAR eligible route.

2.11.2.3 Authorisation for final approach will be in the form of a clearance for the type of approach as shown on the approach chart title. If visual at the minima, the nominated runway then becomes the clearance limit subject to any further ATC instructions and a clearance to land. In the event that the aircraft is unable to land from the instrument approach or loses visual reference whilst circling, the aircraft is cleared to carry out the published missed approach unless ATC directs otherwise. The pilot in command must seek further ATC instructions prior to reaching the end of the missed approach procedure.

2.11.2.4 Where an instrument approach results in the aircraft leaving controlled airspace, the clearance for the approach also provides clearance for the aircraft to re-enter overlying controlled airspace, Restricted or Military Operating Area in the event of a missed approach. ATC should be advised as soon as possible on the missed approach.

2.11.2.5 The chart title for an instrument approach procedure is used for all radiotelephony relating to the procedure (including entry procedures), subject to the following:

a. The word 'APPROACH' is included between 'type of approach' (VOR, ILS, RNP etc. including any procedure suffix) and the runway designator (RUNWAY 01, RWY 33 etc).

b. If multiple approach procedures are on the same chart, e.g. 'NDB-A and VOR-A', only the approach procedure being conducted should be referred to.

c. If the chart title has a parenthetical suffix, e.g. (LNAV/VNAV ONLY), (AR), the text in the parentheses is not included in radiotelephony.

d. A runway designator shown as optional in a radiotelephony phrase (e.g. [RUNWAY (number)]) should only be omitted when there is no possibility of confusion.

e. Other than for circling approaches, a procedure suffix (X, Y, etc.) may be omitted if there is no possibility of confusion.

2.11.2.6 Aircraft may be instructed to track via an instrument approach procedure and a level restriction assigned, if the aircraft is:

a. in VMC conducting instrument approach training; or

b. a military aircraft:

(1) conducting a non-precision approach; or

(2) conducting an approach procedure with vertical guidance or a precision approach provided that clearance for the approach is issued in sufficient time for the aircraft to maintain the required descent rate in accordance with the published procedure.

2.11.3 **Visual Approach (rules related to ATC)**

2.11.3.1 **ATC Authorisation.** Except as detailed in *para 2.11.3.2,* the criteria under which visual approaches may be authorised by ATC are as follows:

a. For an IFR flight:

(1) By day when:

– the aircraft is within 30NM of the aerodrome; and

– the pilot has established and can continue flight to the aerodrome with continuous visual reference to the ground or water; and

– visibility along the flight path is not less than 5,000M, or for helicopters 800M, or the aerodrome is in sight.

(2) By night when:

– the pilot has established and can continue flight to the aerodrome with continuous visual reference to the ground or water; and

– visibility along the flight path is not less than 5,000M; and

– the aircraft is within 30NM of the aerodrome; or

– if being vectored, the flight has been assigned the MVA and given heading or tracking instructions to intercept final or to position the aircraft within the circling area of the aerodrome.

b. For a VFR flight by day and night, the aircraft is within 30NM of the aerodrome.

2.11.3.2 In addition to the requirements of *para 2.11.3.1*, with the exception of Australian and New Zealand operators and aircraft conducting independent visual approaches at Sydney, Super or Heavy jet aircraft will only be assigned a visual approach when:

a. specifically requested by the pilot, and the pilot has reported the landing runway in sight; or

b. the straight-in approach aid is unserviceable.

2.11.3.3 In the case of *sub-para 2.11.3.2b*. above, the aircraft will be:

a. vectored to intercept final no closer than 8NM from the runway threshold, at an altitude not less than 2,500FT Above Aerodrome Level (AAL); and

b. assigned a straight-in visual approach when:

(1) established on final or on a heading to intercept final course at an angle of not more than 30 degrees; and

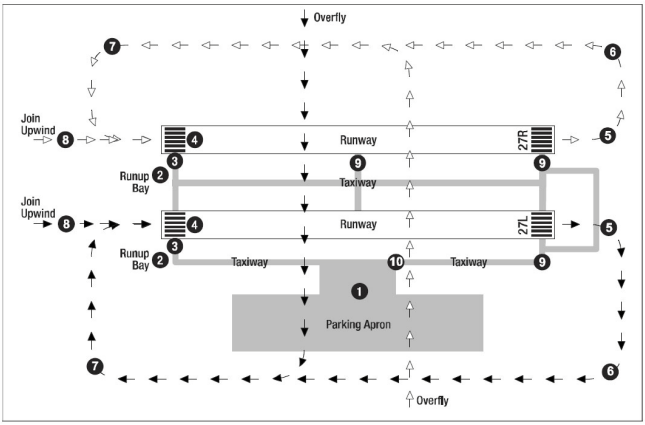
(2) visual glideslope guidance (VASIS/PAPI) is available; and

(3) the pilot has reported the landing runway in sight.

2.11.3.4 **Circuit Joining.** ATC may issue an instruction to join on a leg of the circuit or via:

a. Upwind, which directs circuit entry tracking upwind over the nominated runway centreline at the specified altitude; or

b. Overfly, which directs circuit entry into the opposing circuit by overflying the nominated runway at the specified altitude.



Decode information for the above diagram:

1 Aircraft initiates call to taxi - clearance provided

2 Aircraft vacating the bay give way to aircraft on taxiway

3 Line up or take-off clearance issued here

4 Take-off clearance issued if not issued at position 3

5 Crosswind leg

6 Downwind leg

7 Base leg

8 Final (joining upwind) leg

Landing clearance issued here unless issued on downwind/base leg

9 Taxi instructions, if required

10 Parking information issued if necessary

2.11.3.5 A pilot reporting VISUAL, may initially be given a clearance below the LSALT to a specific altitude in the following terms:

a. by day, “DESCEND TO (level) VISUAL”; or

b. by night, “WHEN ESTABLISHED IN THE CIRCLING AREA, DESCEND TO (level) VISUAL”.

Note: The requirements for pilots and operators relating to the conduct of, or making a request to conduct a visual approach are contained in ENR 1.5 Section 1.14.

2.12 **Landing**

2.12.1 **Provision of Operational Information**

ATC will supply the following information for landing operations:

a. runway or direction;

b. wind direction and speed, QNH and, if required, temperature and/or dew point;

c. known significant weather information, including low cloud and visibility or runway visual range;

d. the crosswind component on the runway to be used, if this equals or exceeds 8KT for single-engined aircraft or 12KT for multi-engined aircraft;

e. the tailwind component

f. aerodrome surface conditions significant to the operation including maintenance work within 23M of the runway side stripe marking;

g. birds or other hazards to aircraft; and

h. cautionary advice of wake turbulence.

2.12.2 **Selection of Landing Direction**

The pilot in command must ensure that the nominated runway or direction is operationally suitable. If the nominated runway or direction is not suitable then ATC must be advised using the phrase “REQUIRE RUNWAY (number)”. Such a request will not result in loss of priority provided it is made:

a. before reaching 80NM (120NM for jets) from a capital city aerodrome (including Essendon) or 30NM from other controlled aerodromes, for arriving aircraft wholly within controlled airspace; or

b. on first contact with ATC for arriving aircraft entering controlled airspace within the distance specified above or a control area step or a control zone.

The decision to land rests solely with the pilot in command.

2.12.3 **Selection of Circuit Direction**

A pilot in command must notify ATC if a particular turn or circuit is essential to the safe operation of the aircraft. The word REQUIRE must be used to enable ATC to identify the safety requirement.

2.12.4 **Downwind Report to Tower**

Unless otherwise instructed by ATC, the pilot of an arriving or circuit training aircraft must report DOWNWIND when starting or entering the downwind leg of the traffic circuit.

If frequency congestion prevents the call being made when starting the downwind leg, the pilot must report MID-DOWNWIND or LATE-DOWNWIND as appropriate.

2.12.5 **Clearances**

A pilot in command must not land unless the specific clearance “CLEARED TO LAND” has been received.

Note: ATC approval must be obtained if asymmetric training is to be carried out within 5NM of a controlled aerodrome (see Section 2.2.10).

2.12.6 **Separation Minima for Landing**

2.12.6.1 The appropriate wake turbulence separation standard will be applied by ATC between landing aircraft, except when a pilot has been assigned responsibility to maintain separation with another aircraft.

2.12.6.2 A landing aircraft will not be permitted to cross the threshold of the runway on its final approach until:

a. a preceding departing aircraft using the same runway:

(1) is airborne, and

– has commenced a turn; or

– is beyond the point on the runway at which the landing aircraft could be expected to complete its landing roll and there is sufficient distance to manoeuvre safely in the event of a missed approach;

(2) is at least 1,000M from the runway threshold, and

– has commenced the take-off run, and

– in the opinion of the controller, no collision risk exists, and

– the aircraft taking off has a MTOW of 7,000KG or less, and

– the landing aircraft is performance Category A and has a MTOW below 3,000KG.

b. a preceding landing aircraft using the same runway:

(1) has vacated it and is taxiing away from the runway; or

(2) has landed and has passed a point at least 1,000M from the threshold of the runway and will vacate the runway without backtracking, and

– in the opinion of the tower controller, no collision risk exists, and

– the preceding landing aircraft has a MTOW of 7,000KG or less, and

– the following landing aircraft is performance Category A and has a MTOW below 3,000KG; or

(3) has landed and has passed a point at least 600M from the threshold of the runway, is in motion and will vacate the runway without backtracking; and

– the preceding landing aircraft has a MTOW of less than 7,000KG, and

– the following landing aircraft has a MTOW of 2,000KG or less, or

(4) in the case where the following landing aircraft is a helicopter, the preceding landing aircraft is at least 300M down the runway from the threshold and ATC is satisfied that no collision risk exists.

c. a preceding aircraft, using a different runway, has crossed or stopped short of the landing aircraft's runway.

In the above situations, a landing clearance may be issued if ATC expect that the required runway separation standard will exist.

2.12.6.3 Other than as specified in *para 7.5.1,* exceptions to separation minima are:

a. aircraft landing in formation with respect to each other; and

b. aircraft operating in different areas or lanes on aerodromes with runways or facilities suitable for simultaneous landings.

2.13 Circuit Operations

2.13.1 Sequencing

2.13.1.1 When appropriate, ATC will issue a sequencing instruction.

2.13.1.2 In sequencing aircraft ATC will indicate the position of the preceding aircraft by reference to a leg of the circuit or as a clock bearing, and describe it either as a specific type or in general terms (e.g. Cessna or Twin).

2.13.1.3 ATC may issue a sequence number. Sequence numbers specify the landing sequence position of an aircraft with respect to any preceding traffic.

2.13.1.4 When issued with a sequencing instruction, a pilot must follow the preceding aircraft *and continue to do so unless otherwise instructed by ATC.*

2.13.1.5 The instruction FOLLOW requires the pilot to sight the preceding aircraft, and regulate circuit speed and approach path to achieve longitudinal separation. If the preceding aircraft cannot be sighted and identified, the pilot must advise ATC.

2.13.2 Non-Standard Circuit Operations

2.13.2.1 Pilots must advise ATC of non-standard circuit operations, e.g. glide and flapless circuits, normally with the DOWNWIND report. This advice will also alert other circuit traffic. ATC must also be advised of single engined aircraft conducting simulated engine failures at the earliest opportunity.

2.13.2.2 Where appropriate, an ATC sequencing instruction will indicate that there are no traffic restrictions precluding the manoeuvre. Subject to traffic, ATC may deny, or apply parameters to such operations for traffic management purposes.

2.13.3 **Parallel Runway Operations at Class D Aerodromes**

2.13.3.1 Where a Class D aerodrome is equipped with parallel runways, ATC may sequence aircraft for simultaneous contra-circuits and may conduct these operations using separate Tower frequencies for each runway. Operations will be regulated independently in each circuit, with an ATC clearance required to enter the opposite circuit or airspace.

2.14 **Go Around and Missed Approach Procedure in VMC**

2.14.1 Except as specified in ERSA for specific locations, an aircraft that is required to go around from a visual approach in VMC must initially climb on runway track, remain visual and await instructions from ATC. If the aircraft can not clear obstacles on runway track, the aircraft may turn.

2.14.2 In the event that an aircraft is unable, or does not wish, to land from an instrument approach in VMC, the aircraft must carry out the published instrument missed approach procedure for the instrument approach being flown, unless ATC directs otherwise.

2.14.3 At Class D aerodromes with parallel runways where contra-rotating circuit operations are in progress, if ATC instructs, or a pilot initiates a go around, the pilot must:

a. commence climb to circuit altitude;

b. position the aircraft on the active side and parallel to the nominated duty runway, while maintaining separation from other aircraft; and

c. follow ATC instructions or re-enter the circuit from upwind.

2.15 **Taxiing After Landing**

2.15.1 A pilot in command must not hold on the runway in use unless ATC has so authorised.

2.15.2 After landing, unless specified otherwise by ATC, an aircraft must comply with the following:

a. Promptly vacate the runway without backtracking.

b. Change from the tower frequency to the ground frequency (where established) when vacating the runway strip, and obtain an ATC taxi instruction.

c. Not cross any runway that intersects the taxi route unless in receipt of a taxi instruction and a "CROSS RUNWAY (number)" instruction from ATC.

d. Taxi to the destination via the most direct taxiway(s) available.

e. Where an apron service is provided on a discrete frequency (see *ERSA*), change to that frequency on entering the apron.

2.15.3 A taxi instruction which contains a taxi limit beyond a runway must include a "CROSS RUNWAY (number)" instruction to cross that runway. When an aircraft is required to hold short of a runway intersecting the taxi route, ATC will issue a taxi instruction limit of the runway-holding position associated with the intersecting runway.

2.15.4 An aircraft which has been issued with a taxi instruction limit of the runway-holding position of a runway intersecting the taxi route, or which has been issued with an instruction to "HOLD SHORT" of that runway, must subsequently be issued with an instruction to "CROSS RUNWAY (number)".

2.15.5 Aircraft required to hold short of a runway must hold at the appropriate runway-holding position for that runway, or the runway strip edge at the intersection of a crossing runway.

2.15.6 When separate frequencies for aerodrome control and surface movement control are in use, the pilot in command, on landing, must change from the aerodrome control frequency to the SMC frequency on vacating the runway strip, and then transmit the aircraft call sign and, if applicable, parking bay number. A pilot in command may "**REQUEST DETAILED TAXI INSTRUCTIONS TO (location)".**

2.15.7 The taxi clearance regulates movement on the manoeuvring area. The separation of aircraft taxiing on the manoeuvring area is a joint pilot and controller responsibility. Taxi clearance shall contain concise instructions and adequate information so as to assist flight crew to follow the correct taxi routes, to avoid collision with other aircraft and objects and to minimise the potential for the aircraft inadvertently entering a runway.

2.15.8 A taxi clearance will not relate to movement on the apron areas. However, available essential information referring to other aircraft entering or leaving the same apron area will be provided.

2.15.9 Radio watch must be maintained on the SMC or tower frequency (where no SMC frequency is provided) until parked.

3. OPERATIONS IN CLASS E AIRSPACE

3.1 ATC Traffic Services

3.1.1 In Class E airspace, IFR and VFR flights are permitted. IFR flights are provided with an ATC service, are separated from other IFR flights, and receive traffic information on VFR flights as far as is practicable. VFR flights receive a Surveillance Information Service (SIS), where available, on request.

3.1.2 Traffic information services provided by ATC do not relieve pilots of their responsibilities for continued vigilance to see-and-avoid other aircraft.

3.1.3 When vectors are provided to IFR flights in Class E airspace, terrain clearance will be provided by ATC. However, in VMC by day, pilots may be assigned responsibility for terrain clearance by use of the phrase “DESCEND TO (level) / CLIMB TO (level)/ TURN RIGHT / TURN LEFT (degrees) VISUAL”.

3.1.4 In Class E airspace, the following also apply:

a. Hazard Alerts will be directed to pilots of IFR flights, and to pilots of known VFR flights.

b. Unless operationally required by a pilot, ATC will only assign IFR levels.

3.2 VFR Flights in Class E Airspace

3.2.1 VFR flights entering Class E airspace do not require a clearance, but may receive a Surveillance Information Service (SIS), where available, on request (see *GEN 3.3 section 3.3*).

3.2.2 VFR flights entering and operating in Class E airspace should:

a. avoid published IFR routes, where possible;

b. monitor the appropriate Class E frequency and announce if in potential conflict; and

c. take appropriate action to avoid potential conflict.

3.2.3 Pilots of VFR flights should avoid IFR holding patterns.

4. NAVIGATION REQUIREMENTS

4.1 Flight under the IFR

4.1.1 An aircraft operating under the IFR must be navigated by:

a. an approved area navigation system that meets performance requirements of the intended airspace or route; or

b. use of a radio navigation system or systems on routes where, after making allowance for possible tracking errors of ±9° from the last positive fix, the aircraft will come within the rated coverage of a radio aid which can be used to fix the position of the aircraft. The maximum time interval between positive fixes must not exceed two (2) hours; or

c. visual reference to the ground or water by day, on route segments where suitable en route radio navigation aids are not available, provided that weather conditions permit flight in VMC and the visual position fixing requirements of *para 4.2.1b*. are able to be met.

4.2 Flight under the VFR

4.2.1 The following apply in respect of flight under the VFR:

a. The pilot in command must navigate the aircraft by visual reference to the ground or water, or by using any of the methods specified in *para 4.1.1*.

b. When navigating by visual reference to the ground or water, the pilot in command must positively fix the aircraft's position by visual reference to features shown on topographical charts at intervals not exceeding 30 minutes. When flying over the sea, visual reference features may include rocks and reefs and fixed man-made objects which are marked on suitable charts and are readily identifiable from the air.

Note: Flight above more than SCT cloud, or over featureless land areas, or over the sea, may preclude visual position fixing at the required intervals and may therefore make visual navigation impracticable.

c. When navigating by visual reference in controlled airspace the pilot must notify ATC if the aircraft's track diverges by more than one (1) nautical mile from the track approved by ATC, or, if navigating by reference to radio navigation aids, by more than the tolerances given in *para 4.6*.

d. VFR flight on top of more than SCT cloud is available provided that:

(1) VMC can be maintained during the entire flight, including climb, cruise and descent.

(2) For VFR flight on top of more than SCT cloud, the visual position fixing requirements of *sub-para b.*, or the other navigational requirements of *section 4.1* must be met.

(3) Prior to conducting a VFR flight on top of more than SCT cloud, the pilot in command must ensure that current forecasts and observations (including those available in-flight observations) indicate that conditions in the area of, and during the period of, the planned descent below the cloud layer will permit the descent to be conducted in VMC.

(4) The position at which descent below cloud is planned to occur must be such as to enable continuation of the flight to the destination and, if required, an alternate aerodrome in VMC (see *Note)*.

e. When navigating by reference to radio navigation aids or GNSS, the pilot in command must obtain positive radio fixes at the intervals and by the methods prescribed in *paras 4.1 and 4.5*.

f. The pilot in command of a VFR flight wishing to navigate by means of radio navigation systems or any other means must indicate in the flight notification only those radio navigation aids with which the aircraft is equipped and the pilot is competent to use under *CASR 61.385.*

Note: Pilots should not initiate VFR flight on top of more than SCT cloud when weather conditions are marginal. Before committing to operate VFR flight on top of more than SCT cloud, pilots should be confident that meteorological information used is reliable and current, and clearly indicates that the entire flight will be able to be conducted in VMC.

4.3 **Time**

4.3.1 During flight, pilots must maintain a time reference accurate to within ± 30 seconds.

4.4 **Track Keeping**

4.4.1 Tolerances are applied to tracks to assess containment areas for the purposes of ensuring navigational integrity, separation from other aircraft, terrain and obstacle clearance, and avoidance of specified airspaces. Although allowing for the errors inherent in the navigation systems used, these tolerances are based on the assumption that the pilot will maintain track as closely as possible.

4.4.2 The pilot in command must, at all times, take positive action to regain track as soon as a deviation from the correct track is recognised.

4.4.3 Aircraft must be navigated by the most precise means of track guidance with which the aircraft is equipped and the pilot is qualified to use.
The order of precision is Localiser, GNSS, VOR, then NDB.

4.5 **Position Fixing**

4.5.1 A positive fix is one determined by:

a. the passage of the aircraft over an NDB, VOR, TACAN, marker beacon or a DME site; or

b. the intersection of two or more position lines which intersect with angles of not less than 45° and which are obtained from NDB, VOR, localisers or DME in any combination. For the purpose of this paragraph, a position line must be within the rated coverage of the aid with the exception that if a fix is determined entirely by position lines from NDB, the position lines must be within a range of 30NM from each of the NDB; or

c. GNSS meeting the equipment requirements of *GEN 1.5 Section 2*.

4.6 **Aircraft Deviations in Controlled Airspace - Advice to ATC**

4.6.1 In controlled airspace, separation standards are based on the pilot maintaining route or track as closely as possible at all times. Corrective action must be taken to regain route or track as soon as any deviation is observed.

4.6.2 Additionally, the pilot must immediately notify ATC for any of the deviations described below:

a. where route or track guidance is provided by a localiser or VOR - half scale deflection or more of the Course Deviation Indicator (CDI);

b. where route or track guidance is provided by NDB - ±5° or more from the specified bearing;

c. where route or track guidance is provided by DME - ±2NM or more from the required arc;

d. where route or track guidance is provided by an area navigation system - when the aircraft cannot be maintained on the desired track plus/minus the prescribed RNP/RNAV value; and

e. when navigating by visual reference to the ground or water - more than 1NM from the cleared track.

Note: The values given above must not be interpreted as defining a sector within which the pilot is permitted to navigate or tolerances within which deviations from route or track without clearance are permitted.

4.7 **Long Over-Water Flights**

4.7.1 If an aircraft on a long over-water flight operating in oceanic Class A airspace has inadvertently deviated from the route specified in its ATC clearance, the pilot must take action to regain the cleared route within 200NM from the position at which the deviation was observed.

4.8 **GNSS - Operations Without RAIM**

4.8.1 ATS services, in particular aircraft separation, are predicated on accurate aircraft navigation and position fixing. If GNSS integrity is not assured, due to loss of RAIM or RAIM ALERT, the navigation system does not meet the required standard for navigation or the application of area navigation based separation standards. When GNSS integrity is not assured, the following procedures must be adopted:

a. Aircraft tracking must be closely monitored against other on-board navigation systems.

b. The pilot in command of an aircraft must advise ATS if any of the following occurs:

(1) during an en route phase of flight — there is RAIM loss or loss of GNSS integrity for more than 5 minutes;

(2) during a terminal phase of flight — there is RAIM loss or loss of GNSS integrity;

(3) when ATS requests the provision of GNSS-derived information — RAIM or GNSS integrity is not available;

(4) when ATS grants a clearance or imposes a requirement based on GNSS-derived information — RAIM or GNSS integrity is not available;

(5) the GNSS receiver is in dead-reckoning mode, or experiences loss of its navigation function, for more than 1 minute.

c. If valid position information is lost, or non- RAIM operation exceeds 5 minutes, the GNSS information is to be considered unreliable and another means of navigation should be used until RAIM is restored and the aircraft is re-established on track.

d. Following re-establishment of RAIM, the appropriate ATS unit should be notified of RAIM restoration prior to using GNSS information. This will allow ATC to reassess the appropriate separation standards.

5. AIR ROUTE SPECIFICATIONS

5.1 Unless otherwise authorised by ATC, when proposing to operate under the IFR on any route segments, or proposing flight in controlled airspace, the pilot in command must plan and conduct a flight in accordance with the:

a. route specifications published in *GEN 3.2* including the relevant en route chart; and

b. applicable flight planning requirements published in ERSA GEN; and

c. published accessibility of airspace.

5.1.1 The pilot in command is responsible for ensuring that the requirements of *Section 4.* can be met.

5.1.2 Where no route specification has been published in the relevant en route chart, a route determined by the pilot in command, and, if in controlled airspace, approved by ATC, will be planned.

5.1.3 Prior ATC approval is required for area navigation tracking on routes other than those published in AIP and the Airservices *"Off Air Route Planning (OARP) Manual"*. Information and rules regarding OARP are available at: www.airservicesaustralia.com/industry-info/flight-briefing/off-air-route-flight-planning-options/

5.1.4 The position reporting points for a route should be separated by a distance approximately 30 minutes or 200NM apart, whichever is least, and, when practicable, should be selected from those shown on en route charts. Otherwise, the position reporting points should be places named on VTC or WAC, and identifiable by radio or visual means. To minimise confusion, when a position is reported over a town which has a nearby aerodrome of the same name, the word 'township' must be used after the name in the text of the report.

5.1.5 For an area-type flight as distinct from route flying, the pilot of an IFR flight or VFR flight in those circumstances identified in *ENR 1.10* may nominate scheduled reporting times. These should be at half-hourly intervals. The pilot must specify the area's boundaries by means of a map provided with the flight notification details.

5.1.6 A pilot in command must make sure, by reference to the forecast, that the route selected for a VFR flight will enable the aircraft to be flown with visual reference to the ground or water for significant portions of the route, and in the vicinity of the destination aerodrome.

6. RADIO COMMUNICATION AND NAVIGATION REQUIREMENTS

6.1 Summary of Report and Broadcast Requirements

6.1.1 In this section:

a. 'Report' means a mandatory radio report from an aircraft to the appropriate ATS unit.

b. 'Broadcast' means a radio broadcast from an aircraft on the appropriate frequency to provide advisory traffic information to other aircraft.

6.1.2 Except in special circumstances (e.g. descent from CTA, formation flights, SAR, police/security), pilots of aircraft are required to comply with the radio communication requirements appropriate for the 'Classes of Airspace - Services and Requirements' table included in *ENR 1.4 Section 4.*

6.1.3 In special circumstances, a pilot may request to change frequency to meet operational report, broadcast, or communication requirements. ATC will facilitate a pilot request for approval to leave a control frequency to make such reports or broadcasts.

When impracticable to approve the frequency change at the time requested due to control requirements, ATC will accommodate the request as soon as possible. In determining when to make reports and broadcasts, pilots should consider the possibility of delays in being released from the ATC frequency.

Requests for frequency change should specify the expected duration when the change required is not permanent.

6.1.4 After any ATS directed frequency change, pilots must advise the last assigned level and, if not maintaining the assigned level, the level maintaining or last vacated level; e.g. "MELBOURNE CENTRE (call sign) CLEARED FLIGHT LEVEL TWO ONE ZERO, LEAVING FLIGHT LEVEL TWO NINER ZERO".

Note: The "last vacated level" may be omitted by identified aircraft squawking pressure altitude derived level information.

6.1.5 Whenever flight rules are changed during flight (i.e. VFR to IFR or IFR to VFR), the pilot must report to ATS at the time the change takes place.

6.1.6 All aircraft departing, arriving or transiting an AFIS broadcast area during AFIS HRS must make broadcasts prior to or as soon as possible after entering the broadcast area.

6.1.7 Pilots of aircraft engaged in parachute operations must:

a. obtain a clearance to drop when the operation is conducted in, or parachutists will enter, a Restricted or Military Operating Area or Classes A, C or D airspace. The drop clearance request must be made at least five (5) minutes before the proposed exit; and

b. in all cases, broadcast intentions on the appropriate area VHF, and/or CTAF, two (2) minutes prior to parachutists exiting the aircraft.

6.1.8 Unless otherwise authorised, gliding operations in controlled airspace (including Class E) must be conducted using the appropriate ATC frequency.

6.1.9 Pilots of IFR flights operating outside controlled airspace who desire to establish communication with a non-ATS station and who will not be able to maintain a listening watch on the ATS frequency must advise ATS of their further SAR requirements before making the frequency change.

6.1.10 Broadcast procedures when ATS is temporarily unavailable is addressed in *Section 11.*

Aircraft in Class A, B, C or D airspace, or IFR aircraft in Class E airspace – prescribed reports	
Situation	**Report**
Ready to Taxi	Report the situation
Airborne	Report the situation
Departure	Report the situation
Position report when required by the ATC service or the route reporting requirements in the authorised aeronautical information	Report the situation
Previously reported position estimate is more than 2 minutes in error	Corrected position estimate
Sustained variation of more than 10KT or Mach 0.02 from any previously notified speed or any standard descent profile agreed between the aircraft operator and ATS	Report the situation
Aircraft performance degraded below: a. the level required for the airspace in which it is operating; or b. the capability of the aircraft reported in the aircraft's flight notification	Report the situation
Leaving a level or reaching an assigned level	Report the situation
Unable to comply with ATC clearances or instructions	Report the situation
Before leaving controlled airspace on descent	Report the situation
Arrival	If cancelling SARWATCH — report cancellation, see *para 10.2.1.*

IFR aircraft in Class G airspace – prescribed reports	
Situation	**Report**
Taxiing	Report the situation
Departure	Report the situation
Reaching cruising level	Report the situation
Position report when required by the ATC service or by the route reporting requirements of the authorised aeronautical information	Report the situation
Previously reported position estimate is more than 2 minutes in error	Report the situation
Before changing level	Report the situation
Before changing frequency	Report the situation
Requiring clearance into controlled airspace	Report the situation
Before changing to CTAF and not monitoring ATS frequency on second COM system	Report the situation
After landing	If cancelling SARWATCH at this time — report the cancellation

VFR aircraft in Class E or G airspace — prescribed reports	
Situation	**Report**
Requiring clearance into controlled airspace	Report the situation
Before, and on completion of, over water stage	Report in accordance with SAR reporting schedules if arranged before the over water stage

Aircraft operating in a mandatory broadcast area – mandatory broadcasts	
Situation	**Broadcast**
Prior to, or immediately after entering an MBA	The pilot's intended use of the MBA
Joining a circuit	Broadcast the situation, and indicate the leg on which the aircraft will join
Conducting a straight-in approach	No later than 3NM from the runway threshold – Broadcast the situation
Passing the final approach fix of an instrument approach procedure	Broadcast the situation
Commencing a missed approach	Broadcast the situation
After landing and clear of the active runway(s)	Broadcast the situation
Starting to taxi	Broadcast the situation and the following information: if the flight is to be conducted under the IFR – this information; either the: (i) planned destination aerodrome for the flight; or (ii) direction in which the pilot intends to fly from the aerodrome; or (iii) nature of operation (e.g. circuits); and the runway proposed to be used for take-off.
Immediately before entering the runway to be used for take-off	Broadcast the following: a statement that the aircraft is entering the runway; and the runway identifier.

Aircraft operating in a mandatory broadcast area - mandatory broadcasts when an SFIS is active	
Situation	**Broadcast**
Taking off from an aerodrome in the MBA Immediately before, or during taxiing: *Note: ATS will issue an SSR code to IFR departures on first contact. Airways clearance may be issued by SFIS where aircraft performance and traffic disposition allow.*	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •if the proposed flight is to be conducted under the IFR a statement to that effect; and •the name of the aerodrome, and a. the proposed flight's intended destination; or b. the direction in which the pilot intends to fly from the aerodrome; or c. airwork intentions (e.g. circuits); and •the runway proposed to be used for take-off. (Aerodrome)
Entering the runway Immediately before entering the runway:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •[BACKTRACKING] and/or (LINING UP) RUNWAY (number). (Aerodrome)

Situation	Broadcast
IFR aircraft departing the aerodrome IFR departure report (non-controlled aerodromes surveillance):	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •*(location reference departure aerodrome)* PASSING **(*current level*)** CLIMBING TO *(intended level)* [ESTIMATING *(first reporting point)* AT *(time)*]. (Aerodrome)
VFR aircraft departing the aerodrome VFR departure report:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •*(location reference departure aerodrome)* PASSING *(current level)* CLIMBING TO *(intended level)*. (Aerodrome)
Changing intentions/track/level When the pilot intends to alter track and/or level given in a previous broadcast of intentions:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •the aircraft's position; and •the aircraft's present level; and •the pilot's intentions in relation to the flight. (Aerodrome)

Situation	Broadcast
Flying through an MBA where SFIS is active without landing The pilot in command of an aircraft intending to fly through the area without landing must, prior to or as soon as possible after the aircraft enters the MBA, make a broadcast that includes the following information:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •the aircraft's position; and •the aircraft's present level; and •the pilot's intentions in relation to the flight. (Aerodrome)
Landing at an aerodrome in the MBA The pilot in command of an aircraft intending to land at an aerodrome in the MBA, must prior to or as soon as possible after the aircraft enters the MBA, make a broadcast that includes the following information:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •the aircraft's radial, bearing or quadrant from, and distance from, the aerodrome at which the pilot proposes to land; and •the aircraft's altitude; and •the pilot's intentions in relation to the flight. (Aerodrome)

Situation	Broadcast
Entering the circuit If and when the aircraft joining the circuit, make a broadcast that consists of:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's type and call sign; and •a statement that the aircraft is joining the circuit; and •the leg on which the aircraft is joining the circuit; or if the pilot intends to make a straight in approach to the landing runway, make a prior broadcast of that intention. (Aerodrome)
Vacating the runway The aircraft is clear of the runway(s): *Note: For IFR aircraft, this would normally be accompanied with request to cancel SARWATCH.*	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's call sign; and •broadcast and report to ATS once established outside the runway strip using the radio telephony 'RUNWAY [number] VACATED'. (Aerodrome)
Operations complete For operations that are not associated with or restricted to runways, when operations within the MBA are complete:	(Aerodrome) INFORMATION AND TRAFFIC •the aircraft's call sign; and •OPERATIONS COMPLETE. (Aerodrome)

6.2 **Inoperative Radio and No Radio Procedures**

6.2.1 *CASR Parts 91, 103* and *131* permit some flights to be conducted without a radio.

6.2.2 A non-radio aircraft may fly in Class G airspace in VMC by day at or below 5,000FT AMSL.

6.2.3 If total or partial failure of the required radio communications equipment occurs before flight commences and repair facilities are available, repairs must be made before the flight proceeds.

6.2.3.1 Where repair facilities are not available, flight to the nearest appropriate repair facility may proceed in Class G airspace in VMC only. If flight to the nearest appropriate repair facility entails flight in controlled airspace, the flight may proceed provided that ATS is advised of the radio failure and a clearance for the flight is obtained from ATC.

6.2.3.2 When arriving at a non-controlled aerodrome where the carriage of radio is required, if a radio failure occurs either en route to or in the circuit of the aerodrome, the pilot may continue to land at that aerodrome provided:

a. where equipped - the aircraft displays its external lights, and its transponder is turned on; and

b. if en route - the pilot uses the overfly joining procedure (*Refer to the graphic at para 9.12.6*).

6.2.3.3 A pilot may depart the aerodrome with an unserviceable radio and fly to another aerodrome for repairs, provided that the aircraft - where equipped - displays its external lights and its transponder turned on.

6.2.3.4 To maximise the safety of commercial passenger transport operations, during a flight to or from an aerodrome for radio repairs, it is requested that pilots plan to avoid arriving or departing from an aerodrome during the known hours of scheduled air transport operations. For aerodromes where there is a UNICOM or CA/GRS, pilots should by alternative means where possible make contact and advise their intentions before conducting operations.

6.2.3.5 An aircraft not equipped with an operative radio may operate at, or in the vicinity of a non-controlled certified or military aerodrome provided that:

a. The aircraft is operated in VMC by day; and

b. The aircraft arrives or departs in the company of another radio-equipped aircraft that is flown by a radio-qualified pilot which will allow the latter to make radio calls on behalf of both aircraft.

Note: The radio-equipped aircraft should be manoeuvred to keep the no radio aircraft at a safe distance and in sight at all times in order to accurately report its position.

6.2.4 Procedures to be adopted when total loss of radio occurs whilst in-flight and within Australian Domestic airspace are contained in *ERSA EMERG*. (For radio failure or no radio procedures at all non-controlled aerodromes refer *ERSA INTRO*).

6.2.5 Procedures to be adopted when total loss of radio occurs whilst in-flight and within Australian administered Oceanic airspace are as follows:

6.2.6 In the event of total loss of communication, an aircraft shall:

a. try to re-establish communication by all other means;

b. if all attempts to re-establish communication with ATC are unsuccessful:

(1) Squawk 7600;

(2) If able, broadcast in the blind at suitable intervals: aircraft identification, flight level, aircraft position (including the ATS route designator or the track code), and intentions on the frequency in use, as well as on frequency 121.5MHz (or, as a back-up, the VHF inter-pilot air-to-air frequency 123.45MHz);

(3) Watch for conflicting traffic both visually and by reference to airborne collision avoidance systems or traffic displays (if equipped);

(4) Turn on all aircraft exterior lights (commensurate with appropriate operating limitations);

(5) Maintain the last assigned speed and level for a period of 60 minutes following the aircraft's failure to report its position over a compulsory reporting point (including ADS-C flights), and thereafter adjust speed and altitude in accordance with the filed flight plan;

Note: In OCA, aircraft experiencing communication failure may also initiate Strategic Lateral Offset Procedures (SLOP) in accordance with ENR 2.2 Section 2.2, including an offset of up to 2NM right of track.

(6) Upon exiting OCA, conform to the relevant State procedures and regulations.

6.2.7 In the event of lost communication, ATC shall maintain separation between the aircraft having the communication failure and other aircraft, based on the assumption that the aircraft having the communication failure will operate in accordance with the procedures in *6.2.6.*

6.3 **GNSS Reporting Requirements and Procedures**

6.3.1 GNSS systems used to provide distance information to ATS units by pilot reports must meet one of the GNSS equipment specifications mentioned in *GEN 1.5 section 2.*

6.3.2 ATC may apply some DME-based separation standards to approved aircraft providing GNSS distance information. Pilots must be familiar with and comply with GNSS reporting requirements and procedures.

6.3.3 When a DME distance is not specifically requested, or when the provision of a DME distance is not possible, distance information based on GNSS-derived information may be provided. When responding to ATC requests for distance information, pilots should:

a. provide either a DME distance or a GNSS distance unless RAIM is currently not available, and has been unavailable for the previous 5 minutes; and

b. include the source and point of reference; e.g. 115 GNSS Melbourne, 79 DME Newman, 257 GNSS BEEZA, etc.

6.3.4 Notwithstanding *para 6.3.3*, if an ATC unit has issued a clearance or restriction based upon GNSS distance (e.g. a restriction to reach a certain level by a GNSS distance), pilots must inform ATC if RAIM is not available.

6.3.5 If a GNSS distance is provided to an ATC unit, and RAIM is not currently available, but has been available in the preceding 5 minutes, the distance report should be suffixed “NEGATIVE RAIM” - e.g. 26 GNSS LT VOR, NEGATIVE RAIM.

6.3.6 Databases sometimes contain waypoint information which is not shown on published AIP charts and maps. Distance information must only be provided in relation to published waypoints unless specifically requested by an ATS unit.

6.3.7 Where GNSS distance is requested or provided from an NDB, VOR, DME, or published waypoint, the geographical coordinates of the navigation aid or waypoint must be derived from a validated database which cannot be modified by the operator or crew.

7. LAND AND HOLD SHORT OPERATIONS (LAHSO)

7.1 Introduction

7.1.1 Notwithstanding the provisions of *paras 2.12.6.2 and 2.12.6.3*, operations by an aircraft landing on one runway and another aircraft either taking off or landing simultaneously on a crossing runway may be permitted subject to the provisions of this section.

7.2 Locations Where LAHSO are Used

7.2.1 LAHSO can be implemented at aerodromes controlled by ATC that have suitable runway configurations, together with taxi markings, signs, runway markings, and lights in accordance with the standards in the *Part 139 MOS.*

7.2.2 LAHSO aerodromes are indicated in *ERSA Aerodrome and Facilities* section and/or *Runway Distance Supplement (RDS)* section by the inclusion in the aerodrome information of a table titled “LDA FOR LAHSO”.

7.3 A Dependent Procedures

7.3.1 LAHSO are to be considered dependent procedures, with participating aircraft classified as either:

a. active - when an aircraft is issued a hold short requirement and is alerted about traffic on a crossing runway; or

b. passive - when an aircraft has unrestricted use of the full runway length and is alerted about traffic on a crossing runway.

7.4 **Participation**

7.4.1 Active participation in LAHSO is available to pilots in each of the following categories:

a. pilots of Australian registered aircraft of performance categories A, B or C engaged in operations conducted under a training and checking organisation, subject to the operator providing information in their operational document suite and certifying participating pilots for LAHSO;

b. pilots of Australian registered aircraft of performance category A, B or C where the pilot has been assessed as competent to conduct LAHSO by a person authorised to conduct LAHSO training;

c. ATC will consider all Australian registered aircraft, operating on a flight number call sign, to be approved active participants, unless advised to ATS as not able to be an active and/or passive participant;

d. pilots of Australian military aircraft in performance categories A, B or C; and

e. pilots of foreign military aircraft in performance categories A, B or C subject to a Letter of Agreement between the relevant military authority and the ATS provider.

7.4.2 Passive participation in LAHSO is available to pilots in each of the following categories:

a. pilots of Australian civil and military aircraft categories A, B and C at pilot discretion;

b. pilots of other civil aircraft, including foreign operators, as approved by CASA;

c. pilots of RAAF Hawk, FA18 and other Australian military aircraft as by the relevant Operational Airworthiness Authority;

d. pilots of foreign military aircraft approved by Defence, operating at Defence aerodromes, subject to a Letter of Agreement; and

e. pilots of foreign military aircraft subject to a Letter of Agreement between the relevant military authority and the civil ATS provider.

(The Letter of Agreement will exclude foreign military aircraft of performance category D operating at civil aerodromes).

7.4.3 Notwithstanding the provisions of *sub-paras 7.4.1a and 7.4.2a* above, pilots of foreign registered civil aircraft and of Australian registered aircraft operating under foreign air carriers' flight number call sign are precluded from participation in either active or passive mode regardless of performance category.

7.4.4 A pilot must not accept a requirement to "HOLD SHORT" unless they are qualified, has situation awareness, and has determined that the LDA is adequate for the prevailing conditions and the status of the aircraft.

7.4.5 Operators of aircraft in any category may elect not to allow their pilots to participate in LAHSO. In these cases, ATS should be advised in writing, specifying the company's withdrawal from active, passive, or both modes of participation.

Note: This notification should be emailed to Airservices: satss@airservicesaustralia.com and/or 44WG: hqsrga7.anspstand@defence.gov.au.

7.5 **Conditions for LAHSO**

7.5.1 LAHSO may be conducted subject to the following conditions:

a. The wind for either the active or passive runway, including gusts, does not exceed:

(1) 20KT crosswind;

(2) 5KT tailwind on a dry runway;

(3) no tailwind when the runway is not dry.

b. A simultaneous take-off and landing is permitted by day only.

c. Simultaneous landings are permitted by day and night.

d. The ceiling is not less than the minimum vectoring altitude (MVA) for the location where LAHSO are being conducted and visibility is not less than 8KM.

e. Visibility may be reduced to 5,000M where ATC are assured of sighting the aircraft prior to a loss of the surveillance standard.

f. Advice to the departing aircraft may be given separately from the take-off clearance.

g. Instructions are issued to prevent a landing aircraft from crossing the Hold-Short Line when the intersecting runway is being used by another aircraft.

h. The distance from the landing threshold to the Hold-Short Line of the intersecting runway is adequate for the performance category of the aircraft being held short.

i. ERSA Aerodromes and Facilities (FAC) and/or Runway Distance Supplement (RDS) show “LDA for LAHSO” information. Pilots must ensure that the aircraft can land safely within the LDA for LAHSO.

j. If a runway is reported as WET, the braking characteristics must have been assessed as GOOD by the pilot of an aircraft in the same performance category prior to the landing aircraft being instructed to hold short. ATC will request pilot assessments of the braking characteristics hourly where weather conditions are deteriorating or remain unchanged.

k. The landing aircraft will not be instructed to hold short when low level wind shear is reported.

l. For active participants ground based visual or electronic glide slope guidance must be available and utilised.

m. After landing, the pilot must inform ATC immediately of any difficulty in complying with the ATC requirement to hold short of a crossing runway strip.

Note to l) above: This requirement does not apply to performance category A and B non-jet aircraft of less than 5,700KG MTOW landing Runway 36 at Darwin.

7.6 **Pilot Advise of LAHSO Approval**

7.6.1 ATC will not intentionally issue, and a pilot must not accept, a clearance for a hold-short landing unless the pilot is LAHSO approved. Pilots who elect to participate actively in LAHSO must obtain the ATIS broadcast as early as possible and if within 200NM of a destination where LAHSO is in progress, immediately advise ATC “LAHSO APPROVED”.

e.g. “MELBOURNE CENTRE, (call sign) DESCENDING TO FLIGHT LEVEL TWO FIVE ZERO, LAHSO APPROVED”.

7.6.2 Pilots of civil aircraft operating under a flight number call sign as advised in flight notification, and pilots of Australian military aircraft, may omit the words “LAHSO APPROVED”. Aircraft of operators who have advised in writing an intention not to participate will not be intentionally sequenced for LAHSO.

Where an aircraft or crew that would normally participate actively or passively in LAHSO does not meet the criteria for participation, this must be communicated to ATC at the earliest opportunity.

7.6.3 Pilots of aircraft not operating under a flight number call sign who will be entering controlled airspace within 120NM of destination must advise ATC “LAHSO APPROVED”.

7.6.4 When crews experience wind shear early advice to ATC is essential to ensure timely information is passed to subsequent aircraft.

7.7 **ATIS Broadcast**

7.7.1 Pilots will be alerted that LAHSO are in progress by a statement on the ATIS;

e.g. “DARWIN TERMINAL INFORMATION BRAVO, RUNWAYS 29 AND 36, LAND AND HOLD SHORT OPERATIONS IN PROGRESS, (wind, temperature, etc)”

7.7.2 Both the active and passive runways will be nominated on the ATIS to aid in crew situational awareness.

Note: The acronym LAHSO may be used at ATC discretion.

7.8 **Directed Traffic Information**

7.8.1 ATC is required to issue directed traffic information to both aircraft participating in LAHSO.

7.9 **Readback Requirements**

7.9.1 In all cases, pilots must readback an ATC-issued requirement to hold short.

7.10 **Landing Distance Assessments**

7.10.1 ATC will normally sequence an aircraft for a runway which requires LAHSO only when the landing distance available for the aircraft is likely to be adequate in accordance with aircraft landing category criteria held by ATC.

7.10.2 ATC may sequence non-jet Category B aircraft below 5,700KG MTOW for LAHSO using the landing distance available from *ERSA*. ATC may sequence an aircraft for LAHSO regardless of category of aircraft where the pilot in command has advised "LAHSO APPROVED". The pilot alone is responsible for ensuring that the LDA is equal to, or better than, that required for the prevailing circumstances.

7.10.3 Pilots should check the *ERSA* entry or ask ATC for landing distance available, and assess their landing distance requirements based on the landing weight and ambient weather conditions. The pilot must ensure that the LDA for LAHSO value for the runway meets or exceeds the relevant landing distance required for their particular flight and operation.

7.11 **Go Around During LAHSO**

7.11.1 It is important for pilots to plan for action in the event of a go around. If a go around does occur, pilots must maintain safe separation from other aircraft, as it may be impractical for ATC to provide standard separation. Nevertheless, ATC will issue traffic information and, if appropriate - based on the relative position of aircraft, instructions for avoiding other aircraft.

7.11.2 When issued with avoiding action instructions, pilots should fly the specified heading without delay.

7.11.3 Regardless of any avoiding action instructions, pilots should always defer to any TCAS RA.

8. VERTICAL SEPARATION IN THE AUSTRALIAN FIR

8.1 **Reduced Vertical Separation Minimum (RVSM)**

8.1.1 **Application of RVSM**

8.1.1.1 Australia applies a 1,000FT reduced vertical separation minimum between approved aircraft operating between FL290 and FL410 inclusive.

8.1.1.2 RVSM does not apply to formation flights and civil formation flights will not be issued clearance to operate between FL290 and FL410 inclusive.

8.2 **RVSM Operations**

8.2.1 Aircraft transiting from adjacent FIR into Australian FIR between FL290 and FL410 inclusive must plan from the waypoint on the FIR boundary using the table of cruising levels at *ENR 1.7 Section 5.*

8.2.2 Aircraft transiting from Australian FIR between FL290 and FL410 inclusive to adjacent FIR must plan until the waypoint on the FIR boundary using the table of cruising levels at *ENR 1.7 Section 5.*

8.2.3 Aircraft that will cross latitude 80°S between FL290 and FL410 inclusive must plan using the table of cruising levels at *ENR 1.7 Section 5.* for operations north of 80°S, and the table of cruising levels at *ENR 1.7 Section 6.* for operations south of 80°S.

8.2.4 To have RVSM applied to their aircraft, operators must be approved by the State of Registry or State of the Operator.

8.2.5 Approved operators must ensure that height-keeping monitoring is under taken at least every two years or within intervals of 1,000 flight hours per aircraft, whichever period is longer, in accordance with the aircraft categories as presented in the current version of the ICAO RVSM Minimum Monitoring Requirements table. The table and further information on monitoring can be obtained from the Australian Airspace Monitoring Agency (AAMA) at www.airservicesaustralia.com/organisations/aama/.

8.2.6 Pilots of aircraft that are not RVSM-approved may plan within the RVSM flight level band (FL290 to FL410 inclusive). However, clearance at RVSM levels is subject to disposition of traffic and RVSM aircraft priority. The conventional vertical separation minimum will be applied between aircraft that are not RVSM-approved and all other aircraft.

8.2.7 Pilots of non RVSM-approved State aircraft will be afforded equal priority with RVSM-approved aircraft. Pilots planning to operate non-RVSM within the RVSM level band must flight plan in accordance with *ENR 1.10.*

8.2.8 Pilots of aircraft that are not RVSM-approved must report “NEGATIVE RVSM” in accordance with the requirements of *GEN 3.4 Sub-section 6.6 Item 2.q*.

8.3 **Operational Procedures Before Entering the RVSM Flight Level Band - RVSM Approved Aircraft**

8.3.1 On each flight before entering the RVSM flight level band, pilots of RVSM-approved aircraft must check to ensure that all of the following minimum mandatory equipment is operating normally:

a. two independent primary altimetry systems,

b. a Mode C-capable SSR transponder,

c. an altitude alert system, and

d. an autopilot with height lock.

8.3.2 If any item of the minimum mandatory equipment listed in *para 8.3.1* is not operating normally, a pilot must notify ATC before entering the RVSM flight level band using the phraseology "NEGATIVE RVSM" (see *GEN 3.4 Sub-section 6.6 Item 2.q.*).

8.4 **Operational Procedures After Entering the RVSM Flight Level Band – RVSM Approved Aircraft**

8.4.1 During changes of level, an aircraft must not overshoot or undershoot its Cleared Flight Level (CFL) by more than 150FT (45M).

8.4.2 **Failure of One Primary Altimetry System.** If one of the primary altimetry systems fails, but the remaining altimetry system is functioning normally, the pilot must:

a. couple that system to the autopilot with height lock;

b. maintain increased vigilance of altitude-keeping; and

c. notify ATC of the failure using the phraseology, "FOR INFORMATION, OPERATING ON ONE PRIMARY ALTIMETER ONLY".

8.4.3 **Failure of All Primary Altimetry Systems.** If all primary altimetry systems fail, or are considered unreliable, the pilot must:

a. maintain the flight level indicated on the standby altimeter (if the aircraft is so equipped) at the time of failure or when considered unreliable;

b. alert nearby aircraft by turning on all exterior lights and, if not in VHF contact with ATC, by broadcasting advice of the failure, position, flight level, and intentions on 121.5MHz;

c. notify ATC of the failure using the phraseology “NEGATIVE RVSM” (see *GEN 3.4 Sub-section 6.6 Item 2.q.*) and the intended course of action.

8.4.4 **In Oceanic Class A Airspace.** If unable to obtain ATC clearance in a timely manner following a failure of all primary altimetry systems in oceanic Class A airspace, the pilot must proceed as follows:

a. If operationally feasible to do so, leave the assigned route or track by turning at least 45° right or left, whenever this is possible, taking account of adjacent routes and descend below FL290.

b. If not operationally feasible to execute this contingency procedure, continue to alert nearby aircraft and coordinate with ATC.

8.4.5 **Divergence in Primary Altimetry Systems’ Indication**. If the primary altimeters diverge by more than 200FT, the pilot must proceed as follows:

a. Attempt to determine the defective system through established “trouble shooting” procedures and/or comparing the primary altimeter displays to the standby altimeter (as corrected by correction card, if required).

b. If the defective system can be determined, couple the functioning altimetry system to the autopilot with height lock and proceed as in *para 8.4.2.*

c. If the defective system cannot be determined, proceed as in *para 8.4.3*.

8.4.6 **Failure of the Mode C-capable SSR Transponder.** If the mode C-capable transponder fails, the pilot must notify ATC of the failure using the phraseology “NEGATIVE RVSM” (see *GEN 3.4 Sub-section 6.6 Item 2.q.)*.

8.4.7 **Failure of the Altitude Alert System.** If the altitude alert system fails, the pilot must notify ATC of the failure using the phraseology “NEGATIVE RVSM” (see *GEN 3.4 Sub-section 6.6 Item 2.q*).

8.4.8 **Failure of the Autopilot with Height Lock.** If the autopilot with height lock fails, the pilot must initiate the following actions sequentially:

a. Maintain CFL.

b. Evaluate the aircraft's capability to maintain altitude through manual control.

c. Assess the situation regarding possible conflicting traffic.

d. Alert nearby aircraft by turning on all exterior lights and, if not in VHF contact with ATC, broadcast advice of failure, position, flight level, and intentions on 121.5MHz.

e. Notify ATC of the failure using the phraseology "NEGATIVE RVSM" (see *GEN 3.4 Sub-section 6.6* Item 2.q.) and the intended course of action.

8.4.9 **In Oceanic Class A Airspace.** Possible courses of action for the pilot following a failure of the autopilot with height lock in oceanic Class A airspace include the following:

a. Provided that the aircraft can maintain CFL, continue at that level.

b. If the aircraft cannot maintain CFL and is unable to obtain a revised ATC clearance, leave the assigned route or track by turning 90° right or left, whenever this is possible, taking account of adjacent routes.

8.5 **ATC Responsibilities**

8.5.1 ATC will apply alternative separation to any aircraft that has reported "NEGATIVE RVSM" in accordance with requirements specified in this Section.

8.6 **Weather and Wake Turbulence and System Alerts**

8.6.1 The pilot of an aircraft operating within the RVSM flight level band that encounters weather turbulence that affects aircraft capability to maintain CFL, or wake turbulence, or experiences distracting aircraft system alerts, must notify ATC and request a revised clearance before deviating from track or CFL.

8.6.2 **Oceanic Control Area (OCA) Only.** If a revised clearance is not possible or practicable, the pilot of an aircraft operating in OCA may initiate the following temporary lateral offset procedure with the intention of returning to the cleared route as soon as possible:

a. If possible, establish contact with other aircraft on the VHF inter-pilot air-to-air frequency 123.45MHz.

b. Initiate a lateral offset (one or both aircraft may initiate) not to exceed 2NM from the cleared route or track, provided that:

(1) as soon as practicable, the pilot(s) of the offsetting aircraft notify ATC that temporary lateral offset action has been taken and the reason for doing so; and

(2) the pilot(s) of the offsetting aircraft notify ATC when the aircraft is re-established on the assigned route(s) or tracks(s).

8.7 **Flight Level Deviation Reporting**

8.7.1 For operations in the Australian FIR, flight crews must report all flight level deviations of 300FT or more from the aircraft's assigned level, irrespective of the cause of the deviation.

8.7.2 In reporting, crews must provide the information in the format detailed in *para 8.7.4*. Reports must be submitted as soon as possible after the occurrence and in writing to:

Australian Airspace Monitoring Agency (AAMA)
Airservices Australia
GPO Box 367
CANBERRA ACT 2601
AUSTRALIA
Fax: +61 2 6268 5695
Email: aama@airservicesaustralia.com

8.7.3 Flight crew may send reports through the airline/operator using its normal reporting procedures.

8.7.4 A report of altitude deviations of 300FT or more, including those due to Traffic Alert and Collision Avoidance System (TCAS), turbulence, and contingency events must use the following format:

1. Reporting Agency:
2. Date and Time:
3. Location of Deviation: (Lat/Long) and indication of the area (e.g. Australian South Pacific airspace/Australian Continental airspace/Australian Indian Ocean airspace)
4. Aircraft Identification and Type:
5. Flight Level Assigned:
6. Observed/Reported (indicate one) Final Flight Level: and indicate whether controller or pilot report
7. Duration at Flight Level:
8. Cause of Deviation:
9. Other Traffic:
10. Crew Comments: (if provided)
11. Remarks: (If the event necessitated contingency action, indicate whether AIP contingency procedures were followed)

9. OPERATIONS IN CLASS G AIRSPACE

9.1 Communications

9.1.1 The pilot in command of an IFR aircraft in Class G airspace must attempt to contact ATS on VHF or HF when taxiing. If the pilot is unable to make contact with the air traffic service in relation to the report required by item 1 of Table 21.06 (1) of the *Part 91 MOS*, the flight may taxi and take-off provided contact is established as soon as possible after take-off and the following conditions are complied with:

a. where the operator of the flight is an AOC holder, aerial work certificate holder or *Part 141* certificate holder — the pilot is assured of radio contact with their operator, or a representative of their operator who has immediate access to a serviceable telephone, until contact is made with the air traffic service; or

b. except for *Part 121* operations conducted using aircraft with a MOPSC greater than 19 seats - a SARTIME for departure, that is a maximum of 30 minutes after commencing to taxi has been established with air traffic services.

Note: Pilots are reminded of their obligations to see and avoid other aircraft and their responsibility for collision avoidance in the vicinity of non-controlled aerodromes using 'see-and-avoid' (CASR 91.325).

9.1.2 The pilot of an IFR flight departing from a non-controlled aerodrome must report "IFR" when making first contact with ATS.

9.1.3 A pilot operating in accordance with *para 9.1.1(b)* may nominate a SARTIME for departure either as part of the arrival report or when submitting flight notification by the phrase "SARTIME FOR DEPARTURE". SAR alerting action will be initiated if a report is not received by the nominated SARTIME for departure.

9.1.4 To achieve the greatest degree of safety *section 21.04* of the *Part 91 MOS* requires pilots of aircraft carrying a serviceable radio which they are qualified to use, to make a broadcast whenever it is reasonably necessary to do so to avoid a collision, or the risk of a collision with another aircraft at a non-controlled aerodrome.

Note: Pilots are reminded of their obligations to see and avoid other aircraft and their responsibility for collision avoidance in the vicinity of non-controlled aerodromes using 'see-and-avoid'.

9.1.5 In Class G airspace, pilots of radio-equipped VFR aircraft should monitor the appropriate VHF frequency and announce if in potential conflict. Pilots intercepting broadcasts from aircraft which are considered to be in potential conflict must acknowledge by transmitting own call sign and, as appropriate, aircraft type, position, actual level and intentions.

9.1.6 For *para 9.1.5*, the appropriate VHF frequency is:

a. when operating in the vicinity of an aerodrome published on aeronautical charts - the CTAF (126.7MHz or the discrete frequency as published); or

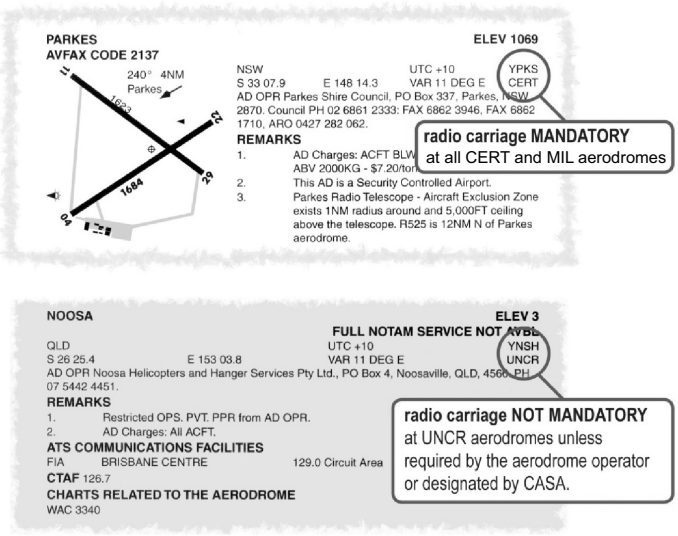
b. when operating within a Broadcast Area - the Broadcast Area CTAF.

Otherwise, it is recommended pilots use the Area VHF. This frequency may provide the best means of gaining assistance from ATC or other pilots in the event of an emergency.

9.1.7 In the vicinity of uncharted aerodromes, pilots have discretion to use the most appropriate frequency that ensures safe operation. This may be 126.7MHz. However, pilots should be aware that transiting aircraft will be monitoring Area VHF. To ensure mutual traffic awareness, it is recommended that pilots using an alternative frequency also monitor Area VHF.

9.1.8 It is required (unless operating in accordance with procedures in the *Part 103 MOS)* that gliders operating above 5,000FT in Class G airspace monitor the Area VHF.

9.1.9 An aircraft is considered in the vicinity of a non-controlled aerodrome if it is within 10NM of the aerodrome and at a height above the aerodrome that could result in conflict with operations at the aerodrome.



9.1.10 Pilots of aircraft transiting in the vicinity of a non-controlled aerodrome should avoid flying over the aerodrome at an altitude that could conflict with operations in the vicinity of the aerodrome.

9.1.11 When a report from an IFR aircraft is made to FLIGHTWATCH on HF, a broadcast on the appropriate CTAF or Area VHF is also required.

9.1.12 If calls are not made clearly and concisely using the standard phraseology, confusion can arise at aerodromes in close proximity that share the same CTAF.

9.1.13 The standard broadcast format is:

a. {Location} Traffic

b. {Aircraft type}

c. {call sign}

d. {Position/level/intentions}

e. {Location}

9.1.14 The following tables provide a summary of recommended broadcasts to be made when operating or arriving at, or departing from, a non-controlled aerodrome. Pilots should also observe local and published noise abatement procedures and curfews.

RECOMMENDED CALLS IN ALL CIRCUMSTANCES	
SITUATION	**BROADCAST**
The pilot intends to take-off	Immediately before, or during taxiing
The pilot is inbound to an aerodrome	10NM from the aerodrome, or earlier, commensurate with aeroplane performance and pilot workload, with an estimated time of arrival (ETA) for the aerodrome
The pilot intends to fly through the vicinity of, but not land at, a non-controlled aerodrome	10NM from the aerodrome, or earlier, commensurate with aeroplane performance and pilot workload, with an estimated time of arrival

RECOMMENDED CALLS DEPENDENT ON TRAFFIC	
SITUATION	**BROADCAST**
The pilot intends to enter a runway	Immediately before entering a runway
The pilot is ready to join the circuit	Immediately before joining the circuit
The pilot intends to make a straight-in approach	On final approach at not less than 3NM from the threshold (see note)
The pilot intends to join on base leg	Prior to joining on base
RECOMMENDED CALLS DEPENDENT ON TRAFFIC	
SITUATION	**BROADCAST**
During an Instrument Approach when: a. departing FAF or established on final approach segment inbound b. terminating the approach, commencing the missed approach	Including details of position and intentions that are clear to all pilots (both IFR and VFR)
The aircraft is clear of the runway(s)	Once established outside the runway strip

Note: Some distances above refer to the runway threshold and others to the ARP. Pilots should be aware that a GNSS indication of 3NM from an aerodrome may not be 3NM to the runway threshold.

9.2 **Circuit Information**

9.2.1 Pilots should fly a circuit commensurate with the aircraft type they are operating. However the use of any circuit procedure does not alter the responsibility of each pilot to see and avoid other aircraft. Pilots operating in the circuit should manoeuvre to follow traffic ahead of them in the circuit.

9.2.2 Left-hand circuits is the standard traffic circuit that must normally be made. Right-hand circuit requirements are listed in *ERSA*.

9.2.3 Aircraft should not be operated in the circuit at an indicated airspeed of more than 200KT.

9.2.4 During the initial climb-out the turn onto crosswind should be made appropriate to the performance of the aircraft, but in any case not less than 500FT so as to be at circuit height when turning downwind.

Note: CASR 91.390(1)(b) - outlines exceptions in this requirement.

9.2.5 Pilots may vary the size of the circuit depending on:

a. the performance of the aircraft;

b. safety reasons; or

c. in accordance with the Aircraft Flight Manual, Pilot's Operating Handbook, or company Standard Operating Procedures.

9.2.6 Pilots are encouraged to turn on aircraft landing lights, anti-collision lights and strobe lights, where fitted, when in the vicinity of a non-controlled aerodrome, until the aircraft has landed.

9.2.7 Transponders can be detected by aircraft equipped with ACAS (TCAS), allowing them to 'see' other aircraft and take evasive action. Pilots of transponder-equipped aircraft should at all times ensure their transponder is switched to ON/ALT (Mode C), including when operating in the vicinity of a non-controlled aerodrome. In the event of a radio failure it is important that pilots select 7600 on their transponder and continue squawking.

9.2.8 So as not to impede commercial aviation, pilots flying recreational or sport aircraft for their own enjoyment, or pilots flying GA aircraft for their own leisure, should consider giving way to aircraft being used for "commerce" provided that the inconvenience to their own operation is not great and it can be done safely. Operators of commercial aircraft should never expect a give way offer to be assumed or automatic. Any offer to give way must be explicit and its acceptance acknowledged.

9.3 **Separation Minima for Take-off**

9.3.1 An aircraft must not commence take-off until:

a. a preceding departing aircraft using the same runway has:

(1) crossed the upwind end of the runway; or

(2) commenced a turn; or

(3) if the runway is longer than 1,800M, become airborne and is at least 1,800M ahead; or

(4) if both aircraft have a MTOW below 2,000KG, the preceding aircraft is airborne and is at least 600M ahead;

b. a preceding landing aircraft using the same runway, has vacated it and is taxiing away from the runway; or

c. a preceding aircraft, using another runway, has crossed or stopped short of the take-off aircraft's runway.

9.3.2 At aerodromes where gliders operate to a common circuit pattern from a parallel strip outside the runway strip, the above separation minima shall apply to aircraft landing or taking off on either runway as if they were a single runway, but aircraft taxiing or stationary on the runway must not affect operations on the other.

9.3.3 Where gliders and glider tugs operate to a contra-circuit, simultaneous operations are permitted.

9.4 **Circuit Height**

9.4.1 When operating at non controlled aerodromes, the following circuit heights are recommended:

a. High performance - includes jets and many turbo-prop aircraft, above approximately 150KT, 1,500FT AGL;

b. Medium performance - includes most piston engine aircraft, between approximately 55KT and 150KT, 1,000FT AGL;

c. Low performance - ultralights and rotary wing with a maximum speed of approximately 55KT, 500FT AGL (*Refer to diagram at 9.12.6*).

9.4.2 Circuit heights for aerodromes which have specific requirements are published in *ERSA*.

9.5 **Taxiing**

9.5.1 Pilots of IFR flights operating from non-controlled aerodromes must report to ATC on taxiing. If unable to establish contact, proceed in accordance with *para 9.1.1*.

9.5.2 Taxiing reports for IFR flights must include the following information:

a. aircraft type;

b. POB (for flights other than air transport operations);

c. IFR;

d. location;

e. destination or departure quadrant or intentions; and

f. runway to be used.

9.5.3 Following the taxi report, a pilot of an IFR flight must report to ATC if changing to a CTAF or Broadcast Area CTAF when the ATS frequency will not, or cannot, be monitored. This report must include the aerodrome location and frequency.

9.6 **Departure Information**

9.6.1 Pilots of departing aircraft should depart by extending one of the standard circuit legs. An aircraft should not execute a turn opposite to the circuit direction unless the aircraft is well outside the circuit area and no traffic conflict exists. This will normally be at least 3NM from the departure end of the runway. The distance may be less for aircraft with high climb performance. The distance should be based on pilots being aware of traffic and the ability of the aircraft to climb above and clear of the circuit area.

Note: Pilots of departing aircraft should be aware of traffic intending to join the circuit by the recommended overfly procedure, as they can be 2,000FT or higher above aerodrome elevation.

9.6.2 A pilot in command of an IFR flight must establish flight on the departure track as soon as practicable after take-off and within 5NM of the aerodrome, except that, at aerodromes which have published standard instrument departure procedures, an IFR aircraft may depart in accordance with those procedures. When established on the departure track, and clear of the circuit traffic, the pilot in command must report departure to ATC unless instructed otherwise.

9.6.3 This report must include the following information:

Non-surveillance	Surveillance
Departure time	Current position
Outbound track in degrees magnetic	Present level
Intended cruising level	Intended cruising level
Estimate for the first en route reporting point	Estimate for the first en route reporting point

9.6.3.1 The departure time must be reported as follows:

a. current time minus an adjustment for the distance from the aerodrome; or

b. when over or abeam the aerodrome.

9.6.4 If the pilot transmits the departure report before intercepting the departure track the report must include advice that the aircraft is manoeuvring to intercept departure track.

9.6.5 When circumstances dictate that flight notification details be submitted in-flight, the above items must be combined with the appropriate items of the flight notification, and prefixed with the words “FLIGHT PLAN”.

9.7 **Climb and Cruise Procedures**

9.7.1 The pilot in command of an IFR flight must notify the intention to amend route, deviate from track or change level in sufficient time for ATS to advise traffic. When a position estimate changes by more than two (2) minutes, the pilot must advise ATS.

9.7.2 Pilots must give ATS notice of an impending position report by use of the word “POSITION”;
e.g. “MELBOURNE CENTRE (call sign) POSITION”.
Pilots must wait for the ATS instruction before reporting position.

9.7.3 Pilots must report maintaining an assigned level.

9.7.4 After any en route frequency change, a pilot in command of an IFR flight must advise present level. If the aircraft is not at its planned cruising level, the pilot must also provide advice of the level to which the aircraft is being climbed.

9.7.5 When operating in Class G airspace in proximity to a controlled aerodrome, pilots should consider monitoring the TWR frequency to enhance situational awareness of traffic entering and leaving the CTR.

9.7.6 A pilot of a flight intending to operate in the vicinity of a non-controlled aerodrome at altitudes used by arriving and departing aircraft should:

a. monitor the appropriate VHF, and broadcast by 10NM or earlier from the aerodrome commensurate with aircraft performance and pilot workload with an ETA (unless otherwise specified in ERSA); and

b. where possible, avoid the circuit area and arrival and departure tracks.

9.8 **Position Reports**

9.8.1 Except when identified, position reporting is mandatory when operating under the IFR and must normally be made at the positions or times notified on the flight notification
(See *ENR 1.10 para 3.5.2* for flight notification requirements).

9.8.2 The position report format is identified at *APPENDIX 1*.

9.8.3 When an aircraft is holding due weather, ATS will nominate scheduled reporting times which will normally be at 15 minute intervals.

9.8.4 IFR aircraft operating area-type flights and nominating schedules reporting times may limit the report to level and the present position or the sector of the survey area in which the aircraft is currently operating.

9.9 **Descent from Controlled Airspace**

9.9.1 Before descending from controlled into Class G airspace and before separation with any aircraft operating near the base of controlled airspace can be compromised, the pilot in command of an IFR flight must report position, level, intentions and estimate for next position/destination to the ATS unit providing services in Class G airspace. If the report is made using HF radio, a broadcast must be made on the appropriate area VHF frequency.

9.10 **Arrival Information**

9.10.1 A pilot of an IFR flight must report when changing to the CTAF when the ATS frequency will not, or cannot be monitored. This report must include the aerodrome location and frequency.

9.10.2 Pilots of IFR flights conducting local training, an instrument approach, or a holding pattern, may extend their SARWATCH by an “OPERATIONS NORMAL” call at scheduled times.

9.11 Landing Manoeuvres

9.11.1 Prior to entering the circuit at non-controlled aerodrome, aircraft should avoid the flow of traffic until established in the circuit. The 'standard aerodrome traffic circuit' facilitates the orderly flow of traffic and is normally a left circuit pattern with all turns to the left, unless published in the AIP (*CASR 91.385 (1)(b)*). Landings should be accomplished on the operational runway most closely aligned to the wind.
Wind and landing direction indicators can be checked while at an altitude of +500FT above the circuit height.

Note: If jet, turbo prop or high performance piston aircraft operate at the aerodrome, 2,000FT AGL is a safer height to remain clear of all circuit traffic.

9.11.2 An aircraft approaching a non-controlled aerodrome for a landing should join the circuit in accordance with *para 9.12* unless it is:

a. following an instrument approach procedure in IMC; or

b. conducting a visual circling procedure after completion of an instrument approach procedure in IMC; or

c. conducting a straight-in approach in accordance with *para 9.13*.

9.11.3 The runway to be used for landing should be:

a. the most closely aligned into-wind runway; or

b. when operational reasons justify, any other available landing direction provided the nominated circuit is executed without conflict to landing or take-off traffic using the most into wind runway; and

c. serviceable, and cleared of ground maintenance equipment and personnel.

9.11.4 Aircraft approaching a non-controlled aerodrome for landing must make all turns to the left except:

a. where right hand circuits are specified for the aerodrome; or

b. when entering the upwind, crosswind or downwind leg; or

c. when following an instrument approach procedure in IMC; or

d. when conducting a visual circling procedure after completion of an instrument approach procedure in IMC.

9.12 **Circuit Entry**

9.12.1 Where a pilot is unfamiliar with the aerodrome layout, or when its serviceability, wind direction, wind speed, or circuit direction cannot be ascertained without prior arrival, the overfly procedure should be used. The pilot should overfly or circle the aircraft at least 500FT above the circuit altitude, usually 2,000FT or more above the aerodrome elevation. When the circuit direction has been determined, the pilot should position the aircraft to a point well clear (normally the non-active side of the circuit) before descending to the circuit altitude that equates to the aircraft's performance. Pilots should not descend into the traffic circuit from directly above the aerodrome. Refer *AC 91.10*.

9.12.2 For low performance ultralight aircraft and rotorcraft with a maximum speed of approximately 55KT, it is recommended that the aircraft overfly midfield at 500FT above aerodrome elevation. This will minimise the risk of conflict with higher or faster traffic.

Note: Ultralight aircraft pilots who choose to use the overfly join procedure above the circuit altitude should be aware:

a. Faster larger aircraft may not be able to see you easily.

b. Faster larger aircraft can create significant wake turbulence.

c. Faster larger aircraft will not be able to slow to the speeds of an ultralight aircraft and follow.

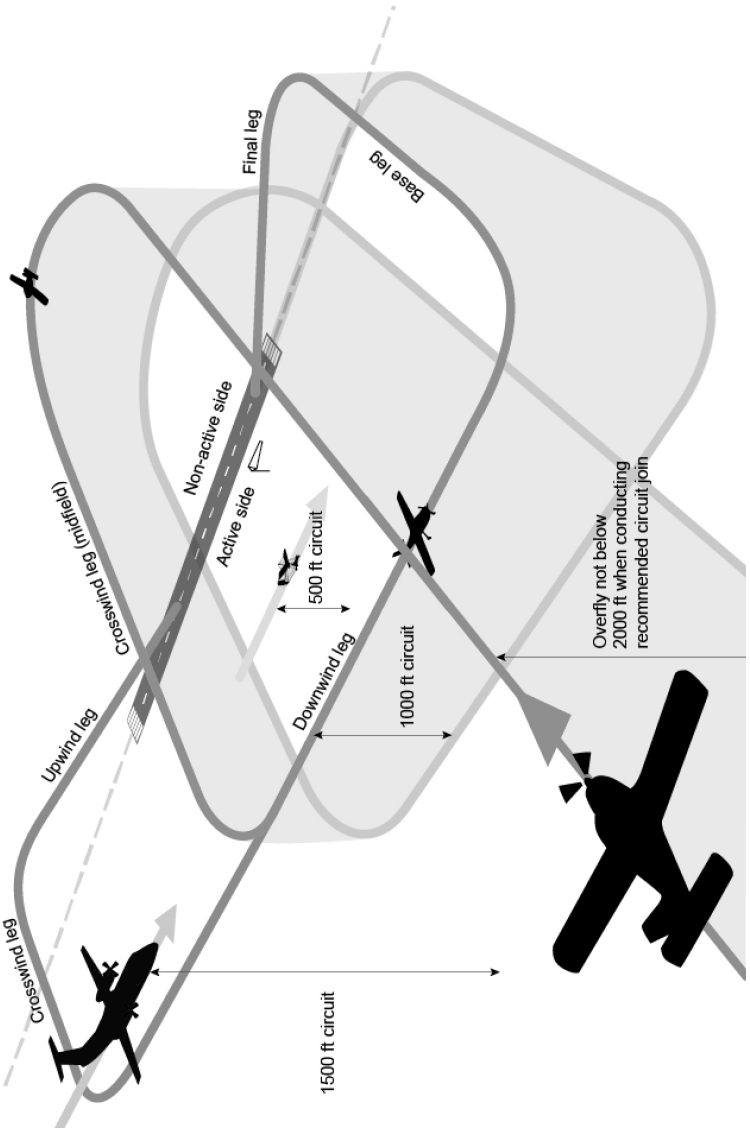
d. Faster larger aircraft - prior to arriving in the circuit and when below 10,000FT - can be at speeds up to 250KT. Therefore although aircraft should be at 200KT maximum in the circuit, an aircraft reporting at 20NM from the aerodrome could be in the vicinity of the circuit within 5 minutes.

9.12.3 For aircraft arriving and intending to join the circuit from overhead, the aircraft should descend on the non-active side of the circuit and be established at its circuit altitude as it crosses the runway centreline on crosswind, at between midfield and the departure end of the runway.

9.12.4 When arriving on the live side, the recommended method is to arrive at the circuit altitude entering midfield at approximately 45 degrees to the downwind leg while giving way to the aircraft already established in the circuit.

9.12.5 On downwind the applicable circuit altitude should be maintained until commencement of the base leg turn. The base leg position is normally when the aircraft is approximately 45 degrees from the reciprocal of the final approach path, measured from the runway threshold. Along the base leg continue to lookout and maintain traffic separation.

9.12.6 When on the final leg, confirm the runway is clear for landing. The turn onto final approach should be completed by a distance and height that is common to the operations at the particular aerodrome and commensurate with the speed flown in the circuit for the aircraft type. In any case, the turn onto final should be completed by not less than 500FT above aerodrome elevation. This should allow sufficient time for pilots to ensure the runway is clear for landing. It will also allow for the majority of aircraft to be stabilised for the approach and landing.


Final leg
Base leg
Non-active side
Active side
Crosswind leg (midfield)
500 ft circuit
Upwind leg
Downwind leg
1000 ft circuit
Overfly not below
2000 ft when conducting
recommended circuit join
Crosswind leg
1500 ft circuit

9.13 **Straight-in Approach**

9.13.1 Straight-in approaches are not a recommended standard procedure. *CASR 91.395* allows pilots to conduct straight-in approaches provided certain conditions are met. Pilots who choose to adopt a straight-in approach should only do so when it does not disrupt or conflict with the flow of circuit traffic. *CASR 91.395 (2)(b)* requires that the pilot conducting a straight-in approach give way to any other aircraft established and flying in the circuit pattern at the aerodrome.

9.13.2 *CASR 91.395 (2)(a)* requires pilots, before commencing a straight-in approach, to determine the wind direction and speed and the runway in use at the aerodrome.

9.13.3 There are several ways to determine the wind direction, speed and runway in use:

a. AWS, AWIS, AAIS, CA/GRS or UNICOM,

b. Radio contact with a ground-based radio communication service, company agent, or aircraft operating at the aerodrome; or

c. Visual indications - if the information cannot be determined by the above means.

9.13.4 When conducting a straight-in approach, the aircraft must be established on final at not less than 3NM from the landing runway's threshold *CASR 91.395 (2)(c)*.

9.13.5 Pilots of aircraft conducting a straight-in approach at a non-controlled aerodrome should observe the following procedures:

a. The pilot in command should not commence a straight-in approach to a runway when the reciprocal runway direction is being used by aircraft already established in the circuit.

b. All manoeuvring to establish the aircraft on final approach must be conducted outside a 3NM radius from the intended landing runway threshold.

Note: Within 3NM, pilots are expected to make only minor corrections to line up accurately on final approach. This will enable pilots conforming to the aerodrome traffic pattern to optimise their visual scan for traffic along the final approach path.

c. The aircraft's transponder should be squawking and its external lights, where fitted, should be illuminated when on final approach. They should remain on until the aircraft has landed and is clear of all runways.

d. An aircraft established on base leg or final approach for any runway has priority over an aircraft carrying out a straight-in approach.

9.14 **Joining on Base**

9.14.1 Joining on base leg, whilst not prohibited, is not a recommended standard procedure. CASA recommends pilots join the circuit on either the crosswind or downwind leg. However, pilots who choose to join on base leg should only do so if they:

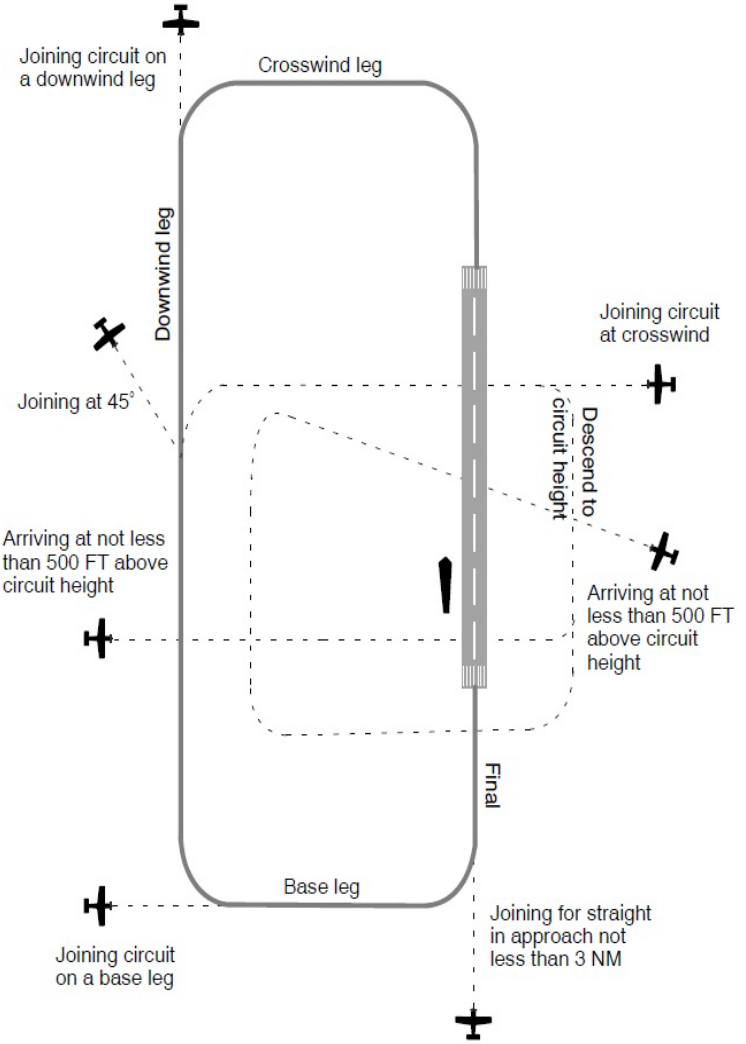
a. have determined the wind direction and speed;

b. have determined the runway in use;

c. give way to other circuit traffic and ensure the aircraft can safely (no traffic conflict likely) join the base leg applicable to the circuit direction in use at the standard height; and

d. broadcast their intentions.

Aerodrome traffic circuit



9.15 **Military Initial and Pitch Circuit Procedures**

9.15.1 For Military fast jet and training aircraft the preferred method of joining the circuit is via a procedure known as Initial and Pitch. The aircraft (or formation) will track to the Initial Point, a point at 5NM downwind of the runway in use displaced to the dead side, and track inbound at high speeds (see diagram below – not to scale).

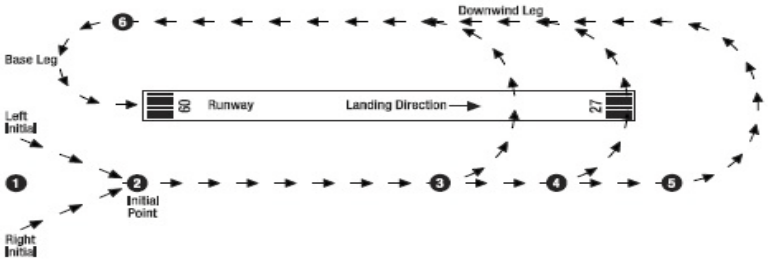
9.15.2 Traffic permitting, Initial and Pitch procedures may be conducted at military, joint user, controlled and non controlled aerodromes. At controlled aerodromes military pilots must comply with ATC circuit entry instructions unless approved for an Initial and Pitch entry.

9.15.3 When conducting this procedure, the height for fast jets is normally 1,500FT AGL and 1,000FT AGL for other aircraft. Aircraft on tactical missions can conduct the initial and pitch at below normal altitudes; this is referred to as a low Initial and Pitch.

9.15.4 At any stage once abeam the threshold of the runway in use, and safe to do so, the aircraft turns (“Pitches”) to join downwind and configures for landing.

9.15.5 Generally pilots conducting this manoeuvre will broadcast their position at the Initial Point and on the base turn.

Military Initial and Pitch



1. First call on run-in to Initial Point	4. Second aircraft in formation pitches out and positions Downwind behind first aircraft.
2. Commence Initial	5. Subsequent aircraft in formation pitch out and positions Downwind behind second aircraft.
3. First aircraft in formation pitches out and positions Downwind	6. First aircraft calls turning on Base leg for the entire formation. Clearances or instructions apply to entire formation. Subsequent aircraft in formation will make independent base calls.
Note: Aircraft conducting a low pitch will be at the low pitch altitude by position 3.	

9.16 **Separation Minima for Landing**

9.16.1 An aircraft must not continue its approach to land beyond the threshold of the runway until:

a. a preceding departing aircraft using the same runway is airborne and:

(1) has commenced a turn; or

(2) is beyond the point on the runway at which the landing aircraft could be expected to complete its landing roll and there is sufficient distance to manoeuvre safely in the event of a missed approach;

b. a preceding landing aircraft using the same runway has vacated it and is taxiing away from the runway;

c. a preceding aircraft using another runway, has crossed or stopped short of the landing aircraft's runway.

9.16.2 At aerodromes where gliders operate to a common circuit pattern from a parallel strip outside the runway strip, the above separation minima shall apply to aircraft landing or taking off on both runways as if they were a single runway, but aircraft taxiing or stationary on one runway must not affect operations on the other. Where gliders and glider tugs operate to a contra-circuit, simultaneous operations are permitted.

Note: Pilots are reminded of their obligations to see and avoid other aircraft (CASR 91.325).

9.17 **The Traffic Mix and Other Hazards at Non-Controlled Aerodromes**

9.17.1 At non-controlled aerodromes, there may be scheduled or non-scheduled air transport flight, gliders, parachutists, helicopters, gyroplanes, ultralights, balloons, general aviation aircraft, and agricultural aircraft operations.

9.17.2 Pilots should consult *AC 91-10*: 'Operations in the vicinity of non-controlled aerodromes', (in conjunction with the *AIP*) for detailed operating procedures when operating in the vicinity of non-controlled aerodromes.

9.17.3 In addition pilots should consult *AC 91-14*: 'Pilots' responsibility for collision avoidance in the vicinity of non-controlled aerodromes using see and avoid'.

9.17.4 The AC mentioned above provide guidance on a code of conduct (good airmanship) which, when followed will provide improved situational awareness and safety for all pilots when flying at, or in the vicinity of, non-controlled aerodromes.

10. OPERATIONAL REQUIREMENTS - GENERAL

10.1 **Taxiing After Landing**

10.1.1 After landing, the runway strip should be vacated as soon as practicable. Aircraft should not stop until clear of the runway strip.

10.2 **SARWATCH**

10.2.1 **Cancellation of SARWATCH other than SARTIME**

10.2.1.1 Pilots wishing to cancel SARWATCH may do so by reporting to ATS.

10.2.1.2 SARWATCH is automatically cancelled without the need to report to ATS when landing:

a. at a controlled aerodrome during tower hours; or

b. at an uncontrolled aerodrome during AFIS hours.

10.2.1.3 When cancelling SARWATCH, pilots must include:

a. the aircraft radio call sign;

b. place of arrival or point from which SARWATCH services are no longer required;

c. the words "CANCEL SARWATCH"; and

d. when communicating with a unit other than that nominated, the name of the ATS unit to which the report shall be relayed.

10.2.1.4 SARWATCH may be cancelled in combination with a pilot report of changing to the CTAF, or in the circuit area, or after landing.

10.2.1.5 When the pilot of an IFR flight elects not to report in the circuit area to cancel SARWATCH and has not reported within 10 minutes of ETA, ATS will commence communications checks to obtain a landing report or an extension of SARWATCH.

10.2.1.6 ATS will acknowledge “CANCEL SARWATCH” reports with a readback of the place of arrival, if appropriate, and the words “SARWATCH TERMINATED”.

10.2.2 **Cancellation of SARTIME**

10.2.2.1 When operating on a SARTIME, the pilot must cancel SARTIME by the time nominated and, during the contact with ATS, include the words “CANCEL SARTIME”.

10.2.2.2 ATS will acknowledge “CANCEL SARTIME” reports with a readback of the place of arrival, if appropriate, and the words “SARTIME CANCELLED”.

10.2.2.3 The preferred method to cancel a SARTIME is via telephone to CENSAR on 1800 814 931. When telephone facilities are not available you may use ATS frequencies.

10.2.2.4 For SARTIME flights, pilots of single VHF radio-equipped aircraft must cancel SARTIME before changing to the CTAF, or after landing.

10.2.3 **SARTIME for Departure**

10.2.3.1 When submitting flight notification, a pilot may nominate a SARTIME for departure for the initial departure aerodrome only. For intermediate departure, it may be nominated by telephone after landing, or as part of the arrival report associated with that aerodrome. Only one SARTIME may be current at any time.

10.2.3.2 The nomination of a SARTIME for departure does not absolve the pilot from complying with the requirements for the carriage of serviceable radio equipment, nor from making the prescribed reports.

10.2.3.3 An IFR departure report is not sufficient to cancel a SARTIME for Departure. Pilots who have nominated a SARTIME for Departure must use the phrase ‘CANCEL SARTIME’ with the departure report.

10.3 **Radio Telephony Requirements Outside Controlled Airspace**

10.3.1 The call sign of the station or service being called must be included at the beginning of each exchange on VHF and HF.

Note: The requirement of para 9.1.11 applies when reporting on HF.

10.3.2 All transmissions between aircraft or when broadcasting intentions must be prefixed with the aircraft call sign.

10.3.3 When requesting operational information on FIS frequencies, the pilot must use the service call sign "**FLIGHTWATCH**" and, when on HF, include the frequency on which they are calling.

10.3.4 Use of the collective "ALL STATIONS" must precede a general information broadcast.

10.4 **Diversion to an Alternate Aerodrome**

10.4.1 The pilot in command is responsible for taking appropriate diversion action based on information received. The pilot must provide the latest diversion time from the destination or from a point en route and, if required, the time interval.

10.5 **Operating Requirements for Transponders**

10.5.1 Pilots of aircraft fitted with a serviceable Mode 3A or Mode S transponder must activate the transponder at all times during flight, and if the Mode 3A transponder is Mode C capable, that mode must also be operated continuously.

10.5.2 Aircraft equipped with a Mode S transponder having an aircraft identification feature shall transmit the aircraft identification as specified in Item 7 of the flight notification or, when no flight notification has been filed, the aircraft registration.

10.5.3 For further information on the operation of transponders, including normal and emergency codes, see *ENR 1.6 Section 7.*

10.6 **Operating Requirements for ADS-B Transmitters**

10.6.1 Pilots of aircraft fitted with a serviceable ADS-B transmitter which has been confirmed suitable to receive ADS-B derived ATS surveillance services in Australia should activate the transmitter at all times during flight.

Note 1: Some ADS-B installations may share controls with the SSR transponder, meaning that independent operation of the two systems is not possible.

Note 2: If it is not possible to comply with a particular instruction the pilot must advise ATC and request alternative instructions.

10.6.2 Aircraft equipped with ADS-B having an aircraft identification feature shall transmit the aircraft identification as specified in the flight notification or, when no flight notification has been filed, the aircraft registration.

10.6.3 For further information on the operation of ADS-B transmitters, including normal and emergency codes, see *ENR 1.6 Section 6.5*.

10.7 **Alternate Aerodromes**

10.7.1 **General**

10.7.1.1 Except for *Part 121* operations, a pilot in command must make provision for flight to a destination alternate aerodrome, when required, in accordance with the paragraphs of this section (i.e. all of *Section 10.7*).

Note: The requirements related to Part 121 alternate aerodromes are contained in Chapter 4 of the Part 121 MOS, with interlinkages to Chapters 2 and 7 of the Part 121 MOS.

10.7.1.2 When a flight is required to provide for a destination alternate aerodrome, any aerodrome may be so nominated for that flight provided:

a. it is suitable as a destination for that flight; and

b. is not an aerodrome for which that flight would require to provide for an alternate aerodrome; and

c. is not a helideck.

10.7.1.3 When the forecasts required by *Section 7.02* of the *Part 91 MOS* for an aerodrome are not available, the pilot in command must make provision for a suitable destination alternate that has the required forecast.

10.7.2 **Weather Conditions**

10.7.2.1 Except when operating an aircraft under the VFR by day within 50NM of the point of departure, the pilot in command must provide for a suitable alternate aerodrome when arrival at the destination will be during the currency of, or up to 30 minutes prior to the forecast commencement of, the following weather conditions:

a. cloud - more than SCT below the alternate minimum (see *paras 10.7.2.10 and 10.7.2.11*); or

Note: In determining requirements for alternate aerodromes, forecast amounts of cloud below the alternate minima are cumulative. For determining requirements, the cumulative cloud amount is interpreted as follows:

FEW plus FEW is equivalent to SCT,

FEW plus SCT is equivalent to BKN,

SCT plus SCT is equivalent to BKN or OVC.

b. visibility - less than the alternate minimum; or

c. visibility - greater than the alternate minimum, but the forecast is endorsed with at least a 30% percentage probability of fog, mist, dust or any other phenomenon restricting visibility below the alternate minimum; or

d. a thunderstorm or associated severe turbulence, or a forecast of at least a 30% probability of such an event; or

e. wind - a crosswind or tailwind component more than the maximum for the aircraft.

Note: Wind gusts must be considered.

10.7.2.2 When operating a helicopter under the VFR, and the use of helicopter VMC is permissible at the destination, the pilot in command must provide for a suitable alternate aerodrome when either of the following conditions is forecast at the destination:

a. cloud - more than SCT below a ceiling of 1,000FT; or

b. visibility - less than 3,000M.

10.7.2.3 When weather conditions at the destination are forecast to be as specified at *para 10.7.2.1*, but are expected to improve at a specific time, provision for an alternate aerodrome need not be made if sufficient fuel is carried to allow the aircraft to hold until that specified time plus 30 minutes.

10.7.2.4 When weather conditions at the destination are forecast to be above the values specified at *para 10.7.2.1*, but, additionally, intermittent or temporary deteriorations in the weather below the values are forecast, provision of an alternate need not be made if sufficient fuel is carried to allow the aircraft to hold for:

a. 30 minutes for intermittent deterioration (INTER); and

b. 60 minutes for temporary deterioration (TEMPO).

10.7.2.5 When thunderstorms or their associated severe turbulence or their probability is forecast at the destination, sufficient fuel must be carried to permit the aircraft to proceed to a suitable alternate or to hold for:

a. 30 minutes when the forecast is endorsed INTER; or

b. 60 minutes when the forecast is endorsed TEMPO.

10.7.2.6 When a forecast has multiple INTER or TEMPO deteriorations and holding fuel will be carried, fuel must be carried to hold for only the most limiting requirement. INTER and TEMPO holding fuel requirements are not cumulative.

10.7.2.7 When TAFs include a FM or a BECMG, causing an operational requirement to either become effective or be removed, the timing for the change in operational requirement is as follows:

a. When the weather following the FM or BECMG is forecast to create an operational requirement, that operational requirement will become effective 30 minutes before the start of the FM time, or 30 minutes before the start of the BECMG period.

b. When the weather following the FM or BECMG is forecast to remove an operational requirement, that operational requirement will remain effective until 30 minutes after the FM time or 30 minutes after the end of the BECMG period.

10.7.2.8 The fuel required by *paras 10.7.2.4* or *10.7.2.5* must be carried when the ETA of the aircraft at its destination or alternate falls within the period 30 minutes before the forecast commencement time to 30 minutes after the expected time of cessation of these deteriorations. If the holding time required by *paras 10.7.2.4* or *10.7.2.5* extends past 30 minutes after the forecast cessation of these deteriorations, the aircraft need only carry sufficient fuel to hold until 30 minutes after the forecast cessation time.

10.7.2.9 At aerodromes receiving a TAF3 service, and only during the first 3 hours of the TAF3 validity, not beyond the end time for the TAF3 service if such a time is specified, the following do not apply:

a. 30 minute buffers required by *paras 10.7.2.7* and *10.7.2.8,*

b. the alternate or holding fuel required by:

(i) *para 10.7.2.1* (c) for reduction in visibility, or

(ii) *para 10.7.2.5* for any PROB30 or PROB40 for TS or associated severe turbulence.

10.7.2.10 For IFR flights, the alternate minima are as follows:

a. For aerodromes with an Instrument Approach Procedure (IAP) that the pilot is able to conduct, the alternate minima published on the chart (see *ENR 1.5, Section 6.*).

b. By day only - for aerodromes without an IAP, or that has an IAP but the pilot is unable to conduct that procedure, the alternate minima is the lowest safe altitude for the final route segment plus 500FT and a visibility of 8KM.

Note: Alternate minima based on weather for night IFR operations is only para (a) above. If the planned destination aerodrome for an IFR flight by night does not have any IAP, or any IAP that the pilot can conduct, then due to para 10.7.3.1, the flight must plan for a destination alternate aerodrome. By virtue of para 10.7.1.2, the destination alternate aerodrome must have an IAP that the pilot is able to conduct, which would then result in para (a) being the applicable alternate weather minima.

10.7.2.11 For flight by aeroplanes under the VFR (day or night) and helicopters operating under the VFR at night, the alternate minima are a ceiling of 1,500FT and a visibility of 8KM.

10.7.2.12 For VFR helicopter operations by day, the alternate minima are the same as for night unless the additional conditions specified in *Section 2.07* of the *Part 91 MOS* are met. When these additional conditions are met, the alternate minima requirements are as shown in *para 10.7.2.2*.

10.7.3 **Radio Navigation Aids**

10.7.3.1 A destination alternate aerodrome must be planned for an IFR flight by night to a planned destination aerodrome that is:

a. not served by an instrument approach procedure; or

b. is served by 1 or more instrument approach procedures, none of which the pilot is able to conduct.

10.7.3.2 A flight operating under the VFR at night (see *ENR 1.2 Section 1.1*) must provide an alternate aerodrome within one (1) hour flight time of the destination unless:

a. the destination is served by a ground based radio navigation aid (NDB/VOR) and the aircraft is fitted with the appropriate radio navigation system capable of using the aid, and the pilot is competent in using the aid, or

b. the aircraft is fitted with an approved GNSS (as defined in the relevant *MOS* for the kind of operation being conducted) and the pilot is competent in using the GNSS.

10.7.3.3 If aircraft navigation is to be conducted using a GNSS receiver certified only to (E)TSO C-129, navigation to a destination alternate aerodrome must be planned using a navigation system other than GNSS.

10.7.4 **Runway Lighting**

10.7.4.1 **Portable Lighting.** When a flight is planned to land at night at an aerodrome where the runway lighting is portable, provision must be made for flight to an alternate aerodrome unless arrangements are made for a responsible person to be in attendance during the period specified in *para 10.8.1.1 (note c),* to ensure that the runway lights are available during that period.

10.7.4.2 **Standby Power.** When a flight is planned to land at night at an aerodrome with electric runway lighting, whether pilot activated or otherwise, but without standby power, provision must be made for flight to an alternate aerodrome unless portable runway lights are available and arrangements have been made for a responsible person to be in attendance during the period specified in *para 10.8.1.1 (note c)*, to display the portable lights in the event of a failure of the primary lighting.

10.7.4.3 **PAL.** When a flight is planned to land at night at an aerodrome with PAL and standby power, provision must be made for a flight to an alternate aerodrome equipped with runway lighting unless a responsible person is in attendance to manually switch on the aerodrome lighting.

10.7.4.4 **Alternate Aerodromes - PAL**. An aerodrome served by PAL may be nominated as an alternate aerodrome.

There is no requirement for a responsible person on the ground to be in attendance, but the aircraft must be equipped with:

a. dual VHF; or

b. single VHF and HF communications and carries 30 minutes holding fuel to allow for the alerting of ground staff in the event of a failure of the aircraft's VHF communication.

10.7.4.5 The alternate requirements of *paras 10.7.4.1 to 10.7.4.4* inclusive need not be applied if the aircraft carries holding fuel for first light plus 10 minutes at the destination.

10.7.4.6 An alternate aerodrome nominated in accordance with the requirements in *paras 10.7.4.2 and 10.7.4.3* need not have standby power or standby portable runway lighting.

10.7.4.7 A responsible person under *para 10.7.4.2* is one who has been instructed in, and is competent to display, the standard runway lighting with portable lights.

10.8 Suitability of Aerodromes

10.8.1 General

10.8.1.1 The following procedures are an acceptable means of compliance with the requirements of *CASR 91.055* in relation to the suitability of an aerodrome:

a. Pilots should verify that the intended aerodrome meets the requirements of *CASR 91.410* as regards to its suitability for the intended operation. Additionally, pilots should confirm with the aerodrome operator that the pavement strength of the aerodrome meets the standards required by the aircraft being operated. Pilots should also be aware that *CASR Parts 121, 133, 135,* and *138* may place additional aerodrome requirements for a particular flight or type of operation.

b. Pilots should ensure that runway edge lighting, threshold lighting, illuminated wind direction indicator, and any obstacle lighting (if required) is serviceable for any planned night operations. Additional lighting may be required for particular operations – refer to the applicable regulation.

c. When aerodrome lighting is required and PAL is not being used, the pilot in command or operator should ensure that arrangements have been made for the runway, obstacle and taxiway lighting to be operating during the following periods:

(i) departure: from at least 10 minutes before departure to at least 30 minutes after take-off;

(ii) arrival: from at least 30 minutes before ETA to the time landing and taxiing has been completed.

Note 1: An operator planning a flight by an aircraft with tyre pressures and/or weight in excess of that permitted by the Part 139 MOS must ensure that a pavement concession is obtained.

Note 2: Emergency Landings. When safety is involved, the nearest aerodrome which will permit a landing without danger to the aircraft may be used, irrespective of the damage that may be caused to the pavement.

Note 3: Aerodrome lighting at an aerodrome where a control tower is operating will be activated by ATC as necessary. Pilots requiring aerodrome lighting outside the control tower's published hours should use PAL, if available, or make appropriate arrangements with ATC. If ATC has already ceased duty, requests should be directed to the local aerodrome operator. Confirmation should be obtained that requests for lighting will be satisfied.

Note 4: A pilot having made arrangements with ATS for night lighting must notify any change in requirements.

Note 5: Additional details on acceptable means of compliance refer to are contained in the Part 91 AMC/GM document entries related to CASR 91.410, and, for Part 121 operations, the Part 121 AMC/GM document entry related to CASR 121.205.

10.8.2 **Runway Width**

10.8.2.1 Specific limitations apply to all *Part 121* operations and certain *Part 135* operations. Refer to *CASR Parts 121* and *135*, and any applicable directions or exemptions.

10.9 Fuel Requirements

10.9.1 General

10.9.1.1 Refer to the relevant *CASR Part 91, 121 133* and *135* rules relating to fuel carriage. *AC 91-15* provides guidance material for these requirements.

10.9.2 Minimum Fuel

10.9.2.1 The pilot in command shall advise ATC of a minimum fuel state by declaring MINIMUM FUEL when, having committed to land at a specific aerodrome, the pilot in command calculates that any change to the existing clearance to that aerodrome may result in landing with less than final reserve fuel.

Note 1: The declaration of MINIMUM FUEL informs ATC that all planned aerodrome options have been reduced to a specific aerodrome of intended landing and any change to the existing clearance may result in landing with less than final reserve fuel. This is not an emergency situation but an indication that an emergency situation is possible should any additional delay occur.

Note 2: Pilots should not expect any form of priority handling as a result of a "MINIMUM FUEL" declaration. ATC will, however, advise the flight crew of any additional expected delays as well as coordinate when transferring control of the aircraft to ensure other ATC units are aware of the flight's fuel state.

10.9.3 Emergency Fuel

10.9.3.1 The pilot in command must declare a situation of emergency fuel by broadcasting MAYDAY MAYDAY MAYDAY FUEL, when the calculated usable fuel predicted to be available upon landing at the nearest aerodrome where a safe landing can be made is less than the final reserve fuel.

Note 1: MAYDAY FUEL declaration is a distress message.

Note 2: In circumstances where a normal approach and landing is expected and the pilot assesses there is no requirement for emergency services, ATS should be so advised as early as possible e.g. "EXPECTING NORMAL APPROACH AND LANDING, EMERGENCY SERVICES NOT REQUIRED".

10.10 **Information by Pilots**

10.10.1 A pilot in command becoming aware of any irregularity of operation of any navigational or communications facility or service or other hazard to navigation must report the details as soon as practicable. Reports must be made to the appropriate ATS unit, except that defects, or hazards on a landing area must be reported to the person or authority granting use of the area.

10.10.2 Pilots are requested to advise ATS about any deterioration or improvement of reported runway surface conditions, deceleration, and/or directional control. See *APPENDIX 1* (AIREP) and *AD 1.2 Section 3* for details about Runway Condition Reports (RCR) and reporting deceleration and directional control.

10.10.3 During a bushfire danger period, pilots in command of aircraft should notify the nearest ATS unit promptly of any evidence of bushfires observed which they believe have not been reported previously.

10.10.4 Australian Customs and Border Protection (Customs) is the government civil surveillance organisation which coordinates aerial patrols of border and off-shore areas to detect breaches of Customs, Fisheries, Quarantine and Immigration legislation.

10.10.5 Pilots are invited to assist Customs by reporting activities which appear to warrant recording or investigation. Reports should be made to the in-contact ATS unit. Observations warranting a report include:

a. observed marine pollution (including oil slicks);

b. shipping abnormalities;

c. other unusual activities within approximately 300NM of the coastline;

d. suspicious activities of unidentified itinerant aircraft, especially if there is a suggestion that they are travelling from or to an overseas destination;

e. aircraft signalling the ground or dropping objects;

f. aircraft operating at night without navigation lights;

g. aircraft operating from non-aerodrome sites, or unexplained or unusual activity at aerodromes or remote strips;

h. possible illegal fishing within the 200NM fishing zone;

i. possible smuggling of drugs or other prohibited goods;

j. possible illegal immigrants entering Australia;

k. unauthorised landings by sea or air;

l. threats to the well being of the Great Barrier Reef or other environmentally significant areas; and

m. unusual activities in remote areas.

10.10.5.1 Such reports should be elaborated on at debriefing.

10.10.6 Observed volcanic activity must be included in AIREP.

10.10.7 All air crews are to report immediately by radio to the appropriate ATC facility, any incidents of unauthorised laser illumination. Reports should include event position, altitude, colour of laser beam(s), originating direction and position, and any other relevant information deemed necessary for ATC and law enforcement action.

10.10.8 Air crews flying in Class G airspace are also requested to immediately broadcast a general laser illumination caution on the appropriate CTAF. This general caution should include the following elements:
Phrase “UNAUTHORISED LASER ILLUMINATION EVENT(s) HAS/HAVE BEEN REPORTED” (general positional information including location and altitude).

10.11 Flights Over Water

10.11.1 There are specific over water flight notification requirements (refer to *CASR 91.240* and the *Part 91 MOS Chapter 9).*

Note: Pilots are reminded of the requirement to not operate an aircraft in a manner which creates a hazard to a person or property (CASR 91.055).

10.11.2 Requirements related to flight over water can be found in the CASR Part relevant to the operation being conducted.

10.11.3 Helicopters must be fitted with an approved emergency flotation system on a *Part 133* AOC operation in accordance with *Chapter 11* of the *Part 133 MOS.*

10.12 **Procedures for Ground Operations of Turbo-Jet Aircraft**

10.12.1 Whenever an engine, other than the APU, of a turbo-jet aircraft is operating on the ground, the aircraft's anti-collision light(s) must be displayed, thereby indicating to pilots of other aircraft to exercise caution. Military turbo-jet aircraft should always be treated with caution as their anti-collision light may not be displayed even though an engine is running.

10.13 **Clearances - Pilot Responsibility**

10.13.1 A clearance issued by an ATS unit is only an authorisation for the pilot in command to proceed in accordance with the terms of the clearance. The clearance is not an authorisation for a pilot to deviate from any regulation, order, operating standard or procedure, or minimum altitude, nor to conduct unsafe operations in their aircraft. Further, the issuance and acceptance of a clearance in no way abrogates or transfers to an ATS unit the responsibilities of the pilot in command.

10.14 **Special Requirements**

10.14.1 **Special Standby of Fire Services**

10.14.1.1 A pilot conducting training in take-offs and landings with a multi engined aircraft may request the airport RFFS to stand by on the field. The request must be made through ATS or direct to the responsible Fire Officer.

10.14.2 **Aeroplane and rotorcraft simulated failures**

10.14.2.1 *CASR 91.730* through to *91.750* contain specific rules relating to certain aeroplane training and checking activities. *CASR 91.755* through to *91.775* contain specific rules relating to certain rotorcraft training and checking activities.

10.14.2.2 For *CASR 91.750(2)(g)*, and for *CASR 91.775(2)(f)(i)*, the circling area mentioned is either:

a. prescribed IFR circling area for the aerodrome associated with an authorised instrument approach procedure, or

b. if there is no prescribed IFR circling area of this kind for the aerodrome, within 3NM of the aerodrome reference point, but only for an aeroplane with MTOW less than or equal to 5,700KG.

Note: The information provided by spot heights on IAL charts must be treated with caution, as they do not necessarily indicate the highest terrain or all obstacles in the circling area. Pilots in command of flights involving simulated engine failures should risk assess their intended flight path options versus aircraft performance capability in this operational configuration.

10.14.3 **Ab Initio Flying Training at an Aerodrome**

10.14.3.1 It is recommended that aerodromes at which ab initio flight training is conducted have the following characteristics:

a. The TODA and LDA for the runway are at least 120% of the distance required by the Aircraft Flight Manual or performance chart;

b. In the case of aeroplanes for which there is no Aircraft Flight Manual or performance chart, the TODA and LDA for the runway are at least 120% of the distance specified in the aeroplane's certificate of airworthiness;

c. There are obstacle-clear approach and take-off slopes of no more than 3.33% to a distance of 1,600M from each end of the runway.

10.14.4 **Fuel Dumping in Flight**

10.14.4.1 Release of fuel from an aircraft is not permitted except in an emergency or non-normal situation.

10.14.4.2 When fuel dumping is required, the pilot in command should request authority from ATC before commencing a fuel dump, and must:

a. notify ATC immediately after an emergency fuel dump;

b. take reasonable precautions to ensure the safety of persons or property in the air and on the ground; and

c. where possible, conduct a controlled dump in clear air above 6,000FT and in an area nominated by ATC.

10.14.4.3 The pilot should advise ATC if radio silence is required during the fuel dumping operation.

10.14.5 **Areas Having Limitations on Access**

10.14.5.1 Although not involving a potential hazard to aircraft, operations over certain areas have limitations placed on them for environmental reasons. Refer to *ERSA GEN* for details.

10.14.6 **Aerial Photography and Survey Operations**

10.14.6.1 Pilots and operators intending to conduct aerial photography or survey operations in controlled airspace should liaise with the ATC unit responsible for the area(s) concerned prior to submitting flight plans. ATC clearance limitations and restrictions on times, tracks and/or levels, which could inhibit the proposed operation(s), may apply in the desired airspace. Preflight approval will enable pilots to plan tasks accordingly, thus minimising disruption to programs.

10.14.7 **Aerial Photography of Military Installations**

10.14.7.1 Pilots or operators intending to photograph military installations or areas which include military installations must contact the appropriate military authority as such photography may require prior approval or not be permissible.

10.14.8 **Police Operations**

10.14.8.1 An aircraft operated by police authorities which requires priority in situations where life is threatened must use the call sign “POLAIR RED” or “FEDPOL RED”. Police must call “POLAIR/ FEDPOL RED PRIORITY” on first contact.

10.14.9 **Military Authority Assumes Responsibility for Separation of Aircraft (MARSA)**

10.14.9.1 MARSA is a procedure which authorises pilots of military aircraft to assume responsibility for separation between their aircraft and other nominated military aircraft, or military contract civil aircraft, in controlled airspace. MARSA can only be used between participating aircraft using the same flight level or altitude, or manoeuvring within the same block of airspace.

10.14.9.2 Operators of foreign military aircraft wishing to participate in MARSA must refer to *RAAF AIP - Flight Information Handbook Australia (FIHA)* for details of the procedure.

10.14.10 **"Due Regard" Operations**

10.14.10.1 Certain operations by State aircraft (usually military), referred to as "Due Regard" operations, cannot be conducted in compliance with normal air traffic rules and procedures. Where these operations are necessary, Article 3 of the *Chicago Convention (1944)* requires contracting States to "have due regard for the safety of navigation of civil aircraft". Safety mitigators may include operations in VMC and/or use of surface and airborne radar.

10.14.11 **Security Awareness**

10.14.11.1 All members of the aviation industry, should be particularly vigilant with regard to any suspicious activity relating to the use, training in, or acquisition of dangerous chemicals, including threats, unusual purchases, and/or unusual contacts with the public.

10.14.11.2 Any suspicious circumstances or unusual behaviour should be immediately reported to the police and the relevant aircraft, airline, or airport operator.

10.14.12 **Fire Operations**

An aircraft operated by fire authorities which requires priority should notify the flight as Fire or Flood Relief (FFR) and use the appropriate special task call sign as per *GEN 3.4 para 5.21*.

10.14.13 **Requirements for Community Service Flights (CSF)**

In addition to any other relevant requirements of the civil aviation legislation, the flight must meet the CSF definition at *GEN 2.2* and satisfy the following requirements (where applicable).

Part 61 licence	PPL (not RPL)	CPL/ATPL
Aeronautical experience, recency and medical	400 hours total aeronautical experience	N/A
	250 hours pilot in command	N/A
	Current Class 1or Class 2 medical (not Basic Class 2)	
	1 landing in the same aircraft class (or type, if type rated aircraft) in the past 30 days	
	25 hours on multi-engine aircraft (if flying multi-engine)	
Flight rules requirements and experience	IFR - 20 hours on aircraft type VFR -10 hours on aircraft type A CSF at night must be conducted under the IFR. NVFR not permitted	
Aircraft	VH-registered aeroplane or helicopter	
	Not amateur built, limited category or experimental certificate	
	Not more than 100 hours or 12 months since last periodic maintenance inspection for those aircraft using the CASA system of maintenance (Schedule 5)	
Passengers	No more than 5 passengers	
	Passengers must be either patients being transported for the purpose of receiving non-emergency medical treatment or services, or persons accompanying the patient to provide support and assistance to the patient	
Flight notification	Full flight notification (IFR or VFR); or SARTIME (VFR)	
	RMK/CSF noted in Item 18 of domestic flight notification	
Record keeping	Flight annotated as CSF in pilot logbook	

Note: The flight notification requirements are also contained in ENR 1.10 section 2.- FLIGHT NOTIFICATION.

10.15 Helicopter Operations - At Aerodromes and Helicopter Access Corridors and Lanes

10.15.1 General

10.15.1.1 The procedures in this section apply to all helicopters operating in the vicinity of aerodromes and in helicopter access corridors and lanes, in accordance with the provisions of C*ASR 91.055, 91.265, 91.267, 91.360* and *91.410.*

10.15.2 Taxiing

10.15.2.1 For all helicopters, maximum use of the “air transit” procedure should be made to expedite traffic movement and flow about an aerodrome.

10.15.2.2 All helicopters may use “air taxiing” procedures as required. However, wheeled helicopters, where practicable, are encouraged to “ground taxi” on prepared surfaces to minimise rotor wash and its effects.

10.15.2.3 At night a helicopter should not taxi via routes which do not meet the physical dimensions and lighting requirements specified in *AC 139.R-01*.

10.15.3 Take-off/Departure

10.15.3.1 At locations within controlled airspace, helicopters may be granted a take-off clearance or instructed to report airborne, as appropriate, from any area nominated by ATC or the pilot, and assessed by the pilot as being suitable as a HLS.

10.15.3.2 Helicopters taking off/departing must proceed in accordance with ATC instructions.

10.15.3.3 Subject to clearance, a turn after take-off may be commenced when the pilot considers that the helicopter is at a safe height to do so.

10.15.3.4 Unless requested by the pilot, a take-off clearance will not be issued for a helicopter if the tailwind component exceeds 5KT.

10.15.3.5 Prescribed exit “gates” and associated standard routes and/or altitudes may be provided to facilitate the flow of helicopter traffic. Procedures for their use will be promulgated in *ERSA*. Use of these “gates” is not mandatory. Helicopters may, subject to an ATC clearance, revert to the standard traffic procedures applicable to aeroplanes. This option may be more appropriate when operating larger helicopters.

10.15.3.6 At night a helicopter should not take-off from other than a site which conforms with the requirements specified in *AC 139.R-01.* Any illuminated runway or illuminated taxiway of dimensions commensurate with the size of the helicopter landing site applicable to the helicopter, in accordance with *AC 139.R-01*, is considered to meet the requirements of *AC 139.R-01*.

10.15.3.7 At a non-controlled aerodrome a pilot may take-off from any area which is assessed as being suitable as a HLS.

10.15.3.8 When the pilot elects to conduct the take-off from outside the flight strip of the runway in use by aeroplanes, the helicopter take-off path must be outside that flight strip.

10.15.3.9 Before take-off, the helicopter is to be positioned to the appropriate side of the runway in use so that the turn after take-off does not cross the extended centre line of that runway. The pre-take-off positioning of the helicopter will be by air transit or by taxiing as appropriate.

10.15.3.10 The turn after take-off onto the desired departure track may be commenced when the pilot considers that the helicopter is at a safe height to do so. If the resultant departure track conflicts with the aeroplane traffic pattern, the helicopter should remain at 500FT above the surface until clear of that circuit pattern. Where this procedure is not practicable on environmental grounds, the helicopter is to adopt the standard departure procedure applicable to aeroplanes.

10.15.3.11 Pilots of radio equipped helicopters must broadcast intentions on the appropriate frequency before take-off.

10.15.4 Helicopter Access Corridors and Lanes

10.15.4.1 The following procedures for operations within promulgated helicopter access corridors and lanes apply:

a. maximum IAS of 120KT;

b. helicopters must operate under VFR, usually not below 500FT above the surface by day, subject to flight over populous area restrictions and the limitations published in *ERSA* for authorised corridors by night;

c. “see and avoid” procedures must be used;

d. formation flights are restricted to line astern with the lead aircraft responsible for maintaining separation from other traffic in accordance with *sub-para c.*;

e. a traffic advisory service is available in access corridors;

f. an ATS Surveillance System advisory service may be given at designated aerodromes;

g. a continuous listening watch on the appropriate ATS frequency in access corridors or broadcast frequency in lanes is mandatory;

h. two-way operations are conducted with all traffic keeping to the right of the central geographical/topographical feature(s) as detailed in *ERSA*;

i. the pilot in command has the responsibility to ensure that operations are confined within the boundaries of the corridor or lane;

j. the limits of corridors and lanes must be adhered to, with any transitional altitude requirements maintained within an accuracy of ±100FT;

k. a helicopter not confining its operations to an access corridor will require ATC clearance and, while outside the corridor, will be subject to separation standards as applied by ATC.

Note: Subject to environmental noise considerations, the imposition of limitations on those types of helicopters which exceed the noise limits specified in ICAO Annex 16 Vol 1 may be necessary.

10.15.5 **Arrivals**

10.15.5.1 At a controlled aerodrome, prescribed entry "gates" and associated standard routes and/or altitudes may be provided to facilitate the flow of helicopter traffic. Procedures for their use will be promulgated in *ERSA*.

Use of these "gates" is not mandatory. Subject to the receipt of an ATC clearance, helicopters may, if required, conform to the standard traffic procedures applicable to aeroplanes. This option may be more appropriate when operating larger helicopters.

10.15.5.2 At locations within controlled airspace, helicopters may be granted a landing clearance or be instructed to report on the ground, as appropriate, at any area nominated by ATC or the pilot, and assessed by the pilot as being suitable as a HLS.

10.15.5.3 Unless requested by the pilot, a landing clearance will not be issued for a helicopter if the tailwind component exceeds 5KT.

10.15.5.4 At night a helicopter should not land at a site other than one which conforms with the requirements specified in the latest issue of *AC 139.R-01.* Any illuminated runway or illuminated taxiway of dimensions commensurate with the size of the helicopter landing site applicable to the helicopter, in accordance with *AC 139.R-01*, is considered to meet the requirements of *AC 139.R-01*.

10.15.5.5 At a non-controlled aerodrome in VMC by day applicable to the aircraft category, helicopters need not join the circuit via standard aeroplane entry procedures, at the pilots discretion.

As an alternative, under such conditions, helicopters may join the circuit area at 500FT above the surface from any direction subject to the normal restrictions of flight over populous areas. Helicopters must avoid other circuit traffic and descend to land at any location assessed by the pilot as being suitable for use as a HLS, provided:

a. the intended landing point is located outside the flight strip of the runway in use;

b. the final approach is clear of the extended centreline of the runway in use;

c. post-landing positioning of the helicopter is by air transit or by taxiing as appropriate.

10.15.5.6 Pilots of radio equipped helicopters must broadcast intentions on the appropriate frequency as specified in *Section 10.3.*

10.15.6 **Circuit Procedures**

10.15.6.1 At controlled aerodromes any specific operating procedures applicable to the helicopter traffic pattern will be detailed in *ERSA*. The following generally applies:

a. where possible, helicopter circuit traffic will be separated from the aeroplane traffic pattern by the use of contra-direction circuits, outside of and parallel to the flight strip of the runway in use, and at a lower altitude than other traffic, but not below 500FT above the aerodrome elevation; or

b. when separate circuit patterns are not practicable, helicopters may utilise the same traffic pattern direction as other traffic, and will normally operate inside and at a lower altitude than that traffic, but not below 500FT above the aerodrome elevation.

10.15.6.2 At non-controlled aerodromes the following circuit operating procedures apply:

a. helicopters may be operated on contra-direction circuits and parallel to the aeroplane traffic pattern at a lower altitude than that traffic, but not below 500FT above the aerodrome elevation. The landing site associated with the helicopter circuit is to be positioned outside the flight strip of the runway in use so that helicopter circuit traffic does not cross the extended centre line of that runway;

b. if the procedure outlined in *sub-para 10.15.6.2a*. is not practicable, the helicopter circuit patterns should be flown inside and parallel to the aeroplane traffic and at lower altitudes, but not below 500FT above the aerodrome elevation. The landing site associated with the helicopter circuit must be positioned outside the flight strip of the runway in use so that helicopter circuit traffic does not cross the extended centre line of that runway; or

c. the helicopter must follow the standard aeroplane traffic pattern and, in this case, may use the flight strip area of the runway in use;

d. the pilots of radio equipped helicopters must broadcast their intentions and listen out for other traffic on the appropriate frequency.

11. PROCEDURES WHEN ATS TEMPORARILY NOT AVAILABLE

11.1 Traffic Information Broadcast by Aircraft (TIBA)

11.1.1 TIBA procedures

11.1.1.2 TIBA procedures are intended to permit reports and relevant supplementary information of an advisory nature to be transmitted by pilots for the information of pilots of other aircraft in the vicinity.

11.1.2 Frequency

11.1.2.2 Aircraft must maintain a listening watch on the appropriate TIBA frequency. Where VHF is used for air-ground communications with ATS and an aircraft has two serviceable VHF sets, one must be tuned to the appropriate ATS frequency and the other to the TIBA frequency.

11.1.2.2 The appropriate TIBA frequencies are:

Flight Profile	TIBA Frequency
At or above FL200	128.95 MHz
Below FL200	
– In continental Class G airspace	Relevant Area VHF
– Otherwise	126.35 MHz

11.2 **Listening Watch**

11.2.1 A listening watch must be maintained on the TIBA frequency 10 minutes before entering the designated airspace until leaving this airspace. For an aircraft taking off from an aerodrome located within 10 minutes flying time of that airspace, listening watch must start as soon as practicable after take-off.

11.3 **Time of Broadcasts**

11.3.1 Broadcasts must be made:

a. 10 minutes before entering the designated airspace or, for an aircraft taking off from an aerodrome located with 10 minutes flying time of the airspace, as soon as practicable after take-off;

b. 10 minutes prior to crossing a reporting point;

c. 10 minutes prior to crossing or joining an ATS contingency route;

d. at 20-minute intervals between distant reporting points;

e. 2 to 5 minutes, where possible, before a change in flight level;

f. at the time of a change in flight level; and

g. at any other time considered necessary by the pilot.

11.4 **Acknowledgment of Broadcasts**

11.4.1 Broadcasts should not be acknowledged unless a potential collision risk exists.

11.5 **Changes of Cruising Level**

11.5.1 Cruising level changes should not be made within the designated airspace, unless considered necessary by pilots to avoid traffic conflicts, for weather avoidance or for other valid operational reasons.

11.5.2 When changes to cruising level are unavoidable, all available aircraft lighting which would improve the visual detection of the aircraft must be displayed while changing levels.

11.5.3 When a change of level is anticipated or initiated, a change of level report must be made. When the new level is reached, a report advising that the aircraft is maintaining the new level must be made.

11.6 **Collision Avoidance**

11.6.1 If, on receipt of a traffic information broadcast from another aircraft, a pilot decides that immediate action is necessary to avoid an imminent collision risk to the aircraft, and this cannot be achieved in accordance with the right of way provisions or TCAS resolution, the pilot should:

a. unless an alternative manoeuvre appears more appropriate, immediately descend 1,000FT if above FL410, or 500FT if at or below FL410;

b. display all available aircraft lighting which would improve the visual detection of the aircraft;

c. as soon as possible, reply to the broadcast advising action being taken;

d. notify the action taken on the appropriate TIBA frequency; and

e. as soon as practicable, resume normal flight level, notifying the action on the appropriate TIBA frequency.

11.7 **Position Reporting**

11.7.1 Normal position reporting procedures should be continued at all times, regardless of any action taken to initiate or acknowledge a traffic information broadcast.

11.7.2 A position report must be made on the next CTA/Area VHF 15 minutes prior to leaving airspace in which TIBA procedures apply to obtain a clearance or re-establish SARWATCH on the appropriate ATS frequency.

12. MANDATORY BROADCAST PROCEDURES (ATS TEMPORARILY NOT AVAILABLE)

12.1 When ATS is temporarily not available, mandatory broadcast procedures may be specified in addition to TIBA broadcasts.

12.2 When arriving or departing from an aerodrome where mandatory broadcast procedures apply, pilots must monitor the appropriate mandatory broadcast frequency. Broadcasts must be made as follows:

Situation	Phrase
1. Broadcasts When a pilot broadcasts intentions.	**ALL STATIONS (location) (appropriate information)**
2. Taxi Taxiing at an aerodrome.	**(aircraft type) TAXIING (location) RUNWAY (number) FOR (destination, or departure quadrant or intention)**
3. About to Commence Take-off	**LINING UP/ROLLING (runway number) TURNING (left/right) TRACKING (quadrant) CLIMBING TO (level)**
4. Departing	**DEPARTED (location) TRACKING (degrees magnetic) CLIMBING TO (level) FOR (destination)**
5. Inbound When inbound - before crossing the boundary of the area in which mandatory broadcasts apply.	**(Aircraft type) (position reported as either the radial, bearing or quadrant from the aerodrome) (level) (intentions)**
6. Joining the Circuit	**(Aircraft type) JOINING (position in circuit) RUNWAY (number)**

12.3 Pilot discretion should be used in making other than the prescribed calls to assist other traffic; e.g. executing a missed approach, or position in the circuit area, or leaving levels designated on TMA routes.

APPENDIX 1

POSITION REPORTS, AIREP SPECIAL, AND VOLCANIC ASH REPORTS

1. **Position reports**

Refer to *FORM 1: ROUTINE POSITION AND AIREP SPECIAL REPORTS*.

Section 1 is obligatory.

Section 2 must be added only when requested by the operator or deemed necessary by the pilot in command.

In the en route phase, section 3 (other than braking action) should be added:

a. by AMDAR equipped aircraft only — at designated compulsory MET reporting points, and

b. when requested by ATC or MET.

At a controlled aerodrome, section 3 (braking action) should be reported when encountered runway braking action is not as good as reported.

2. **AIREP Special**

To be made whenever any of the phenomena listed under item 9 are observed or encountered. Items 1-4 and the appropriate phenomena specified in item 9 are the minimum required.

3. **Detailed reporting instructions for selected items**

Item 1: Use aircraft call sign as per flight plan or as requested by ATC.

Item 3: Time must be the actual time of the aircraft at the position, not the time of the message. Time shall be expressed in hours and minutes UTC.

Item 4: Aircraft on a block level, or climb, or descent clearance are to report actual flight level or altitude at the position plus level climbing/descending to.

Item 5: 'Next position' shall only be a compulsory reporting point or waypoint, unless ATC requests an estimate for a different place. Time shall be expressed in hours and minutes UTC.

Item 6: 'Ensuing significant point' shall be the next compulsory or non-compulsory reporting point or waypoint after the 'Next position'. This item is only required if requested or necessary to confirm route to be followed.

Item 9:

- Turbulence: The following specifications apply:
  - Moderate: Changes to accelerometer readings of between 0.5 g and 1.0 g at the aircraft's centre of gravity. Moderate changes to aircraft attitude and/or altitude may occur but aircraft remains under positive control. Usually small changes in airspeed. Difficulty in walking. Loose objects move about.
  - Severe: Changes to accelerometer readings greater than 1.0 g at the aircraft's centre of gravity. Abrupt changes to aircraft attitude and/or altitude may occur; aircraft may be out of control for short periods. Usually large changes of airspeed. Loose objects tossed about.
- Icing: The following specifications apply:
  - Moderate: Conditions in which a change of heading and/or altitude may be considered desirable.
  - Severe: Conditions in which immediate change of heading and/or altitude is considered essential.
- MOUNTAINWAVE SEVERE means conditions in which the downdraft is 600FT/MIN or more and/or severe turbulence is encountered.
- Thunderstorms: Only report those thunderstorms which are:
  - obscured in haze; or
  - embedded in cloud; or
  - widespread; or
  - forming a squall-line.

- Pre-eruption volcanic activity: In an AIREP, this means unusual and/or increasing volcanic activity which could presage a volcanic eruption.

Note: In case of volcanic ash cloud, pre-eruption volcanic activity or volcanic eruption, a post-flight report shall also be made on the special air-report of volcanic activity form (Model VAR - see example in FORM 2: VOLCANIC ACTIVITY FORM (MODEL VAR)).

- Braking action: The following specifications apply:
  - GOOD — Braking deceleration is normal for the wheel braking effort applied AND directional control is normal.
  - GOOD TO MEDIUM — Braking deceleration OR directional control is between Good and Medium.
  - MEDIUM — Braking deceleration is noticeably reduced for the wheel braking effort applied OR directional control is noticeably reduced.
  - MEDIUM TO POOR — Braking deceleration OR directional control is between Medium and Poor.
  - POOR — Braking deceleration is significantly reduced for the wheel braking effort applied OR directional control is significantly reduced.
  - LESS THAN POOR — Braking deceleration is minimal to non-existent for the wheel braking effort applied OR directional control is uncertain.

4. **Volcanic Ash Reports**

Refer to *FORM 2: VOLCANIC ACTIVITY FORM (MODEL VAR)*.

AIREP are critically important for assessing the hazards of volcanic ash cloud to aircraft operations. Information recorded on the volcanic activity reporting form (Model VAR) is not for transmission by RTF but, on arrival at an aerodrome, is to be delivered without delay by the operator or a flight crew member to the aerodrome meteorological office. If such an office is not easily accessible, the completed form shall be delivered in accordance with local arrangements made between the meteorological and ATS authorities and the operator.

FORM 1: ROUTINE POSITION AND AIREP SPECIAL REPORTS			
	ITEM	**PARAMETER**	**TRANSMIT IN TELEPHONY as appropriate**
		Message-type designator: – Routine position-report – Special air-report	 (call sign) POSITION [AIREP] SPECIAL
Section 1	1	Aircraft identification	*(aircraft identification)*
	2	Position	POSITION *(latitude and longitude)* OVER *(significant point)** ABEAM *(significant point)* *(significant point) (bearing) (distance)*
	3	Time	*(time)*
	4	Level	FLIGHT LEVEL *(number)* or *(number)* FEET* CLIMBING TO FLIGHT LEVEL *(number)* or *(number)* FEET* DESCENDING TO FLIGHT LEVEL *(number)* or *(number)* FEET*
	5	Next position and estimated time over	*(position) (time)*
	6	Ensuing significant point (only if requested or necessary to confirm route to be followed	*(position)* NEXT
Section 2	7	Estimated time of arrival	*(aerodrome) (time)*
	8	Endurance	ENDURANCE *(hours and minutes)*

	ITEM	PARAMETER	TRANSMIT IN TELEPHONY as appropriate
Section 3	9	Phenomenon encountered or observed, prompting a special air-report:	
		– Moderate turbulence	TURBULENCE MODERATE
		– Severe turbulence	TURBULENCE SEVERE
		– Moderate icing	ICING MODERATE
		– Severe icing	ICING SEVERE
		– Severe mountain wave	MOUNTAINWAVE SEVERE
		– Thunderstorms without hail	THUNDERSTORMS
		– Thunderstorms with hail	THUNDERSTORMS WITH HAIL
		– Heavy dust/sandstorm	DUSTSTORM *or* SANDSTORM HEAVY
		– Volcanic ash cloud	VOLCANIC ASH CLOUD
		– Pre-eruption volcanic activity or volcanic eruption	PRE-ERUPTION VOLCANIC ACTIVITY *or* VOLCANIC ERUPTION
		– On climb out or approach:	(Plain language description)
		– Cloud - unexpected significant variations to amount, base or tops (by reference to QNH);	
		– Visibility - reduced due fog, mist, hail, rain, snow or dust, or improvement observed;	
		– Wind - significant variation to forecast;	
		– Other Phenomena - incidence of severe or moderate turbulence, thunderstorms, moderate or severe icing, hail, line squalls, standing waves or winds of 40KT or more within 2,000FT of ground level.	

Section 3	9	Runway braking action:	BRAKING ACTION:
		– Good	– GOOD
		– Good to Medium	– GOOD TO MEDIUM
		– Medium	– MEDIUM
		– Medium to Poor	– MEDIUM TO POOR
		– Poor	– POOR
		– Less than Poor	– LESS THAN POOR

*In Australian domestic airspace, the words “OVER” and “FEET” may be omitted.

FORM 2: VOLCANIC ACTIVITY FORM (MODEL VAR)		
OPERATOR:	ACFT IDENTIFICATION: (per flight plan)	PIC:
DEP FROM:	DATE:	TIME (UTC):
ARR AT:	DATE:	TIME (UTC):
ADDRESSEE:		AIREP SPECIAL:

Items 1-8 are to be reported immediately to the ATS unit that you are in contact with.

1) Aircraft identification:	2) Position:
3) Time:	4) FL or ALT:
5) VOLCANIC ACTIVITY OBSERVED AT: (Position or bearing, estimated level of ash cloud and distance from aircraft)	
6) AIR TEMPERATURE:	7) SPOT WIND:
8) SUPPLEMENTARY INFORMATION: (Brief description of activity especially vertical and lateral extent of ash cloud and, where possible, horizontal movement, rate of growth, etc.) SO2 detected Yes ☐ No ☐ Ash encountered Yes ☐ No ☐	

After landing complete items 9-16, then deliver to the aerodrome meteorological office or deliver in accordance with local arrangements made between the Meteorological, ATS authorities and the operator.

9) Density of ash cloud:	(a) Wispy ☐	(b) Moderate dense ☐	(c) Very dense ☐
10) Colour of ash cloud:	(a) White ☐	(b) Light grey ☐	(c) Dark grey ☐
	(d) Black ☐	(e) Other ☐	
11) Eruption:	(a) Continuous ☐	(b) Intermittent ☐	(c) Not visible ☐
12) Position of activity:	(a) Summit ☐	(b) Side ☐	(c) Single ☐
	(d) Multiple ☐	(e) Not observed ☐	
13) Other observed features of eruption:	(a) Lightning ☐	(b) Glow ☐	(c) Large rocks ☐
	(d) Ash fallout ☐	(e) Mushroom cloud ☐	(f) All ☐
14) Effect on aircraft:	(a) Communication ☐	(b) Navigation systems ☐	(c) Engines ☐
	(d) Pitot static ☐	(e) Windscreen ☐	(f) Windows ☐
15) Other effects:	(a) Turbulence ☐	(b) St. Elmo's Fire ☐	(c) Other fumes ☐
16) Other information: (Any information considered useful)			

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ENR 1.2 VISUAL FLIGHT RULES

1. FLIGHT RULES

1.1 The Visual Flight Rules (VFR)

1.1.1 A VFR flight may only be conducted:

a. in VMC;

b. provided that, when not navigating by visual reference to the ground or water, the pilot in command must comply with the requirements of *Chapter 14* of the *Part 91 MOS*, as if the flight were an IFR flight;

c. at sub-sonic speeds; and

d. in accordance with the airspace speed limitations specified in *ENR 1.4.*

1.1.2 Unless the pilot in command is authorised under *CASR Part 61* to conduct a flight under the IFR or at night under the VFR and the aircraft is appropriately equipped for flight at night or under the IFR, a VFR flight must not be conducted at night.

1.1.3 For pilots not authorised to fly at night, it is recommended that they plan to arrive at the later of the destination aerodrome or alternate aerodrome at least 10 minutes before last light (allowing for any required holding).

1.2 Special VFR

1.2.1 By day, when VMC do not exist, the ATC unit responsible for a CTR or CTA may issue, at pilot request, and provided an IFR flight will not be unduly delayed, a Special VFR clearance for flight:

a. in the CTR; or

b. in a CTA next to the CTR for the purpose of entering or leaving the CTR.

1.2.2 When operating under a Special VFR clearance, pilots are responsible for ensuring that:

a. the flight is conducted clear of cloud;

b. the visibility is not less than:

(1) for aeroplanes, 1,600M;

(2) for helicopters, 800M; or

(3) for balloons, 100M below 500FT AGL and 1,600M at and above 500FT AGL;

c. a helicopter is operated at such a speed that the pilot has adequate opportunity to observe any obstructions or other traffic in sufficient time to avoid a collision; and

D

1.2.3 Special VFR is not permitted in Class E airspace.

2. VISUAL METEOROLOGICAL CONDITIONS (VMC) - TAKE-OFF, EN ROUTE AND LANDING

2.1 The cloud and visibility criteria for VMC, including specific additional requirements, are contained in *section 2.07* of the *Part 91 MOS*.

D

ENR 1.3 INSTRUMENT FLIGHT RULES

1. THE INSTRUMENT FLIGHT RULES (IFR)

1.1 The IFR must be used by flights conducted in circumstances other than those specified in *ENR 1.2 Section 1.1*.

1.2 IFR flights must be conducted in accordance with the airspace speed limitations specified in *ENR 1.4*.

2. FLIGHT RULES NOMINATION

2.1 The nomination of a flight as IFR or VFR determines:

a. the flight notification requirements;

b. separation requirements in Classes A, C and D airspace;

c. separation requirements for aircraft in receipt of an airways clearance in Class E airspace; and

d. traffic information requirements in classes E and G airspace.

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ENR 1.4 ATS AIRSPACE CLASSIFICATION

1. CONTROLLED AIRSPACE

1.1 General

1.1.1 Controlled airspace is defined as “airspace of defined dimensions within which ATC services are provided to IFR flights and to VFR flights in accordance with the airspace classification”.

1.1.2 Controlled airspace is a generic term which, in Australia, covers ATS airspace classes A, C, D and E.

1.1.3 Controlled airspace is established generally on the basis of the kinds of operations and considerations of flight procedures used. Such airspace does not necessarily cover routes to alternate aerodromes.

1.1.4 Controlled airspace within the Brisbane and Melbourne FIR is generally established as follows:

Class of Airspace	Application
Class A	• within radar coverage - lower limit above FL180 and upper limit FL600; • outside radar coverage - lower limit FL245 and upper limit FL600; • an area extending from 90NM south of Melbourne to Launceston and Hobart, lower limit FL180 and upper limit FL600.
Class C	• within radar coverage south of Sydney, lower limit FL125 and upper limit FL180 under Class A airspace; • in the control area steps associated with controlled aerodromes, excluding control area steps classified as Class D airspace; • in control zones of defined dimensions.
Class D	Control zones of defined dimensions, and associated control area steps, upper limit 4,500FT.

Class of Airspace	Application
Class E	• within radar coverage:
	– south of Sydney, lower limit 8,500FT and upper limit FL125 under Class C airspace; – north of Sydney, lower limit 8,500FT and upper limit FL125; • in the vicinity of Williamtown: coincident with the lateral limits of R578A-E above A045 - when R578 is not active; • continental Australia, lower limit FL125 and upper limit FL245 under Class A airspace; • in the control area steps associated with Class D controlled aerodromes excluding Class D or C airspace: • Karratha Class E lower limit 5,500FT to upper limit FL125 • Broome Class E lower limit 1,200FT AGL to upper limit FL125 • Avalon Class E lower limit 700FT AGL to upper limit 4,500FT; • in the control area steps associated with Class C controlled aerodromes excluding Class C airspace: • Perth Class E lower limit 8,500FT to upper limit FL125.

1.1.5 Operations in control areas and control zones must be conducted in accordance with the published procedures and requirements for that specific airspace and air traffic clearances. Special procedures may also be specified for an aerodrome within a control zone.

1.1.6 The extent of controlled airspace is promulgated in Airservices *Aeronautical Charts, NOTAM, AIP SUP* and *DAH*.

1.1.7 The hierarchy of airspace from most restrictive to least restrictive is as follows:

a. Prohibited, Restricted or Military Operating Areas;

b. Class A airspace;

c. Class C airspace;

d. Class D airspace;

e. Class E airspace;

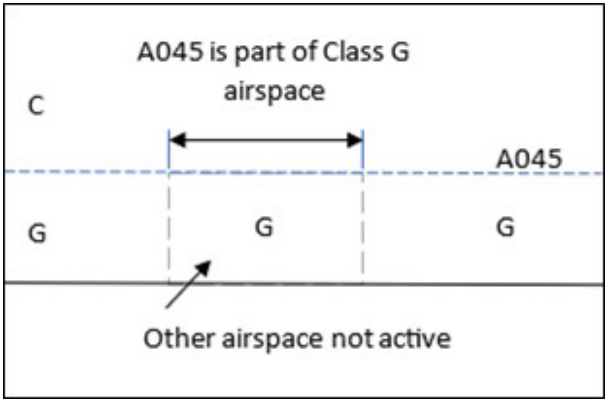
f. Class G airspace.

1.1.7.1 When airspaces adjoin vertically (one above the other), flight at the common level must comply with the requirements of and will be provided the services applicable to, the less restrictive class of airspace.

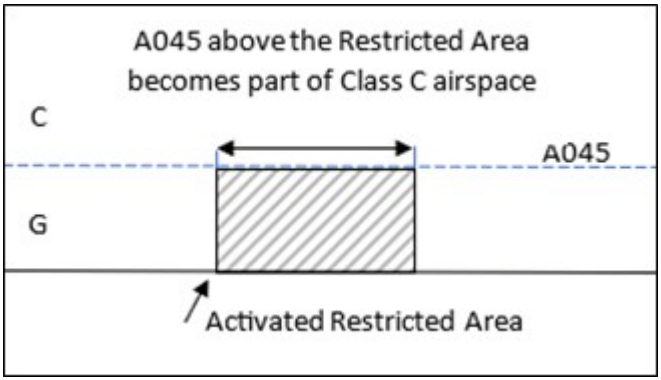
1.1.7.2 Where a non-continuous airspace vertically adjoins controlled airspace (one above the other), the common level between airspaces becomes part of controlled airspace upon activation and is subject to an ATC clearance.

Airspace examples:

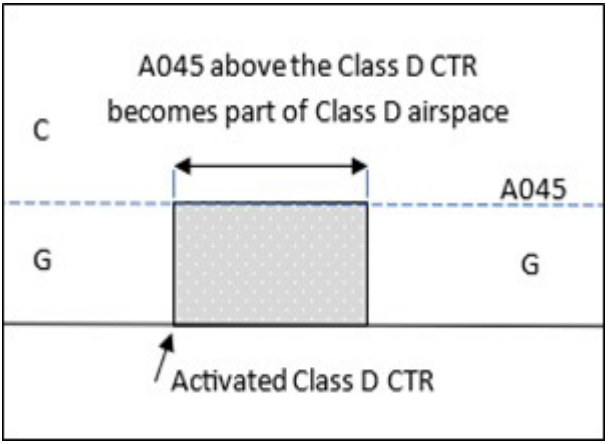
a. When a Restricted Area or Class D control zone is not active, the common boundary at A045, between Class C and Class G airspace, is part of Class G airspace (the less restrictive airspace) and receives a Class G service (clearance not required):



b. When a Restricted Area becomes active below Class C airspace, the common boundary at A045, between the Restricted Area and Class C airspace, becomes part of Class C airspace (the less restrictive airspace), and receives Class C services (clearance required):



c. When a Class D control zone becomes active below Class C airspace, the common boundary at A045, between Class C and Class D airspace, becomes Class D airspace (the less restrictive airspace) with a Class D service (clearance required):

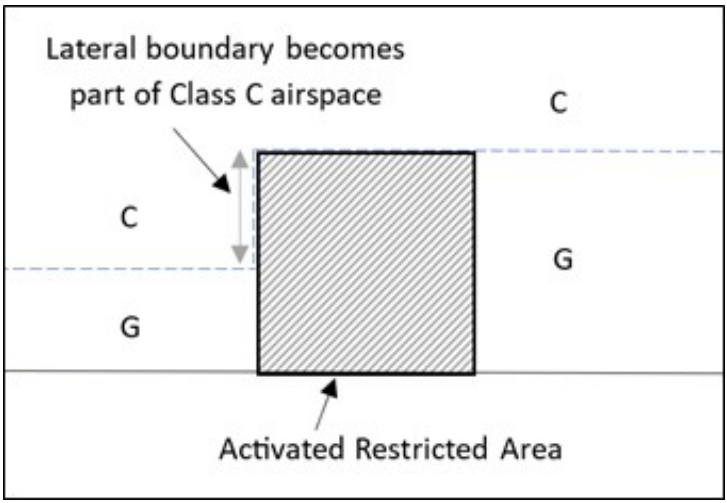


1.1.7.3 When airspaces adjoin laterally (one beside the other), flights at the common boundary must comply with, and will be provided the services applicable to, the less restrictive class of airspace.

1.1.7.4 Where a non-continuous airspace laterally adjoins controlled airspace, the common side boundary between airspaces becomes part of controlled airspace upon activation and is subject to an ATC clearance.

Airspace example:

When a Restricted Area activates adjacent to a control area step, the common boundary between the Restricted Area and Class C airspace, becomes Class C airspace (the less restrictive airspace) with a Class C service (clearance required):



1.2 Control Areas

1.2.1 A control area (CTA) is defined as “a controlled airspace extending upwards from a specified limit above the earth.” Control areas normally operate continuously. Areas identified by the word NOTAM may be activated, or additional areas may be established to meet temporary requirements, by the issuance of a NOTAM or SUP.

1.3 **Control Zone**

1.3.1 A control zone (CTR) is defined as “a controlled airspace extending upwards from the surface of the earth to a specified upper limit.” CTR surround controlled aerodromes and are designated as follows:

a. **Civil CTR**: A CTR administered by a civil air navigation service provider, other than a military CTR. Class C or Class D procedures and services apply.

b. **Military CTR**: A CTR administered by the Australian Defence Force. Class C procedures and services apply to civil flights.

1.3.2 CTR are active during the hours of the operation of the control tower as published in *ERSA* or as varied by NOTAM.

Note: MIL CTR may be activated at short notice. Pilots should plan their operations on the basis that CTR are active unless advised to the contrary.

1.4 **Airspace Reservation**

1.4.1 A designated airspace or portion thereof under the control of another authority may be reserved to allow the following:

a. flights of special military significance requiring the use of controlled airspace, which would be subject to unacceptable restrictions if normal procedures applied;

b. civil flights requiring passage through a military airspace when weather conditions or other factors make flight on the normal air route inadvisable, or impossible, and when other routes are unavailable, or the use of such routes would impose severe economic penalties on the operation of the aircraft.

1.4.2 There are two types of airspace reservations: fixed defined areas and “mobile” (e.g. aerial refuelling, en route formation flights, etc). Such reservations are normally only applied during limited periods. A designated airspace or portion thereof under the control of a Military ATC may also be reserved to confine particular activities. In such airspace, Military ATC is responsible for the separation of transiting civil or military aircraft from areas reserved or restricted for air defence operations.

1.4.3 Airspace reservations do not alter the underlying established airspace classification or the level of ATS. In controlled airspace or military airspace, the reservation provides assurance that ATC will accommodate the activity for the published duration.

1.5 **Authorisation of Civil Flights in Military Airspace**

1.5.1 Civil flights in military airspace may be authorised subject to military activity, weather and any special procedures for the control of civil aircraft which have been published.

1.5.2 Unless otherwise specified, civil aircraft operating in military airspace will receive the following level of service:

Airspace	Level of service for civil flight
Military airspace controlled/ administered by Military ATC	Approved aircraft - equivalent to Class A and/or Class C airspace, as applicable
Restricted Areas and Military Operating Areas controlled/ administered by other military units e.g. Navy or Army	No ATS *Note: ATC may provide limited FIS and SAR services in accordance with ENR 1.1 Section 2.2*
Note: Military ATC areas are identified in DAH as controlled/ administered by 452SQN or 453SQN. *See ENR 1.1 Section 2.2.4*	

1.6 **Release of Control Areas, Control Zones, Restricted or Military Operating Areas**

1.6.1 The responsibility for a Control Area or Control Zone (civil or military), either wholly or in part, may be released to another ATS unit (civil or military). The airspace remains active and a clearance is required to enter.

1.6.2 The responsibility for a Restricted or Military Operating Area, either wholly or in part, may be transferred to an ATS unit. The released airspace remains active and a clearance or approval, as appropriate, is required to enter.

1.6.3 The service provided to aircraft within released airspace will be:

a. for Restricted or Military Operating Areas - in accordance with the established airspace classification depicted on *AIP MAP or in DAH* (i.e. the airspace classification when the area is not active).

b. For control areas or control zones - in accordance with the airspace classification for the airspace when it is active.

1.6.4 ATC will advise pilots of the level of service they will receive when granting approval or clearance to enter a released area from Class E or G airspace, or if the level of service will be in accordance with Class E or G within the released area.

1.7 **Deactivation of Control Zones or Special Use Airspace**

1.7.1 The published cessation time of a control zone or SUA may be amended:

a. without issue of a NOTAM provided the new cessation time is within one hour prior to the original published cessation time, or

b. with subsequent issue of a NOTAM when the deactivation is one hour or more prior to the original published cessation time.

1.7.2 Any amendments to the published cessation time of a control zone or SUA will be notified to affected pilots in accordance with *GEN 3.3 Section 3.3*.

1.7.3 Non-controlled aerodrome procedures apply to all military towered aerodromes when the CTR is deactivated.

2. PROVISION OF SEPARATION IN CONTROLLED AIRSPACE

2.1 **General**

2.1.1 In Class A airspace, IFR flights only are permitted. All flights are provided with an ATC service and are separated from each other.

2.1.2 In Class C airspace, IFR and VFR flights are permitted. All flights are provided with an ATC service and IFR flights are separated from other IFR, Special VFR, and VFR flights. VFR flights are separated from IFR flights and receive traffic information in respect of other VFR flights. Special VFR flights are separated from other Special VFR flights when visibility is less than VMC.

2.1.3 In Class D airspace, IFR and VFR flights are permitted and all flights are provided with an ATC service. IFR flights are separated from other IFR and Special VFR flights, and receive traffic information in respect of VFR flights. VFR flights receive traffic information in respect of all other flights. Special VFR flights are separated from other Special VFR flights when visibility is less than VMC.

2.1.4 In Class E airspace, IFR and VFR flights are permitted. IFR flights are provided with an ATC service, are separated from other IFR flights, and receive traffic information on VFR flights as far as is practicable. VFR flights receive a Surveillance Information Service (SIS) on request.

2.1.5 At controlled aerodromes, ATC provides runway separation to all aircraft.

2.2 **Special Provisions**

2.2.1 Not withstanding the general provisions of *section 2.1,* the following also apply:

a. The separation of aircraft taxiing on the manoeuvring area (which does not include apron and parking areas) is a joint pilot and controller responsibility. The pilot must maintain separation while complying with clearances and instructions.

b. In the traffic circuit, pilots are required to position their aircraft in such a manner that, while complying with clearances and instructions from ATC, they maintain the necessary separation from other traffic.

c. Separation is not normally provided within a training area in controlled airspace.

d. Under certain conditions, the pilot of one aircraft may be given the responsibility for separation with other aircraft. In this circumstance:

(1) the pilot is also responsible for the provision of wake turbulence separation;

(2) the pilot must advise ATC when they are unable to maintain, or have lost, sight of the other aircraft;

(3) where an aircraft has been instructed to maintain own separation from an IFR aircraft, ATC will issue traffic information to the pilot of the IFR aircraft, including advice that responsibility for separation has been assigned to the other aircraft; and

(4) aircraft flying in formation will not be provided with separation with respect to other aircraft of the same formation, including for take-off and landing.

(5) aircraft flying as part of an in-company flight will not be provided with separation with respect to other aircraft of the same in-company flight whilst airborne. Runway separation will continue to be provided.

e. ATC will consider a formation of aircraft broken and will therefore process aircraft individually from the time the formation aircraft are:

(1) cleared to carry out touch-and-go landings;

(2) required to go around; or

(3) cleared to carry out individual activities.

Note: A group of civil aircraft conducting the same flight (e.g. an air safari), which require the aircraft to operate at separation distances greater than those specified for formation flights will be considered to be separate aircraft when applying separation.

2.3 **Performance Based Navigation (PBN) for separation**

All Australian administered airspace is designated PBN airspace, generally to facilitate the use of ICAO RNP-based aircraft separation. A specific PBN capability is not required unless a particular route or procedure specifies a particular RNP or RNAV value (e.g. RNP 10, 4, 2, 1 etc). However, an aircraft flight planning its full PBN capability may benefit from more efficient ATC separation.

2.4 **Wake Turbulence Separation**

2.4.1 ATC is not required to apply wake turbulence separation in the following situations:

a. when a MEDIUM fixed-wing aircraft of less than 25,000KG MTOW precedes a LIGHT aircraft;

b. when an aircraft is landing behind another aircraft that is taking off on the same runway;

c. subject to *para 2.4.2*, if a pilot has initiated a waiver of the relevant departure wake turbulence separation minimum;

d. when a VFR aircraft is in flight and is:

(1) operating directly behind a preceding HEAVY or MEDIUM aircraft; or

(2) landing on the same runway as a preceding HEAVY or MEDIUM aircraft; or

(3) landing on a parallel runway separated by less than 760M from the runway of a preceding HEAVY or MEDIUM aircraft;

e. when an IFR aircraft is in flight and the pilot has:

(1) reported the preceding aircraft in sight; and

(2) accepted responsibility to follow, or maintain their own separation with that aircraft.

Note 1: For paragraphs (d) and (e), the pilot in command of the aircraft is responsible for ensuring that the spacing from a preceding aircraft of a heavier wake turbulence category is acceptable. If it is determined that additional spacing is required, the flight crew may inform ATC accordingly, stating their requirements.

Note 2: Super, Heavy or Medium Wake Turbulence category aircraft may be operating near the base or boundaries of controlled airspace. Aircraft operating in Class G airspace in the vicinity of controlled airspace may be affected by wake turbulence from aircraft operating within controlled airspace.

2.4.2 **Pilot-initiated waiver of wake turbulence separation**

The pilot of a departing aircraft may ask ATC to waive the application of wake turbulence separation with a preceding aircraft. A waiver can only be requested in VMC by day. A pilot requesting the waiver must be keenly aware that this makes the requesting pilot responsible for avoiding or mitigating the effects of wake turbulence from preceding aircraft.

Note 1: ATC will not waive wake turbulence separation if the preceding aircraft is a heavy or super wake turbulence category aircraft (e.g. Airbus A330 or larger).

Note 2: More information about wake turbulence can be found in Advisory Circular (AC) 91-16, available on the CASA website.

3. CLASS G AIRSPACE

3.1 Flight Information Areas

3.1.1 Non-controlled airspace in the Brisbane FIR and Melbourne FIR is classified as Class G airspace.

3.1.2 North of 65° South, Class G airspace is divided into designated Flight Information Areas (FIAs) within which a Flight Information Service (FIS) and SAR alerting services are provided by an ATS unit.

3.1.3 On and north of 65° South, in Class G airspace, IFR and VFR flights are permitted. IFR flights receive traffic information and a flight information service. VFR flights receive a flight information service and may receive a surveillance information service if requested (ATC workload permitting).

3.1.4 South of 65° South, in Class G airspace, IFR and VFR flights are permitted and all flights receive a flight information service on request.

3.2 Broadcast Areas

3.2.1 Broadcast Areas are defined airspace volumes that form part of a Flight Information Area and have a discrete frequency (CTAF).

Note: The conditions described at paras 3.1.2 and 3.1.3 apply.

3.2.2 The vertical boundaries of a Broadcast Area are:

a. Surface to 5,000FT AMSL (default); or

b. Surface to the base of CTA if 8,500FT or less; or

c. Surface to a nominated level.

3.2.3 The lateral and vertical boundaries are defined in *AIP DAH*.

3.2.4 Broadcast Areas may be subject to mandatory broadcast requirements, these requirements are defined in the *ERSA.*

4. CLASSES OF AIRSPACE-SERVICES AND REQUIREMENTS

4.1 The following table summarises the services and requirements for the various classes of airspace used in Australian FIR.

Class	Type of Flight	Separation Provided	Service Provided	Airspace Speed Limitation	Radio COM RQMNTS	SUBJ ATC CLR
A	IFR	All aircraft	ATC service	N/A	Continuous two-way	Yes
	VFR not permitted					
C	IFR	IFR from IFR, IFR from VFR IFR from Special VFR	ATC service	N/A	Continuous two-way	Yes
	VFR	VFR from IFR	1. ATC service for separation from IFR 2. VFR/VFR traffic INFO (and traffic avoidance advice on request).	250KT IAS below 10,000FT AMSL	Continuous two-way	Yes
	Special VFR	Special VFR from Special VFR, when VIS does not meet VMC	ATC service		Continuous two-way	Yes
D	IFR	IFR from IFR IFR from Special VFR	ATC service, traffic information about VFR flights.	200KT IAS at or below 2,500FT AAL within 4NM of the primary Class D aerodrome (Note 3) 250KT IAS - in the remaining Class D airspace	Continuous two-way	Yes
	VFR	Nil	ATC service, traffic INFO on all other flights.		Continuous two-way	Yes
	Special VFR	Special VFR from Special VFR when visibility is less than VMC	ATC service		Continuous two-way	Yes

Class	Type of Flight	Separation Provided	Service Provided	Airspace Speed Limitation	Radio COM RQMNTS	SUBJ ATC CLR
E	IFR	IFR from IFR	ATC service and traffic info on VFR flights as far as is practicable	250KT IAS below 10,000FT AMSL.	Continuous two-way	Yes
	VFR	Nil	FIS SIS - flight following O/R, (ATC workload permit)	250KT IAS below 10,000FT AMSL.	Continuous two-way	No
G On & North of 65° South	IFR	Nil	FIS	250KT IAS below 10,000FT AMSL.	Continuous two-way	No
	VFR	Nil	FIS SIS - flight following O/R, (ATC workload permit)	250KT IAS below 10,000FT AMSL. 250KT IAS below 10,000FT AMSL.	VHF radio required for OPS above 5,000FT AMSL and at aerodromes where carriage and use of radio is required. VHF radio required for OPS in reduced VMC.	No No
G South of 65° South	IFR	Nil	FIS O/R	250KT IAS below 10,000FT AMSL.	Continuous two-way	No
	VFR	Nil	FIS O/R	250KT IAS below 10,000FT AMSL.	Nil	No

Note 1: Pilots must comply with airspace speed limitation unless specifically cancelled by ATC.

Note 2: Speed limitations are not applicable to military aircraft, except as specified in ERSA.

Note 3: If traffic conditions permit, ATC may approve a pilot's request to exceed the 200KT speed limit to a maximum limit of 250KT unless the pilot informs ATC a higher minimum speed is required.

Note 4: VMC minima are detailed in Section 2.07 of the Part 91 MOS.

4.2 Separation is not provided between aircraft within controlled airspace and any aircraft in close proximity but remaining outside controlled airspace.

Note: Aircraft within controlled airspace or SUA may be operating up to the lateral boundary of the airspace.

5. LANES OF ENTRY (LOE)

5.1 Lanes of entry are established to permit passage to and from specified Class D CTR without entering an adjacent civil or military controlled airspace. The vertical limits of the LOE provide separation from overlying control areas and military airspace.

5.1.1 When using these lanes, pilots must:

a. operate under the VFR;

b. conform with the general flight rules regarding terrain clearance, flight over populous areas, and low level restricted areas;

c. operate not higher than the altitude specified as the upper limit in the section being flown; and

d. keep to the right.

6. REGULATION OF FLIGHT - ASSESSMENT OF PRIORITIES

6.1 Subject to the duty to facilitate and maintain the safe, orderly and expeditious flow of air traffic, ATC will apply priorities in the following order:

a. An aircraft in an emergency, including being subjected to unlawful interference, will be given priority in all circumstances.

b. A multi-engined aircraft which has suffered the loss of an engine and has not been subject to a SAR phase, or has not been considered under the provision of *sub-para a*. above, shall be granted priority for landing.

c. An aircraft which has suffered radio communications failure will be granted priority for landing.

d. An aircraft participating in a Search and Rescue (SAR), Medical (MEDEVAC), or Fire and Flood Relief (FFR) flights shall be granted priority as necessary.

e. An aircraft operating under police call sign “POLAIR RED” or “FEDPOL RED” engaged in operations where life is at risk.

f. An aircraft engaged in the personal transport of Heads of State or of Government, or other selected dignitaries on official visits to Australia, or the personal transport of the Governor-General or the Prime Minister.

g. State aircraft special requirements flights where clearance has been prearranged.

h. Aircraft directed by Defence to participate in significant aerial displays.

i. Navaid and instrument flight procedure checks where prior arrangement has been made for aircraft engaged in these activities.

6.2 Subject to the priorities of *para 6.1*, an aircraft first able to use the manoeuvring area or desired airspace in the normal course of its operations will be given priority except:

a. an aircraft landing or taking off will be given priority over taxiing aircraft;

b. a landing aircraft will have priority over a departing aircraft if the latter cannot take-off with prescribed separation standards;

c. aircraft operating to an approach minima where a successful landing can be achieved will be prioritised over aircraft unable to make a successful approach based on the weather conditions at the time.

d. for flights in Class C terminal control areas associated with Brisbane, Melbourne, Perth and Sydney, ATC will apply priorities in the following order;

(i) long haul flights from the Americas or Europe and the United Kingdom will be offered priority where practical when holding exceeds 10 minutes;

(ii) with equal priority, flights compliant with their ATFM requirements, flights exempt from ATFM measures and Medical Aircraft (HOSP) operations; and

(iii) flights not compliant with their ATFM requirements;

(iv) all other aircraft.

Note: Further information about ATFM procedures at Australian airports is available at ENR 1.9.

e. for flights in other Class C terminal control areas, ATC will apply priorities in the following order:

(i) with equal priority flights with a Calculated Off Blocks Time (COBT), scheduled air transport operations, State aircraft (other than training flights) and Medical Aircraft (HOSP) operations; and

(ii) all other aircraft

f. RVSM-approved aircraft will be given priority for level requests between FL290 and FL410 inclusive over aircraft not RVSM-approved;

g. within ATS surveillance system coverage, identified aircraft may be given priority over non-identified aircraft;

h. inside military airspace surrounding a military aerodrome, priorities will be determined by the controlling or administrating authority published in DAH. Military aerodromes do not include Darwin or Townsville;

i. for training flights;

(i) training flights operating in the traffic pattern in general use will be given priority over other training flights desiring to operate in conflicting patterns for training purposes; and

(ii) when a training instrument approach is approved, priority will be given to that aircraft from the time it commences its final approach until the approach is completed.

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ENR 1.5 HOLDING, APPROACH AND DEPARTURE PROCEDURES

1. HOLDING AND APPROACH TO LAND PROCEDURES

1.1 General

1.1.1 Unless otherwise specified, the provisions of this section apply to aircraft operating under the IFR.

1.1.2 The arrival, holding, approach and departure procedures are developed in accordance with the criteria contained in *ICAO DOC 8168: Procedures for Air Navigation Services - Operations (PANS-OPS), ICAO DOC 9905: Required Navigation Performance Authorization Required (RNP AR) Procedure Design Manual*, and the *Part 173 MOS* and other design criteria as approved by CASA under *CASR Part 173*.

1.2 Aircraft Performance Category

1.2.1 The following categories, based upon V_{at} (except for CAT H), determine landing minima for aircraft:

CAT: A speeds up to 90KT IAS.

B speeds from 91KT to 120KT IAS.

C speeds from 121KT to 140KT IAS.

D speeds from 141KT to 165KT IAS.

E speeds from 166KT to 210KT IAS.

H (helicopters) see *section 1.3*.

Note: V_{at} is the indicated airspeed at the threshold which is equal to the stalling speed V_{so} multiplied by 1.3 or the stalling speed V_{s1g} multiplied by 1.23. Both V_{so} and V_{s1g} apply to aircraft in the landing configuration at the maximum certificated landing weight. If both V_{so} and V_{s1g} are available for an aircraft, the higher resulting V_{at} must be used.

1.2.2 An aircraft must fit into and be operated in accordance with the requirements of only one category. An aircraft:

a. may not reduce category because of reduced operating weight, but

b. must increase category when actual handling speeds are in-excess of those for category (based on V_{at}) detailed at *section 1.17*.

1.2.3 Provided an aircraft can be operated within the limits of the handling speeds (detailed at *section 1.16*) for a lower category than the category determined by V_{at}, and subject to approval by CASA under *CASR 91.320*, an aircraft type may be operated at the lower category.

1.3 **Helicopters**

1.3.1 The following criteria apply to helicopter-specific instrument approach procedures and operations:

a. the stall speed method of calculating aircraft category does not apply to helicopters;

b. where helicopters are operated similarly to aeroplanes, they may be classified as CAT A;

c. procedures developed for the specific use of helicopters are:

(1) designated “CAT H”, and

(2) promulgated on separate charts; i.e. they are not included on charts containing procedures for other aircraft categories.

1.4 **Minimum Route Altitudes**

Except when complying with the requirements for a visual approach, when conforming to a published DME or GNSS Arrival Procedure, or when identified and assigned an altitude by ATC, an aircraft approaching an aerodrome must not descend below the LSALT or the MSA for the route segment being flown (see *para 2.3*) until it has arrived over the IAF or facility.

1.5 **Procedure Entry**

Having arrived over the IAF or facility, and except as provided for in *para 2.5*, further descent must be made in accordance with the entry and holding procedures to the specified altitude for commencing the approach and, subsequently, in accordance with the approved instrument approach procedure.

1.6 **Circling Approaches and Visual Circling**

1.6.1 A circling area is defined to facilitate a circling approach which provides for visual circling of the aerodrome, to align with the runway prior to landing. The lateral boundary of the circling area determines the geographical area subject to obstacle assessment.

1.6.2 A circling approach is an extension of an instrument approach to the published circling approach minima with the intent or requirement to visually manoeuvre the aircraft to align with the runway for a landing. Each circling situation is different due to variations in runway layout, final approach track, wind velocity and meteorological conditions. Therefore, there can be no single procedure designed that will cater for the conduct of a circling approach in every situation.

1.6.3 The circling approach minima is specifically designed to ensure a safe approach and landing when conducting circling manoeuvres. Pilots should always exercise caution and undertake thorough preparation before conducting a circling approach, especially for unfamiliar locations. While spot heights on approach charts provide valuable information, they may not encompass all potential obstacles including the highest points in the relevant circling area. As such, the information provided by spot heights on instrument approach procedure charts must be treated with caution.

1.6.4 The lateral dimensions for circling area obstacle assessment areas are developed by procedure designers in accordance with the criteria contained in *ICAO DOC 8168: Procedures for Air Navigation Services – Operations (PANS-OPS) Vol II*. The circling area is determined by drawing an arc centred on the threshold of each usable runway and joining these arcs by tangents, see *Figure 1.1* for an example of obstacle assessment areas within the circling area at an aerodrome. Pilots must be fully aware of the maximum circling IAS for each aircraft category when performing a circling approach. The maximum IAS values are provided in *paragraph 1.17.1, Table 1.2*. The radius includes the following parameters:

a. the aircraft TAS at maximum IAS for circling for an aircraft performance category, calculated at 1000FT above the aerodrome elevation (therefore AD ELEV + 1000FT),

b. a tail wind of 25KT throughout the turn,

c. 20° average achieved bank angle or the bank angle producing a turn rate of 3° per second, whichever is the lesser bank angle, and

d. ISA +15°.

1.6.5 Australia has no aerodromes with an elevation below sea level and therefore the smallest circling area radii in Australia are for aerodromes at sea level. Aerodromes at higher elevations will have larger circling areas due to increased TAS, bank angles and turn radius parameters. The larger circling areas ensure adequate obstacle clearance during the visual manoeuvre, approach and landing phases. Pilots must be mindful of these factors and adjust their manoeuvres accordingly to ensure safety during flight operations in such environments. The table below (*Table 1.1*) presents example radius values for circling areas for reference only. These values should not be interpolated. Pilots are reminded to maintain visual reference with the runway environment and to remain within or as close as possible to the circuit area, ensuring the aircraft does not approach the circling area boundary:

AD ELEVATION	CIRCLING ALTITUDE	CAT A	CAT B	CAT C	CAT D
0FT	1000FT	1.67NM	2.59NM	4.11NM	5.15NM
1000FT	2000FT	1.69NM	2.65NM	4.21NM	5.28NM
2000FT	3000FT	1.70NM	2.71NM	4.31NM	5.40NM
3000FT	4000FT	1.74NM	2.77NM	4.41NM	5.54NM
4000FT	5000FT	1.77NM	2.83NM	4.52NM	5.67NM
5000FT	6000FT	1.81NM	2.90NM	4.63NM	5.82NM

Table 1.1 Example radius values for circling areas

1.6.6 Published circling approach minima include the necessary minimum obstacle clearance for conducting a circling approach at a particular aerodrome. These minima are based on various factors such as the surrounding terrain, obstacles, elevation of the aerodrome, the category of the aircraft and the relevant parameters mentioned in *paragraph 1.6.4*. Therefore, visual circling conducted at or above the published circling approach minima will provide protection from obstacles within the circling area, reduce the likelihood of Controlled Flight Into Terrain (CFIT) accidents and enhance overall flight safety.

The minimum obstacle clearance required above the highest obstacle within the circling area is listed below. These values are rounded for simplicity and may differ from those calculated by the procedure designer:
Categories A and B - 300FT
Categories C and D - 400FT

1.6.7 Restrictions on Visual Circling

1.6.7.1 The runway environment includes features such as the runway threshold or approach lighting aids or other markings identifiable with the runway. Refer to *Figure 1.2* for the recommended typical procedure for executing the visual circling circuit manoeuvre for landing on the opposite runway.

1.6.7.2 After establishing visual contact, the pilot should manoeuvre the aircraft to remain within the circling area and in such way that visual contact with the runway of intended landing or runway environment is maintained at all times. During visual circling, descent below the promulgated circling approach minima should not be made until:

a. required visual reference has been established and can be maintained throughout the manoeuvre;

b. the pilot has the landing threshold in sight; and

c. the required obstacle clearance can be maintained and the aircraft is in a position to carry out a landing using normal rates of descent and angles of bank.

1.6.7.3 Once descent is initiated below the circling approach minima, the obstacle protection offered by visual circling at the circling minima ends and pilots are responsible for ensuring the required clearance from obstacles is maintained visually.

1.6.7.4 Where a prominent obstacle or obstacles within the circling area prevent visual circling the sector in which the obstacles are located may be eliminated from the visual circling area. Sectors which have been eliminated from the visual circling area are annotated No Circling (*see Figure 1.3*).

1.6.7.5 Visual circling is prohibited in No Circling sectors.

1.6.7.6 Obstacle clearance is assured in those cases where the published final approach track overflies the no circling area while tracking to the MAPt and/or the MAPt is inside the No Circling area, provided the published tracks and altitudes are being complied with. Deviating from the published procedure can compromise obstacle clearance and safety during the circling approach.

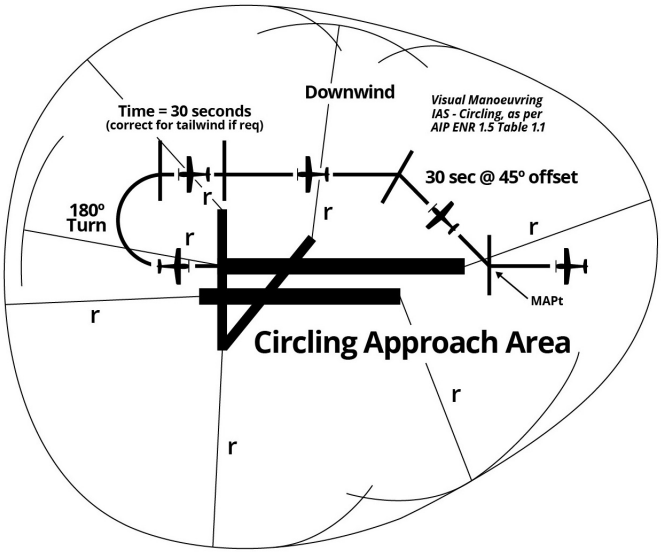


Figure 1.1 - Example Visual Circling Obstacle Assessment Area

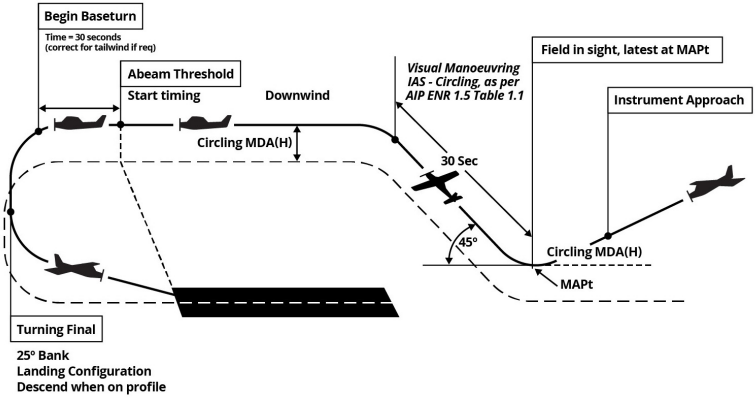


Figure 1.2 - Recommended Typical Visual Circling Circuit Manoeuvre

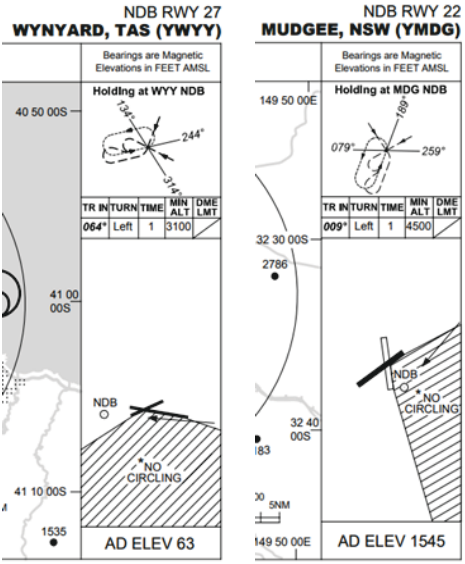


Figure 1.3 - Examples of No Circling Areas

1.7 Visual Manoeuvring (non-Circling) Subsequent to Non-Precision Approaches (NPA) and Approaches with Vertical Guidance (APV)

1.7.1 Straight-in NPA and APV do not normally require visual circling. In those circumstances where the NPA or APV does not serve the landing runway, the provisions of *para 1.6* apply.

1.7.2 **Descent Below the Straight-in MDA.** Descent below the straight-in MDA or continuation of the approach below the DA during APVs, may only occur when:

- visual reference can be maintained;
- all elements of the meteorological minima are equal to or greater than those published for the aircraft performance category (see *para 5.1.1*); and
- the aircraft is continuously in a position from which a descent to a landing on the intended runway can be made at a normal rate of descent using normal flight manoeuvres that will allow touchdown to occur within the touchdown zone of the runway of intended landing.

1.7.3 **NPA and APV Alignment.** APV are aligned with the runway centreline. Straight-in NPA may be aligned with the runway centreline or may be offset by up to 15°(Category C & D) or 30° (Category A & B) see Note 1.

1.7.4 **Alignment with the Runway Centreline.** Manoeuvring to align the aircraft with the runway centreline can be undertaken when:

- within the circling area,
- visual reference can be maintained,
- continuously in sight of ground or water.

Note 1: Procedures with offset angles greater than 5° are designed such that aircraft cross the runway centreline no closer than 1,400M to the threshold.

For offset angles equal to or less than 5°, the final approach track is designed to be within 150 metres of the runway centreline at 1,400M. Some older procedures may use 900M in place of 1,400M.

Note 2: For the purpose of this section 'visual reference' means the runway threshold, or approach lights or other markings identifiable with the landing runway clearly visible to the pilot and a flight visibility not less than that specified for the procedure.

1.8 Visual Approach Area - Helicopter

1.8.1 Helicopter GNSS instrument approach procedures may include a Visual Approach Area–Helicopter (VAA–H). These procedures are annotated in the minima box with the term 'VAA'. The VAA-H extends from the Missed Approach Waypoint (MAWP) to the Helicopter Landing Site (HLS). Obstacle clearance at MDA is assured within a VAA–H.

1.8.2 The VAA–H comprises a 1NM wide corridor centred on the track from the MAWP to the HLS, plus the area beyond the HLS contained within a 0.5NM radius centred on the HLS.

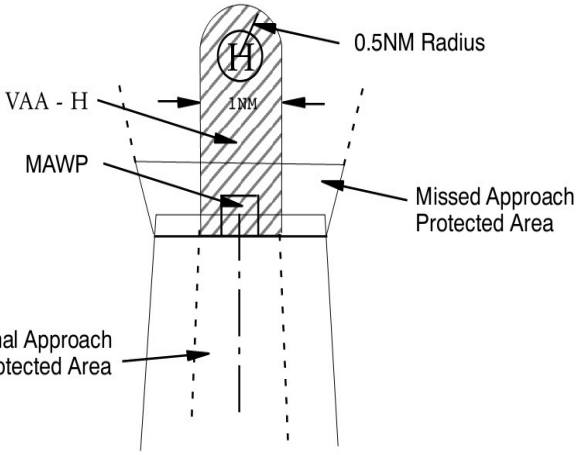


Figure 1.4 Visual Approach Area - Helicopter

1.8.3 Flight within a VAA–H is a visual flight manoeuvre. After visual contact is established, and after passing the MAWP, the helicopter is manoeuvred within the VAA–H, at an altitude not below the MDA, utilising key lead-in points until the HLS is sighted.

1.8.4 Descent below MDA may only occur when the pilot:

a. maintains the helicopter within the VAA–H; and

b. maintains visibility along the intended flight path:

(1) by day, not less than that specified for helicopter VMC, and

(2) at night, not less than the published minimum visibility for the procedure; and

c. maintains visual contact with key lead-in points (i.e. lighting or other prominent identifiable features) or the HLS; and

d. while complying with a, b, and c, intercepts a normal approach path to the HLS for the particular helicopter and a landing is assured.

D

1.9 Missed Approach - Standard Procedures

1.9.1 A missed approach must be executed if:

a. during the final segment of an instrument approach, the aircraft is not maintained within the applicable navigation tolerance for the aid in use; or

b. during an instrument approach and below MSA (as specified on the IAL chart) the performance of the radio aid becomes suspect, or the radio aid fails; or

c. visual reference is not established at or before reaching the MAPT or DA/RA Height from which the missed approach procedure commences; or

d. a landing cannot be effected from a runway approach, unless a circling approach can be conducted in weather conditions equal to or better than those specified for circling; or

e. visual reference is lost while circling to land from an instrument approach.

Note 1: For the purpose of this paragraph "visual reference" means the runway threshold, or approach lights or other markings identifiable with the landing runway clearly visible to the pilot, and either:

a. for circling approaches, clear of cloud, in sight of the ground or water and with a flight visibility not less than the minimum specified for circling; or

b. for runway approaches, a flight visibility not less than that specified for the procedure.

Note 2: In IAL procedures, the missed approach is designed to provide a minimum obstacle clearance of 100FT to an aircraft climbing along the specified missed approach path at a gradient of 2.5% (152FT/NM) from the MAPT or DA/RA Height from which the missed approach procedure commences. If this missed approach climb gradient cannot be achieved the DA, MDA or RA Height should be increased, or other action taken to achieve the required obstacle clearance along the specified missed approach flight path.

1.9.2 In executing a missed approach, pilots must follow the missed approach procedure specified for the instrument approach flown. In the event that a missed approach is initiated prior to arriving at the MAPT, pilots must fly the aircraft to the MAPT and then follow the missed approach procedure. The MAPT in a procedure may be:

a. the point of intersection of an electronic glide path with the applicable DA; or

b. a navigation facility; or

c. a fix; or

d. a specified distance from the Final Approach Fix (FAF).

1.9.3 When a missed approach is required from visual circling, the expectation is that the pilot will make an initial climbing turn toward the landing runway and overhead the aerodrome where the pilot will establish the aircraft climbing on the missed approach track. In as much as the circling manoeuvre may be accomplished in more than one direction, different patterns will be required to establish the aircraft on the prescribed missed approach course depending on its position at the time visual reference is lost.

1.10 Missed Approach Tracking

1.10.1 A missed approach procedure may, or may not, specify lateral guidance. In either situation the expectations of the pilot will vary as follows depending whether the procedure is based on a radio navaid or GNSS.

a. When lateral guidance is specified with reference to a radio navaid (i.e. a VOR radial, an NDB bearing) the expectation is that the pilot will intercept the nominated track. Where an intercept is required it will be both stated in the missed approach procedure's text and shown in the plan view on the procedure plate. The text will take the form of: *"At the NDB (or VOR), Turn Left (or Right) to intercept xxx° (ZZ NDB or VOR). Climb to..."*

b. When lateral guidance is specified based on GNSS the expectation of the pilot will take one of two forms:

- For a straight missed approach or a turn at the MAPt, the pilot is expected to follow the GNSS navigation commands to the next waypoint. The text will take the form of: *'Turn Left (or Right), Track DCT XXXXX. Climb to...'*
- For a turn after the MAPt where there is not a subsequent waypoint the pilot is expected to make-good the nominated track on the chart using the GNSS for navigation. The text will take the form of: *'Turn Left (or Right), Track xxx°. Climb to...'*

c. When the instrument procedure is based on a radio navaid but the missed approach does not specify lateral guidance the expectation is that the pilot will use DR to achieve the nominated track. Allowance for wind must be made to make-good this nominated track. The radio navaid may be used to supplement track keeping during the missed approach when it is a straight continuation of the final track, however guidance is not mandatory. The missed approach procedure's text will take the form of: *'Turn Left (or Right), Track xxx°. Climb to...'*

1.11 **Missed Approach Requirements - GNSS**

1.11.1 If a loss of RAIM or RAIM warning is indicated at any time after passing the Initial Approach Fix, the pilot must immediately carry out a missed approach in accordance with published procedures.

1.11.2 Provided the RAIM warning ceases when the missed approach is selected on the GNSS equipment, it may be used for missed approach guidance.

1.11.3 Should the RAIM warning remain when the missed approach is selected, or should there be any doubt about the accuracy of the GNSS, then an alternative means of guidance or dead reckoning must be used to fly the missed approach.

1.12 **Missed Approach - Helicopter Procedures**

1.12.1 Pilots flying a helicopter instrument approach procedure, or flying visually within a VAA–H, must execute a missed approach if:

a. during the instrument approach and below MSA (as specified on the IAL chart) the performance of the navigation aid becomes suspect, or the navigation aid fails; or

b. visual reference is not established at or before reaching the MAWP from which the published missed approach procedure commences; or

c. visual reference is lost within the VAA–H; or

d. a landing at the HLS is not assured.

Note 1: For the purpose of this paragraph "visual reference" means:

a. the key lead-in points or HLS are clearly visible to the pilot; and

b. clear of cloud, in sight of ground or water and with a flight visibility:

(1) by day, not less than that specified for Helicopter VMC, and

(2) at night, not less than the published minimum visibility for the procedure.

Note 2: The missed approach is designed to provide a minimum obstacle clearance of 100FT to a helicopter climbing at a gradient of 4.2% (255FT/NM) from the MDA at, or before, the MAWP or from any point within the VAA–H, to the Missed Approach Turning Waypoint (MATWP) or Missed Approach Holding Waypoint (MAHWP), as applicable. If this missed approach climb gradient cannot be achieved, the MDA should be increased, or other action taken, to achieve the required obstacle clearance along the missed approach flight path.

1.12.2 If executing a missed approach from within the VAA–H of a helicopter GNSS approach, pilots must immediately track towards the MATWP or the MAHWP, as required by the particular procedure.

1.13 **Visual Segments**

When an instrument approach procedure specifies a visual segment from the point where the MDA is reached to the circling area of the aerodrome, a missed approach shall be executed unless the visual segment can be flown clear of cloud and in sight of the ground or water in accordance with the altitude and visibility specified for circling.

1.14 **Visual approach requirements for all flights**

1.14.1 A pilot conducting a visual approach in controlled airspace may be assigned the responsibility to follow another arriving aircraft which they have reported sighting. When assigned this responsibility, the pilot must maintain separation from and not overtake that aircraft. In this circumstance, the pilot is also responsible for providing their own wake turbulence separation. If sighting is subsequently lost, advise ATC immediately.

1.14.2 A pilot who is unable to continue a visual approach which has been authorised by ATC must immediately advise ATC.

1.14.3 When conducting a visual approach in controlled airspace, a pilot in command must not climb above an altitude reported to ATC as having been reached or left, unless authorised to do so.

1.15 Visual approach requirements for IFR flights

1.15.1 The requirements of this section are the visual approach procedures applicable to IFR flights under *CASR 91.305(3)(b)(i)*.

1.15.2 A pilot in command operating under the IFR in controlled airspace must be satisfied that the visual approach requirements of *para 1.15.3* can be met before requesting a visual approach from ATC. The pilot must report 'VISUAL' to signify that these requirements can be met and maintained as part of any request to ATC for a visual approach.

1.15.3 Subject to the requirements of *paras 1.6, 1.9 and 1.13*, the pilot need not commence or may discontinue the approved instrument approach procedure to that aerodrome when:

a. **By Day.** Within 30NM of that aerodrome at an altitude not below the LSALT/MSA for the route segment, the appropriate step of the DME or GNSS Arrival Procedure, or the MDA for the procedure being flown, the aircraft is established;

(1) clear of cloud;

(2) in sight of ground or water;

(3) with a flight visibility not less than 5,000M or, in the case of a helicopter, is able to proceed under helicopter VMC, or the aerodrome is in sight; and

(4) subsequently can maintain (1), (2) and (3) at an altitude not less than

(i) if in controlled airspace – 500FT above the lower limit of the CTA unless a clearance is received from ATC to depart and re-enter controlled airspace during the descent; and

(ii) the minimum height prescribed by *CASR 91.265* or *91.267* as relevant to the location of the aircraft.

b. **By Night.** At an altitude not below the LSALT/MSA for the route segment, the appropriate step of the DME or GNSS Arrival Procedure, or the MDA for the procedure being flown, the aircraft is established:

(1) clear of cloud;

(2) in sight of ground or water;

(3) with a flight visibility not less than 5,000M; and

(4) subsequently can maintain (1), (2) and (3) at an altitude not less than:

(i) in controlled airspace – 500FT above the lower limit of the CTA unless a clearance is received from ATC to depart and re-enter controlled airspace during descent; and

(ii) one of the following:

– route segment LSALT/MSA; or

– the appropriate step of the DME/GNSS Arrival Procedure; or

– if being vectored – the last assigned altitude;

(5) until the aircraft is:

– for an aerodrome with an authorised instrument approach procedure that the flight crew members of the aircraft are capable of using – within the prescribed circling area for the category of aircraft or a higher category, where the limitations of the higher category are complied with, or VAA–H, as applicable and the aerodrome is in sight; or

– for an aerodrome without an authorised instrument approach procedure that the flight crew members of the aircraft are capable of using – within 3NM of the aerodrome reference point, and the aerodrome is in sight; or

– within 5NM (7NM for a runway equipped with an ILS/GLS) of the aerodrome, aligned with the runway centreline and established not below “on slope” on the T-VASIS or PAPI; or

– within 10NM (14NM for Runways 16L and 34L at Sydney) of the aerodrome, established not below the ILS/GLS glide path with less than full scale azimuth deflection.

Note: Reference to circling area in this section includes the circling area for the category of aircraft or a higher category where the limitations of the higher category are complied with.

c. If in controlled airspace:

(1) a clearance is received from ATC to conduct a visual approach; and

(2) when tracking via a STAR and subsequently cleared for visual approach, the pilot continues to follow the lateral profile of the STAR, including any visual or instrument termination route; and

(3) except when on a STAR, the pilot maintains track/heading on the route progressively authorised by ATC until:

(i) by day, within 5NM of the aerodrome; or

(ii) by night, the aerodrome is in sight and the aircraft is within the prescribed circling area for an IFR flight;

Note: ATC will provide directions to the aircraft regarding how to join the circuit for the nominated runway from these positions.

1.16 **Visual approach requirements for VFR flights**

1.16.1 If in controlled airspace and a clearance is received from ATC to conduct a visual approach, the pilot in command of an aircraft operating under the VFR must:

a. maintain a level 500FT above the lower limit of CTA unless a clearance is received from ATC to depart and re-enter controlled airspace;

b. maintain track/heading on the route progressively authorised by ATC until

(i) by day, within 5NM of the aerodrome; or

(ii) by night, the aerodrome is in sight and the aircraft is within 3NM of the aerodrome reference point.

Note: ATC will provide directions to the aircraft regarding how to join the circuit for the nominated runway from these positions.

1.17 Handling Speeds

1.17.1 The handling speeds for aircraft categories during IAL procedures are as follows:

SPEEDS FOR PROCEDURE IN KNOTS IAS					
ACFT CAT	**V_{at}**	**Range of Speeds for Initial and Intermediate Approach**	**Range of Final Approach Speeds**	**Max Speeds for Visual Manoeuvring (Circling)**	**Max Speeds for Missed Approach**
A	< 91	90 – 150 (110*)	70 – 100	100	110
B	91 – 120	120 – 180 (140*)	85 – 130	135	150
C	121 – 140	160 – 240	115 –160	180	240
D	141 – 165	185 – 250	130 – 185	205	265
E	166 – 210	185 – 250	155 – 230	240	275
H	N/A	70 – 120	60 – 90	N/A	90
* Max speed for reversal procedures					

Table 1.2

Note 1: On reversal procedures (see section 2.8) for which a FAF is not published, final approach speed should be obtained before descending on the inbound track.

Note 2: Speed reduction below the initial segment speed range is permitted to enable the final approach speed to be achieved prior to the commencement of the final segment.

1.18 Speed Restrictions

1.18.1 IAL charts may have speed restrictions identified on the chart when:

a. handling speeds, below the maximum values specified in *Table 1.2* above, are required to contain the aircraft within the procedure design area; or

b. the speeds are used for ATC flow management (ATC APCH speeds).

1.18.2 Where ATC APCH speeds do not align with the IAL procedure handling speeds, either at *Table 1.2* or published on the chart, pilots must comply with the IAL procedure handling speeds. Notification to ATC is not required.

1.18.3 The following may amend the ATC APCH speed depicted on a STAR or IAL chart without approval or notification to ATC:

a. Performance Category B aircraft may cross 5NM from THR at 145-160KT; or

b. Performance Category A and H aircraft may fly lower speeds if required.

1.19 **Obstacle Clearance Altitude (OCA)**

1.19.1 Obstacle clearance altitude is:

a. in a precision approach procedure, the lowest altitude at which a missed approach must be initiated to ensure compliance with the appropriate obstacle clearance criteria; or

b. in a non-precision runway approach procedure, the lowest altitude below which the aircraft can not descend without infringing the appropriate obstacle clearance criteria; or

c. in a visual (circling) procedure, the lowest altitude above the aerodrome elevation in accordance with obstacle clearance criteria.

1.20 **Aerodrome Operating Minima (AOM)**

1.20.1 Landing minima are published on Australian approach charts as MDA/H or DA/H. Obstacle Clearance Altitude/Height is not published. Landing minima are the basis for determining AOM.

1.20.2 Operators must establish AOM for each aerodrome to be used for operations. After consideration of the factors listed below, operators may determine that their AOM should be higher than the published landing minima:

a. The type, performance and handling characteristics of the aeroplane.

b. The composition, experience and competence of the flight crew.

c. The means used to determine and report meteorological conditions.

1.20.3 **Partial Runway Lighting Failure**. At a controlled aerodrome, in the event of failure of one electrical circuit on a runway equipped with interleaved circuitry lighting, pilots will be notified of a doubled spacing of runway edge lights; i.e. from 60M to 120M spacing. When such a failure occurs at night, pilots must apply the following requirements to an approach to land:

a. In VMC:

No restriction.

b. In Less Than VMC:

The prevailing visibility must be equal to, or greater than, the published minimum for the instrument approach procedure being used for an aircraft's arrival multiplied by a factor of 1.5.

1.21 **Descent Gradients**

1.21.1 Procedures are designed with the following descent gradients:

SEGMENT	GRADIENT	
	NORMAL	**MAXIMUM**
Arrival	As required	As required
Initial	4%	8%
Intermediate	Level	5%
Final: non-precision	5.2%	6.5%
precision	3°	Not applicable

Table 1.3

Note 1: The chart will indicate when other than a normal gradient is used in the final segment.

Note 2: For procedures published with a distance/altitude scale, a 3° glideslope is used in calculating the descent data.

1.21.2 Aircraft may commence a segment in excess of the specified commencement altitude provided that any upper altitude limitation is observed. However, rate of descent after the FAF should not normally exceed 1,000FT per minute.

1.22 **Descent**

1.22.1 For a straight approach (no reversal procedure), the aircraft must:

a. for a radio navaid-based approach, cross the fix or facility, or

b. for an area navigation-based approach, pass the waypoint, and when established on the specified track, descend to not below the specified altitude.

1.22.2 For an approach which incorporates a reversal procedure, if an outbound descent is specified, the descent to the specified altitude may be commenced after the aircraft has crossed the fix or facility and is established on the specified track or has turned to a heading to intercept the specified outbound track. The reversal procedure must be completed, again descending to any lower altitude specified. Further descent, after the reversal procedure, must not be started until established on the inbound track. For approaches without a FAF, the final segment commences at the completion of the reversal procedure.

Note: "Established" means being within half full scale deflection for the ILS, VOR and GNSS, within ±5° of the required bearing for the NDB, or within ±2NM of the DME arc.

1.23 **Wind Effect**

1.23.1 For all flights, during any holding or IAPs, allowance should be made in heading and timing to compensate for the effects of wind. Full use should be made of indications available from the aid and estimated or known winds.

1.24 **Bank Angle**

1.24.1 Procedures are based on a bank angle of 25°, or a bank angle which will produce a Rate One turn, whichever is less.

2. APPROACH PROCEDURES

2.1 Operations On Parallel Runways At Class C Aerodromes

2.1.1 Parallel runways may be used for the following simultaneous operations:

a. dependent parallel approaches;

b. independent parallel approaches;

c. dependent visual approaches;

d. independent visual approaches;

e. segregated parallel operations; or

f. Simultaneous Opposite Direction Parallel Runway Operations (SODPROPS).

2.1.2 Refer to the relevant section of *AIP DAP* for location-specific procedures and guidance.

2.2 Use of Navigation Aids

2.2.1 Instrument approach procedures are based on specific navigation aids, with the applicable navigation tolerances associated with the aids being used in the development of the procedure's obstacle protection surfaces. The navigation aid, or aids, upon which the procedure is based is/are identified on each instrument approach chart. Only the navigation aid, or aids, included in the chart title or identified on the instrument approach chart as suitable may be used to fly the procedure. Use of a non-specified aid (e.g. another DME located on the aerodrome) is prohibited as it may jeopardise the integrity of the instrument approach procedure.

2.3 Minimum Sector Altitude

2.3.1 The 25NM and 10NM MSA provide 1,000FT obstacle clearance. An aircraft within the applicable 25NM Sector MSA or 10NM MSA of the nominated significant point, the ARP or the HRP may use the applicable MSA, and deviation from the track being flown is permitted to facilitate entry to the instrument approach. In instances where the 25NM MSA has been divided into sectors, and the appropriate 25NM Sector MSA is lower than the 10NM MSA (as a result of the 10NM MSA not being sectorised), then the 25NM Sector MSA may be used for tracking to the nominated significant point, ARP or HRP provided aircraft tracking can be maintained within the sector.

2.4 Approach Design Concept

2.4.1 **Definitions.** “Segment Minimum Safe Altitude” and “Procedure Altitude” are defined at *GEN 2.2 - Definitions and Abbreviations.*

2.4.2 Approach procedures are designed to facilitate descent from a Procedure Altitude to an altitude from which either a straight-in landing or a circling procedure can be conducted. Approach procedures are classified as PA, APV or NPA. NPA fall into two categories: those with distance measuring information (e.g. VOR/DME, LOC/DME, GNSS) and those without (e.g. NDB and VOR).

2.4.3 **Vertical Profile - NPA <u>with</u> Distance Measuring.** NPAs with distance measurement are designed to provide a Constant Descent Final Approach (CDFA) path from the procedure altitude to a point 50FT above the threshold (or to the circling altitude for circling-only procedures). The CDFA path is shown on the profile diagram with the descent angle annotated in degrees and an altitude/distance scale. At each fix on approach, an advisory crossing altitude is shown on the profile diagram to assist in maintaining the descent path.

Each segment of an NPA also specifies a Segment Minimum Safe Altitude identified by shading on the profile diagram. In conducting a CDFA, the pilot should follow the descent profile but must always ensure that the aircraft remains at or above each Segment Minimum Safe Altitude. Descent below the CDFA profile to the Segment Minimum Safe Altitude (sometimes called “dive and drive”) is permitted at pilot’s discretion, but is not recommended.

2.4.4 **NPA as a 2D or 3D operation.** An NPA procedure flown using the charted altitude/distance scale to determine the aircraft's rate of descent is considered a 2D instrument approach operation. Where the advisory vertical guidance is calculated and provided by on-board navigation equipment, the approach can be flown as a 3D instrument approach operation. Advisory vertical guidance is generated by aircraft navigation systems to assist pilots in managing vertical navigation but provides no assurance of compliance with the descent limitations specified for the NPA procedure. Accordingly, it is vital that pilots conform with Segment Minimum Safe Altitude and MDA requirements regardless of any advisory vertical guidance information provided by the aircraft's system. NPA procedures using advisory vertical guidance will not be charted as VNAV capable.

When flying an NPA as a 3D operation (e.g. CDFA), pilots should initiate any missed approach at an altitude above the MDA to ensure the aircraft does not descend below the published MDA.

2.4.5 **Vertical Profile - NPA <u>Without</u> Distance Measuring.** As these types of procedures are time based, a 3° profile cannot be published. Accordingly, a Procedure Altitude will be published to establish the top of descent at the beginning of the inbound leg. This altitude will provide the necessary obstacle clearance in addition to keeping the outbound and inbound rates of descent within the PANS-OPS limits.

2.5 **Procedure Entry**

2.5.1 An aircraft which is not required to hold or to lose height in a holding pattern may commence the approach without entering the holding pattern if:

a. in controlled airspace, ATC has cleared the aircraft for the approach;

b. in any airspace, for procedures using navaids:

(1) the reversal procedure entry requirements of *para 2.8* are satisfied; or

(2) the DME arc entry requirements of *para 2.9* are satisfied; or

(3) the en route track to the procedure's commencement fix or facility is within 30° either side of the first track of the procedure.

c. for procedures using GNSS:

(1) in any airspace, the aircraft is tracking to an initial approach waypoint from within the capture region (see *Figure 2.1 and Figure 2.2)* for that waypoint; or

(2) in controlled airspace, the aircraft is being vectored to intercept the initial approach segment or is tracking direct to the intermediate fix.

Note to c.(1): The first track of a GNSS procedure must be joined using the tracking guidance provided by the GNSS receiver.

Note to c.(2): "direct to" clearances may be requested to the intermediate fix (IF) provided that the resultant track change at the IF does not exceed 45°.

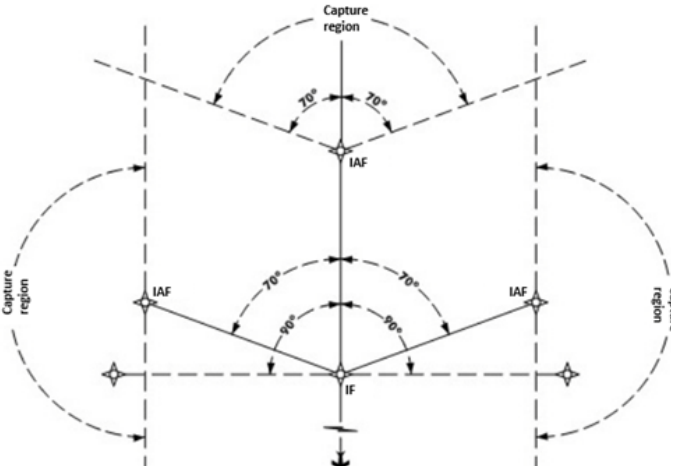


Figure 2.1 - RNP APCH Capture Regions (Three Initial Approach Fixes)

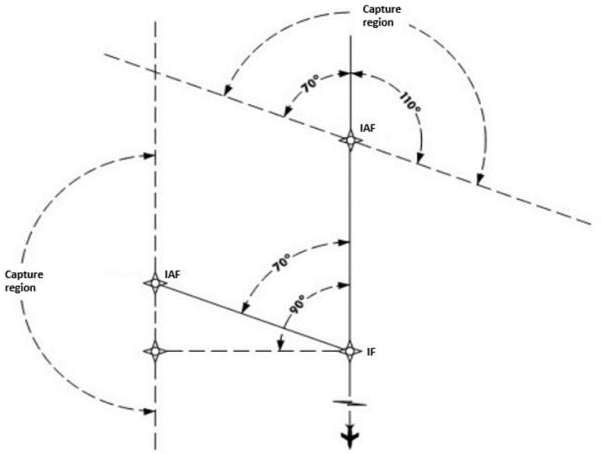


Figure 2.2 - RNP APCH Capture Regions (Two Initial Approach Fixes)

2.6 **Approach Segments and Fixes**

2.6.1 Fixes associated with segments occur at the beginning of their segment. The initial, final and missed approach fixes are shown. In addition, an area for circling the aerodrome in visual conditions is considered.

2.6.2 **Segment Parameters.** Segment parameters are defined in *ICAO PANS OPS.* Parameters significant to flight handling are as follows:

SEGMENT	PARAMETER	VALUE/COMMENT
Arrival	**As for en route flight**	**Nil**
Initial	Descent Gradient and Rate	See *para 1.21*
Intermediate	Bank Angle	See *para 1.24*
Final	Speed Navigation Aid Tracking	See *Table 1.2* See *para 1.22*
Missed Approach	Nominal Path Gradient	2.5%
	Average Achieved Bank Angle	15°
	Speed	See *Table 1.2*

Table 2.1

2.7 **Missed Approach Procedure**

2.7.1 The MAPT in a procedure may be:

a. the point of intersection of an electronic glide path with the applicable DA/RA Height; or

b. a navigational facility; or

c. a fix; or

d. a specified distance from the FAF, or

e. a waypoint.

Note: If the MAPT is defined by distance, a distance/groundspeed/time table will be provided to enable pilots to establish the MAPT by DR from the FAF.

2.7.2 If on reaching the MAPT, the required visual reference is not established, the pilot must immediately initiate the published missed approach procedure or, where applicable, comply with alternative ATC instructions. The phrase 'or as directed by ATC' is included in published missed approach instructions for applicable instrument approach procedures.

2.7.3 A published missed approach procedure must not be flown unless commenced at the MAPT. If a missed approach climb is initiated before the MAPT, the aircraft must track to the MAPT before commencing the missed approach procedure.

2.8 **Reversal Procedures**

2.8.1 **General.** Reversal procedures are used to establish the aircraft inbound on an intermediate or final approach track at the desired altitude. A reversal procedure consists of an outbound track followed by a turning manoeuvre in order to reverse direction onto the inbound track. The procedure can be a procedure turn or a base turn.

2.8.2 **Types.** Reversal procedures are illustrated at *Figure 2.3* and described below:

a. **Procedure Turn** (45°/180°), consisting of a specified outbound track and timing from the facility or fix, a 45° turn away from the outbound track for 1 minute from the start of turn for categories A and B aircraft (1 minute 15 seconds from the start of turn for categories C, D and E aircraft), followed by a 180° turn in the opposite direction to intercept the inbound track [see *Figure 2.3* (a)]. The 45°/180° procedure turn is an alternative to the 80°/260° procedure turn [paragraph (b) below] unless specifically excluded.

Note: Some instrument approach procedures require a procedure turn after passing over a navigation aid or fix. Where this requirement exists, the turn must be initiated immediately after passing over the navigation aid or fix.

b. **Procedure Turn** (80°/260°), consisting of a specified outbound track and timing from the facility or fix, an 80° turn away from the outbound track, followed by a turn of 260° in the opposite direction to intercept the inbound track [see *Figure 2.3* (b)].The 80°/260° procedure turn is an alternative to the 45°/180° procedure turn unless specifically excluded.

Note: Some instrument approach procedures require a procedure turn after passing over a navigation aid or fix. Where this requirement exists, the turn must be initiated immediately after passing over the navigation aid or fix

c. **Base Turn**, consisting of a specified outbound track and timing or DME distance from a facility, followed by a turn to intercept the inbound track (see *Figure 2.3* (c)). The outbound track and/or time may be different for differing aircraft performance categories.

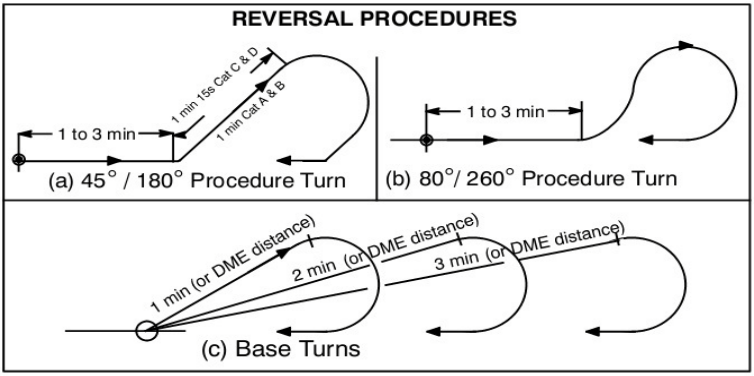


Figure 2.3

2.8.3 **Entry.** Reversal procedures must be entered from a track within ±30° of the outbound track of the reversal procedure (see Figure 2.4). However, for base turns, where the ±30° direct entry sector does not include the reciprocal of the inbound track, the entry sector is expanded to include it (see *Figure 2.5*). Where entry is required from tracks outside these limits, manoeuvring to establish the aircraft onto the outbound track must be in accordance with the entry procedures associated with the holding pattern

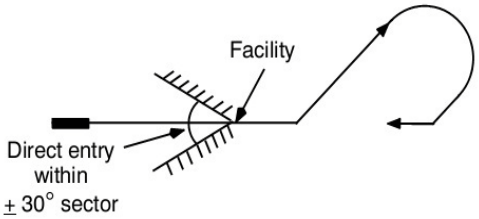


Figure 2.4 - Direct Entry to Procedure Turn

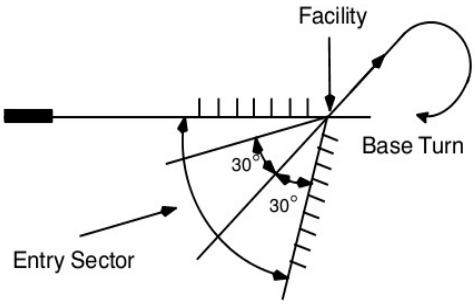


Figure 2.5 - Direct Entry to Base Turn

2.9 **DME Arcs**

2.9.1 DME arcs must be joined at or before an IAF and at an altitude not below the relevant MSA or the appropriate sector DME Arrival step.

3. HOLDING PROCEDURES

3.1 **General**

3.1.1 When holding is required in a specified pattern the procedures set out in this section must be used.

3.1.2 Shape and terminology associated with a standard holding pattern are given in *Figure 3.1*.

3.1.3 Right turns holding patterns are standard holding patterns and must be flown unless the AIP chart depicts, or ATC directs, otherwise.

3.1.4 Some area navigation systems are unable to fly non-area navigation holding patterns with strict compliance with the *PANS-OPS, Volume II,* assumptions. These systems may not be used operationally unless the operator has obtained approval in writing that CASA is satisfied that the area navigation system commands will contain the aircraft within the basic holding area defined by *PANS-OPS, Volume II*, for the environmental conditions assumed by those criteria. Where approval has been given, the pilot must verify over flight of the stipulated fixes by means of the reference facility.

3.2 **Holding in controlled airspace**

3.2.1 Pilots instructed to hold by ATC must hold at the designated location until further cleared.

3.2.2 ATC will normally assign aircraft estimated to arrive first over a holding fix, or first able to commence an approach, the lowest available level for assignment.

3.2.3 Where a delay of six minutes or more is expected, ATC will advise an expected approach time or expected landing time.

3.2.4 When operationally necessary, a pilot holding must advise ATC of the latest divert time.

3.2.5 When an aircraft is holding because weather conditions are worse than the prescribed landing minima, ATC will nominate scheduled reporting times, normally at 15 minute intervals.

3.2.6 At the time or position advised, the pilot must depart from the hold. A pilot should leave the holding fix on time, or up to one (1) minute ahead of time, and unless identified, report leaving the holding fix.

3.3 **Limitations**

3.3.1 Unless otherwise specified, holding procedures are subject to the following limitations:

a. **Speed.** Indicated speed must not exceed

(1) up to and including FL140

– 230KT, or

– 170KT for holding where the approach is limited to Cat A and B aircraft only;

(2) above FL140 up to and including FL200, 240KT; and

(3) above FL200, 265KT.

Note: Above the highest MSA in turbulent conditions, speeds may be increased to the lesser of 280KT or M0.8 subject to ATC approval in CTA.

b. **Outbound timing** begins abeam the fix or on attaining the outbound heading, whichever comes later.

c. **Time/Distance outbound.** The outbound leg must be no longer than:

(1) up to and including FL 140 – 1 minute or the time or distance limit specified on the chart;

(2) above FL 140 – 1.5 minutes or the time or distance limit specified on the chart.

d. **Turns.** All turns in nil wind should be at a bank angle of 25° or Rate One, whichever requires the lesser bank.

e. **Wind allowance.** Allowance should be made in heading and timing to compensate for the effects of wind to ensure the inbound track is regained before passing the holding fix inbound. Full use should be made of indications available from the aid and estimated or known winds.

f. **Exiting.** For ATC traffic management, jet aircraft in CTA must leave an en route holding pattern at 250KT IAS, unless otherwise published or advised by ATC. Pilots may request a variation to this requirement.

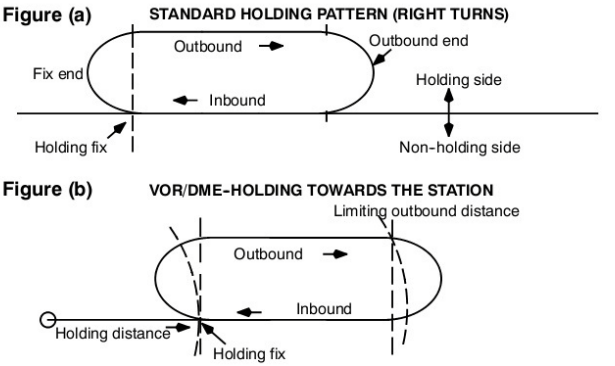


Figure 3.1 - Shape and Terminology Associated With Right Turn Holding Patterns.

3.4 **Entry Into the Holding Pattern**

3.4.1 The entry into the holding pattern must be according to heading in relation to the three entry sectors shown in *Figure 3.2 a and b,* recognising a zone of flexibility of 5° on either side of the sector boundaries. For holding on a VOR intersection, the entry track is limited to the radials forming the intersection. For holding on a VOR/DME fix the entry track is limited to either the VOR radial, DME arc or alternatively along the entry radial to a VOR/DME fix at the end of the outbound leg, as published.

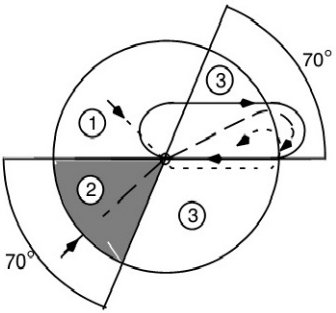


Figure 3.2 a Right Turn Holding Procedure

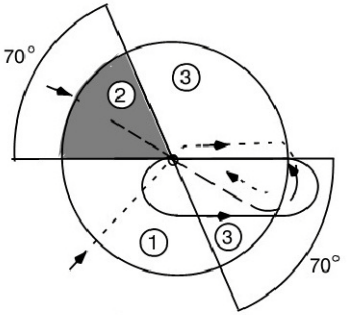


Figure 3.2 b Left Turn Holding Procedure

3.4.2 **Entry from Sector 1 (Parallel entry):**

a. On reaching the holding fix, the aircraft is turned onto an outbound heading (to track parallel with the inbound track) for the appropriate period of time (taken from over or abeam the holding fix whichever is later), or until reaching the limiting DME distance if earlier; then

b. the aircraft is turned onto the holding side to intercept the inbound track or to return to the fix; and then

c. on the second arrival over the holding fix, the aircraft is turned to follow the holding pattern.

3.4.3 **Entry from Sector 2 (Offset entry):**

a. On reaching the holding fix, the aircraft is turned onto a heading to make good a track making an angle of 30° from the reciprocal of the inbound track on the holding side; then

b. flown outbound:

(1) for the appropriate period of time from the holding fix, where timing is specified, up to a maximum of 1 minute 30 seconds; or, if earlier,

(2) until the appropriate limiting DME distance is attained, where distance is specified; then

c. the aircraft is turned in the direction of the holding pattern to intercept the inbound holding track; then

d. on second arrival over the holding fix, the aircraft is turned to follow the holding pattern.

3.4.4 **Entry from Sector 3 (Direct entry).** On reaching the holding fix, the aircraft is turned to follow the holding pattern. Outbound timing begins abeam the fix or, when the abeam position cannot be determined, from completion of the outbound turn.

3.4.5 **DME Arc Entry.** Having reached the fix, the aircraft must enter the holding pattern in accordance with either the Sector 1 or Sector 3 entry procedure.

3.5 **Standard Holding Pattern**

3.5.1 When flying the standard holding pattern, an aircraft must:

a. follow the prescribed track inbound to the holding fix;

b. execute a 180° turn in the direction specified, so as to fly outbound a track parallel to the inbound track;

c. continue outbound to the earlier of the time, or the DME limit specified; and

d. execute a 180° turn to realign the aircraft on the inbound track.

3.6 **DME limit**

3.6.1 The “DME Limit”, where prescribed for holding patterns, is the DME distance at which the outbound leg of the holding pattern must be terminated and the turn to the reciprocal track commenced.

3.7 **Shortening**

3.7.1 The pilot may shorten the holding pattern to leave the holding fix at a specified time. For prolonged holding at a level not limited by obstacles, the length of the pattern may be increased, subject to ATC approval where appropriate.

3.8 **Descent in Holding Pattern**

3.8.1 Subject to ATC approval, where appropriate, aircraft may descend as required.

4. AERODROME METEOROLOGICAL MINIMA

4.1 **Ceiling and Visibility Minima**

4.1.1 The ceiling and visibility minima prescribed in this part are the meteorological conditions under which an aircraft may take-off or land at an aerodrome. The meteorological conditions for a particular aerodrome are below the minima for the aerodrome when, in the airspace encompassing the intended flight path:

a. the total cloud amount below the ceiling minimum specified is continuously greater than SCT; or

b. the visibility is continuously below the visibility specified.

Note: MDA equals ceiling minimum plus the elevation of the aerodrome.

4.2 **Runway Visual Range and Runway Visibility**

4.2.1 In Australia, Runway Visual Range (RVR) observations are based solely on the information provided by instrumented systems such as transmissometers. RVR observations representative of the touchdown, midpoint and rollout/stop end zones are automatically displayed in the local ATC unit. At locations where RVR information is accessible to the Bureau of Meteorology, the RVR is included in METAR and SPECI reports.

4.2.2 At places not equipped with RVR sensors or where one or more RVR sensors are unserviceable, a Runway Visibility (RV) assessment may be provided instead. An RV assessment is a report on the visibility in the touchdown and midpoint zones of a runway, and is assessed by a ground observer counting visible runway lights or visibility markers.

4.2.3 An RV assessment is NOT a substitute for a required RVR observation and CANNOT be used:

a. for SA CAT I, SA CAT II, CAT II and CAT III precision approaches, or

b. for CAT I approaches when the visibility is less than 800M, or

c. for low visibility take-offs where the visibility is less than 350M.

An RV assessment is a subset of a general visibility observation and is intended to provide visibility information specific to a particular runway; which may be more useful to a pilot than the overall ground visibility.

4.2.4 Pilots will be notified by ATIS broadcast or directed transmission if RVR/RV is not available when visibility is less than 800M.

4.2.5 See *GEN 3.4 para 6.9 Meteorological Information* for the relevant RVR/RV phraseologies.

4.3 **IFR Take-off and Landing Minima**

4.3.1 See *Chapter 15* of the *Part 91 MOS* for IFR take-off and landing minima. Specifically:

– section 15.02 for definition of qualifying multi-engine aeroplane

– section 15.03 for take-off minima requirements

– section 15.04 for take-off minima for low-visibility operations

– section 15.05 for take-off minima for qualifying multi-engine aeroplanes

– section 15.06 for take-off minima for other aeroplanes

– section 15.07 for take-off minima for qualifying multi-engine rotorcraft

– section 15.08 for take-off minima for other rotorcraft

– section 15.09 for landing minima requirements

– section 15.10 for landing minima.

4.4 Correction of Instrument Procedure Minima for non-standard temperatures

Pressure altimeters are calibrated to indicate true altitude under ISA standard conditions. Any deviation from ISA will result in an erroneous altimeter reading. In cold conditions the true altitude will be lower than the indicated altitude and will reduce the obstacle clearance margins incorporated into instrument procedures. Published landing minima do not make any allowance for non-standard temperatures at the QNH source (usually the aerodrome of destination).

This effect, and various methods to address it, is discussed in *ICAO Doc 8168 (PANS-OPS), Vol III, Section 2, Chapter 4*. A correction must be added to the published MDA or DA and procedure altitudes when the temperature at the aerodrome of landing is less than ISA -15° C. Altitude corrections can be determined by charts *DAP 2-2* and 2-3 and the worked example at *DAP 1-1, para 1.5* as detailed in the *Departure and Approach Procedures (DAP)* publications.

5. APPLICATION OF AERODROME METEOROLOGICAL MINIMA

5.1 Pilot Responsibilities

5.1.1 Prior to take-off and when an aircraft reaches the DA, MDA or RA Height, the pilot in command is responsible for assessing whether the meteorological conditions are equal to or better than the minimum prescribed for take-off or landing as applicable. A pilot must not take-off or, except in an emergency, land or continue an approach below the prescribed DA, MDA or RA Height for the approach being conducted when any element of the prescribed meteorological criteria is continuously less than the minima for the aircraft performance category (*CASR 91.307*).

5.2 ATC Assessment

5.2.1 Whilst the decision to operate is solely that of the pilot in command, ATC will provide the pilot in command with an assessment of ceiling and/or visibility as follows:

a. **Take-off.** Ceiling and visibility will be assessed in the airspace enclosing the expected path of the aircraft during take-off and initial climb.

b. **Landing.** Ceiling and visibility will be assessed in the airspace enclosing the expected final approach path and runway to be used.

5.3 **QNH Sources**

5.3.1 In accordance with *subsection 14.03(1)* of the *Part 91 MOS* prior to passing the IAF, pilots are required to set either:

a. the actual aerodrome QNH from an approved source, or

b. the forecast Aerodrome (TAF) QNH, or

c. the forecast area QNH

5.3.2 Where instrument approach charts are identified by a shaded background to either the minima titles for IAL charts or the published minima for DME or GNSS Arrival Procedures, landing, circling and alternate minima have been calculated assuming the use of Aerodrome Forecast (TAF) QNH. These minima may be reduced by 100FT whenever an actual aerodrome QNH is set. Approved sources of actual QNH are ATC and ATIS except when the aerodrome forecast QNH is provided, AWIS and Bureau of Meteorology (BoM) accredited meteorological observers. An actual aerodrome QNH obtained from an approved source is valid for a period of 15 minutes from the time of receipt.

Note: METAR QNH does not meet this requirement.

5.3.3 When the actual aerodrome QNH is not available, ATC will report the Aerodrome Forecast (TAF) QNH on the ATIS. The ATIS will include information in the format “ACTUAL QNH NOT AVAILABLE, AERODROME FORECAST QNH...”

Note: Forecast QNH reported by ATC or on the ATIS is not an approved source of actual QNH.

5.3.4 Where the forecast area QNH is used, the minima used must be increased by 50FT.

6. ALTERNATE WEATHER MINIMA

6.1 **IFR Flights**

6.1.1 Each approach chart shows the ceiling and visibility minima to be compared with the meteorological forecasts and reports to determine both the need to provide for an alternate aerodrome and the suitability of an aerodrome as an alternate.

Note: Requirements for an aerodrome without an instrument approach procedure are detailed at ENR 1.1 sub-para 10.7.2.10c.

6.2 **Special Alternate Weather Minima**

6.2.1 Special alternate weather minima are available for specified approaches at some aerodromes for use by aircraft with dual ILS/VOR approach capability. Dual ILS/VOR approach capability must include:

a. duplicated LOC, and

b. duplicated GP, and

c. duplicated VOR; and

d. either:

(i) duplicated DME, or

(ii) duplicated GNSS; or

(iii) single DME and single GNSS.

6.2.2 Special alternate weather minima are identified on applicable instrument approach charts by a double asterisk adjacent to the ALTERNATE title and a note detailing the special minima.These special alternate minima will not be available (minima will revert to the standard alternate minima) during periods when:

a. local METAR/SPECI or forecasting services are not available; or

b. an aerodrome control service is not provided.

The non-availability of MET or ATS services will be notified by NOTAM.

6.2.3 Where:

a. there is a protracted unserviceability (i.e. more than seven days) of any one VHF approach aid, or

b. facilities required for the conduct of a VHF-based instrument approach and landing are unserviceable or not available,

Airservices Australia will, if necessary, advise the non-availability of, or any revision to, special alternate minima by NOTAM.

7. PRECISION APPROACH OPERATIONS

7.1 General

7.1.1 Precision approach operations involve the use of either ILS or GLS facilities.

7.1.2 An ILS supports all types of precision approach operations. The ground facilities comprise localiser equipment, glide path equipment and marker beacons, usually supported by an NDB or dedicated DME.

7.1.3 A GLS currently supports precision approach operations with minima as low as CAT I, but with the future potential for supporting CAT II and III operations. A GLS consists of a GBAS ground station located on or in the vicinity of one or more aerodromes and an aircraft subsystem. The GBAS provides data and corrections for the GNSS ranging signals over a digital VHF data broadcast to the aircraft subsystem. The aircraft subsystem translates the position signal into flight guidance similar to that provided for an ILS.

ILS Caution:

1. False courses may exist or course reversals may occur outside the sector 35° (or 20° at certain aerodromes specified in ERSA) either side of an ILS localiser course.

2. Back beam radiation of a ILS LOC can be received and displayed on aircraft navigation instrumentation. Pilots should be alert to this possibility at locations providing ILS/LOC approaches on reciprocal runways.

3. A severe and sudden pitch-up upset can occur in cases when the aircraft:

a. intercepts an ILS glidepath from above; or

b. during an ILS approach, deviates significantly above the normal glidepath angle.

Caution should be exercised in such situations particularly for autopilot coupled approaches. See AIC 14/14.

7.2 Failures

a. **NDB.** In the event of failure of an associated NDB, aircraft must join the ILS outside the outer marker as directed by NOTAM or ATC.

b. **Glide path.** For ILS operations where the glide path fails, only the localiser procedure is available.

c. **Markers.** Where marker beacons are not available, aircraft may use the ILS if the alternate fixes nominated on the IAL chart or by NOTAM are used for altimeter checks.

d. **GBAS.** If GBAS fails, GLS approaches are not available.

7.3 Altimeter Checks and Flight Tolerances

7.3.1 The final approach segment contains a fix at which the glide path/altimeter relationship should be verified. If the check indicates an unexplained discrepancy, the ILS/GLS approach should be discontinued. Pilots must conform to the following flight tolerances:

a. To ensure obstacle clearance, both LOC/GLS final approach course and glideslope should be maintained within half scale deflection (or equivalent on expanded scale).

b. If, at any time during the approach after the FAP, the LOC/GLS final approach or glideslope indicates full scale deflection a missed approach should be commenced.

7.4 Protection of GLS Critical and Sensitive Areas

7.4.1 There are no GLS critical and sensitive areas.

Note: A CAT I GLS is not required to support autoland operations. Pilots are responsible for obtaining information necessary to make operational decisions to conduct a GLS autoland.

8. STANDARD INSTRUMENT DEPARTURES

8.1 General

8.1.1 The pilot must advise ATC if cleared via a SID which requires the use of navigation aids not available to the aircraft.

8.1.2 SID procedures assume that pilots will not compensate for wind effects when being radar vectored, but will compensate for known or estimated wind effects when flying departure routes which are expressed as tracks.

8.1.3 SID procedures may be flown by aircraft already airborne provided that, before commencing a SID, the pilot visually positions the aircraft over the runway centre line so that all tracking and altitude restrictions can be met.

8.1.4 Each SID procedure specifies the minimum design climb gradient that ensures obstacle clearance. Where the initial required climb gradient exceeds 3.3%, the altitude at which a 3.3% climb gradient may be flown is also shown. A gradient shown in brackets specifies the climb gradient required to remain inside controlled airspace.

8.1.5 For aircraft on a radar SID, ATC will assign a departure heading (or track - *see para 8.1.6*) to be flown after the initial take-off phase. The pilot is not to commence the take-off without having obtained the assigned departure heading or track and should advise ATC if the heading or track is unacceptable.

8.1.6 Instead of assigning a departure heading, ATC may instruct an aircraft to track the extended runway centre line for departure - if the first and only track on a radar SID is aligned with the runway bearing.

8.1.7 The climb gradient shown on a radar SID chart provides obstacle clearance up to the MSA/LSALT. If a SID chart has multiple climb gradient sectors, ATC will not issue heading instructions to an airborne aircraft that would require the pilot to adopt a higher climb gradient than the gradient specified for the initial departure heading.

8.1.8 When the aircraft is above the MVA, any subsequent changes of headings are ATC vectors and ATC will issue instructions that ensure prescribed obstacle clearance will exist at all times.

8.1.9 The climb gradient requirements of a radar SID cease when the aircraft reaches the MSA/LSALT as applicable.

8.2 **SID Procedures**

8.2.1 Unless explicitly cancelled or amended by ATC, the pilot must follow the vertical and lateral profile of the SID and comply with any published speed restrictions.

8.2.2 The use of a SID designator without a cleared level does not authorise the pilot to climb on the SID vertical profile.

8.2.3 A level restriction depicted on a SID chart does not authorise a pilot to climb to meet that restriction. ATC will assign climb to permit compliance with vertical navigation restrictions. Pilots must inform ATC if a level restriction cannot be met.

8.2.4 ATC level change instructions to aircraft on a SID will indicate if published level and/or speed restrictions are to be followed or are cancelled.

8.2.5 When conducting a SID, the priority is to meet the vertical navigation restrictions of the SID. When speed restrictions do not enable the aircraft to meet a SID level restriction, the pilot must advise ATC of any speed deviation requirement at ACD stage or as soon as the situation is identified. Pilots must advise ATC when able to resume the SID speed restrictions.

8.2.6 For ATC traffic management: unless varied by ATC, DAP or ERSA, at or before 3,000FT AGL or at the completion of a noise abatement procedure, jet aircraft departing Class C aerodromes must:

a. commence acceleration to 250KT IAS; and

b. maintain 250KT until leaving 10,000FT AMSL

The pilot must advise ATC, preferably at ACD stage, if the aircraft will be unable to comply.

8.2.7 Cancellation of 'published speed restrictions' cancels all speeds published on the SID chart. Cancellation of 'ATC-issued speed control instructions' cancels any speed control instructions issued by ATC. Airspace speed limitations must be complied with unless specifically cancelled.

8.2.8 When a departing aircraft is cleared to proceed direct to a published waypoint on the SID, the speed and level restrictions associated with the bypassed waypoints are cancelled. The pilot must comply with any published SID speed and level restrictions, at and after the waypoint where the SID is rejoined. An aircraft cleared to bypass one or more waypoints on a SID will not receive a specific instruction to rejoin the SID.

8.2.9 When a departing aircraft is vectored or cleared to proceed away from the SID, all the published speed and level restrictions of the SID are cancelled. ATC will notify the pilot if there is an expectation the aircraft will subsequently rejoin the SID.

Note: Unless specifically cancelled by ATC, any ATC traffic management speed specified in ERSA will apply to aircraft when vectored or cleared away from a SID.

8.2.10 ATC instructions to rejoin a SID will specify any transition restrictions that must be complied with up to, but not including the waypoint where the SID is rejoined. The pilot must comply with any published SID speed and level restrictions, at and after the waypoint where the SID is rejoined.

8.2.11 In a surveillance environment prior to take-off, ATC may cancel a procedural SID and:

a. issue a radar SID; or

b. require the aircraft to depart on runway track using the climb gradient specified in the cancelled SID. In this case, ATC will use the phrase “CANCEL SID, TRACK EXTENDED CENTRE LINE *(three digits)* DEGREES”.

Note: For the application of this procedure, the runway and radar SID tracks must be coincident up to the MVA.

8.2.12 In VMC by day, the pilot may request, or ATC may offer a visual departure.

8.2.13 When a departure report is required during a SID, the SID designator must be included in the report.

8.2.14 For a radar SID, the direction of turn and assigned heading must be advised in the airborne report.

9. NOISE ABATEMENT PROCEDURES

9.1 Application

9.1.1 Noise Abatement Procedures (NAP) normally apply to all jet-propelled aircraft and other aircraft having a MTOW exceeding 5,700KG.

Note: A subsonic jet-propelled aircraft will not be permitted to operate in Australia unless it meets the requirement of ICAO ANNEX 16, VOL 1, Chapter 3.

9.1.2 Where noise abatement procedures are prescribed, and ATC traffic management permits, the runway nomination provisions of DAP NAP will be applied. Not withstanding this, noise abatement will not be a determining factor in runway selection under the following circumstances (unless required by Noise Abatement legislation):

a. in conditions of low cloud, thunderstorms and/or poor visibility;

b. for runway conditions that are completely dry:

(1) when the crosswind component, including gusts, exceeds 20KT;

(2) when the tailwind component, including gusts, exceeds 5KT;

c. for runway conditions that are not completely dry:

(1) when the crosswind component, including gusts, exceeds 20KT;

(2) when there is a tailwind component;

d. when wind shear has been reported;

e. when, in the opinion of the pilot in command, safety would be prejudiced by runway conditions or any other operational consideration.

9.1.3 Preferred flight paths for arriving and departing aircraft have been determined for particular locations. For departing aircraft they may be in the form of a SID. Arriving aircraft must not make approaches to land below the visual or electronic glide paths for the runway in use. The requirement to follow the noise abatement flight paths shall be subject to a specific ATC clearance or instruction, and may be varied by ATC for operational reasons, e.g. weather, traffic complexity.

9.1.4 Aircraft operating outside tower hours of operation (at locations which do not have continuous tower services) must comply with relevant noise abatement procedures only where they do not conflict with circuit direction requirements detailed in the *ERSA* entry for that location.

9.1.5 Noise abatement departure procedures will be developed by the operator for each aeroplane type in accordance with the requirements of *ICAO Procedures for Air Navigation Services - Aircraft Operations (PANS-OPS) Vol. I, Part V, Chapter 3*, and are subject to approval by CASA.

9.1.6 Noise abatement departure procedures must be used by jet-propelled aircraft from the locations and runways identified under the NAPs published in *DAP East and West*. The departure procedure to be used on a specific departure should satisfy the noise abatement objectives of the aerodrome operator in alleviating noise either close to the aerodrome or distant from the aerodrome. Examples of such procedures are given in *PANS - OPS Vol. I, Part 1, Section 7, Chapter 3 (NADP 1 and NADP 2*).

Note 1: NADP 1 and NADP 2 are EXAMPLES only. The actual procedures developed by the operator for a specific aircraft type may vary from these examples provided the minimum requirements of the procedures are met.

Note 2: The power settings to be used subsequent to the failure or shutdown of an engine or any other apparent loss of performance, at any stage in the take-off or noise abatement climb, are at the discretion of the pilot in command, and noise abatement considerations no longer apply.

9.1.7 As an alternative to the procedures detailed in *para 9.1.6*, operators of aircraft which have engines with a by-pass ratio greater than 3.5 may use the procedure detailed below:

a. climb at V_2 +10KT to V_2 +20KT – or body angle limit speed; and

b. maintain take-off power to a height above the aerodrome of 1,000FT:

c. then maintaining a positive rate of climb, accelerate to zero flap minimum safe manoeuvring speed (V_{ZF}) retracting flap on schedule;

d. then reduce to normal climb power/thrust; and

Note: For aeroplanes with slow flap retraction, reduce power/ thrust at an intermediate flap setting.

e. continue climb at not greater than V_{ZF} +10KT to a height above the aerodrome of 3,000FT:

f. accelerate smoothly to en route climb speed; and

g. maintain runway heading unless required to do otherwise in accordance with a SID or specific ATC instruction.

9.2 **Curfews**

9.2.1 There are curfews on some operations at Adelaide, Gold Coast, Essendon and Sydney airports. For details, see *DAP East/West NAP* for those airports.

9.3 **Flying Considerately**

9.3.1 General aviation pilots and operators are encouraged to consider opportunities to minimise aircraft noise and other impacts on local communities where operationally appropriate and consistent with safety requirements.

9.3.2 The *Flying Considerately Guidance* provides a single national reference for voluntary good practices that pilots and operators can adopt to minimise the impact of general aviation activity on local communities.

9.3.3 The *Flying Considerately Guidance* is available from the Department of Infrastructure, Transport, Regional Development, Communications, Sport and the Arts website at: https://www.infrastructure.gov.au/sites/default/files/documents/flying-considerately-guidance.pdf

10. STANDARD INSTRUMENT ARRIVALS (STAR)

10.1 General

10.1.1 The pilot must advise ATC if cleared via a STAR which requires the use of navigation aids not available to the aircraft.

10.1.2 When a STAR includes more than one instrument termination procedure, pilots must plan to fly the procedure listed first on the chart, for that runway. If the listed termination procedure is not available, e.g. the ILS is not available, pilots must plan for the next listed procedure.

10.1.3 An operational requirement or pilot request for an alternative instrument termination procedure should be made prior to the STAR being issued.

10.1.4 Unless the pilot requests an alternative approach, flights that have included PBN/T1 in Field 18 of the flight notification form will normally be issued a STAR with an RNP AR termination (where published) or an expectation of an RNP AR approach.

Note: At some locations traffic complexity may prevent allocation.

10.1.5 When a clearance for the termination procedure is authorised e.g. visual approach, the published STAR speed restrictions still apply unless specifically cancelled.

10.2 STAR Procedures

10.2.1 Unless explicitly cancelled or amended by ATC, the pilot must follow the vertical and lateral profile of the STAR and comply with any published speed restrictions.

10.2.2 The use of a STAR designator without a cleared level does not authorise the pilot to descend on the STAR vertical profile.

10.2.3 A level restriction depicted on a STAR chart does not authorise a pilot to descend to meet that restriction. ATC will assign descent to permit compliance with vertical navigation restrictions. Pilots must inform ATC if a level restriction cannot be met.

10.2.4 ATC level change instructions to aircraft on a STAR will indicate if published level and/or speed restrictions are to be followed or are cancelled.

10.2.5 Cancellation of 'published speed restrictions' cancels all speeds published on the STAR chart. Cancellation of 'ATC-issued speed control instruction' cancels any speed control instructions issued by ATC. Airspace speed limitation must be complied with unless specifically cancelled.

10.2.6 When an arriving aircraft is cleared to proceed direct to a published waypoint on the STAR, the speed and level restrictions associated with the bypassed waypoints are cancelled. The pilot must comply with any published STAR speed and level restrictions at and after the waypoint where the STAR is rejoined. An aircraft cleared to bypass one or more waypoints on a STAR will not receive a specific instruction to rejoin the STAR.

10.2.7 When an arriving aircraft is vectored or cleared to proceed away from the STAR, all the published speed and level restriction of the STAR are cancelled. ATC will notify the pilot if there is an expectation the aircraft will subsequently rejoin the STAR.

Note: Unless specifically cancelled by ATC, any ATC traffic management speed specified in ERSA will apply to aircraft when vectored or cleared away from a STAR.

10.2.8 ATC instructions to rejoin a STAR will specify any transition restrictions that must be complied with up to, but not including the waypoint where the STAR is rejoined. The pilot must comply with any published STAR speed and level restrictions, at and after the waypoint where the STAR is rejoined.

10.2.9 Following holding, pilots can expect to continue the previously issued STAR. ATC will indicate if published level and/or speed restrictions are to be followed or are cancelled.

10.2.10 Where a STAR incorporates circuit legs to a runway, pilots of aircraft not equipped with a flight management system may accept the STAR clearance and request vectors when contacting Approach Control.

11. DME OR GNSS ARRIVAL PROCEDURES

11.1 General

11.1.1 The DME or GNSS Arrival Procedure is an instrument approach procedure that provides descent guidance along a specified track or sector, to the visual circling area of an aerodrome. Azimuth guidance is required from the specified radio navigation aid. The requirements of *subsections 1.6, 1.9 and 1.13* apply.

11.1.2 Descent is not permitted until the aircraft is established within the appropriate sector or on the specified inbound track.

11.1.3 If manoeuvring within a sector is required, the pilot must ensure that the aircraft is contained within the sector, at or above the appropriate segment minimum safe altitude. Manoeuvring within a sector after passing the final approach fix is prohibited.

11.2 **Operations in Controlled Airspace**

11.2.1 The clearance "CLEARED DME (or GNSS) ARRIVAL" constitutes a clearance for final approach and authorises an aircraft to descend to the minimum altitude specified in the appropriate DME or GNSS Arrival Procedure. ATC is not permitted to impose any altitude restriction on such a clearance.

11.2.2 When cleared for a DME or GNSS Arrival in controlled airspace an aircraft must not orbit, enter a holding pattern, or use holding pattern entry procedures. ATC will not issue a clearance for a DME or GNSS Arrival that involves the use of a holding pattern entry procedure.

11.2.3 When ATC cannot issue a clearance for an unrestricted DME or GNSS Arrival, the phrase "DESCEND TO (level) NOT BELOW DME (or GNSS) STEPS" may be used. Such an instruction authorises descent in accordance with the DME or GNSS steps only to the specified altitude.

11.2.4 ATC may clear an aircraft to intercept the final approach segment of another instrument approach procedure. When clearing an aircraft for such a procedure, ATC will use the phrase "DESCEND TO (level) NOT BELOW DME (or GNSS) STEPS" and will issue further instructions prior to the aircraft's reaching the cleared level.

11.2.5 Nothing in these procedures absolves the pilot in command from their responsibilities to maintain the aircraft on the authorised track or within the defined sector.

Note 1: Where the track being flown is not aligned with the landing runway, a clearance for a DME or GNSS Arrival includes a clearance to manoeuvre within the circling area to position the aircraft on final for landing.

Note 2: Where possible, DME and GNSS Arrival Procedures are designed to contain the aircraft within controlled airspace and provide 500FT separation from the CTA lower limit. However, there are locations where the procedure commences in Class G airspace, or which can take aircraft into Class G airspace on descent. Pilots should check procedures to ensure that aircraft are contained in CTA where required.

12. SIGNALS FOR THE CONTROL OF AERODROME TRAFFIC

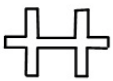
12.1 Light Signals to Aircraft

Light Signal	Meaning in Flight	Meaning on Aerodrome
Steady Green	Authorised to land if pilot satisfied no collision risk exists.	Authorised to take-off if pilot satisfied no collision risk exists.
Steady Red	Give way to other aircraft and continue circling.	Stop.
Green Flashes	Return for landing.	Authorised to taxi if pilot satisfied that no collision risk exists.
Red Flashes	Aerodrome unsafe - do **not** land.	Taxi clear of landing area in use.
White Flashes	No Significance	Return to starting point on Aerodrome.

12.2 **Light Signals to Vehicles and Pedestrians**

Light Signal	**Meaning**
Green Flashes	Permission to cross landing area or to move onto taxiway.
Steady Red	Stop.
Red Flashes	Move off the landing area or taxiway and watch out for aircraft.
White Flashes	Vacate the manoeuvring area in accordance with local instructions.
Note: In emergency conditions or if the above signals are not observed, the following meaning will be indicated by the use of the runway or taxiway lighting:	
Flashing runway or taxiway lighting	Vacate the runway or taxiway and observe the tower for light signal.

12.3 **Ground Signals to Aircraft**

GROUND SIGNAL	DESCRIPTION	WHERE DISPLAYED	MEANING
	Horizontal white dumb-bell	Adjacent to wind direction indicator.	Use only hard surface movement areas. Where there are sealed and gravel manoeuvring areas, use only the sealed surfaces. Where there are constructed gravel and natural surface manoeuvring areas, use only the gravel surfaces. (see *ERSA FAC* for any local information relating to the dumb-bell signal).
	White Cross	(i) Adjacent to wind direction indicator. (ii) On manoeuvring area.	(i) Aerodrome completely unserviceable. (ii) An area marked by a cross or crosses with the limit delineated by markers is unfit for use by aircraft.
	White Double Cross.	Adjacent to wind direction indicator	Gliding operations in progress.

ENR 1.6 ATS SURVEILLANCE SERVICES AND PROCEDURES

1. RADIO COMMUNICATIONS PROCEDURES

1.1 Pilots requesting an ATS surveillance service should address their request to the ATS unit with which they are communicating.

1.2 Where an Area Approach Control Centre (AACC) is not established, the pilot will be advised the time or place to transfer to a control frequency.

1.3 Where an AACC is established, procedural and ATS surveillance services may be provided on a common frequency. The call sign identifies the service being provided – e.g. ...CENTRE... APPROACH...DEPARTURES.

2. IDENTIFICATION PROCEDURES

2.1 Before providing an ATS surveillance service there will be positive identification of the aircraft concerned. However, control services will not be provided until the aircraft is within controlled airspace.

3. VECTORING PROCEDURES

3.1 On receipt of heading instructions the pilot must, unless otherwise instructed, immediately commence a rate 1 turn, or the standard rate of turn for the aircraft type, and then maintain the heading given.

3.2 Aircraft will normally be vectored on routes along which the pilot can monitor navigation.

3.3 ATC are not permitted to vector Special VFR flights, unless warranted by emergency conditions.

3.4 When an aircraft is given a vector which will take it off an established route, the pilot will be advised of the reason for the vector, unless it is self-evident.

3.5 When an aircraft reports unreliable directional instruments, the pilot will be requested, prior to the issuance of manoeuvring instructions, to make all turns at an agreed rate and to carry out the instructions immediately on receipt.

3.6 When aircraft are being vectored, the controller will assign altitudes which allow for terrain clearance. However, in VMC by day, an aircraft may be permitted to arrange its own terrain clearance. In such instances, the aircraft will be instructed to [TURN LEFT (*or* RIGHT) HEADING (*heading*)] [CLIMB (or DESCEND) TO (level)] VISUAL.

3.7 Pilots being vectored will be routinely advised of their position to enable pilot navigation in the event of communication or ATS surveillance system failure.

3.8 The interval between ATC transmissions will be kept short to enable the pilot to quickly recognise a communication failure. When aircraft are on headings that could infringe terrain clearance or separation standards, the interval between transmissions will not exceed 30 seconds.

3.9 Before take-off, ATC may assign a heading for a departing aircraft to assume after take-off, followed by frequency change instructions if appropriate. Headings, other than those assigned for a radar SID, will only be issued for a visual departure by day in VMC.

3.10 Arriving aircraft may be vectored to:

a. establish for a radar or pilot-interpreted approach;

b. a position from which a visual approach can be made;

c. avoid areas of hazardous weather or severe turbulence;

d. expedite traffic flow or conform to noise abatement requirements.

3.11 For a pilot-interpreted approach, aircraft will be vectored to be established on final track at least two (2) miles prior to commencement of final approach. The final intercept heading will normally intercept the final approach track at an angle of 45 ° or less. When an aircraft is vectored for a shortened instrument approach, the final approach point is the interception of the prescribed descent profile.

3.12 Should the aircraft have to be vectored through the final approach track, the controller will advise the pilot.

3.13 When the aircraft is provided with the vector to intercept final for a pilot-interpreted approach, the pilot will be:

a. advised of range from the aerodrome, or position reference the final approach point;

b. informed that the vector is to intercept the approach aid;

c. provided with a clearance for the approach, when such a clearance has been authorised; and

d. instructed to report when established on the final approach track.

Note: When ILS/GLS is used for final approach, pilots should report when established on the localiser or final approach course and not delay this report until the glide path is intercepted.

3.14 When the pilot reports established on final, they shall be instructed when to transfer to the tower frequency.

3.15 Unless otherwise instructed, the pilot in command should change automatically to tower frequency, provided that:

a. the aircraft is established on the final approach track and has been cleared for final approach; and

b. air-ground communications congestion or failure has not allowed the pilot to report ESTABLISHED, or obtain a clearance to transfer to tower; and

c. transfer to tower shall not be prior to 4NM from touchdown.

3.16 A vectoring service will not normally be terminated until the aircraft is established within the navigation tolerance of its cleared route, except on transfer to tower. However, subject to identification, a vectoring service will be continued, if requested.

3.17 When a vectoring service is terminated, the pilot will be:

a. provided with position information including, if applicable, displacement from the nominated track; and

b. provided with a heading or track clearance to intercept the nominated track for the pilot-interpreted navigation aid; or

c. provided with a track clearance direct to a waypoint to intercept the nominated track (for an RNAV or RNP capable aircraft).

3.18 Position information will be passed to aircraft in one of the following forms:

a. a bearing and distance (using points of the compass) from the ARP, a navigation aid, or a known position;

b. a heading and distance to the appropriate reporting point, en route navigation aid, or approach aid;

c. over a well known geographical position; or

d. a distance to the runway touchdown (as track miles to run).

3.19 An aircraft under ATS surveillance service control will be advised of its position in the following circumstances:

a. on identification, unless the identification is established:

(1) based on the pilot's report of the aircraft position, or within 1NM of the runway on departure, if the observed position on the situation display is consistent with the aircraft's time of departure; or

(2) by use of ADS-B aircraft identification, SSR Mode S aircraft identification or assigned discrete SSR codes if the location of the observed position indication is consistent with the current flight plan of the aircraft; or

(3) by transfer of identification;

b. when pilot requests position information;

c. when the pilot's position or estimate differs significantly from the controller's estimate based on the observed position;

d. when the pilot is instructed to resume own navigation after vectoring, if the current instructions had diverted the aircraft from a previously assigned route;

e. immediately before termination of ATS surveillance service, if the aircraft is observed to deviate from its intended route;

f. as soon, after first contact with approach radar control, as a distance to run to touchdown becomes evident;

g. when a regular circuit pattern is used to vector on to the final approach path (at least once on each leg);

h. when a straight-in approach is provided.

4. ATC RESPONSIBILITIES IN RESPECT OF UNIDENTIFIED AIRCRAFT

4.1 ATC has no responsibility to initiate avoiding action for aircraft in controlled airspace in respect of unidentified aircraft which can reasonably be assumed to be outside controlled airspace.

4.2 If an aircraft is likely to be a hazard to controlled aircraft receiving an ATS surveillance service, the controller will take appropriate action to preserve the safety of the controlled aircraft.

4.3 Where there is an ATS surveillance service in non-controlled airspace, identified IFR aircraft and VFR aircraft receiving a SIS will be provided with traffic information about known conflicting air craft, unless it is impracticable. If requested by the pilot and if possible, a course of avoiding action will be suggested.

4.4 Traffic information in respect of an unidentified aircraft will normally take the following form:

a. relative position of the unidentified aircraft to aircraft track in terms of the 12 hour clock except that, if the identified aircraft is turning, relative position will be specified by reference to compass points;

b. distance from the unidentified aircraft in miles;

c. direction in which the unidentified aircraft appears to be proceeding.

5. SPEED CONTROL (ARRIVING AIRCRAFT)

5.1 To facilitate the provision of ATS surveillance services in controlled airspace, a pilot of a controlled flight may expect the application of speed control. ATC-issued speed control instructions refer to indicated airspeed or Mach number.

5.2 The pilot must request an alternative when an ATC-issued speed control instruction is unacceptable on operational grounds.

5.3 When the application of speed control can be foreseen, a pilot will be advised of future intentions.

5.4 An ATC-issued speed control instruction, whilst in force, explicitly cancels published speed restrictions.

Note: Airspace speed limitations still apply.

5.5 A pilot will be advised when a specific ATC-issued speed control instruction is no longer necessary. Unless otherwise stated, an ATC-issued speed control instruction applies until the aircraft reaches the point in the descent profile where the speed would normally be reduced below that assigned by ATC. Except for a STAR, a DME arrival, or unless otherwise specified, a clearance for final approach or a clearance for a visual approach terminates speed control.

6. EMERGENCY PROCEDURES

6.1 General

All possible assistance will be given to aircraft in distress.

6.2 **Radio Failure Procedure**

6.2.1 When an aircraft is being vectored the interval between radio transmission is short. Pilots should make a radio check if no transmission is heard after a reasonable interval.

6.2.2 In the event of failure of two-way communications while receiving an ATS surveillance service, the pilot must change to the alternative frequency and request instructions.

6.2.3 If unable to make contact on the alternative frequency, the pilot must comply with standard radio failure procedures.

6.2.4 If able to receive but not transmit, the pilot must remain on the assigned frequency and comply with instructions issued which are designed to establish that the aircraft is receiving. If this is established, further instructions will be issued.

6.3 **ATS Surveillance System - Failure Procedure**

In the event of ATS surveillance system failure, or loss of identification, appropriate instructions will be issued.

6.4 **SSR Emergency Codes**

6.4.1 The pilot of an aircraft encountering an emergency in flight, other than loss of two-way communications, should select code 7700 unless they have specific reason to believe that maintaining the assigned code would be the better course of action.

6.4.2 The pilot of an aircraft subject to unlawful interference should select code 7500. On receipt of this code the controller will:

a. request confirmation of the setting of the assigned code as follows: “CONFIRM SQUAWKING ASSIGNED CODE”. (The absence of a reply in these circumstances shall be regarded as positive evidence of the emergency);

b. provide the aircraft with priority in all respects;

c. transmit all useful information pertinent to the conduct of the flight without expecting a reply from the aircraft;

d. avoid references to the nature of the emergency except if it is first referred to by the pilot;

e. monitor and plot the progress of the flight;

f. coordinate transfer of control, as appropriate, without requiring responses from the aircraft, unless communication remains normal; and

g. relay messages as required between the aircraft and appropriate authorities.

6.4.3 The pilot of an aircraft losing two-way communication must set the transponder to code 7600.

6.4.4 A controller observing a 7600 code shall request the pilot to operate the identification (SPI) function. If the identification signal is received, further control of the aircraft will be continued using the identification transmission to acknowledge receipt of instructions issued.

6.4.5 If the identification signal is not received, the aircraft must continue with the transponder on code 7600 and follow radio failure procedures.

6.4.6 When an RPAS experiences a lost link between the RP and the RPA, the SSR code to be selected or automatically enabled is 7400.

6.5 ADS-B Emergency Codes

6.5.1 Due to the ADS-B emergency processing limitations, if a generic ADS-B emergency indication is received from an aircraft outside of radar coverage and the flight crew does not verbally communicate the nature of the emergency, the controller will use the procedures detailed in *para 6.4.2 a*.

7. AIRCRAFT TRANSPONDER

7.1 Operation of SSR Transponders

7.1.1 Except as indicated below, ATS will assign a temporary discrete code for each flight sector for aircraft operating in controlled airspace, and for aircraft participating in Surveillance Information Service (SIS).

7.1.2 Unless advised otherwise by ATC, pilots of Mode 3A or Mode S transponder equipped aircraft operating in Australian airspace must activate their transponders, and where a Mode C capability is also available it must be activated simultaneously with Mode 3A.

Note: Pilots must ensure that transponders and ADS-B transmitters are activated and the altitude function is selected as:

a. primary radar coverage only exists within 50NM of major airports and the remainder of the ATS surveillance system relies on SSR transponder and ADS-B transmitter information, and

b. TCAS relies on transponder information for its pilot alerting and collision avoidance functions.

7.1.3 Consistent with ICAO Regional (Asia & Pacific – APAC) SSR code management code continuity objectives, Australia's ATM system has been configured to maximise retention of the discrete code assigned on departure to international flights inbound to, or overflying, Australia. This retention normally relies on code assignment notified via the DEP message, and is principally enabled for departures from other APAC Region States. When a departure or other code assigned to a flight cannot be retained in Australian airspace, pilots will be assigned a new SSR code. ATC procedures may also require that pilots be asked to squawk the code being retained.

7.1.4 When operating in Australian airspace, or on reaching the Australian FIR boundary if inbound to Australia, pilots of Mode 3A transponder equipped aircraft must squawk the assigned temporary discrete code for that flight sector, or if not assigned a temporary discrete code, the appropriate non-discrete code from the following listing, unless advised otherwise by ATS:

a. Civil flights in classes A, C and D airspace, or IFR flights in Class E airspace	3000
b. Civil IFR flights in Class G airspace	2000
c. Civil VFR flights in classes E or G airspace	1200
d. Military flights in classes A, C, D or E airspace	5000
e. Military flights in Class G airspace	6000
f. Civil flights in Class G over water at a distance greater than 15NM from shore.	4000
g. Civil flights engaged in littoral surveillance	7615
h. Ground testing by aircraft maintenance staff	2100
i. Flights operating at aerodromes (in lieu of a., b., or c. when assigned by ATC)	0100
j. RPAS in all classes of airspace and when instructed to enable transponder.	7000

7.1.5 Pilots of flights that will require a SIS and/or a clearance into controlled airspace, and for which a discrete code has already been coordinated, must select that code immediately prior to making their SIS/clearance request.

7.1.6 A pilot must not operate the identification function (SPI) unless requested by ATC.

7.1.7 Flights assigned a temporary discrete SSR code by ATS must squawk that code until termination of the flight sector, unless advised otherwise by ATS. If not assigned a discrete code, the appropriate generic code must be used.

7.1.8 A pilot operating a Mode 3A/C transponder at a radar controlled aerodrome must:

a. on departure, leave the transponder selected to STANDBY until entering the departure runway; and

b. on arrival, select the transponder to STANDBY or OFF as soon as practicable after landing.

7.1.9 A pilot operating a Mode S transponder must:

a. Enter the aircraft's identification that corresponds exactly to the Aircraft Identification shown in Item 7 of the flight notification filed with ATC for the flight; for those aircraft that are capable of reporting Aircraft Identification. The ICAO defined format for entry of the Aircraft Identification shall be used except for domestic operations when VH is not to be entered on the flight notification. (e.g. VOZ123D, REX638, QFA842, VHQFO (international), FDA...)

b. On receipt of ATC clearance, or requesting the earlier of pushback or taxi, select TA/RA/XPDR/ON AUTO as applicable.

Note 1: If AUTO mode is not available Select ON (e.g. XPDR) and assigned Mode A code.

Note 2: Australia does not require TA/RA to be de-selected while aircraft is on ground.

c. When parked and shutting down engines, select STANDBY.

d. For Mode S equipped aircraft taxiing without flight plan, the appropriate Mode A code according to *para 7.1.4* should be selected and the aircraft identification entered exactly as the call sign used in flight.

7.1.10 Pilots must select the transponder to STANDBY before effecting an SSR code change and returning the transponder to ON/ALT.

Note: This action is required to prevent possible loss of displayed aircraft position/label information and possible misidentification of aircraft in automated Australian ATC systems due to temporary selection (while effecting the change) of a code already in use.

7.1.11 When acknowledging code setting instructions or changes to settings, the pilot must read back the code to be set.

7.1.12 To facilitate harmonisation with the air traffic management systems used in the FIRs of adjacent ICAO states (other than the Mauritius FIR), ATC will allocate civil international flights a discrete code from Australia's ICAO international reservation for use from the time of their departure.

7.1.13 Unless instructed otherwise by ATS, pilots of military international flights are required to set code 5000 before departure from an Australian airport.

ENR 1.7 ALTIMETER SETTING PROCEDURES

1. PRE-FLIGHT ALTIMETER CHECK

1.1 General

1.1.1 Whenever an accurate QNH is available and the aircraft is at a known elevation, pilots must conduct an accuracy check of the aircraft altimeter(s) at some point prior to take-off.

Note: Where the first check indicates that an altimeter is unserviceable, the pilot is permitted to conduct a further check at another location on the same airfield; for example, the first on the tarmac and the second at the runway threshold (to determine altimeter serviceability).

1.2 IFR Altimeters

1.2.1 With an accurate QNH set, the altimeter(s) should read the nominated elevation to within 60FT. If an altimeter has an error in excess of ± 75FT, the altimeter must be considered unserviceable.

1.2.2 When two altimeters are required for the category of operation, one of the altimeters must read the nominated elevation to within 60FT. When the remaining altimeter has an error between 60FT and 75FT, flight under the IFR to the first point of landing, where the accuracy of the altimeter can be re-checked, is approved. In the event that the altimeter shows an error in excess of 60FT on the second check, the altimeter must be considered unserviceable for flight under the IFR.

1.2.3 An aircraft fitted with two altimeters but requiring only one for the category of operation may continue to operate under the IFR provided one altimeter reads the nominated elevation to within 60FT. Should the remaining altimeter have an error in excess of 75FT that altimeter must be placarded unserviceable and the maintenance release appropriately endorsed.

1.2.4 When an aircraft is fitted with only one altimeter and that altimeter has an error between 60FT and 75FT, flight under the IFR to the first point of landing, where the accuracy of the altimeter can be re-checked, is approved. In the event that the altimeter shows an error in excess of 60FT on the second check the altimeter is to be considered unserviceable for flight under the IFR.

1.3 **VFR Altimeters**

1.3.1 With an accurate QNH set, a VFR altimeter(s) should read site elevation to within 100FT (110FT at test sites above 3,300FT) to be accepted as serviceable by the pilot. If an aircraft fitted with two VFR altimeters continues to fly with one altimeter reading 100FT (110FT) or more in error, the faulty altimeter must be placarded unserviceable and the error noted in the maintenance release.

1.3.2 VFR altimeters are not permitted for aeroplane operations above FL200. VFR flights operating above FL200 must be equipped with an altimeter calibrated to IFR standards.

1.4 **Accurate QNH and Site Elevation**

1.4.1 A QNH can be considered accurate only if it is provided by one of the following:

a. AAIS;

b. ATC;

c. ATIS;

d. AWIS;

e. CA/GRS; or

f. WATIR.

Note: QNH contained in an authorised weather forecast must not be used for checking the accuracy of a pressure altitude system.

1.4.2 Site elevation must be derived from aerodrome survey data that is authorised in writing by either CASA or an NAA, or supplied in writing by the relevant aerodrome operator.

2. BASIC ALTIMETER SETTING PROCEDURES

2.1 **Transition Altitude, Transition Layer and Transition Level**

2.1.1 The transition altitude in all Australian FIR is 10,000FT.

2.1.2 The system of altimetry in Australia provides a transition layer of at least 1,000FT between the transition altitude and the transition level. This means the transition level varies from FL110 to FL130 depending on QNH (see *Figure 1*).

2.1.3 Cruising within the transition layer is not permitted.

2.1.4 For an operation at or below the transition altitude, the altimeter setting must be:

a. the current local QNH (either an accurate QNH or forecast QNH) of a station along the route within 100NM of the aircraft; or

b. the current forecast Area QNH.

Note: QNH is available from a reporting station (AAIS, ATIS, AWIS, CA/GRS, VOLMET or WATIR), a TAF, the Area QNH forecast, or from ATS (ATC or FIS).

2.1.5 For an operation above the transition altitude, the altimeter setting must be 1013.2 hPa.

2.1.6 On climb, the altimeter setting must be changed from QNH to 1013.2 hPa after passing 10,000FT and before levelling off.

2.1.7 On descent, and just before passing the transition layer, the altimeter setting must be changed from 1013.2 hPa to the relevant altimeter setting stated in *para 2.1.4*.

2.2 **Area QNH**

2.2.1 Area QNH is a forecast value which is valid for a period of 3 hours and normally applies throughout an Area QNH Zone (AQZ).

2.2.2 Area QNH Zones will be subdivided, if necessary, to meet the following standards of accuracy:

a. Area QNH forecasts are to be within ± 5 hPa of the actual QNH at any low-level point (below 1,000FT AMSL) within or on, the boundary of the appropriate area during the period of validity of the forecasts.

b. Area QNH must not differ from an adjoining Area QNH by more than 5 hPa.

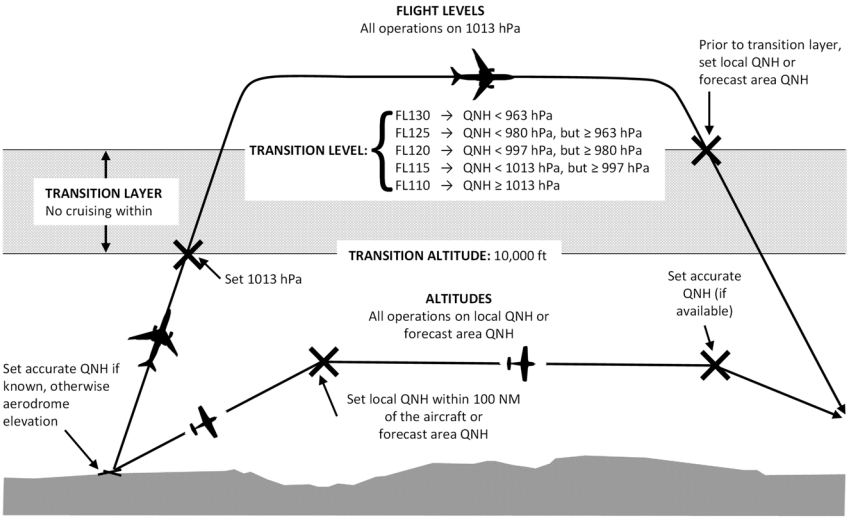


Figure 1 - Altimeter Settings

3. CRUISING LEVELS

3.1 Selection of Levels

3.1.1 Flights must be planned in accordance with levels selected from the tables at *Section 5.* Any part of a flight that will take place south of 80°S must be planned in accordance with levels selected from the tables at *Section 6.*

3.1.2 Within controlled airspace, ATC may assign and pilots may request a level that does not accord with the tables in *Section 5.*

3.1.2.1 Pilots must only request a level not conforming to the table of cruising levels when it is determined by the pilot in command to be essential to the safety of the flight and its occupants. In such circumstances, the phrase “DUE OPERATIONAL REQUIREMENT” must be included with the level change request.

3.1.3 ATC will only assign cruising levels not conforming to these tables when traffic or other operational circumstances require.

3.1.4 Subject to ATC instructions, a VFR flight must be flown at a cruising level appropriate to its magnetic track according to *Section 5. Table B* or *Section 6. Table B*:

a. whenever the flight is conducted at a height of 3,000FT AMSL or more; or

b. if the flight is conducted at a height of less than 3,000FT AMSL whenever practicable *(CASR 91.275)*.

3.2 An IFR flight must be flown:

a. within controlled airspace at a cruising level authorised for the flight by ATC; or

b. outside controlled airspace at a cruising level appropriate to its magnetic track according to *Section 5, Table A* or *Section 6, Table A (CASR 91.290)*.

3.2.1 When an IFR flight operating outside controlled airspace is unable to comply with the Table of Cruising Levels, the pilot must:

a. notify the appropriate ATS unit of the intended change in operating level, and any subsequent changes; and

b. in the event of conflict with another aircraft complying with the Table of Cruising Levels, give way to that aircraft or assume a cruising level in accordance with the Table of Cruising Levels until the aircraft with which it is in conflict is past and clear *(CASR 91.295* and *91.300).*

Note: At pilot request, ATC may assign to aircraft a level for cruise within a control area which does not provide the prescribed separation from the lower or upper limit of the control area.

3.3 **Block Levels**

3.3.1 On request from the pilot, a flight may be cleared to operate within controlled airspace within a Block Level provided that other aircraft are not denied the use of that airspace contained within that block.

3.3.2 Civil IFR flights will not be allocated block levels in Class E airspace.

3.3.3 The pilot shall have complete freedom to change levels within the block, provided that the upper and lower limits are not exceeded. However, a clearance to operate within a Block Level shall be cancelled or amended if another aircraft requests the use of a level within the block.

3.3.4 When cancelling or amending a Block Level clearance, the aircraft operating in a Block Level shall be instructed to climb or descend to an appropriate level or block level in order to provide vertical separation from the other aircraft requesting one of the levels.

3.3.5 Aircraft at standard flight levels will be afforded priority over aircraft using non-standard flight levels.

3.3.6 Mach number technique separation will not be applied to aircraft using block level clearances.

3.3.7 Aircraft operating within a block level must report the upper and lower block levels in all positions and frequency change reports.

Note: As most altitude alerting systems do not provide protection for both upper and lower assigned levels, flight crews are reminded to be vigilant in monitoring the aircraft altitude when operating within a Block Level.

4. CHANGE OF LEVELS

4.1 ATC Approval Required

4.1.1 The pilot in command must commence a change of level as soon as possible, but not later than one (1) minute after receiving that instruction from ATC, unless that instruction specifies a later time or place.

4.1.2 ATC may require that an assigned level must be reached by a specific time, distance or place. If a pilot in command doubts that the restriction can be met, ATC must be advised immediately.

4.1.3 ATS advised expectation of a level restriction does not authorise a pilot to climb or descend to meet that restriction.

4.1.4 An expectation of a level restriction is not required to be read back.

4.1.5 A requirement to report at a time or place given in the same clearance as a descent/climb instruction does not require the new level to be reached by the specified time or place.

4.1.6 The pilot in command of an aircraft, receiving an instruction from ATC to change level, must report:

a. when the aircraft has left a level at which level flight has been conducted in the course of climb, cruise or descent; and

b. when the aircraft leaves a level for which ATC has requested a report.

4.1.7 During a change of level, the pilot in command must advise ATC if they maintain an unassigned interim level.

4.1.8 ATC may provide vertical separation between two climbing aircraft, not otherwise separated, by means of a step-climb. Pilots in command, who are subjected to a step-climb, must adopt the following procedure:

a. The pilot in command of the lower aircraft must report approaching each assigned level in the sequence.

b. The pilot in command of the higher aircraft, on hearing the lower aircraft report approaching each assigned level, must report the last vacated level.

4.1.9 Step-descents reverse the above *para 4.1.8* procedure.

4.1.10 ATC may specify a rate of climb or descent. Other considerations are as follows:

a. The phrase “STANDARD RATE”, when included in a clearance, specifies a rate of climb or descent of not less than 500FT per minute, except that the last 1,000FT to an assigned level must be made at 500FT per minute.

b. In the case of a step-climb or descent, the specified rate will be applicable to all level clearances issued in the course of the step climb or descent. If unable to comply with the prescribed rate, the pilot in command must advise ATC.

4.1.11 Cruise Climb is not used in Australian administered airspace. Where possible, block level clearances will be issued upon request.

4.2 **ATC Approval Not Required**

4.2.1 In airspace where ATC approval is not required to change level, the pilot of an IFR flight must report present position and intention to ATC approximately one (1) minute prior to making any change.

5. TABLES OF CRUISING LEVELS (NORTH OF 80° S)

TABLE A - IFR				
Magnetic Tracks	**From 000° through East to 179°**		**From 180° through West to 359°**	
Cruising Altitudes (Area QNH)	3,000	7,000	2,000	8,000
	5,000	9,000	4,000	10,000
			6,000	
Cruising Flight Levels (1013HPA)	110*	290	120*	300
	130	310	140	320
	150	330	160	340
	170	350	180	360
	190	370	200	380
	210	390	220	400
	230	410	240	430
	250	450	260	470
	270	490	280	510
Note FL110 is not available for level flight when the Area QNH is less than 1013HPA.* *FL120 is not available for level flight when the Area QNH is less than 980HPA.*				

TABLE B - VFR				
Magnetic Tracks	**From 000° through East to 179°**		**From 180° through West to 359°**	
Cruising Altitudes (Area QNH)	1,500	7,500	2,500	8,500
	3,500	9,500	4,500	
	5,500		6,500	
Cruising Flight Levels (1013HPA)	115*	195	125*	205
	135	215	145	225
	155	235	165	245
	175		185	
Note FL115 is not available for level flight when the Area QNH is less than 997 HPA.* *FL125 is not available for level flight when the Area QNH is less than 963 HPA.*				

Note 1: Pilots should be aware that VFR aircraft outside controlled airspace might be operating at random levels below 3,000FT AMSL (see para 3.1.4).

6. TABLES OF CRUISING LEVELS (SOUTH OF 80° S)

TABLE A - IFR				
Grid Tracks	**From 000° through East to 179°**		**From 180° through West to 359°**	
Cruising Altitudes (Area QNH)	3,000	7,000	2,000	8,000
	5,000	9,000	4,000	10,000
			6,000	
Cruising Flight Levels (1013HPA)	110*	250	120*	260
	130	270	140	280
	150	290	160	310
	170	330	180	350
	190	370	200	390
	210	410	220	430
	230	etc.	240	etc.
Note FL110 is not available for level flight when the Area QNH is less than 1013HPA.* *FL120 is not available for level flight when the Area QNH is less than 980HPA.*				

TABLE B - VFR				
Grid Tracks	**From 000° through East to 179°**		**From 180° through West to 359°**	
Cruising Altitudes (Area QNH)	1,500	7,500	2,500	8,500
	3,500	9,500	4,500	
	5,500		6,500	
Cruising Flight Levels (1013HPA)	115*	255	125*	265
	135	275	145	285
	155	300	165	320
	175	340	185	360
	195	380	205	400
	215	420	225	440
	235	460	245	480
Note FL115 is not available for level flight when the Area QNH is less than 997HPA.* *FL125 is not available for level flight when the Area QNH is less than 963HPA.*				

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ENR 1.8 REGIONAL SUPPLEMENTARY PROCEDURES

1. There are no notifiable regional supplementary procedures applicable to Australia.

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ENR 1.9 AIR TRAFFIC FLOW MANAGEMENT

1. FLOW MANAGEMENT STAGES

1.1 At major airports within Australia, Air Traffic Flow Management (ATFM) procedures are applied to manage demand and capacity at specific airports. These procedures are defined in three phases:

a. **Strategic** - Generally occurs more than one day prior to the day of operation. This is known as schedule coordination. The relevant airport operator manages this service.

b. **Pre-tactical** - Occurs on the day prior to operation through the implementation of traffic management initiatives such as a Ground Delay Program (GDP) and Airport Collaborative Decision Making (A-CDM). Airservices Australia National Operations Management Centre (NOMC) manages these services.

c. **Tactical** - Occurs on the day of operation and uses real time traffic information to sequence traffic to the destination airport. ATC manage this service.

1.2 Tactical flow management takes precedence over pre tactical air traffic flow management which in turn takes precedence over strategic air traffic flow management.

2. STRATEGIC - SCHEDULE COORDINATION

2.1 An airline has the responsibility to obtain permission to operate services from the airport owner and/or operator.

2.2 In addition, all aircraft operators (excluding emergency and state aircraft) must request time-slots (slots) in advance of the operation from the relevant slot management organisation in accordance with the following table:

Airport	Type of operation	Slot Manager
Sydney/Kingsford Smith	All arrivals and departures	Airport Coordination Limited (ACL-APAC)
Brisbane and Perth	All arrivals and departures	Capacity Optimisation Group (COG)
Adelaide and Darwin	All international and scheduled domestic flights	
Cairns, Gold Coast and Melbourne	All international flights	

2.3 Contact details are:

2.3.1 Slot Manager: Capacity Optimisation Group (COG)
MON-FRI 2200-0600 UTC

Ph: 1800 449 355

Email: slots@cog.aero

Web: www.cog.aero

2.3.2 Slot Manager: Airport Coordination Limited - Asia Pacific (ACL-APAC)
MON-FRI 2130-0600 UTC

Ph: +61 2 9667 9016

Email: slots@acl-international.com
(for SRC requests) or
syd@acl-international.com
(non-formatted requests)

Web: www.acl-apac.com.au/airport-information/sydney

Mail or in person: Level 1, Nigel Love Building,
10 Arrivals Court, Sydney/Kingsford Smith Airport.

2.4 Notification of changes to slots allocated to existing scheduled flights should be advised to the relevant slot manager in accordance with the requirements of the appropriate traffic management scheme.

2.5 COG or ACL-APAC slots may be obtained outside office hours for short notice non-scheduled flights from the NOMC on 1800 020 626. These slots will be allocated from the available pool.

2.6 Allocated slots may be subject to change by ATFM due to operational constraints. *(See Section 3. and 5.*)

3. PRE-TACTICAL - GROUND DELAY PROGRAM

3.1 General

3.1.1 The NOMC publishes GDP for arrivals to Sydney/Kingsford Smith (YSSY), Brisbane (YBBN), Melbourne (YMML) and Perth (YPPH) Airports.

Note: Additional operating procedures are contained in ERSA FAC for the specified airport.

3.1.2 The NOMC considers all available meteorological data, network capacity and aerodrome information to forecast the available flow rate into GDP ports. Notified Flight Plan ETD and type-specific EET information is then used to calculate aircraft landing times and assertain the arrival port demand versus predicted capacity. A Calculated Off-Block Time (COBT) and a Calculated Take-Off Time (CTOT) are then formulated for the departure port to reduce any delays which may otherwise be encountered airborne.

3.1.3 Unless instructed by ATC, pilots should maintain normal or specified climb, cruise and descent profiles.

3.2 Calculated Off Blocks Time (COBT)

3.2.1 Pilots must obtain an Air Traffic Flow Management COBT for operations at a GDP airport. Pilots of scheduled flights will receive their COBT through their operator. Other flights may obtain a COBT through the NOMC by email: atfmu@airservicesaustralia.com or phone: 1800 020 626 (H24) (recorded line).

3.2.2 Unless afforded priority in accordance with *ENR 1.4 Section 6.* all aircraft are required to operate within the compliance window for their allocated COBT. The compliance window is from –5 MIN to +15 MIN of the COBT.

3.2.3 Aircraft unable to operate within the compliance window are to obtain a new COBT through their operator or the NOMC (as per *para 3.2.1*). ATC are not able to provide new or amended COBT.

3.2.4 If a new COBT has been issued after receiving airways clearance, pilots are to advise ATC of the amended COBT when requesting start-up, pushback, or taxi clearance.

3.2.5 Failure to obtain a COBT and/or submit a flight plan for a flight to a GDP program airport will result in the flight being considered early non-compliant.

3.2.6 For flights with a COBT that are early non-compliant, ATC will only issue a clearance to pushback or taxi for a significant ground-based operational requirement or if there is a reasonable expectation that, due to taxi or runway holding position delays, the required CTOT will be achieved.

3.2.7 Notwithstanding actions taken under *3.2.2* and *3.2.6* to achieve compliance, flights departing non-compliant can expect delays en route. Non-compliant flights will be allocated the next available slot time up to a maximum delay as follows:

a. Early non-compliant - 60 MIN; or

b. Late non-compliant - published traffic holding delay.

3.3 **Non-scheduled flights**

3.3.1 Prior to submitting a flight plan, pilots of non-scheduled flights intending to operate into a GDP airport during the hours of program operation:

a. should, if required, obtain a strategic time-slot from the relevant slot management organisation (*see section* 2.) prior to contacting the NOMC; and

b. must contact the NOMC for a COBT and, if unable to obtain prior, a strategic time-slot.

3.3.2 Where possible pilots should contact the NOMC prior to 0800 UTC the day before to ensure their flight is included in the GDP run for the following day. Operators who contact the NOMC after the GDP has been run will be allocated the next available COBT.

3.3.3 Pilots must provide the following information to the NOMC at least one hour prior to the proposed operation. Any changes must be notified to the NOMC prior to departure.

(1) Aircraft call sign

(2) Aircraft registration

(3) Aircraft type

(4) Departure aerodrome

(5) Destination aerodrome

(6) ETD (UTC time only)

(7) ETA (UTC time only)

(8) COBT notification email/mobile phone number

3.3.4 Notification of flight details to the NOMC is additional to all existing flight plan notification requirements.

3.3.5 Pilots of non-scheduled flights must check their COBT for any amendments prior to flight by:

a. being able to receive a message from the NOMC via their mobile phone/email; or

b. contacting the NOMC within one hour of the flight; or

c. where no communication facilities are available, contacting the appropriate HF frequency.

3.4 **GDP run times**

3.4.1 GDP for the following day's operations are run on an on-demand basis. Following assessment (see para 3.1.2), GDPs deemed to be required are run in the order of Perth, Melbourne, Brisbane, Sydney in the timeslots of 0815 UTC, 0845 UTC, 0915 UTC and 1000 UTC. E.g. if only the Melbourne and Sydney GDPs are necessary, they would be run at 0815 UTC and 0845 UTC. D

3.5 **GDP revision**

3.5.1 When unforeseen circumstances significantly reduce the capacity of an airport, a GDP revision may be initiated and pilots must obtain a new COBT. Tower ATC may stop departures to the GDP airport to facilitate the revision.

3.5.2 There are three levels of revision:

a. Level 1 – compliance with the new COBT will commence in 30 MIN; or

b. Level 2 – immediate compliance with the new COBT should be observed, however flights that have already manoeuvred to depart may continue; or

c. Level 3 – immediate compliance with the new COBT should be observed by all flights.

Note: Level 2 and 3 revisions will not be applied to flights departing Perth, Darwin, Karratha, Port Hedland or Broome, for Brisbane, Sydney or Melbourne.

3.5.3 The NOMC will advise pilots and operators when a revision occurs. This advice may be provided through ATS when required. When a level 2 or 3 GDP revision occurs, ATS will advise pilots subject to immediate compliance.

4. AIRPORT COLLABORATIVE DECISION MAKING (A-CDM)

4.1 General

4.1.1 A-CDM procedures apply H24 to all IFR fixed wing aircraft operating from an A-CDM airport - refer to *ERSA* for A-CDM locations. The NOMC will set departure rates, then A-CDM will generate Target Off-Block Time (TOBT) and Target Start-up Approval Time (TSAT).

4.1.2 At aerodromes where A-CDM is implemented, start-up approval is required for all aircraft with ground power. Aircraft without ground power may start-up at the discretion of the pilot in command, however ATC should be advised when start-up is complete. Refer *ERSA* for the applicable frequency.

4.1.3 ATC will instruct aircraft to stand by on ground frequency once compliant with ATFM procedures. Ground will approve start-up or pushback when able.

4.2 TOBT and TSAT

4.2.1 All non-exempt aircraft must have a TOBT to depart.

4.2.2 The TOBT compliance window is from –5 MIN to +5 MIN of TOBT.

4.2.3 Aircraft must request start-up or pushback within their TOBT compliance window. Aircraft unable to request start-up or pushback within their TOBT compliance window must contact their company or ground handling agent for a new TOBT. ATC are not able to provide a new or amended TOBT.

4.2.4 TSAT compliance window commences from TSAT –5 MIN and aircraft will receive start-up or pushback approvals as operations permit. A-CDM TSAT incorporates GDP CTOT requirements.

4.3 **GDP and A-CDM compliance**

4.3.1 For TOBT/TSAT early non-compliant flights, ATC will only approve start-up or pushback for a significant ground-based operational requirement.

4.3.2 The TSAT prevents early non-compliance with GDP CTOT. In the case of late non-compliance with GDP CTOT resulting from an operator TOBT update, operators must obtain a revised GDP CTOT which will generate a new TSAT.

Note: TSAT late non-compliance may result from traffic management or ATC operational requirements.

4.4 **GDP revisions at A-CDM airports**

4.4.1 When unforeseen circumstances significantly reduce the capacity of a destination airport, a GDP revision may be initiated. Tower ATC may stop departures to the GDP airport to facilitate the revision.

4.4.2 Aircraft departing from an A-CDM airport to a destination subject to a GDP revision, will be processed as follows:

a. Level 1 and 2 – TSAT compliance continues;

b. Level 3 – immediate compliance with the new TSAT should be observed by all flights. Flights that have manoeuvred to depart may be subject to ground delay.

5. TACTICAL

5.1 **Aircraft sequencing near ATFM Airports**

5.1.1 Due to terminal area traffic density, pilots may expect airborne traffic delays for arrival at locations adjacent to or within Class C control zones.

5.1.2 When sequencing arriving aircraft to controlled aerodromes, ATC may apply one or more of the following:

a. Enroute holding procedures;

b. Allocate a waypoint crossing time to the pilot; or

c. Tactically apply delaying action such as speed control or vectoring.

5.1.3 When ATC allocates a waypoint crossing time, aircraft must adjust speed to cross the waypoint at the specified time or up to 30 seconds early. Speed adjustment to meet a waypoint crossing time only applies to the specified waypoint. Unless otherwise published or instructed by ATC, pilots must cross the waypoint at the lesser of 250KT or profile speed.

5.1.4 Aircraft unable to meet the waypoint crossing time must inform ATC as early as possible.

ENR 1.10 FLIGHT PLANNING

1. FLIGHT PLAN PREPARATION

1.1 Before beginning a flight, a pilot in command must study all available information appropriate to the intended operation and must make a careful study of:

a. current weather reports and forecasts for the route to be flown and the aerodromes to be used;

b. the airways facilities available on the route to be flown and the condition of those facilities;

c. the condition of aerodromes to be used and their suitability for the aircraft to be used;

d. the ATC rules and procedures appertaining to the particular flight; and

e. all Head Office and FIR NOTAM applicable to the en route phase of flight, and location-specific NOTAM for aerodromes.

The pilot must then plan the flight in relation to the information obtained.

Note: Full details on the services provided by the briefing office are available through the Preflight Information and Flight Planning Manual, available online: www.airservicesaustralia.com/aip/aip.asp and www.airservicesaustralia.com/industry-info/flight-briefing/,

1.2 **Forecasts**

1.2.1 Forecast information must include:

a. an aerodrome forecast for the:

(i) departure

(ii) destination; and

(iii) when required, any alternate aerodromes; and

b. one of the following:

(i) a flight forecast; or

(ii) a GAF (at and below A100); or

(iii) a SIGWX forecast (above A100); and

c. a wind and temperature forecast

For a flight to a destination for which a prescribed instrument approach procedure does not exist, the minimum requirement is a GAF.

Note: A wind and temperature forecast may be obtained from Wind and Temperature Charts, Grid Point Wind and Temperature Charts, Route Sector Winds and Temperatures Forecasts, a NAIPS Wind and Temperature Profile (applicable for the flight), as well as from approved flight planning systems that derive data from the Bureau of Meteorology or the WAFS.

1.2.2 For flights for which a forecast is required and cannot be obtained, the flight is permitted to depart provided the pilot is satisfied that the weather at the departure point will permit the safe return of the flight within one hour of departure. The flight is permitted to continue provided:

a. for *Part 121* operations – the authorised weather forecasts for the route, destination and any planned alternate aerodromes, are obtained within 30 minutes after departure.

b. for all other operations - the authorised weather forecast for the destination aerodrome is obtained within 30 minutes after departure.

1.2.3 For flights to a destination for which an aerodrome forecast is required and cannot be obtained, the flight is permitted to depart provided an alternate aerodrome meeting all the requirements specified in *ENR 1.1 Section 10.7* is provided.

1.2.4 A pilot in command must ensure that the forecasts cover the period of the flight and that the aerodrome forecasts for the destination and alternate aerodromes, to be nominated in the flight plan, are valid for a period of not less than 30 minutes before and 60 minutes after the planned ETA.

1.2.5 When a flight is delayed so that the meteorological and operational information does not cover the period of flight, updates must be obtained as necessary, to allow the flight to be concluded safely.

1.2.6 A series of flights may be included on the one flight plan provided that:

a. the meteorological forecast will cover all the flights; and

b. relevant AIS information is available at flight planning.

1.2.7 When preflight briefing is obtained more than one hour prior to EOBT, pilots should obtain an update before each departure to ensure that the latest information available can be used for the flight. The update should be obtained by NAIPS pilot access, telephone, or, when this is impracticable, by radio.

1.3 **GNSS Prediction Analysis - Flight in Oceanic and Remote Areas**

1.3.1 A requirement for flight in oceanic and remote areas using GNSS is that an appropriate en route GNSS prediction analysis be conducted prior to each flight. For details see *ENR 2.2 Section 4.*

2. **FLIGHT NOTIFICATION**

2.1 Flight notification requirements are divided into two specific categories:

a. those affecting IFR flights, and

b. those affecting VFR flights.

2.2 IFR flights require the submission of comprehensive flight notification and the transmission of in-flight progress reports at regular intervals. SARWATCH is based primarily on the receipt of these reports by ATS. (See also *GEN 1.5 Section 1.*).

2.3 Pilots of VFR flights nominating a SARTIME to ATS, and those intending to operate in controlled airspace (except for VFR flights in Class E airspace) must submit flight details to ATS.

2.4 The order of preference for pilots to submit a comprehensive flight notification is:

a. via pilot access to NAIPS (via the Internet),

b. in writing,

c. by telephone, or

d. by radio to ATS.

2.5 Pilots should submit details required for flight in controlled airspace at least 30 minutes before the expected time of entry. Flight details submitted with less than the 30 minutes notification may be subject to delay.

2.6 Pilots submitting SARTIME flight notifications by fax must confirm receipt of the notification with the briefing office. Further, Airservices strongly recommends that when any flight notification is submitted by fax, the pilot or operator telephones the briefing office before departure to confirm that the fax has been received.

2.7 Abbreviated details for operations in controlled airspace may be advised by radio if the flight is to operate locally, or operations will be for a brief duration. However, prior contact with ATC may avoid delays. Pilots may submit details by radio to ATS when associated with a clearance request, or to nominate a SARTIME.

2.8 When submitting flight notification by radio, pilots should be mindful of the need to minimise frequency congestion and transmit only that information required by ATS for the current flight stage. Acceptance is subject to ATS workload and may be delayed.

2.9 Submission of comprehensive travel flight notification by radio is not a preferred method of notification and should not be used when submission by some other means is available. Flight notification by radio for travel flights requiring the submission of comprehensive details will not be accepted at controlled aerodromes.

2.10 Pilots of VFR flights wishing to operate in other than classes C or D airspace, and who wish to nominate a SARTIME, may submit details in the NAIPS SARTIME flight notification format (via the internet). If submitting the flight notification by fax or via telephone, the only form available is the Australian Domestic Flight Notification form.

2.11 Pilots conducting a Community Service Flight as defined in *GEN 2.2* are required to submit a flight notification (*see para 2.21* and *ENR 1.1 para 10.14.13*) and identify the flight as a community service flight by entering the acronym “CSF” in the remarks section of the flight notification. If the flight notification is submitted by radio, then the pilot is required to request ATS to annotate the flight as a CSF.

Note: Pilots are also required to annotate their personal logbook entry for the flight (required by regulation 61.345) to reflect that the flight was conducted as a CSF.

2.12 If a VFR flight is one of the following:

a. a flight conducting an air transport operation; or

b. a flight over water that is conducted beyond a distance from land greater than which would allow the aircraft to reach land with an engine inoperative; or

c. a flight in a designated remote area; or

d. a flight at night proceeding beyond 120NM from the aerodrome of departure;

Then the pilot in command must ensure one of the following has occurred:

a. submission of a flight plan, or

b. nomination of a SARTIME for arrival, or

c. leave a flight note with a responsible person.

2.13 VFR flights which are required to, or wish to, use a SARTIME may do so by providing ATS with the following details:

a. call sign;

b. aircraft type;

c. departure point;

d. route to be flown;

e. destination;

f. POB; and

g. SARTIME.

Note: Only one SARTIME may be current at any time. To prevent the existence of multiple SARTIMEs for aircraft used by more than one pilot, SARTIMEs should be nominated immediately before the start of each flight.

2.14 VFR flights operating on SARTIME are requested to include contact telephone details for the pilot or company at the destination where available.

2.15 VFR flights may operate on reporting schedules in the following circumstances:

a. flood, fire or famine relief flights;

b. search and rescue flights;

c. overwater flights; and

d. military flights.

2.16 When the pilot of a flight wishes to indicate a variation of SAR requirements, this must be indicated in Item 8 - Flight Rules, amplified in Item 15 (Route) by the position at which the change will occur, followed by the new Flight Rules.

2.17 Submission of flight details at least 30 minutes before EOBT is recommended.

2.18 Where notification of flight details, or changes to details, are submitted less than 30 minutes before EOBT, delays will be encountered when an ATC unit requires that the data be programmed into the computerised SSR Code/Call sign Management System.

2.19 The preferred method to cancel a SARTIME is via telephone to CENSAR on 1800 814 931. When telephone facilities are not available you may use ATS frequencies.

2.20 SARTIME are managed on a national basis by the central SARTIME management database, CENSAR.

2.21 The following table identifies flight notification options for the various classes and types of operations when flying IFR or VFR:

Flight Rules	Class of Airspace	Type of Operation	Summary of Flight Notification Options
IFR	All Classes	All Operations	Submit a Flight Plan
VFR	Class C and D	All Operations	Submit a Flight Plan
VFR	Any	Air transport operation; or Over water flights; or In designated Remote Areas; or At night proceeding beyond 120NM from the aerodrome of departure	Submit a Flight Plan; or Nominate a SARTIME; or Leave a Flight Note
VFR	Any	CSF	Submit a Flight Plan; or Nominate a SARTIME
VFR	Class E and G	Any other operations not mentioned above	Submit a Flight Plan; or Nominate a SARTIME; or Leave a Flight Note; or No notification

2.22 Pilots not formally required to submit flight notification, or leave a flight note as defined in the preceding paragraphs, are nevertheless encouraged to leave a flight note as defined in *AIP GEN.*

3. FLIGHT NOTIFICATION/NOTE CONTENTS

3.1 Forms

3.1.1 An example of the Australian Domestic Flight Notification form is at *APPENDIX 1*. Instructions for completion of the Australian Domestic flight notification form for both IFR and VFR flights are contained at *APPENDIX 2*. In a number of cases, particularly in Item 19, completion is recommended as good practice. If mandatory Items are left incomplete, delays may occur.

Note: The reverse side of the Australian Flight Notification Form contains a "flight log/template" to assist pilots in planning and navigation. It is not intended to be mandatory or prescriptive, and pilots may use any template, or other device, of their choice.

3.1.2 The flight notification forms are available from the Airservices website: www.airservicesaustralia.com/flight-briefing/.

3.1.3 The suggested format for a Flight Note is at *APPENDIX 3.* This form is available from the CASA website: www.casa.gov.au.

3.2 Flight Rules

3.2.1 Flight rules must be indicated in any flight notification, except for VFR flights operating wholly outside controlled airspace nominating a SARTIME.

3.2.2 Flights which, within a single stage, will be flown under both the IFR and VFR must indicate:

a. in the Flight Rules section of the flight notification, the flight rules applicable to the first route segment of the flight plan, Y to indicate IFR first followed by one or more changes of flight rules, or Z to indicate VFR first followed by one or more changes of flight rules;

b. in Field 15 of the flight notification, the position at which the change of flight rules will occur; and

c. for each subsequent stage, the flight rules applicable to that stage, with a change in Field 15 if applicable.

Note: The use of Y or Z must not be used to indicate a variation of flight rules between individual stages. Where the flight rules field is left blank on a multistage flight, the previous flight rule will apply.

3.3 **PBN Notification**

3.3.1 No indication on the flight notification form is required for Visual Navigation or DR Substitute applications of GNSS.

3.3.2 Notification of PBN capabilities requires a combination of entries in Item 10 (Equipment and Capabilities) and Item 18 of the ATS Flight Notification form. Guidance is provided in *Appendix 2* to this section.

3.4 **Military Flights**

3.4.1 Aircraft operating on LJR for any part of a flight must submit flight notification to ATS, regardless of flight rules.

3.4.2 LJR notification is required for flights planned below 5,000FT AGL within Class G airspace for aircraft that:

a. operate with a TAS above 250KT

b. have nominated NOCOM for any portion of the flight; or

c. are unable to meet reporting requirements.

Note: LJR notification is not required if the MLJ is operating within a Danger Area established for that purpose.

3.4.3 Military flights with an LJR component are required to provide EETs for all points nominated in Item 15 of their flight notification to enable ATS units to provide an effective service.

3.4.4 Military flights carrying out specific operations notified in the remarks section of the flight notification form, together with a level at or below 1,000FT will be operating with reference to ground level.

3.4.5 Pilots of military aircraft that are not RVSM-approved, but require priority in the RVSM flight level band for operational reasons, must enter STS/NONRVSM and RMK/MIL SPEC REQ in Item 18.

3.4.6 Formation flights of State aircraft must not insert the letter W in Item 10 of the ICAO flight plan form, regardless of the RVSM approval status of the aircraft concerned.

3.5 **General**

3.5.1 Pilots of flights operating IFR under a Private IFR rating must include this advice when submitting flight notification. Flight procedure Authorisations (FPAs) applicable to flight within controlled airspace must also be included.

3.5.2 For flights not operating along an ATS route, reporting points should be provided in Item 15 for locations approximately 30 minutes or 200NM apart.

3.6 **Location data**

3.6.1 Any location abbreviations used should be authorised abbreviations (e.g. published in *AIP*).

3.6.2 If a common name is entered into NAIPS in lieu of an aerodrome abbreviation or navigational air/waypoint, the flight notification output will assume that the aircraft is tracking over a navigational aid/waypoint and not aerodrome; e.g. the location HOLBROOK will translate to HBK, not YHBK.

3.6.3 Some locations with abbreviations may not have fixed positions e.g. HLS associated with a mobile oil platform or ship. These location codes are linked to fixed geographical coordinates in systems and datasets. To ensure correct provision of ATS, in the event a platform is relocated or is in the process of relocating the NOTAM office must be immediately advised. A NOTAM will be issued to suspend use of the four (4) letter location code in flights plans. Pilots must then use “ZZZZ” procedure as specified in *Appendix 2 Item 13 and Item 16.*

When the NOTAM office is notified that the mobile oil platform/ ship is back in its original position the NOTAM will be cancelled, and use of the four (4) letter location code in flight plans may resume.

3.6.4 Pilots entering geographical coordinates must adhere to the correct format e.g. 2730S15327E.

4. FLIGHT NOTIFICATION AMENDMENT

4.1 When flight notification details have been submitted and amendment is necessary, advise ATS of the following items as soon as possible:

Item	Details	1. All IFR 2. VFR in CTR/CTA	VFR wholly OCTA nominating a SARTIME
7	Aircraft ident and/or registration	X	X
8	Flight rules to which flight will be operating	X	
10	Serviceability of equipment carried	X	
13	DEP aerodrome and EOBT if the change exceeds 30 minutes	X	X (DEP aerodrome only)
15 16	Route, landing points or alternates	X	X
15	Cruising level	X	
15	Speed and estimated total elapsed time	X	
18	Any change to: STS/PBN/NAV/ RMK/(includes SARTIME)	X	X
19	POB	X	

4.2 If advising ATS of a change of aircraft ident and/or registration, pilots of SARTIME flights must also advise, prior to take-off, that the flight is subject to a SARTIME.

4.3 To assist in managing the airways system, pilots should always warn ATS of any flight notification amendments by utilising appropriate alerting phraseologies; e.g.

"MELBOURNE CENTRE, DELTA MIKE GOLF, IFR FLIGHT PLAN AMENDMENT"
or
"FLIGHTWATCH, DELTA MIKE GOLF, SARTIME FLIGHT PLAN AMENDMENT".

5. CARRIAGE OF FLIGHT DOCUMENTATION

5.1 Pilots conducting certain flights (see the relevant rules in *CASR Parts 91, 103, 121, 131, 133* and *135* relating to the carriage of documents) are required to carry, and have readily accessible in the aircraft, the latest editions of the aeronautical maps, charts and other aeronautical information and instructions, published:

a. in AIP, or

b. in NOTAM, or

c. authorised aeronautical information by a data service provider, that are relevant to the route to be flown, and any probable diversionary route that may be flown, on that flight; or

d. for a flight in a foreign country:

   (i) in the document that in that country is equivalent to the AIP; or

   (ii) by an organisation approved to publish aeronautical information by the national aviation authority of that country.

APPENDIX 1

Australian – Domestic Flight Notification Form

airservices australia

7. Aircraft Identification

8. Flight Rules: I V Y Z

Type of Flight: S N G M

9. No | Type | Wake Turb Cat: J H M L

10. Equipment: N or S

And / Or: A B C D E1 E2 E3 F G H I J1 J2 J3 J4 J5 J6 J7 K L M1 M2 M3 O R T U V W X Y Z

SSR: L E H S I P X C A N

ADS-B: B1 B2 V1 V2 U1 U2

ADS-C: D1 G1

13. DEP Aerodrome	EOBT	15. Cruising Speed N M	Level A F	16. DEST Aerodrome	Total EET HR MIN	ALTN Aerodrome

15. Route

18.

(Stage 2)

8.	13. DEP Aerodrome	EOBT	15. Cruising Speed N M	Level A F	16. DEST Aerodrome	Total EET HR MIN	ALTN Aerodrome
I							

V 15. Route

Y 18. (Info relevant to Stage 2)

Z

(Stage 3)

8.	13. DEP Aerodrome	EOBT	15. Cruising Speed N M	Level A F	16. DEST Aerodrome	Total EET HR MIN	ALTN Aerodrome
I							

V 15. Route

Y 18. (Info relevant to Stage 3)

Z

18. (Information relevant to all stages)

DOF/ REG/VH

PER/

RMK / SARTIME	To ATS Unit	Location	DEST Tel No:
Date/Time Arr / Dep			ORGN/

19. Supplementary Information (optional)

ENDURANCE HR MIN | PERSONS ON BOARD

E/ P/

E/ P/

E/ P/

EMERGENCY RADIO: R/ UHF [U] VHF [V] ELT [E]

SURVIVIAL EQUIPMENT: S/ POLAR [P] DESERT [D] MARITIME [M] JUNGLE [J]

DINGHIES: D/ NUMBER CAPACITY COVER [C] COLOUR

JACKETS: J/ LIGHT [L] FLUORES [F] UHF [U] VHF [V]

A/ AIRCRAFT COLOUR AND MARKINGS

N/ REMARKS

C/ PILOT-IN-COMMAND | PHONE | MOBILE | EMAIL | COMPANY

PSN	LSALT	FL or ALT	TAS	TR(M)	WIND	HDG (M)	G/S	DIST	ETI	EET	PLN EST	REV EST	ATA ATD

Item	Fuel Calculation	Min	lbs, L or kg	Min	lbs, L or kg
a	Taxi fuel				
b	Trip fuel				
c	Contingency fuel (_____% of b)				
d	Alternate fuel				
e	Final reserve fuel				
f	Additional fuel				
g	Holding fuel				
h	Fuel required (a+b+c+d+e+f+g)				
i	Discretionary fuel				
j	Margin fuel				
k	Endurance (h+i+j)				
	From				

Pilot Notes

CALL SIGN:	PHONE NO:	FAX NO:

BRIEFING 1800 805 150 FAX 1800 805 150 CENSAR 1800 814 931

APPENDIX 2

ATS FLIGHT NOTIFICATION - USER GUIDE

The Australian Domestic Flight Notification Form provides a modified ICAO flight plan form for Australian requirements and to allow entry of multiple stages of flight.

Item 7 – Aircraft Identification

Enter — Aircraft registration/flight number. ZZZZ and TBA cannot be accepted.

Requirements — One call sign per flight notification.

For VH registered aircraft conducting a domestic flight (i.e. within Australian FIR), enter the three letters after the prefix only e.g. for VH-ZFR enter ZFR. For international flights, enter the full registration e.g. VHZFR.

For flight numbers, and other approved call signs, enter a mixture of figures and letters that do not exceed seven alphanumeric characters and without hyphens or symbols; e.g. QFA611.

For unmanned aircraft except when assigned a flight number call sign or another approved call sign:

a. enter the prefix UX then at least two characters of the aircraft model e.g. UXSCE4
b. enter the full radiotelephony call sign in Item 18 after RMK/RTF e.g. UNMANNED SCAN EAGLE FOUR

Item 8 (a) – Flight Rules

Circle

- I — if the entire flight will be operated under the Instrument Flight Rules (IFR)
- V — if the entire flight will be operated under the Visual Flight Rules (VFR)
- Y — if the flight will be operated initially under the IFR followed by one or more changes of flight rules
- Z — if the flight will be operated initially under the VFR followed by one or more changes of flight rules

Requirements — If Y or Z is circled, an entry in Item 15 must specify where the change of flight rules will occur; e.g. YBAF VFR.

Type of Flight

Circle	S	for scheduled air service
	N	for non-scheduled air service
	G	for general aviation
	M	for military
	X	if other than any of the defined categories above

Item 9 – Number of Aircraft

Enter	Number of aircraft where there are more than one, otherwise leave blank.

Type

Enter	Aircraft type. Where more than one aircraft type is included in a formation, enter the type of the lowest performance aircraft. Additional details regarding the formation must be inserted at Item 18.
Requirements	Use the two to four letter ICAO approved aircraft type abbreviation. For aircraft type abbreviations not approved by ICAO, enter ZZZZ and specify the type of aircraft in Item 18 preceded by TYP/.

Wake Turbulence Category

Circle	J	SUPER, to indicate an A388 aircraft type
	H	HEAVY, for aircraft 136,000KG MTOW or more, except for A388 aircraft type
	M	MEDIUM, for aircraft between 7,000 and 136,000KG MTOW
	L	LIGHT, for aircraft 7,000KG MTOW or less.

Item 10 – Equipment and Capabilities

Circle to indicate the presence of serviceable equipment that the pilot is qualified to use and where applicable, has authorisations from the State of Registry:

N	no COM/NAV/Approach Aid equipment for the route to be flown or the equipment is unserviceable.

S	standard COM/NAV/Approach Aid equipment of VHF/ILS/VOR.
A	GBAS Landing System
B	LPV (APV with SBAS)
C	LORAN C
D	DME
E1	FMC WPR ACARS
E2	D-FIS ACARS
E3	PDC ACARS
F	ADF
G	GNSS
H	HF RTF
I	Inertial NAV
J1	CPDLC ATN VDL Mode 2
J2	CPDLC FANS 1/A HFDL
J3	CPDLC FANS 1/A VDL Mode A
J4	CPDLC FANS 1/A VDL Mode 2
J5	CPDLC FANS 1/A SATCOM (INMARSAT)
J6	CPDLC FANS 1/A SATCOM (MTSAT)
J7	CPDLC FANS 1/A SATCOM (Iridium)
K	MLS
L	ILS
M1	ATC SATVOICE (INMARSAT)
M2	ATC SATVOICE (MTSAT)
M3	ATC SATVOICE (Iridium)
O	VOR
P1	CPDLC RCP 400
P2	CPDLC RCP 240
P3	SATVOICE RCP 400
P4-P9	Reserved for RCP
R	PBN Approved
T	TACAN

U	UHF RTF
V	VHF RTF
W	RVSM Approved
X	MNPS
Y	VHF with 8.33 kHz channel spacing capability
Z	other equipment or capabilities (see Note 1).

Note 1: If the letter Z is used, specify the other equipment carried or other capabilities in Item 18, preceded by COM/, NAV/, and/or DAT/, as appropriate.

Note 2: If the letter R is used, specify the performance based navigation levels that can be met in Item 18 following the indicator PBN/.

Note 3: The NAIPS interface does not currently support the use of P1, P2 and P3. Operators may only have to declare the RCP capability for flights that will operate in airspace administrated by States that require it.

Enter 'G' (GNSS) and 'R' (PBN capability) in Item 10 for aircraft equipped with a GNSS enabled area navigation system with additional entries as appropriate. The correlation between Item 10 and Item 18 entries for common PBN approvals is summarised below:

	PBN Capability	**Item 10**	**Item 18**
Oceanic	RNAV10 (RNP10)	GR and I (if appropriate)	PBN/A1
	RNP4	GR	PBN/L1
Continental	RNP2	GZ	NAV/RNP2
Terminal	RNP1, all permitted sensors	GRDI	PBN/O1
	RNP1, GNSS	GR	PBN/O2
Approach	RNP APCH	GR	PBN/S1
	RNP APCH with Baro-VNAV	GR	PBN/S2
	RNP AR APCH with RF	GRI	PBN/T1
Precision Approach	GLS	AGZ	NAV/GLS

For the majority of Australian IFR operations the appropriate field 10 navigation entries will be:

S Standard COM/NAV/Approach Aid combination of VHF/VOR/ILS, and

R PBN capable, and

G GNSS, and

Z other equipment or capabilities (required to enable nomination of NAV/RNP2 in Item 18.

Surveillance Equipment

Circle N for Nil, or

Aircraft with ADS-B capability:

Notes:

1. *ADS-B capability indicated in a domestic flight notification is only for a capability suitable for ATC service. ADS-B equipment outputting a Source Integrity Level (SIL) of 1 (SIL=1) (e.g. TABS devices and EC devices) is not suitable for ATC service. Therefore an aircraft fitted with ADS-B equipment outputting SIL=1 should not enter an ADS-B code in Field 10b. See later note about indicating transponder capability.*
2. *Light Sport Aircraft (LSA), experimental aircraft and other eligible aircraft fitted non-TSO ADS-B equipment eligible for and outputting SIL=2 or SIL=3 may indicate a relevant ADS-B code in Field 10b.*

Enter up to two ADS-B codes: either 'L' or 'E' and 'B1' or 'B2'.

L SSR Transponder Mode S, including aircraft identification, pressure altitude, ADS-B Out and enhanced surveillance capability.

E SSR Transponder Mode S, including aircraft identification, pressure altitude and ADS-B Out capability.

B1 ADS-B "Out" capability using 1090MHz extended squitter

B2 ADS-B "Out" and "In" capability using 1090MHz extended squitter

Note: Enhanced surveillance capability is the ability of the aircraft to down-link aircraft derived data via a Mode S transponder.

Use the following table to determine the Field 10b entries for ADS-B transponder (use only one entry)

Mode S transponder with ADS-B

Field 10b Entry	Transponder Capability					
	Mode S (ADS-B)	Aircraft ID	Pressure Altitude	Enhanced Surveillance	ADS-B 1090 OUT	ADS-B 1090 IN
LB2	X	X	X	X	X	X
EB2	X	X	X		X	X
LB1	X	X	X	X	X	
EB1	X	X	X		X	

Aircraft without ADS-B capability:

Note: Aircraft fitted with a transponder together with a TABS device or EC device outputting SIL=1 should only enter a code in Field 10b appropriate for the transponder fitted to the aircraft.

Enter one SSR code representing the highest level of non-ADS-B surveillance capability available (in order highest is H then S, I, P, X, C and A is lowest).

H SSR Transponder Mode S, including aircraft identification, pressure altitude and enhanced surveillance capability.

S SSR Transponder Mode S, including both pressure altitude and aircraft identification capability.

I SSR Transponder Mode S, including aircraft identification, but no pressure altitude capability.

P SSR Transponder Mode S, including pressure altitude, but no aircraft identification capability.

X SSR Transponder Mode S with neither aircraft identification nor pressure altitude capability.

C SSR Transponder Mode C

A SSR Transponder Mode A

Note: Enhanced surveillance capability is the ability of the aircraft to down-link aircraft derived data via a Mode S transponder.

Use the following table (listed in order of highest to lowest capability) to determine to correct Field 10b entry for non-ADS-B transponder (use only one entry).

Mode S transponder without ADS-B

Field 10b Entry	Transponder Capability			
	Mode S (non-ADS-B)	Aircraft ID	Pressure Altitude	Enhanced Surveillance
H	X	X	X	X
S	X	X	X	
I	X	X		
P	X		X	
X	X			

Aircraft with ADS-C capability:

Enter up to two ADS-C codes: 'D1' and/or 'G1'

D1 ADS-C with FANS 1/A capabilities

G1 ADS-C with ATN capabilities

Note: The RSP specification(s), if applicable, will be listed in Item 18 following the indicator SUR/. Operators may only have to declare the RSP capability for flights that will operate in airspace administered by State that require it.

Item 13 – Departure Aerodrome

Item 16 – Destination Aerodrome and Total Estimated Elapsed Time

– Alternate Aerodrome

Enter Aerodrome abbreviation in four letters.

Requirements Enter the four letter authorised abbreviation then, without a space, the total estimated elapsed time as four figures in hours and minutes; e.g. 0340. Include any aerial work delay noted as DLE in Item 18.

For aerodromes without an authorised abbreviation, enter ZZZZ. In Item 18 write DEP/ (or as applicable "DEST/ ALTN/") followed by either the latitude and longitude of the aerodrome or bearing and distance from a location with an authorised abbreviation or, the first point of the route or the marker radio beacon if the aircraft has not taken off from the aerodrome.

In Item 18, enter the common name of the alternate location after RMK/.

Note 1: For bearing and distance, enter the designator of the location followed by three figures in degrees magnetic followed by three figures in nautical miles; e.g. BN270120 is a position 120NM, 270 degrees from Brisbane.

Note 2: Use of authorised aerodrome abbreviations for mobile locations may be suspended by NOTAM when not in the normal location. Pilots must use ZZZZ and provide location details when the aerodrome abbreviation is suspended.

Total EET

Enter — Total estimated elapsed time of the flight as four figures in hours and minutes; e.g. 0340 and include any aerial work delay noted as DLE in Item 18.

AFIL

AFIL (Flight Notification Filed in the Air) can be used instead of the departure aerodrome abbreviation when ATS services are only required for entry to, or to cross controlled airspace. (Estimated Off Blocks Time becomes the estimate for the point where the ATS service is to commence).

Note: For a flight plan received from an aircraft in flight, the total estimated elapsed time is the estimated time from the first point of the route to which the flight plan applies to the termination point of the flight plan.

Estimated Off Blocks Time

Enter — Estimated off blocks time (EOBT), or the estimate for the point where the ATS service is to commence (applicable for use with AFIL - as referred to above in the departure aerodrome section), in four figure UTC.

Requirements — Provide an EOBT for every flight stage as HHMM. All flights must also include DOF/ followed by the date of flight as YYMMDD at Item 18, even if the date of flight is the current day. EOBT/DOF more than 120 hours (5 days) in advance of the time of notification cannot be accepted. A change more than 30 minutes to a submitted EOBT must be advised to ATS or through NAIPS.

Item 15 – Cruising Speed

Enter	Enter TAS in knots or enter Mach number.
Requirements	Circle N, then enter zero and three figures for knots; e.g. 0180. Circle M, then enter zero and two figures for mach number to the nearest hundredth of a unit; e.g. 082.

Level

Enter	First planned cruising level.
Requirements	Enter either "A" followed by three figures to indicate altitude in hundreds of feet up to and including 10,000FT; e.g. A085; or, "F" followed by three figures to indicate flight levels above 10,000FT; e.g, F350.

Item 15 – Route

Enter	Details of the planned route, change of level, flight rules, and cruise climb.
Requirements for locations/ waypoints	For an aerodrome, use authorised abbreviation; e.g. YMBL for Marble Bar. For a navaid identifier, use published two or three letter abbreviation; e.g. CDU for Ceduna NDB. For a latitude and longitude identification, use degrees and minutes in an eleven character group; e.g. 2730S15327E. For a waypoint use assigned designator; e.g. DORSU. For bearing and distance, enter the identification of the significant point followed by three figures in degrees magnetic followed by three figures in nautical miles; e.g. BN270120 is a position 120NM, 270° from Brisbane.
Requirements for route	Check *AIP charts* and *DAH* for full route details and *ERSA Flight Planning Requirements* for specific route requirements/restrictions and city pair options. Where specific route requirements/restrictions are not specified, route details may be entered according to the following rules: a. Route details must start and end with DCT (direct); b. DCT must be following or preceded by one of the following points: (i) Navaid (ii) Waypoint; or

(iii) ARP, that is not the departure or destination location (unless a DLE is planned at the location).

c. Subsequent points should be described by ATS route designators where defined.

When planning via an intersection waypoint (black square) to change from one route to another, flight plan via:

a. the air route to the waypoint short of the intersection waypoint, then
b. direct to the intersection waypoint, then
c. direct to the first waypoint on the second air route, then
d. via the new air route.

ROUTE TYPE	EXAMPLE ENTRY
Flights outside designated ATS routes:	
Direct from departure point to destination without the use of navaids.	*For YAUR-YPMP:* DCT
Direct from departure point to destination with the use of navaids	*For YROM-YCMU* DCT ROM CMU DCT
From departure point to destination via published or non-published points	*For YBDV-YLRE* DCT BDV BDV062150 LRE DCT *or* DCT BDV 2440S14147E LRE DCT *or* DCT BDV YMOO LRE DCT
For survey work, include the points where the aircraft will enter and exit the survey area. *(See Note 2)*	*For YGLA-YGLA (via survey area):* DCT GLA BUD YGYM 2500S15100E GLA DCT

Flights on designated ATS routes:	
To or from locations with or without navigation aids.	*For YPAD-YLLE:* DCT AD H246 OOM DCT *For YSSY-YLHI:* DCT TESAT B450 LHI DCT
Via the SID or STAR transition point of the route. (*See Note 3*)	*For YBBN-YSSY* DCT SANEG H91 IGDAM H12 BOREE DCT *Where SANEG is the SID transition point from Brisbane and BOREE is the STAR transition point to Sydney* *For YSSY-YLHI:* DCT NOBAR B450 LHI DCT
Changing routes at a waypoint intersection	L503 IGEVO DCT LEKET DCT VIMAV N759

Note 1: Pilots should refer to ENR 1.1 para 5., "Air Route Specifications" and ENR 1.1 para 4. "Navigation Requirements" when planning a route.

Note 2: When planning to conduct survey work, a map of the survey area must be provided to ATS with the flight notification.

When planning survey work, include in Item 18 the expected delay en route (DLE) at the commencement of survey; e.g. DLE/GYM0130 indicates a delay at Gympie for 90 minutes.

Note 3: SID/STAR designators and instrument approach fixes/waypoints for Australian airports must not be entered. Designated ATS routes and published location identifiers or waypoints must be used instead.

Requirements for change of speed/level

Enter the significant point at which a change of speed (5% TAS or 0.01 Mach or more) or a change of level is planned to commence, followed by an oblique stroke and both the cruise speed and the level without a space between them; e.g. AYE/N0130A080, AS/M082F350. Both cruise speed and level must be entered even when only one of these quantities will be changed.

Requirements for change of flight rules	Enter details of a change to flight rules following the entry in Item 8 of Y or Z. Enter the location where the change will occur followed by a space and VFR or IFR; e.g. YBAF VFR. A change in level may also be included; e.g. ROM/ N0180A090 IFR.
Requirements for cruise climb/ block level reservation	Enter the letter C followed by an oblique stroke, the point at which the cruise climb or block level is planned to start, an oblique stroke, the speed to be maintained during the cruise climb or block level, AND the two levels defining the layer to be occupied during the cruise climb or block level, OR one level and the word PLUS; e.g. C/FERET/N0380F370F390, or C/FERET/N0380F370PLUS.

Note: Cruise Climb is not used in Australian administered airspace. Level clearances will be issued upon request, subject to availability.

Item 18

Enter Other information such as navaid training, block surveys and other plain language remarks of significance. Whenever possible, limit item 18 to 250 characters or less.

Note 1: Flight plans with excessive Item 18 data require manual processing to address an ATM system limitation, which may cause delay and an increased risk of processing error.

Note 2: ACARS and TCAS or ACAS are not required to be included in the flight notification.

Enter information in the sequence shown below:

STS/ Use for special aircraft handling, followed by one or more of the indicators below separated by a space e.g. STS/ MEDEVAC NONRVSM;

ALTRV – flight operated in accordance with an altitude reservation

ATFMX - flight approved for exemption from ATFM measures by ATC

FFR – fire-fighting

FLTCK – flight check for calibration of navaids

HAZMAT – flight carrying hazardous material

HEAD – flight engaged in, or positioning for, the transport of dignitaries with Head of State status

HOSP – medical flight declared by medical authorities

HUM – flight operating on a humanitarian mission

MARSA – flight for which a military entity assumes responsibility for separation of military aircraft

MEDEVAC – life critical medical emergency evacuation

NONRVSM – non RVSM-capable flight intending to operate in RVSM airspace

SAR – flight engaged in a search and rescue mission; and

STATE – for a flight engaged in domestic or international military services; or international customs or police services.

Note: Other reasons for special handling by ATS may be denoted under the designator RMK/.

PBN/ Followed by PBN capabilities. R must have been entered in Item 10. Include as many of the descriptors below, as apply to the flight without spaces e.g. PBN/A1L1T1. The field capacity is 16 characters only i.e. 8 entries. In order to make efficient use of the available capacity to present relevant aircraft capability use the following guidance:

- Only include one of the RNP APCH entries S1 or S2, not both
- Only include one of the RNP AR APCH entries T1 or T2, not both
- If RNAV 5 and B2, B3, B4 and B5 are applicable use B1, All Sensors. LORAN C (B6) is not required in Australia to qualify for B1.
- If a DME/DME/IRU specification is filed (C3, D4 or O4) do not file DME/DME (C3, D3 or O3) as well.

D

Descriptor	RNAV SPECIFICATION
A1	RNAV 10 (or RNP 10)
B1	RNAV 5 all permitted sensors (except LORANC)
B2	RNAV 5 GNSS
B3	RNAV 5 DME/DME
B4	RNAV 5 VOR/DME
B5	RNAV 5 INS or IRS
B6	RNAV 5 LORANC
C1	RNAV 2 all permitted sensors
C2	RNAV 2 GNSS
C3	RNAV 2 DME/DME
C4	RNAV 2 DME/DME/IRU

Descriptor	RNAV SPECIFICATION
D1	RNAV 1 all permitted sensors
D2	RNAV 1 GNSS
D3	RNAV 1 DME/DME
D4	RNAV 1 DME/DME/IRU
	RNP SPECIFICATION
L1	RNP 4
O1	Basic RNP 1 all permitted sensors
O2	Basic RNP 1 GNSS
O3	Basic RNP 1 DME/DME
O4	Basic RNP 1 DME/DME/IRU
S1	RNP APCH
S2	RNP APCH with BARO-VNAV
T1	RNP AR APCH with RF
T2	RNP AR APCH without RF

Note: RNP2 has not yet been allocated a PBN code. Enter RNP2 in NAV/ with G, R and Z in Field 10.

NAV/ Followed by navigation equipment or capabilities other than those listed for Item 10 or under PBN/ e.g. NAV/ RNP2. Z must have been entered in Item 10.

COM/ Followed by communication equipment and capabilities other than those listed for Item 10a. Use when Z has also been entered in Item 10a; e.g. COM/HF3452.

DAT/ Followed by data communication equipment and capabilities not specified in 10a. Use when Z has also been entered in Item 10a.

SUR/	Indicate surveillance equipment and capabilities not specified in 10b. Indicate as many RSP specification(s) as apply to the flight, using designator(s) with no space. Multiple RSP specifications are separated by a space. Example: RSP180 RSP400.
DEP/	when ZZZZ has been entered in Item 13 followed by latitude and longitude or bearing and distance from a location with an authorised abbreviation; e.g. DEP/ BN090120.
DEST/	when ZZZZ has been entered in Item 16 followed by latitude and longitude or bearing and distance from a location with an authorised abbreviation; e.g. DEST/ 2730S15327E.
DOF/	Followed by YYMMDD to indicate the date of flight. e.g. DOF/121115
REG/	Followed by the full aircraft registration; e.g. REG/ VHZFR.
EET/	For international flights that enter or leave the Australian FIR use EET/ to indicate the estimated elapsed time to the FIR boundary. Enter EET/ followed by the FIR boundary indicator and the estimated elapsed time in hours and minutes; e.g. EET/YMMM0130.
SEL/	Followed by the SELCAL Code, for aircraft so equipped.
TYP/	When an approved aircraft type designator has not been assigned and ZZZZ has been entered in Item 9, enter TYP/ followed by the aircraft type; e.g. TYP/Echo Mk1.
CODE/	Aircraft address (optional). Expressed in the form of an alphanumerical code of six hexadecimal characters e.g. CODE/7C0001.
DLE/	followed by the point where the aircraft will be operating and the estimated time in hours and minutes as a four figure group; e.g. DLE/MDG0030 RMK/MDG NDB indicates that the aircraft will be delayed at Mudgee for 30 minutes training on the NDB.
OPR/	Name of the aircraft operating agency, if different from the aircraft identification in item 7.

ORGN/	Followed by the originator's 8 letter AFTN address or other appropriate contact details such as a contact phone number when submitting a SARTIME.
PER/	Followed by the aircraft performance category as described in ENR 1.5 para 1.2; e.g. PER/B. IFR aircraft arriving at a controlled aerodrome must insert their performance category.
ALTN/	when ZZZZ has been entered in Item 16 followed by latitude and longitude or bearing and distance from a location with an approved abbreviation; e.g. ALTN/ 2700S15320E.
RMK/	When any other plain language remarks are required or deemed necessary. Where applicable, followed by one or more of the indicators below: **SARTIME**, followed by FOR ARR (for arrival) or FOR DEP (for departure), date/time as a six figure group, the authority (TO CENSAR) and location as an authorised aerodrome abbreviation, navaid identifier or latitude/ longitude. ZZZZ cannot be accepted for the location. Only one SARTIME per flight notification may be entered. If more than one SARTIME is required, then TBA can be entered, e.g. RMK/SARTIME FOR ARR 080430 TO CENSAR YROM or RMK/SARTIME FOR DEP TBA TO CENSAR YBMV. Pilots are also requested to submit contact telephone details under ORGN/ when available. **ATC APPROVED NIL ADSB,** insert if in receipt of an approval issued by Airservices in accordance with *GEN 1.5 Section 7.2.* **ADSB EXEMPT,** if in receipt of an individual CASA exemption or authorisation. **CSF,** if conducting a community service flight as defined in *GEN 2.2*. **FLT** Insert if flight numbers are used either in RTF phraseologies or for traffic sequencing, and are not entered in Item 7.

FORM Insert details of the aircraft taking part in a formation flight if more than one aircraft type or different RVSM approval is included in the formation. The number, type and wake turbulence category and RVSM approval of the second and subsequent types of aircraft are entered, separated by a plus sign; e.g. RMK/FORM 2PC9+4F18 M OPS IN R577, or RMK/FORM 2F18+2F18 W.

PIFR Insert PIFR as the first element of RMK/ to indicate that the pilot is rated to Private IFR. Include relevant FPAs applicable to flight within controlled airspace as per the table below:

FPA	Abbreviation		Example/Notes
	Prefix	Suffix	
Navigation Only	NAV		Enter equipment as per item 10 and RMK/PIFR NAV in item 18.
Night Flying	NGT		RMK/PIFR NGT
Instrument Departures	IDEP	SID	RMK/PIFR IDEP RMK/PIFR IDEP SID
Instrument Approaches (Single or Multi-engine as applicable to the aircraft being flown)	IAL	NDB, VOR, DME, DMEGNSS, RNAVGNSS, ILS, LOC	RMK/PIFR IAL NDB RMK/PIFR IAL DMEGNSS RMK/PIFR IAL RNAVGNSS RMK/PIFR IAL VOR, ILS
Visual circling approach	VSA		RMK/PIFR VSA Not required where other IAL FPA are also listed.
STAR	STAR	NDB, VOR, GNSS, DME	RMK/PIFR STAR GNSS
Holding	HLDG	NDB, VOR, GNSS, DME	RMK/PIFR HLDG VOR
Multiple FPA			RMK/PIFR NAV IAL RNAVGNSS HLDG VOR GNSS

Item 19 - Supplementary Information

Enter	Additional information relevant to the flight for search and rescue purposes (optional).	
E/	Endurance - Enter a 4-figure group giving fuel endurance in hours and minutes for each stage of flight.	
P/	Persons on board – Enter the total number of persons on board (passengers and crew) for each stage of flight. Enter TBN if the total number of persons is not known at the time of filing.	
R/	Emergency radio – Circle the following if carried:	
	U	UHF radio on 243.0MHz
	V	VHF radio on 121.5MHz
	E	ELT
D/	Dinghies – Enter the following:	
	NUMBER	Total number of dinghies carried.
	CAPACITY	Total capacity, in persons, of all dinghies.
	COVER	Circle if dinghies are covered.
	COLOUR	Colour of dinghies.
S/	Survival Equipment – Circle the following if carried:	
	P	Polar
	D	Desert
	M	Maritime
	J	Jungle
	Note: See ERSA – EMERGENCY PROCEDURES for further information	
J/	Jackets – Circle if life jackets carried and circle if equipped with the following:	
	L	Lights
	F	Fluorescein
	U	UHF radio on 243.0MHz
	V	VHF radio on 121.5MHz

A/ Aircraft colour and markings is used to record predominate colour and significant markings of the aircraft.

N/ Remarks – Indicate any other survival equipment carried and any other remarks regarding survival equipment.

C/ Pilot in command - Include telephone, mobile and email address and company name (if applicable).

Military Supplement

Item 18

Enter The following list of abbreviations shows those approved for use by the Military. These abbreviations must be used to indicate the type of flying activity to be conducted. No other abbreviations are to be used. All levels specified with these abbreviations indicate operations at or below that level. Levels below 1,000FT are to be treated as AGL.

EET/ All fighter/strike operations must enter EET/ for the LJR component of the flight. EET/is followed by the designator and the elapsed time in hours and minutes from the departure point to the significant point; e.g. EET/CG0108 2726S15333E0116 BN0120 indicates an elapsed time to Gold Coast of 68 minutes, Point Lookout 76 minutes and Brisbane 80 minutes.

Any en route delay time (DLE) must be accumulated with the estimated elapsed time associated with the route segment from the airwork position; e.g. EET/CG0108 2726S15333E0116 BN0140 DLE/2726S15333E 0020 indicates an estimated elapsed time to Gold Coast of 68 minutes, Point Lookout 76 minutes and Brisbane 100 minutes (including the 20 minutes airwork).

RMK/ followed by one or more of the indicators below:

TFR followed by a level to indicate Terrain Following Radar; e.g. RMK/TFR003.

LLN followed by a level to indicate Low Level Navigation; e.g. RMK/LLN010.

LLO followed by a level to indicate Low Level Operations; e.g. RMK/LLO030.

SVY followed by a level to indicate Aerial Survey; e.g. RMK/SVY050.

NVG followed by a level to indicate Night Vision Goggle exercise; e.g. RMK/NVG008.

AVM followed by a significant point and upper level of operation to indicate Abrupt Vertical Manoeuvres; e.g. RMK/AVM2515S14330EA090.

NOCOM followed by (time after ATD) + (time after ATD) CNL (agency) (frequency), to indicate that communications will be non-continuous for the specified period: e.g. RMK/NOCOM 10+34 CNL WLM APP 135 7, indicates that the aircraft will be NOCOM from 10 minutes after ATD until 34 minutes after ATD and will cancel NOCOM with Williamtown Approach on 135.7 MHz.

Note: There may be more than one NOCOM period annotated.

MILSPECREQ To indicate special requirements flights for military aircraft.

Pilots must include the reason for MILSPECREQ in:

a. STS/ for NONRVSM

b. RMK/ for:

(1) LTD COMNAV;

(2) LTD FUEL ENDCE;

(3) TO LAND BY TIME;

(4) AAR MARSA; or

(5) other purpose as decided by the military authority.

MARSA followed by the call sign of the aircraft or formation with whom MARSA will apply e.g. RMK/ MARSA PSTL. Use when STS/MARSA has also been entered in Item 18.

AAR followed by RVCP or anchor point, track designator or MAAA (as applicable) and MARSA call sign e.g. RMK/ AAR AMX100 MARSA BUCK4; or RMK/AAR CARBN W946 MARSA COLT.

APPENDIX 3
FLIGHT NOTE

FLIGHT NOTE

The holder of this Flight Note should alert/contact **JRCC Australia on 1800 815 257** if the pilot has not contacted the holder, to confirm their safety, prior to the **Alert Authorities Time** below.
Any delay could be crucial to the safety of the occupants of the aircraft.
Note: All times are local at each location

Final Destination:	**Alert Authorities Time:** (Local Time)	**Date:**

By supplying all available details below, search and rescue will be more efficient, potentially saving lives, time and cost.

Call-sign:	**Type:**	**Aircraft colour/markings:**	**Navaids:** (Carried & used, include GNSS)	**TAS:**

Pilot's Name:	**Mobile Ph:**	**Alternative Ph (if any):**

Emergency/Secondary/After Hours Contact (Name/Company/Location/Ph)**:**

Note: Complete a separate line for each flight sector

DEP AD/Point & Ph	**EOBT** (Local time)	**Route** (Turning points)	**DEST & Ph**	**POB**	**Endurance HR|MIN**	

Remarks (if any): (Other useful information to aid Search and Rescue - Mobile phone number of passengers/registration if different from call-sign)						
Emergency Equipment (tick box as appropriate)						
Survival Equipment	Life raft ☐ Capacity & colour:		First Aid ☐	Water ☐	Lifejackets ☐	Emergency Rations ☐
ELT/PLB/EPIRB	Fixed ☐	Portable ☐	**Insert HEX ID/UIN if known:**			
Flight monitoring/aircraft tracking		Fitted ☐	**Type:**			Nil ☐
Emergency recovery system		Parachute ☐	**Other:**			Nil ☐
Other signalling/Life-saving devices						

ENR 1.11 ADDRESSING OF FLIGHT PLAN MESSAGES

1. GENERAL

1.1 Flight plans are submitted to the briefing office in Melbourne, or transmitted directly via NAIPS or the AFTN (see AFTN address list at *GEN 3.4, APPENDIX 1*.)

1.2 Flight movement messages relating to traffic into or via the Brisbane and Melbourne FIRs should be addressed in accordance with the following table:

FIR or controlled aerodrome	Message address
Brisbane FIR	YBBBZQZX
Melbourne FIR	YMMMZQZX
Departure or destination aerodrome	[ICAO location code] ZTZX (see *GEN 3.4 APPENDIX 1* controlled aerodrome location codes)

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ENR 1.12 INTERCEPTION OF CIVIL AIRCRAFT

1. INTERCEPTION PROCEDURES

The following procedures and visual signals apply over the territory and territorial waters of Australia in the event of interception of an aircraft.

1.1 Action by intercepted aircraft

1.1.1 An aircraft which is intercepted by another aircraft must immediately:

a. follow the instructions given by the intercepting aircraft, interpreting and responding to visual signals in accordance with the tables in *Sections 1.3 and 1.4;*

b. notify, if possible, the appropriate ATS unit;

c. attempt to establish radio communication with the intercepting aircraft, or with the appropriate intercept control unit, by making a general call on the emergency VHF frequency 121.5MHz and repeating this call on the emergency UHF frequency 243.0MHz, if practicable, giving the identity and position of the aircraft and nature of the flight;

d. if equipped with SSR transponder, select code 7700, unless otherwise instructed by the appropriate ATS unit; and

e. if equipped with ADS-B or ADS-C, select the appropriate emergency functionality, if available, unless otherwise instructed by the appropriate ATS unit.

1.2 Radio communications during interception

1.2.1 If radio contact is established during interception but communication in a common language is not possible, attempts must be made to convey instructions, acknowledgment of instructions and essential information by using the following phrases and pronunciations and transmitting each phrase twice.

Phrases for use by INTERCEPTED aircraft		
Phrase	**Pronunciation**[1]	**Meaning**
CALL SIGN (call sign)[2]	**KOL** SA-IN (call sign)	My call sign is (call sign)
WILCO	**VILL**-KO	Understood. Will comply.
CAN NOT	**KANN** NOTT	Unable to comply.
REPEAT	REE-**PEET**	Repeat your instruction.
AM LOST	**AM LOSST**	Position unknown.
MAYDAY	**MAYDAY**	I am in distress.
HIJACK[3]	**HI-JACK**	I have been hijacked.
LAND (place name)	**LAAND** (place name)	I request to land at (place name).
DESCEND	DEE-**SEND**	I require descent.

1. *Syllables to be emphasized are printed in bold letters.*
2. *The call sign required to be given is that used in radiotelephony communications with ATS units and corresponding to the aircraft identification in the flight notification.*
3. *Circumstances may not always permit, nor make desirable, the use of the phrase "HIJACK".*

1.2.2 The phrases shown in the table below should be used by the intercepting aircraft and transmitted twice in the circumstances described in *para 1.2.1.*

1.2.3 If any instructions received by radio from any sources conflict with those given by the intercepting aircraft by visual signals, the intercepted aircraft should request immediate clarification while continuing to comply with the visual instructions given by the intercepting aircraft.

1.2.4 If instructions received by radio from any sources conflict with those given by the intercepting aircraft by radio, the intercepted aircraft should request immediate clarification while continuing to comply with the radio instructions given by the intercepting aircraft.

1.2.5 The visual signals for use in the event of interception are detailed in *Sections 1.3 and 1.4*.

Phrases for use by INTERCEPTING aircraft		
Phrase	**Pronunciation**[1]	**Meaning**
CALL SIGN	**KOL** SA-IN	What is your call sign?
FOLLOW	**FOL**-LO	Follow me.
DESCEND	DEE-**SEND**	Descend for landing.
YOU LAND	**YOU LAAND**	Land at this aerodrome.
PROCEED	PRO-**SEED**	You may proceed.

1. Syllables to be emphasized are printed in bold letters.

1.3 Signals initiated by intercepting aircraft and responses by intercepted aircraft

Series	INTERCEPTING Aircraft Signals	Meaning	INTERCEPTED Aircraft Response	Meaning
1	**DAY or NIGHT**- Rocking aircraft and flashing navigational lights at irregular intervals (and landing lights in the case of a helicopter) from a position slightly above and ahead of, and normally to the left of, the intercepted aircraft (or to the right if the intercepted aircraft is a helicopter) and, after acknowledgement, a slow level turn, normally to the left (or to the right in the case of a helicopter) on the desired heading. *Notes:* *1. Meteorological conditions or terrain may require the intercepting aircraft to reverse the positions and direction of turn given above in Serial 1.* *2. If the intercepted aircraft is not able to keep pace with the intercepting aircraft, the latter is expected to fly a series of race-track patterns and to rock the aircraft each time it passes the intercepted aircraft.*	You have been intercepted. Follow me.	**DAY or NIGHT** - Rocking aircraft, flashing navigational lights at irregular intervals and following.	Understood, will comply.

Series	INTERCEPTING Aircraft Signals	Meaning	INTERCEPTED Aircraft Response	Meaning
2	**DAY or NIGHT** - An abrupt break-away manoeuvre from the intercepted aircraft consisting of a climbing turn of 90 degrees or more without crossing the line of flight of the intercepted aircraft.	You may proceed.	**DAY or NIGHT -** Rocking the aircraft.	Understood, will comply.
3	**DAY or NIGHT**- Lowering landing gear (if fitted), showing steady landing lights and overflying runway in use, or if the intercepted aircraft is a helicopter, overflying the helicopter landing area. In the case of helicopters, the intercepting helicopter makes a landing approach, coming to hover near to the landing area.	Land at this aerodrome	**DAY or NIGHT** - Lowering landing gear, (if fitted), showing steady landing lights and following the intercepting aircraft and, if, after overflying the runway in use or helicopter landing area, landing is considered safe, proceeding to land.	Understood, will comply.

1.4 **Signals initiated by intercepted aircraft and responses by intercepting aircraft**

Series	INTERCEPTED Aircraft Signals	Meaning	INTERCEPTING Aircraft Response	Meaning
4	**DAY or NIGHT** - Raising landing gear (if fitted) and flashing landing lights while passing over runway in use of helicopter landing area at a height exceeding 300M (1,000FT) but not exceeding 600M (2,000FT) (in the case of a helicopter, at a height exceeding 50M (170FT) but not exceeding 100M (330FT)) above the aerodrome level, and continuing to circle runway in use of helicopter landing area. If unable to flash landing lights, flash any other lights available.	Aerodrome you have designated is inadequate	**DAY or NIGHT** - If it is desired that the intercepted aircraft follow the intercepting aircraft to an alternate aerodrome, the intercepting aircraft raises its landing gear (if fitted) and uses the *Series 1* signals prescribed for intercepting aircraft. If it is decided to release the intercepted aircraft, the intercepting aircraft uses the *Series 2* signals prescribed for intercepting aircraft.	Understood, follow me. Understood, you may proceed.
5	**DAY or NIGHT -** Regular switching on and off of all available lights but in such a manner as to be distinct from flashing lights.	Cannot comply	**DAY or NIGHT** - Use *Series 2* signals prescribed for intercepting aircraft.	Understood.
6	**DAY or NIGHT** - Irregular flashing of all available lights.	In distress.	**DAY OR NIGHT**- Use *Series 2* signals prescribed for intercepting aircraft.	Understood.

ENR 1.13 UNLAWFUL INTERFERENCE

1. PILOT ACTIONS

1.1 An aircraft which is being subjected to unlawful interference must endeavour to inform ATS of this fact, along with any deviation from the current flight plan and any other significant factors affecting the operation. SSR-equipped aircraft should use an appropriate code. Information pertinent to the safe conduct of the flight will continue to be transmitted by ATS and appropriate action taken to expedite the conduct of the flight.

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ENR 1.14 AIR TRAFFIC INCIDENTS

1. AUSTRALIAN TRANSPORT SAFETY BUREAU (ATSB)

1.1 About the ATSB

1.1.1 The ATSB is an independent Commonwealth government statutory agency. The ATSB is governed by a Commission and is entirely separate from transport regulators, policy makers and service providers.

1.1.2 The ATSB's function is to improve safety and public confidence in the aviation, marine and rail modes of transport through excellence in:

a. independent investigation of transport accidents and other safety occurrences;

b. safety data recording, analysis and research; and

c. influencing safety action, including through public communications, education and safety recommendations.

1.1.3 The ATSB is established by the *Transport Safety Investigation Act 2003 (TSI Act)* and conducts its investigations in accordance with the provisions of the Act. The fundamental objective of transport safety investigations is the prevention of accidents and incidents. Under the *TSI Act*, it is not a function of the ATSB to apportion blame or provide a means for determining liability. The ATSB does not investigate for the purpose of taking administrative, regulatory or criminal action.

1.1.4 The ATSB publishes all transport safety investigations, including identified safety issues and related safety actions, on the ATSB website www.atsb.gov.au.

1.2 Contact information

1.2.1 Reporting reportable matters

a. Immediately reportable matters via phone 1800 011 034 (International: +61 2 6230 4470).

b. Written reports: www.atsb.gov.au/mandatory/

1.2.2 General enquiries

a. Phone: 1800 020 616 (International: +61 2 6122 1600).

b. Email: atsbinfo@atsb.gov.au

1.2.3 REPCON voluntary confidential reporting scheme

a. Written reports: www.atsb.gov.au/voluntary/

b. Phone: 1800 020 505 (International: +61 2 6230 5135).

1.2.4 Witness report of an accident or incident

a. Contact via www.atsb.gov.au/witness/

b. Phone: 1800 992 986 (International: +61 2 6122 1600, Option1).

2. REPORTING OF ACCIDENTS AND INCIDENTS

2.1 Introduction

2.1.1 The *TSI Act* specifies that certain accidents and incidents, known as reportable matters, must be reported to the ATSB by responsible persons. Depending on the severity of the accident or incident and the type of operation, reportable matters are defined in the *Transport Safety Investigation Regulations 2021* (TSI Regulations) as either immediately reportable matters or routine reportable matters.

2.1.2 The ATSB is the repository for Australia's official aviation safety occurrence database. While the ATSB does not investigate all accidents and incidents, it still needs to be notified of safety occurrences so that the information can be used for safety research and analysis. The ATSB publishes a number of statistical and research reports based on reported occurrences.

2.1.3 The ATSB publishes a searchable database of de-identified aviation accidents and incidents since July 2003. The database can be used for research and analysis, and is available at www.atsb.gov.au/avdata/

2.2 Aircraft that must report reportable matters

2.2.1 All Australian aircraft operating inside and outside of Australia, including:

a. Australian-registered (VH) aircraft on the Civil Aviation Safety Authority (CASA) civil aircraft register.

b. Aircraft listed with sport aviation bodies.

c. Type 1 and Type 2 Remotely Piloted Aircraft (RPA) (type certified or an RPA greater than 250g that is not an excluded RPA as defined by the Civil Aviation Safety Regulations 1998 (CASR) Part 101).

2.2.2 All foreign civilian-registered aircraft operating inside Australia.

2.2.3 The following aircraft are excluded from reporting:

a. Australian Defence aircraft – being aircraft used by the Australian Defence Force (other than aircraft registered on the CASA civil aircraft register).

b. Exempt foreign aircraft – being aircraft used by the military, customs or police services of a foreign country;

c. An RPA that is not Type 1 RPA or Type 2 RPA (this means an excluded RPA as defined by *CASR Part 101*, or micro RPA below 250g).

d. Model aircraft.

e. Small (uncrewed) balloons and light (uncrewed) balloons.

f. Tethered balloons with no persons on board.

g. Kites.

h. Rockets (this is not considered an aircraft).

2.3 **Responsible persons for reporting safety matters to the ATSB**

2.3.1 Reportable matters known by responsible persons must be reported to the ATSB by that person unless that person believes, on reasonable grounds, that one or more other responsible persons:

a. have already reported the matter; or

b. will as soon as is reasonably practicable report the matter.

2.3.2 However, the ATSB still encourages all responsible persons with information about a reportable matter to report that information to the ATSB.

2.3.3 The following are responsible persons:

a. A crew member of the aircraft concerned.

b. The owner or operator of the aircraft.

c. A person performing an air traffic control service in relation to the aircraft.

d. A person performing a dedicated aerodrome rescue or firefighting service in relation to the aircraft.

e. A person who is licensed as an aircraft maintenance engineer and does any work in relation to the aircraft.

f. A member of the ground handling crew in relation to the aircraft.

g. A member of the staff of CASA.

h. The operator of an aerodrome.

i. A sport aviation body that administers aviation activities in relation to the aircraft.

2.4 **Reporting of accidents and incidents**

2.4.1 Immediately reportable matters must be reported by phone to the ATSB on 1800 011 034 (+61 2 6230 4470 international) as soon as is reasonably practical.

2.4.2 Immediately reportable matters also require a more detailed written report to be submitted within 72 hours. Written reports can be submitted via the ATSB's online reporting system www.atsb.gov.au/mandatory/ or via a data transfer from a company safety management system database.

2.4.3 Routine reportable matters only require a written report to be submitted within 72 hours. Written reports can be submitted via the ATSB's online reporting system www.atsb.gov.au/mandatory or via a data transfer from a company safety management system database.

3. REPORTABLE MATTERS

3.1 **Introduction**

3.1.1 Reportable matters are defined in the *TSI Regulations* via a combination of the type of safety occurrences (*see 3.1.2*) and the type of operation (*see 3.1.3*). The combination determines whether a safety occurrence is an immediately reportable matter, routine reportable matter, or not reportable.

3.1.2 The TSI Regulations refer to the following types of safety occurrences:

a. An aircraft accident.

b. A reportable serious aircraft incident.

c. A loss of a separation standard between aircraft.

d. A declaration of an emergency in relation to an aircraft.

e. A serious property damage incident (external).

f. An aircraft incident.

g. An aircraft incident (external).

3.1.3 The TSI Regulations refer to four categories of aircraft operations, each with different reporting obligations:

a. Category A (passenger transport) aircraft operation.

b. Category B (commercial non-passenger) aircraft operation.

c. Category C (non-commercial) aircraft operation.

d. Category D (type 2 RPA and certain uncrewed balloons) aircraft operation.

Note: The full title of Categories A to C (as shown in 3.1.3) primarily relate to crewed aircraft operations. Uncrewed aircraft, including remotely piloted aircraft (RPA), fit into these categories as follows:

a. Category A: RPA conducting passenger transport aircraft operations (future).

b. Category B: type 1 RPA operations (heavier than 25KG or with a type certificate) that is not an excluded RPA as per CASR Part 101.

c. Category C: no uncrewed aircraft operations.

d. Category D: type 2 RPA (at least 250g and under 25KG) that is not an excluded RPA as per CASR Part 101, and uncrewed balloons operations (payload of least 4KG).

3.1.4 See *Section 6.* for details of reportable matters required to be reported.

3.2 **Reportable matters for each type of operation category**

Category A operations have the most reporting requirements and Category D have the least reporting requirements. The below table shows the types of safety occurrences that are either immediately reportable matters (telephone as soon as reasonably practicable and written report within 72 hours), or routine reportable matters (written report within 72 hours), for each type of operation category.

Category	Immediately reportable matters (IRM)	Routine reportable matters (RRM)
Category A (passenger transport) aircraft operations	– Aircraft accident – Reportable serious aircraft incident – Loss of separation standard between aircraft – Declaration of an emergency in relation to the aircraft – Serious property damage incident (external)	– Aircraft incident – Aircraft incident (external)
Category B (commercial non-passenger) aircraft operations	– Aircraft accident – Loss of separation standard between aircraft – Serious property damage incident (external)	– Reportable serious aircraft incident – Declaration of an emergency in relation to the aircraft – Aircraft incident (external)
Category C (non-commercial) aircraft operations	– Aircraft accident – limited to fatal aircraft-related injuries, serious aircraft-related injuries or missing aircraft – Loss of separation standard between aircraft – Serious property damage incident (external)	– Aircraft accident other than reportable as an IRM – Reportable serious aircraft incident – Declaration of an emergency in relation to the aircraft – Aircraft incident (external)

Category	Immediately reportable matters (IRM)	Routine reportable matters (RRM)
Category D (type 2 RPA and uncrewed balloons) aircraft operations	– Aircraft accident – limited to fatal aircraft-related injuries or serious aircraft-related injuries – Serious property damage incident (external)	– Aircraft accident other than reportable as an IRM – Loss of a separation standard between aircraft

3.3 **Category A (passenger transport) aircraft operations**

3.3.1 Passenger transport operation, including the following when they are commercial operations:

a. Air transport operations (scheduled or non-scheduled).

b. Balloon transport operations.

c. Mining fly-in-fly-out operations.

d. Scenic flights/joy flights (where it is not a cost sharing flight).

e. Parachuting operations (where it is not a cost sharing flight).

f. RPA passenger carrying operations and advanced air mobility (future).

g. Aerial work operations that carry passengers who are not crew members (for example, marine pilot transfers).

h. Medical transport operation, including transporting medical patients, medical personnel, or supplies (blood, tissue etc).

i. Repositioning flights prior to conducting a substantive Category A operation.

3.4 **Category B (commercial non-passenger) aircraft operations**

3.4.1 Operation conducted for a commercial purpose – any non-passenger carrying crewed aircraft operation that is conducted for a commercial purpose, including:

a. Aerial work operations such as surveying, spotting, surveillance, aerial application, mustering, aerial photography, media operations, external load operations, search and rescue operations.

b. Flying training activities (such as training under a *CASR Part 141* or *Part 142* training organisation).

3.4.2 Crewed cargo transport operation.

3.4.3 Operating Type 1 remotely piloted aircraft (medium or heavy RPA (heavier than 25KG) or any RPA with a type certificate) that is not an excluded RPA as defined by *CASR Part 101*.

3.4.4 Repositioning flights prior to conducting a substantive Category B operation.

3.5 **Category C (non-commercial) aircraft operations**

3.5.1 All crewed non-commercial operations generally. This includes:

a. Recreational, sport, travel and pleasure flying.

b. Flights where the pilot shares equally in costs with passengers (cost sharing).

c. Community service flights.

d. An aircraft with a special certificate of airworthiness.

e. Operation (where the registered operator is an individual) without payment or reward.

3.6 **Category D (Type 2 RPA and certain uncrewed balloons) aircraft operations**

3.6.1 Operating Type 2 (very small or small) RPA (at least 250g and under 25KG) that is not an excluded RPA as defined by *CASR Part 101*.

3.6.2 Operating an uncrewed free medium or heavy balloon (payload at least 4KG).

4. DEFINITIONS AND EXAMPLES OF TYPES OF REPORTABLE MATTERS

4.1 Introduction

4.1.1 Reportable matters are safety occurrences that involve the operation of an aircraft, including preparation for departure and during disembarkation.

4.1.2 Specifically, in relation to a particular crewed aircraft, reportable matters must occur during the period:

a. Beginning when the aircraft is being prepared for take-off; and

b. ending when all passengers and crew members have disembarked after the flight.

4.1.3 For uncrewed aircraft, reportable matters must occur during the period:

a. Beginning when the aircraft is ready to move with the purpose of flight; and

b. ending when the aircraft comes to rest at the end of the flight and the primary propulsion system is shut down. In the case of an uncrewed aircraft without a primary propulsion system (such as a gas balloon), the end of flight is when the aircraft comes to rest at the end of the flight.

4.2 Emergency declaration

4.2.1 A declaration of an emergency, in relation to an aircraft, means:

a. A declaration, by a flight crew member of the aircraft, of an alert phase (PAN PAN) or a distress phase (MAYDAY); or

b. a declaration by an air traffic service provider of a distress phase (DETRESFA) in relation to the aircraft.

4.3 Loss of separation standard between aircraft

4.3.1 A situation where the recognised separation standard (vertical, lateral or longitudinal) between aircraft that are being provided with an Air Navigation Service Provider (ANSP) separation service is infringed. This includes any of the following:

a. Loss of procedural or surveillance separation standards.

b. Loss of prescribed runway or wake turbulence separation standards.

c. Visual reference is lost during visual separation by a pilot or air traffic controller in controlled airspace.

4.4 **Serious property damage incident (external)**

4.4.1 In relation to the operation of an aircraft, damage to property outside the aircraft that would cost at least $25,000 to repair or replace, that is caused by:

a. Contact with any part of the aircraft, including anything that is attached to the aircraft or that has become detached from the aircraft; or

b. direct exposure to jet blast, propeller wash or rotor downwash from the aircraft.

4.5 **Aircraft accident**

4.5.1 A person suffers a fatal aircraft-related injury (*see 5.1*) in relation to the operation of the aircraft; or

4.5.2 A person suffers a serious aircraft-related injury (*see 5.2*) in relation to the operation of the aircraft; or

4.5.3 The aircraft sustains damage or structural failure (*see 5.3*), or there are reasonable grounds for believing that the aircraft has sustained damage or structural failure, which:

a. Adversely affects the structural strength, performance or flight characteristics of the aircraft; and

b. would normally require major repair or replacement of the affected component.

Except for any of the following:

(i) Engine failure;

(ii) engine damage limited to a single engine (including damage to its cowlings or accessories);

(iii) damage to propellers, wing tips, antennas, probes, vanes, tyres, brakes, wheels, fairings, panels, landing gear doors or windscreens;

(iv) damage such as small dents or puncture holes to the aircraft skin;

(v) minor damage to main rotor blades, tail rotor blades or landing gear;

(vi) minor damage resulting from hail or bird strike (including holes in the RADOME); or

4.5.4 the aircraft is missing; or

4.5.5 the aircraft is completely inaccessible.

4.6 **Reportable serious aircraft incident**

4.6.1 A serious aircraft incident is an incident involving:

a. Circumstances indicating that there was a high probability of an aircraft accident (that is, a high probability of a fatal or serious aircraft-related injury and/or the aircraft sustaining damage or structural failure); or

b. an incident in the following table of incidents that have the potential to be classified as a serious incident.

Note: The International Civil Aviation Organization (ICAO) Annex 13 Attachment C list of examples of serious incidents is consistent with the TSI Regulations examples.

Note: A reportable serious aircraft incident includes a serious aircraft incident (as per 4.6.1) even when not in the list of examples in the following table, as well as any incident in the list of examples below (with clarifying notes) even if it might not meet the definition of a serious aircraft incident.

4.6.2 List of examples of reportable serious aircraft incidents

Reportable serious aircraft incident as defined by the TSI Regulations	Clarifying notes
A near collision requiring an avoidance manoeuvre to avoid a collision or an unsafe situation, or when an avoidance action would have been appropriate.	– Immediate evasive action was required or should have been taken to avoid a collision. – Includes during flight and on the ground.
A collision with anything other than an animal or a bird.	Unless the resulting damage or injury qualifies it as an accident, an aircraft collides with another aircraft or collides with terrain (including water, trees and wires), a person, structure or object.
Where controlled flight into terrain was only marginally avoided.	– Includes any legitimate terrain avoidance system warning ‘pull up’ type annunciation. – Excludes anticipated or spurious annunciations that occur in VMC.

Reportable serious aircraft incident as defined by the TSI Regulations	Clarifying notes
An aborted take-off on a closed, engaged or unassigned runway, or on a taxiway (other than an authorised operation by a helicopter).	
A take-off from a closed, engaged or unassigned runway, or from a taxiway (other than an authorised operation by a helicopter).	
A landing or attempted landing on a closed, engaged or unassigned runway, on a taxiway (other than an authorised operation by a helicopter), or on an unintended landing location (such as a road).	Excludes helicopter landings on taxiways by authorised helicopter operations.
The retraction of a landing gear leg during landing, or a wheels up landing.	Unless the resulting damage or injury qualifies it as an accident.
The dragging, during landing, of a wing tip, engine pod or any other part of the aircraft.	Unless the resulting damage or injury qualifies it as an accident. Includes during a go-around. Dragging includes any contact with the surface.
A significant failure to achieve predicted performance during take-off or initial climb.	
Fire or smoke in the cockpit, the passenger compartment or a cargo compartment, or engine fire, even if the fire was extinguished by the use of extinguishing agents.	– Includes explosions. – Excludes events involving fumes only.
An event requiring the emergency use of oxygen by a flight crew member.	Includes oxygen use for fumes, depressurisation.

Reportable serious aircraft incident as defined by the TSI Regulations	Clarifying notes
Aircraft structural failure, engine disintegration or uncontained turbine engine failure.	Significant structural airframe failures, excluding dents, missing panels and minor skin damage.
Multiple malfunctions of one or more aircraft systems seriously affecting the operation of the aircraft.	
Incapacitation of a flight crew member: – during a single pilot operation; or – during a multi pilot operation, if the safety of the operation is compromised because of a significant increase in workload for the remaining flight crew members.	– Incapacitation for any reason to the extent that their ability to perform any flight management role is significantly impaired. – Includes remote pilots.
Fuel quantity level or distribution situations (such as insufficient fuel, fuel exhaustion, fuel starvation, or inability to use all usable fuel on board) requiring the declaration of an emergency by the pilot.	Includes when declaration of emergency would be expected but not done.
A runway incursion where a collision is narrowly avoided.	Only includes runway incursions classified with severity index A as per the *ICAO Manual on the Prevention of Runway Incursions (Doc 9870)*

Reportable serious aircraft incident as defined by the TSI Regulations	Clarifying notes
A take-off or landing incident such as under shooting, overrunning or running off the side of a runway.	– Over-running refers to either a landing or rejected take-off where the aircraft continues beyond the effective operational length available for use by aircraft for landing at certified or registered aerodromes or the distance available for landing on an aircraft landing area. – Under-shoot refers to a landing that touches down prior to the designated landing area on a runway within the aerodrome perimeter.
Any of the following which caused, or could have caused, difficulties controlling the aircraft: – system failures (including loss of power or thrust) – weather phenomena – operations outside the approved flight envelope – any other occurrence.	– Relates only those circumstances that require immediate intervention. – Includes system alerts such as engine indications requiring inflight shutdowns, stall warnings during critical phases of flight. – Includes complete or partial loss of engine power.
Failure of more than one redundant system mandatory for flight guidance and navigation.	
The unintentional or emergency release of a slung load or any other load carried external to the aircraft.	

4.7 **Aircraft incidents**

4.7.1 Any event that is associated with the operation of an aircraft and affects, or could affect, the safety of the operation of the aircraft.

4.7.2 The degree to which an occurrence “affects or could affect” the safety of the operation of the aircraft should be understood to mean occurrences that, if not corrected, could endanger the aircraft or its occupants. To be clear, a responsible person is required to report an aircraft incident whether or not it was actually corrected or able to be corrected.

4.7.3 If an event, without correction, does not endanger the aircraft or its occupants, then the ATSB would consider that it has not affected or could not affect the safe operation of the aircraft (and therefore is an event that does not need to be reported).

4.7.4 Below are typical examples of aircraft incidents. For completeness, the list also includes occurrences that are aircraft accidents and/or reportable serious aircraft incidents, with notes to indicate this. However, the list is not exhaustive, and other operational events not in the list that meet the definition of an aircraft incident must be reported as aircraft incidents.

4.7.5 Operational related aircraft incident examples:

Incident	Definition	Clarifying note
Aircraft control incidents		
Hard landing	The vertical deceleration operational limits for the aircraft set out in the aircraft's flight manual are exceeded during the landing.	Note damage criteria for accidents.
Airframe overspeed	The airspeed limit has been exceeded for the current aircraft configuration as published in the aircraft manual.	– General airframe limits such as VNE; Extension speeds for flaps, slats, spoilers; Undercarriage extension speed. – Minor overspeeds are not incidents. In determining whether the overspeed is minor, both the degree and duration of the overspeed event should be taken into account.

Incident	Definition	Clarifying note
Stall warning	Any cockpit warning or alert that indicates the aircraft is approaching an aerodynamic stall.	– Warnings and alerts that are reportable include: – aural stall warnings – stick shaker activations – stick pusher activations – alpha protection or alpha floor activations. – Stall warnings that are intentionally generated during flying training or flight testing operations are not reportable. – Stall warnings are not reportable if the crew conclusively establish that the indication was false and provided that the false warning did not result in difficulty controlling the aircraft or a hazard arising from the response to the warning. – Note criteria for reportable serious aircraft incidents for stall warnings during critical phases of flight.

Incident	Definition	Clarifying note
Incorrect configuration	Where an aircraft system is incorrectly set for the current and/or intended phase of flight.	– Landing gear not extended in preparation for landing; – Inadvertent retraction of landing gear after landing; – Incorrectly setting the flaps or slats; – Incorrect application of carburettor heat (when carburettor icing occurred or was likely to have occurred). – Incorrectly setting the auto flight system mode; – Raising the flaps instead of the landing gear after becoming airborne. – Minor configuration issues that are not reportable include: – Momentary Enhanced Ground Proximity Warning System (EGPWS) flap and gear warnings related to incorrect settings in the EGPWS – Configuration warnings on the application of take-off power that are resolved by the crew and the aircraft subsequently departs without incident.

Incident	Definition	Clarifying note
Control issues	Occurrences where there were difficulties controlling the aircraft either airborne or on the ground.	– Minor control issues arising from: – Weather phenomenon (icing, severe turbulence, significant wind shear, thunderstorm encounter); – Wake turbulence; – Minor technical issues. – Note criteria for reportable serious aircraft incidents.
Weather events	– Icing issue that affect the performance of the aircraft; – Lightning strikes; – Turbulence or windshear that affect aircraft performance;	Note criteria for reportable serious aircraft incidents.

Incident	Definition	Clarifying note
Unstable approach	Flight crew continues an approach to landing when unstable (rather than conducting a go-around) when: a. there is an approved operator-specific stabilised approach criteria and that criteria is not met; or b. there is no approved operator-specific stabilised approach criteria, and one of the clarifying notes is not met under the following conditions: (i) IMC at or below 1,000FT AGL; (ii) VMC at or below 500FT AGL.	– An approach is unstable when one or more of the following criteria is not met: – the aircraft is on the correct flight path; – only small changes in heading/pitch are necessary to maintain the correct flight path; – the airspeed is not more than VREF + 20KT indicated speed and not less than VREF; – the aircraft is in the correct landing configuration; – sink rate is no greater than 1,000FT/minute; – power setting is appropriate for the aircraft configuration and is not below the minimum power for the approach as defined by the operating manual; – all briefings and checklists have been conducted; – instrument landing system approaches flown within one dot of the glideslope and localiser;

Incident	Definition	Clarifying note
		– a Category II or III instrument approach flown within the expanded localiser band; – a circling approach is flown with wings level on final approach when the aircraft reaches 300 feet above airport elevation. – Does not include when the approach was discontinued due to being unstable in the vicinity of 1,000FT (IMC) or 500FT (VMC).
Wheels-up landing	An aircraft contacts the intended landing area with the landing gear retracted.	– Includes intentional and unintentional wheels-up landing. – Includes amphibious aircraft landing on water with landing gear not retracted. *Note: All wheels-up landings are either accidents or reportable serious aircraft incidents.*

Incident	Definition	Clarifying note
Aircraft loading incidents		
Loading-related	Incorrect loading of an aircraft if the loading adversely affected, or could have affected, any of the following: a. the aircraft's weight; b. the aircraft's balance; c. the aircraft's structural integrity; d. the aircraft's performance; e. the aircraft's flight characteristics.	– Incorrect load sheet provided to flight crew; – Incorrect weight data input into flight computers; – Incorrect passenger numbers or seating; – Incorrect freight or baggage or incorrect loading; – Freight shifting inflight or unrestrained freight; – Incorrect fuel quantity or tank usage. – Does not include events that were detected and corrected before flight.
Cabin safety incidents		
Flight crew incapacitation	A flight crew member becomes incapacitated, such that their ability to perform normal flight duties is impaired.	– Incapacitation may be due to illness, injury, physiological or psychological factors, or environmental or other factors. Incapacitation's may be short term or persist for the duration of the flight, and includes both partial and total incapacitation. – Note criteria for reportable serious aircraft incidents.

Incident	Definition	Clarifying note
Depressurisation	Air pressure inside the cabin of a pressurised aircraft reduces to an extent that requires action by the flight crew.	Note criteria for reportable serious aircraft incidents.
Cabin injuries	Crew and passenger injuries and incapacitation's sustained as a direct result of an aircraft operation.	– Injuries to flight crew, cabin crew or passengers if they are the result of: – the movement of the aircraft due to a weather phenomenon like windshear or turbulence; – an abrupt aircraft manoeuvre, either airborne or on the ground – unrestrained objects. – Does not include workplace health and safety related injuries like slips, trips, falls, spillage of hot beverages, bumping head on overhead lockers or passenger illness unless the injury or illness is a direct result of the operation of the aircraft.

Incident	Definition	Clarifying note
Unrestrained occupants/objects	Aircraft occupants, equipment or objects are not appropriately restrained for the aircraft operation or phase of flight.	Includes crew or passengers standing during take-off or landing, passenger not wearing seat belts when required, and unrestrained galley equipment during critical flight phases.
Fire, smoke, and fumes		
Fumes	Smells, odours or odourless fumes not generally associated with normal aircraft operations.	– Includes fumes from: – dangerous goods – post compressor wash – oil / electrical smells – carbon monoxide from engine combustion or heating. – Excludes (if no other consequences): – fumes from galley oven contents or residual cleaning products – bird ingestion through air conditioning – passenger hand luggage contents. – Note criteria for reportable serious aircraft incidents when incapacitation results from fumes. – Note separate reporting requirements for declaration of an emergency.

Incident	Definition	Clarifying note
Fire	Any fire that has been detected and confirmed in relation to an aircraft operation.	All fire events are reportable serious aircraft incidents.
Smoke	Smoke is reported to be emanating from: a. inside the aircraft; or b. an external component of the aircraft.	– All smoke related events are reportable serious aircraft incidents. – Excludes passengers smoking inside the aircraft when the safety of the aircraft was not compromised.
Flight preparation / Navigation incidents		
Lost/unsure of position	Uncertainty by flight crew in relation to an aircraft's position where the flight crew request assistance from an external source.	Applies to pilots that request navigational assistance from ATC, other aircraft, or other person outside the aircraft in determining their current position.
Pre-flight planning and preparation	Inadequate or incorrect pre-flight planning or preparation of an aircraft for flight that affected, or if not corrected, could have affected the safety of the operation of the aircraft.	– Inadequate or incorrect fuel planning; – Navigation/flight planning issues including flight management computer data entry errors; – Deficiencies or erroneous data in navigation databases; – Inadequate pre-flight aircraft inspection.

Incident	Definition	Clarifying note
Flight below safe altitude	An aircraft is operated below the designated or planned lowest safe altitude for the in-flight conditions and phase of flight.	– Crew error to descend below the lowest safe altitude in IMC; – Aircraft operating below lowest safe altitude without knowledge of terrain in the vicinity; – ATC instruction to descend or operate below the lowest safe altitude; – Aircraft that continue the approach below minima with no visual reference to the runway; – Intentional unauthorised low flight.
VFR into IMC	An aircraft operating under the visual flight rules enters instrument meteorological conditions.	Note criteria for reportable serious aircraft incidents.
Fuel related incidents		
Fuel contamination	The presence of a foreign substance in fuel loaded into an aircraft.	– Fuel that is manufactured outside the technical specifications for the fuel grade or type; – Contamination of fuel in aircraft fuel tanks or aircraft fuel systems; – Incorrect fuel type for aircraft.
Fuel leak or venting	Unplanned loss of fuel from a fuel tank or fuel system.	Includes missing or insecure fuel cap.

Incident	Definition	Clarifying note
Fuel starvation	Fuel supply to the engine(s) is interrupted although there is usable fuel on board the aircraft.	– Mismanagement of the fuel system by the flight crew; – Mechanical failure involving the fuel system; – Unporting of the fuel standpipes during an aircraft manoeuvre. – Note criteria for reportable serious aircraft incidents.
Low fuel	The aircraft's supply of fuel becoming so low that the safety of the aircraft is compromised.	– Any occurrence where final reserve fuel is compromised. – Note the separate reporting requirements for a declaration of an emergency.
Fuel exhaustion	When the engine stops because the aircraft has become completely devoid of useable fuel.	All fuel exhaustion events are reportable serious aircraft incidents.

Incident	Definition	Clarifying note
Ground Proximity Warning		
TAWS/GPWS	A Terrain Avoidance and Warning System (TAWS) warning or alert.	– TAWS warning or alert, such as from a Ground Proximity Warning System (GPWS) and EGPWS – Excludes: – expected or spurious terrain warnings in VMC – momentary EGPWS flap and gear warnings related to incorrect settings in the EGPWS – momentary EGPWS glideslope warnings where a safe approach is continued. – Note criteria for reportable serious aircraft incidents.

Incident	Definition	Clarifying note
Ground operations' incidents		
Near collision on ground	An aircraft has a near collision with another aircraft, vehicle, structure, person or object while it is operating on the ground or water.	– Near collision on taxiway or apron with another aircraft, vehicle, person or object. *Note: A near collision on the runway strip is a reportable serious aircraft incident.* – Note criteria for reportable serious aircraft incidents for near collisions. *Note: All actual collisions are either an aircraft accident or a reportable serious aircraft incident depending on the level of damage.*
Foreign object damage/debris	Loose objects on a runway or in an aircraft that have caused, or have the potential to cause, damage to an aircraft.	– Aircraft panels/parts that have dislodged from aircraft or vehicles and are a potential hazard to other aircraft; – Tools or equipment left in an engine or avionics bay (found during preflight preparation); – Loose objects in the cockpit/aircraft that result a hazardous condition. – Excludes foreign objects on a runway that have had no interaction with an aircraft.

Incident	Definition	Clarifying note
Ground handling	Aircraft ground handling and aircraft servicing that have caused, or have the potential to cause, damage to the aircraft or injury.	– Any hazardous condition such as vehicles colliding with a stationary aircraft or fuel spillages resulting from ramp operations (after the aircraft is being prepared for flight and before all passengers and crew have disembarked). – Note separate reporting requirement for Aircraft loading incidents.
Jet blast/prop wash/rotor wash	Air disturbance from a ground-running aircraft propeller, rotor, or jet engine that have caused or have the potential to cause damage or injury.	Jet blast or propeller wash that has the likelihood of causing injury to persons, or damage to aircraft or other objects. Also includes instances of helicopter rotor down wash where helicopters are hover taxiing or flying at low level.
Runway events		
Depart, approach, or lands on wrong runway	An aircraft approaches an area other than that authorised or intended for landing or departure.	*Note: All take-off, aborted take-off, landing and attempted landings on closed or engaged runway, on a taxiway, or an unassigned runway, or roadway, are reportable serious aircraft incidents.*

Incident	Definition	Clarifying note
Runway incursions	Incorrect presence of an aircraft, vehicle or person on the protected area of a surface designated for the landing and take-off of aircraft.	Incorrect presence means: a. anything within the confines of the runway strip, irrespective of having an appropriate clearance, which hinders the operation of an arriving or departing aircraft; or b. an aircraft, vehicle or person entering the confines of the flight path without a clearance to do so, regardless of other aircraft operations. *Note: A runway incursion where a collision is narrowly avoided is a reportable serious aircraft incident.*
Runway excursion	An aircraft veers off the side of the runway or overruns the runway threshold.	Over-running refers to either a landing or a rejected take-off where the aircraft continues beyond the effective operational length available for use by aircraft for landing at certified or registered aerodromes or the distance available for landing on an aircraft landing area. *Note: A runway veer-off or overrun incident during take-off or landing is a reportable serious aircraft incident.*

Incident	Definition	Clarifying note
Runway under shoot	An aircraft attempting a landing touches down prior to the designated landing area on a runway within the aerodrome perimeter.	Includes aircraft landing on the runway surface before a displaced runway threshold. *Note: An aircraft touching down prior to the runway surface is a reportable serious aircraft incident.*
Terrain collisions		
Collision with terrain and near collision with terrain	Any collision or near collision with terrain or water, including wirestrikes, or where controlled flight into terrain is only narrowly avoided	*Note: All collisions and near collisions (where avoidance manoeuvre was required or appropriate) are aircraft accidents or reportable serious aircraft incidents.*
Ground strike	Part of the aircraft drags on, or strikes, the ground or water in an unintended manner during take-off or landing.	– A rotor or propeller makes contact with the ground during take-off or landing; – An engine pod, wingtip, or tail contacts the ground during take-off or landing. – Note criteria for accidents and reportable serious aircraft incidents.

Incident	Definition	Clarifying note
Miscellaneous warning devices		
Warning device other	An aural or visual aircraft warning device activates to alert the flight crew to a situation requiring immediate or prompt corrective action.	Includes chip detectors, propeller/rotor low RPM horns, fire warning devices, carbon monoxide detection. *Note: Separate reporting criteria for stall warnings, EGPWS/TAWS, TCAS/ ACAS, abnormal engine indications or low fuel warnings.*

4.7.6 Mechanical related aircraft incident examples:

Incident	Definition	Inclusions/exclusions
Airframe related incidents		
Doors/exits	An aircraft door (passenger, cargo, or emergency), or its component parts, has exhibited damage or has failed.	Excludes internal doors, like cockpit doors or lavatory doors.
Landing gear	An aircraft's landing gear, brakes, their component parts or tyres have failed.	– Landing gear collapse due to mechanical malfunction; – Use of emergency gear extension; – Tyre deflation; – Overheated or smoking brakes; – Faults with float type undercarriages; – Faults with emergency flotation devices in helicopters. – Excluded: – failure of landing gear indication bulbs – flat tyres while standing or taxiing. – Note criteria for accidents and reportable serious aircraft incidents for landing gear collapse, and smoke and fire.

Incident	Definition	Inclusions/exclusions
Windows	A window of the aircraft has exhibited damage or has failed.	– Separation of windows from the aircraft in flight; – Shattering, cracking, crazing, or delamination of a window; – Window heat arcing.
Wing/fuselage/ empennage	Part of the fuselage, wing, or empennage has structurally failed.	– Cracks; – Debonding; – Delamination. – Note criteria for reportable serious aircraft incidents for near collisions.
Objects falling from aircraft	Objects that are inadvertently detached or dropped from an airborne aircraft.	– Detached aircraft parts; – Inadvertent release of towed banner – Cameras or phones lost during open door operations. – Excludes objects deliberately detached or dropped from an aircraft. – Note criteria for reportable serious aircraft incidents loss of a slung load or any other load carried external to the aircraft.

Incident	Definition	Inclusions/exclusions
Power plant/propulsion related incidents		
Abnormal engine indications	Any indications that an engine is malfunctioning or operating outside normal parameters.	– Abnormal engine instrument readings, such as engine power output or temperature, oil pressure or temperature, fuel pressure; – Observation of abnormal sights or sounds by a crew member; – Engine overspeed or over-torque warnings.
Engine failure or malfunction	An engine malfunction that results in a total engine failure, a loss of engine power or rough running engine.	– Partial power loss (loss of RPM, surging, coughing); – Inflight shutdown of a failing engine; – Full power loss to an individual engine. – Note criteria for accidents and reportable serious aircraft incidents.
Propellers and rotor malfunctions	A failure or malfunction of any part of a propeller, helicopter rotor, or associated components.	– Failure of associated propeller accessories, such as feathering mechanisms, constant speed units, and reduction gearboxes; – General reports of damage to a propeller or rotor including delamination.
Transmission and gearboxes	The failure or malfunction of an aircraft transmission or gearbox or associated components.	

Incident	Definition	Inclusions/exclusions
Systems related incidents		
Systems failure	An aircraft system failure. This includes the following systems: a. air/pressurisation b. avionics/flight Instruments c. electrical d. fire protection e. flight controls f. fuel g. hydraulics h. anti-ice protection i. datalink (remotely pilot aircraft)	– Includes only when the operation of the aircraft was compromised or had the potential to compromise safety. – Note criteria for reportable serious aircraft incidents for failure of more than one redundant system.

4.7.7 Airspace related aircraft incident examples:

Incident	Definition	Inclusions / Exclusions
Aircraft separation related incidents		
Collision or near collision	A collision or near collision between aircraft, either airborne or on the runway strip.	– Near collision is where immediate evasive action was required or should have been taken. *Note: All aircraft collisions and near collisions are either accidents or reportable serious aircraft incidents.*
Airborne Collision Alert System warnings	An airborne collision avoidance system resolution advisory alert or equivalent.	– Includes all systems, such as TCAS. – Excludes traffic advisories.

Incident	Definition	Inclusions / Exclusions
Loss of separation	A situation where the recognised separation standard (vertical, lateral or longitudinal) between aircraft that are being provided with an ANSP separation service is infringed.	Note separate reporting requirements for all loss of separation incidents.
Loss of separation assurance	A separation standard existed, however, planned separation was not provided by the ANSP separation service.	– LOSA is an occurrence where separation existed but: – The potential conflict was not identified; or – Separation was not planned or was inappropriately planned; or – The separation plan was not executed or was inappropriately executed; or – Separation was not monitored or was inappropriately monitored.
Other separation issues	Aircraft separation is a concern but does not meet the definition of near collision.	Includes separation issues inside and outside controlled airspace.

4.8 **Aircraft incident (external)**

4.8.1 An aircraft incident (any occurrence associated with the operation of an aircraft that affects or could affect the safety of the operation of the aircraft) that originates from any of the following outside the aircraft:

a. Infrastructure

b. Flying and other objects

c. Animals or birds.

4.8.2 The following table lists typical examples of aircraft incidents (external). The list also notes when they meet additional requirements to be considered as reportable serious aircraft incidents or aircraft accidents. However, the list is not exhaustive, and other events not in the list that meet the definition of an aircraft incident (external) will also need to be reported.

4.8.3 Aerodrome related aircraft incident (external) examples:

Incident	Definition	Inclusions/exclusions
Aerodrome related	Where aircraft safety has been compromised due to the failure or inadequacy of any aerodrome infrastructure used in conjunction with aircraft operations, including: a. Runway lighting (including approach and slope guidance lighting); b. Runway, taxiway or apron surface areas; c. Signs and markings.	Must impact on the operation of an aircraft to be reportable.
Airways facility	Where aircraft safety has been compromised due to the failure or inadequacy of a facility used in connection with an aircraft operation, including: a. A navigation aid; b. Communications; c. Radar/surveillance (including ADS-B); d. Air traffic services; or e. General operational services (e.g. briefing, UNICOM, etc).	– Must impact on the operation of an aircraft to be reportable. – Navigation aids include ground-based and satellite-based aids.

4.8.4 Environment related aircraft incident (external) examples:

Incident	Definition	Inclusions/exclusions
Birdstrike or animal strike	A collision between an aircraft and an animal or a bird.	– Includes birdstrikes (including when the pilot suspects a birdstrike) where the aircraft is in flight, or taking off or landing anywhere. – Birdstrike and animal strikes where a carcass is found on a runway. – Excludes near strikes.
Collision or near encounter with flying object	Collision or near encounter with a flying object when the object interrupts flight or is sighted in the proximity of an aircraft.	– Includes collision or near encounter with any RPA or model aircraft or parachute. *Note: If an RPA is known to be a Type 1 or Type 2 RPA, then reporting requirements for collision and near collision apply.*
Interference from ground	– Near encounter between an airborne aircraft and an object when the object interrupts the aircraft's flight path, or – a laser or spotlight being directed at an airborne aircraft that affects the flight; or – any unauthorised communication, signal or system interference directed at an aircraft, air traffic control or air navigation aid.	– Must impact on the operation of an aircraft to be reportable. – Includes interference from laser pointer lights, kites, yacht masts, weather balloons.

Incident	Definition	Inclusions/exclusions
Other	Other environmental issues that affect the safety of a flight.	Includes insect nests or bodies, or dirt/sand blocking pitot tubes.

5. GUIDANCE FOR DETERMINING INJURY AND DAMAGE

5.1 Fatal aircraft-related injury

5.1.1 A fatal aircraft related injury is one where the person dies as a result of the injury within 30 days after the injury occurs, and the person suffers the injury as a result of:

a. Being in the aircraft during the operation; or

b. direct contact during the operation with any part of the aircraft, including parts which have become detached from the aircraft; or

c. direct exposure to jet blast during the operation.

5.1.2 Fatal aircraft related injuries do not include:

a. The injury results from natural causes.

b. The injury is intentionally self-inflicted.

c. The injury is intentionally caused by another person.

d. The injury is to a person who is a stowaway in a part of the aircraft that is not usually accessible to crew members or passengers after take-off.

5.2 Serious aircraft-related injury

5.2.1 A serious aircraft related injury is when a person suffers a serious injury as a result of:

a. Being in the aircraft during the operation; or

b. direct contact during the operation with any part of the aircraft, including parts which have become detached from the aircraft; or

c. direct exposure to jet blast during the operation.

5.2.2 A serious injury is defined as an injury where any of the following apply:

a. The injury requires, or would usually require, admission to hospital, for more than 48 hours, within 7 days after the day when the injury is suffered.

b. The injury involves a fracture of any bone (other than a simple fracture of fingers, toes or nose).

c. The injury involves lacerations which cause severe hemorrhage or severe nerve, muscle or tendon damage.

d. The injury involves injury to any internal organ.

e. The injury involves second or third degree burns, or any burns affecting more than 5% of the body surface.

f. The injury involves exposure to hazardous chemicals, infectious substances or injurious radiation.

5.2.3 Serious aircraft related injuries do not include when:

a. The injury results from natural causes.

b. The injury is intentionally self-inflicted.

c. The injury is intentionally caused by another person.

d. The injury is to a person who is a stowaway in a part of the aircraft that is not usually accessible to crew members or passengers after take-off.

5.3 **Guidance for the determination of aircraft damage**

5.3.1 *ICAO Annex 13 Attachment E* provides guidance for when determining if aircraft damage is considered to be an accident, including:

a. If an engine separates from an aircraft, the event is categorised as an accident even if damage is confined to the engine.

b. A loss of engine cowls (fan or core) or reverser components which does not result in further damage to the aircraft is not considered an accident.

c. Occurrences where compressor or turbine blades or other engine internal components are ejected through the engine tail pipe are not considered accidents.

d. A collapsed or missing RADOME is not considered an accident unless there is related substantial damage in other structures or systems.

e. Occurrences of missing flaps, slats and other lift augmenting devices, winglets, etc., that are permitted for dispatch under the Configuration Deviation List (CDL) are not considered accidents.

f. Retraction of a landing gear leg or wheels-up landing, resulting in skin abrasion only, when the aircraft can be safely dispatched after minor repairs or patching, and subsequently undergoes more extensive work to effect a permanent repair, would not be classified as an accident.

g. If the structural damage is such that the aircraft depressurises, or cannot be pressurised, the occurrence is categorised as an accident.

h. The removal of components for inspection following an occurrence, such as the precautionary removal of an undercarriage leg following a low-speed runway excursion, while involving considerable work, is not considered an accident unless significant damage is found.

i. Occurrences that involve an emergency evacuation are not counted as accidents unless someone receives serious injuries or the aircraft has sustained significant damage.

Note 1: Regarding aircraft damage which adversely affects the structural strength, performance or flight characteristics, the aircraft may have landed safely, but cannot be safely dispatched on a further sector without repair.

Note 2: If the aircraft can be safely dispatched after minor repairs and subsequently undergoes more extensive work to effect a permanent repair, then the occurrence would not be classified as an accident. Likewise, if the aircraft can be dispatched under the CDL with the affected component removed, missing or inoperative, the repair would not be considered as a major repair and consequently the occurrence would not be considered an accident.

Note 3: The cost of repairs, or estimated loss, such as provided by insurance companies may provide an indication of the damage sustained but should not be used as the sole guide as to whether the damage is sufficient to count the occurrence as an accident. Likewise, an aircraft may be considered a 'hull loss' because it is uneconomic to repair, without it having incurred sufficient damage to be classified as an accident.

6. DETAILS TO BE REPORTED

6.1 Introduction

6.1.1 The TSI Regulations outline details of immediately and routine reportable matters that are required to be reported to the ATSB. Practically, the below details should be reported when they are known to the reporter.

6.1.2 Investigation decision making and research using details of occurrences included in the ATSB database benefit from reporting as much detail about the reportable matters as is known. Reporters are encouraged to report all details known and to provide detailed descriptions.

6.2 Telephone notification of Immediately reportable matters

a. The type, model, nationality, registration marks and flight number (if any) of the aircraft the subject of the immediately reportable matter.

b. The kind of aircraft operation (with reference to the CASR Part number), and aircraft activity, that the aircraft was engaged in at the time of the immediately reportable matter.

c. The name and contact details of the operator of the aircraft.

d. As much detail as is known about the nature of the immediately reportable matter.

e. A description of any damage to the aircraft or any other property.

f. A description of any dangerous goods on board the aircraft.

g. Whether a person died, or was seriously injured.

h. Where the immediately reportable matter occurred (including a description of the location, or the geographical coordinates).

i. The aircraft's place of departure and destination.

j. The day and local time when the immediately reportable matter occurred.

k. A description of the following, in as much detail as is known:

(i) what happened;

(ii) how and why it happened.

l. In relation to the responsible person reporting the matter, their name and a method of contacting the person that will enable the person to be promptly contacted for ATSB to conduct further enquiries into the matter.

6.3 **Written reports (Immediately and Routine reportable matters) excluding bird/animal strikes**

Note: For bird/animal strikes see para 6.4.

a. The name and contact details of the person making the report.

b. The person's role in relation to the aircraft concerned.

c. The type, model, nationality, registration marks and flight number (if any) of the aircraft.

d. The name of the owner of the aircraft.

e. The name and contact details of the operator of the aircraft.

f. If the aircraft was under hire when the reportable matter occurred, the name of the hirer.

g. The name and nationality of each flight crew member, and the type and licence number of the licence held by each of them (for example, a Recreational Pilot Licence Number).

h. The Aviation Reference Number of each flight crew member if one has been issued by CASA.

i. The day and local time when the reportable matter occurred.

j. The place where the flight started (or intended to start); and the place where the flight was intended to end, and if different, the actual place the flight ended.

k. The purpose of the flight.

l. Unless the reportable matter occurred at an airport, the location of the aircraft immediately after the occurrence of the reportable matter, including the geographical coordinates of that location.

m. The number of persons on board the aircraft when the reportable matter occurred (separately for crew and passengers).

n. The nature of the reportable matter, including:
   (i) its outcome or effect on the flight of the aircraft;
   (ii) the phase of the aircraft's flight when the matter occurred;
   (iii) the weather conditions;
   (iv) the airspace designation;
   (v) the altitude at which the matter occurred;
   (vi) if the matter occurred at, or in relation to, an airport—the name of the airport;
   (vii) if the matter occurred on, or in relation to, a runway—the runway number;
   (viii) the causes of the occurrence (if known), including any human performance issues;
   (ix) any safety action carried out to prevent a recurrence of the matter; and
   (x) the nature and extent of any damage to the aircraft.

o. The physical characteristics of the area where the reportable matter occurred (e.g. the terrain, vegetation cover, and existence and location of any buildings, runways or aerodromes).

p. The flight rules under which the aircraft was operating at the time of the reportable matter.

q. The kind of aircraft operation the aircraft was engaged in at the time of the reportable matter (including the CASR Part number and general activity).

r. If the matter resulted in a death or serious injury, and the aircraft carried an emergency locator transmitter—whether the emergency locator transmitter was fixed or portable and whether it was activated at the time the immediately reportable matter occurred.

s. If the aircraft's pilot has died (and if available, for all accidents):
   (i) the pilot's date of birth; and
   (ii) the pilot's total flying hours on all aircraft and flying hours on the same type of aircraft.

t. If any crew members, passengers and/or other persons have been fatally or seriously injured:

(i) how many;

(ii) their names and nationalities; and

(iii) descriptions of their injuries.

6.4 **Written reports – aircraft incidents (external) bird/animal strikes**

a. The name and contact details of the person making the report.

b. The day and local time when the reportable matter occurred.

c. The nature of the reportable matter, including:

(i) if the matter occurred at, or in relation to, an airport, the name of the airport, and if it occurred on, or in relation to, a runway, the runway number; and

(ii) the nature and extent of any damage to the aircraft.

d. Any other information that the person making the report considers appropriate such as the species, number of birds/animals seen and struck, weather conditions.

7. PROTECTING OF EVIDENCE FOR INVESTIGATIONS

7.1 The *TSI Act* provides powers to ATSB transport safety investigators to obtain information necessary to conduct investigations into reportable matters. However, investigations always seek, where possible, to obtain information in cooperation with the owner.

7.2 The *TSI Act* makes provision for an offence if a person engages in reckless conduct that adversely affects an investigation that is either being conducted at that time or could be later conducted into an immediately reportable matter. Practically, this means that potential evidence relating to all immediately reportable matters must be protected, including recorded data (flight recorder, cockpit voice recorder, air traffic communications etc), wreckage and damaged parts, and related documents, unless authorised by the ATSB or informed that the ATSB will not be investigating the matter.

7.3 Similarly, the ATSB may issue a ‘protection order’ for the purpose of protecting evidence that might be relevant to an investigation. This may direct that specified things, or things in a specified class of things, must not be removed or interfered with except with the permission of the ATSB.

7.4 These provisions do not apply if the conduct was necessary for any of:

a. to ensure the safety of persons, animals or property

b. to remove deceased persons or animals from an accident site

c. to move a transport vehicle, or the wreckage of a transport vehicle, to a safe place

d. to protect the environment from significant damage or pollution.

ENR 2. AIR TRAFFIC SERVICES AIRSPACE

ENR 2.1 FIR, UIR, TMA

1. A full description of Australia's FIRs and TMAs is contained in AIP DAH. Further, diagrammatic presentation of Australia's TMAs is contained in aeronautical charts.

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ENR 2.2 OTHER REGULATED AIRSPACE

1. OPERATIONS IN OCEANIC CONTROLLED AIRSPACE

1.1 Clearance Requirements

1.1.1 Aircraft not in receipt of an airways clearance in another FIR are required to obtain an airways clearance prior to entering Australian administered Class A airspace. Clearances may be issued by the primary guard station prior to FIR entry. If a clearance is not received 15 minutes prior to entry, it may be obtained directly by one of the following methods:

a. Voice clearances may be obtained from Brisbane on INO-1, SEA-3, SP-6 or other advised frequencies as appropriate.

b. CPDLC clearances may be obtained from Brisbane (YBBB) or Melbourne (YMMM), as appropriate.

1.2 Mach Number Technique

1.2.1 Mach Number Technique (MNT) is the term used to describe the method of clearing successive jet aircraft, operating along the same track, to maintain specified mach numbers in order to maintain longitudinal separation.

1.2.2 The MNT may be used by ATC in the application of longitudinal separation standards on routes within oceanic controlled airspace. Pilots of jet aircraft must include the planned true Mach Number in their flight plans.

1.2.3 Pilots are required to readback and maintain an assigned Mach Number. ATC approval must be obtained before making any change. If an immediate temporary Mach Number change is essential (e.g. due to turbulence), ATC must be notified as soon as possible that such a change has been made.

1.2.4 MNT may also be applied by ATC in other Australian airspace.

2. SPECIAL PROCEDURES FOR IN-FLIGHT CONTINGENCIES

2.1 Introduction

2.1.1 Although all possible contingencies cannot be covered, the procedures in *paras 2.2*, *2.3* and *2.4* provide for the more frequent cases such as:

a. the inability to comply with assigned clearance due to meteorological conditions (*para 2.4* refers);

b. en route diversion across the prevailing traffic flow (for example, due to medical emergencies (*paras 2.2* and *2.3* refer)); and

c. the loss of, or significant reduction in, the required navigation capability when operating in an airspace where the navigation performance accuracy is a prerequisite to the safe conduct of flight operations, or pressurisation failure (*paras 2.2* and *2.3* refer).

2.1.2 The pilot must take action as necessary to ensure the safety of the aircraft, and the pilot's judgement must determine the sequence of actions to be taken, having regard to the prevailing circumstances. ATC will render all possible assistance.

2.2 General procedures

2.2.1 If an aircraft is unable to continue the flight in accordance with its ATC clearance, a revised clearance must be obtained, whenever possible, prior to initiating any action.

2.2.2 If prior clearance cannot be obtained, the following contingency procedures should be employed until a revised clearance is received. In general terms, the aircraft should be flown at an offset level and on an offset track where other aircraft are less likely to be encountered. Specifically, the pilot should:

a. leave the cleared track or ATS route by initially turning at least 30 degrees to the right or to the left, in order to establish and maintain a parallel, same direction track or ATS route offset by 5NM. The direction of the turn should be based on one or more of the following factors:

(1) Aircraft position relative to any organised track or ATS route system;

(2) The direction of flights and flight levels allocated on adjacent tracks;

(3) The direction to an alternate airport;

(4) Any strategic lateral offset being flown; and

(5) Terrain clearance;

b. maintain a watch for conflicting traffic both visually and by reference to ACAS (if equipped), leaving ACAS in RA mode at all times, unless aircraft operating limitations dictate otherwise;

c. turn on all aircraft exterior lights (commensurate with appropriate operating limitations);

d. keep the SSR transponder on at all times and, when able, squawk 7700 as appropriate and, if equipped with ADS-B or ADS-C, select the appropriate emergency functionality;

e. as soon as practicable, advise ATC of any deviation from their assigned clearance;

f. use means as appropriate (i.e. voice and/or CPDLC) to communicate during a contingency or emergency;

Note 1: When emergency situations are communicated via CPDLC, the controller may respond via CPDLC. However, the controller may also attempt to make voice contact with the aircraft.

Note 2: Guidance on emergency procedures for controllers, radio operators, and flight crew in data link operations can be found in the Global Operational Data Link (GOLD) Manual (Doc 10037).

g. if voice communication is used, use the radiotelephony distress signal (MAYDAY) or urgency signal (PAN PAN) preferably spoken three times, as appropriate; and

h. establish communications with and alert nearby aircraft by broadcasting at suitable intervals on the frequency in use and on 121.5MHz (or, as a backup, on the inter-pilot air-to-air frequency 123.45MHz):

(1) aircraft identification;

(2) flight level;

(3) position including the ATS route designator or the track code, as appropriate;

(4) the nature of the distress condition; and

(5) intentions.

Note: ATC will attempt to determine the nature of the emergency and ascertain any assistance that may be required. Subsequent ATC action with respect to that aircraft will be based on the intentions of the pilot and overall traffic situation.

2.3 **Actions to be taken once offset from track**

Note: The pilot's judgement of the situation and the need to ensure the safety of the aircraft will determine the actions outlined to be taken. Factors for the pilot to consider when deviating from the cleared track or ATS route or level without an ATC clearance include, but are not limited to:

a. operation within a parallel track system;

b. the potential for User Preferred Routes (UPR) parallel to the aircraft's track or ATS route;

c. the nature of the contingency (e.g. aircraft system malfunction); and

d. weather factors (e.g. convective weather at lower flight levels).

2.3.1 If possible, maintain the assigned flight level until established on the 5NM parallel, same direction track or ATS route offset. If unable, initially minimise the rate of descent to the extent that is operationally feasible.

2.3.2 Once established on a parallel, same direction track or ATS route offset by 5NM, either:

a. descend below FL 290, and establish a 500FT vertical offset from those flight levels normally used, and proceed as required by the operational situation or if an ATC clearance has been obtained, in accordance with the clearance; or

Note 1: Flight levels normally used are those contained in ENR 1.7 Section 5.

Note 2: Descent below FL 290 is considered particularly applicable to operations where there is a predominant traffic flow (e.g. east-west) or parallel track system where the aircraft's diversion path will likely cross adjacent tracks or ATS routes. A descent below FL 290 can decrease the likelihood of conflict with other aircraft, ACAS RA events and delays in obtaining a revised ATC clearance.

b. establish a 500FT vertical offset (or 1,000FT vertical offset if above FL 410) from those flight levels normally used, and proceed as required by the operational situation, or if an ATC clearance has been obtained, in accordance with the clearance.

Note: Altimetry System Errors (ASE) may result in less than 500FT vertical spacing (less than 1,000FT above FL410) when the above contingency procedure is applied.

2.4 **Weather deviation procedures**

2.4.1 **General**

2.4.1.1 When weather deviation is required, the pilot should communicate with ATC via voice or CPDLC. A rapid response may be obtained by either:

a. stating “WEATHER DEVIATION REQUIRED” to indicate that priority is desired on the frequency and for ATC response; or

b. requesting a weather deviation using a CPDLC lateral downlink message.

Note: Pilots are advised to contact ATC as soon as possible with requests for weather deviations to provide adequate time for the request to be assessed and acted upon.

2.4.1.2 When requesting a weather deviation or offset clearance, the flight crew should specify the distance off route with respect to the cleared route of the aircraft.

2.4.1.3 If the flight crew has received an off-route clearance and then requests and receives a subsequent off-route clearance, the new clearance supersedes the previous clearance (i.e. only the most recent clearance is valid).

Note: All clearances to deviate left or right of route (including multiple clearances) are by reference to the cleared route before any deviation commenced.

2.4.1.4 When necessary, the pilot should initiate the communications using the urgency call “PAN PAN” (preferably spoken three times) or by using a CPDLC urgency downlink message.

2.4.1.5 The pilot must inform ATC when weather deviation is no longer required, or when a weather deviation has been completed and the aircraft has returned to its cleared route. D

2.4.1.6 After communicating with ATC, the pilot:

a. must comply with the ATC clearance issued; or

b. when ATC are unable to issue a clearance for the requested deviation, should advise ATC of intentions and execute the procedures detailed in the following *para 2.4.2.1*.

2.4.2 **Actions to be taken if a revised ATC clearance cannot be obtained**

2.4.2.1 If the aircraft is required to deviate from track or ATS route to avoid adverse meteorological conditions and prior clearance cannot be obtained, an ATC clearance must be obtained at the earliest possible time. Until an ATC clearance is received, the pilot should take the following actions:

a. If possible, deviate away from an organised track or ATS route system;

b. Establish communications with and alert nearby aircraft by broadcasting, at suitable intervals on the frequency in use and on 121.5MHz (or, as a back-up, on the inter-pilot air-to-air frequency 123.45MHz):

(1) aircraft identification;

(2) flight level;

(3) position including ATS route designator or the track code; and

(4) intentions.

c. Watch for conflicting traffic both visually and by reference to ACAS (if equipped);

d. Turn on all exterior lights (commensurate with appropriate operating limitations);

e. For deviations of less than 5NM from the originally cleared track or ATS route, remain at a level assigned by ATC;

f. For deviations greater than or equal to 5NM from the originally cleared track or ATS route, when the aircraft is approximately 5NM from track, initiate a level change in accordance with the following table:

Originally cleared track or ATS route centre line	Deviations >5 NM	Level Change
EAST (000° – 179° magnetic)	LEFT RIGHT	DESCEND 300FT CLIMB 300FT
WEST (180° – 359° magnetic)	LEFT RIGHT	CLIMB 300FT DESCEND 300FT

g. If the pilot receives clearance to deviate from cleared track or ATS route for a specified distance and, subsequently, requests, but cannot obtain a clearance to deviate beyond that distance, the pilot should apply an altitude offset in accordance with the table above before deviating beyond the cleared distance;

h. When returning to track or ATS route, be at its assigned flight level when the aircraft is within approximately 5NM of the centre line; and

i. If contact was not established prior to deviating, continue to attempt to contact ATC to obtain a clearance. If contact was established, continue to keep ATC advised of intentions and obtain traffic information.

Note: If, as a result of actions taken under the provisions of this paragraph, the pilot determines that there is another aircraft at or near the same flight level with which a conflict may occur, then the pilot is expected to adjust the path of the aircraft, as necessary, to avoid conflict.

3. STRATEGIC LATERAL OFFSET PROCEDURES (SLOP) IN OCA

3.1 Aircraft operating in OCA within Australian administered airspace are authorised to use SLOP in accordance with the requirements detailed in *para 3.2.*

3.2 The following requirements apply to the use of SLOP:

a. The offset must only be applied by aircraft with automatic offset tracking capability.

b. The offset must be established in tenths of a nautical mile up to a maximum of 2NM to the RIGHT of track relative to the direction of flight.

Note: Offsets to the left of track are not permitted.

c. The offset must only be applied during the en route phase of flight.

d. The offset may only be used in OCA. Pilots must fly the route centreline for any portion of their route within CTA. Pilots must return to centreline before leaving OCA or, where the subsequent state does not allow SLOP, prior to leaving Australian administered airspace.

e. The offset must not be used in addition to diversions or other offsets; e.g. weather or wake turbulence.

f. The offset must not be applied at levels where obstacle clearance would be affected.

g. Identified aircraft:

(1) may continue an offset in OCA; and

(2) must advise ATC prior to initiating or changing an offset.

3.3 The decision to apply SLOP is the responsibility of the pilot in command - a clearance is not required. Except when an identified aircraft initiates or changes a lateral offset, pilots are not required to notify ATC that SLOP are being applied.

3.4 The use of SLOP is recommended in OCA for aircraft cruising at levels not in compliance with the table of cruising levels specified at *ENR 1.7 Section 5.*

3.5 OCA is depicted on Australian AIP charts as follows:

a. From 150NM SSE of YBBN south to 200NM NNE of YSSY – the blue line depicting the Class C airspace boundary.

b. South of 150NM SE of YSSY – the FIR boundary.

c. Remainder – the brown line depicting the Class E airspace boundary.

4. USE OF GNSS IN OCEANIC AND REMOTE AREAS

4.1 Australia has approved the use of GNSS as a primary means of navigation for oceanic/remote areas. Aircraft operators intending to utilise GNSS as a primary means of navigation in these areas must be approved by the State of Registry or State of the Operator, as appropriate.

4.2 To ensure navigation integrity, an appropriate en route GNSS prediction analysis, using the software provided by the GNSS manufacturer, must be conducted prior to each flight. For this analysis, the following parameters, or equivalents, must be used:

a. the route or airspace RNP, where published; or

b. a centreline space of:

(1) 20NM for flight in CTA, and

(2) 50NM for flight in OCA.

4.3 Aircraft meeting the requirements for the use of GNSS as a primary means of navigation in oceanic/remote continental airspace must indicate the approval in the flight notification. Such aircraft may flight plan on designated Area Navigation routes within Australian FIRs.

5. OPERATIONS IN OCEANIC AIRSPACE REQUIRING PERFORMANCE-BASED COMMUNICATION AND SURVEILLANCE (PBCS) AUTHORISATION

5.1 ICAO have introduced new provisions on PBCS calling for States to prescribe Required Communication Performance (RCP) and Required Surveillance Performance (RSP) specifications in their airspace as appropriate to the level of air traffic services provided. RCP 240 and RSP 180 are being applied in addition to Required Navigation Performance (RNP) specifications to certain separation minima in oceanic airspace by some States.

5.2 Australia has filed a difference with ICAO and has not yet implemented RCP or RSP.

5.3 Flights proceeding outside Australia should be aware that some States have prescribed RCP 240 and RSP 180 specifications in their administered airspace. Those States may require operators to be authorised by CASA to declare their RCP and RSP capabilities and for entering the respective descriptors in their flight plans.

5.4 CASA has issued an Instrument, number CASA 33/18 - *Required Communication Performance and Required Surveillance Performance (RCP 240 and RSP 180) Capability Declarations - Direction 2018*, which states the requirements for Australian operators to assess their compliance to RCP 240 and RSP 180 specifications and declare their readiness by entering the respective RCP and RSP descriptors in their flight plans.

5.5 Guidance on PBCS and the subject CASA Instrument is provided in *Advisory Circular (AC) 91-06 V1.0*.

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ENR 3. ATS ROUTES

ENR 3.1 LOWER ATS ROUTES

1. Details regarding lower ATS routes in Australia FIR can be found in the *DAH.*

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ENR 3.2 UPPER ATS ROUTES

1. Details regarding upper ATS routes in Australian FIRs can be found in *AIP DAH*.

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ENR 3.3 AREA NAVIGATION ROUTES

1. Details regarding Area Navigation routes in Australian FIR can be found in *AIP DAH*.

2. Information regarding which GNSS equipment can be used for different navigation specifications (i.e. RNP or RNAV) can be found in the the *CASR Part 91*, acceptable means of compliance and guidance material (AMC/GM) is available on the CASA website.

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ENR 3.4 HELICOPTER ROUTES

1. Details regarding helicopter routes and lanes-of-entry to specified airports can be found in *ERSA FAC*.

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ENR 3.5 OTHER ROUTES

1. Details of all routes in Australian FIRs can be found in *AIP DAH*. Information concerning flight planning restrictions can be found in *ERSA GEN*.

2. CONDITIONAL ROUTES (CDR)

2.1 CDR are promulgated to facilitate the flexible use of SUA.

2.2 CDR and associated conditions are published in *ERSA GEN FPR*. Conditions may include time, level and/or flight planning restrictions. Changes to the availability of a CDR (full route or portion) will be notified by NOTAM.

2.3 CDR are divided into categories according to their availability and flight planning requirements. A CDR may be established in one or more of the following categories:

a. **Category one (CDR1) - standard plannable CDR**
CDR1 are normally available for flight planning during times published in the *ERSA GEN FPR* and may be restricted to fixed altitude and/or flight level bands.

b. **Category two (CDR2) - non-standard plannable CDR**
CDR2 are normally available for flight planning when CDR1 are not available, or to address specific ATS conditions.

c. **Category three (CDR3) - non-plannable CDR**
CDR3 are not available for flight planning. CDR3 may be available from ATC at short notice when the published activity in the relevant SUA has paused, ceased, or for addressing specific ATS conditions.

2.4 Conditional route categories and Restricted Area (RA) conditional status should align. Where the flight planning allowance for a CDR category is different to the published RA conditional status, contact the National Airspace Management Office (NAMO) to ascertain the correct flight planning requirement.

2.5 For enquiries regarding CDR, contact the NAMO by emailing: namo@airservicesaustralia.com.

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ENR 3.6 EN ROUTE HOLDING

1. En route holding positions are identified on AIP aeronautical charts.

2. Further details regarding:

   a. En route holding may be found in *ENR 1.5 Section 3.*

   b. Holding Fuel requirements are contained in *ENR 1.1 Section 10.*

   c. ATFM procedures are contained in *ENR 1.9*.

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ENR 4. RADIO NAVIGATION AIDS/SYSTEMS

ENR 4.1 RADIO NAVIGATION AIDS - EN ROUTE

1. En route radio navigation aids for all ATS routes are identified in *AIP DAH* and on aeronautical charts.

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ENR 4.2 SPECIAL NAVIGATION SYSTEMS

1. There are no special navigation systems in existence in Australian FIRs.

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ENR 4.3 GLOBAL NAVIGATION SATELLITE SYSTEM (GNSS)

1. No SBAS signal should be used for navigation in Australian airspace until the Southern Positioning Augmentation Network (SouthPAN) is commissioned for safety of life applications including aeronautical navigation.

2. SouthPAN (Service Provider Identifier 8) is radiating test configuration SBAS L1 and DFMC services. Both services are not available for aeronautical navigation, and this is indicated by transmission of Message Type 0. A pilot in command becoming aware of irregular operation of a navigation function due to SouthPAN testing should report to the appropriate ATS unit in accordance with *ENR 1.1 para 10.10.1*. The keyword 'SouthPAN' should be included in any report.

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ENR 4.4 NAME-CODE DESIGNATORS FOR SIGNIFICANT POINTS

1. The five-letter codes and respective geographical coordinates for IFR waypoints in Australian FIR are listed in *ERSA GEN* and identified on en route charts.

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ENR 4.5 AERONAUTICAL GROUND LIGHTS – EN ROUTE

1. Aeronautical ground lights may indicate visual lanes of entry at some Class D aerodromes. If present, these lights are identified on Visual Terminal Charts (VTC).

2. Aerodrome beacons and hazard beacons for particular locations are identified in *ERSA FAC*.

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ENR 5. NAVIGATION WARNINGS

ENR 5.1 PROHIBITED, RESTRICTED, DANGER AND MILITARY OPERATING AREAS

1. GENERAL

1.1 Airspace in which a potential hazard to aircraft operations may exist, and all areas over which the operation of civil aircraft may be restricted are promulgated as follows:

a. Prohibited Area;

b. Restricted Area;

c. Danger Area; or

d. Military Operating Area (MOA).

1.2 Details of PRD and MOA are promulgated in the *DAH, ERSA* and/or NOTAM, and depicted on *AIP aeronautical charts*. PRD and MOA are identified using:

a. a letter, as follows

(i) P - Prohibited Area

(ii) R - Restricted Area

(iii) D - Danger Area

(iv) M - Military Operating Area

b. a number, to identify the specific area.

1.3 Temporary Restricted, Danger and Military Operating Areas, such as for military exercises, air shows and special events, are promulgated by AIP SUP, or as Brisbane FIR (YBBB) or Melbourne FIR (YMMM) NOTAM as appropriate for the location. Temporary areas are identified using:

a. a temporary airspace identifier:

(i) TRA – temporary Restricted Area

(ii) TDA – temporary Danger Area

(iii) TM – temporary Military Operating Area, and

b. an additional airspace identifier such as:

(i) the airspace name e.g. ‘Coral South’, and/or

(ii) numbers from the 900 series, i.e. numbers that start with a nine e.g. TRA940.

1.4 Unless otherwise specified, vertical limits are promulgated as AMSL when at or below the transition altitude, or as a flight level when above the transition altitude. The abbreviation “SFC” means the surface of the ground or water. “NOTAM” indicates that the vertical limits or hours of activation will be notified by NOTAM.

1.5 The promulgated vertical limits of Prohibited, Restricted and Military Operating Areas include all the buffers necessary for the protection of aircraft operating outside these areas. Therefore, the promulgated levels may be used by aircraft avoiding the areas, except where the vertical limit abuts controlled airspace, in which case, a clearance is required.

2. FLIGHT WITHIN PRD AND MOA

2.1 Flight within a Prohibited Area is not permitted in any circumstances.

2.2 Flight within active Restricted Areas is subject to the conditions published in *AIP, ERSA, DAH* and NOTAM. To obtain access to a Restricted Area, pilots must request approval from the controlling authority (see *ERSA*). When an ATC service is available within that airspace, approval may be requested from ATC directly, in the same manner as a clearance request to enter controlled airspace.

Note: Clearances may be withheld when activities hazardous to the aircraft are taking place, or when those activities require absolute priority.

2.3 All Restricted Areas have been allocated an RA conditional status to indicate the likelihood of obtaining a clearance to fly through the area.

Conditional Status	Meaning
RA1	Pilots may flight plan through the Restricted Area and, under normal circumstances, expect a clearance from ATC.
RA2	**Pilots must not flight plan through the Restricted Area** unless on a route specified in *ERSA GEN FPR* or under agreement with the controlling authority. Even so, a clearance from ATC is not assured. Other tracking may be offered through the Restricted Area on a tactical basis.
RA3	**Pilots must not flight plan through the Restricted Area** and clearances will not be available.
Note 1: NOTAM may be issued to indicate changes to the conditional status, which should be checked prior to flight planning.	
Note 2: In a declared emergency, every effort will be made to obtain approval to transit a Restricted Area, irrespective of its conditional status.	
Note 3: See ENR 3.5 para 2 Conditional Routes.	

2.4 If the conditional status is uncertain, treat the Restricted Area as conditional status RA3 and avoid the area.

2.5 Approval for flight within an active Danger Area outside controlled airspace is not required. However, it is the responsibility of the pilot in command to be aware of the dangerous activity and take appropriate precautions.

2.6 MOA are generally established to encompass intensive military activities, including live firing. For non-participating aircraft, flight within active MOA is generally only approved in exceptional circumstances.

MOA have the same entry approval requirements as Restricted Areas:

a. For all aircraft within Australian territory (including Australian territorial waters which are generally up to 12NM offshore); and

b. Only Australian registered aircraft outside Australian territory.

2.6.1 Outside Australian territory, foreign registered aircraft are not subject to MOA entry control, however flights electing to transit an active MOA without approval will not receive an air traffic control service and protection from dangerous activities occurring in the MOA cannot be provided. A limited FIS and SAR service will be available from civil ATS, in accordance with the relevant flight rules.

The Pilot in Command should take appropriate precautions against any safety risks that could arise from the flight, including contacting the MOA administering authority and identifying their operation. Where contact cannot be made with the MOA administering authority, squawk 7700 and broadcast at suitable intervals on 121.5 MHz including aircraft identification, position, flight level, and intentions.

2.7 Restricted, Danger and Military Operating Areas may be activated or deactivated at short notice. Access to a Restricted or Military Operating Area may be available if the activity for which it has been activated has ceased (early deactivation). It is a pilot responsibility to check current status with ATS.

ENR 5.2 MILITARY EXERCISE AND TRAINING AREAS AND AIR DEFENCE IDENTIFICATION ZONE (ADIZ)

1. MILITARY SENSITIVE AREAS

Note 1: These areas are applicable to military activities only.

Note 2: See also ENR 5.6.

Note 3: Levels are AGL unless specified otherwise.

1.1 Brisbane FIR

1.1.1 YB/S1- Katherine Gorge

Overfly at or above 2,500FT AMSL or remain outside the area bounded by 141554S 1322406E - 141454S 1323706E - 141954S 1323606E - 141954S 1322406E - 141554S 1322406E.

1.1.2 YB/S2 - Newcastle/Stockton

Overfly at or above 2,000FT or remain outside the area bounded by 325118S 1514218E - south bank of Hunter River 325224S 1514300E - 325224S 1514836E - 325942S 1514836E clockwise around arc radius 5NM centre 325554S 1514218E - 325118S 1514218E.

1.1.3 YB/S3 - Singleton City

Overfly at or above 2,000FT or remain outside a circle radius 1NM centre 322354S 1511106E.

1.1.4 YB/S4 - Legges Camp / Bombah Point

Overfly at or above 2,000FT or remain outside a circle radius 1NM centre 322954S 1521806E.

1.1.5 YB/S5 - Largs

Overfly at or above 2,000FT or remain outside a circle radius 1NM centre 324154S 1513606E.

1.1.6 YB/S6 - Siding Spring Observatory

Aircraft operating outside controlled airspace are to remain outside the following circular areas centre 311630S 1490406E.

Aircraft operating in controlled airspace are to remain on track unless otherwise instructed by ATC.

SFC to 10,000FT AMSL	3NM radius
10,000FT to FL200	5NM radius
FL200 to FL300	8NM radius
FL300 to FL450	12NM radius
above FL450	17NM radius

1.1.7 **YB/S7 - Leadville**

Overfly at or above 1,000FT or remain outside a circle radius 3NM centre 320100S 1493018E.

1.1.8 **YB/S8 - Saxonvale Mine**

All military aircraft are advised caution is to be exercised flying over coal mines adjacent to west boundary Singleton firing ranges/training areas.

Mine operator advises vertical danger to unspecified heights exists during regular explosive blasting in pits. Aircraft should avoid overflying pits, but no restrictions exist overflying mine buildings, or other parts of mine operation.

Saxonvale is the southern mine of the two.

1.1.9 **YB/S9 - Pallarenda**

Overfly at or above 2,000FT or remain outside a circle radius 1NM centre 191206S 1464624E. Aircraft departing/arriving Townsville are to comply with avoidance instructions issued by Townsville ATC.

1.1.10 **YB/S10 Great Barrier Reef Marine Park (Far Northern Section)**

Overfly at or above 3,000FT or remain outside the following circular areas radius 3NM and centres:

Ashmore Banks	114754S	1433818E
	115118S	1433536E
	115354S	1433906E
Boydong Islet and Reef	112854S	1430106E
	112954S	1430506E
Bird Islet	114554S	1430536E
Burkitt Island	135654S	1434518E
Bushy Island and Reef	111454S	1425306E
	113454S	1425406E
	114354S	1425806E
Cairncross Island and Reef	111354S	1425506E
Cholmondeley Islet and Reef	112154S	1430306E
Clack Island and Reef	140354S	1441506E
Combe Island and Reef	142412S	1445424E
Coquet Island	143224S	1445936E
Davie Reef	135842S	1442654E
Douglas Island and Reef	111354S	1425906E
Fife Island and Reef	133854S	1434306E
Hannah Island	135154S	1434306E
Hay Island	134006S	1434136E
Houghton Island	143124S	1445836E
Howick Island	143006S	1445836E
Ingham Island	142506S	1445318E
King Island and Reef	140554S	1442006E
Lloyd Island and Reef	124554S	1432406E
Lowrie Islet	131624S	1433548E
MacArthur Islands	114354S	1425906E
MacLennan Reef	112412S	1434636E
Magra Islet and Reef	115154S	1431706E
Morris Island and Reef	132854S	1434306E
Moulter Reef	112442S	1435936E
Newton Island	143024S	1445506E

Night Island	131054S	1433436E
Pelican Island and Reef	135442S	1435006E
Pipon Island and Reef	140654S	1443106E
Quoin Island	122418S	1432936E
Raine Island	113606S	1440154E
Reef	132612S	1435818E
Reef	132136S	1435736E
Saunders Islet	114154S	1431106E
	114254S	1431206E
Sinclair Island and Reef	143254S	1445406E
Sinclair Islet and Reef	110654S	1430106E
Sir Charles Hardy Island	115354S	1432806E
Stainer Islet	135724S	1435006E
Tydeman Reef	135900S	1443018E
Wallace Island and Reef	112654S	1430206E
West Hannibal Lighthouse	113554S	1425706E
Wilke Island	134618S	1433830E

1.1.11 **YB/S11 Great Barrier Reef Marine Park (Cairns Section)**

Overfly at or above 3,000FT or remain outside the following circular areas radius 3NM and centres:

Agincourt Reefs	155948S	1455000E
Beaver Reef	175006S	1462906E
Eagle Island and Reef	144154S	1452306E
Green Island	164530S	1455806E
Hastings Reef	163112S	1460124E
Hope Islands	154454S	1452706E
Low Islets	162306S	1453418E
Low Wooded Island and Reef	150554S	1452306E
Mackay Reef	160254S	1453906E
Michaelmas Reef	163618S	1455900E
Nymph Island	143854S	1451506E
Rocky Islets	145136S	1452900E

Stephens Island and Reef	174354S	1461006E
Taylor Reef	175006S	1463342E
Three Isles	150654S	1452536E
Turtle Island	144312S	1451130E
Two Islands	150112S	1452654E
Undine Reef	160654S	1454006E

1.1.12 **YB/S12 - Great Barrier Reef Marine Park (Central Section)**

Overfly at or above 3,000FT or remain outside the following circular areas radius 3NM and centres:

Brook Islands	180836S	1461718E
Dunk Island	175636S	1460924E
Eshelby Island	200112S	1483736E
Family Islands	180218S	1461048E
Hardy Reef	194542S	1491354E
John Brewer Reef	183806S	1470348E

1.1.13 **YB/S13 - Great Barrier Reef Marine Park (Mackay/Capricorn Section)**

Overfly at or above 3,000FT or remain outside the following circular areas radius 3NM and centres:

Bell Cay	214848S	1511454E
Bushy Island	205730S	1500506E
Bylund Cay	214712S	1522442E
Gannett Cay	215854S	1522830E
Frigate Cay	214424S	1522506E
Price Cay	214712S	1522636E
Redbill Island	205824S	1500506E
Thomas Cay	213854S	1522136E

1.1.14 **YB/S14 - Great Barrier Reef Marine Park (Southern Section)**

Overfly at or above 3,000FT or remain outside the following circular areas radius 3NM and centres:

Fairfax Islands	235124S	1522224E
Heron Island	232630S	1515448E
Hoskyn Islands	234812S	1521748E
Lady Elliot Island	240654S	1524236E
Lady Musgrave Island	235424S	1522336E
Masthead Islet	233212S	1514436E
North Reef	231100S	1515424E
North West Island	231748S	1514224E
One Tree Island	233024S	1520530E
Tryon Island	231448S	1514636E
Wilson Island	231818S	1515454E
Wreck Island	231954S	1515718E

1.1.15 **YB/S15 - Pelican Rock/Akens Island**

During the months October to March, the rock is a nesting area for pelicans. Pilots of low flying aircraft are requested to avoid the area over and near 222024S 1501536E as far as practicable during this period.

1.1.16 **Restricted and Military Operating Areas withing Marine Parks**

1.1.16.1 The existing military Restricted and Military Operating Areas that are located within a marine park have been identified in the Queensland State Marine Parks Act Zoning Plans. The Great Barrier Reef Marine Park Regulations 2019, as well as other Marine Park Zoning Plans, contain restrictions regarding aircraft operations in the vicinity of whales. Although the conduct of military operations within Restricted and Military Operating Areas is not constrained, military pilots are requested to comply with the following requirements based on the legislation and zoning plans, particularly during the migratory period between 1 August and 30 November:

a. not operate an aircraft below 1,000FT within 300M of a whale, and

b. not operate a helicopter below 2,000FT within 1KM of a whale.

1.1.16.2 Whenever possible, prior notification of military flights within a marine park should be provided to the Assistant Director Marine Parks, Queensland Department of Environment and Heritage, PO Box 155 North Quay Queensland 4002.

1.2 Melbourne FIR

1.2.1 YM/S1 - Altona

Overfly at or above 500FT and preferably by 1,500FT or remain outside a circle radius 0.5NM centre 375106S 1444824E

1.2.2 YM/S2 - Sarsfield

Overfly at or above 1,000FT or remain 5NM outside a circle radius 5NM centre ESL 040034 (374454S 1474248E).

1.2.3 YM/S3 - Clyde River Valley

Overfly at or above 4,000FT or remain outside an area bounded by a semicircle of 2NM radius centred on Nelligen (353854S 1500806E) thence 2NM either side of the Clyde river, Northbound to join a semicircle radius 2NM centre on Brooman (352754S 1501424E)

1.2.4 YM/S4 - Mt Stromlo Observatory

Aircraft operating outside controlled airspace are to remain outside the following circular areas centre 351912S 1490024E.

Aircraft operating in controlled airspace are to remain on track unless otherwise instructed by ATC.

SFC to 10,000FT AMSL	1NM radius
10,000FT to FL200	5NM radius
FL200 to FL300	8NM radius
FL300 to FL450	12NM radius
Above FL450	17NM radius

1.3 **Permanent Drilling Rigs**

1.3.1 Oil and gas platforms (land and sea) are declared sensitive areas. They are to be avoided by an airspace of 1,000FT vertically and 1NM horizontally. Permanent Drilling Rigs are listed in below.

Barracouta	381749S	1474034E
Whiting	381424S	1475226E
Snapper	381139S	1480132E
Marlin	381350S	1481315E
Tuna	381012S	1482510E
West Tuna	381133S	1482319E
Flounder	381841S	1482622E
Halibut	382416S	1481913E
Fortescue	382427S	1481641E
Cobia	382659S	1481833E
Mackerel	382846S	1482034E
Kingfish B	383550S	1481117E
Kingfish A	383548S	1480841E
West Kingfish	383536S	1480620E
Bream B	383102S	1475020E
Bream A	383000S	1474620E
Dolphin	382914S	1472239E
Perch	383409S	1471921E

2. MILITARY AAR AND AEW&C AIRSPACE

2.1 **General**

The conduct of Air to Air Refuelling (AAR) and Airborne Early Warning and Control (AEW&C) operations is based on the strict requirement that participating aircraft remain within specifically designated airspace. In general AAR and AEW&C operations in Australian airspace are conducted on tracks and in anchor patterns published in this document. Where mission requirements necessitate operations outside of published areas refer to Australian Defence Force Flight Information Handbook Australia (ADF FIHA) for planning information.

AAR operations will be conducted under instrument flight rules. The tanker aircraft is responsible for requesting altitude clearance and routing, if different than flight plan routing, for the receiver(s) and tanker beyond the AR Exit point. Throughout the refuelling operation, controller initiated heading assignments may not be effected without a clearance from the tanker. Each aircraft must receive a specific clearance prior to leaving the refuelling track/anchor if outside Military Restricted airspace.

2.2 **Explanation of terms**

2.2.1 **Refuelling tracks**

a. **RVIP – Rendezvous Initial Point.** A planned geographic point prior to the RVCP to which tankers and receivers time independently to effect an arrival at the RV control time.

b. **RVCP – Rendezvous Control Point.** The planned geographic point over which the receiver(s) arrive in the observation/astern position with respect to the assigned tanker.

c. **Navigation Checkpoints.** Designated points where required to provide a means for adequate navigation along track.

d. **Exit.** The point at which the refuelling track terminates.

e. **Communication/Rendezvous Plan (C/R Plan):**

(i) Primary UHF

(ii) Backup UHF

(iii) APN 69/134/135 Settings

(iv) APX 78/Encode/Decode

(v) TACAN Channels Receiver/Tanker

(vi) N/R = Not required

f. **Refuelling Altitudes.** The block of airspace within which refuelling operations may be conducted.

g. **Scheduling Unit.** The military unit responsible for scheduling AAR operations along the route.

h. **Assigned Centre.** Primary airspace control authority.

i. **SODAR.** Simultaneous Opposite Direction Air Refuelling.

2.2.2 **Refuelling Anchors**

a. **Entry Points.** Designated points where tanker aircraft may enter the anchor area without the assistance of radar. When tanker is actively controlled by ATC or GCI the tanker may proceed to the anchor point without crossing an Entry point.

b. **Anchor Point/Inbound Course/Leg length.** The geographic point upon which the anchor pattern is oriented and the inbound course to the anchor point and the length of the inbound leg.

c. **Anchor Pattern**. A left hand pattern with the four defining points of the pattern.

d. **Exit Point**. Designated points where tanker and receiver aircraft may depart the anchor area after refuelling is complete. Under positive radar control tanker and receiver aircraft may exit anchor pattern without crossing an exit point.

2.3 Amberley AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
AM11									
262906.0S 1540454.0E	262906.0S 1540454.0E	260606.0S 1544800.0E (AM11C) 048° 45NM	260606.0S 1544800.0E 254854.0S 1543636.0E 261148.0S 1535330.0E 262906.0S 1540454.0E	261148.0S 1535330.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M640C/D active, Brisbane (As assigned) if M640C/D deactivated	NOTAM
AIRSPACE: 261542S 1532954E (AMX100) - 244942S 1550206E (AMX101) - 253930S 1555400E (AMX102) - 264642S 1535330E (AMX103) - 261542S 1532954E (AMX100).									
REMARKS: Pattern size: 45NM x 20NM. Pattern exists within M640C and D (5000-NOTAM).									
AM13									
273136.0S 1551154.0E	273136.0S 1551154.0E	270654.0S 1555418.0E (AM13C) 045° 45NM	270654.0S 1555418.0E 265006.0S 1554200.0E 271448.0S 1545936.0E 273136.0S 1551154.0E	271448.0S 1545936.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M649B active, Brisbane (As assigned) if M649B deactivated	NOTAM
AIRSPACE: 274630.0S 1544430.0E (AMX104) - 271312.0S 1544430.0E (AMX105) - 263324.0S 1554654.0E (AMX106) - 261548.0S 1563030.0E (AMX107) - 271636.0S 1573106.0E (AMX108) - 285012.0S 1563754.0E (AMX109) - 274630.0S 1544430.0E (AMX104).									
REMARKS: Pattern size: 45NM x 20NM. Pattern exists within M649B (5000-NOTAM).									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
AM15									
300436.0S 1541700.0E	300436.0S 1541700.0E	292430.0S 1544042.0E (AM15C) 015° 45NM	292430.0S 1544042.0E 291524.0S 1542018.0E 295530.0S 1535630.0E 300436.0S 1541700.0E	295530.0S 1535630.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M661A active, Brisbane (As assigned) if M661A deactivated	NOTAM
AIRSPACE: 285212.0S 1534942.0E (AMX110) - 282912.0S 1535736.0E (AMX111) - 283348.0S 1540918.0E (AMX112) - 291324.0S 1550712.0E (AMX113) - 302812.0S 1542554.0E (AMX114) - 301854.0S 1532930.0E (AMX115) - 293924.0S 1533736.0E (AMX116) then along the minor arc of a circle radius 30.00NM centre 291154.0S 1532342.0E - 285212.0S 1534942.0E (AMX110).									
REMARKS: Pattern size: 45NM x 20NM. Pattern exists within M661A (5000-NOTAM).									
AM17									
290512.0S 1503006.0E	290512.0S 1503006.0E	283500.0S 1505924.0E (AM17C) 030° 45NM	283500.0S 1505924.0E 282206.0S 1504218.0E 285218.0S 1501254.0E 290512.0S 1503006.0E	285218.0S 1501254.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R639D active, Brisbane (As assigned) if R639D deactivated	NOTAM
AIRSPACE: 292212.0S 1500918.0E (AMX117) - 282830.0S 1495324.0E (AMX125) - 280830.0S 1510336.0E (AMX126) - 281942.0S 1510848.0E (AMX127) - 283530.0S 1512036.0E (AMX128) - 284430.0S 1512942.0E (AMX129) - 292218.0S 1500918.0E (AMX117).									
REMARKS: Pattern size: 40NM x 20NM. Pattern exists within R639D (10000-NOTAM).									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
AM19									
281136.0S 1501548.0E	281136.0S 1501548.0E	274112.0S 1504512.0E (AM19C) 030° 40NM	274112.0S 1504512.0E 272812.0S 1502806.0E 275830.0S 1495842.0E 281136.0S 1501548.0E	275830.0S 1495842.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R639C active, Brisbane (As assigned) if R639C deactivated	NOTAM
AIRSPACE: 281000.0S 1494800.0E (AMX118) - 265736.0S 1494406.0E (AMX130) - 271742.0S 1505342.0E (AMX131) - 272830.0S 1505306.0E (AMX132) - 274500.0S 1505536.0E (AMX133) - 280106.0S 1510012.0E (AMX134) - 280830.0S 1510336.0E (AMX135) - 282830.0S 1495324.0E (AMX136) - 281000.0S 1494800.0E (AMX118).									
REMARKS: Pattern size: 40NM x 20NM. Pattern exists within R639C (10000-NOTAM).									
AM22									
241348.0S 1543754.0E	241348.0S 1543754.0E	245112.0S 1551424.0E (AM22C) 125° 50NM	245112.0S 1551424.0E 243754.0S 1553048.0E 240030.0S 1545418.0E 241348.0S 1543754.0E	240030S 1545418E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647		NOTAM
AIRSPACE: 250654.0S 1551954.0E (AMX200) - 244942.0S 1550206.0E (AMX201) - 240824.0S 1541948.0E (AMX202) - 234554.0S 1544730.0E (AMX203) - 244542.0S 1554554.0E (AMX204) - 250654.0S 1551954.0E (AMX200).									
REMARKS: Pattern size: 50NM x 20NM. Airspace size: 80NM x 30NM. No current procedure available for anchor AAR operations outside Military controlled airspace.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
AM24									
262412.0S 1564712.0E	262412.0S 1564712.0E	270106.0S 1572500.0E (AM24C) 125° 50NM	270106.0S 1572500.0E 264730.0S 1574130.0E 261036.0S 1570342.0E 262412.0S 1564712.0E	261036S 1570342E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647		NOTAM
AIRSPACE: 271636.0S 1573106.0E (AMX205) - 261648.0S 1563124.0E (AMX206) - 255612.0S 1565630.0E (AMX207) - 265506.0S 1575700.0E (AMX208) - 271636.0S 1573106.0E (AMX205).									
REMARKS: Pattern size: 50NM x 20NM. Airspace size: 80NM x 30NM. No current procedure available for anchor AAR operations outside Military controlled airspace.									
AM26									
304348.0S 1560036.0E	304348.0S 1560036.0E	300312.0S 1562306.0E (AM26C) 017° 45NM	300312.0S 1562306.0E 295430.0S 1560218.0E 303506.0S 1553942.0E 304348.0S 1560036.0E	303506S 1553942E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647		TBA
AIRSPACE: 295142.0S 1563542.0E (AMX209) - 305936.0S 1555818.0E (AMX210) - 304630.0S 1552654.0E (AMX211) - 293842.0S 1560436.0E (AMX212) - 295142.0S 1563542.0E (AMX209).									
REMARKS: Pattern size: 45NM x 20NM. Airspace size: 75NM x 30NM. No current procedure available for anchor AAR operations outside Military controlled airspace.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
AM28									
283530.0S 1494700.0E	283530.0S 1494700.0E	274606.0S 1493748.0E (AM28C) 344° 50NM	274606.0S 1493748.0E 274918.0S 1491536.0E 283848.0S 1492436.0E 283530.0S 1494700.0E	283848.0S 1492436.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL200/ FL300	33SQN OPS 0753629647		NOTAM
AIRSPACE: 273512.0S 1490724.0E (AMX213) - 272942.0S 1494548.0E (AMX214) - 281000.0S 1494800.0E (AMX118) - 284900.0S 1495924.0E (AMX215) - 285424.0S 1492142.0E (AMX216) - 273512.0S 1490724.0E (AMX213).									
REMARKS: Pattern size: 50NM x 20NM. Airspace size: 80NM x 30NM. No current procedure available for anchor AAR operations outside Military controlled airspace.									
AM83									
254830.0S 1564106.0E	254830.0S 1564106.0E	250012.0S 1555254.0E (AM83C) 305° 70NM	250012.0S 1555254.0E 251336.0S 1553624.0E 260200.0S 1562442.0E 254830.0S 1564106.0E	260200.0S 1562442.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 244542.0S 1554554.0E (AMX900) - 255612.0S 1565630.0E (AMX901) - 261648.0S 1563124.0E (AMX902) - 261548.0S 1563030.0E (AMX107) - 253930.0S 1555400.0E (AMX102) - 250654.0S 1551954.0E (AMX903) - 244542.0S 1554554.0E (AMX900).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
AM84									
291012.0S 1495330.0E	291012.0S 1495330.0E	271754.0S 1493242.0E (AM84C) 340° 70NM	271754.0S 1493242.0E 272106.0S 1491030.0E 291324.0S 1493100.0E 291012.0S 1495330.0E	291324.0S 1493100.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 292806.0S 1492754.0E (AMX905) - 270348.0S 1490148.0E (AMX137) - 265736.0S 1494406.0E (AMX138) - 281000.0S 1494800.0E (AMX118) - 292212.0S 1500918.0E (AMX117) - 292806.0S 1492754.0E (AMX905).									
REMARKS: Pattern size: 113NM x 20NM. Airspace size: 146NM x 30NM.									

2.4 Darwin AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
DN81									
102630.0S 1331236.0E	102630.0S 1331236.0E	094524.0S 1321530.0E (DN81C) 302° 70NM	094524.0S 1321530.0E 100142.0S 1320336.0E 104248.0S 1330042.0E 102630.0S 1331236.0E	104248.0S 1330042.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 093230.0S 1320612.0E (DNX900) - 103118.0S 1332754.0E (DNX901) - 110436.0S 1330330.0E (DNX902) then along the minor arc of a circle radius 150.00NM centre 122524.0S 1305424.0E (DN/DME) - 100254.0S 1314354.0E (DNX903) - 093230.0S 1320612.0E (DNX900).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
DN82									
145848.0S 1291712.0E	145848.0S 1291712.0E	140900.0S 1282600.0E (DN82C) 312° 70NM	140900.0S 1282600.0E 142312.0S 1281130.0E 151306.0S 1290236.0E 145848.0S 1291712.0E	151306.0S 1290236.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 141600.0S 1275654.0E (DNX904) - 135024.0S 1282318.0E (DNX905) then along the minor arc of a circle radius 170.00NM centre 122524.0S 1305424.0E (DN/DME) - 145912.0S 1293836.0E (DNX906) - 152718.0S 1290954.0E (DNX907) - 141600.0S 1275654.0E (DNX904).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									
DN83									
131742.0S 1275854.0E	131742.0S 1275854.0E	120730.0S 1275230.0E (DN83C) 352° 70NM	120730.0S 1275230.0E 120918.0S 1273212.0E 131930.0S 1273830.0E 131742.0S 1275842.0E	131930.0S 1273830.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 115442.0S 1272548.0E (DNX908) - 115118.0S 1280418.0E (DNX909) then along the minor arc of a circle radius 170.00NM centre 122524.0S 1305424.0E (DN/DME) - 133136.0S 1281354.0E (DNX910) - 133500.0S 1273442.0E (DNX911) - 115442.0S 1272548.0E (DNX908).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

2.5 EAXA AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
NW81									
365906.0S 1521524.0E	365906.0S 1521524.0E	360924.0S 1531654.0E (NW81C) 030° 70NM	360924.0S 1531654.0E 355518.0S 1525918.0E 364454.0S 1515742.0E 365906.0S 1521524.0E	364454.0S 1515748.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 360212.0S 1533412.0E (NWX900) - 371312.0S 1520630.0E (NWX901) - 364618.0S 1513318.0E (NWX902) then along the minor arc of a circle radius 120.00NM centre 345700.0S 1503200.0E (UXUTI) - 352906.0S 1525306.0E (NWX903) - 360212.0S 1533412.0E (NWX900).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
NW82									
364212.0S 1503706.0E	364212.0S 1503706.0E	353224.0S 1502930.0E (NW82C) 342° 70NM	353224.0S 1502930.0E 353406.0S 1500506.0E 364354.0S 1501218.0E 364212.0S 1503706.0E	364354.0S 1501218.0E	N/A	FL260/ FL320	2SQN	Melbourne	TBA
AIRSPACE: 351700.0S 1503412.0E (NWX904) - 365642.0S 1504500.0E (NWX905) - 365918.0S 1500736.0E (NWX906) - 351930.0S 1495730.0E (NWX907) - 351700.0S 1503412.0E (NWX904).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

2.6 Edinburgh AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
ED01									
314500.0S 1444600.0E	314500.0S 1444600.0E	303500.0S 1444500.0E (ED01C) 350° 70NM	303500.0S 1444500.0E 303448.0S 1441030.0E 314436.0S 1441106.0E 314500.0S 1444600.0E	314436.0S 1441106.0E	N/A	FL190/ FL240	33SQN	Melbourne	NOTAM
AIRSPACE: 301818.1S 1432613.4E (EDX100) - 302123.9S 1452143.3E (EDX101) - 320135.6S 1451905.1E (IBODA/WPT) - 315823.9S 1432132.8E (EDX102)									
REMARKS: Pattern size: 70NM x 30NM. Airspace size: 100NM x 100NM.									

2.7 Tindal AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
TN11									
143812.0S 1323918.0E	143812.0S 1323918.0E	141530.0S 1321900.0E (TN11C) 315° 30NM	141530.0S 1321900.0E 142524.0S 1320718.0E 144806.0S 1322736.0E 143812.0S 1323918.0E	144806.0S 1322736.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL160/ FL190	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R249 active, Brisbane (As assigned) if R249 Deactivated	NOTAM
AIRSPACE: A circle radius 27.50NM centre 143118.0S 1322242.0E (YPTN/AD).									
REMARKS: Pattern exists within R249 (10000-NOTAM) remaining 2.5NM clear of boundary. Pattern size: 30NM x 15NM.									
TN12									
133530.0S 1312124.0E	133530.0S 1312124.0E	133542.0S 1304524.0E (TN12C) 266° 35NM	133542.0S 1304524.0E 135348.0S 1304530.0E 135348.0S 1312130.0E 133530.0S 1312124.0E	135348.0S 1312130.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL160/ FL190	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R225A&B active, Brisbane (As assigned) if R225A&B Deactivated	NOTAM
AIRSPACE: 140000.0S 1302224.0E (TNX101) - 132524.0S 1303542.0E (TNX102) then along the minor arc of a circle radius 62.50NM centre 122524.0S 1305423.0E (DN/DME) -132136.0S 1312254.0E (TNX103) - 140000.0S 1314942.0E (TNX104) - 140000.0S 1302224.0E (TNX101).									
REMARKS: Pattern exists within R225A&B (9500-NOTAM). Pattern size: 35NM x 18NM. Tanker to remain North of latitude S14 and South of 62.5DME DN.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
TN81									
144848.0S 1350618.0E	144848.0S 1350618.0E	155506.0S 1344218.0E (TN81C) 195° 70NM	155506.0S 1344218.0E 160148.0S 1350154.0E 145512.0S 1352554.0E 144848.0S 1350618.0E	145512.0S 1352554.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 144230.0S 1353554.0E (TNX900) - 161736.0S 1350142.0E (TNX901) - 160500.0S 1342412.0E (TNX902) then along the minor arc of a circle radius 150.00NM centre 143118.0S 1322242.0E (YPTN/AD) - 143000.0S 1345718.0E (TNX903) - 144230.0S 1353554.0E (TNX900).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									
TN82									
165724.0S 1311348.0E	165724.0S 1311348.0E	161354.0S 1301612.0E (TN82C) 304° 70NM	161354.0S 1301612.0E 162942.0S 1300318.0E 171312.0S 1310100.0E 165724.0S 1311348.0E	171312.0S 1310100.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 162418.0S 1294754.0E (TNX904) - 155342.0S 1301248.0E (TNX905) then along the minor arc of a circle radius 150.00NM centre 143118.0S 1322242.0E (YPTN/AD) - 165454.0S 1313542.0E (TNX906) - 172624.0S 1311012.0E (TNX907) - 162418.0S 1294754.0E (TNX904).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
TN83									
144142.0S 1295812.0E	144142.0S 1295812.0E	133548.0S 1302330.0E (TN83C) 017° 70NM	133548.0S 1302330.0E 132842.0S 1300418.0E 143436.0S 1293854.0E 144142.0S 1295812.0E	143424.0S 1293854.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 131254.0S 1300454.0E (TNX908) - 132324.0S 1303342.0E (TNX909) - 145730.0S 1295730.0E (TNX910) - 144654.0S 1292836.0E (TNX911) - 131254.0S 1300454.0E (TNX908).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									
TN84									
170811.0S 1300552.0E	170811.0S 1300552.0E	172433.0S 1300532.0E (TN84C) 087° 110NM	172433.0S 1300532.0E 172453.0S 1315950.0E 170724.0S 1315948.0E 170811.0S 1300552.0E	170724.0S 1315948.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 170146.0S 1294313.0E (TNX912) - 170153.0S 1322236.0E (TNX913) - 173154.0S 1322236.0E (TNX 914) - 173154.0S 1294313.0E (TNX915) - 170146.0S 1294313.0E (TNX912).									
REMARKS: Pattern size: 110NM x 20NM. Airspace size: 153NM x 30NM									

2.8 Townsville AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
TL11									
193136.0S 1454730.0E	193136.0S 1454730.0E	193612.0S 1450524.0E (TL11C) 256° 40NM	193612.0S 1450524.0E 195606.0S 1450748.0E 195130.0S 1454948.0E 193136.0S 1454730.0E	195130.0S 1454948.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	Within NOTAM vertical limits	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R738 active, Brisbane (As assigned) if R738 Deactivated	NOTAM

AIRSPACE: 201154.0S 1445342.0E (TLX100) - 192642.0S 1443918.0E (TLX101) - 192154.0S 1453154.0E (TLX102) - 191900.0S 1460142.0E (TLX103) - 193500.0S 1460430.0E (TLX104) - 194506.0S 1461236.0E (TLX105) - 195200.0S 1463700.0E (TLX106) - 200112.0S 1455000.0E (TLX107) - 201154.0S 1445342.0E (TLX100).

REMARKS: Pattern exists within R738A and R738B (7000-NOTAM). Pattern size: 40NM x 20NM. Check status of R741B (5000-NOTAM).

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
TL12									
185530.0S 1454742.0E	185530.0S 1454742.0E	183406.0S 1451201.0E (TL12C) 295° 40NM	183406.0S 1451206.0E 185100.0S 1450048.0E 191230.0S 1453624.0E 185530.0S 1454747.0E	191230.0S 1453624.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	Within NOTAM vertical limits	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R738 active, Brisbane (As assigned) if R738 Deactivated	NOTAM
AIRSPACE: 192642.0S 1443918.0E (TLX101) - 181430.0S 1445612.0E (TLX108) - 181812.0S 1451224.0E (TLX109) - 183836.0S 1454206.0E (TLX110) - 190600.0S 1461000.0E (TLX111) - 191900.0S 1460142.0E (TLX103) - 192154.0S 1453154.0E (TLX102) - 192642.0S 1443918.0E (TLX101).									
REMARKS: Pattern exists within R738C and R738D (7000-NOTAM). Pattern size: 40NM x 20NM.									
TL81									
194512.0S 1443454.0E	194512.0S 1443454.0E	183506.0S 1443000.0E (TL81C) 349° 70NM	183506.0S 1443000.0E 183624.0S 1440900.0E 194630.0S 1441342.0E 194512.0S 1443454.0E	194630.0S 1441342.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 200200.0S 1440930.0E (TLX900) - 182130.0S 1440242.0E (TLX901) - 181754.0S 1445400.0E (TLX902) then along the minor arc of a circle radius 120.00NM centre 191442.0S 1464530.0E (TL/DME) - 195930.0S 1444730.0E (TLX903) - 200200.0S 1440930.0E (TLX900).									
REMARKS: Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

2.9 Williamtown AAR and AEW&C airspace

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM11									
310448.0S 1544818.0E	310448.0S 1544818.0E	305542.0S 1535124.0E (WM11C) 277° 50NM	305542.0S 1535124.0E 311530.0S 1534700.0E 312442.0S 1544348.0E 310448.0S 1544818.0E	312442.0S 1544348.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL130/ FL330	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M550D active, Brisbane (As assigned) if M550D Deactivated	NOTAM
AIRSPACE: 304333.0S 1532427.0E (WMX300) - 310443.0S 1553803.0E (WMX301) - 320101.0S 1550606.0E (WMX302) - 311724.0S 1530823.0E (WMX303) - 304333.0S 1532427.0E (WMX300)									
REMARKS: Pattern exists within M550D. Pattern size: 50NM x 20NM.									
WM18									
315400.0S 1541742.0E	315400.0S 1541742.0E	313400.0S 1532330.0E (WM18C) 281° 50NM	313400.0S 1532330.0E 315224.0S 1531406.0E 321218.0S 1540806.0E 315400.0S 1541742.0E	321218.0S 1540806.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL130/ FL330	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M550C active, Brisbane (As assigned) if M550C Deactivated	NOTAM
AIRSPACE: 311724.0S 1530823.0E (WMX303) - 320101.0S 1550606.0E (WMX302) - 325503.0S 1543446.0E (WMX304) - 322917.0S 1532440.0E (WMX305) - 321258.0S 1524130.0E (WMX306) - 311724.0S 1530823.0E (WMX303)									
REMARKS: Pattern exists within M550C. Pattern size: 50NM x 20NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM19									
320754.0S 1531318.0E	320754.0S 1531318.0E	314006.0S 1532642.0E (WM19C) 010° 30NM	314006.0S 1532642.0E 313424.0S 1531030.0E 320206.0S 1525700.0E 320754.0S 1531318.0E	320206.0S 1525700.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL130/ FL330	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M550C active, Brisbane (As assigned) if M550C Deactivated	NOTAM
AIRSPACE: 311724.0S 1530823.0E (WMX303) - 320101.0S 1550606.0E (WMX302) - 325503.0S 1543446.0E (WMX304) - 322917.0S 1532440.0E (WMX305) - 321258.0S 1524130.0E (WMX306) - 311724.0S 1530823.0E (WM303)									
REMARKS: Pattern exists within M550C. Pattern size: 30NM x 15NM.									
WM20									
332048.0S 1524624.0E	332048.0S 1524624.0E	323918.0S 1530700.0E (WM20C) 010° 45NM	323918.0S 1530700.0E 323324.0S 1525042.0E 331500.0S 1522954.0E 332048.0S 1524624.0E	331500.0S 1522954.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL130/ FL330	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M550A active, Brisbane (As assigned) if M550A Deactivated	NOTAM
AIRSPACE: 321258.0S 1524130.0E (WMX306) - 322917.0S 1532440.0E (WMX305) - 334906.0S 1524538.0E (WMX307) - 334854.0S 1520641.0E (WMX308) - 333926.0S 1520206.0E (WMX309) - 331154.0S 1515814.0E (WMX310) - Then along the counter clockwise arc of a circle radius 25.00NM centre WLM TAC 324750.0S 1514959.0E (WMX311) - 324219.0S 1521854.0E (WMX312) - 324038.0S 1522734.0E (WMX313) - 321258.0S 1524130.0E (WMX306)									
REMARKS: Pattern exists within M550A. Pattern size: 45NM x 15NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM21									
333654.0S 1535724.0E	333654.0S 1535724.0E	330518.0S 1541512.0E (WM21C) 012° 35NM	330518.0S 1541512.0E 325848.0S 1535906.0E 333030.0S 1534112.0E 333654.0S 1535724.0E	333030.0S 1534112.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL130/ FL330	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if M550B active, Brisbane (As assigned) if M550B Deactivated	NOTAM
AIRSPACE: 322917.0S 1532440.0E (WMX305) - 325503.0S 1543446.0E (WMX304) - 334855.0S 1540250.0E (WMX314) - 334906.0S 1524538.0E (WMX307) - 322917.0S 1532440.0E (WMX305)									
REMARKS: Pattern exists within M550B. Pattern size: 35NM x 15NM.									
WM22									
322200.0S 1500812.0E	322200.0S 1500812.0E	315736.0S 1503730.0E (WM22C) 034° 35NM	315736.0S 1503730.0E 314436.0S 1502248.0E 320906.0S 1495318.0E 322200.0S 1500812.0E	320906.0S 1495318.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	A100/ FL220	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R560 active, Brisbane (As assigned) if R560 Deactivated	NOTAM
AIRSPACE: 320236.0S 1490547.0E (WMX401) - 321329.0S 1492125.0E (WMX402) - 322745.0S 1494203.0E (WMX403) - 324315.0S 1495932.0E (WMX404) - 320827.0S 1504205.0E (WMX405) - 320243.0S 1505208.0E (WMX406) - 315350.0S 1505254.0E (WMX407) - 314710.0S 1504945.0E (WMX408) - 313029.0S 1504156.0E (WMX409) - 312442.0S 1503707.0E (WMX410) - 310811.0S 1502326.0E (WMX411) - 314535.0S 1494048.0E (WMX412) - 320236.0S 1490547.0E (WMX401)									
REMARKS: Pattern exists within R560A (8500-FL240) and R560B. Pattern size: 35NM x 18NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM23									
311024.0S 1492500.0E	311024.0S 1492500.0E	304542.0S 1500148.0E (WM23C) 041° 40NM	304542.0S 1500148.0E 303006.0S 1494718.0E 305436.0S 1491036.0E 311024.0S 1492500.0E	305436.0S 1491036.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL120/ FL330	33SQN OPS 0753629647	RAAF CRU (FREQ TBA) if R570A active, Brisbane (As assigned) if R570A Deactivated	NOTAM
AIRSPACE: 304714.0S 1485813.0E (WMX413) - 300647.0S 1500449.0E (WMX414) - 310811.0S 1502326.0E (WMX411) - 314535.0S 1494048.0E (WMX412) - 315822.0S 1491434.0E (WMX415) - 315126.0S 1490439.0E (WMX416) - 304714.0S 1485813.0E (WMX413)									
REMARKS: Pattern exists within R570A. Pattern size: 40NM x 20NM.									
WM87									
323212.0S 1551712.0E	323212.0S 1551712.0E	312848.0S 1555206.0E (WM87C) 012° 70NM	312848.0S 1555206.0E 312012.0S 1553100.0E 322336.0S 1545548.0E 323212.0S 1551712.0E	322336.0S 1545548.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 310443.0S 1553803.0E (WMX926) - 311153.0S 1561323.0E (WMX927) - 324648.0S 1552030.0E (WMX910) - 323824.0S 1544512.0E (WMX911) - 310443.0S 1553803.0E (WMX926)									
REMARKS: Pattern size: 70NM X 20NM. Airspace size: 100NM x 30NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM88									
294324.0S 1490242.0E	294324.0S 1490242.0E	294242.0S 1494236.0E (WM88C) 078° 35NM	294242.0S 1494236.0E 292242.0S 1494206.0E 292318.0S 1490206.0E 294324.0S 1490242.0E	292318.0S 1490206.0E	N/A	FL260/ FL320	33SQN OPS 0753629647	Brisbane	TBA
AIRSPACE: 294813.0S 1485603.0E (WMX931) - 291720.0S 1485356.0E (WMX932) - 291703.0S 1494911.0E (WMX933) - 294725.0S 1495046.0E (WMX934) - 294813.0S 1485603.0E (WMX931)									
REMARKS: Pattern size: 35NM x 20NM. Airspace size: 50NM x 30NM.									
WM89									
332200.0S 1544912.0E	332200.0S 1544912.0E	320036.0S 1553442.0E (WM89C) 012° 90NM	320036.0S 1553442.0E 315200.0S 1551330.0E 331324.0S 1542742.0E 332200.0S 1544912.0E	331324.0S 1542742.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 313951.0S 1551809.0E (WMX929) - 315154.0S 1555732.0E (WMX930) - 334036.0S 1545524.0E (WMX923) - 332742.0S 1541612.0E (WMX924) - 313951.0S 1551809.0E (WMX929)									
REMARKS: Pattern size: 90NM x 20NM. Airspace size: 118NM x 33NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM92									
304406.0S 1534318.0E	304406.0S 1534318.0E	305642.0S 1550200.0E (WM92C) 098° 70NM	305642.0S 1550200.0E 303654.0S 1550518.0E 302418.0S 1534642.0E 304406.0S 1534318.0E	302418.0S 1534642.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 304330.0S 1532430.0E (WMX904) - 301854.0S 1532930.0E (WMX905) - 302812.0S 1542554.0E (WMX906) - 303824.0S 1553124.0E (WMX907) - 310142.0S 1551812.0E (WMX908) - 304330.0S 1532430.0E (WMX904). **REMARKS:** Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									
WM93									
334936.0S 1543812.0E	334936.0S 1543812.0E	324612.0S 1551336.0E (WM93C) 012° 70NM	324612.0S 1551336.0E 323736.0S 1545212.0E 334100.0S 1541630.0E 334936.0S 1543812.0E	334100.0S 1541630.0E	N/A	FL260/ FL320	2SQN	Brisbane	TBA
AIRSPACE: 322145.0S 1545452.0E (WMX928) - 323454.0S 1552642.0E (WMX914) - 340454.0S 1543412E (WMX915) - 335130S 1540154E (WMX916) - 322145.0S 1545452.0E (WMX928) **REMARKS:** Pattern size: 70NM x 20NM. Airspace size: 100NM x 30NM.									

ENTRY POINTS	RVIP	ANCHOR PT/ IB COURSE/ PATTERN LEN	ANCHOR PATTERN	EXIT POINTS	CR PLAN	REFUEL ALTS	SCHED UNIT	ASSIGNED CENTRE	TIMES OF OPS
WM94									
315348.0S 1524248.0E	315348.0S 1524248.0E	311836.0S 1525936.0E (WM94C) 010° 38NM	311836.0S 1525936.0E 311336.0S 1524536.0E 314848.0S 1522842.0E 315348.0S 1524248.0E	314848.0S 1522842.0E	N/A	FL260/ FL340	2SQN	Brisbane	TBA

AIRSPACE: 310014.0S 1524826.0E (WMX315) - 310021.0S 1531630.0E (WMX316) - 311724.0S 1530823.0E (WMX303) - 320152.0S 1524655.0E (WMX317) - 320215.0S 1522513.0E (WMX318) - 320219.0S 1522105.0E (WMX319) - 320323.0S 1521746.0E (WMX320) - 313320.0S 1522706.0E (WMX321) - 310014.0S 1524826.0E (WMX315)

REMARKS: Pattern exists within R586C (FL125-FL600 NOTAM). Pattern size: 38NM x 13NM.

3. MILITARY AAR AND AEW&C AIRSPACE ANCHOR WAYPOINTS

3.1 Amberley

WPT	LAT	LONG	WPT	LAT	LONG
AMX100	261542.0S	1532954.0E	AMX200	250654.0S	1551954.0E
AMX101	244942.0S	1550206.0E	AMX201	244942.0S	1550206.0E
AMX102	253930.0S	1555400.0E	AMX202	240824.0S	1541948.0E
AMX103	264642.0S	1535330.0E	AMX203	234554.0S	1544730.0E
AMX104	274630.0S	1544430.0E	AMX204	244542.0S	1554554.0E
AMX105	271312.0S	1544430.0E	AMX205	271636.0S	1573106.0E
AMX106	263324.0S	1554654.0E	AMX206	261648.0S	1563124.0E
AMX107	261548.0S	1563030.0E	AMX207	255612.0S	1565630.0E
AMX108	271636.0S	1573106.0E	AMX208	265506.0S	1575700.0E
AMX109	285012.0S	1563754.0E	AMX209	295142.0S	1563542.0E
AMX110	285212.0S	1534942.0E	AMX210	305936.0S	1555818.0E
AMX111	282912.0S	1535736.0E	AMX211	304630.0S	1552654.0E
AMX112	283348.0S	1540918.0E	AMX212	293842.0S	1560436.0E
AMX113	291324.0S	1550712.0E	AMX213	273512.0S	1490724.0E
AMX114	302812.0S	1542554.0E	AMX214	272942.0S	1494548.0E
AMX115	301854.0S	1532930.0E	AMX215	284900.0S	1495924.0E
AMX116	293924.0S	1533736.0E	AMX216	285424.0S	1492142.0E
AMX117	292212.0S	1500918.0E	AMX900	244542.0S	1554554.0E
AMX118	281000.0S	1494800.0E	AMX901	255612.0S	1565630.0E
AMX125	282830.0S	1495324.0E	AMX902	261648.0S	1563124.0E
AMX126	280830.0S	1510336.0E	AMX903	250654.0S	1551954.0E
AMX127	281942.0S	1510848.0E	AMX905	292806.0S	1492754.0E
AMX128	283530.0S	1512036.0E	AM11C	260606.0S	1544800.0E
AMX129	284430.0S	1512942.0E	AM13C	270654.0S	1555418.0E
AMX130	265736.0S	1494406.0E	AM15C	292430.0S	1544042.0E
AMX131	271742.0S	1505342.0E	AM17C	283500.0S	1505924.0E
AMX132	272830.0S	1505306.0E	AM19C	274112.0S	1504512.0E
AMX133	274500.0S	1505536.0E	AM22C	245112.0S	1551424.0E
AMX134	280106.0S	1510012.0E	AM24C	270106.0S	1572500.0E
AMX135	280830.0S	1510336.0E	AM26C	300312.0S	1562306.0E
AMX136	282830.0S	1495324.0E	AM28C	274918.0S	1491536.0E
AMX137	270348.0S	1490148.0E	AM83C	250012.0S	1555254.0E
AMX138	265736.0S	1494406.0E	AM84C	291324.0S	1493100.0E

3.2 Darwin

WPT	LAT	LONG	WPT	LAT	LONG
DNX900	093230.0S	1320612.0E	DNX908	115442.0S	1272548.0E
DNX901	103118.0S	1332754.0E	DNX909	115118.0S	1280418.0E
DNX902	110436.0S	1330330.0E	DNX910	133136.0S	1281354.0E
DNX903	100254.0S	1314354.0E	DNX911	133500.0S	1273442.0E
DNX904	141600.0S	1275654.0E	DN81C	094524.0S	1321530.0E
DNX905	135024.0S	1282318.0E	DN82C	140900.0S	1282600.0E
DNX906	145912.0S	1293836.0E	DN83C	120730.0S	1275230.0E
DNX907	152718.0S	1290954.0E			

3.3 EAXA

WPT	LAT	LONG	WPT	LAT	LONG
NWX900	360212.0S	1533412.0E	NWX905	365642.0S	1504500.0E
NWX901	371312.0S	1520630.0E	NWX906	365918.0S	1500736.0E
NWX902	364618.0S	1513318.0E	NWX907	351930.0S	1495730.0E
NWX903	352906.0S	1525306.0E	NW81C	360924.0S	1531654.0E
NWX904	351700.0S	1503412.0E	NW82C	353224.0S	1502930.0E

3.4 Edinburgh

WPT	LAT	LONG	WPT	LAT	LONG
EDX100	301818.1S	1432613.4E	ED01C	303500.0S	1444500.0E
IBODA	320135.6S	1451905.1E	EDX102	315823.9S	1432132.8E
EDX101	302123.9S	1452143.3E			

3.5 Tindal

WPT	LAT	LONG	WPT	LAT	LONG
TNX101	140000.0S	1302224.0E	TNX906	165454.0S	1313542.0E
TNX102	132524.0S	1303542.0E	TNX907	172624.0S	1311012.0E
TNX103	132136.0S	1312254.0E	TNX908	131254.0S	1300454.0E
TNX104	140000.0S	1314942.0E	TNX909	132324.0S	1303342.0E
TNX900	144230.0S	1353554.0E	TNX910	145730.0S	1295730.0E
TNX901	161736.0S	1350142.0E	TNX911	144654.0S	1292836.0E
TNX902	160500.0S	1342412.0E	TNX912	170146.0S	1294313.0E
TNX903	143000.0S	1345718.0E	TNX913	170153.0S	1322236.0E
TNX904	162418.0S	1294754.0E	TNX914	173154.0S	1322236.0E
TNX905	155342.0S	1301248.0E			

WPT	LAT	LONG	WPT	LAT	LONG
TNX915	173154.0S	1294313.0E	TN81C	155506.0S	1344218.0E
TN11C	141530.0S	1321900.0E	TN82C	161354.0S	1301612.0E
TN12C	133542.0S	1304524.0E	TN83C	133548.0S	1302330.0E

3.6 Townsville

WPT	LAT	LONG	WPT	LAT	LONG
TLX100	201154.0S	1445342.0E	TLX110	183836.0S	1454206.0E
TLX101	192642.0S	1443918.0E	TLX111	190600.0S	1461000.0E
TLX102	192154.0S	1453154.0E	TLX900	200200.0S	1440930.0E
TLX103	191900.0S	1460142.0E	TLX901	182130.0S	1440242.0E
TLX104	193500.0S	1460430.0E	TLX902	181754.0S	1445400.0E
TLX105	194506.0S	1461236.0E	TLX903	195930.0S	1444730.0E
TLX106	195200.0S	1463700.0E	TL11C	193612.0S	1450524.0E
TLX107	200112.0S	1455000.0E	TL12C	183406.0S	1451206.0E
TLX108	181430.0S	1445612.0E	TL81C	183506.0S	1443000.0E
TLX109	181812.0S	1451224.0E			

3.7 Williamtown

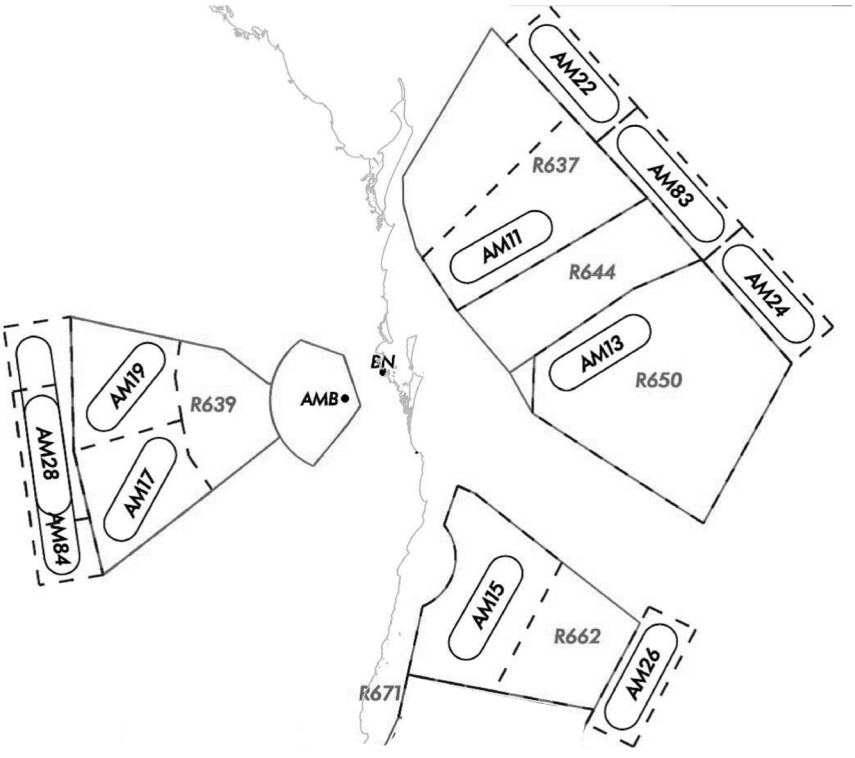
WPT	LAT	LONG	WPT	LAT	LONG
WM11C	305542.0S	1535124.0E	WMX306	321258.0S	1524130.0E
WM18C	313400.0S	1532330.0E	WMX307	334906.0S	1524538.0E
WM19C	314006.0S	1532642.0E	WMX308	334854.0S	1520641.0E
WM20C	323918.0S	1530700.0E	WMX309	333926.0S	1520206.0E
WM21C	330518.0S	1541512.0E	WMX310	331154.0S	1515814.0E
WM22C	315736.0S	1503730.0E	WMX311	324750.0S	1514959.0E
WM23C	304542.0S	1500148.0E	WMX312	324219.0S	1521854.0E
WM87C	312848.0S	1555206.0E	WMX313	324038.0S	1522734.0E
WM88C	294242.0S	1494236.0E	WMX314	334855.0S	1540250.0E
WM89C	320036.0S	1553442.0E	WMX315	310014.0S	1524826.0E
WM92C	305642.0S	1550200.0E	WMX316	310021.0S	1531630.0E
WM93C	324612.0S	1551336.0E	WMX317	320152.0S	1524655.0E
WM94C	311836.0S	1525936.0E	WMX318	320215.0S	1522513.0E
WMX300	304333.0S	1532427.0E	WMX319	320219.0S	1522105.0E
WMX301	310443.0S	1553803.0E	WMX320	320323.0S	1521746.0E
WMX302	320101.0S	1550606.0E	WMX321	313320.0S	1522706.0E
WMX303	311724.0S	1530823.0E	WMX401	320236.0S	1490547.0E
WMX304	325503.0S	1543446.0E	WMX402	321329.0S	1492125.0E
WMX305	322917.0S	1532440.0E	WMX403	322745.0S	1494203.0E

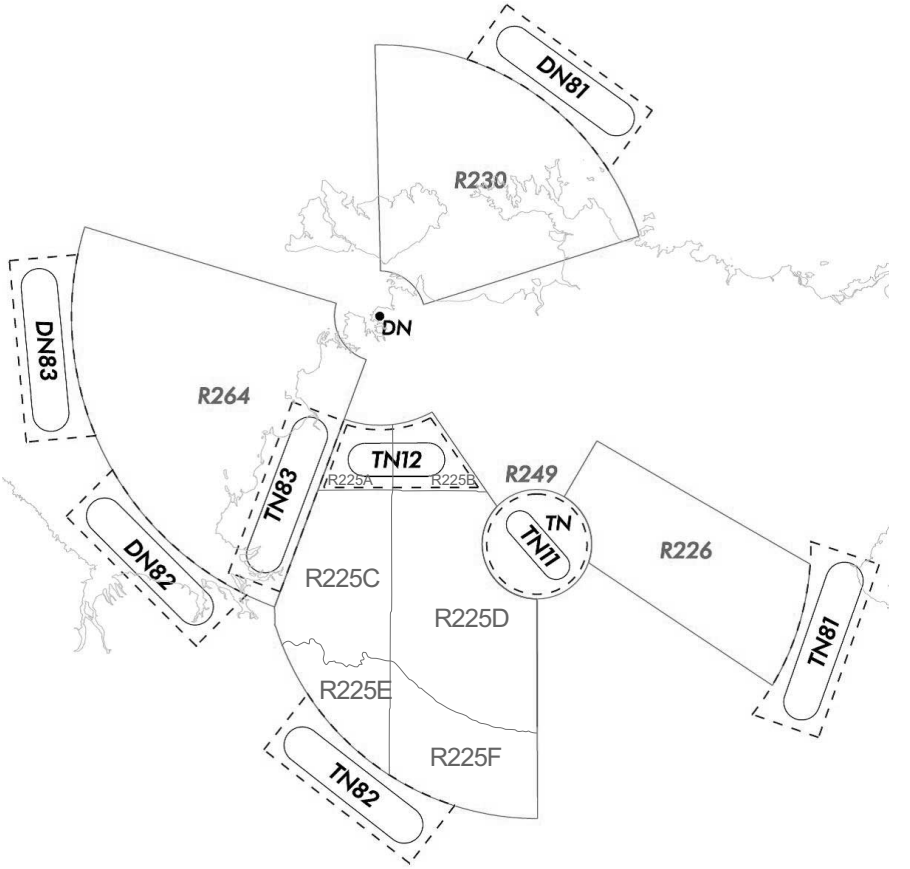
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WMX404	324315.0S	1495932.0E	WMX906	302812.0S	1542554.0E
WMX405	320827.0S	1504205.0E	WMX907	303824.0S	1553124.0E
WMX406	320243.0S	1505208.0E	WMX908	310142.0S	1551812.0E
WMX407	315350.0S	1505254.0E	WMX910	324648.0S	1552030.0E
WMX408	314710.0S	1504945.0E	WMX911	323824.0S	1544512.0E
WMX409	313029.0S	1504156.0E	WMX914	323454.0S	1552642.0E
WMX410	312442.0S	1503707.0E	WMX915	340454.0S	1543412.0E
WMX411	310811.0S	1502326.0E	WMX916	335130.0S	1540154.0E
WMX412	314535.0S	1494048.0E	WMX923	334036.0S	1545524.0E
WMX413	304714.0S	1485813.0E	WMX924	332742.0S	1541612.0E
WMX414	300647.0S	1500449.0E	WMX926	310443.0S	1553803.0E
WMX415	315822.0S	1491434.0E	WMX927	311153.0S	1561323.0E
WMX416	315126.0S	1490439.0E	WMX928	322145.0S	1545452.0E
WMX900	320400.0S	1521700.0E	WMX929	313951.0S	1551809.0E
WMX901	310412.0S	1524706.0E	WMX930	315154.0S	1555732.0E
WMX902	311112.0S	1530624.0E	WMX931	294813.0S	1485603.0E
WMX903	321106.0S	1523630.0E	WMX932	291720.0S	1485356.0E
WMX904	304330.0S	1532430.0E	WMX933	291703.0S	1494911.0E
WMX905	301854.0S	1532930.0E	WMX934	294725.0S	1495046.0E

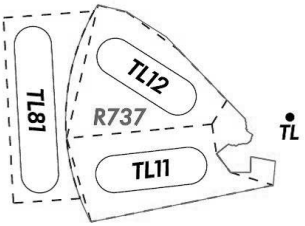
4. MILITARY AAR TRACK WAYPOINTS

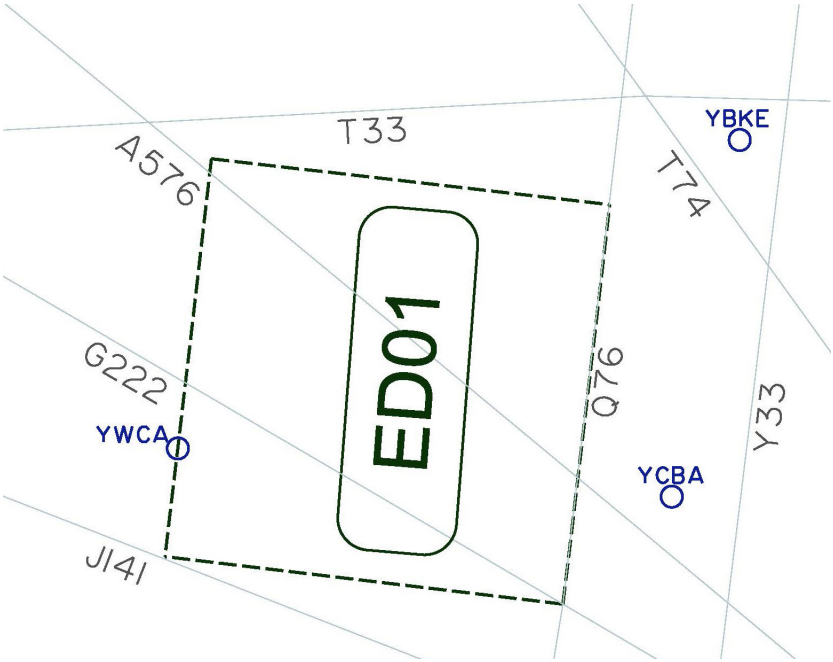
WPT	LAT/LONG	TR OUT (ºM)	DIST (NM)
W946			
AGETA	300143.1S 1480742.3E	301	
CARBN	290721.8S 1465807.3E	303	81
CMU	280205.2S 1453725.5E	313	96
EXXON	254636.0S 1433906.0E	314	172
VALDZ	232700.0S 1414148.0E	316	175
MA	203952.6S 1392908.3E	305	207
TEXAN	190042.0S 1372924.0E	306	150
PETRL	171216.0S 1352134.5E	308	163
GUKVI	161454.0S 1341536.0E		85
W951			
ROM	263235.1S 1484654.1E	285	
ESLES	255831.1S 1472326.5E	286	82
LIDBU	244524.5S 1443301.5E	288	171
VALDZ	232700.0S 1414148.0E		175
W952			
AMOCO	240000.0S 1521230.0E	328	
KELPI	223410.2S 1513511.6E	332	92
BOOMA	200000.0S 1504112.0E	333	162
LAMEK	180916.6S 1500235.5E	334	116
CALTX	170000.0S 1493913.7E		73
W953			
NOBIP	170000.0S 1501754.0E	154	
ORIGA	180918.0S 1504206.0E	154	73
GASSO	200000.0S 1512024.0E	152	116
VEGDI	223418.0S 1521850.0E	151	163
DOLKO	240000.0S 1525218.0E		91
W954			
CFS	301910.2S 1530715.2E	093	
GALON	303804.7S 1543200.4E	092	76
POSUM	310553.3S 1564700.5E	090	119
LHI	313144.1S 1590422.6E	088	120
LADUR	314400.0S 1601430.0E		61

5. MILITARY AAR AND AEW&C PATTERNS




DN81
R230
DN
DN83
R264
TN12
R225A
R225B
R249
TN
TN11
R226
TN83
DN82
R225C
R225D
R225E
R225F
TN81
TN82


TL12
TL81
R737
TL
TL11


T33
A576
YBKE
T74
ED01
G222
Q79
Y33
YWCA
YCBA
J141

6. REFUELLING TRACKS

RVIP	RVCP	NAV CHK POINTS	EXIT	CR PLAN	REFUELLING ALTITUDES	SCHED UNIT	ASSIGNED CENTRE
W946 (North)							
AGETA 300143.1S 1480742.3E	CARBN 290721.8S 1465807.3E	CMU 280205.2S 1453725.5E EXXON 254636.0S 1433906.0E VALDZ 232700.0S 1414148.0E MA 203952.6S 1392908.3E TEXAN 190042.0S 1372924.0E PETRL 171216.0S 1352134.5E	GUKVI 161454.0S 1341536.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Simultaneous Opposite Direction Refuelling operations are authorised on W946.							

RVIP	RVCP	NAV CHK POINTS	EXIT	CR PLAN	REFUELLING ALTITUDES	SCHED UNIT	ASSIGNED CENTRE
W946 (South)							
GUKVI 161454.0S 1341536.0E	PETRL 171216.0S 1352134.5E	TEXAN 190042.0S 1372924.0E MA 203952.6S 1392908.3E VALDZ 232700.0S 1414148.0E EXXON 254636.0S 1433906.0E CMU 280205.2S 1453725.5E CARBN 290721.8S 1465807.3E	AGETA 300143.1S 1480742.3E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Simultaneous Opposite Direction Refuelling operations are authorised on W946.							

RVIP	RVCP	NAV CHK POINTS	EXIT	CR PLAN	REFUELLING ALTITUDES	SCHED UNIT	ASSIGNED CENTRE
W951 (West)							
ROM 263235.1S 1484654.1E	ESLES 255831.1S 1472326.5E	LIDBU 244524.5S 1443301.5E	VALDZ 232700.0S 1414148.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Simultaneous Opposite Direction Refuelling operations are authorised on W951. Track refuelling may continue on W946 if required.							
W951 (East)							
VALDZ 232700.0S 1414148.0E	LIDBU 244524.5S 1443301.5E	ESLES 255831.1S 1472326.5E	ROM 263235.1S 1484654.1E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Simultaneous Opposite Direction Refuelling operations are authorised on W951.							

RVIP	RVCP	NAV CHK POINTS	EXIT	CR PLAN	REFUELLING ALTITUDES	SCHED UNIT	ASSIGNED CENTRE
W952 (North)							
AMOCO 240000.0S 1521230.0E	KELPI 223410.2S 1513511.6E	BOOMA 200000.0S 1504112.0E LAMEK 180916.6S 1500235.5E	CALTX 170000.0S 1493913.7E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Track runs parallel to W953.							
W952 (South)							
CALTX 170000.0S 1493913.7E	LAMEK 180916.6S 1500235.5E	BOOMA 200000.0S 1504112.0E KELPI 223410.2S 1513511.6E	AMOCO 240000.0S 1521230.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Track runs parallel to W953.							

RVIP	RVCP	NAV CHK POINTS	EXIT	CR PLAN	REFUELLING ALTITUDES	SCHED UNIT	ASSIGNED CENTRE
W953 (South)							
NOBIP 170000.0S 1501754.0E	ORIGA 180918.0S 1504206.0E	GASSO 200000.0S 1512024.0E VEGDI 223418.0S 1521850.0E	DOLKO 240000.0S 1525218.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Track runs parallel to W952.							
W953 (North)							
DOLKO 240000.0S 1525218.0E	VEGDI 223418.0S 1521850.0E	GASSO 200000.0S 1512024.0E ORIGA 180918.0S 1504206.0E	NOBIP 170000.0S 1501754.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Track runs parallel to W952.							

RVIP	RVCP	NAV CHK POINTS	EXIT	CR PLAN	REFUELLING ALTITUDES	SCHED UNIT	ASSIGNED CENTRE
W954 (East)							
CFS 301910.2S 1530715.2E	GALON 303804.7S 1543200.4E	POSUM 310553.3S 1564700.5E LHI 313144.1S 1590422.6E	LADUR 314400.0S 1601430.0E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Simultaneous Opposite Direction Refuelling operations are authorised on W954.							
W954 (West)							
LADUR 314400.0S 1601430.0E	LHI 313144.1S 1590422.6E	POSUM 310553.3S 1564700.5E GALON 303804.7S 1543200.4E	CFS 301910.2S 1530715.2E	a. 301.500 b. TBA c. reserved d. reserved e. 33/96	FL250/FL300	33SQN OPS 0753629647	Brisbane (FREQ as assigned)
REMARKS: Simultaneous Opposite Direction Refuelling operations are authorised on W954.							

7. PROCEDURES FOR AIRCRAFT OPERATING IN AN AIR DEFENCE IDENTIFICATION ZONE

7.1 General

7.1.1 The following general rules and procedures apply to enable identification of air traffic entering any designated Air Defence Identification Zone (ADIZ) under the control of Australia.

7.1.2 An ADIZ is airspace of defined dimensions within which identification of all aircraft is required.

7.1.3 When a flight is intended to operate within an ADIZ, the pilot, unless exempted in accordance with *para 7.1.4*, must:

a. lodge a flight notification covering flight within the ADIZ with the appropriate ATS unit at least 60 minutes before entry into the ADIZ;

b. report position to ATS when passing each position reporting point within the ADIZ;

c. report position to ATS at ADIZ boundary with a geographical reference (e.g. 15NM east of...) or, if the departure point is within 100NM of the ADIZ boundary, report departure;

d. report departure if departing from a point in the ADIZ;

e. maintain a continuous listening watch on the communications frequency of the appropriate ATS unit or on another frequency as directed until the flight is through the ADIZ;

f. not deliberately deviate from tracks and altitudes filed in the flight notification unless prior ATC clearance is obtained, or, outside controlled airspace, notification is given to the appropriate ATS unit; and

g. activate the aircraft transponder when within 100NM of the ADIZ and when operating within the ADIZ.

7.1.4 The following flights over Australia and its territorial waters are exempted from compliance with the requirements of *para 7.1.3:*

a. a flight originating within an ADIZ which maintains a steady outbound track;

b. a flight which remains within 10NM of the point of departure;

c. aircraft performing published approach, holding or recovery procedures; and

d. a flight conducted in accordance with special procedures arranged with the Regional Air Defence Commander.

7.1.5 Flight notifications lodged in accordance with *para 7.1.3* must include details of:

a. tracks and altitudes to be flown while operating in the ADIZ;

b. estimated elapsed times for each route segment in the ADIZ, including the segment in which the ADIZ boundary is crossed;

c. position reporting points, departure and landing points; and

d. estimated time at the commencing point of the first route segment for which details are required in accordance with *sub-para b.*

7.1.6 Reporting points published in aeronautical charts must be used plus those required by the Regional Air Defence Commander.

7.1.7 Pilots must immediately notify ATS of any deviation from flight notification beyond the following tolerances:

a. estimated time of commencing the ADIZ route segments – ± 5 minutes;

b. over land area – ± 10NM from track;

c. over oceanic areas - ± 20NM from track.

Note: The 5 minutes expressed in sub-para a. will be used in considering an appropriate response, but pilots must report predicted deviations of greater than two (2) minutes.

7.1.8 In the event of failure of two-way radio communication, the pilot must proceed in accordance with the normal radio failure procedures.

7.2 **Special Requirements**

7.2.1 Special requirements may be published relative to a particular ADIZ. Flights exempted in accordance with *para 7.1.4* will not be exempted from the special requirements unless so specified.

7.3 **Non-Compliance**

7.3.1 Significant deviations from the requirements for flight in an ADIZ must be reported immediately to ATS and details and reasons for the deviation must be reported at the first point of landing, for transmission to the Regional Air Defence Commander.

7.4 Diversion of Aircraft for Defence Operations

7.4.1 The Regional Air Defence Commander may, through ATS, direct the flight of aircraft in the interests of national security. Messages initiating such requirements will be prefaced by ‘MILITARY OPERATIONS REQUIRE...’

ENR 5.3 OTHER ACTIVITIES OF A DANGEROUS NATURE AND OTHER POTENTIAL HAZARDS

1. Other general activities of a dangerous nature.

Lateral limits Coordinates	Vertical limits	Advisory measures	Authority responsible for INFO	Remarks Time of ACT
1	2	3	4	5
BUSHFIRES 5NM from any bushfire	3,000FT AGL SFC	Avoid area due to potential aerial firefighting traffic	NOTAM may be issued by the relevant State Fire Authority	Any time a bushfire is present
WARSHIPS 5NM from any warships	2,000FT AMSL SFC	Avoid area due to potential military aerial traffic	Traffic alerts and warnings may be passed on FREQ 121.5MHz	Any time warships are present

2. Other potential hazards

Lateral limits Coordinates	Vertical limits	Advisory measures	Authority responsible for INFO	Remarks Time of ACT
1	2	3	4	5
VOLCANIC ASH Subject to prevailing winds	FL600 SFC	Avoid flying through volcanic ash. Remain clear of visible volcanic ash and reports of ash clouds	SIGMET report may be issued by the Bureau of Meteorology	Any time volcanic ash is present. See *AC 91-33*

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ENR 5.4 AIR NAVIGATION OBSTACLES - EN ROUTE

1. MAN-MADE OBSTACLES GREATER THAN 360 FEET

1.1 In Australia, man-made obstacles greater than 360FT are required to be reported. Where required for navigation purposes, these obstacles are identified on aeronautical maps and charts, including WAC. Unreported obstacles up to 360FT may exist in navigation tolerance areas. Pilots are, therefore, required to take this into account when calculating LSALT.

2. MARKING OF POWER LINES AND OTHER OVERHEAD CABLES

2.1 The standards for marking power lines and other overhead cables with long spans are addressed by Standards Australia in the following documents:

a. AS3891.1 – 1991, Air navigation – Cables and their supporting structures – Mapping and marking – Permanent marking of overhead cables and their supporting structures; and

b. AS3891.2 – 1992, Air navigation – Cables and their supporting structures – Mapping and marking – Marking of overhead cables for low flying.

3. NVIS AND OBSTACLE LIGHTING

3.1 Some LED lighting systems, clearly visible to the naked eye, fall outside the combined visible and near-infrared spectrum of NVIS and therefore will not be visible to operating crew using NVIS.

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ENR 5.5 AERIAL SPORTING AND RECREATIONAL ACTIVITIES

1. GLIDING OPERATIONS

1.1 General

1.1.1 For rules relating to gliding operations refer to *CASR Part 103* and the *Part 103 MOS.*

1.1.2 Pilots should take extra care when operating at an aerodrome where gliding operations are in progress. Gliding operations are indicated by the “gliding operations in progress” ground signal displayed next to the primary wind direction indicator. Pilots should also establish whether the gliders are being launched by wire or aerotow, or both.

1.1.3 Where aerotowing is in progress, pilots should remain well clear of gliders under tow. If wire launching is used, pilots should establish the locations of either the winch or tow car and the cable, and remain well clear. Over-flying the runway below 2,000FT AGL is not advised, nor is landing without first ascertaining that the cable is on the ground and not across the landing path. Aerotow and winch launching are possible up to 4,000FT AGL, but launches to 1,500FT or 2,000FT AGL are normal.

1.1.4 In Class G airspace, gliders may be operating no-radio, on Area VHF or on frequencies 122.5, 122.7 or 122.9MHz.
Radio-equipped gliders at, or in the vicinity of, non-controlled aerodromes make broadcasts in accordance with the table at *ENR 1.1 para 9.1.14*.

1.2 Operations at Certified Aerodromes

1.2.1 Gliding operations may be conducted from:

a. a glider runway strip within the runway strip (single runway), using a common circuit direction;

b. a glider runway strip adjacent to the existing runway strip (dual runways), using a common circuit direction; or

c. a separate glider runway strip parallel to and spaced away from the existing runway strip (parallel runways), using contra-circuit procedures.

1.2.2 Details of the gliding operation are published in the ERSA entry for the aerodrome. When procedures are changed for intensive short-term gliding activity, a NOTAM will be issued.

1.2.3 Where dual or parallel runways are established, the glider runway strip will conform to normal movement area standards, but will be marked by conspicuous markers of a colour other than white. Glider runway strips must not be used except by gliders, tug aircraft and other authorised aircraft.

1.2.4 Where a single runway is established and gliders operate within the runway strip, the runway strip markers may be moved outwards to incorporate the glider runway strip. Glider movement and parking areas are established outside of the runway strips. When the glider runway strip is occupied by a tug aircraft or glider, the runway is deemed to be occupied. Aircraft using the runway may, however, commence their take-off run from a position ahead of a stationary glider or tug aircraft.

1.2.5 Except for gliders approaching to land, powered aircraft have priority in the use of runways, taxiways and aprons where a single runway or dual runway operation is established.

1.2.6 At the locations where parallel runways exist and contra-circuit procedures apply, operations on the two parallel runways by aircraft below 5,700KG MTOW may be conducted independently in VMC by day. Aircraft must not operate within the opposing circuit area below 1,500FT AGL. Pilots should ascertain the runways in use as early as possible and conform to that circuit. A crossing runway should only be used when operationally necessary, and traffic using the crossing runway should avoid conflicting with the established circuit.

1.2.7 At aerodromes other than for which contra-circuits are prescribed, gliders are generally required to conform to the established circuit direction. However, unforeseen circumstances may occasionally compel a glider to execute a non-standard pattern, including use of the opposite circuit direction in extreme cases.

1.2.8 A listening watch on the appropriate VHF frequency must be maintained while operating at or in the vicinity of non-controlled aerodromes by the tug pilot. The winch or tow-vehicle driver should also maintain a listening watch during wire launching. The tug pilot or winch/car driver may be able to advise glider traffic information to inbound or taxiing aircraft.

Note: The appropriate VHF frequency is as described at ENR 1.1 subsection 9.1.

1.2.9 Where wire launching is used, launching will cease and the wire will be retracted or moved off the strip when another aircraft joins the circuit or taxies, or a radio call is received indicating this. A white strobe light is displayed by a winch, or a yellow rotating beacon by a tow-car or associated vehicle, whenever the cable is deployed.

2. PARACHUTING OPERATIONS

2.1 General

2.1.1 For rules relating to parachuting operations refer to *CASR Part 105* and the *Part 105 MOS.*

2.2 Conflicting Traffic

2.2.1 ATC will provide separation between parachutists and non-parachuting aircraft in Class A, C and D airspace, and provide traffic information to pilots of aircraft engaged in parachuting operations on known or observed traffic in Class E and Class G airspace.

2.3 Additional Requirements in Controlled Airspace

2.3.1 ATC base separation on the assumption that the parachutist will be dropped within 1NM of the target. If an extension of this area is necessary, the pilot must advise ATC of the direction and distance required.

2.4 Additional Requirements for Operations Above 10,000FT AMSL

2.4.1 Pilots should refer to *Division 26.11* of the *Part 91 MOS* for the requirements relating to oxygen usage for high altitude flights.

3. BALLOON OPERATIONS

3.1 General

3.1.1 For rules relating to ballooning operations refer to *CASR Part 131* and the *Part 131 MOS*. Specific guidance on a limited number of operationally focused balloon matters is contained in *AC 131-02; 'Manned free balloons – Operations'*.

3.1.2 Pilots are reminded of the requirement to not operate an aircraft in a manner which creates a hazard to a person or property, (*CASR 91.055*). Outside of a populous area or public gathering, balloons do not need to maintain a minimum height AGL. However, this does not absolve pilots from any responsibility not to cause a hazard to landholders, stock, persons or property. Pilots planning to conduct balloon operations in an unfamiliar area are recommended to contact local area balloon operators for landowner information, and to check for compliance with State, Territory or Federal laws and the rights of landowners and occupiers.

ENR 5.6 BIRD MIGRATION AND AREAS WITH SENSITIVE FAUNA

1. In Australia, bird migration areas are not identified. However, at locations where birds may pose a particular hazard, a note describing the hazard at a particular location will be contained in *ERSA FAC*.

2. Some areas which may have special significance because of sensitive fauna or other ecological considerations are identified as a “Fly Neighbourly Advice” area. Details on these areas are contained in *ERSA GEN* under the heading SPECIAL PROCEDURES (NOT ASSOCIATED WITH AN AERODROME).

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ENR 6. EN ROUTE CHARTS

ENR 6.1 AUSTRALIAN AIP AERONAUTICAL CHARTS

1. Details on all aeronautical charts produced in Australia are contained in *GEN 3.2.*

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PART 3 - AERODROMES (AD)

AD 0.

AD 0.1 TABLE OF CONTENTS

AD 1. AERODROMES/HELIPORTS - INTRODUCTION ..... AD 1.1 - 1
AD 1.1 AERODROMES/HELIPORTS AVAILABILITY AND CONDITIONS OF USE ..... AD 1.1 - 1
1. GENERAL CONDITIONS ..... AD 1.1 - 1
2. CIVIL USE OF MILITARY AIR BASES ..... AD 1.1 - 1
3. LOW VISIBILITY OPERATIONS ..... AD 1.1 - 2
3.1 Approval required ..... AD 1.1 - 2
3.2 Low Visibility Procedures ..... AD 1.1 - 2
3.3 Protection of ILS critical and sensitive areas ..... AD 1.1 - 2
3.4 Radio Altimeter Requirements ..... AD 1.1 - 4
4. TEMPORARY REDUCTIONS IN DECLARED RUNWAY DISTANCES ..... AD 1.1 - 4
5. VISUAL DOCKING GUIDANCE SYSTEMS ..... AD 1.1 - 4
5.3 Nose in Guidance (NIG) System ..... AD 1.1 - 5
5.4 Aircraft Positioning and Information System (APIS) ..... AD 1.1 - 6
5.5 Safegate Docking Guidance System (DGS) ..... AD 1.1 - 8
AD 1.2 RESCUE AND FIRE FIGHTING SERVICES AND SNOW PLAN ..... AD 1.2 - 1
1. RESCUE AND FIRE FIGHTING SERVICES ..... AD 1.2 - 1
1.1 General ..... AD 1.2 - 1
1.2 Levels of Emergency Response at Aerodromes ..... AD 1.2 - 1
1.3 RFFS Emergency Frequency ..... AD 1.2 - 2
2. SNOW PLAN ..... AD 1.2 - 2
3. RUNWAY SURFACE CONDITION ASSESSMENT AND REPORTING ..... AD 1.2 - 2
3.2 Assessment of Runway Surface Condition ..... AD 1.2 - 3
3.3 Reporting of Runway Surface Condition ..... AD 1.2 - 5
3.4 Pilot reports about braking action ..... AD 1.2 - 6
3.5 Communicating RCR ..... AD 1.2 - 6
3.6 Surface condition assessments for unpaved runways ..... AD 1.2 - 7
AD 1.3 INDEX TO AERODROMES AND HELIPORTS ..... AD 1.3 - 1
AD 1.4 GROUPING OF AERODROMES/HELIPORTS ..... AD 1.4 - 1
AD 1.5 STATUS OF CERTIFICATION OF AERODROMES ..... AD 1.5 - 1
AD 2. AERODROMES ..... AD 2. - 1
AD 3. HELIPORTS ..... AD 3. - 1

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AD 1. AERODROMES/HELIPORTS - INTRODUCTION

AD 1.1 AERODROMES/HELIPORTS AVAILABILITY AND CONDITIONS OF USE

1. GENERAL CONDITIONS

1.1 Information concerning aerodromes within Australia is published by CASA in the *Manual of Standards Part 139 – Aerodromes* and associated guidance material available on CASA website: www.casa.gov.au/search-centre/rules/part-139-casr-aerodromes

1.2 Aerodrome requirements for *Part 121* operations require that an aerodrome must be suitable for the aeroplane to take-off and land, and either be a certified aerodrome that complies with the standards mentioned in the *Part 139 Manual of Standards* or an aerodrome for which the operator's exposition includes procedures to determine the matters required by *CASR 121.210*.

1.3 The aerodrome directory is published as *En Route Supplement Australia (ERSA) Facilities (FAC)* which lists certified aerodromes along with their availability and conditions of use.

1.4 Details of uncertified aerodromes and helicopter landing sites are also included in *ERSA FAC* when requested by the aerodrome operator. It is the responsibility of pilots and air operators intending to conduct operations at uncertified aerodromes or helicopter landing sites to obtain current information on that location from the aerodrome owner/operator prior to flight planning and to assess the suitability of the aerodrome/heliport for the intended operation.

2. CIVIL USE OF MILITARY AIR BASES

2.1 Civil aircraft may only operate at a military aerodrome (labelled as MIL in the *AIP-ERSA*) or nominate a military aerodrome as an alternate with prior permission issued by the relevant Military Authority. Any conditions imposed on the operation of the aircraft at the military aerodrome by the relevant Military Authority, whether conditions or instructions specified in the relevant *ERSA* entry for the military aerodrome, or in AIP SUP and NOTAM for temporary military activities, or instructions to the specific operator or pilot, must be followed by both the operator and pilot in command of the aircraft unless those requirements are contrary to the civil aviation legislation.

2.2 When conducting operations at a joint military/civilian aerodrome (labelled as JOINT in the *AIP-ERSA*), the operator and the pilot in command of the aircraft must each comply with the applicable requirements for the aerodrome contained in the AIP unless those requirements are contrary to the civil aviation legislation.

3. LOW VISIBILITY OPERATIONS

3.1 Approval required

3.1.1 Aircraft operators may conduct a low visibility operation (LVO) only if specifically approved by CASA. Approvals for LVO are granted in the form of an exemption to the standard IFR take-off and approach minima.

3.1.2 *AC 91-11* contains specific information and guidance on gaining approval to conduct LVO operations.

3.2 Low Visibility Procedures

3.2.1 Low Visibility Procedures (LVPs) are applied at controlled aerodromes for protecting aircraft operations during conditions of reduced visibility or low cloud. LVPs are initiated when the visibility on an aerodrome becomes insufficient for ATC to control aerodrome traffic by visual surveillance. Low visibility protection measures are progressively implemented at the weather deteriorates.

3.2.2 Pilots will be notified that low visibility procedures are in force by ATIS broadcast or directed transmissions.

3.3 Protection of ILS critical and sensitive areas

3.3.1 ATC provides differing levels of protection for ILS critical or sensitive areas depending on the type of approach or departure, the position of an approaching aircraft, and the prevailing weather conditions at the time the approach is commences. The different scenarios and protection levels are described in the following table:

Type of operation	Weather Conditions	Level of ILS critical or sensitive area protection
Any ILS approach	– Cloud ceiling > 600FT or – Visibility > 2,000M	ILS critical and sensitive areas **are not protected.**
Any ILS approach	– Cloud ceiling ≤ 600FT, but ≥ CAT I minimums; or – Visibility ≤ 2,000M but ≥ CAT I minimums.	ILS critical area is protected except: – When a preceding aircraft enters the critical area while landing or vacating; or – For the time prior to the approaching aircraft passing the ILS OM, or, if no OM, is within 4NM of threshold. Sensitive area is not protected.
SA CAT I, SA CAT II, CAT II and CAT III	Cloud ceiling or visibility < CAT I minimums.	– ILS critical area is protected when the arriving aircraft has passed the OM, or if no OM, is within 4NM of touchdown. – ILS sensitive area is protected when the aircraft is within 2NM of touchdown.
Localiser guided take-off	Visibility < 550M	– Localiser critical and sensitive area protected until the aircraft has completed its take-off.

3.3.2 **Caution:** Pilots may experience the ILS beam bends and other interference in the circumstances where ATC is not protecting the ILS critical area or ILS sensitive area.

3.3.3 In weather conditions where the ceiling and/or visibility are above CAT I minima, pilots should inform ATC about any intention to conduct:

a. an approach with minima less than standard CAT I; or

b. an autoland procedure.

This information must not be taken as a request for or expectation of the protection of the ILS but to enable ATC to inform the flight crew of any known or anticipated disturbance.

3.3.4 ATC will inform pilots that the relevant protection measures are in place by reporting ‘LVP in force’ on the ATIS or by direct advice.

3.3.5 At start-up, pilots should inform ATC about any intention to conduct a guided take-off that requires guidance provided by an ILS localiser.

Note: Localiser guided take-offs are generally supported on runways which have published CAT III approaches.

3.4 **Radio Altimeter Requirements**

3.4.1 Radio altimeters fitted to aircraft used for the conduct of low-visibility approaches need to comply with relevant FAA airworthiness directives unless an exemption applies. The relevant aircraft equipment requirements are found in *GEN 1.5 Section 6.*

4. TEMPORARY REDUCTIONS IN DECLARED RUNWAY DISTANCES

4.1 Declared runway distances may be varied during aerodrome works, either planned or emergency, or during temporary obstructions to the aerodrome Obstacle Limitation Surface (OLS). Operators of certified aerodromes will promulgate any variation to declared distances via NOTAM.

4.2 At Controlled aerodromes when reduced runway distances are in operation for the runway in use, ATC will:

a. Provide advice via ATIS: 'REDUCED RUNWAY LENGTH(S) IN OPERATION [RUNWAY (*number*)], SEE NOTAM (*number*)';

b. Provide directed advice with the last taxi clearance to the holding point for the departure runway if a reduced length is in operation from the departure holding point; and

c. Provide directed advice on tower frequency prior to the landing clearance being issued.

Note: Except at Class D aerodromes, the runway number will be included in directed advice where more than one runway is being nominated for use.

5. VISUAL DOCKING GUIDANCE SYSTEMS

5.1 Visual Docking Guidance Systems (VDGS) used in Australia include:

– The generic Nose in Guidance (NIG) system

– Aircraft Positioning and Information System (APIS)

– Safegate Docking Guidance System

5.2 An aerodrome's DAP Apron Chart specifies the bays/stands equipped with VDGS and the particular system installed.

5.3 **Nose in Guidance (NIG) System**

5.3.1 This system is identified on DAP charts either as 'NIG' or 'Centreline+Sidemarker'. It includes the following elements:

- Position Identification Light
- Aerobridge Retracted Indicator
- Centreline Guidance Light unit
- One or more Side Marker Light units.

5.3.2 The following is a brief description of the system:

a. The Position Identification Light indicates the number of the docking position and is white numerals on a dark background (illuminated at night).

b. The Aerobridge Retracted Indicator consists of two lights. The green light indicates the Aerobridge is in the fully retracted position. The red light indicates that the Aerobridge is not fully retracted or that an element of the visual guidance docking system is unserviceable.

c. The Centreline Guidance Light provides azimuth information and is aligned with the left pilot position. The unit emits RED/GREEN light beams and the signals are interpreted as shown in *Figure 1*.



Figure 1
Centreline Guidance Light Unit

d. One or more Side Marker Light units – with relevant aircraft types marked on the unit – indicate the stopping position as described below:

(1) Approaching the position, a preliminary dull GREEN light will show through the arrow-shaped aperture which also exhibits a cross bar.

(2) As the aircraft moves forward, the intensity of the green light increases until it becomes a bright arrow-head.

(3) As the aircraft continues, the arrow-head starts to reduce in size.

(4) When the arrow-head disappears, two white bars appear, one above the other, indicating the stopping position. In some installations, two sets of bars will appear.

(5) If the stopping position is passed, then a single RED bar appears.



Figure 2
Side Marker Lights

5.4 **Aircraft Positioning and Information System (APIS)**

5.4.1 APIS is based on a centreline guidance sub-display. The steering and stop indication is provided from a display unit mounted on a pole in front of the cockpit in line with the left hand pilot seat. The parking bay position identification is mounted on top of the guidance pole.

5.4.2 On approach to the parking position, the pilot will see the display box face showing two rows of yellow alpha-numeric characters on a black background across the top, an illuminated closing-rate 'thermometer' at lower left, and an illuminated azimuth guidance display at lower right. The alpha-numeric characters on the top row should be flashing (see diagram at *Figure 3*).

5.4.3 The following is the sequence of APIS operation from initial approach to STOP:

a. Identify the correct parking bay position.

b. Ensure that the aerobridge retraction light indicates green.

c. Follow the taxi-in line and watch the centreline beacon.

d. Check that the correct aircraft type is flashing and that the door number is shown (where applicable).

e. About 20M before STOP, the aircraft type display goes steady and the door number disappears.

f. Follow the azimuth guidance display. The black arrow heads indicate which direction to steer for the centreline. When the aircraft is properly aligned in azimuth, the black vertical bar will be displayed.

g. The full closing rate 'thermometer' indicates at least 13M to STOP.

h. When the aircraft reaches 13M to STOP, the 'thermometer' bar lights begin to move from bottom to top.

i. The deletion of each 'thermometer' bar indicates about one half metre progression.

j. When the STOP position is reached, all the closing rate 'thermometer' lights extinguish and the lower display indicates STOP. If the aircraft is parked correctly, the top display indicates OK.

k. If the aircraft overshoots the limit for correct parking, the top display indicates TOO FAR (alternating TOO then FAR).

l. The entire display automatically shuts down after some seconds.

Note: When the last row of lights of the closing rate 'thermometer' is extinguished and the word STOP is displayed, the aircraft should be at a standstill.

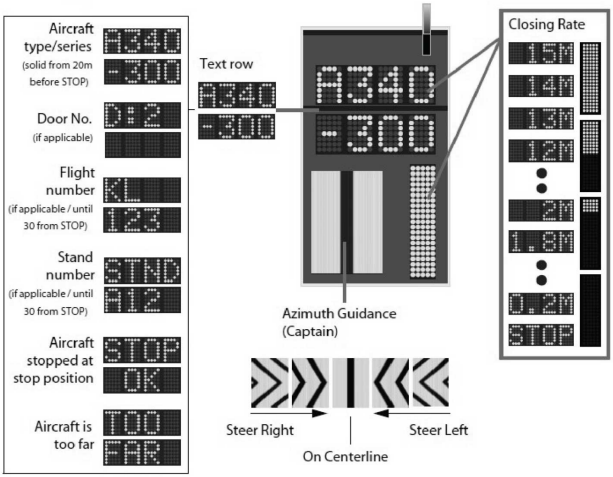


Figure 3
APIS++ VDGS – typical configuration

Note: Some APIS++ installations have a single row of text information.

5.5 Safegate Docking Guidance System (DGS)

5.5.1 The Safegate Docking Guidance System (DGS) is the most common VDGS in use in Australia. The complete system consists of the following three elements:

a. Position Identification Unit (Bay Marker)

b. Aerobridge Retracted Indicator Light, and

c. DGS NIG Unit.

5.5.2 The Position Identification Unit gives clear indication of the parking bay for the aircraft. It consists of large white numerals on a dark background (illuminated at night).

5.5.3 The Aerobridge Retraction Indicator Light, mounted on the aerobridge, gives an early warning of the state of aerobridge location. Green indicates a fully retracted aerobridge position or a safe pre-parked position; red indicates that the aerobridge is out of position and the pilot should not proceed with parking the aircraft.

5.5.4 The NIG unit, mounted on the Terminal wall, consists of two components which supply the following information to the pilot:

a. The top alphanumeric information display which shows aircraft type designation, and other message information as necessary in yellow.

b. The azimuth and centreline guidance displays in red and yellow and the Closing Rate Bar in yellow.

5.5.5 The following is the sequence of system operation from initial approach to STOP:

a. The pilot identifies the correct parking bay position.

b. The pilot ensures that the aerobridge retraction light is green.

c. The pilot observes that the rising vertical yellow arrows are indicating the system is activated and searching for the approaching aircraft.

Note: The pilot must not enter the stand area unless the rising vertical arrows are displayed.

d. The pilot follows the taxi-in line and checks that the correct aircraft type is displayed in yellow.

Note: The pilot must not enter the stand area unless the correct aircraft type is displayed.

e. On successful capture of the aircraft, the vertical arrows are replaced by the yellow T-shaped Closing Rate Bar.

Note: The pilot must not proceed to the bridge unless the arrows have been superseded by the Closing Rate Bar.

f. A vertical yellow arrow shows the aircraft position in relation to the centreline.

g. A flashing red arrow indicates the direction to turn to return to the centreline.

Note: If the aircraft is approaching faster than the accepted speed, the system will show SLOW DOWN as a warning.

h. The display of the yellow digital closing rate countdown will start when the aircraft is 20 metres from the STOP position.

Note: If the detected aircraft is lost prior to 12 metres to STOP, the display will show WAIT. The docking will continue as soon as the system detects the aircraft again.

i. When the aircraft is 12 metres from the STOP position, the Closing Rate Bar will decrease in size from the bottom by one row of lights per 0.5 metres closing rate.

Note: If the detected aircraft is lost after 12 metres to STOP, the display will show STOP and ID FAIL. Assistance must then be sought from the ground engineers.

j. When the correct STOP position is reached, the display shows STOP and red lights will be lit.

k. When the aircraft has parked, OK will be displayed.

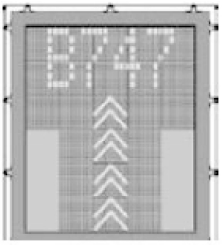
l. If the aircraft has overshot the position, TOO FAR will be displayed.

m. When ground engineers have placed the chocks at the nosewheel, they will manually change the display to CHOCK ON.

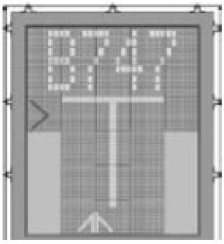
n. During heavy rain or fog, the visibility for the docking system might be reduced. When the system is activated and in capture mode, the display will deactivate the rising vertical arrows and show DOWN GRADE. This text will be superseded by the Closing Rate Bar once the aircraft is detected.

Note 1: The pilot must not continue the approach to the bridge unless the DOWN GRADE text has been superseded by the Closing Rate Bar.

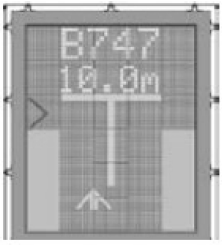
Note 2: Ground engineers have access to emergency push-buttons to deactivate the system. When an emergency stop is activated, the display will show STOP. The ground engineers will then be required to complete the docking manually once the emergency situation is cleared.



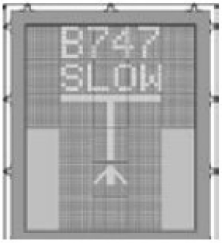
CAPTURE
Searching for aircraft



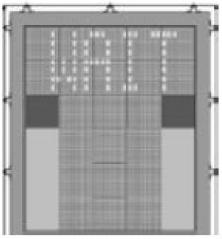
TRACKING AIRCRAFT
Aircraft left of centreline



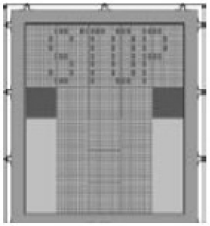
CLOSING RATE



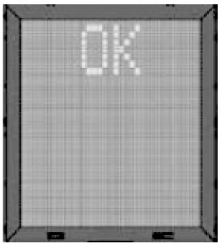
SLOW (DECREASE SPEED)



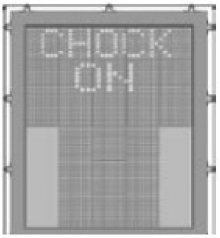
WAIT



STOP POSITION REACHED



DOCKING COMPLETE



CHOCKS ON

Figure 4
Typical Safegate indications – normal operations

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AD 1.2 RESCUE AND FIRE FIGHTING SERVICES AND SNOW PLAN

1. RESCUE AND FIRE FIGHTING SERVICES

1.1 General

1.1.1 Subpart 139.H (Aerodrome rescue and firefighting services) of *CASR 1998* and *Part 139H Manual of Standards*, which are available via the CASA website, prescribe the standards for RFFS in Australia. RFFS categories are based on the rescue and firefighting standards contained in *ICAO Annex 14, Vol I, Aerodrome Design and Operations, Chapter 9*.

1.1.2 RFFS is provided at certain aerodromes according to the type of aircraft normally using that aerodrome and the frequency of aircraft operations. Details of the RFFS available at Australian aerodromes, including hours of operation, can be found in the ERSA.

1.2 Levels of Emergency Response at Aerodromes

1.2.1 This section summarises the levels of emergency response that are normally provided under an AEP.

1.2.2 **Local Standby**

1.2.2.1 A Local Standby involves the activation of the airport-based responding agencies. Outside services do not normally respond for a Local Standby. At an aerodrome without RFFS facilities, the response (if any) to a declaration of a Local Standby will be set out in the AEP.

1.2.2.2 A Local Standby will be the normal response when an aircraft approaching an airport is known or is suspected to have developed some defect or operational issue, but the trouble is not such as would normally involve any serious difficulty making a safe landing.

1.2.2.3 Unless the pilot requests otherwise, at least a Local Standby will be initiated in response to a PAN call.

1.2.3 **Full Emergency**

1.2.3.1 A Full Emergency involves the activation of both airport and off-airport responding agencies.

1.2.3.2 A Full Emergency will be the normal response when an aircraft approaching the airport is known or suspected to be in such trouble that there is danger of an accident.

1.2.3.3 Unless the pilot requests otherwise, a Full Emergency will be initiated in response to a MAYDAY call.

1.2.3.4 A crash on the airport will initiate a Full Emergency response.

1.2.4 **Aerodromes where no RFFS is Provided/in Operation**

1.2.4.1 For an aerodrome where there is no on-airport fire service or when the airport fire service has been stood down, ATS will ask the pilot to advise services required on arrival. ATS will then activate the AEP at the appropriate level.

1.3 RFFS Emergency Frequency

1.3.1 RFFS emergency frequency 131.0MHz is available for direct communication between the Fire Commander and affected flight crew during an airport emergency at selected airports (identified in ERSA FAC under Rescue and Firefighting Services). ATC must be advised of an intention to operate on the frequency.

1.3.2 All communication with ATC during an airport emergency must be conducted on the normal ATC frequencies.

2. SNOW PLAN

2.1 Only a small number of aerodromes and heliports within Australia are normally subject to snow conditions. Consequently there is no national snow plan, or any specific regulation for the clearance of snow and ice from runways and manoeuvring areas.

2.2 For aerodromes and heliports likely to be affected, pilots and operators are advised to contact the aerodrome operator directly for information about the relevant aerodrome snow plan.

3. RUNWAY SURFACE CONDITION ASSESSMENT AND REPORTING

3.1 This section covers the methodology for assessing and reporting runway surface conditions in accordance with the ICAO Global Reporting Format (GRF). Aeroplane operators should utilise the information in conjunction with the performance data provided by the aircraft manufacturer to determine if landing or take-off operations can be conducted safely, or if additional landing or take-off distances are required. Further information on the GRF as well as the assessment and reporting of runway surface conditions can be found in: www.casa.gov.au/sites/default/files/2024-02/multi-part-advisory-circular-91-32-and-139-22-global-reporting-format-runway-surface-condition.pdf.

3.2 Assessment of Runway Surface Condition

3.2.1 GRF is applicable at all certified aerodromes and only to sealed runways. Runway surface conditions are assessed and reported whenever water or contaminants are present on an operational runway. A Runway Condition Report (RCR) is disseminated when there are significant changes in the runway surface condition.

3.2.2 The RCR contains a Runway Condition Code (RWYCC) and information that describes the runway surface condition, i.e. type of surface contaminants, depth and coverage for each runway third.

3.2.3 The predominant runway surface conditions in Australia are due to water contamination of a runway i.e. ‘wet’ or ‘slippery wet’ runways or runways with ‘standing water’. Guidance for winter runway surface conditions is provided in:

www.casa.gov.au/sites/default/files/2024-02/multi-part-advisory-circular-91-32-and-139-22-global-reporting-format-runway-surface-condition.pdf.

3.2.4 The initial RWYCC is assigned based on the runway surface description:

Runway surface description	Applicable RWYCC
• DRY	6
• WET (The runway surface is covered by any visible dampness or water up to and including 3mm depth)	5
• WET (“slippery wet” runway)	3
• STANDING WATER (depth of more than 3mm)	2

3.2.5 The surface friction characteristics of a runway, or a portion of it, can become degraded due to rubber deposits (e.g. in the touchdown zone), surface polishing, poor drainage or other factors. The determination that a runway is 'slippery wet' stems from various methods used solely or in combination. These methods may include functional friction measurements or using a continuous friction measuring device which are available to the aerodrome operator. Other ways for the aerodrome operator to become aware that a runway is 'slippery wet' is by receiving two or more consecutive pilot reports or relayed reports from ATC of a reduced braking action for a 'wet' runway that is 'MEDIUM' instead of 'GOOD'. To alert pilots to the particular hazards associated with slippery wet runways, the surface description of ‘SLIPPERY WET’ is used instead of ‘WET’.

3.2.6 After the initial assignment of a RWYCC, an aerodrome operator can downgrade a RWYCC based on 2 or more consecutive pilot reports of a braking action less than that allocated for the RWYCC. The pilot reports can be provided by ATC to the aerodrome operator or directly from the pilot. The correlation of RWYCC with braking action is as follows:

Pilot report of runway braking action	Description	RWYCC
Not applicable		6
GOOD	Braking deceleration is normal for the wheel braking effort applied AND directional control is normal	5
GOOD TO MEDIUM	Braking deceleration OR directional control is between good and medium	4
MEDIUM	Braking deceleration is noticeably reduced for the wheel braking effort applied OR directional control is noticeably reduced	3
MEDIUM TO POOR	Braking deceleration OR directional control is between medium and poor	2
POOR	Braking deceleration is significantly reduced for the wheel braking effort applied OR directional control is significantly reduced	1

Pilot report of runway braking action	Description	RWYCC
LESS THAN POOR	Braking deceleration is minimal to non-existent for the wheel braking effort applied OR directional control is uncertain	0

3.3 Reporting of Runway Surface Condition

3.3.1 The RWYCC is reported for each third of the runway assessed. The RCR includes:

a. Aerodrome location indicator

b. Date and time of assessment

c. Runway designator:

(i) at non-controlled aerodromes, the runway designator promulgated by NOTAM will be the lower runway designation number

(ii) at controlled aerodromes, the runway in use as promulgated by ATIS

d. RWYCC for each runway third. The direction for listing the runway thirds shall be in the direction as seen from the runway designation number

e. Percentage coverage of each runway third for slippery wet runways

f. Contaminant depth, if available

g. Surface description for each runway third.

Note 1: The runway designator is prefixed by 'RWY' to indicate to pilots that the RCR or NOTAM is critical to safety and to aid NOTAM filtering, where available.

Note 2: If 25% or less of a runway third has standing water or is otherwise contaminated it is to be assigned a RWYCC of 5 and a surface description of WET.

Note 3: NOTAM are not issued for WET (RWYCC = 5) or DRY (RWYCC = 6) runways.

3.4 Pilot reports about braking action

3.4.1 If the braking action experienced on a runway at a controlled aerodrome is not as good as that reported, pilots are required to submit an AIREP SPECIAL. If the braking action experienced on a runway at a non-controlled aerodrome is different to that reported, pilots should inform ATC directly or submit an AIREP SPECIAL. Pilots may also be asked by the ATC to report their assessment of the braking action. Pilot reports directly to ATC should use the terms listed in *para 3.2.6* or the AIREP template in *ENR 1.1 APPENDIX 1*. At non-controlled certified aerodromes and controlled aerodromes where ATC is not present (i.e. not a 24/7 tower) pilots are required to report braking action, not as good as that reported, directly to the aerodrome operator. Contact details for aerodrome operators are in *AIP-ERSA*.

3.5 Communicating RCR

3.5.1 ATIS

3.5.1.1 ATIS is the primary means for communicating runway surface conditions at controlled aerodromes. The RCR will describe conditions for each runway third in the direction of landing/take-off.

3.5.2 Air-ground voice communications

3.5.2.1 ATC will only provide RCR information through voice communications to inform about changes to the information provided via ATIS, or when specifically requested by a pilot. Normally, only the change in RWYCC or surface description will be communicated.

3.5.3 NOTAM

3.5.3.1 Australia does not issue SNOWTAM. Instead, a NOTAM similar in content to a SNOWTAM may be issued for a runway that is contaminated and expected to remain that way for some time. The RCR content will be in Item E) of the NOTAM.

3.5.4 FIS

3.5.4.1 When a RCR NOTAM is issued for a non-controlled aerodrome, or when ATC is not present at a controlled aerodrome e.g. not a 24/7 tower, ATC (Area Controllers) will provide a Flight Information Service (FIS) and relay the NOTAM to affected pilots.

3.5.5 **UNICOM and CA/GRS**

3.5.6 Where an aerodrome has an established UNICOM or CA/GRS which is operating during the period a RCR NOTAM is issued, it will be relayed to pilots on the appropriate frequency.

3.6 Surface condition assessments for unpaved runways

3.6.1 The GRF runway surface condition assessment and reporting format does not apply to unpaved runways and no RCR will be issued.

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AD 1.3 INDEX TO AERODROMES AND HELIPORTS

1. A complete list of codes for aerodromes and helicopter landing sites in Australia is available in *ERSA GEN*.

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AD 1.4 GROUPING OF AERODROMES/HELIPORTS

1. Australian aerodromes and heliports are not grouped.

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AD 1.5 STATUS OF CERTIFICATION OF AERODROMES

1. The status of certification of aerodromes is available at the following website:
https://www.casa.gov.au/search-centre/aerodromes

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AD 2. AERODROMES

1. Details of aerodromes in Australia and its territories are contained in *ERSA FAC*.

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AD 3. HELIPORTS

1. Details of domestic heliports in Australia and its territories are contained in *ERSA FAC*.

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